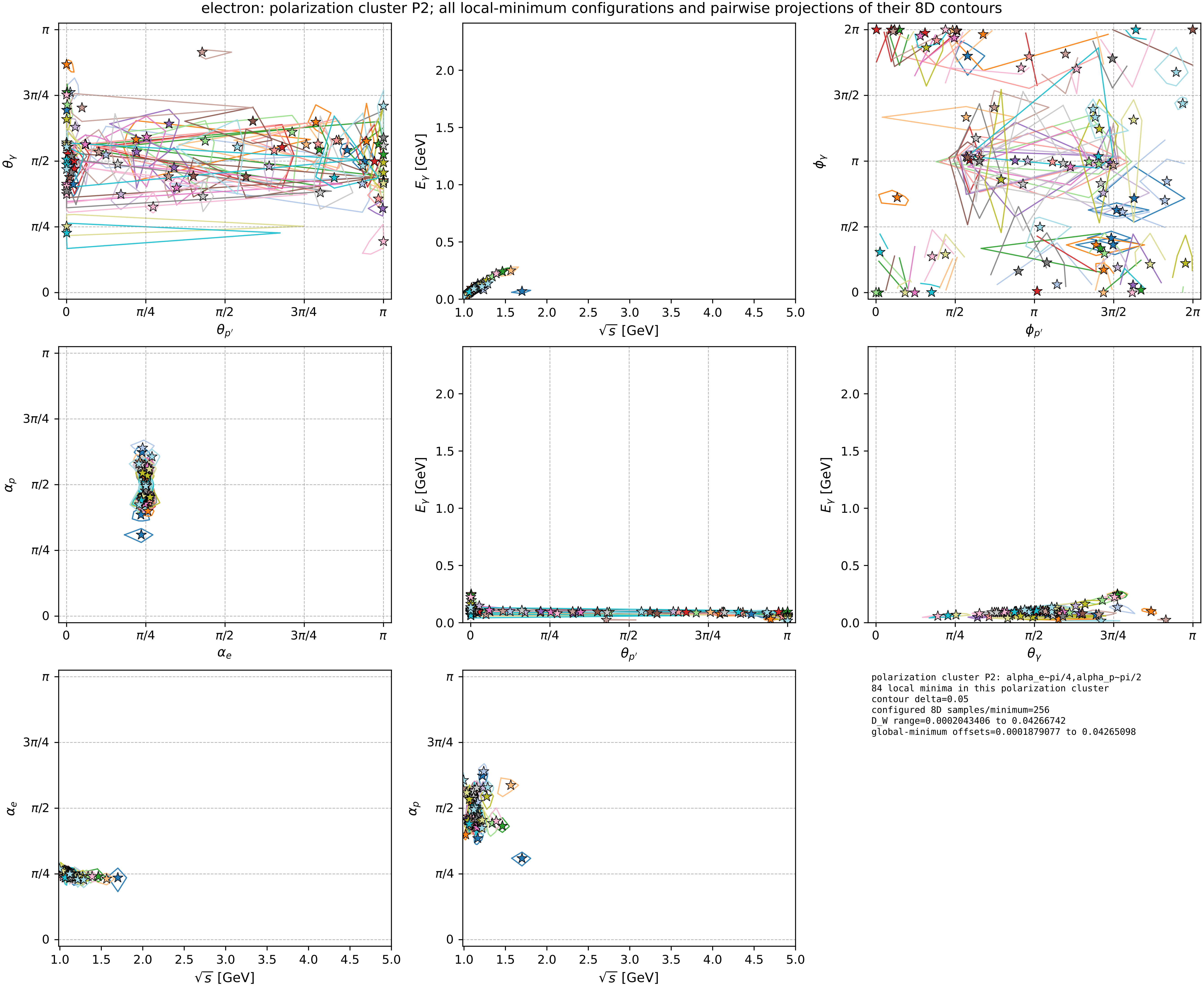
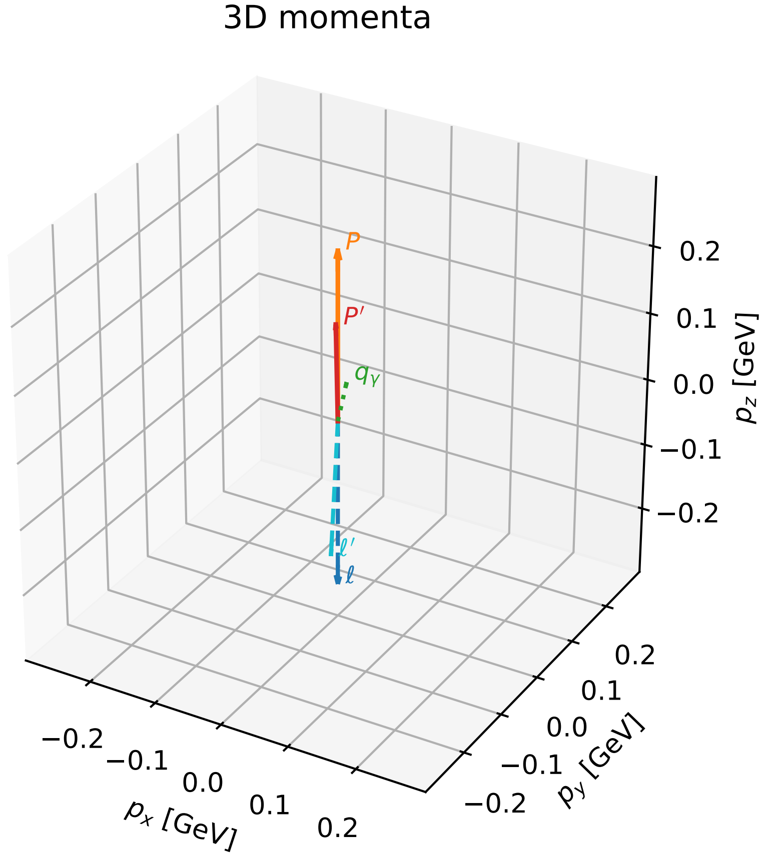

electron: polarization cluster P2; all local-minimum configurations and pairwise projections of their 8D contours
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
E_γ [GeV]
$\theta_{p'}$
E_γ [GeV]
θ_γ
α_e
$\sqrt{s}$ [GeV]
α_p
$\sqrt{s}$ [GeV]
polarization cluster P2: alpha_e~pi/4,alpha_p~pi/2
84 local minima in this polarization cluster
contour delta=0.05
configured 8D samples/minimum=256
D_W range=0.0002043406 to 0.04266742
global-minimum offsets=0.0001879077 to 0.04265098

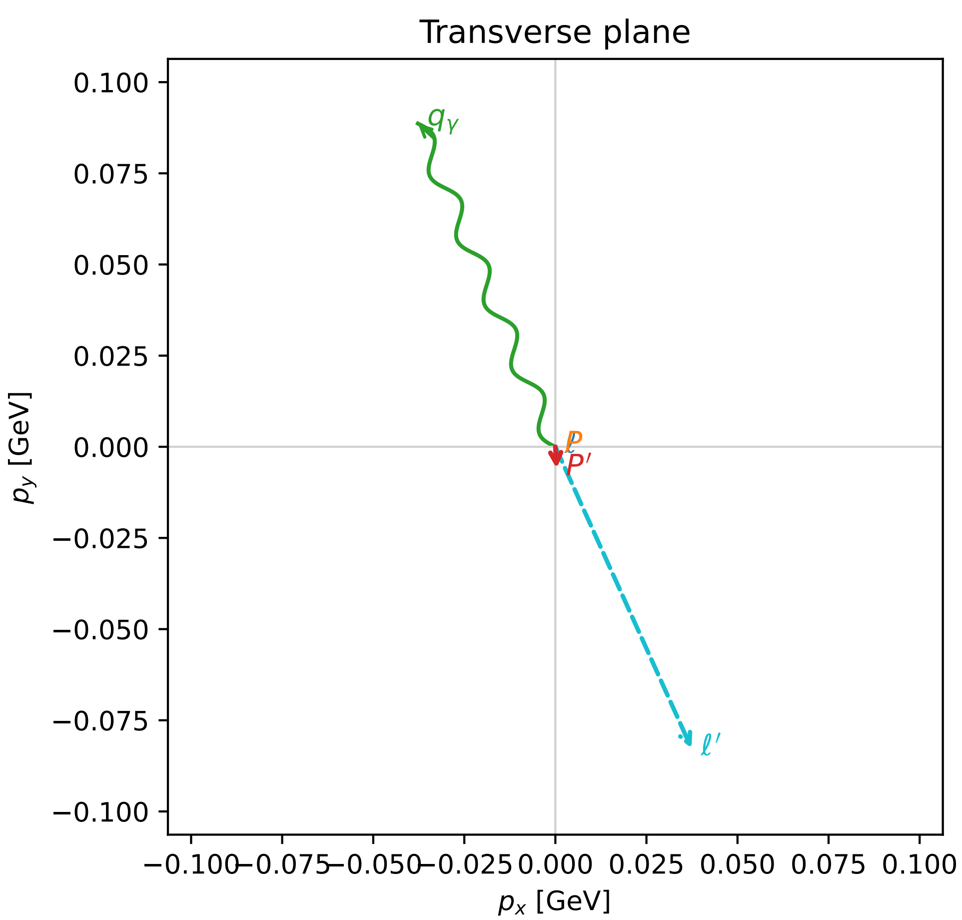
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

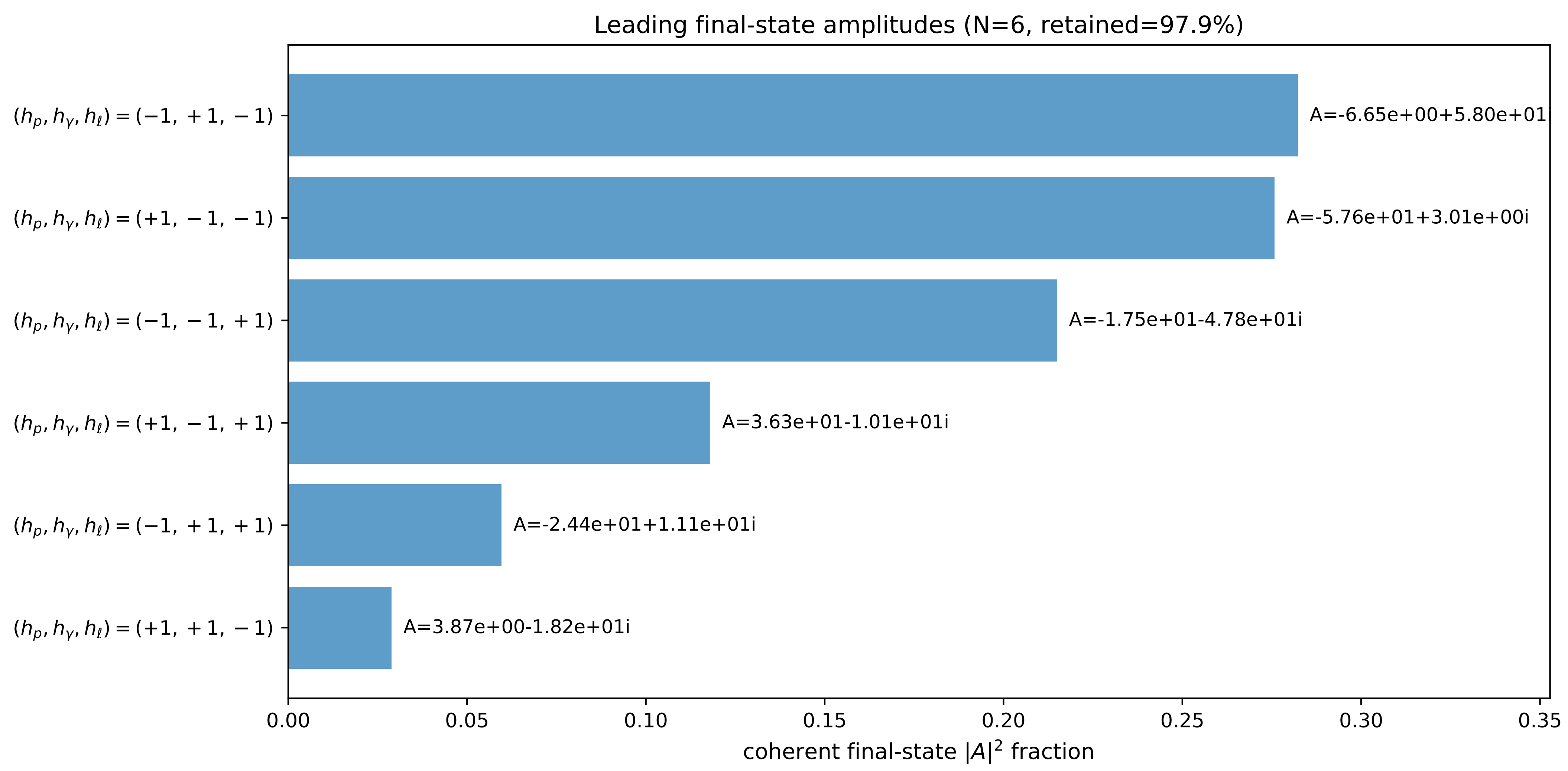


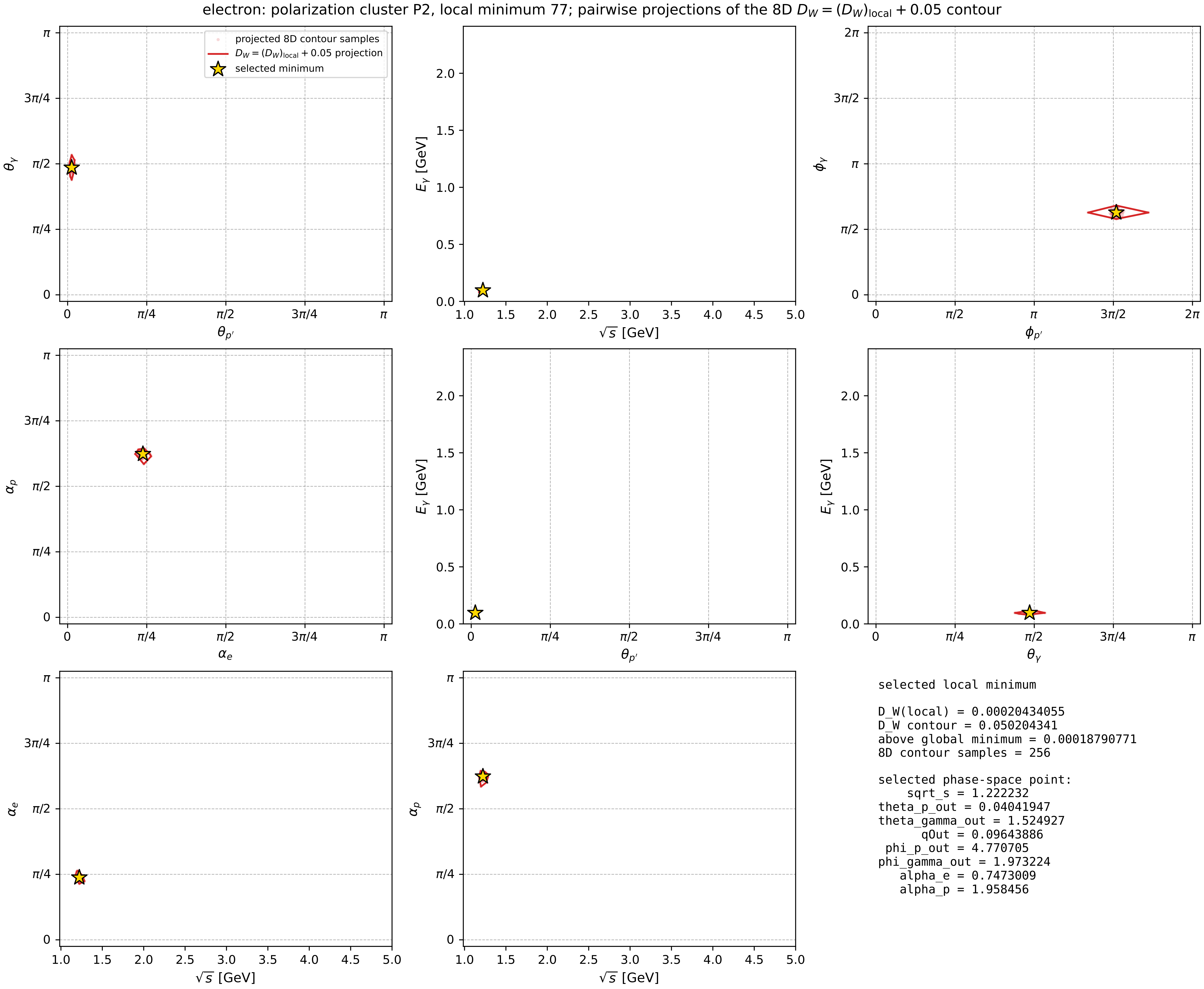

```
dw_mixing_angles_polarization_02_local_minimum_77  D_W=0.000204341
region: polarization_cluster_02_local_minimum_77
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7473009 rad
incoming lepton: 0.733526|+> +0.679661|->
proton mixing angle: alpha_p=1.9584559 rad
incoming proton: -0.378023|+> +0.925796|->
```

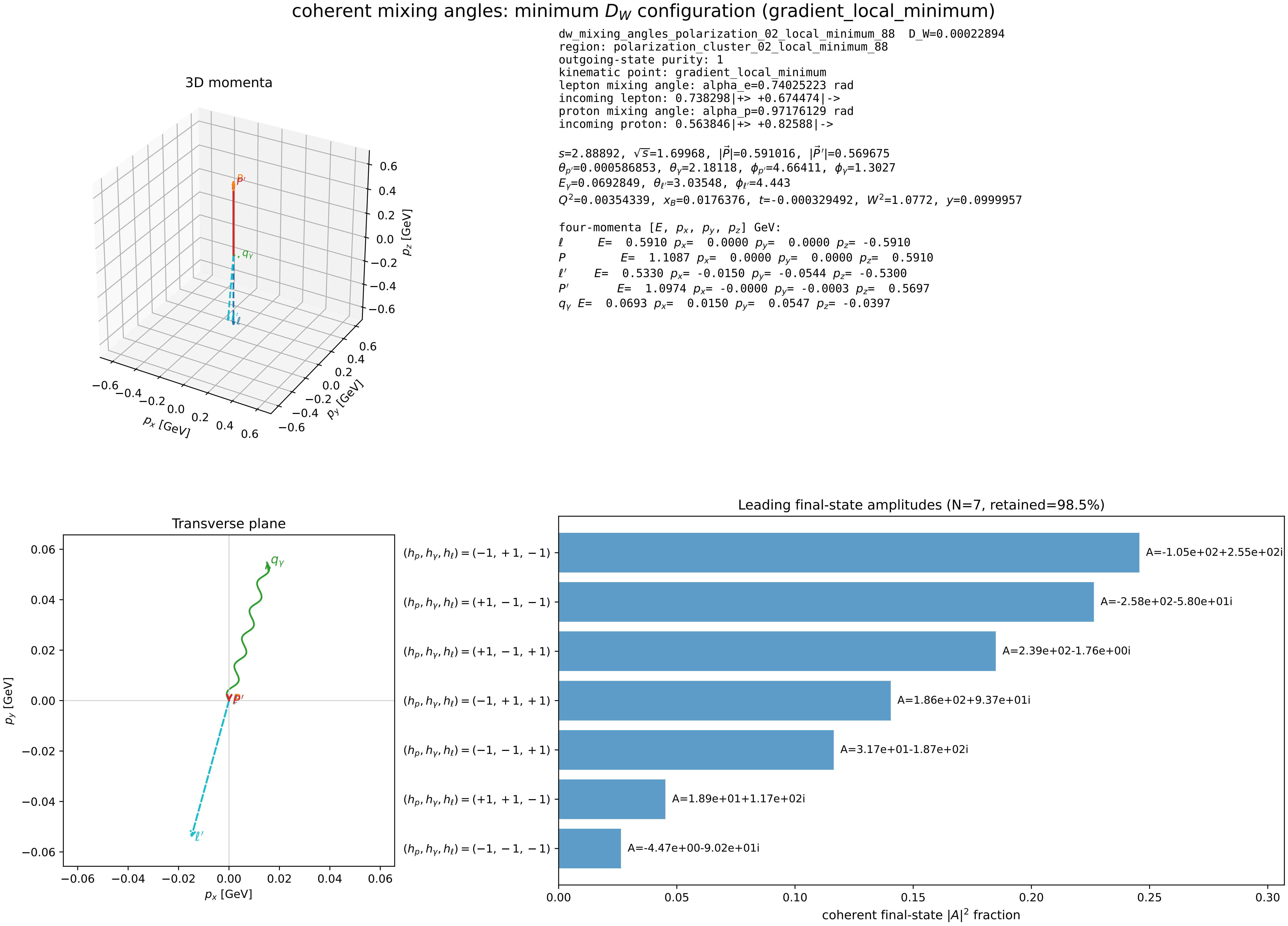
s=1.49385, $\sqrt{s}$=1.22223, $|\vec{P}|$=0.251182, $|\vec{P}'|$=0.146903
$\theta_{p'}$=0.0404195, θ_γ=1.52493, $\phi_{p'}$=4.7707, ϕ_γ=1.97322
E_γ=0.0964389, $\theta_{\ell'}$=2.60094, $\phi_{\ell'}$=5.13688
Q^2=0.0126364, x_B=0.0646231, t=-0.0104672, W^2=1.06275, y=0.318465

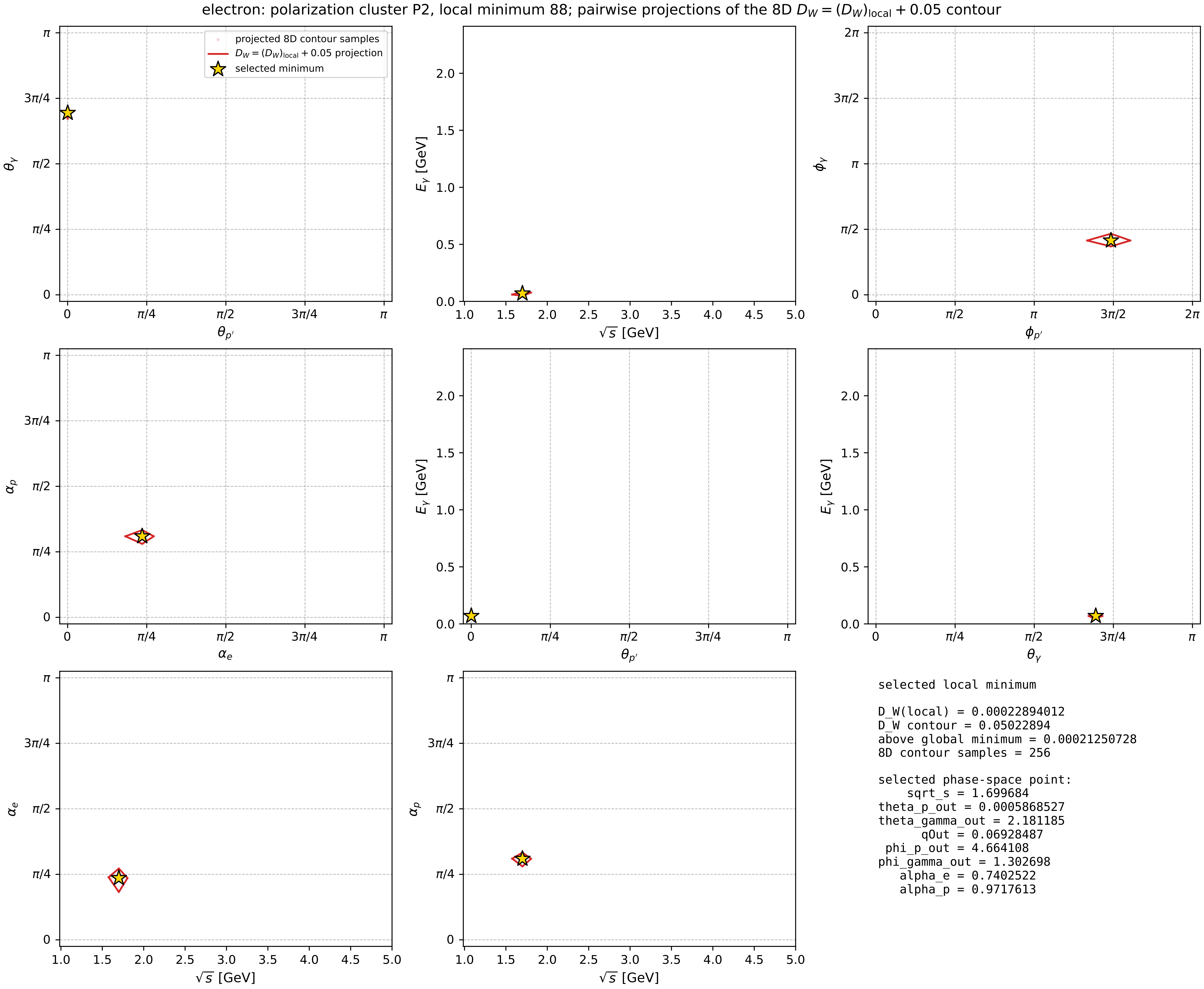
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2512 p_x= 0.0000 p_y= 0.0000 p_z= -0.2512
P E= 0.9710 p_x= 0.0000 p_y= 0.0000 p_z= 0.2512
ℓ' E= 0.1764 p_x= 0.0374 p_y= -0.0827 p_z= -0.1512
P' E= 0.9494 p_x= 0.0003 p_y= -0.0059 p_z= 0.1468
q_γ E= 0.0964 p_x= -0.0377 p_y= 0.0886 p_z= 0.0044



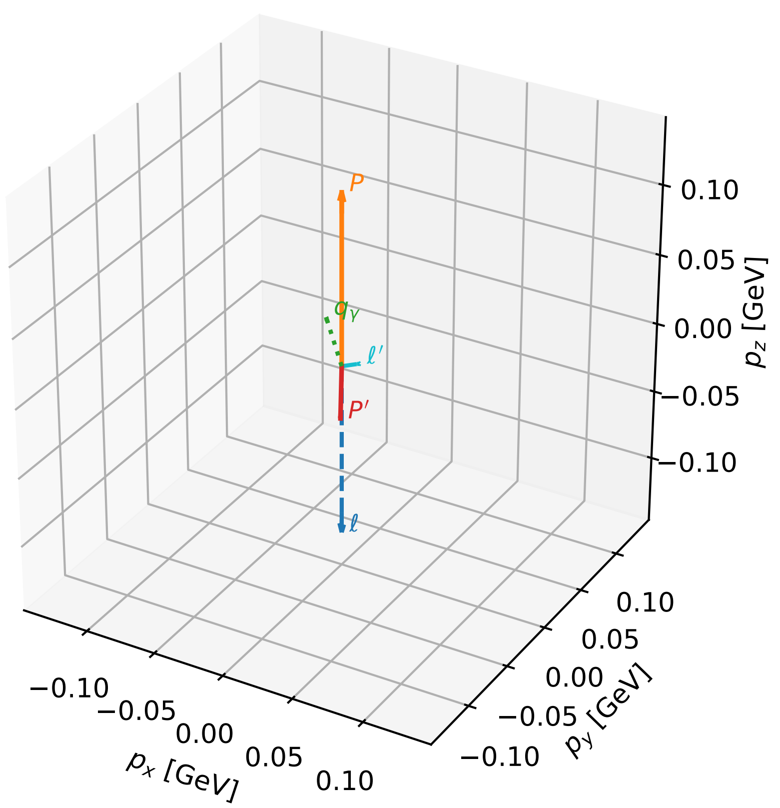



electron: polarization cluster P2, local minimum 77; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.00020434055
D_W contour = 0.050204341
above global minimum = 0.00018790771
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.222232
theta_p_out = 0.04041947
theta_gamma_out = 1.524927
qOut = 0.09643886
phi_p_out = 4.770705
phi_gamma_out = 1.973224
alpha_e = 0.7473009
alpha_p = 1.958456


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_88 D_W=0.00022894
region: polarization_cluster_02_local_minimum_88
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.74025223 rad
incoming lepton: 0.738298|+> +0.674474|->
proton mixing angle: alpha_p=0.97176129 rad
incoming proton: 0.563846|+> +0.82588|->
s=2.88892, $\sqrt{s}$=1.69968, $|\vec{P}|$=0.591016, $|\vec{P}'|$=0.569675
$\theta_{p'}$=0.000586853, θ_γ=2.18118, $\phi_{p'}$=4.66411, ϕ_γ=1.3027
E_γ=0.0692849, $\theta_{\ell'}$=3.03548, $\phi_{\ell'}$=4.443
Q^2=0.00354339, x_B=0.0176376, t=-0.000329492, W^2=1.0772, y=0.0999957
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5910 p_x= 0.0000 p_y= 0.0000 p_z= -0.5910
P E= 1.1087 p_x= 0.0000 p_y= 0.0000 p_z= 0.5910
ℓ' E= 0.5330 p_x= -0.0150 p_y= -0.0544 p_z= -0.5300
P' E= 1.0974 p_x= -0.0000 p_y= -0.0003 p_z= 0.5697
q_γ E= 0.0693 p_x= 0.0150 p_y= 0.0547 p_z= -0.0397
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
ℓ'
ℓ
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=7, retained=98.5%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.05e+02+2.55e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-2.58e+02-5.80e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=2.39e+02-1.76e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=1.86e+02+9.37e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=3.17e+01-1.87e+02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=1.89e+01+1.17e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-4.47e+00-9.02e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 88; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.00022894012
D_W contour = 0.05022894
above global minimum = 0.00021250728
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.699684
theta_p_out = 0.0005868527
theta_gamma_out = 2.181185
qOut = 0.06928487
phi_p_out = 4.664108
phi_gamma_out = 1.302698
alpha_e = 0.7402522
alpha_p = 0.9717613

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



dw_mixing_angles_polarization_02_local_minimum_379 D_W=0.00121395

region: polarization_cluster_02_local_minimum_379

outgoing-state purity: 1

kinematic point: gradient_local_minimum

lepton mixing angle: alpha_e=0.79117094 rad

incoming lepton: 0.703013|+> +0.711177|->

proton mixing angle: alpha_p=1.7484036 rad

incoming proton: -0.176675|+> +0.984269|->

s=1.14298, $\sqrt{s}$=1.0691, $|\vec{P}|$=0.123061, $|\vec{P}'|$=0.0348211

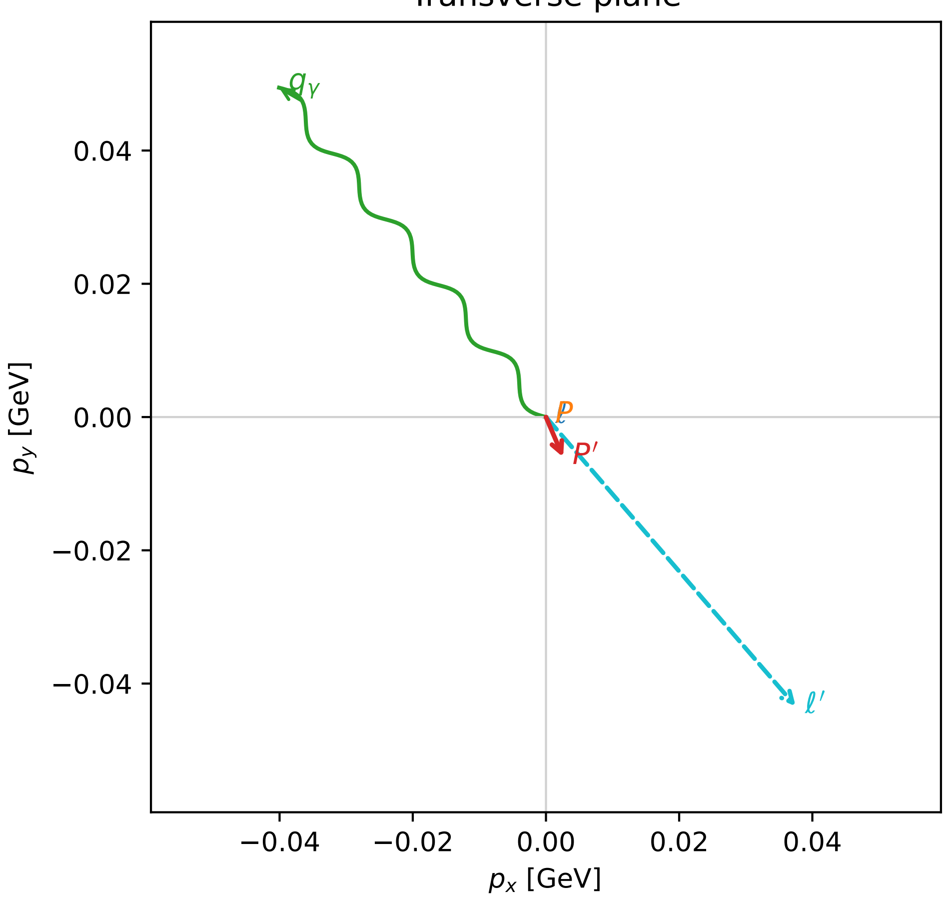
$\theta_{p'}$=2.95231, θ_γ=1.5721, $\phi_{p'}$=5.11662, ϕ_γ=2.25205

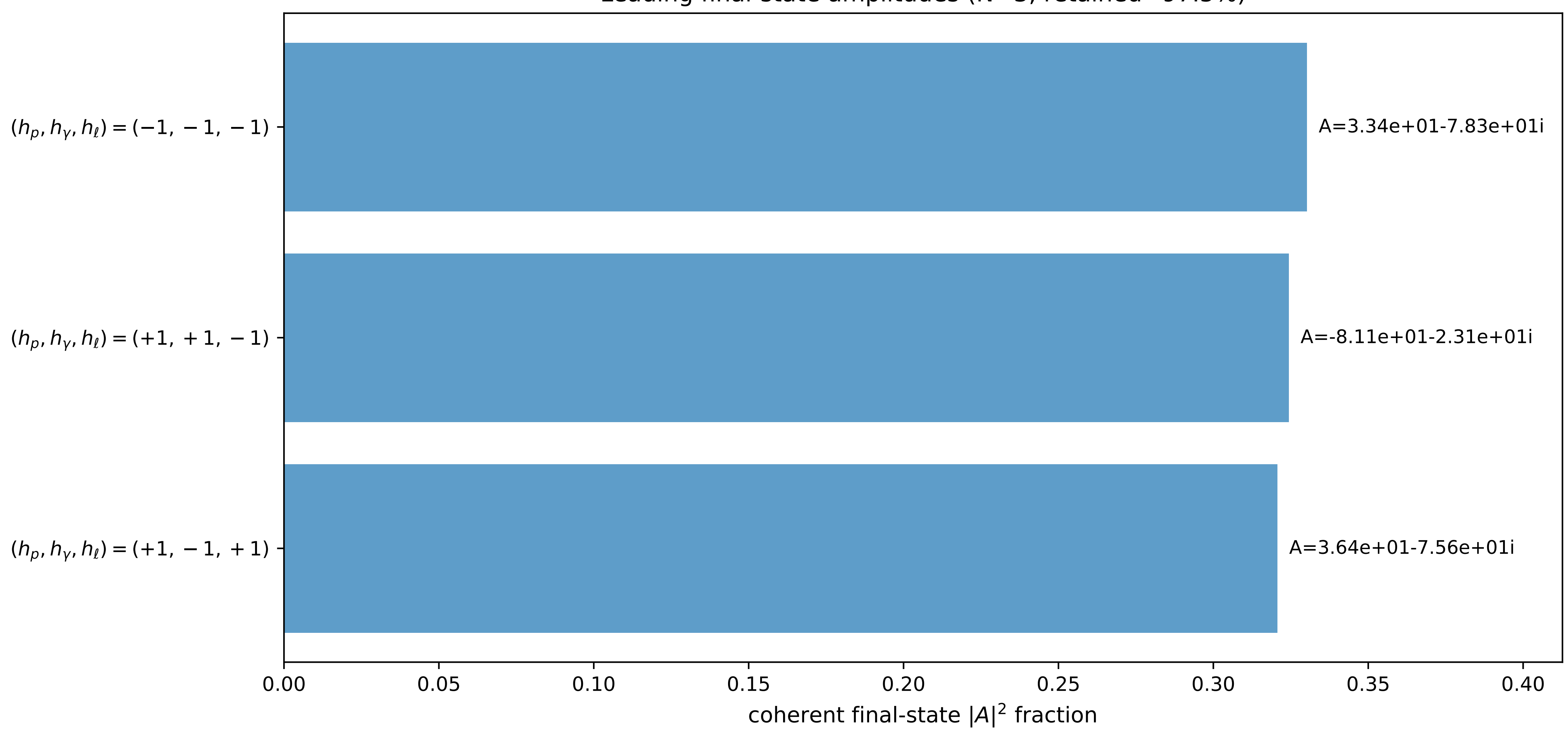
E_γ=0.063631, $\theta_{\ell'}$=1.03207, $\phi_{\ell'}$=5.42489

Q^2=0.0248839, x_B=0.171454, t=-0.0247191, W^2=1.00009, y=0.551566

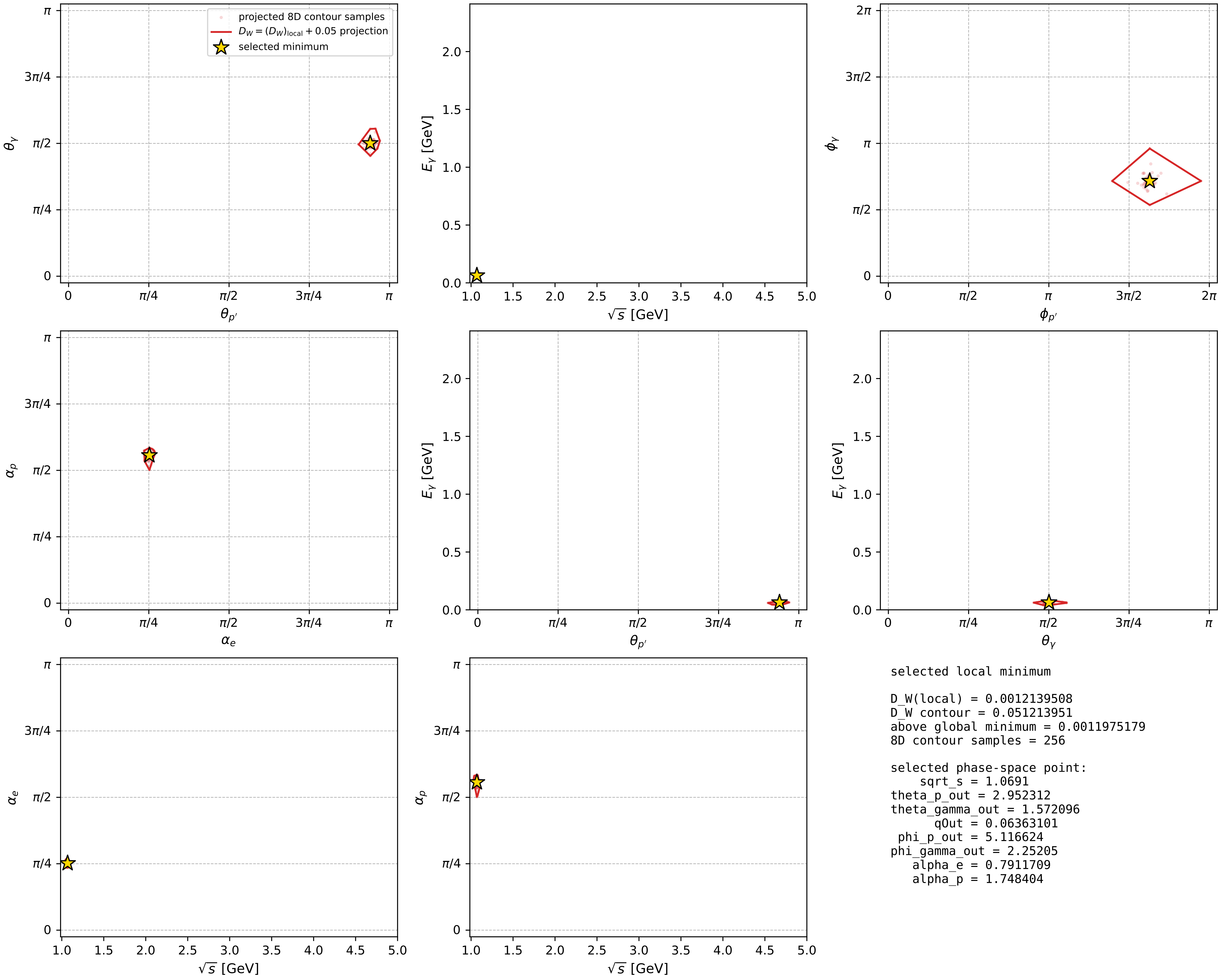
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1231	p_x= 0.0000	p_y= 0.0000	p_z= -0.1231
P	E= 0.9460	p_x= 0.0000	p_y= 0.0000	p_z= 0.1231
ℓ'	E= 0.0668	p_x= 0.0375	p_y= -0.0434	p_z= 0.0343
P'	E= 0.9386	p_x= 0.0026	p_y= -0.0060	p_z= -0.0342
q_γ	E= 0.0636	p_x= -0.0401	p_y= 0.0494	p_z= -0.0001

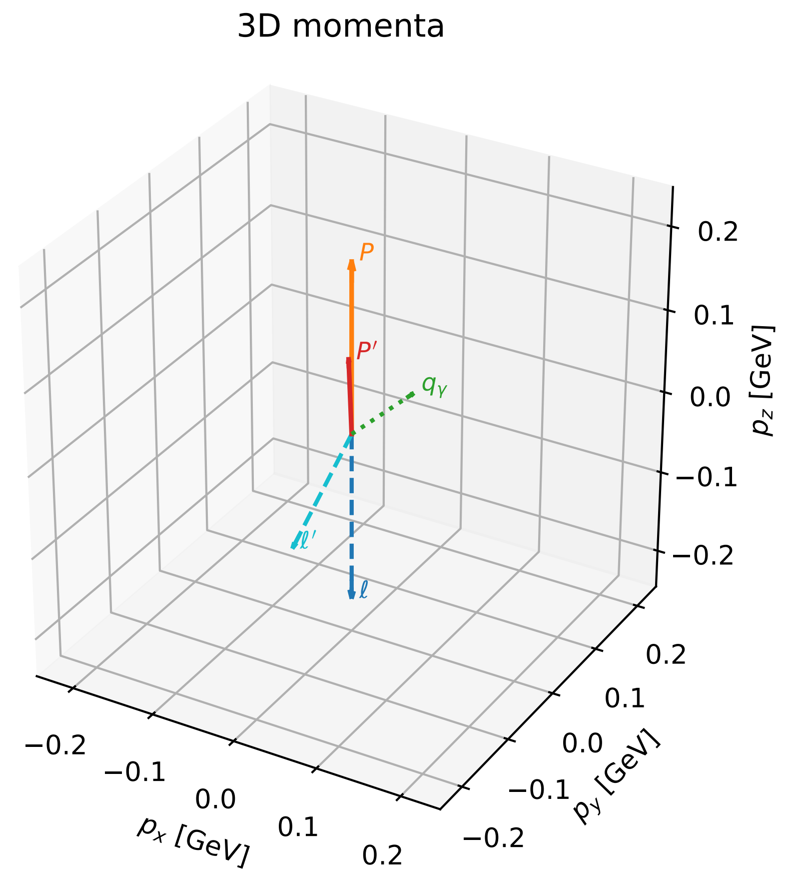




electron: polarization cluster P2, local minimum 379; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



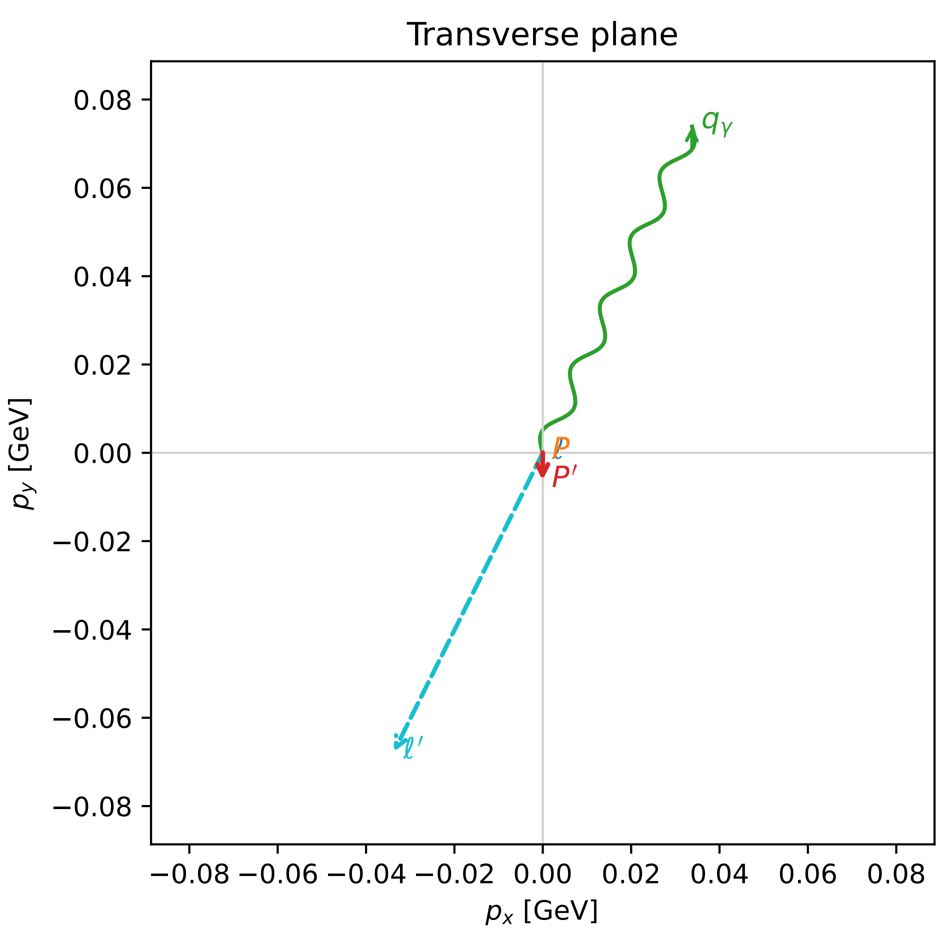
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

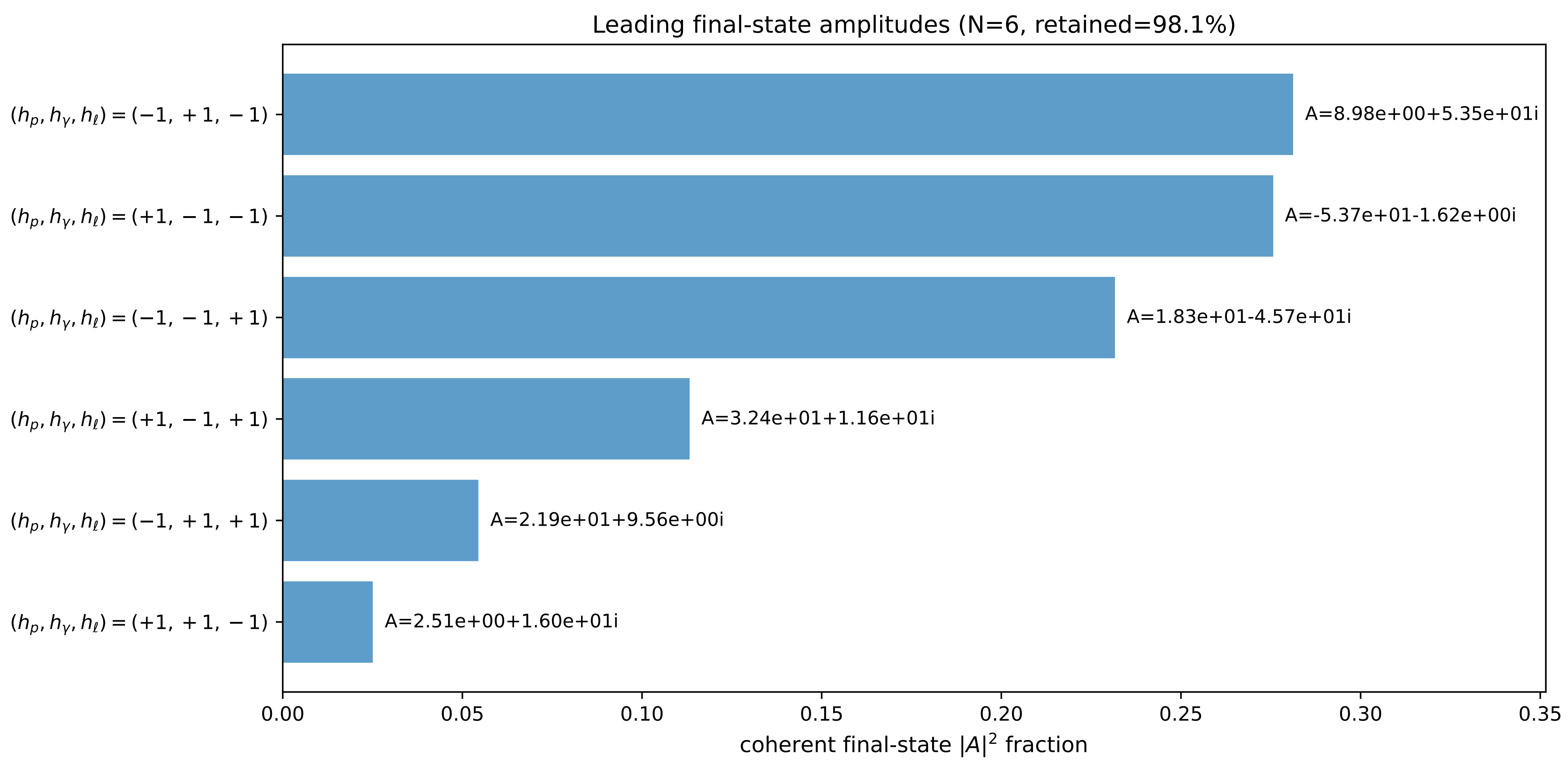


dw_mixing_angles_polarization_02_local_minimum_401 D_W=0.00133038
region: polarization_cluster_02_local_minimum_401
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.73677849 rad
incoming lepton: 0.740637|+> +0.671905|->
proton mixing angle: alpha_p=1.2109638 rad
incoming proton: 0.352117|+> +0.935956|->

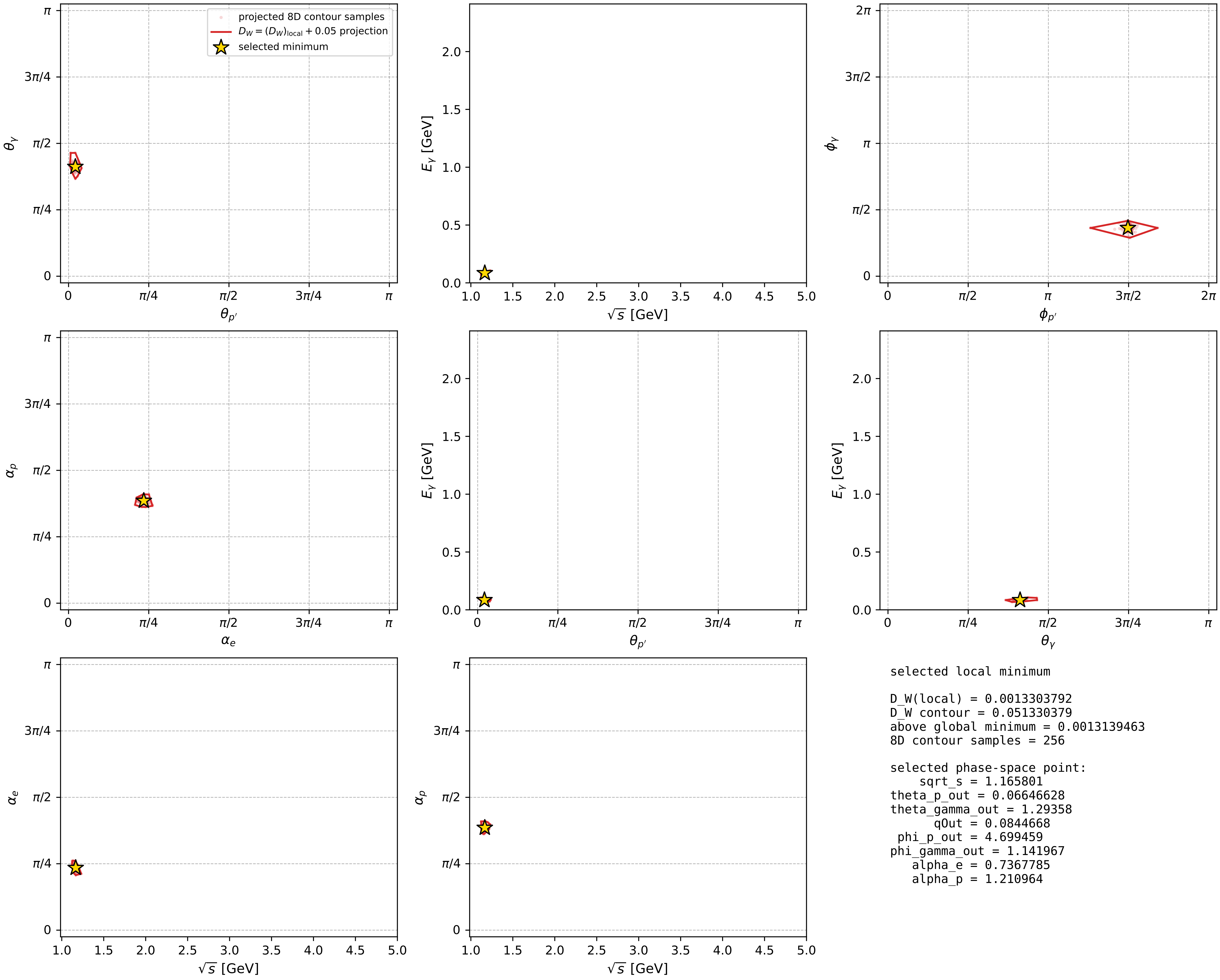
s=1.35909, $\sqrt{s}$=1.1658, $|\vec{P}|$=0.205544, $|\vec{P}'|$=0.0933652
$\theta_{p'}$=0.0664663, θ_γ=1.29358, $\phi_{p'}$=4.69946, ϕ_γ=1.14197
E_γ=0.0844668, $\theta_{\ell'}$=2.56503, $\phi_{\ell'}$=4.25042
Q^2=0.00921745, x_B=0.055838, t=-0.0123583, W^2=1.0357, y=0.344446

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2055 p_x= 0.0000 p_y= 0.0000 p_z= -0.2055
P E= 0.9603 p_x= 0.0000 p_y= 0.0000 p_z= 0.2055
ℓ' E= 0.1387 p_x= -0.0337 p_y= -0.0677 p_z= -0.1163
P' E= 0.9426 p_x= -0.0001 p_y= -0.0062 p_z= 0.0932
q_γ E= 0.0845 p_x= 0.0338 p_y= 0.0739 p_z= 0.0231

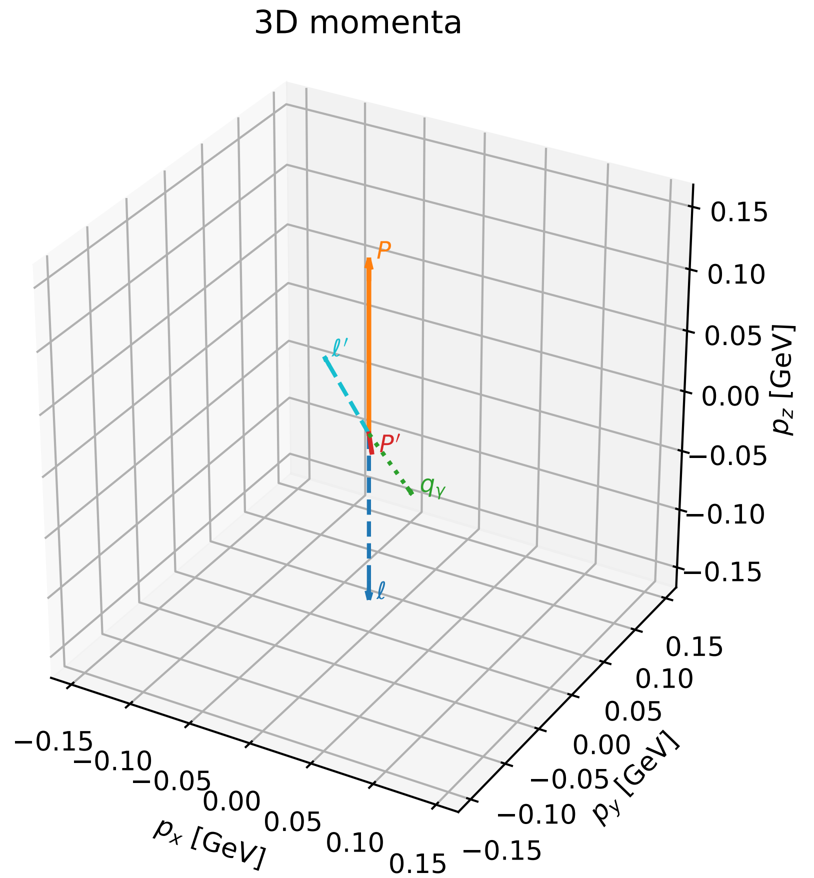




electron: polarization cluster P2, local minimum 401; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



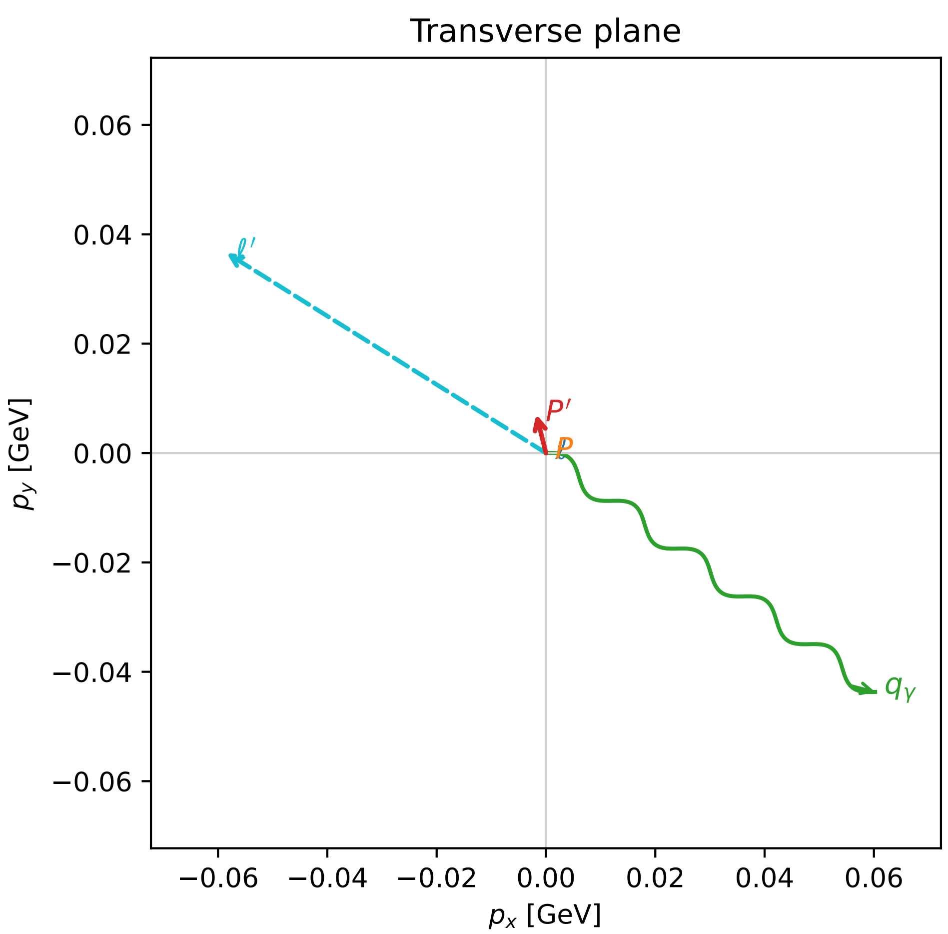
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

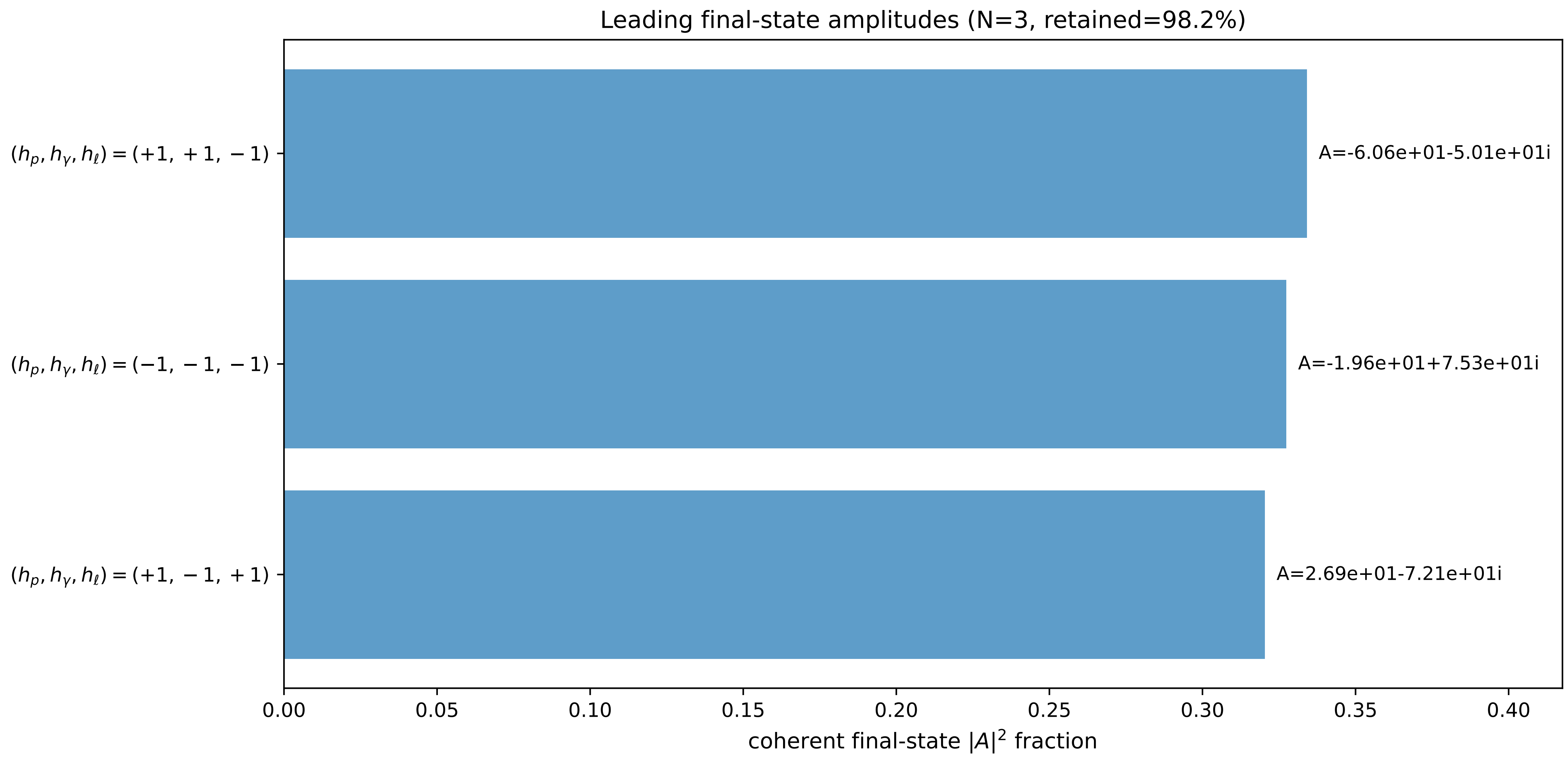


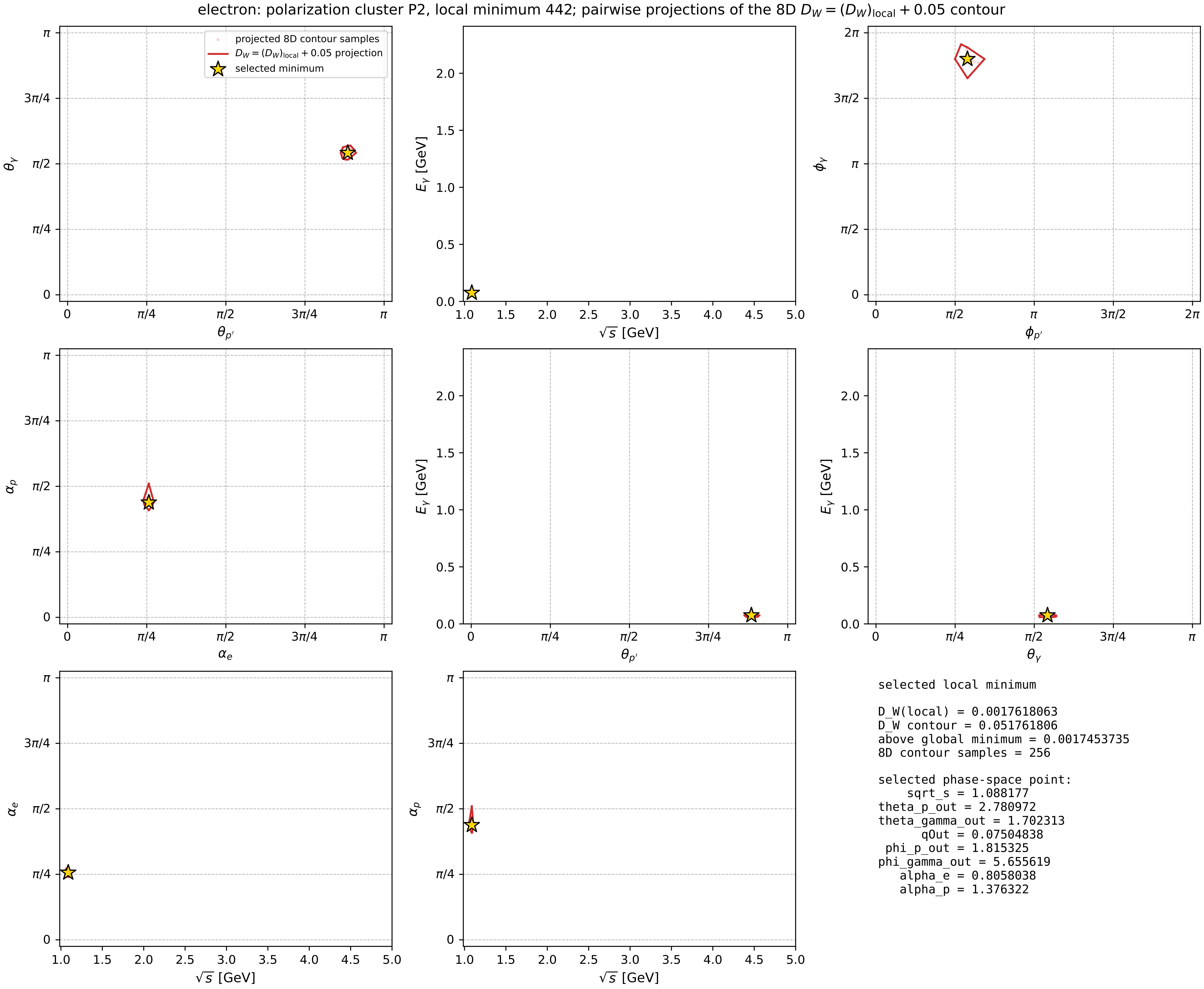

```
dw_mixing_angles_polarization_02_local_minimum_442  D_W=0.00176181
region: polarization_cluster_02_local_minimum_442
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80580383 rad
incoming lepton: 0.692532|+> +0.721388|->
proton mixing angle: alpha_p=1.3763217 rad
incoming proton: 0.193251|+> +0.981149|->
```

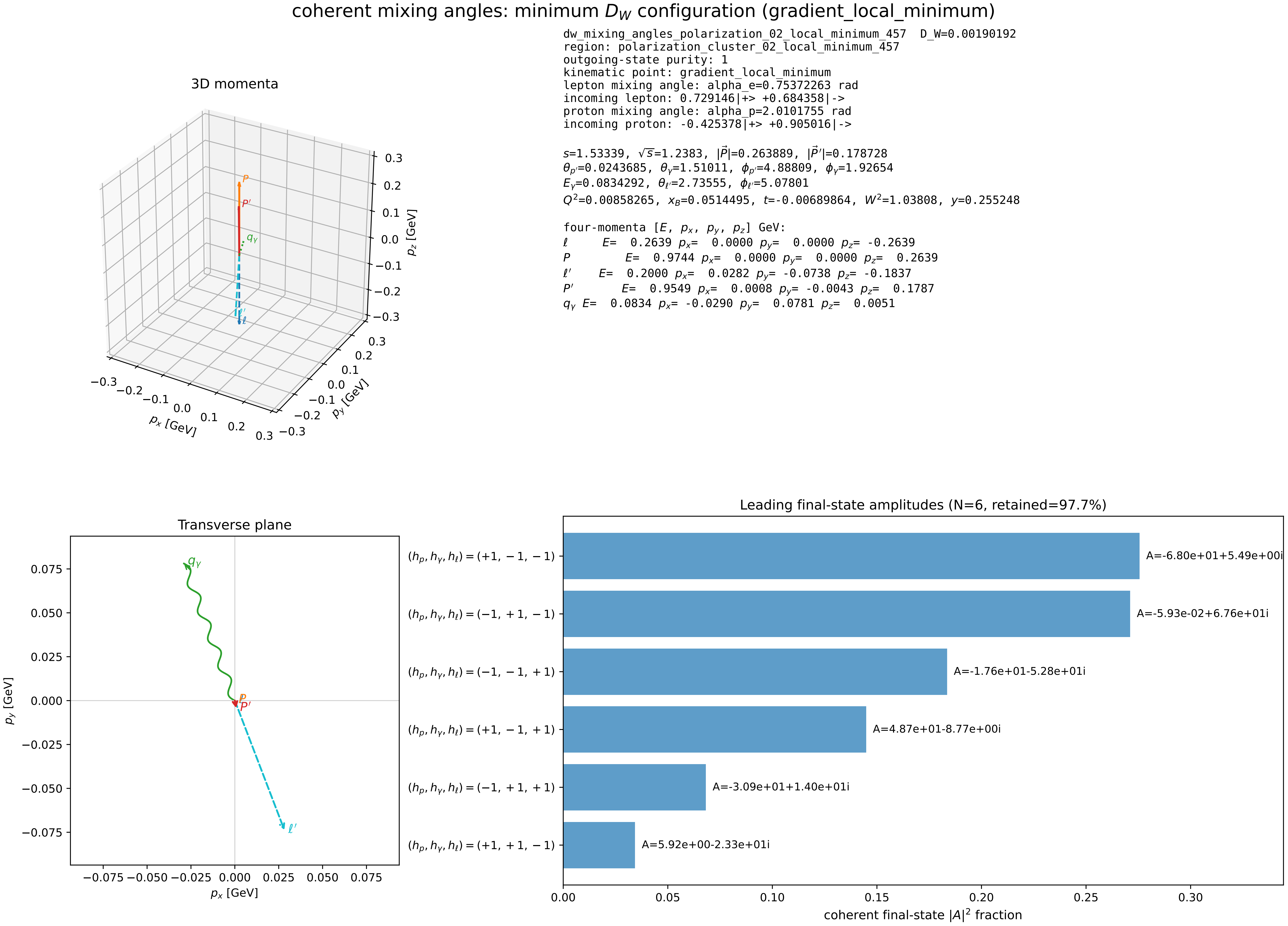
s=1.18413, $\sqrt{s}$=1.08818, $|\vec{P}|$=0.139813, $|\vec{P}'|$=0.0206987
$\theta_{p'}$=2.78097, θ_γ=1.70231, $\phi_{p'}$=1.81532, ϕ_γ=5.65562
E_γ=0.0750484, $\theta_{\ell'}$=1.17019, $\phi_{\ell'}$=2.5822
Q^2=0.0291113, x_B=0.170854, t=-0.0252891, W^2=1.02112, y=0.55996

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1398 p_x= 0.0000 p_y= 0.0000 p_z= -0.1398
P E= 0.9484 p_x= 0.0000 p_y= 0.0000 p_z= 0.1398
ℓ' E= 0.0749 p_x= -0.0585 p_y= 0.0366 p_z= 0.0292
P' E= 0.9382 p_x= -0.0018 p_y= 0.0071 p_z= -0.0194
q_γ E= 0.0750 p_x= 0.0602 p_y= -0.0437 p_z= -0.0098



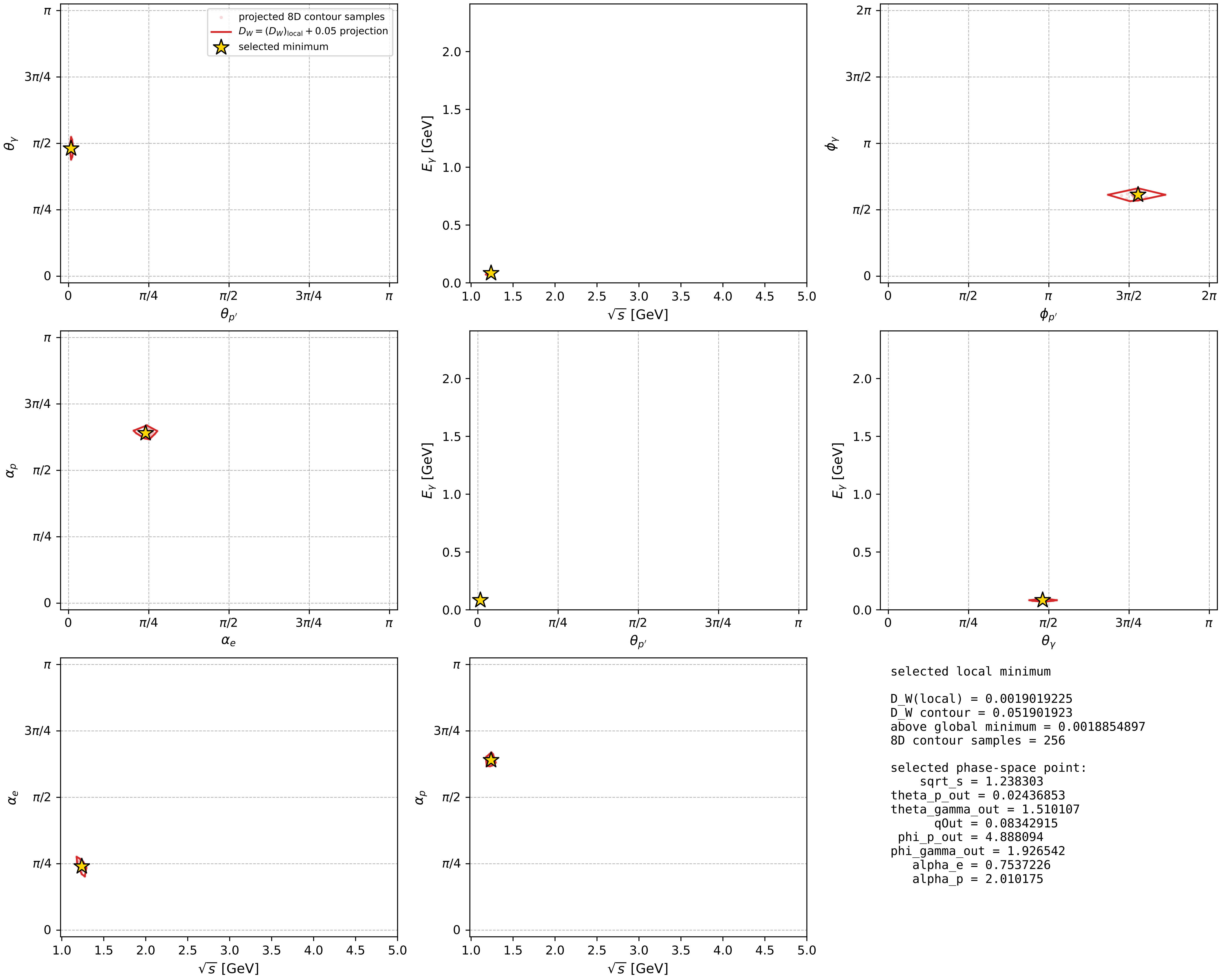



electron: polarization cluster P2, local minimum 442; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0017618063
D_W contour = 0.051761806
above global minimum = 0.0017453735
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.088177
theta_p_out = 2.780972
theta_gamma_out = 1.702313
qOut = 0.07504838
phi_p_out = 1.815325
phi_gamma_out = 5.655619
alpha_e = 0.8058038
alpha_p = 1.376322

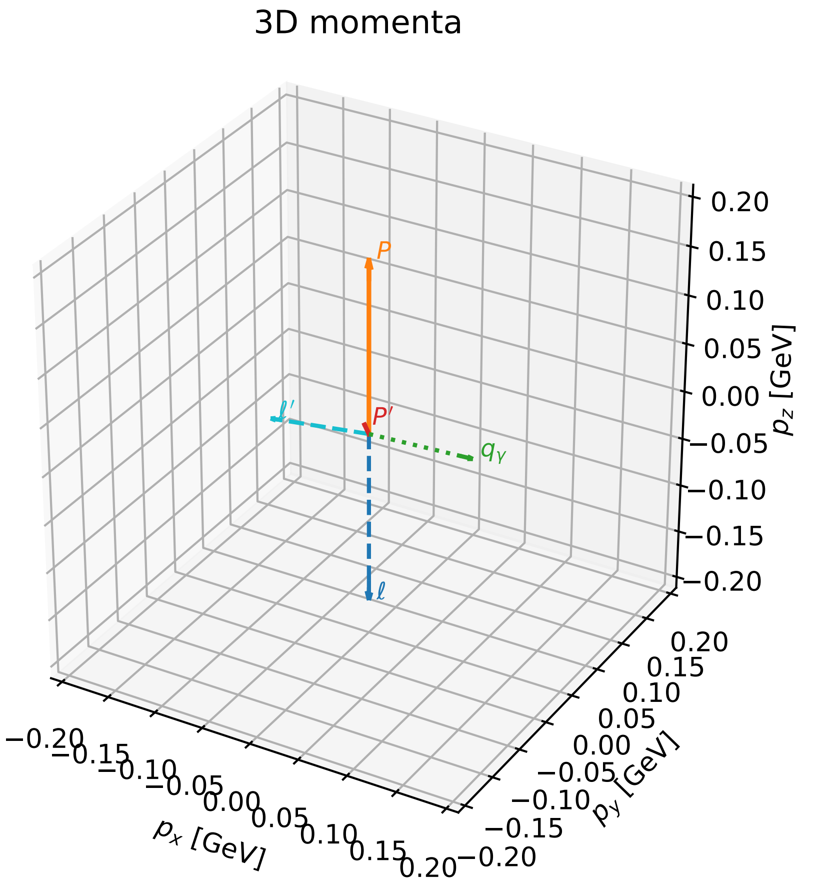


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_02_local_minimum_457 D_W=0.00190192
region: polarization_cluster_02_local_minimum_457
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75372263 rad
incoming lepton: 0.729146|+> +0.684358|->
proton mixing angle: alpha_p=2.0101755 rad
incoming proton: -0.425378|+> +0.905016|->
s=1.53339, $\sqrt{s}$=1.2383, $|\vec{P}|$=0.263889, $|\vec{P}'|$=0.178728
$\theta_{p'}$=0.0243685, θ_γ=1.51011, $\phi_{p'}$=4.88809, ϕ_γ=1.92654
E_γ=0.0834292, $\theta_{\ell'}$=2.73555, $\phi_{\ell'}$=5.07801
Q^2=0.00858265, x_B=0.0514495, t=-0.00689864, W^2=1.03808, y=0.255248
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2639 p_x= 0.0000 p_y= 0.0000 p_z= -0.2639
P E= 0.9744 p_x= 0.0000 p_y= 0.0000 p_z= 0.2639
ℓ' E= 0.2000 p_x= 0.0282 p_y= -0.0738 p_z= -0.1837
P' E= 0.9549 p_x= 0.0008 p_y= -0.0043 p_z= 0.1787
q_γ E= 0.0834 p_x= -0.0290 p_y= 0.0781 p_z= 0.0051
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=6, retained=97.7%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-6.80e+01+5.49e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-5.93e-02+6.76e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-1.76e+01-5.28e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=4.87e+01-8.77e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-3.09e+01+1.40e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=5.92e+00-2.33e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 457; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



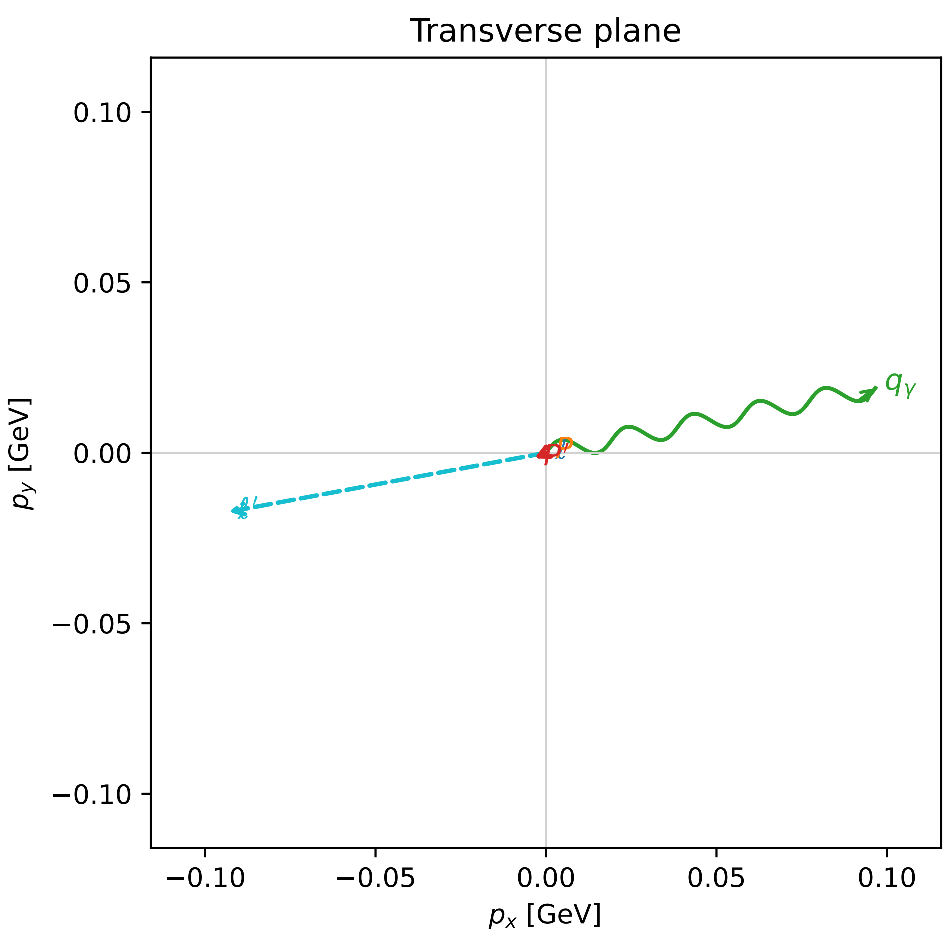
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

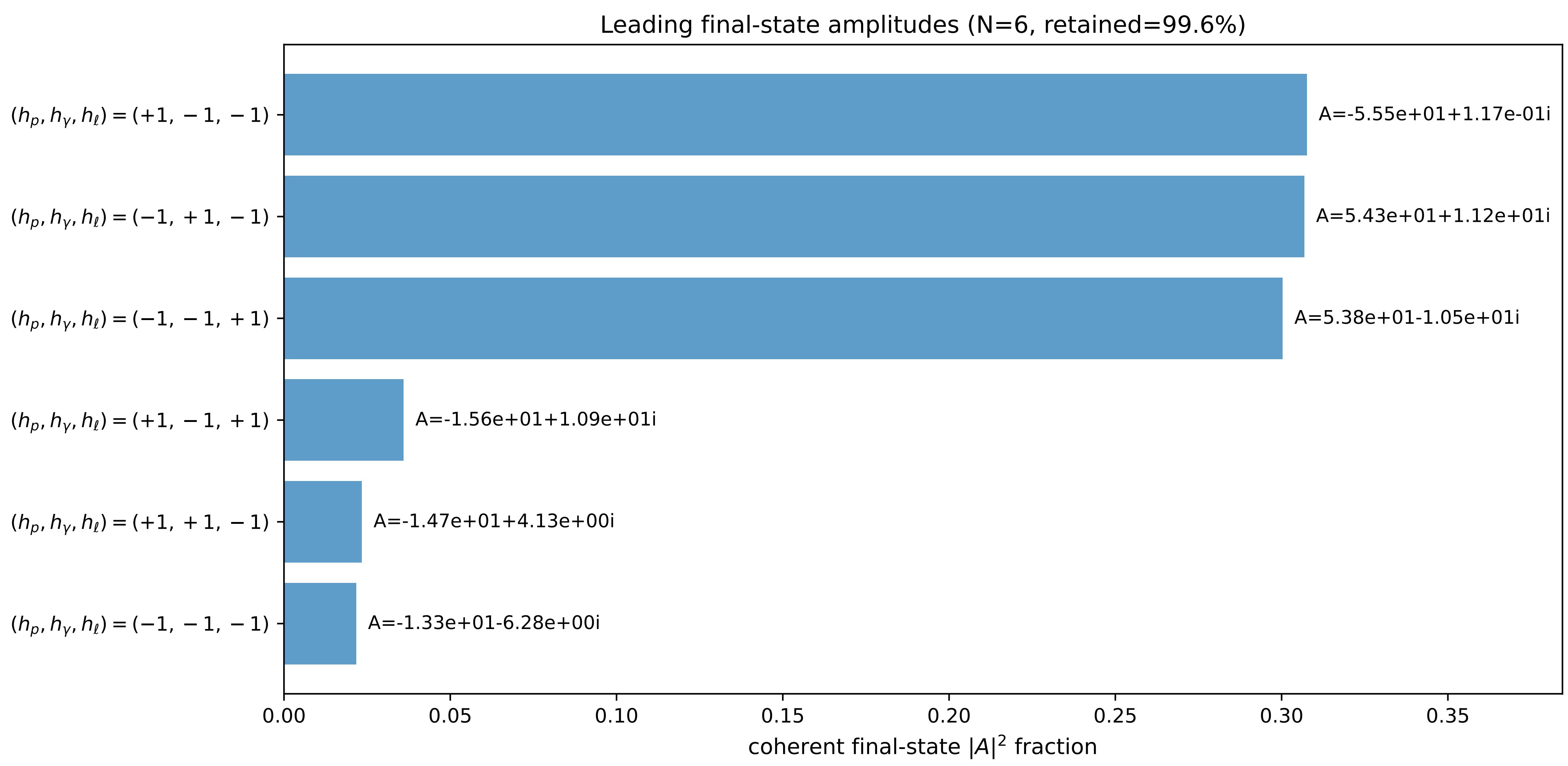


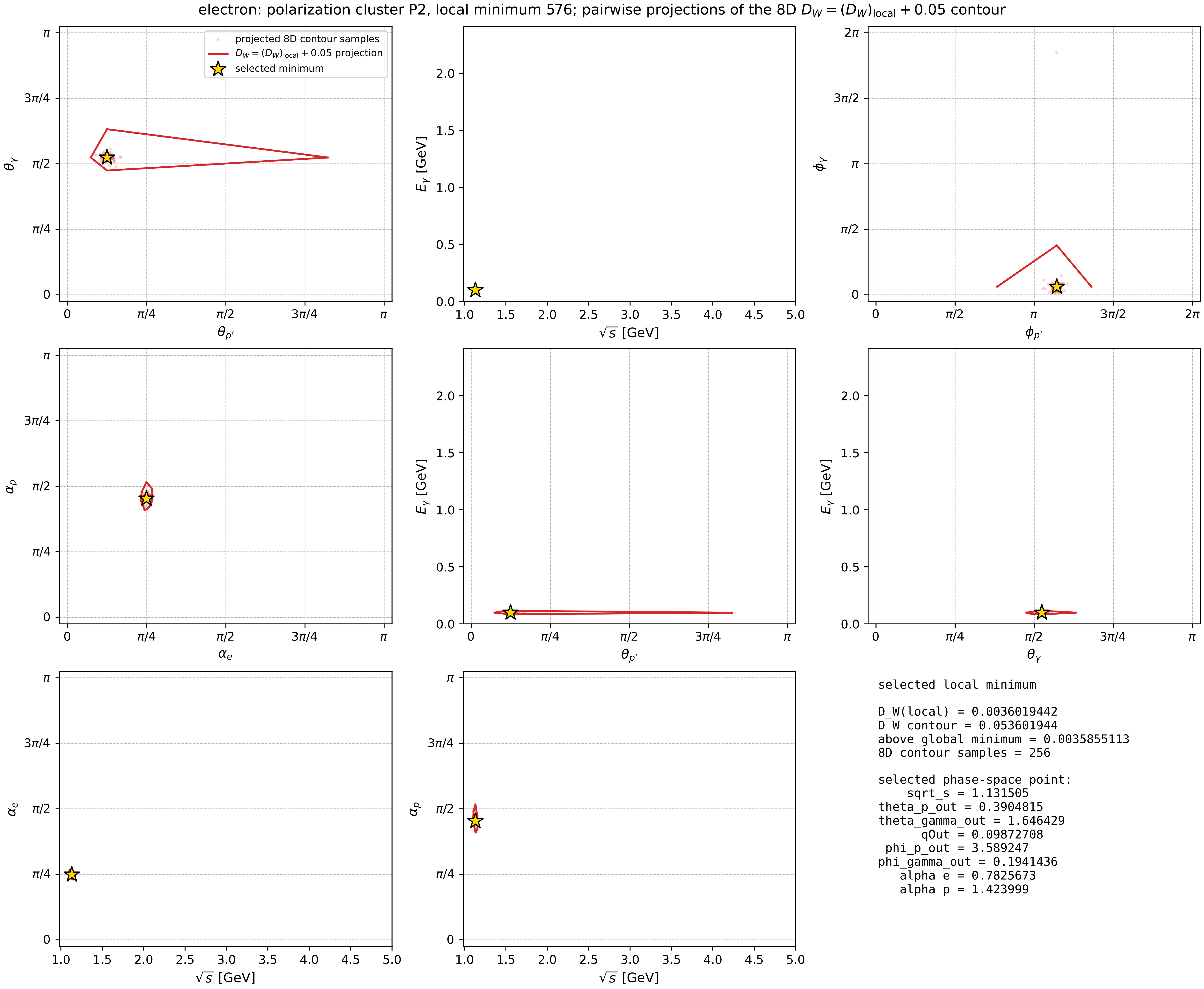
dw_mixing_angles_polarization_02_local_minimum_576 D_W=0.00360194
region: polarization_cluster_02_local_minimum_576
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78256727 rad
incoming lepton: 0.709106|+> +0.705102|->
proton mixing angle: alpha_p=1.4239986 rad
incoming proton: 0.146271|+> +0.989245|->

s=1.2803, $\sqrt{s}$=1.13151, $|\vec{P}|$=0.176958, $|\vec{P}'|$=0.0101727
$\theta_{p'}$=0.390482, θ_γ=1.64643, $\phi_{p'}$=3.58925, ϕ_γ=0.194144
E_γ=0.0987271, $\theta_{\ell'}$=1.59135, $\phi_{\ell'}$=3.32548
Q^2=0.0328345, x_B=0.149973, t=-0.0278165, W^2=1.06595, y=0.546711

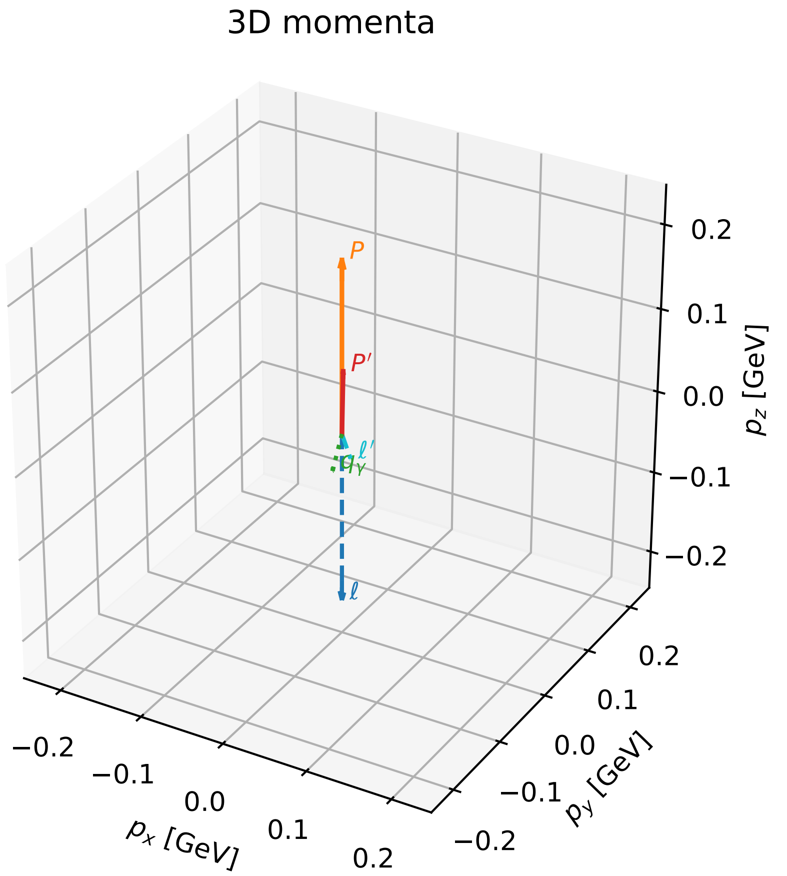
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1770 p_x= 0.0000 p_y= 0.0000 p_z= -0.1770
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1770
ℓ' E= 0.0947 p_x= -0.0931 p_y= -0.0173 p_z= -0.0019
P' E= 0.9381 p_x= -0.0035 p_y= -0.0017 p_z= 0.0094
q_γ E= 0.0987 p_x= 0.0966 p_y= 0.0190 p_z= -0.0075






electron: polarization cluster P2, local minimum 576; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0036019442
D_W contour = 0.053601944
above global minimum = 0.0035855113
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.131505
theta_p_out = 0.3904815
theta_gamma_out = 1.646429
qOut = 0.09872708
phi_p_out = 3.589247
phi_gamma_out = 0.1941436
alpha_e = 0.7825673
alpha_p = 1.423999

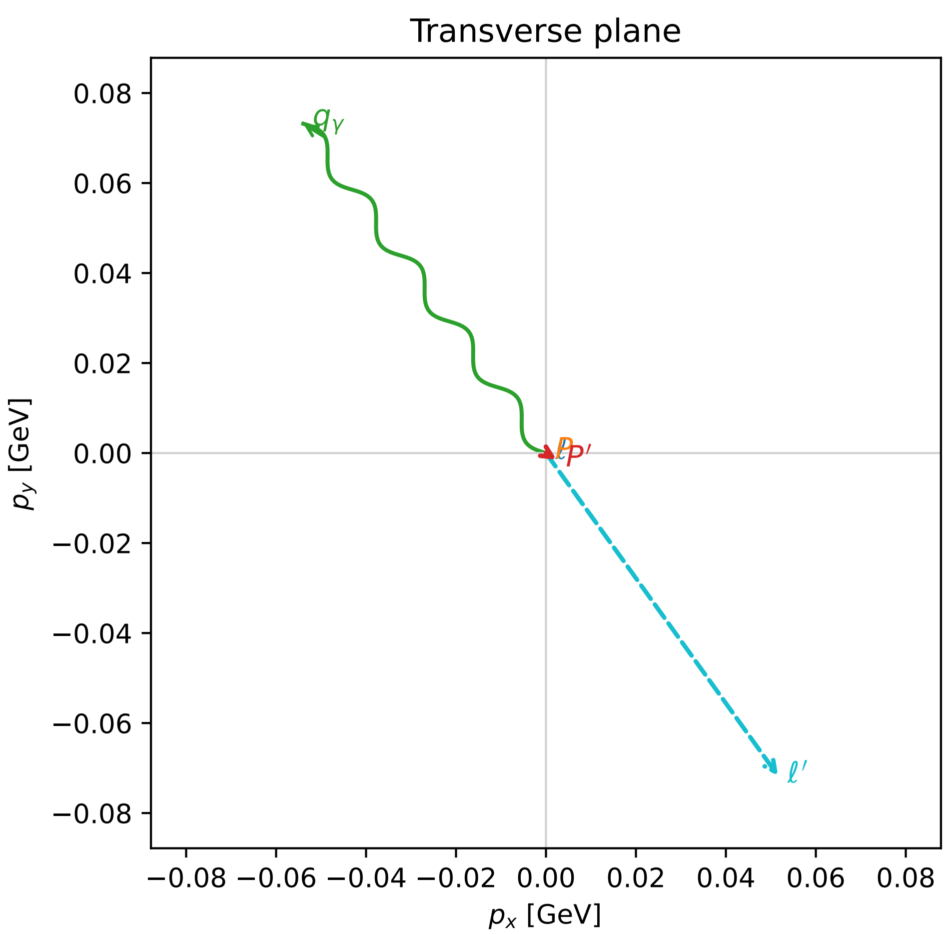
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

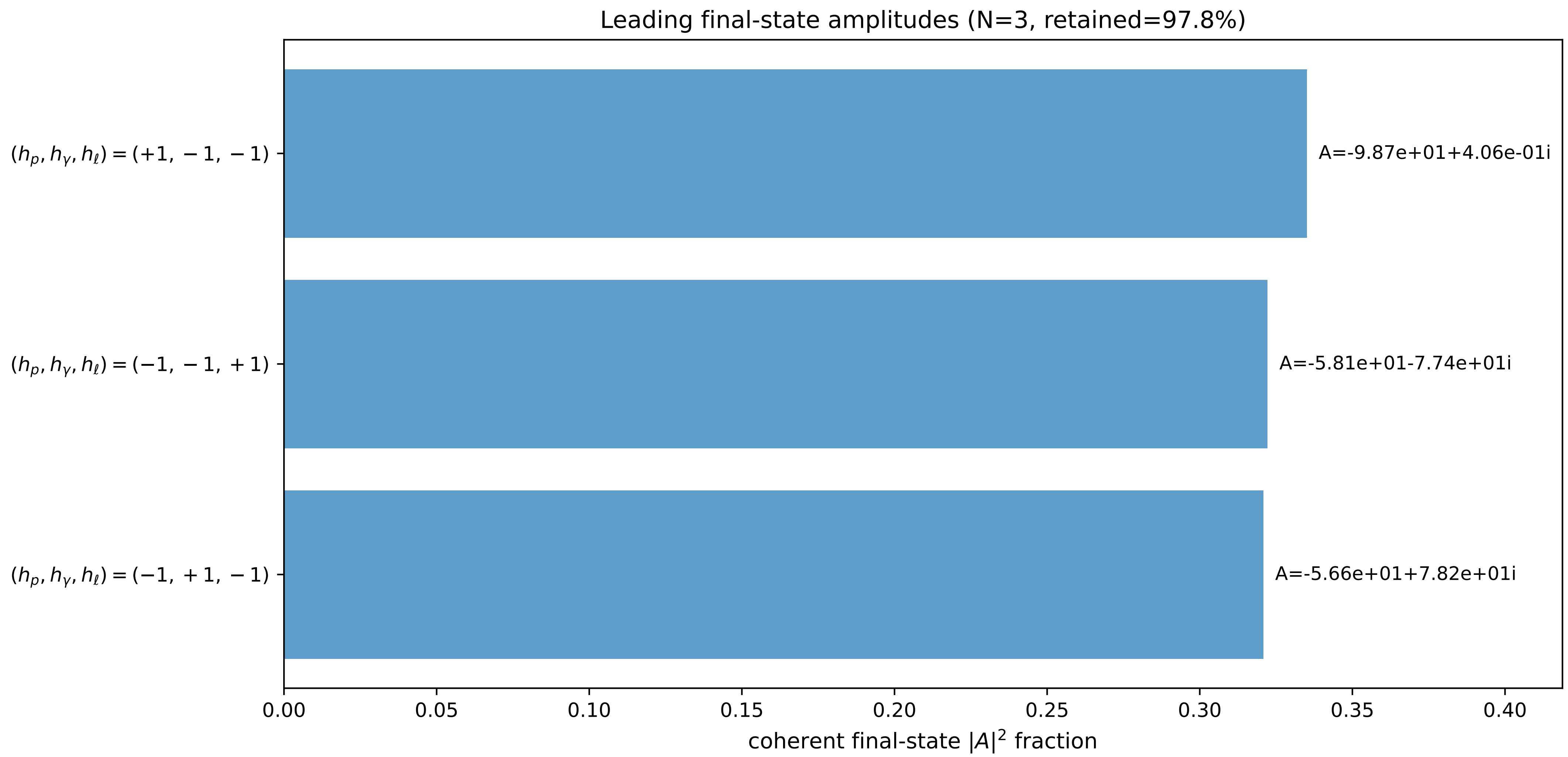



```
dw_mixing_angles_polarization_02_local_minimum_590  D_W=0.00392354
region: polarization_cluster_02_local_minimum_590
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78087704 rad
incoming lepton: 0.710296|+> +0.703903|->
proton mixing angle: alpha_p=1.6435266 rad
incoming proton: -0.0726662|+> +0.997356|->
```

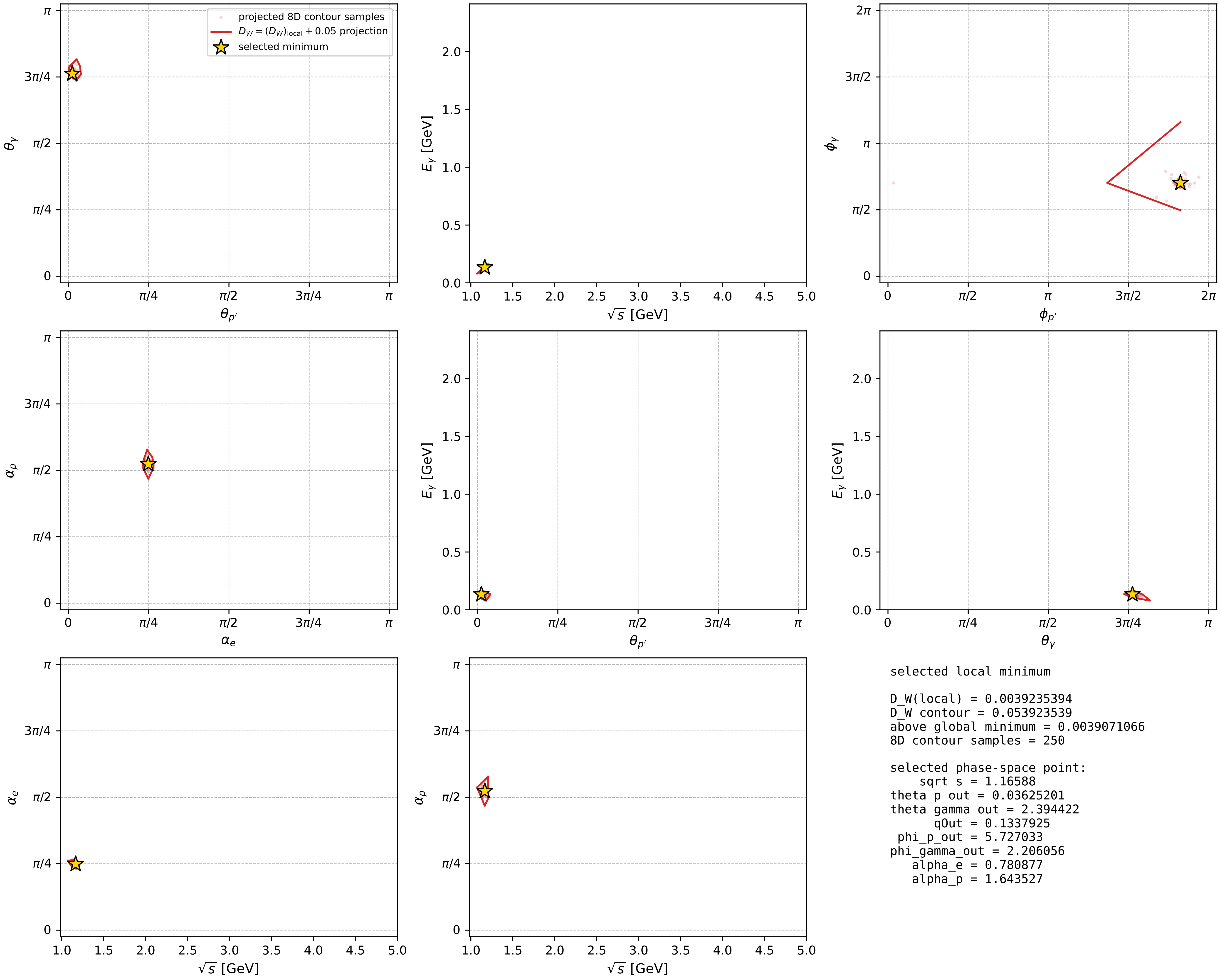
s=1.35928, $\sqrt{s}$=1.16588, $|\vec{P}|$=0.205609, $|\vec{P}'|$=0.0765314
$\theta_{p'}$=0.036252, θ_γ=2.39442, $\phi_{p'}$=5.72703, ϕ_γ=2.20606
E_γ=0.133793, $\theta_{\ell'}$=1.33026, $\phi_{\ell'}$=5.33602
Q^2=0.0463199, x_B=0.147689, t=-0.0163149, W^2=1.14715, y=0.654171

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2056 p_x= 0.0000 p_y= 0.0000 p_z= -0.2056
P E= 0.9603 p_x= 0.0000 p_y= 0.0000 p_z= 0.2056
ℓ' E= 0.0910 p_x= 0.0516 p_y= -0.0717 p_z= 0.0217
P' E= 0.9411 p_x= 0.0024 p_y= -0.0015 p_z= 0.0765
q_γ E= 0.1338 p_x= -0.0540 p_y= 0.0732 p_z= -0.0982



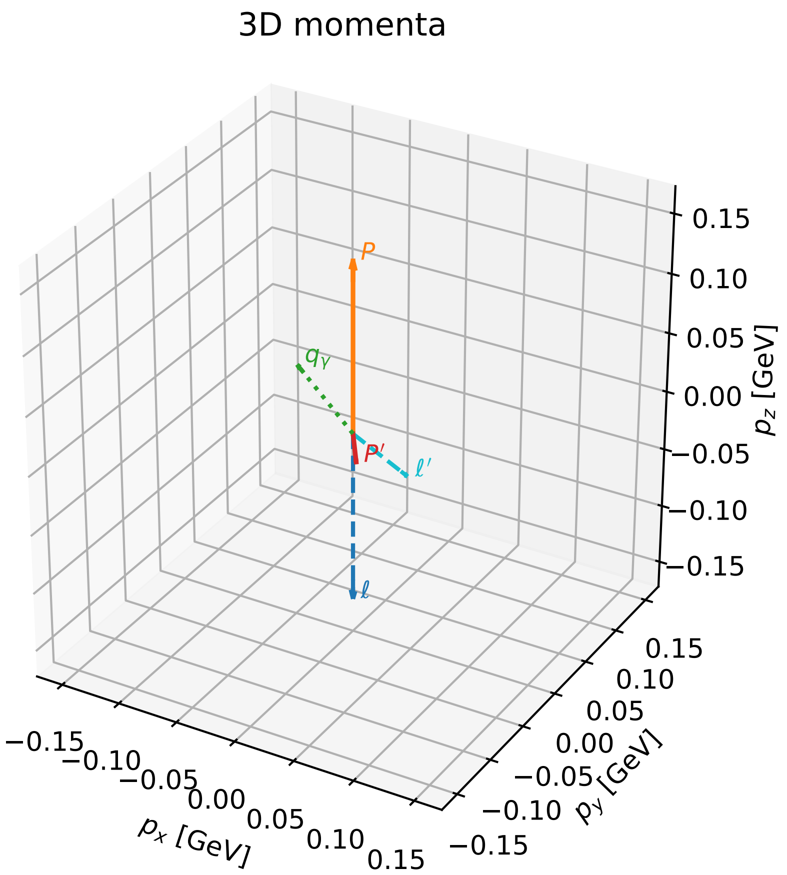


electron: polarization cluster P2, local minimum 590; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.0039235394
D_W contour = 0.053923539
above global minimum = 0.0039071066
8D contour samples = 250

selected phase-space point:
    sqrt_s = 1.16588
theta_p_out = 0.03625201
theta_gamma_out = 2.394422
      qOut = 0.1337925
 phi_p_out = 5.727033
phi_gamma_out = 2.206056
   alpha_e = 0.780877
   alpha_p = 1.643527
```

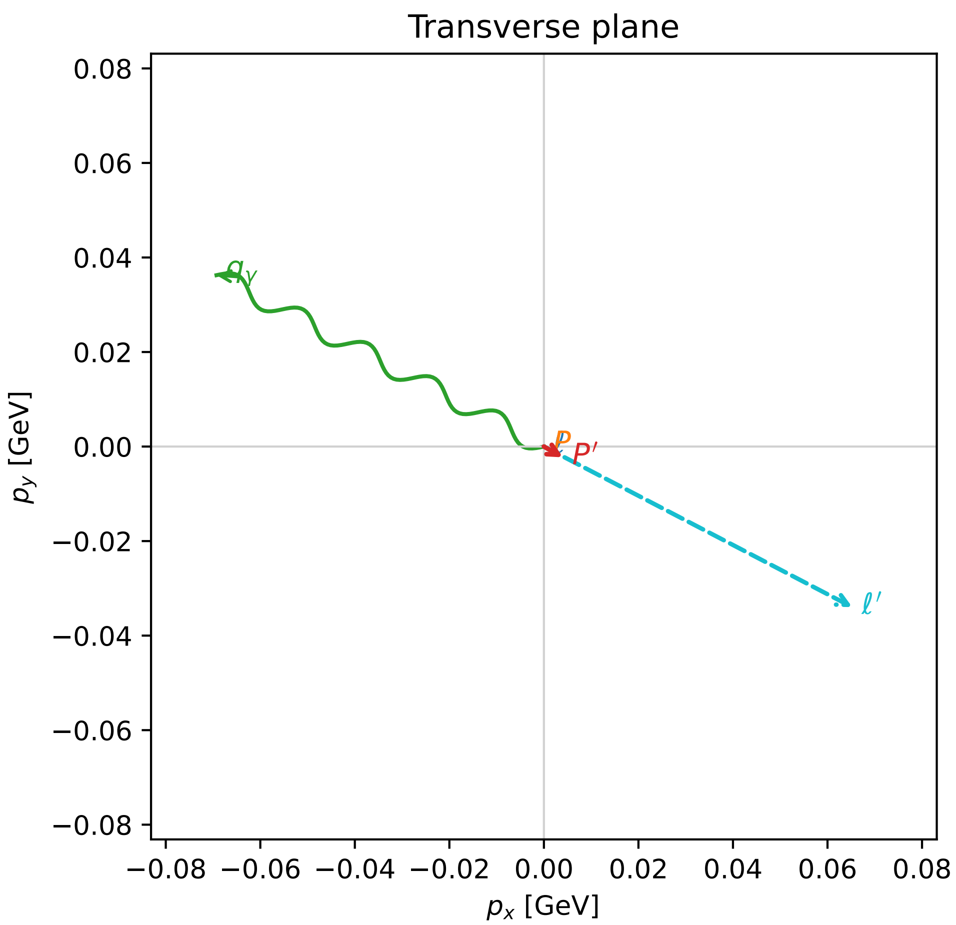
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

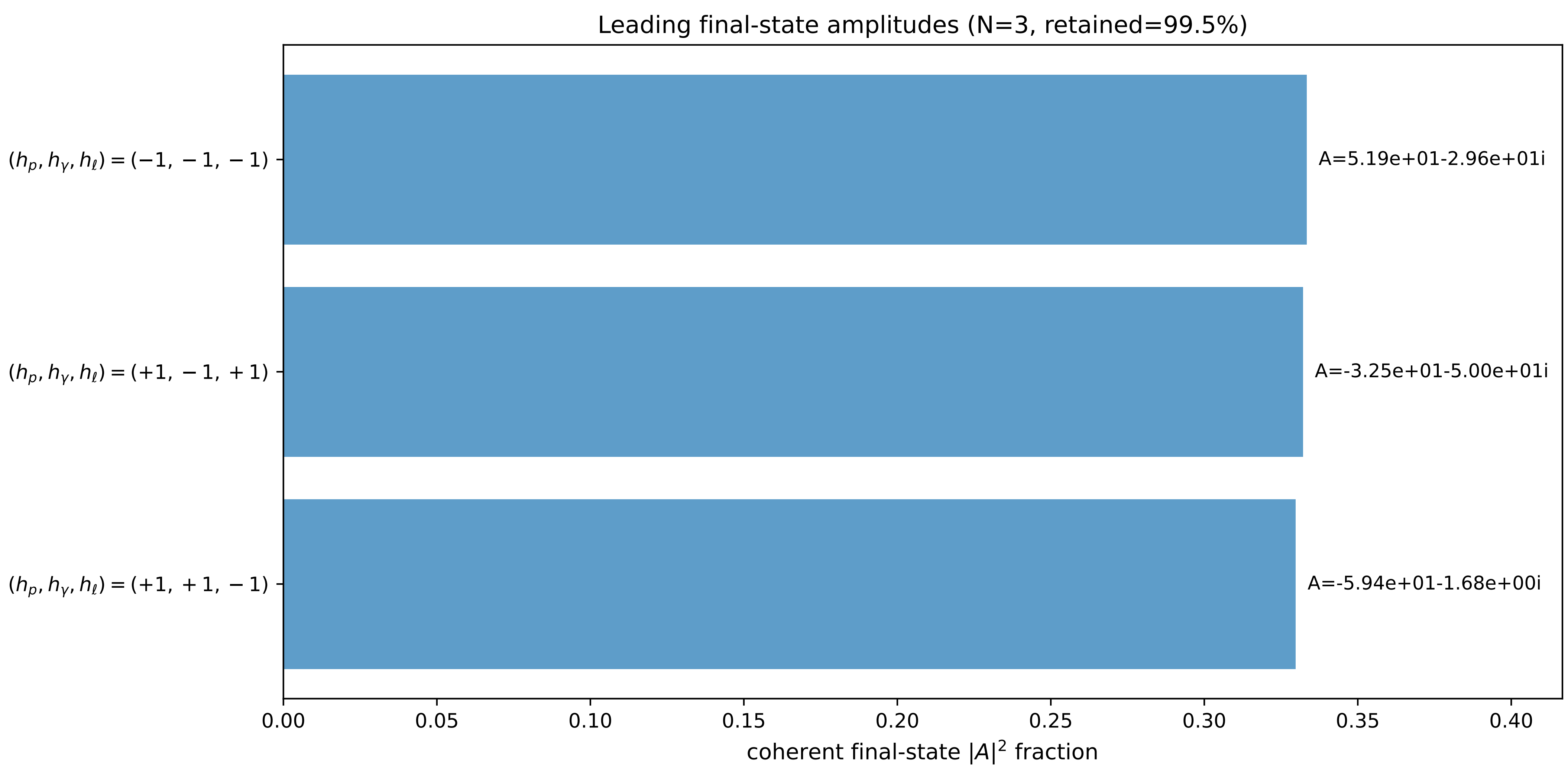


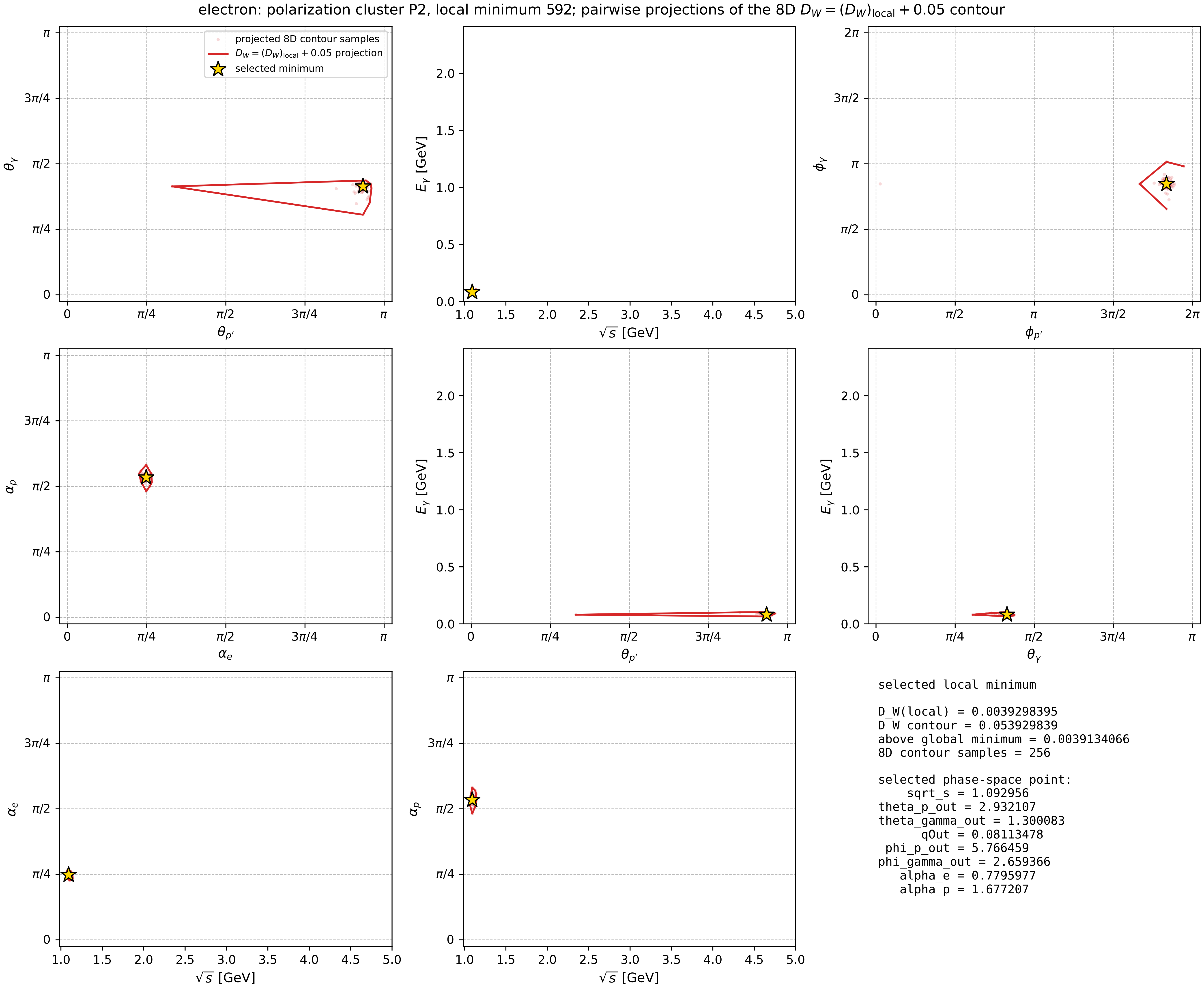
dw_mixing_angles_polarization_02_local_minimum_592 D_W=0.00392984
region: polarization_cluster_02_local_minimum_592
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77959771 rad
incoming lepton: 0.711196|+> +0.702993|->
proton mixing angle: alpha_p=1.6772067 rad
incoming proton: -0.10621|+> +0.994344|->

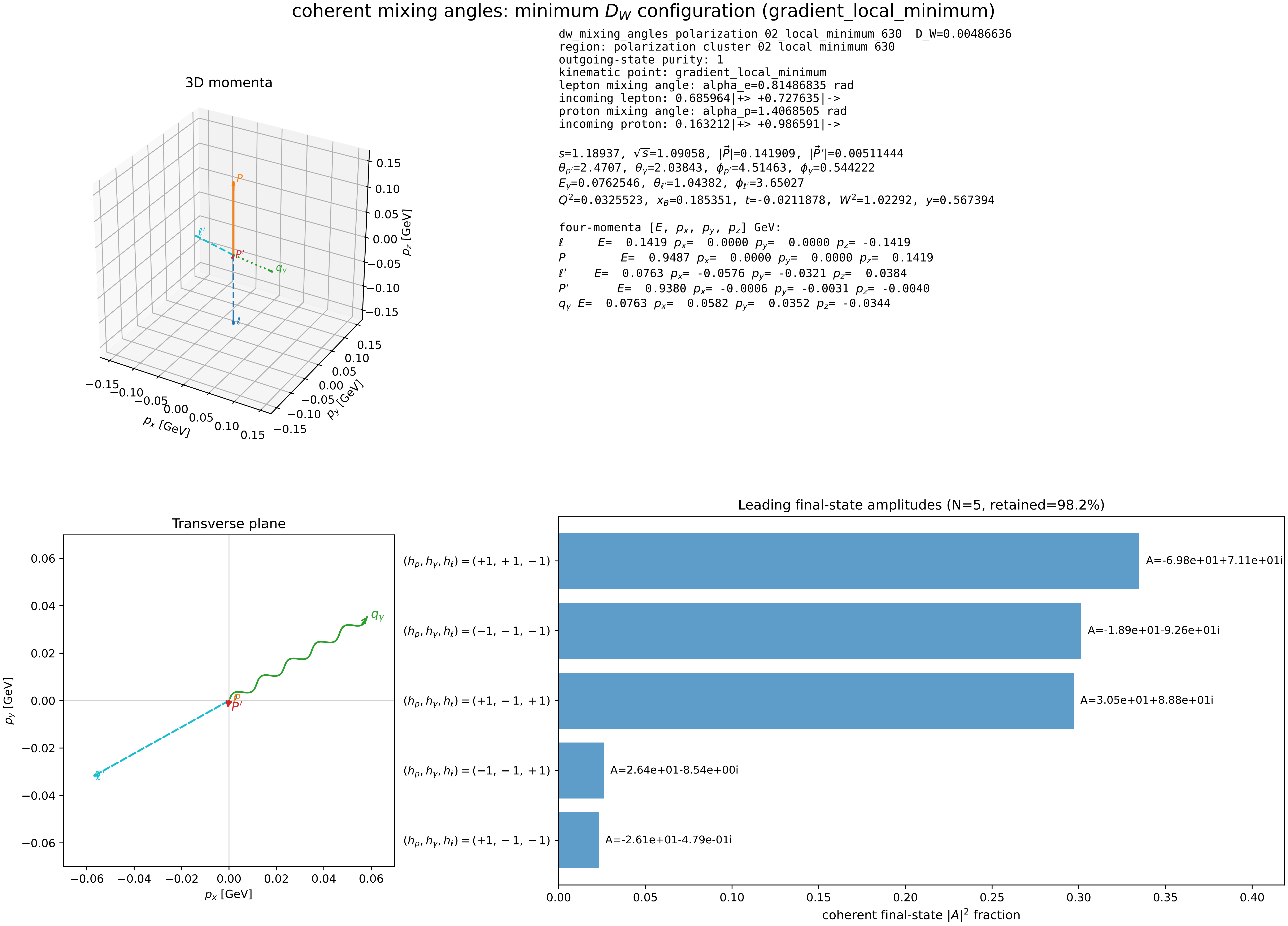
s=1.19455, $\sqrt{s}$=1.09296, $|\vec{P}|$=0.14397, $|\vec{P}'|$=0.022253
$\theta_{p'}$=2.93211, θ_γ=1.30008, $\phi_{p'}$=5.76646, ϕ_γ=2.65937
E_γ=0.0811348, $\theta_{\ell'}$=1.56985, $\phi_{\ell'}$=5.80313
Q^2=0.0211997, x_B=0.121059, t=-0.0273752, W^2=1.03376, y=0.55645

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1440 p_x= 0.0000 p_y= 0.0000 p_z= -0.1440
P E= 0.9490 p_x= 0.0000 p_y= 0.0000 p_z= 0.1440
ℓ' E= 0.0736 p_x= 0.0652 p_y= -0.0340 p_z= 0.0001
P' E= 0.9383 p_x= 0.0040 p_y= -0.0023 p_z= -0.0218
q_γ E= 0.0811 p_x= -0.0693 p_y= 0.0363 p_z= 0.0217



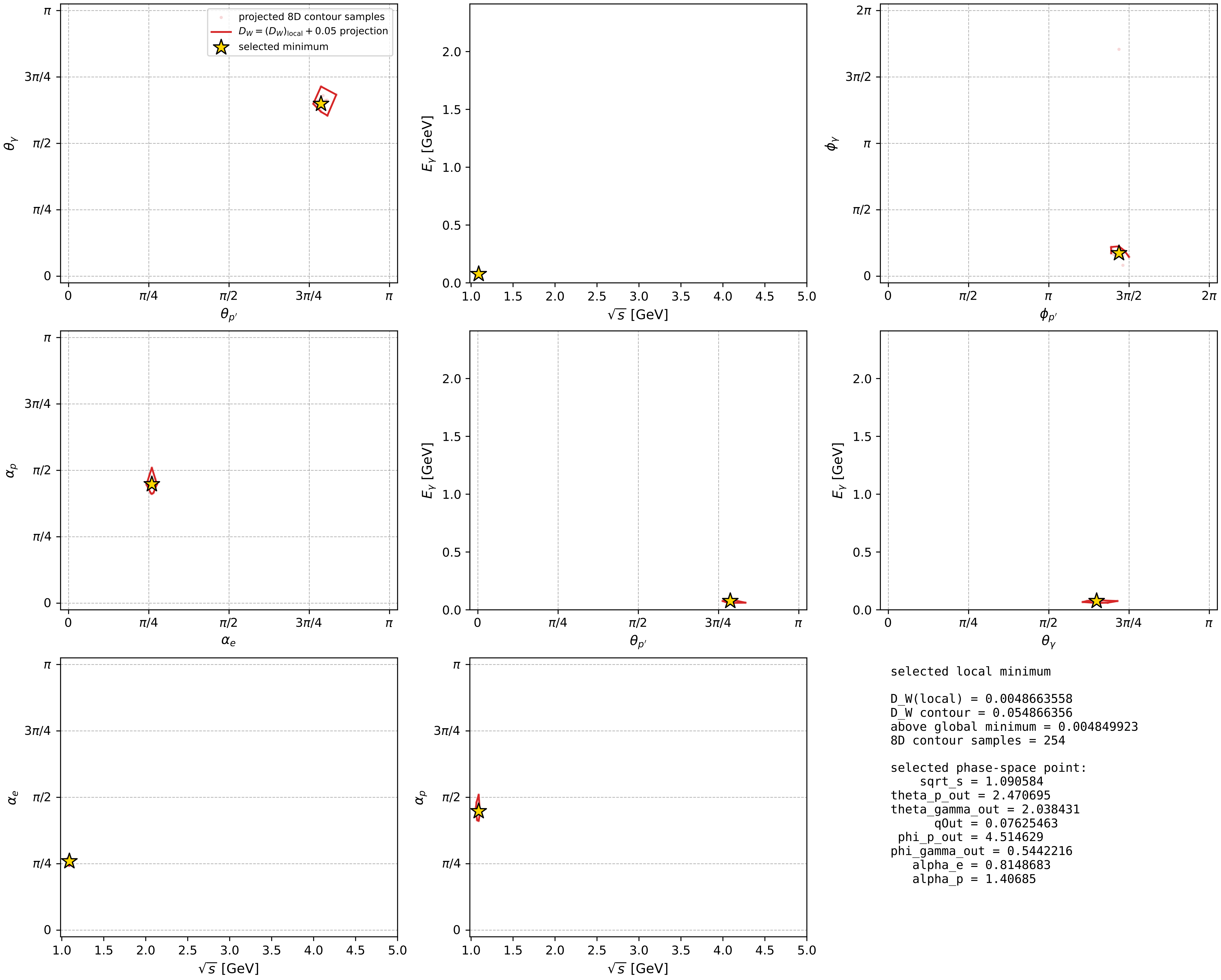


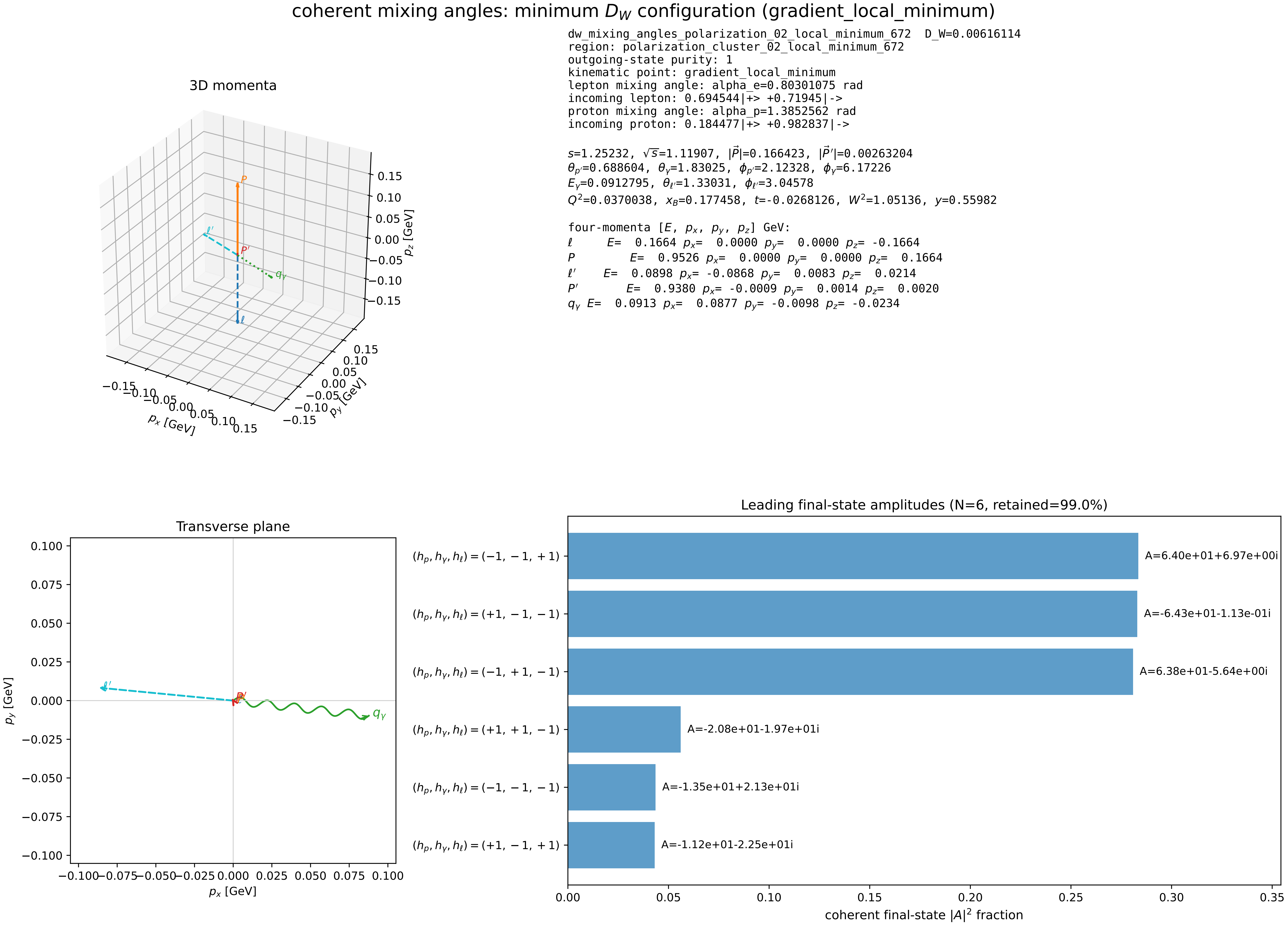

electron: polarization cluster P2, local minimum 592; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0039298395
D_W contour = 0.053929839
above global minimum = 0.0039134066
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.092956
theta_p_out = 2.932107
theta_gamma_out = 1.300083
qOut = 0.08113478
phi_p_out = 5.766459
phi_gamma_out = 2.659366
alpha_e = 0.7795977
alpha_p = 1.677207

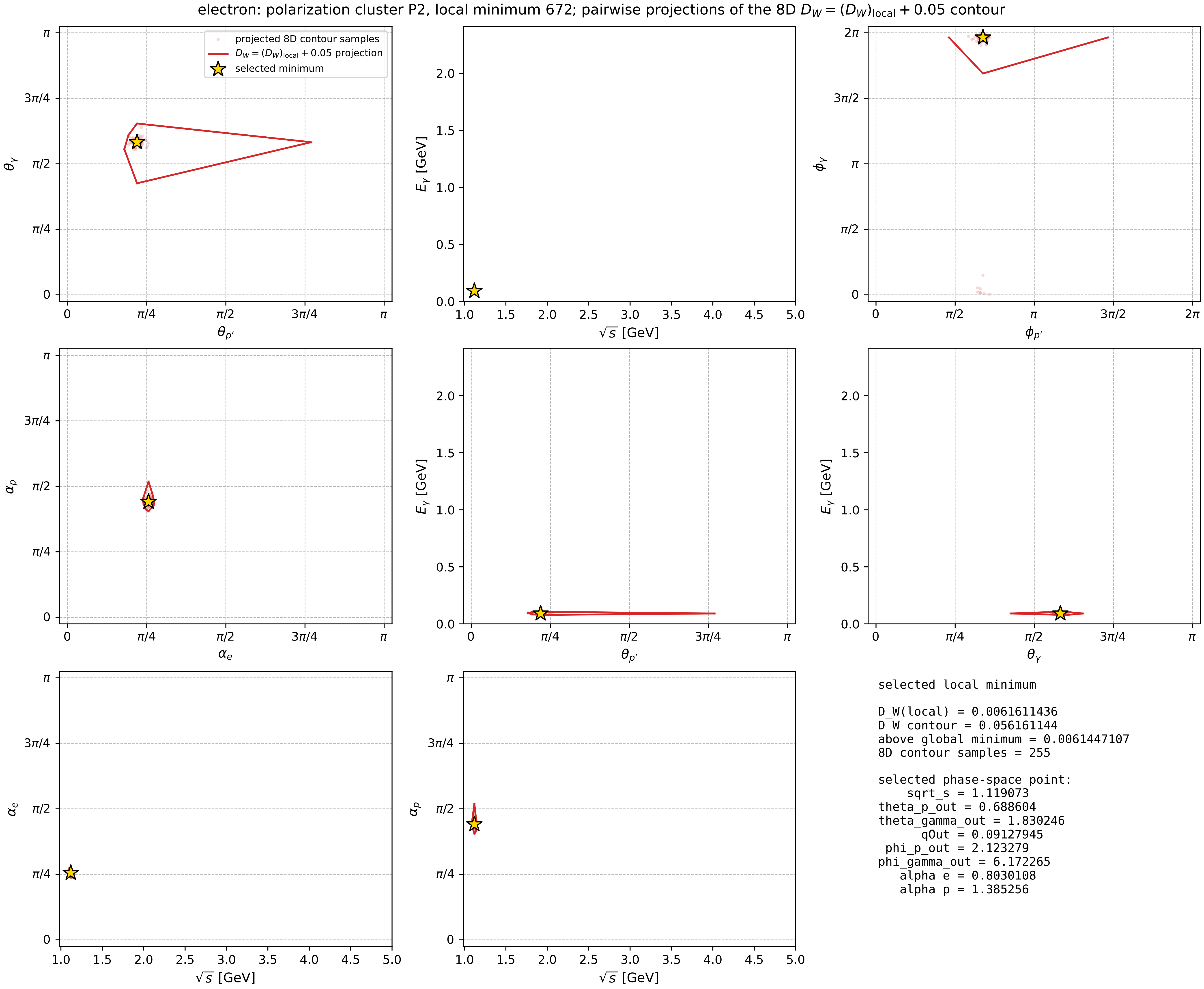


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_630 D_W=0.00486636
region: polarization_cluster_02_local_minimum_630
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.81486835 rad
incoming lepton: 0.685964|+> +0.727635|->
proton mixing angle: alpha_p=1.4068505 rad
incoming proton: 0.163212|+> +0.986591|->
s=1.18937, $\sqrt{s}$=1.09058, $|\vec{P}|$=0.141909, $|\vec{P}'|$=0.00511444
$\theta_{p'}$=2.4707, θ_γ=2.03843, $\phi_{p'}$=4.51463, ϕ_γ=0.544222
E_γ=0.0762546, $\theta_{\ell'}$=1.04382, $\phi_{\ell'}$=3.65027
Q^2=0.0325523, x_B=0.185351, t=-0.0211878, W^2=1.02292, y=0.567394
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1419 p_x= 0.0000 p_y= 0.0000 p_z= -0.1419
P E= 0.9487 p_x= 0.0000 p_y= 0.0000 p_z= 0.1419
ℓ' E= 0.0763 p_x= -0.0576 p_y= -0.0321 p_z= 0.0384
P' E= 0.9380 p_x= -0.0006 p_y= -0.0031 p_z= -0.0040
q_γ E= 0.0763 p_x= 0.0582 p_y= 0.0352 p_z= -0.0344
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Transverse plane
p_x [GeV]
p_y [GeV]
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
Leading final-state amplitudes (N=5, retained=98.2%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-6.98e+01+7.11e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
A=-1.89e+01-9.26e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=3.05e+01+8.88e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=2.64e+01-8.54e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-2.61e+01-4.79e-01i
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 630; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

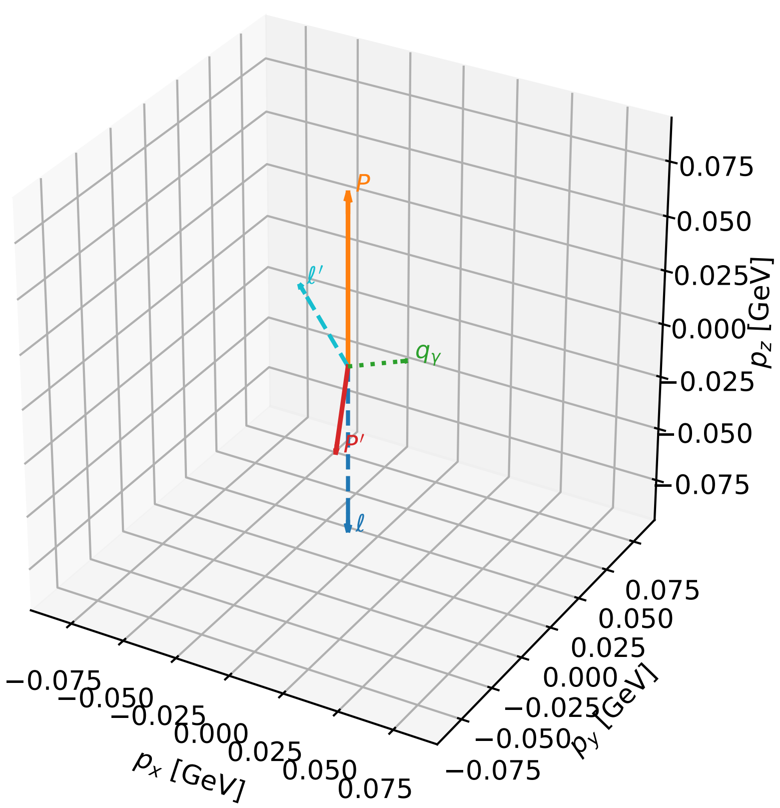



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
ℓ'
q_γ
ℓ
dw_mixing_angles_polarization_02_local_minimum_672 D_W=0.00616114
region: polarization_cluster_02_local_minimum_672
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80301075 rad
incoming lepton: 0.694544|+> +0.71945|->
proton mixing angle: alpha_p=1.3852562 rad
incoming proton: 0.184477|+> +0.982837|->
s=1.25232, $\sqrt{s}$=1.11907, $|\vec{P}|$=0.166423, $|\vec{P}'|$=0.00263204
$\theta_{p'}$=0.688604, θ_γ=1.83025, $\phi_{p'}$=2.12328, ϕ_γ=6.17226
E_γ=0.0912795, $\theta_{\ell'}$=1.33031, $\phi_{\ell'}$=3.04578
Q^2=0.0370038, x_B=0.177458, t=-0.0268126, W^2=1.05136, y=0.55982
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1664 p_x= 0.0000 p_y= 0.0000 p_z= -0.1664
P E= 0.9526 p_x= 0.0000 p_y= 0.0000 p_z= 0.1664
ℓ' E= 0.0898 p_x= -0.0868 p_y= 0.0083 p_z= 0.0214
P' E= 0.9380 p_x= -0.0009 p_y= 0.0014 p_z= 0.0020
q_γ E= 0.0913 p_x= 0.0877 p_y= -0.0098 p_z= -0.0234
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=99.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=6.40e+01+6.97e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-6.43e+01-1.13e-01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=6.38e+01-5.64e+00i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.08e+01-1.97e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-1.35e+01+2.13e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-1.12e+01-2.25e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 672; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0061611436
D_W contour = 0.056161144
above global minimum = 0.0061447107
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.119073
theta_p_out = 0.688604
theta_gamma_out = 1.830246
qOut = 0.09127945
phi_p_out = 2.123279
phi_gamma_out = 6.172265
alpha_e = 0.8030108
alpha_p = 1.385256

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta

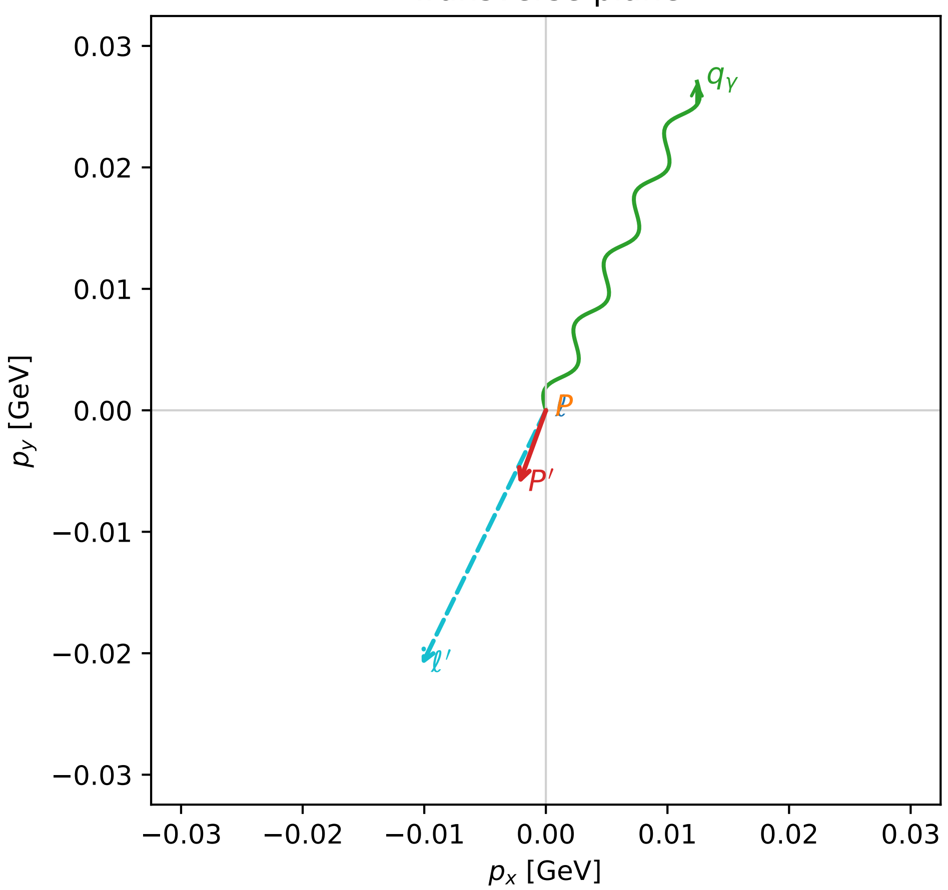


dw_mixing_angles_polarization_02_local_minimum_674 D_W=0.00620347
region: polarization_cluster_02_local_minimum_674
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80636526 rad
incoming lepton: 0.692126|+> +0.721776|->
proton mixing angle: alpha_p=1.2549437 rad
incoming proton: 0.310627|+> +0.950532|->

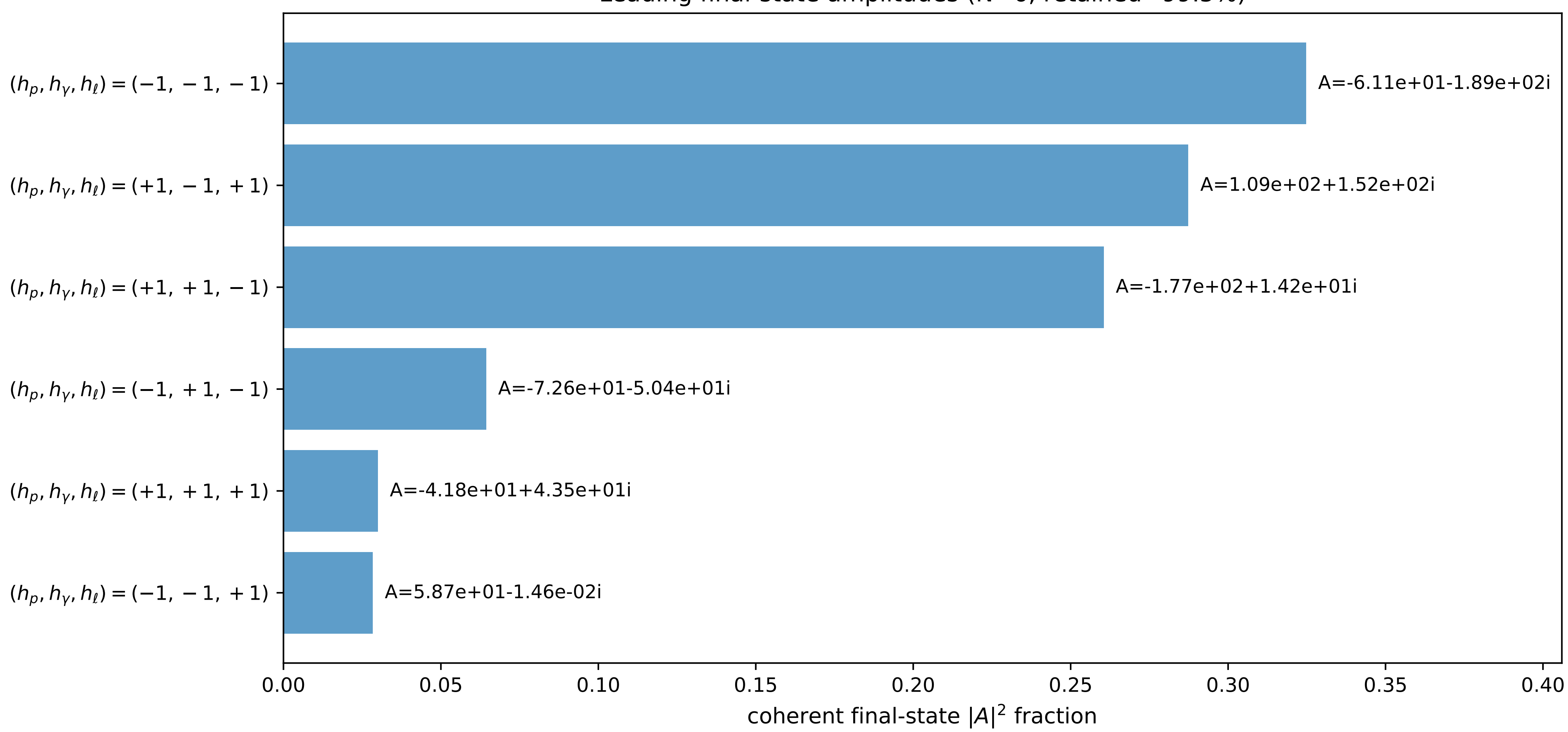
s=1.04109, $\sqrt{s}$=1.02034, $|\vec{P}|$=0.0790131, $|\vec{P}'|$=0.0385693
$\theta_{p'}$=2.97301, θ_γ=1.80878, $\phi_{p'}$=4.36215, ϕ_γ=1.14001
E_γ=0.0306365, $\theta_{\ell'}$=0.47605, $\phi_{\ell'}$=4.25929
Q^2=0.0151943, x_B=0.209427, t=-0.0137328, W^2=0.937202, y=0.449954

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0790 p_x= 0.0000 p_y= 0.0000 p_z= -0.0790
P E= 0.9413 p_x= 0.0000 p_y= 0.0000 p_z= 0.0790
ℓ' E= 0.0509 p_x= -0.0102 p_y= -0.0210 p_z= 0.0452
P' E= 0.9388 p_x= -0.0022 p_y= -0.0061 p_z= -0.0380
q_γ E= 0.0306 p_x= 0.0124 p_y= 0.0271 p_z= -0.0072

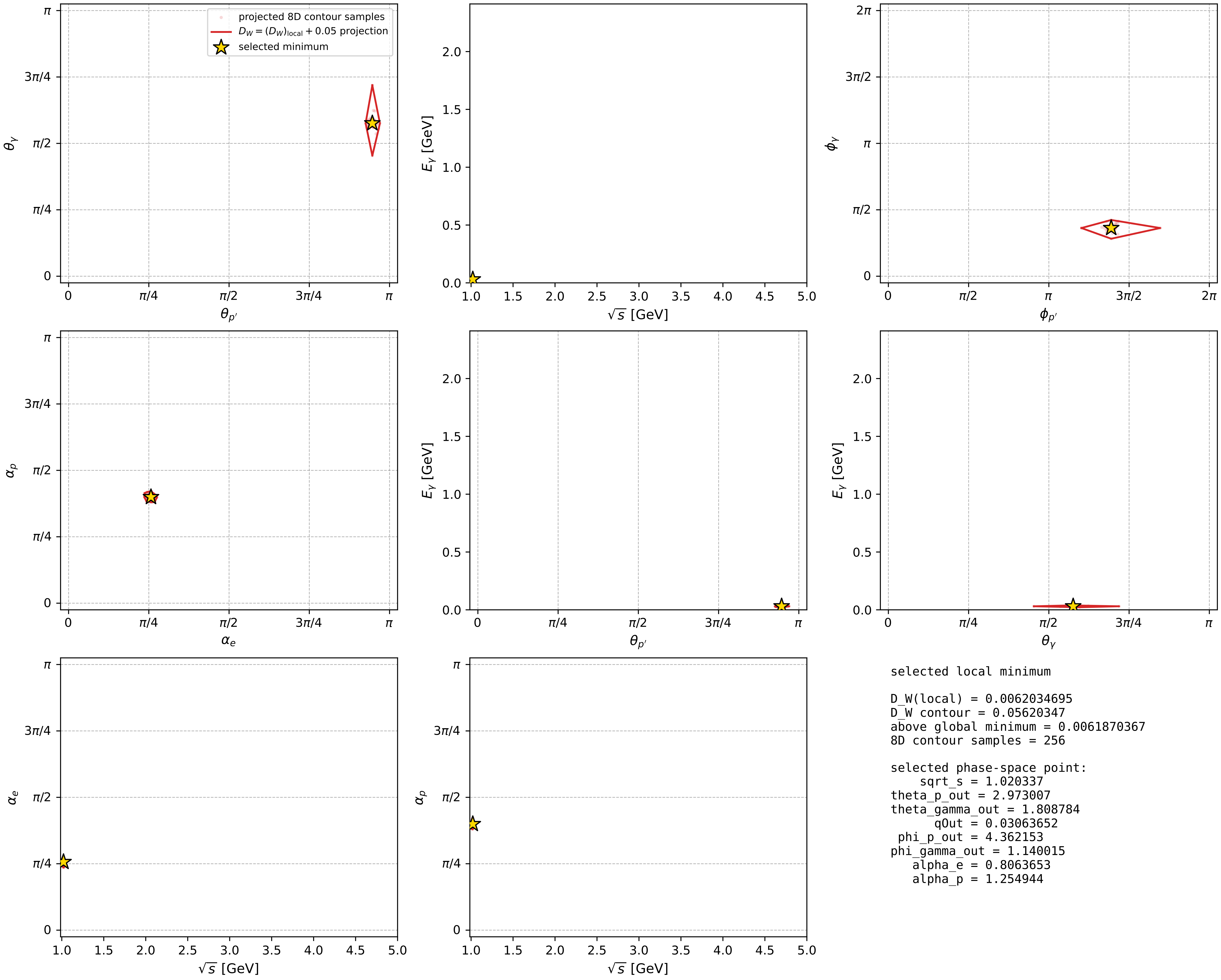
Transverse plane

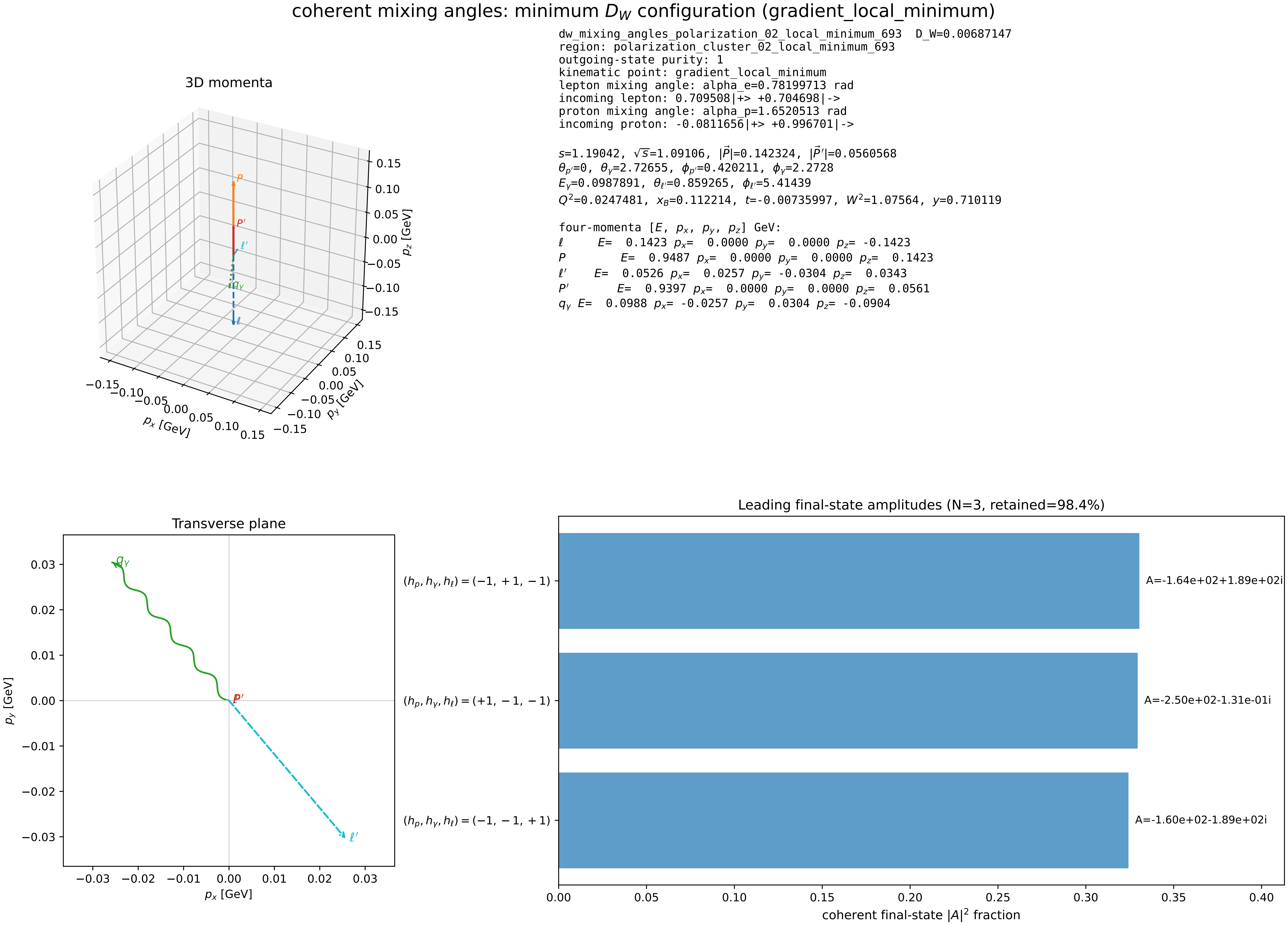


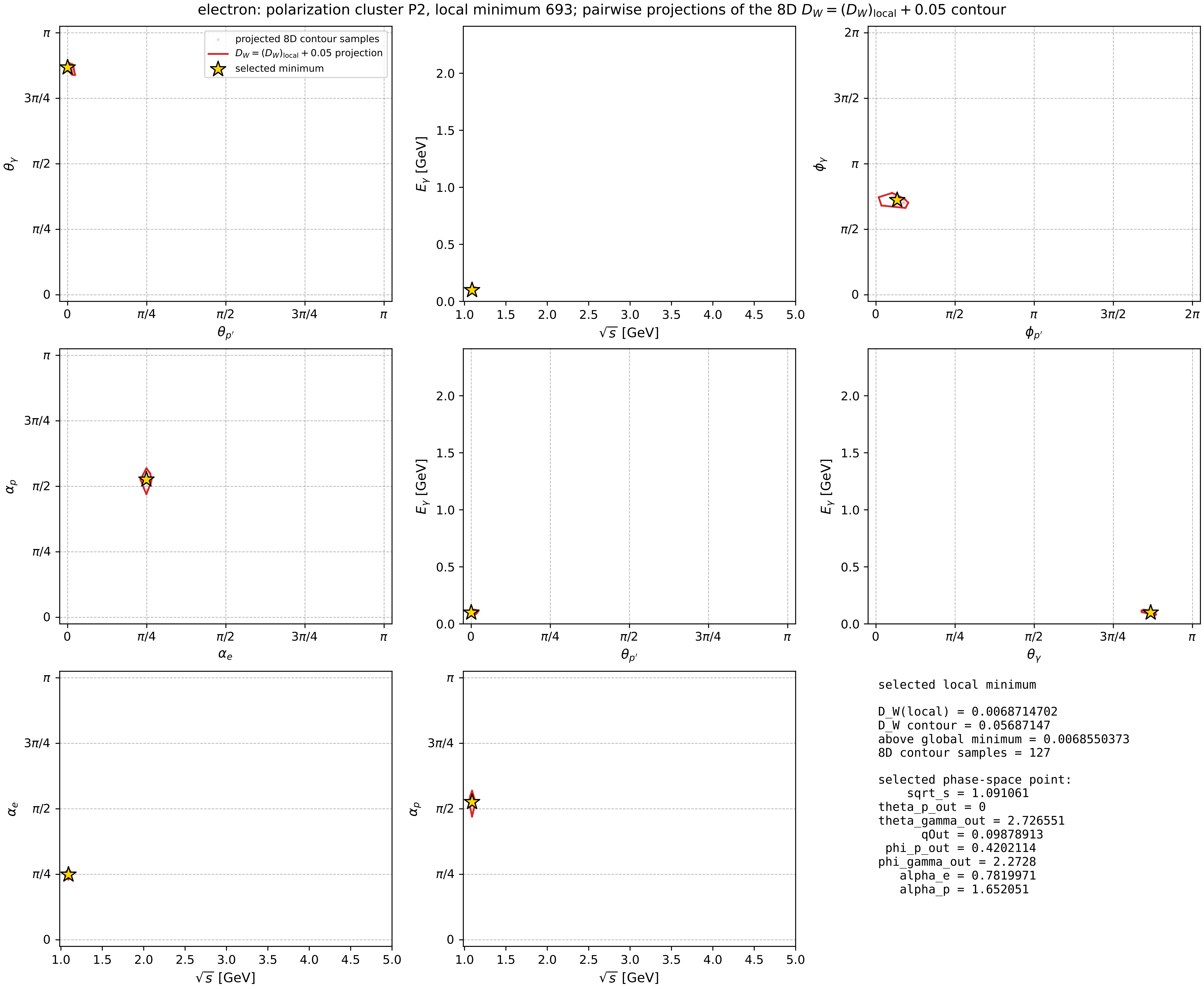
Leading final-state amplitudes (N=6, retained=99.5%)



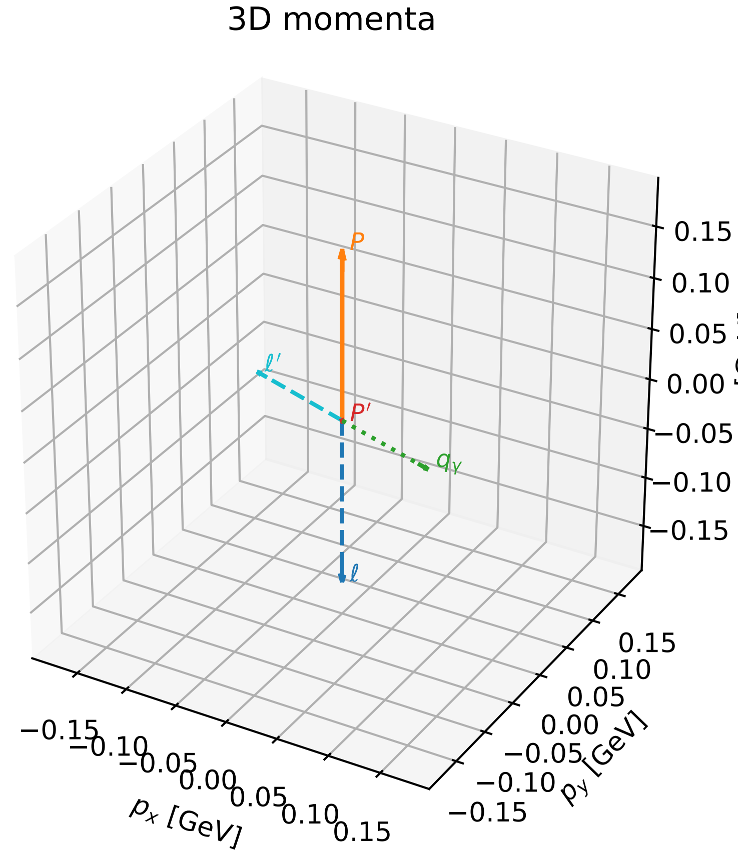
electron: polarization cluster P2, local minimum 674; pairwise projections of the 8D $D_W = (D_W)_{\rm local} + 0.05$ contour




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_693 D_W=0.00687147
region: polarization_cluster_02_local_minimum_693
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78199713 rad
incoming lepton: 0.709508|+> +0.704698|->
proton mixing angle: alpha_p=1.6520513 rad
incoming proton: -0.0811656|+> +0.996701|->
s=1.19042, $\sqrt{s}$=1.09106, $|\vec{P}|$=0.142324, $|\vec{P}'|$=0.0560568
$\theta_{p'}$=0, θ_γ=2.72655, $\phi_{p'}$=0.420211, ϕ_γ=2.2728
E_γ=0.0987891, $\theta_{\ell'}$=0.859265, $\phi_{\ell'}$=5.41439
Q^2=0.0247481, x_B=0.112214, t=-0.00735997, W^2=1.07564, y=0.710119
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1423 p_x= 0.0000 p_y= 0.0000 p_z= -0.1423
P E= 0.9487 p_x= 0.0000 p_y= 0.0000 p_z= 0.1423
ℓ' E= 0.0526 p_x= 0.0257 p_y= -0.0304 p_z= 0.0343
P' E= 0.9397 p_x= 0.0000 p_y= 0.0000 p_z= 0.0561
q_γ E= 0.0988 p_x= -0.0257 p_y= 0.0304 p_z= -0.0904
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=98.4%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.64e+02+1.89e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-2.50e+02-1.31e-01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-1.60e+02-1.89e+02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 693; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0068714702
D_W contour = 0.05687147
above global minimum = 0.0068550373
8D contour samples = 127
selected phase-space point:
sqrt_s = 1.091061
theta_p_out = 0
theta_gamma_out = 2.726551
qOut = 0.09878913
phi_p_out = 0.4202114
phi_gamma_out = 2.2728
alpha_e = 0.7819971
alpha_p = 1.652051

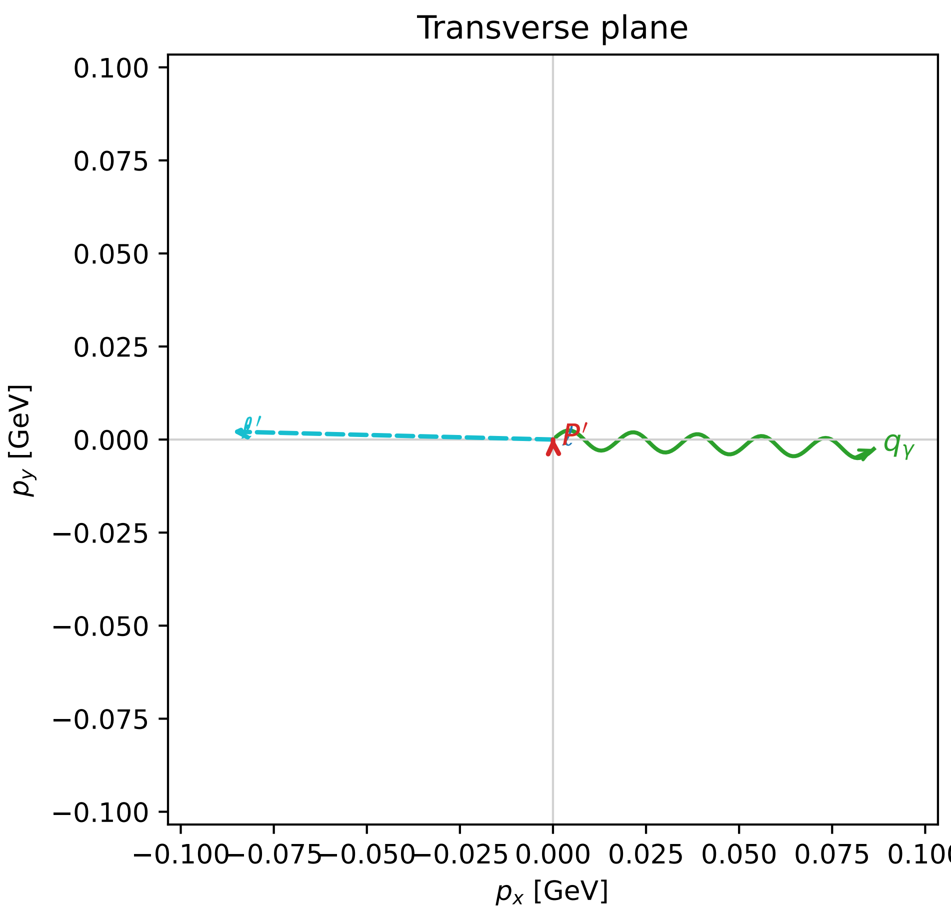
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

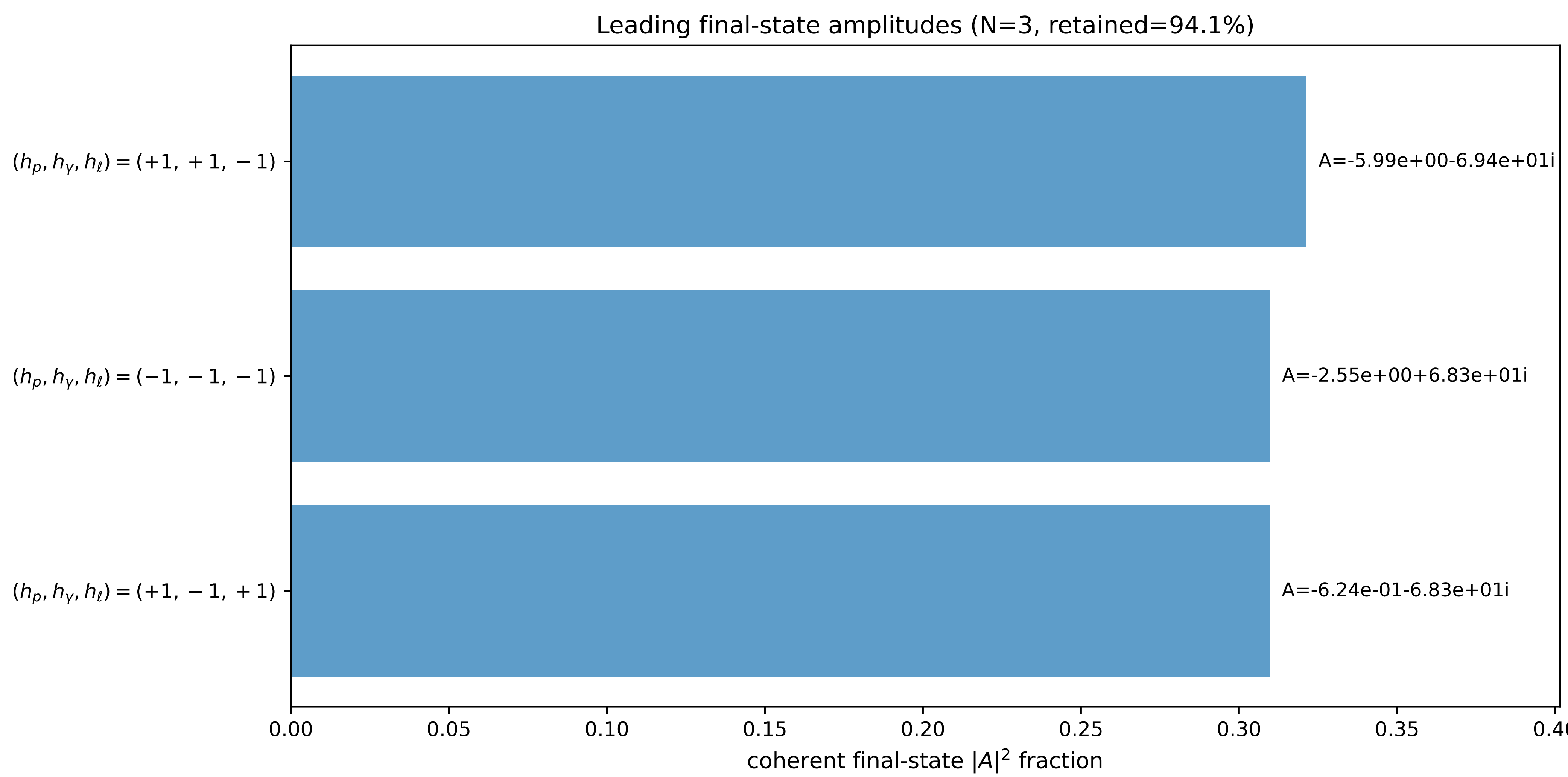


dw_mixing_angles_polarization_02_local_minimum_713 D_W=0.00788043
region: polarization_cluster_02_local_minimum_713
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80739396 rad
incoming lepton: 0.691384|+> +0.722488|->
proton mixing angle: alpha_p=1.3762109 rad
incoming proton: 0.19336|+> +0.981128|->

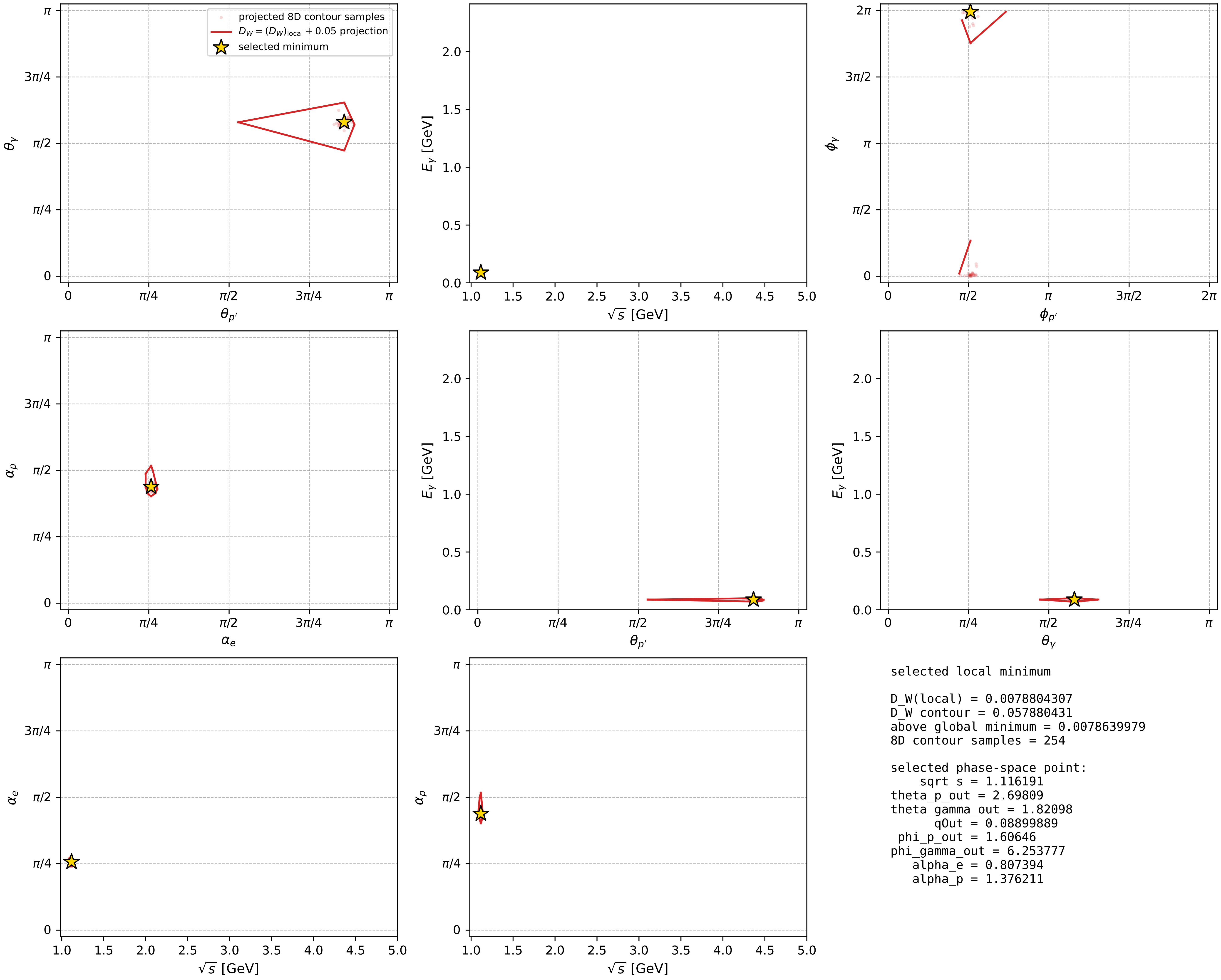
s=1.24588, $\sqrt{s}$=1.11619, $|\vec{P}|$=0.163967, $|\vec{P}'|$=0.000947889
$\theta_{p'}$=2.69809, θ_γ=1.82098, $\phi_{p'}$=1.60646, ϕ_γ=6.25378
E_γ=0.0889989, $\theta_{\ell'}$=1.31124, $\phi_{\ell'}$=3.11689
Q^2=0.0367552, x_B=0.180452, t=-0.0269645, W^2=1.04677, y=0.556456

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1640 p_x= 0.0000 p_y= 0.0000 p_z= -0.1640
P E= 0.9522 p_x= 0.0000 p_y= 0.0000 p_z= 0.1640
ℓ' E= 0.0892 p_x= -0.0862 p_y= 0.0021 p_z= 0.0229
P' E= 0.9380 p_x= -0.0000 p_y= 0.0004 p_z= -0.0009
q_γ E= 0.0890 p_x= 0.0862 p_y= -0.0025 p_z= -0.0220

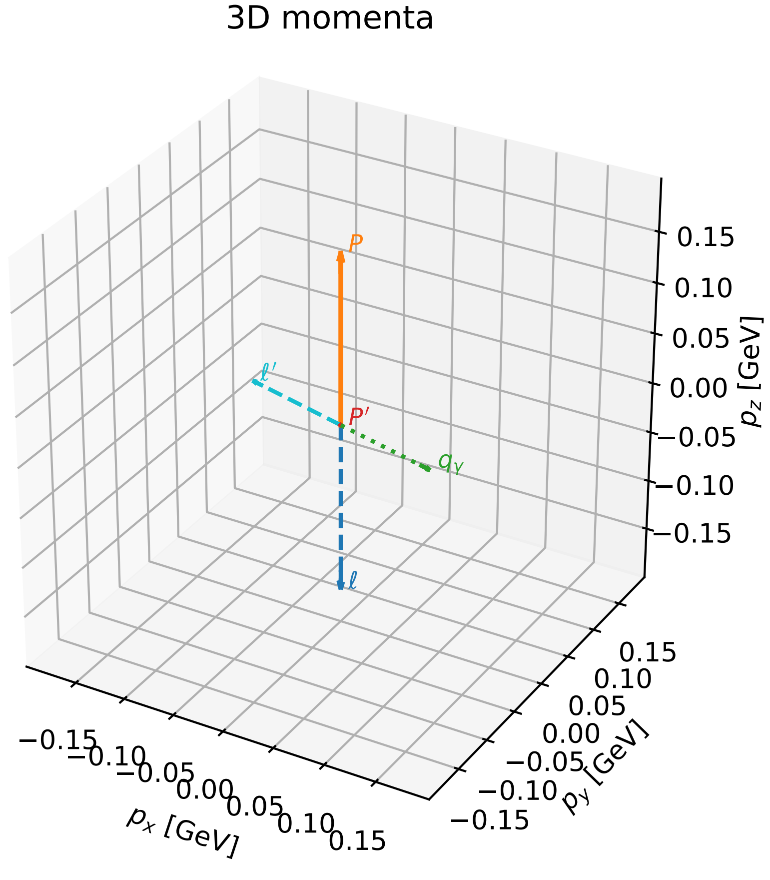




electron: polarization cluster P2, local minimum 713; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



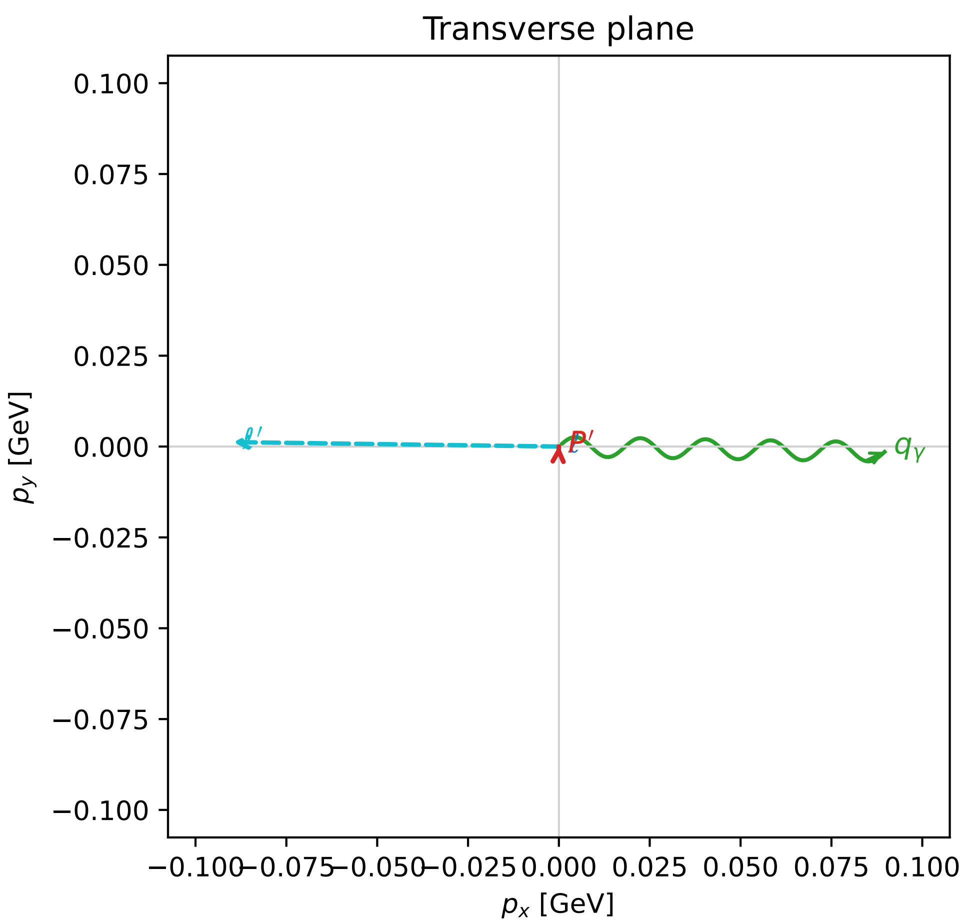
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

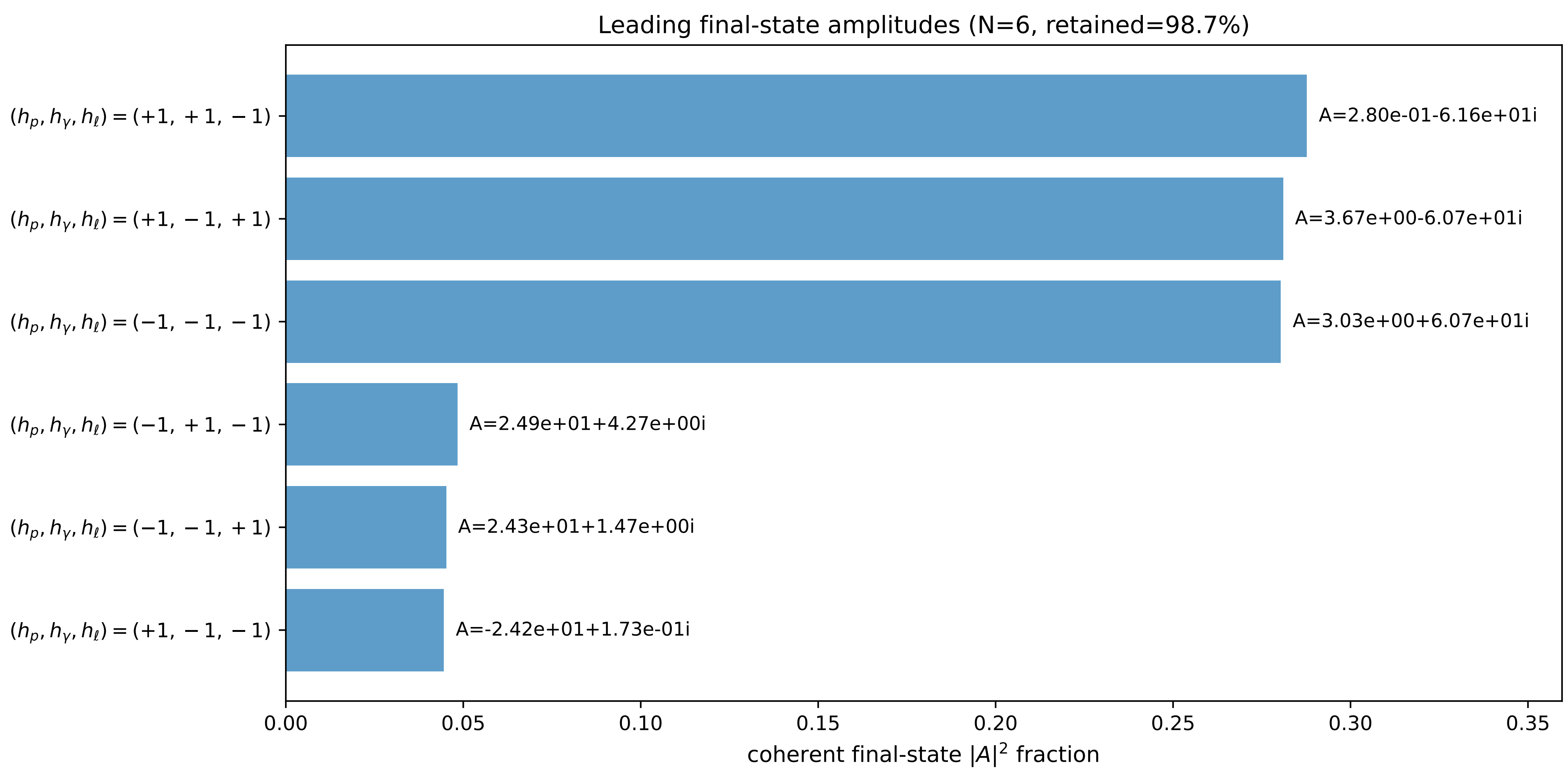



```
dw_mixing_angles_polarization_02_local_minimum_717  D_W=0.00798736
region: polarization_cluster_02_local_minimum_717
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80197977 rad
incoming lepton: 0.695285|+> +0.718734|->
proton mixing angle: alpha_p=1.3702503 rad
incoming proton: 0.199204|+> +0.979958|->
```

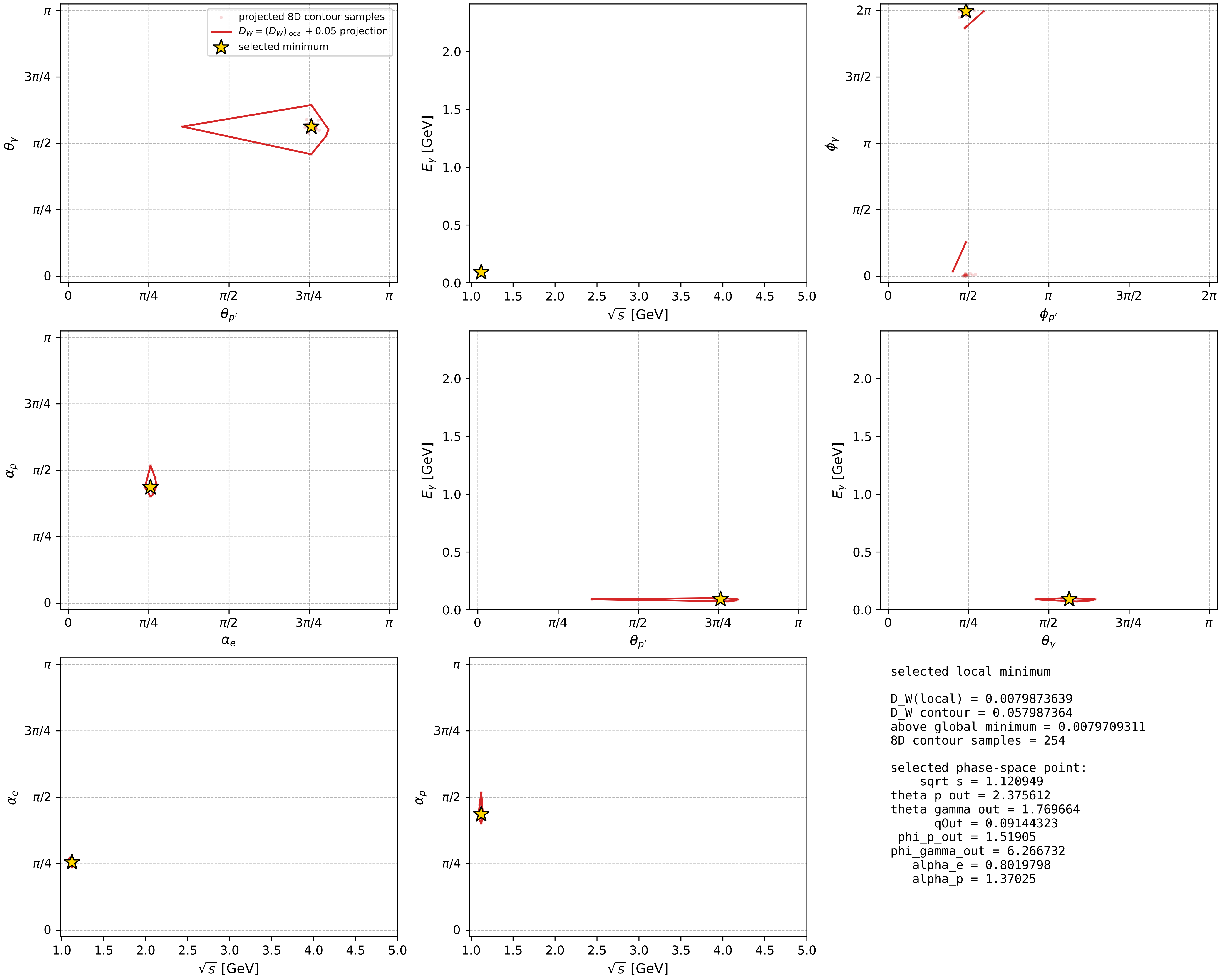
s=1.25653, $\sqrt{s}$=1.12095, $|\vec{P}|$=0.168019, $|\vec{P}'|$=0.000362875
$\theta_{p'}$=2.37561, θ_γ=1.76966, $\phi_{p'}$=1.51905, ϕ_γ=6.26673
E_γ=0.0914432, $\theta_{\ell'}$=1.36915, $\phi_{\ell'}$=3.12794
Q^2=0.0369076, x_B=0.177062, t=-0.0280955, W^2=1.05138, y=0.553368

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1680 p_x= 0.0000 p_y= 0.0000 p_z= -0.1680
P E= 0.9529 p_x= 0.0000 p_y= 0.0000 p_z= 0.1680
ℓ' E= 0.0915 p_x= -0.0896 p_y= 0.0012 p_z= 0.0183
P' E= 0.9380 p_x= 0.0000 p_y= 0.0003 p_z= -0.0003
q_γ E= 0.0914 p_x= 0.0896 p_y= -0.0015 p_z= -0.0181





electron: polarization cluster P2, local minimum 717; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

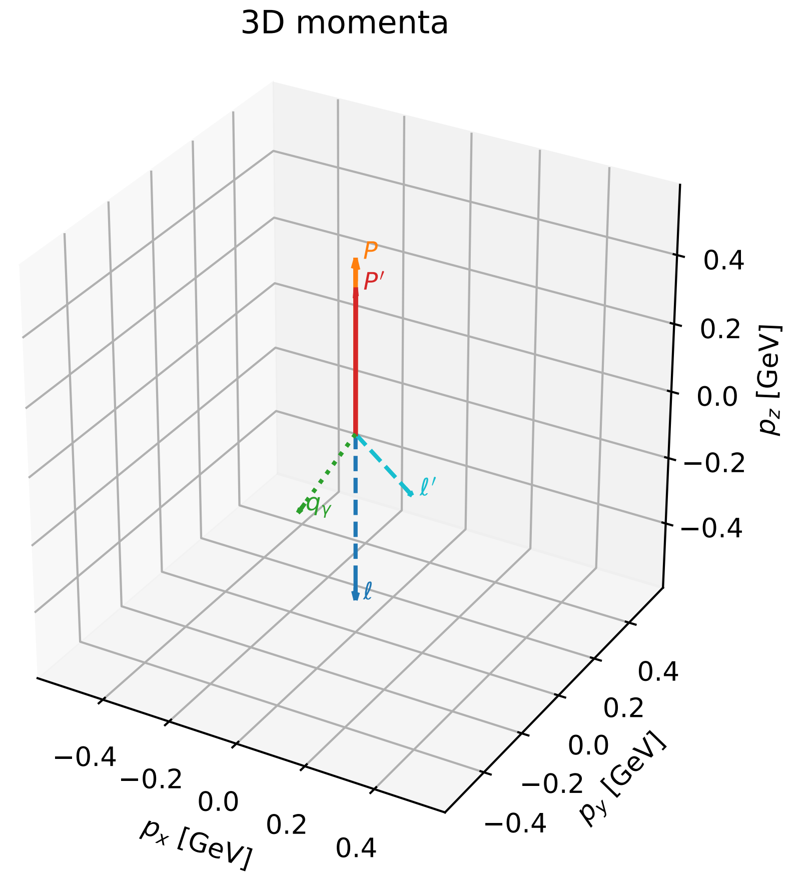


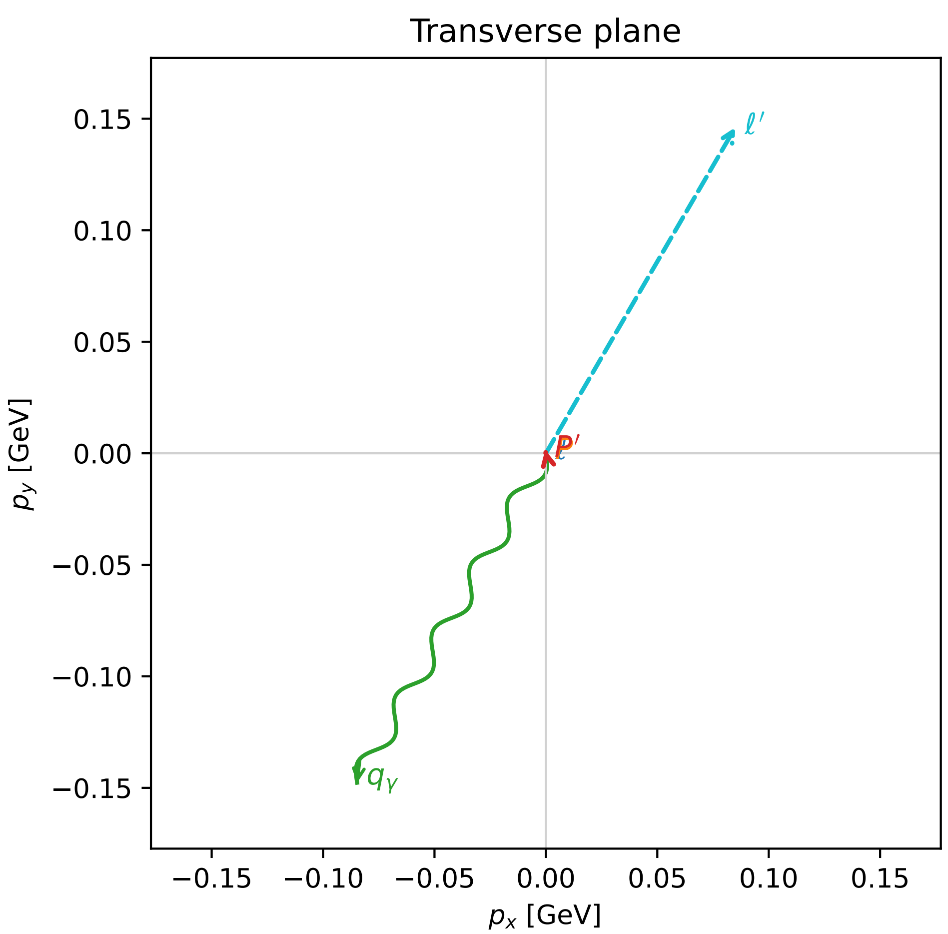
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

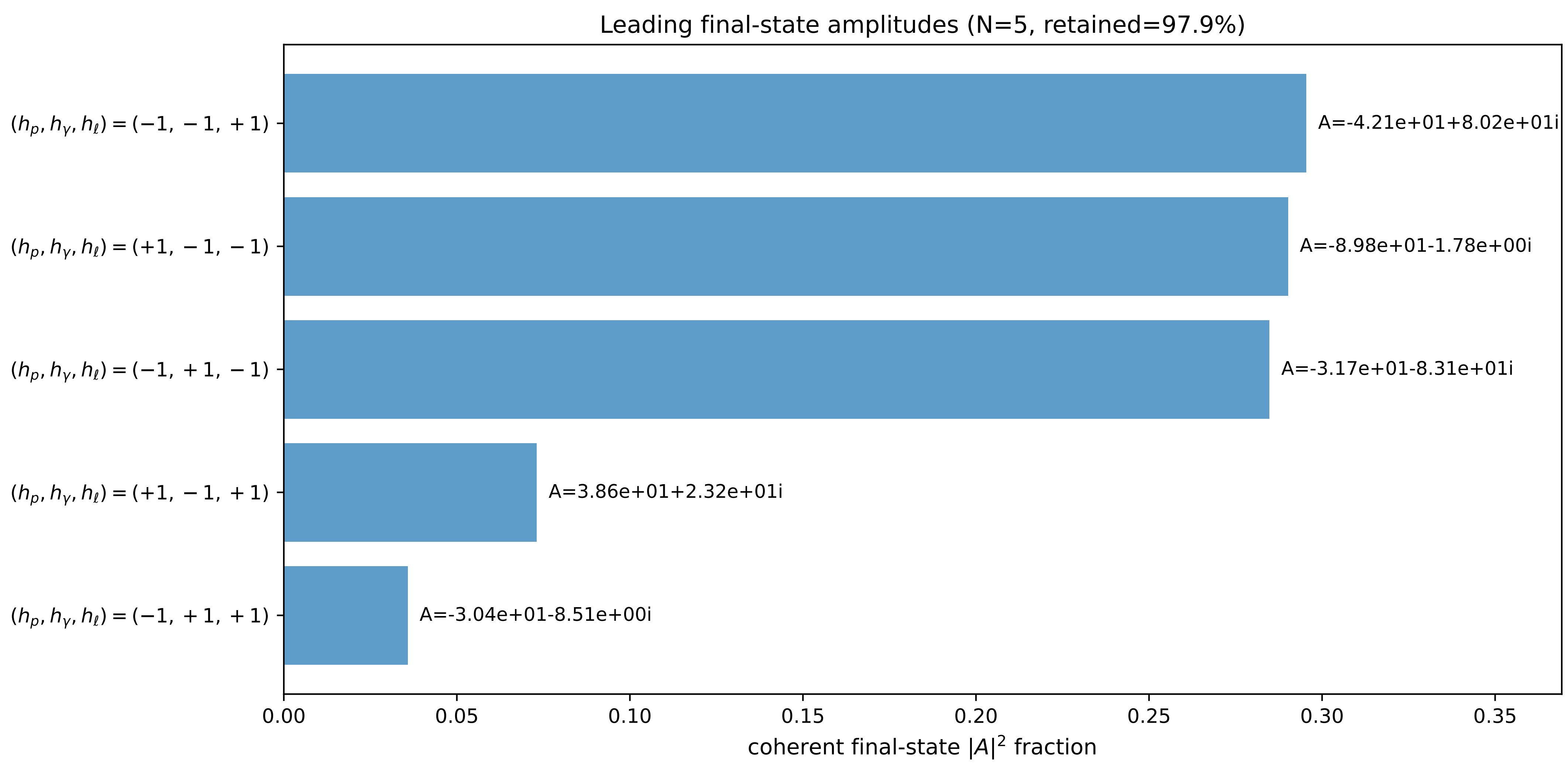
```
dw_mixing_angles_polarization_02_local_minimum_724  D_W=0.00814214
region: polarization_cluster_02_local_minimum_724
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.72681034 rad
incoming lepton: 0.747298|+> +0.664489|->
proton mixing angle: alpha_p=1.8452709 rad
incoming proton: -0.271041|+> +0.962568|->
```

s=2.44745, $\sqrt{s}$=1.56443, $|\vec{P}|$=0.501015, $|\vec{P}'|$=0.416405
$\theta_{p'}$=0.00378515, θ_γ=2.39332, $\phi_{p'}$=1.78681, ϕ_γ=4.19168
E_γ=0.250265, $\theta_{\ell'}$=2.51375, $\phi_{\ell'}$=1.04382
Q^2=0.0550144, x_B=0.0762143, t=-0.00578196, W^2=1.54667, y=0.460471

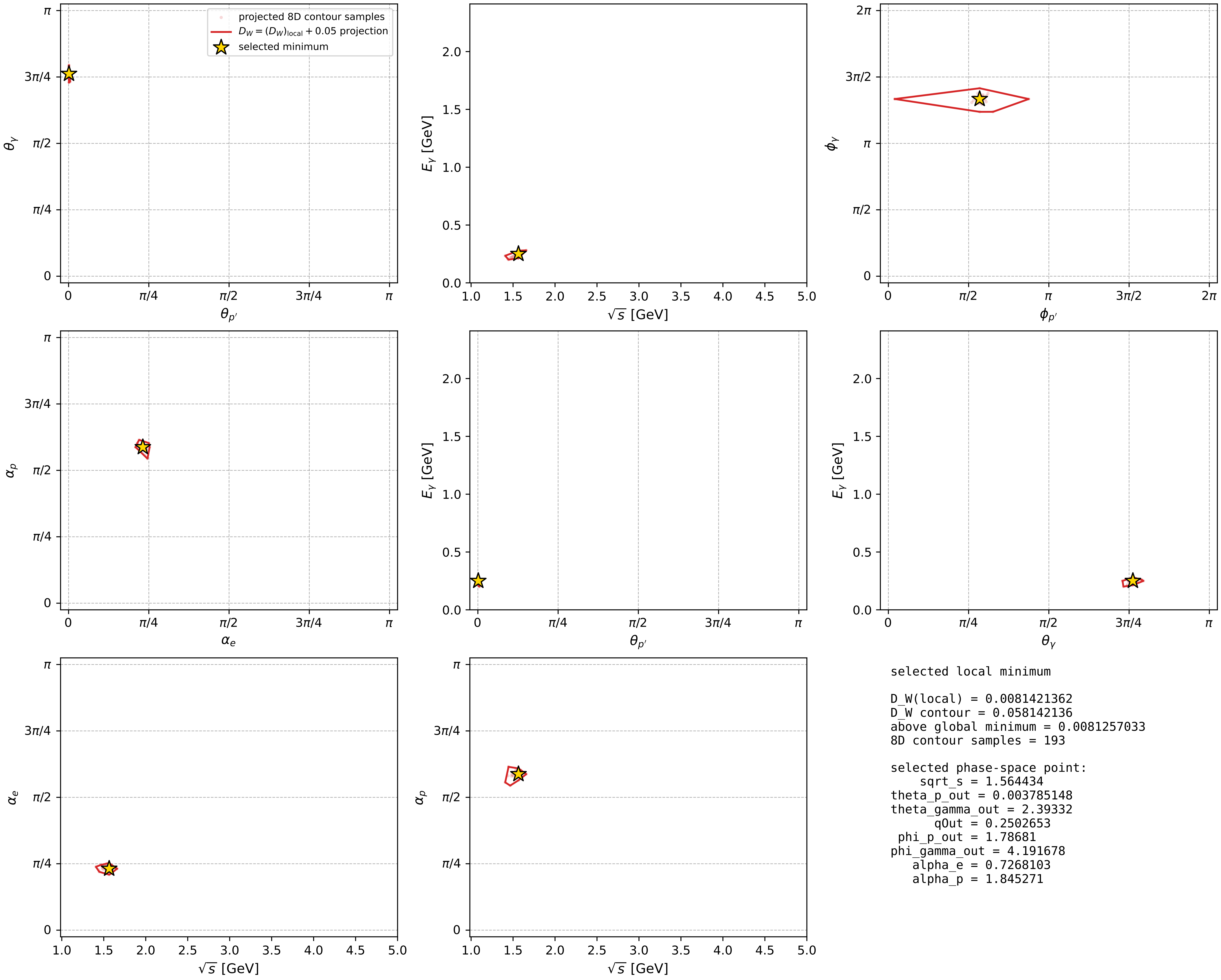
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5010 p_x= 0.0000 p_y= 0.0000 p_z= -0.5010
P E= 1.0634 p_x= 0.0000 p_y= 0.0000 p_z= 0.5010
ℓ' E= 0.2879 p_x= 0.0850 p_y= 0.1462 p_z= -0.2330
P' E= 1.0263 p_x= -0.0003 p_y= 0.0015 p_z= 0.4164
q_γ E= 0.2503 p_x= -0.0847 p_y= -0.1477 p_z= -0.1834







electron: polarization cluster P2, local minimum 724; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



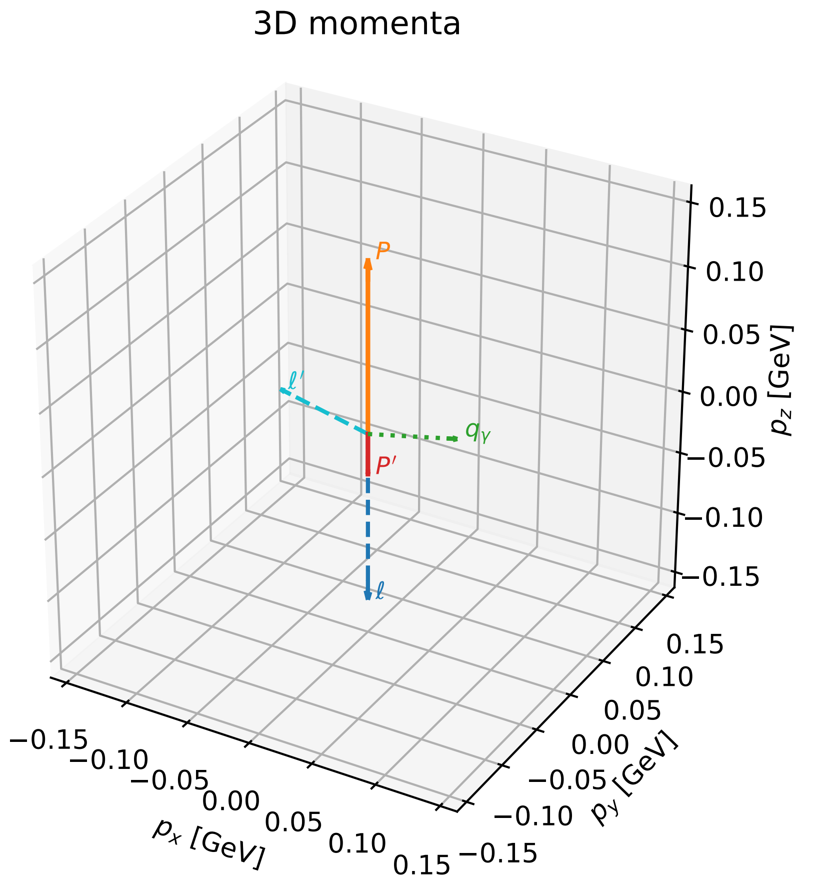
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

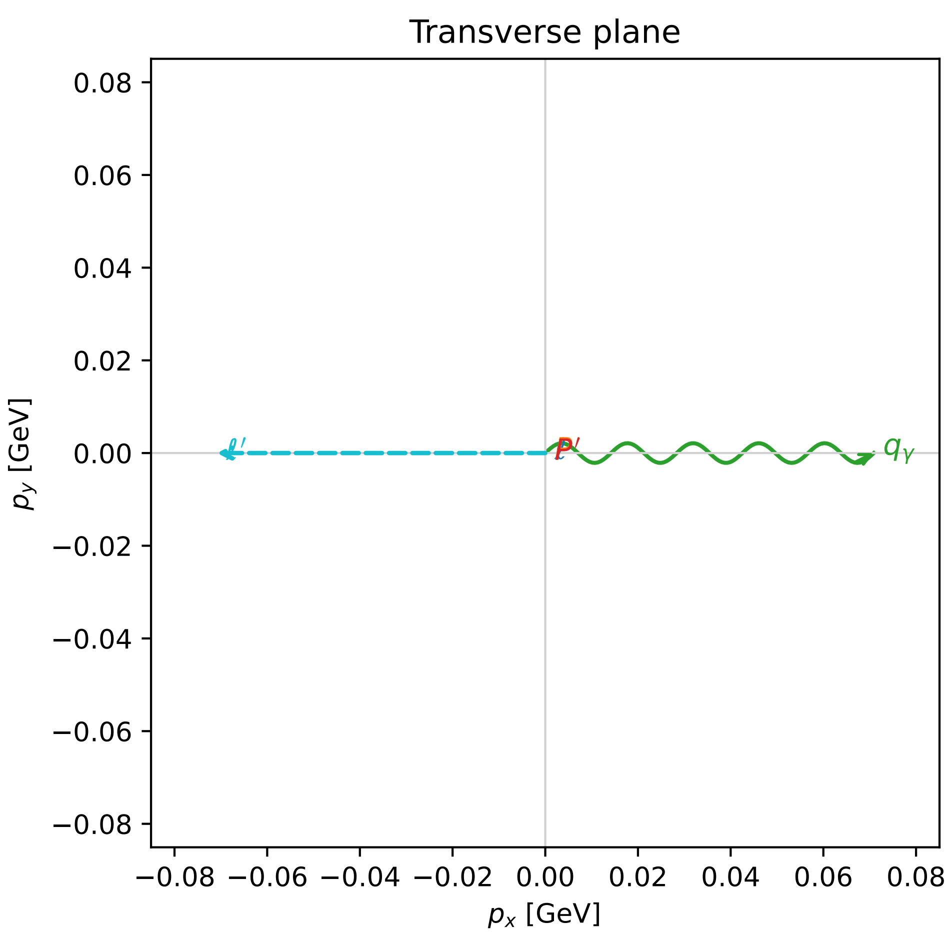
```
dw_mixing_angles_polarization_02_local_minimum_726  D_W=0.00815666
region: polarization_cluster_02_local_minimum_726
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80690904 rad
incoming lepton: 0.691734|+> +0.722153|->
proton mixing angle: alpha_p=1.3699192 rad
incoming proton: 0.199529|+> +0.979892|->
```

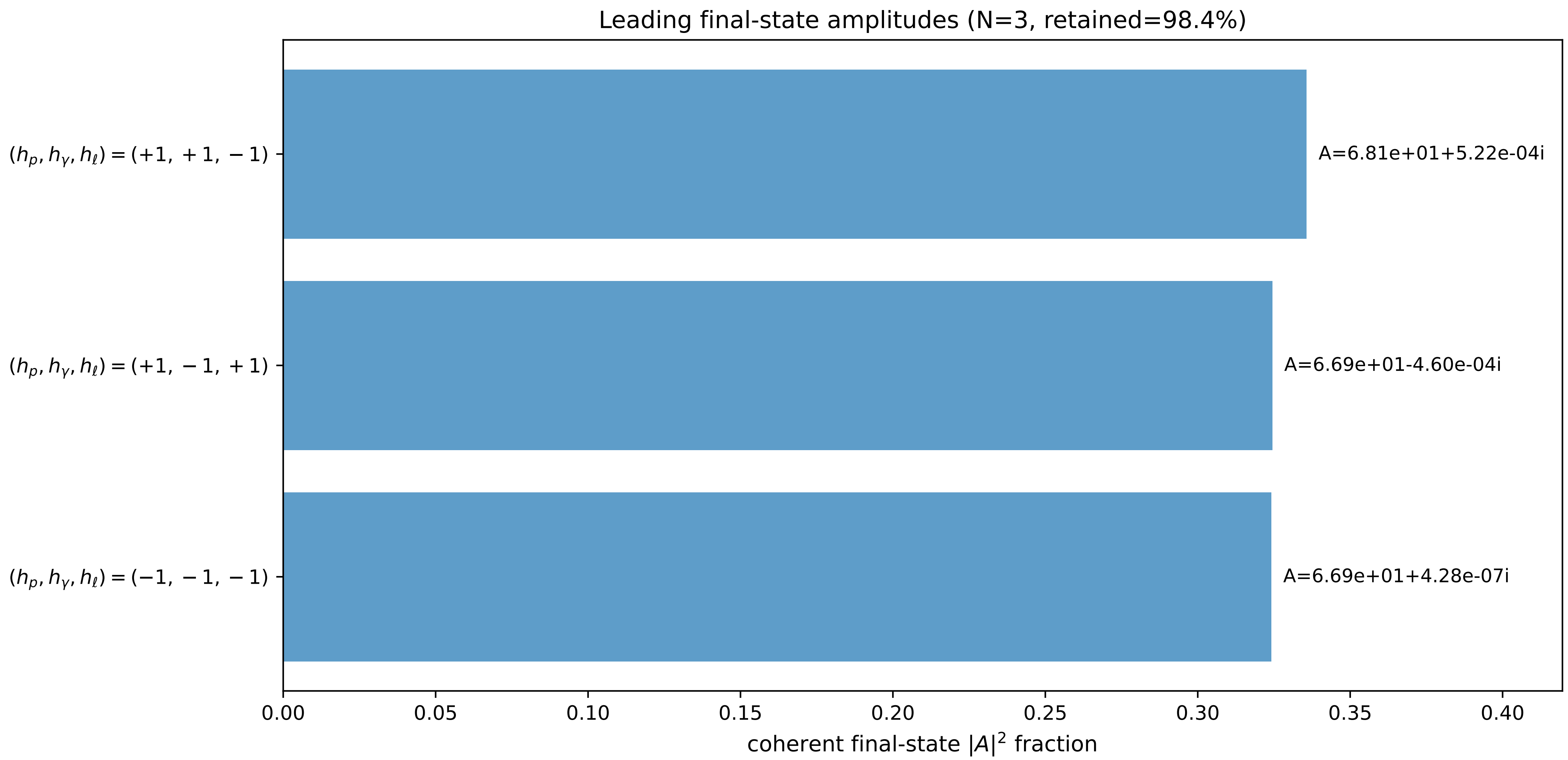
s=1.17502, $\sqrt{s}$=1.08398, $|\vec{P}|$=0.136152, $|\vec{P}'|$=0.0323792
$\theta_{p'}$=3.14159, θ_γ=1.34233, $\phi_{p'}$=4.50934, ϕ_γ=7.32617e-06
E_γ=0.0727764, $\theta_{\ell'}$=1.35019, $\phi_{\ell'}$=3.1416
Q^2=0.0241106, x_B=0.149026, t=-0.0283168, W^2=1.01752, y=0.548109

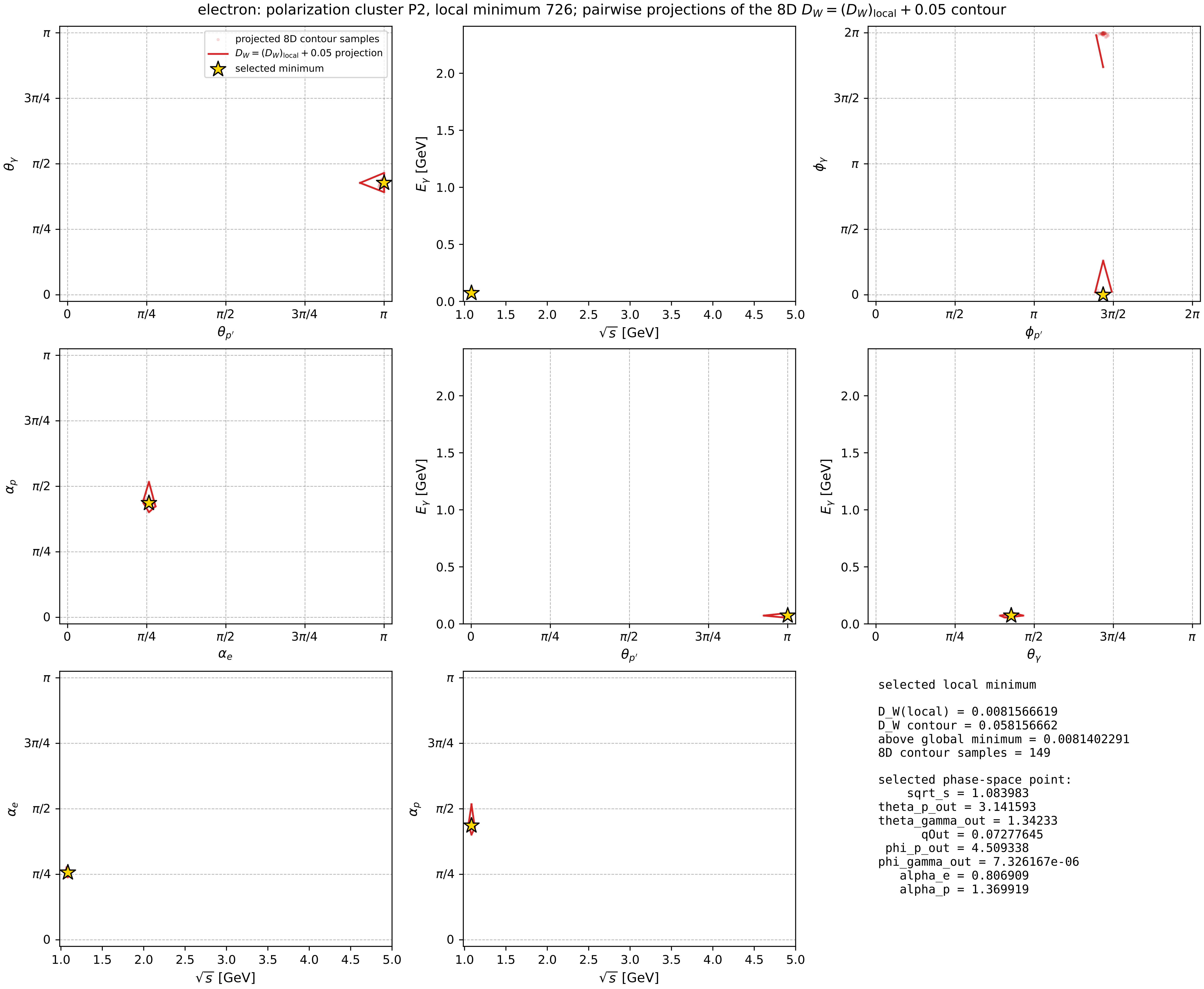
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1362	p_x= 0.0000	p_y= 0.0000	p_z= -0.1362
P	E= 0.9478	p_x= 0.0000	p_y= 0.0000	p_z= 0.1362
ℓ'	E= 0.0726	p_x= -0.0709	p_y= -0.0000	p_z= 0.0159
P'	E= 0.9386	p_x= -0.0000	p_y= -0.0000	p_z= -0.0324
q_γ	E= 0.0728	p_x= 0.0709	p_y= 0.0000	p_z= 0.0165

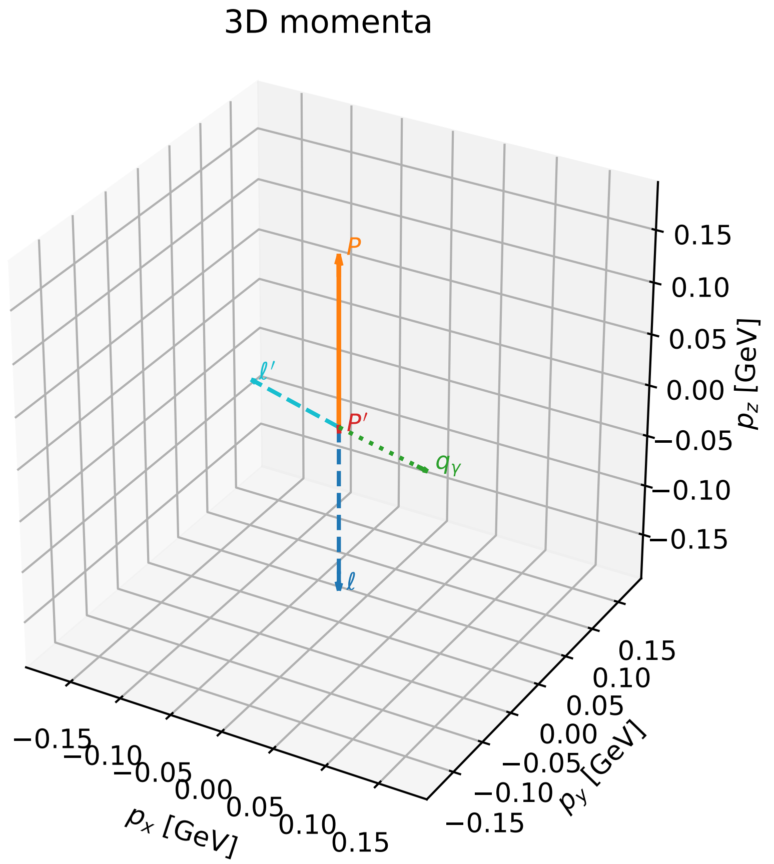







electron: polarization cluster P2, local minimum 726; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0081566619
D_W contour = 0.058156662
above global minimum = 0.0081402291
8D contour samples = 149
selected phase-space point:
sqrt_s = 1.083983
theta_p_out = 3.141593
theta_gamma_out = 1.34233
qOut = 0.07277645
phi_p_out = 4.509338
phi_gamma_out = 7.326167e-06
alpha_e = 0.806909
alpha_p = 1.369919

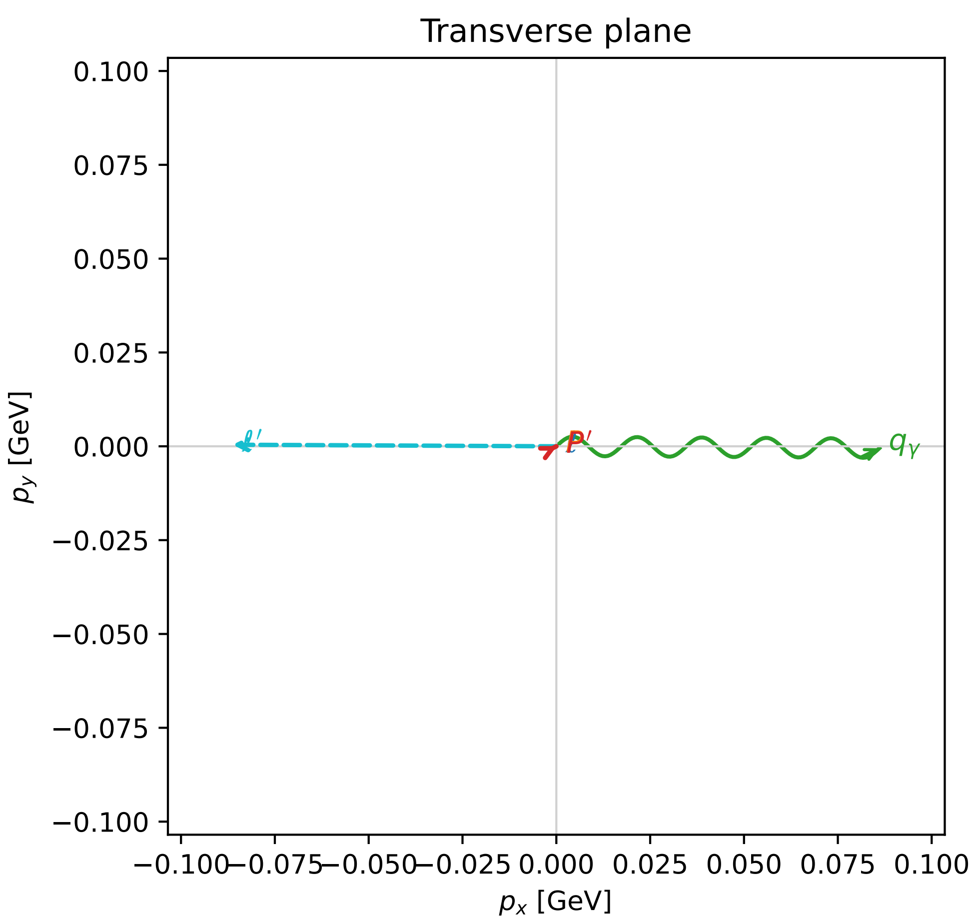
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

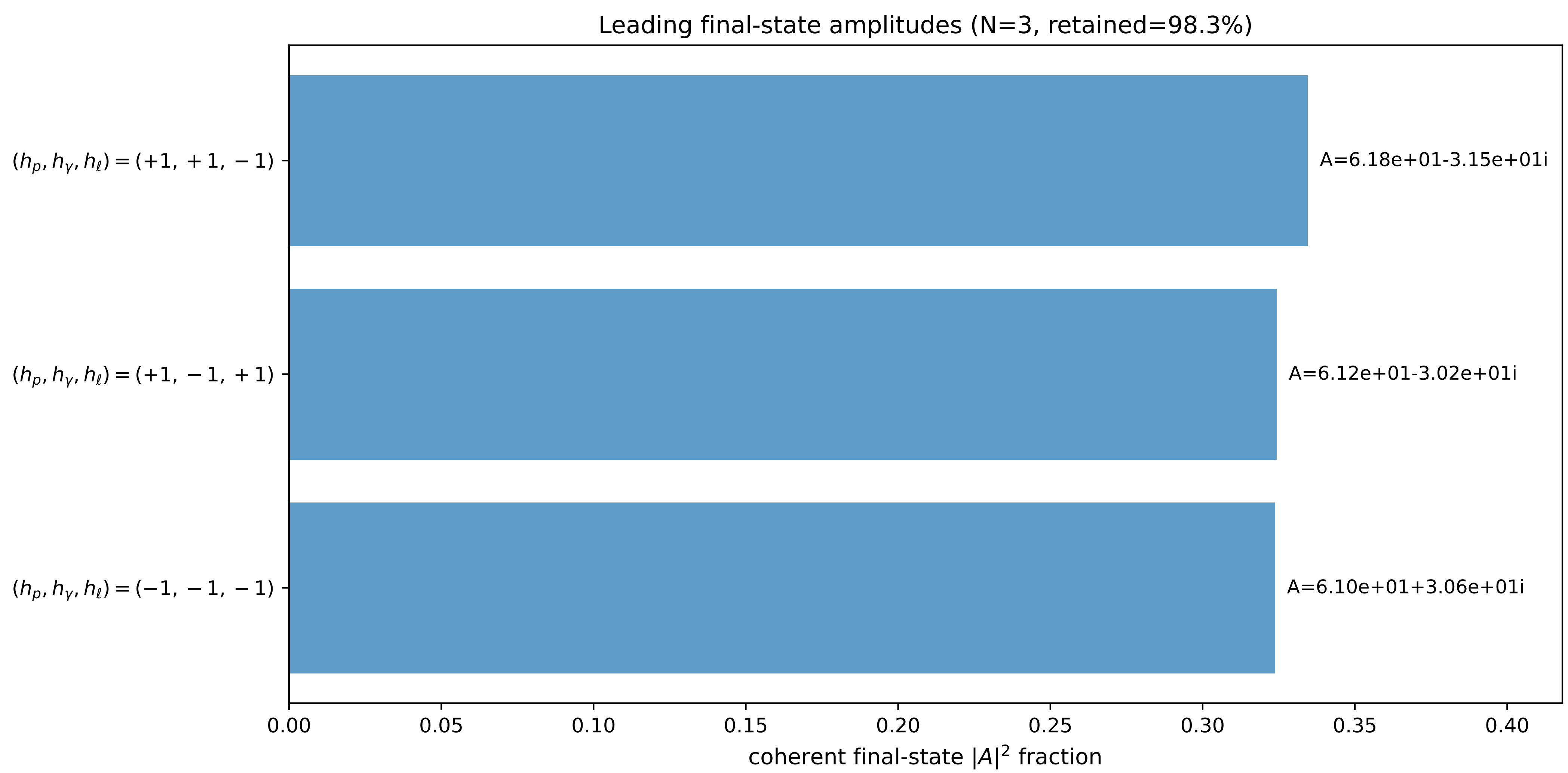



```
dw_mixing_angles_polarization_02_local_minimum_729  D_W=0.00833546
region: polarization_cluster_02_local_minimum_729
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80706385 rad
incoming lepton: 0.691622|+> +0.72226|->
proton mixing angle: alpha_p=1.3715292 rad
incoming proton: 0.197951|+> +0.980212|->
```

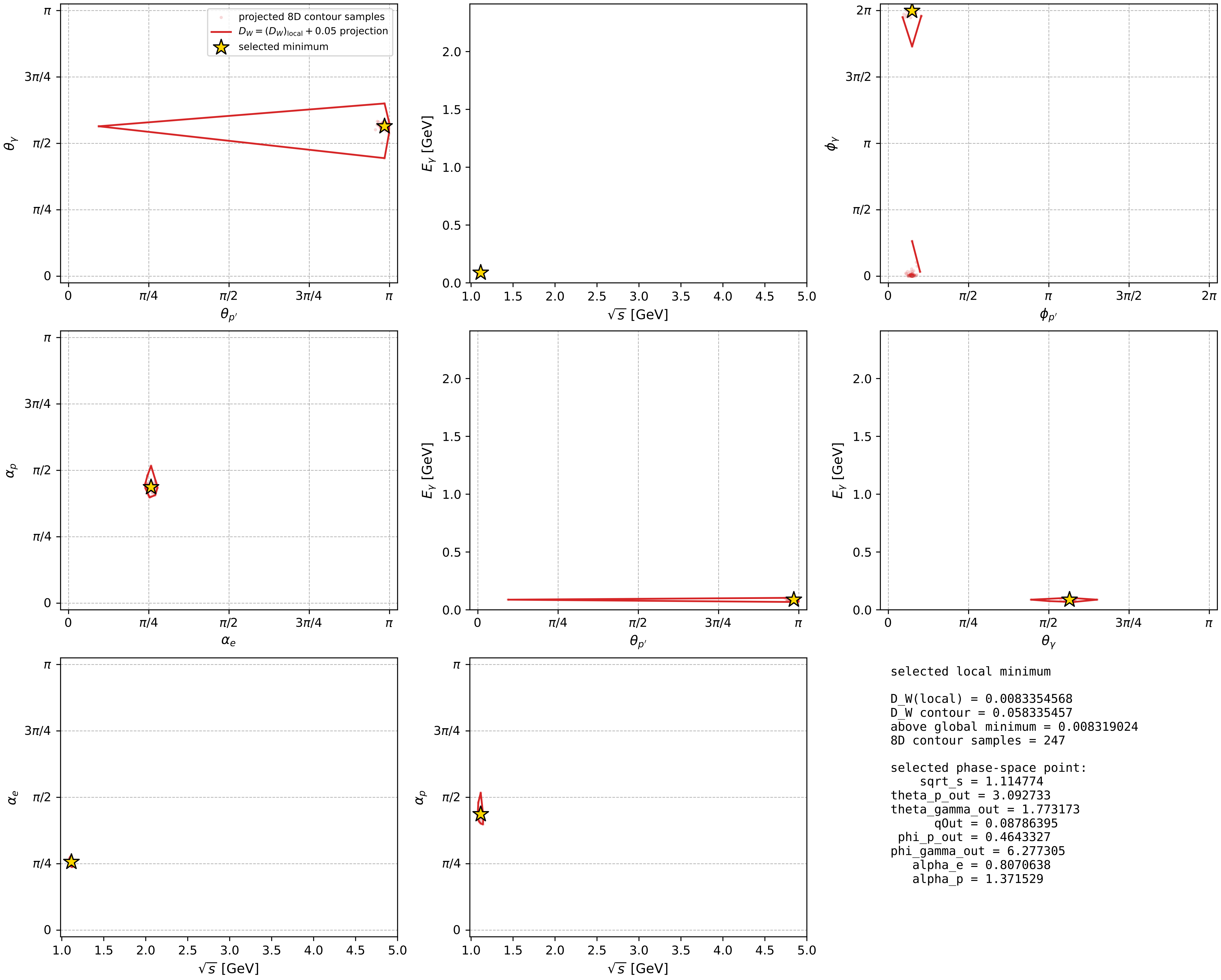
s=1.24272, $\sqrt{s}$=1.11477, $|\vec{P}|$=0.162758, $|\vec{P}'|$=0.00392398
$\theta_{p'}$=3.09273, θ_γ=1.77317, $\phi_{p'}$=0.464333, ϕ_γ=6.27731
E_γ=0.087864, $\theta_{\ell'}$=1.32561, $\phi_{\ell'}$=3.13672
Q^2=0.0359632, x_B=0.179252, t=-0.027585, W^2=1.04451, y=0.552885

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1628 p_x= 0.0000 p_y= 0.0000 p_z= -0.1628
P E= 0.9520 p_x= 0.0000 p_y= 0.0000 p_z= 0.1628
ℓ' E= 0.0889 p_x= -0.0862 p_y= 0.0004 p_z= 0.0216
P' E= 0.9380 p_x= 0.0002 p_y= 0.0001 p_z= -0.0039
q_γ E= 0.0879 p_x= 0.0861 p_y= -0.0005 p_z= -0.0177

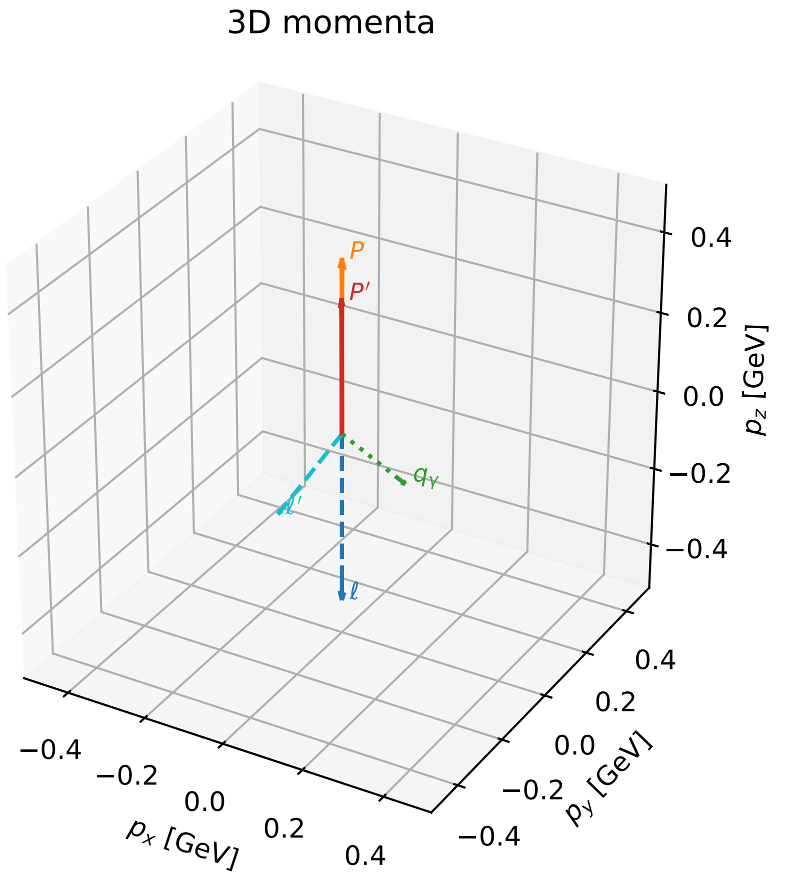




electron: polarization cluster P2, local minimum 729; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




```
dw_mixing_angles_polarization_02_local_minimum_732  D_W=0.00844907
region: polarization_cluster_02_local_minimum_732
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7549277 rad
incoming lepton: 0.728321|+> +0.685236|->
proton mixing angle: alpha_p=1.3584312 rad
incoming proton: 0.210773|+> +0.977535|->
```

s=2.14128, $\sqrt{s}$=1.46331, $|\vec{P}|$=0.431021, $|\vec{P}'|$=0.333537

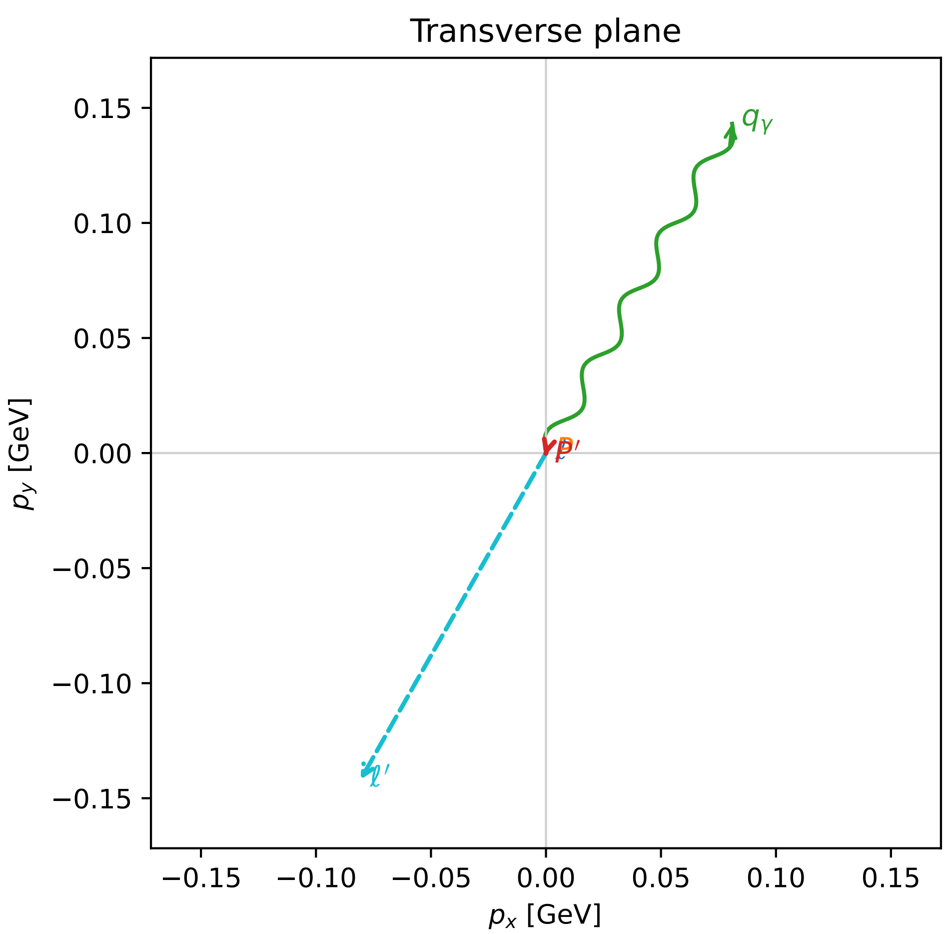
$\theta_{p'}$=0.00360127, θ_γ=2.39419, $\phi_{p'}$=4.45002, ϕ_γ=1.05615

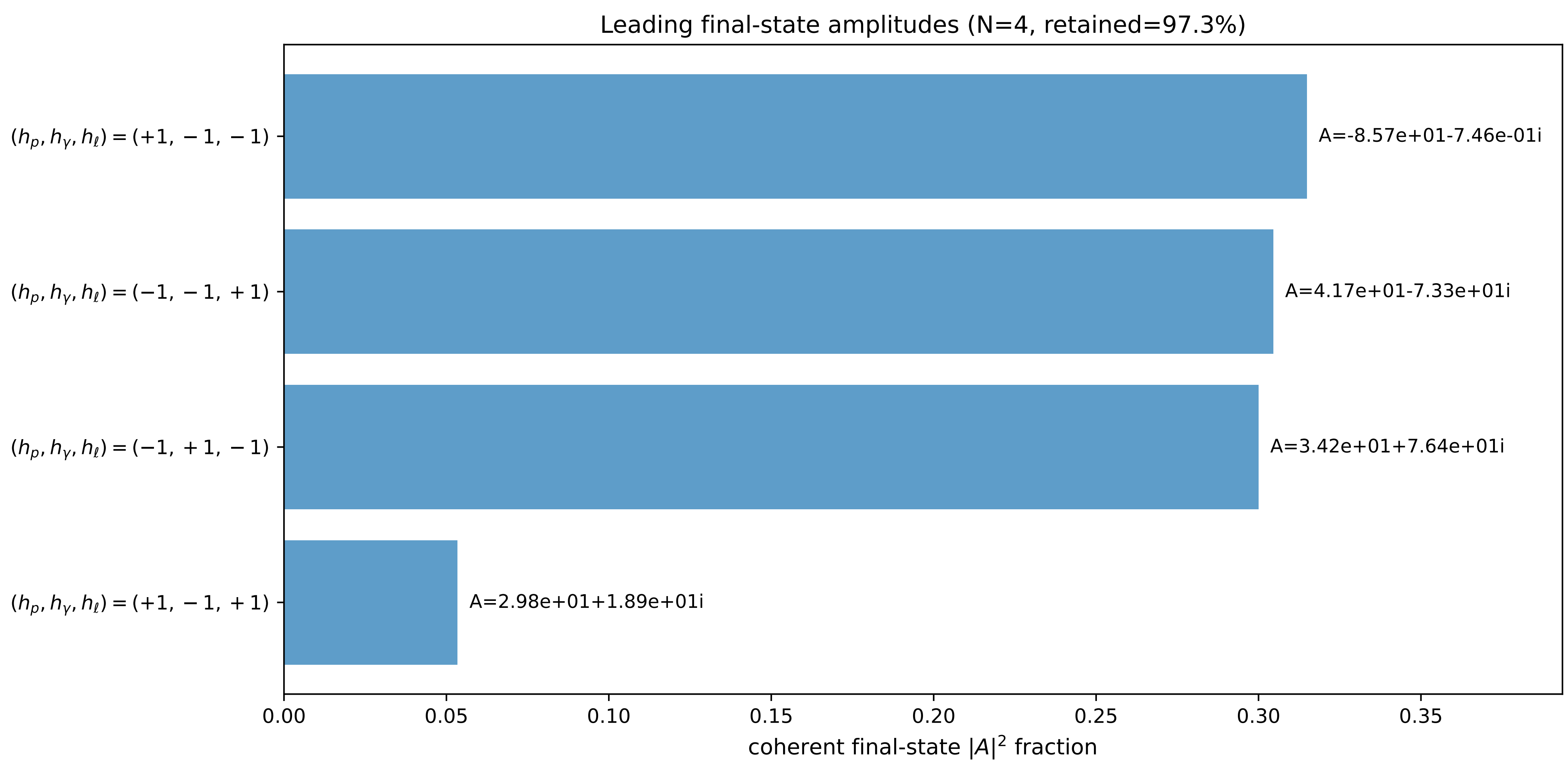
E_γ=0.241886, $\theta_{\ell'}$=2.33386, $\phi_{\ell'}$=4.1959

Q^2=0.0601432, x_B=0.0910592, t=-0.00815409, W^2=1.48019, y=0.523597

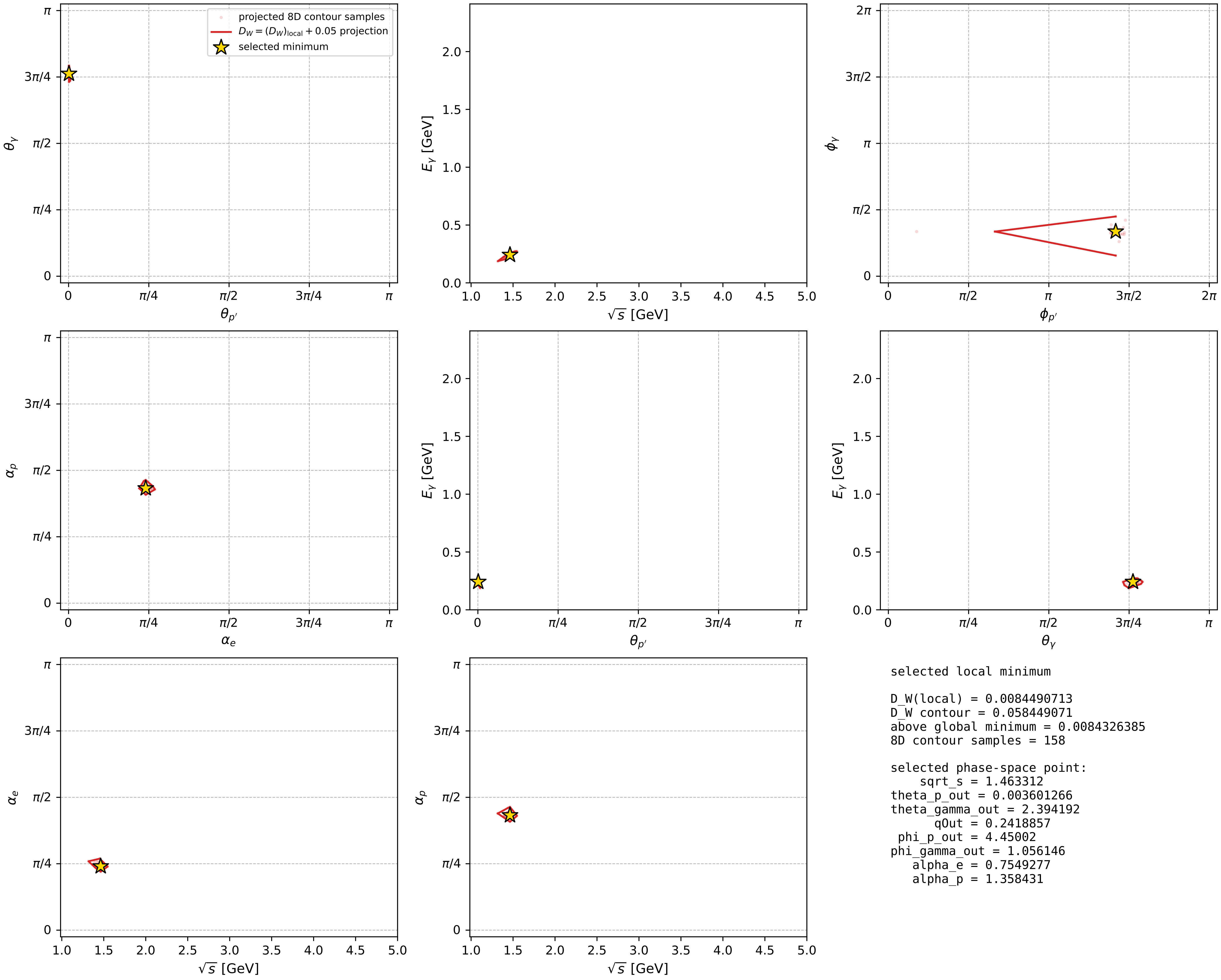
four-momenta [E, p_x, p_y, p_z] GeV:

	E	p_x	p_y	p_z
ℓ	0.4310	0.0000	0.0000	-0.4310
P	1.0323	0.0000	0.0000	0.4310
ℓ'	0.2259	-0.0806	-0.1420	-0.1561
P'	0.9955	-0.0003	-0.0012	0.3335
q_γ	0.2419	0.0809	0.1431	-0.1774

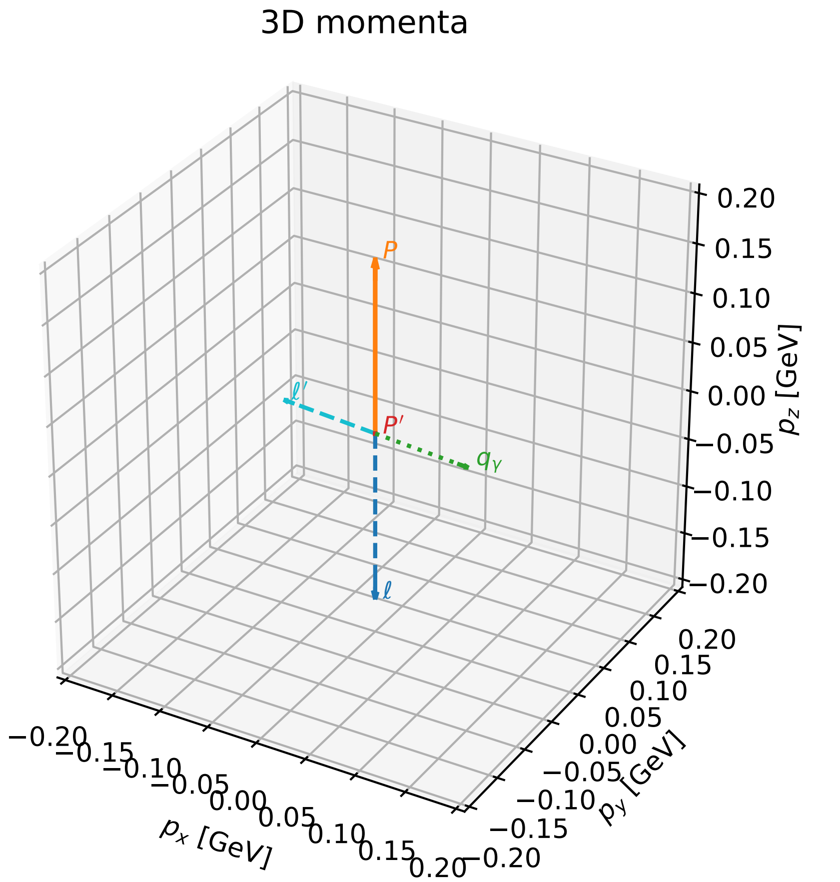




electron: polarization cluster P2, local minimum 732; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



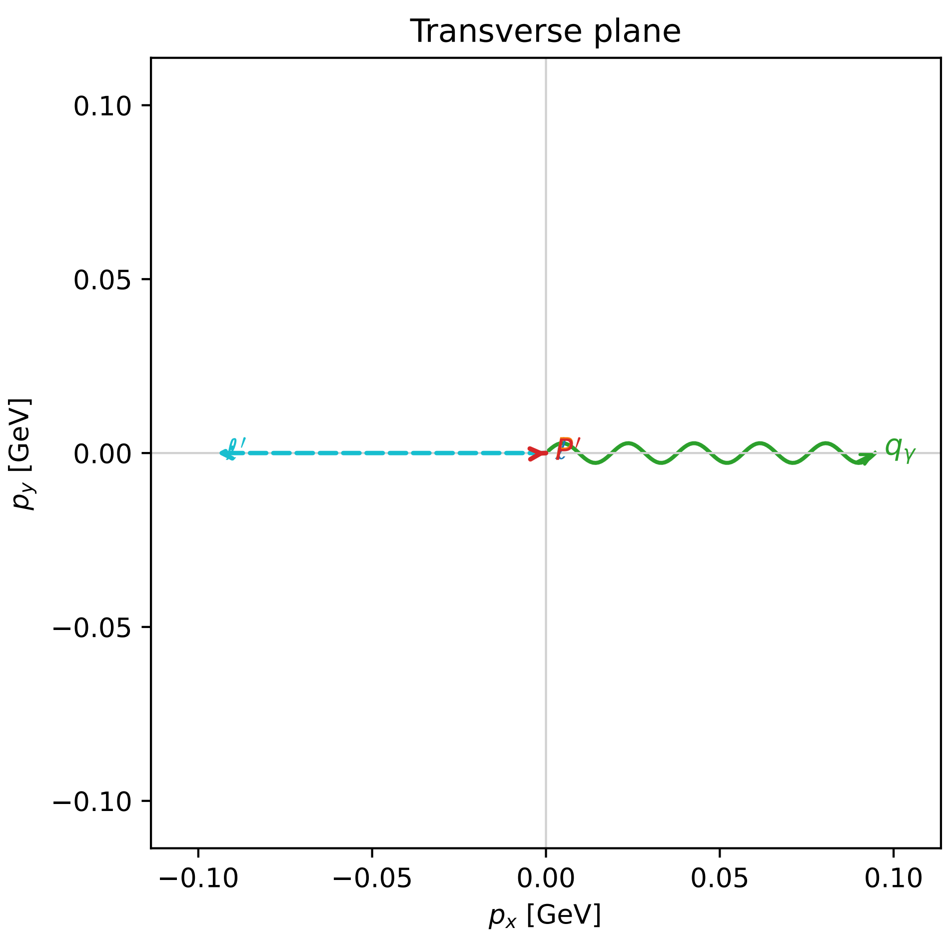
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

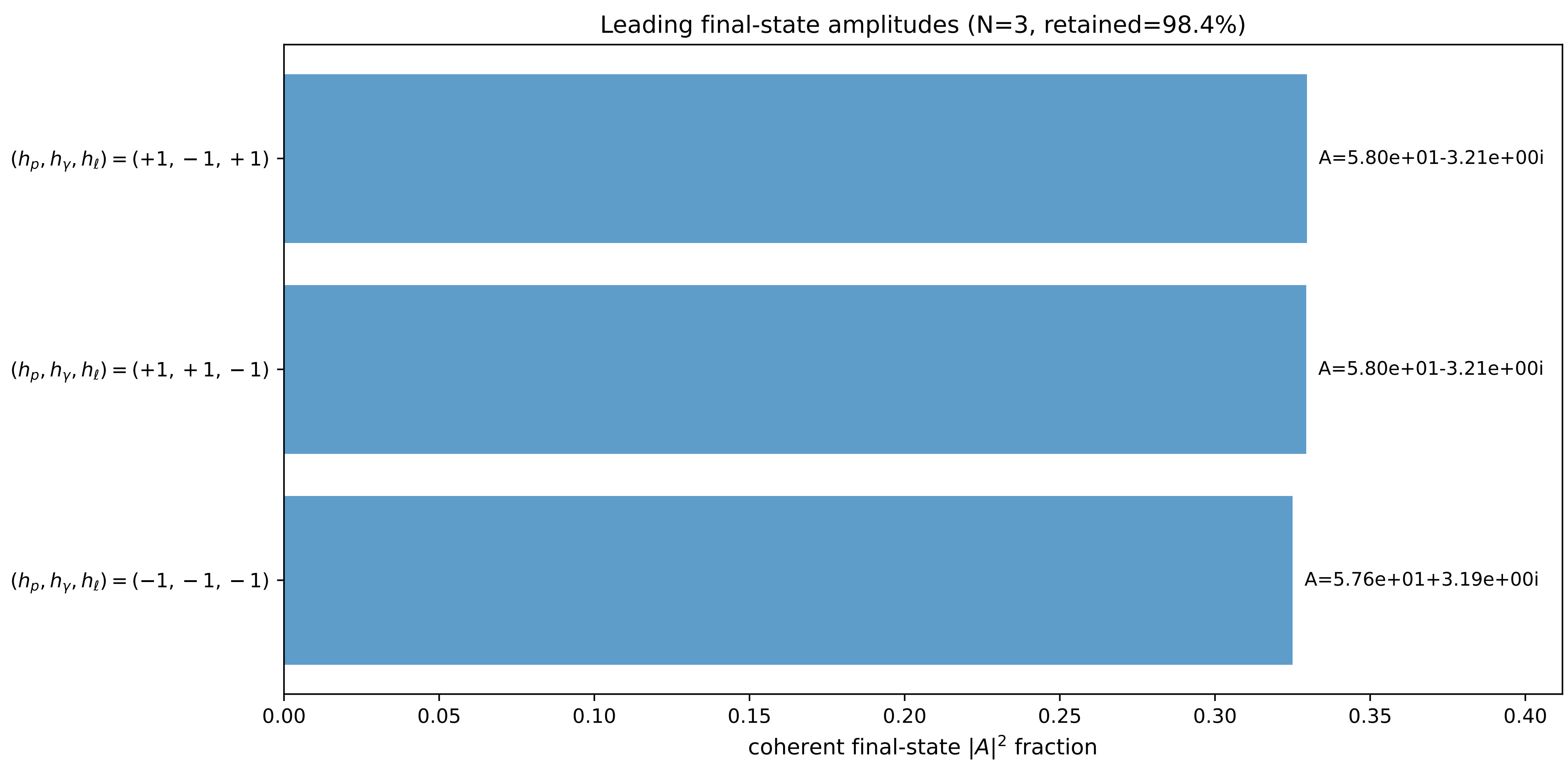


dw_mixing_angles_polarization_02_local_minimum_739 D_W=0.0085628
region: polarization_cluster_02_local_minimum_739
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7888655 rad
incoming lepton: 0.704651|+> +0.709554|->
proton mixing angle: alpha_p=1.3637025 rad
incoming proton: 0.205617|+> +0.978633|->

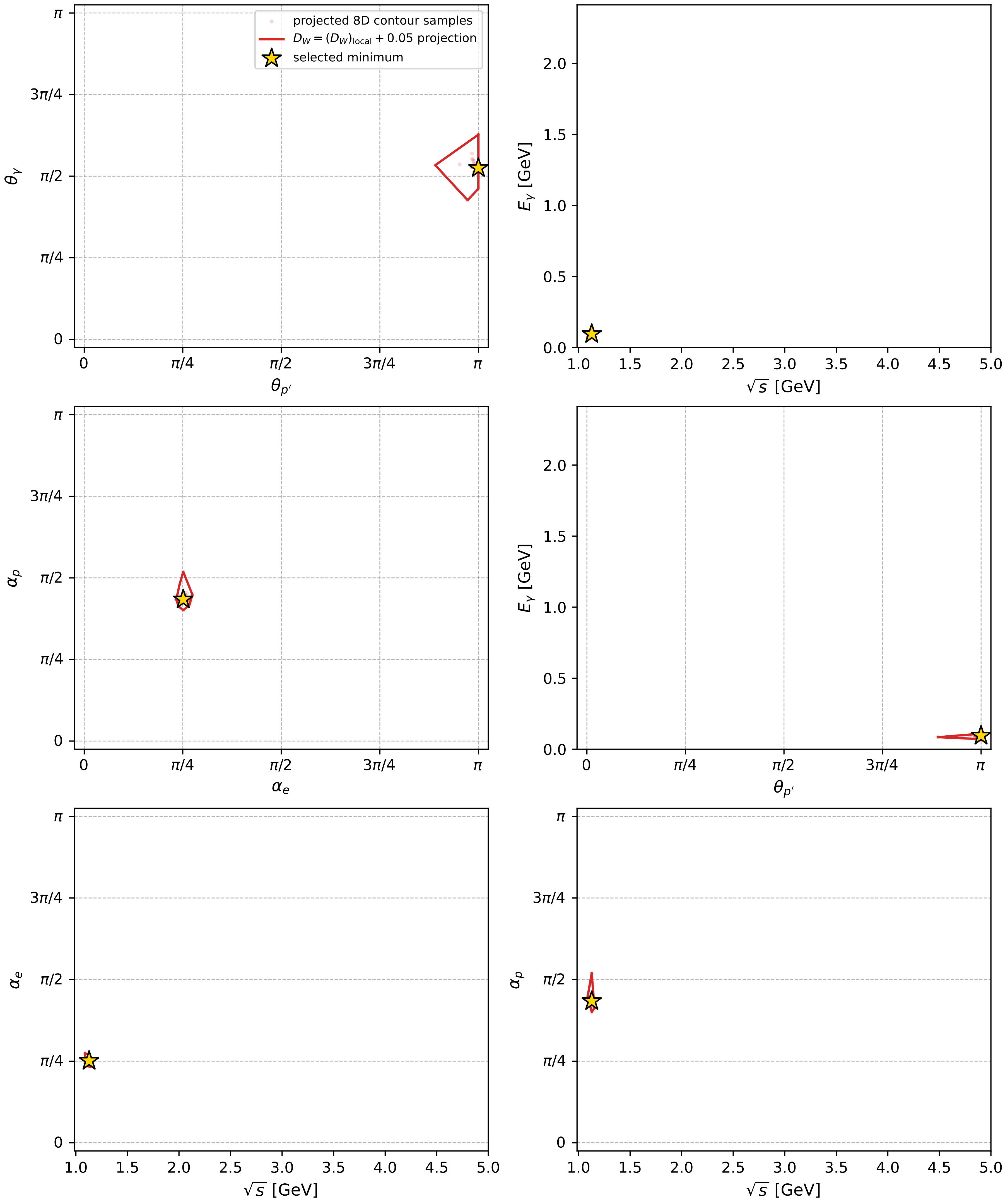
s=1.27231, $\sqrt{s}$=1.12797, $|\vec{P}|$=0.173971, $|\vec{P}'|$=0.000134712
$\theta_{p'}$=3.14126, θ_γ=1.64947, $\phi_{p'}$=0.0552426, ϕ_γ=1.07418e-06
E_γ=0.0949782, $\theta_{\ell'}$=1.49071, $\phi_{\ell'}$=3.14159
Q^2=0.0356947, x_B=0.166898, t=-0.0300569, W^2=1.05802, y=0.54494

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1740 p_x= 0.0000 p_y= 0.0000 p_z= -0.1740
P E= 0.9540 p_x= 0.0000 p_y= 0.0000 p_z= 0.1740
ℓ' E= 0.0950 p_x= -0.0947 p_y= -0.0000 p_z= 0.0076
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0001
q_γ E= 0.0950 p_x= 0.0947 p_y= 0.0000 p_z= -0.0075



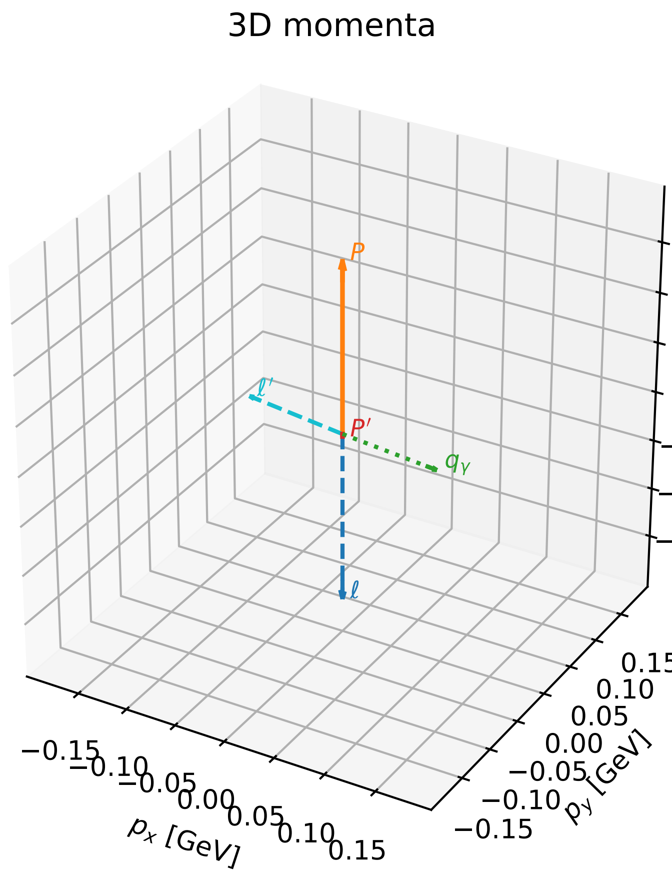


electron: polarization cluster P2, local minimum 739; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.008562795
D_W contour = 0.058562795
above global minimum = 0.0085463622
8D contour samples = 188

selected phase-space point:
    sqrt_s = 1.127969
theta_p_out = 3.141261
theta_gamma_out = 1.649466
       qOut = 0.09497824
 phi_p_out = 0.05524255
phi_gamma_out = 1.074179e-06
   alpha_e = 0.7888655
   alpha_p = 1.363703
```

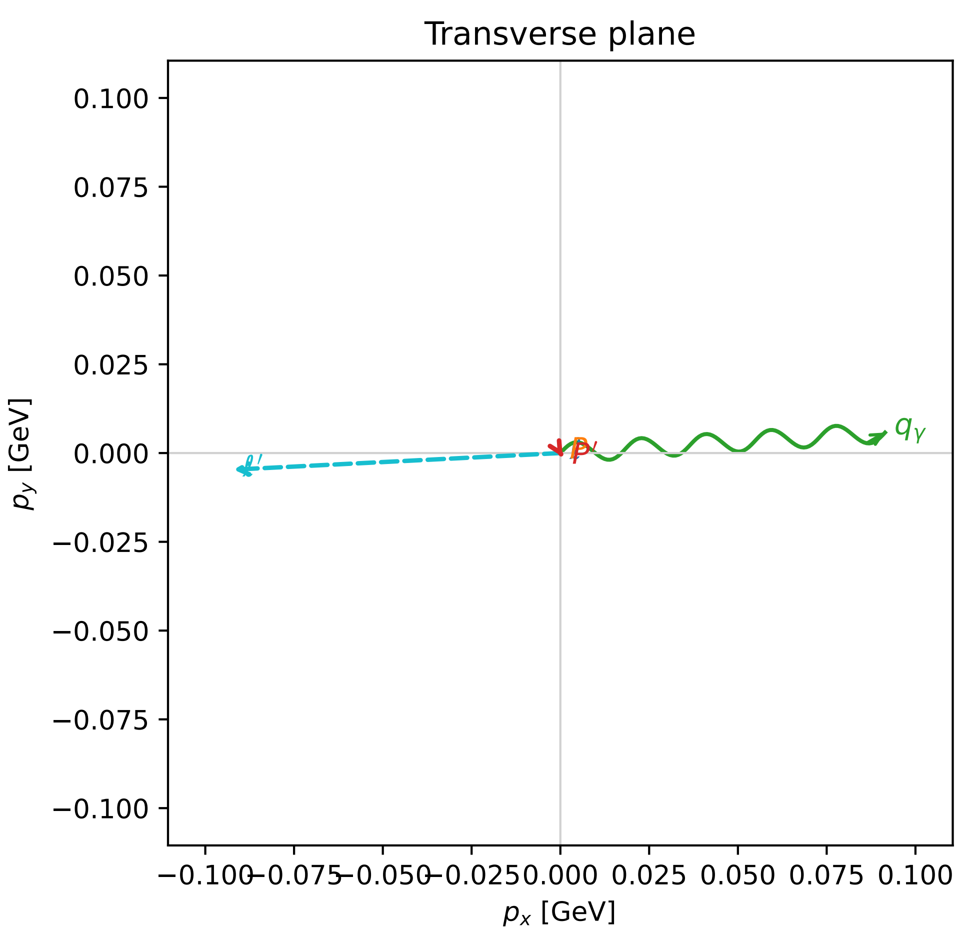
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

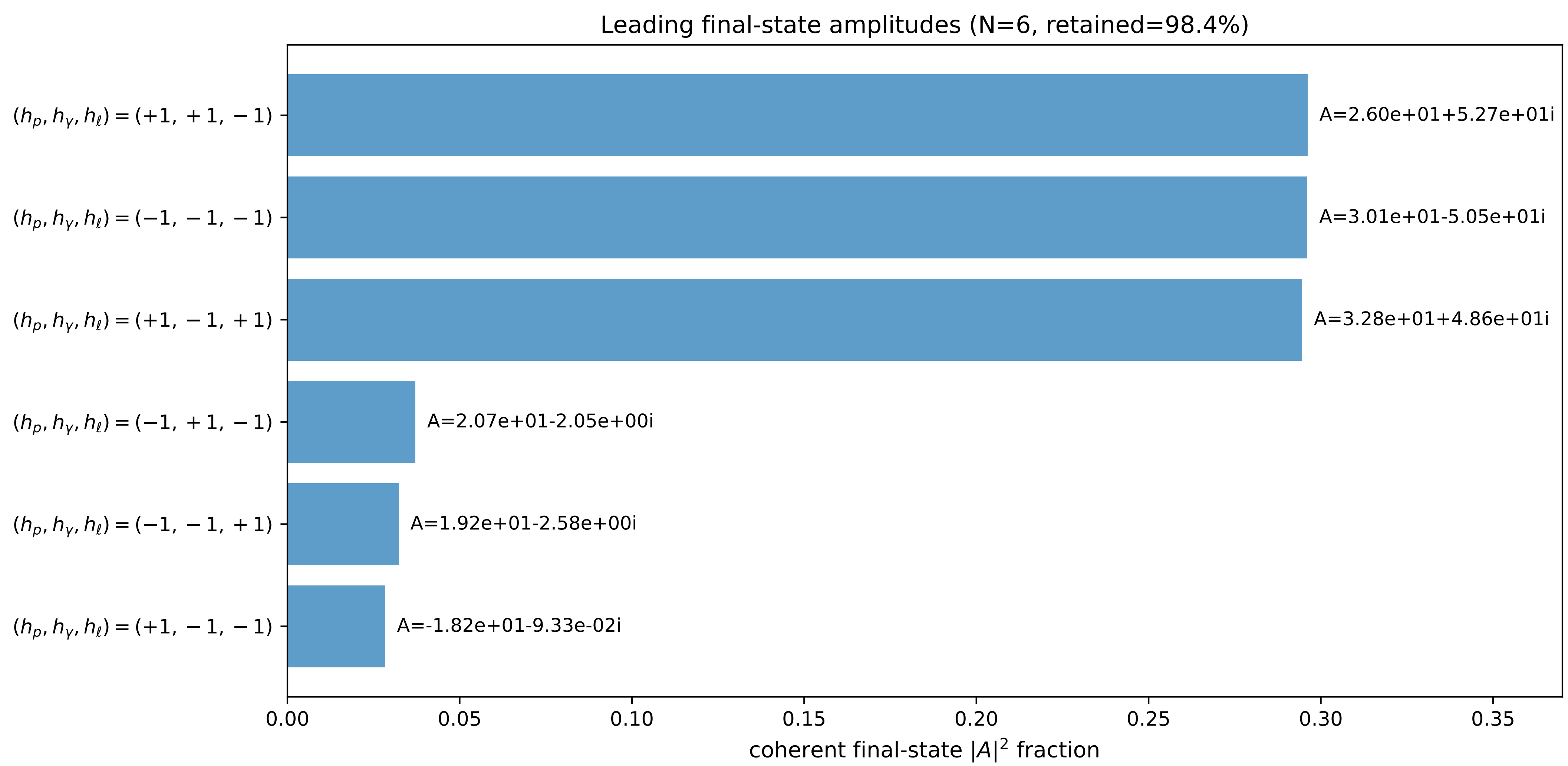


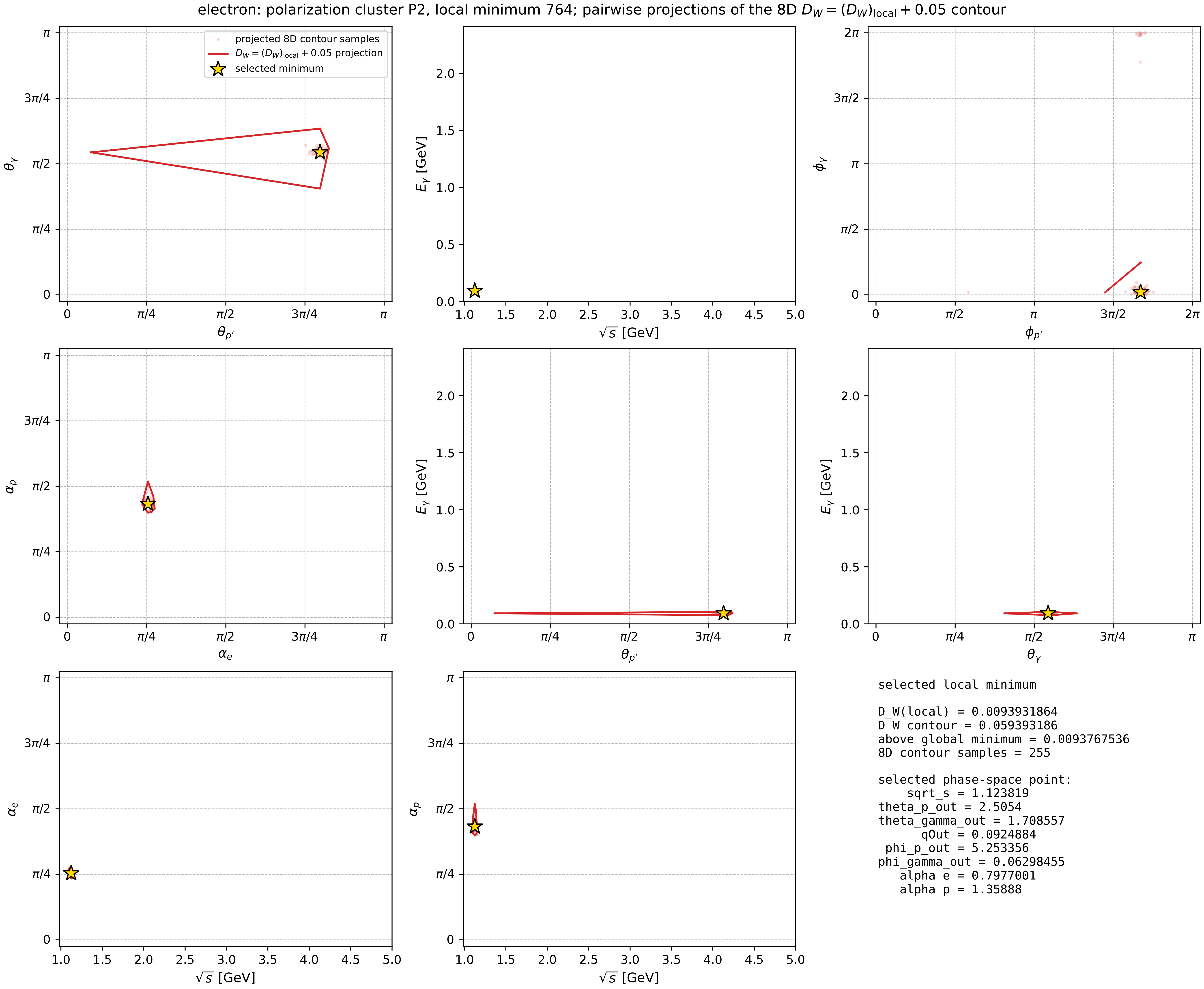
dw_mixing_angles_polarization_02_local_minimum_764 D_W=0.00939319
region: polarization_cluster_02_local_minimum_764
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7977001 rad
incoming lepton: 0.698355|+> +0.715752|->
proton mixing angle: alpha_p=1.3588802 rad
incoming proton: 0.210334|+> +0.97763|->

s=1.26297, $\sqrt{s}$=1.12382, $|\vec{P}|$=0.170456, $|\vec{P}'|$=0.0021435
$\theta_{p'}$=2.5054, θ_γ=1.70856, $\phi_{p'}$=5.25336, ϕ_γ=0.0629846
E_γ=0.0924884, $\theta_{\ell'}$=1.41561, $\phi_{\ell'}$=3.19231
Q^2=0.0367339, x_B=0.174847, t=-0.0294117, W^2=1.0532, y=0.548364

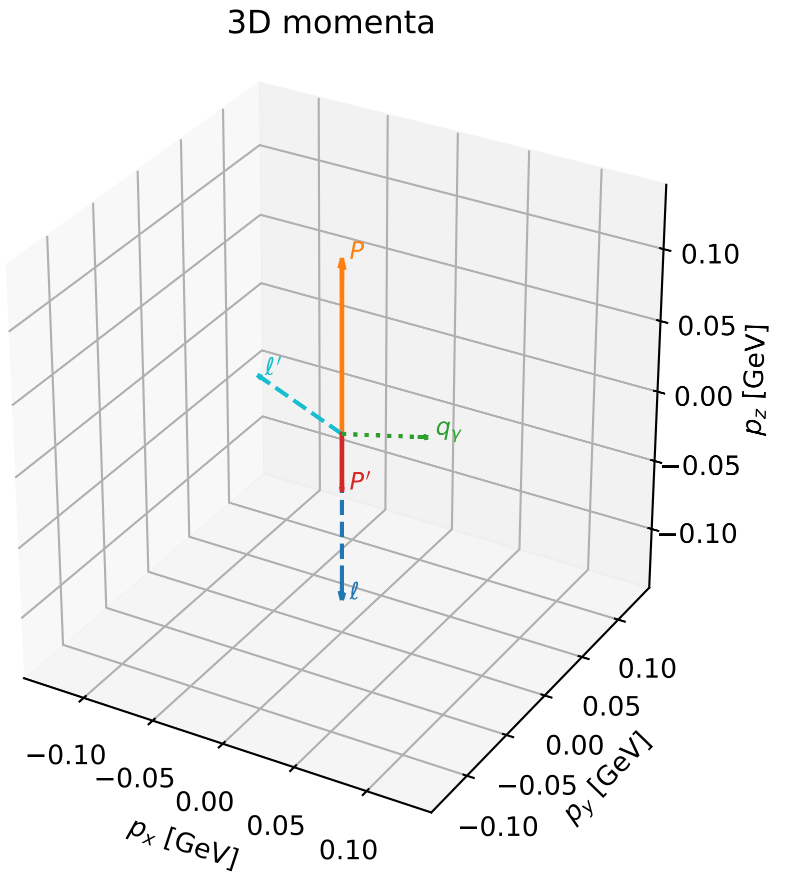
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1705 p_x= 0.0000 p_y= 0.0000 p_z= -0.1705
P E= 0.9534 p_x= 0.0000 p_y= 0.0000 p_z= 0.1705
ℓ' E= 0.0933 p_x= -0.0921 p_y= -0.0047 p_z= 0.0144
P' E= 0.9380 p_x= 0.0007 p_y= -0.0011 p_z= -0.0017
q_γ E= 0.0925 p_x= 0.0914 p_y= 0.0058 p_z= -0.0127






electron: polarization cluster P2, local minimum 764; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0093931864
D_W contour = 0.059393186
above global minimum = 0.0093767536
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.123819
theta_p_out = 2.5054
theta_gamma_out = 1.708557
qOut = 0.0924884
phi_p_out = 5.253356
phi_gamma_out = 0.06298455
alpha_e = 0.7977001
alpha_p = 1.35888

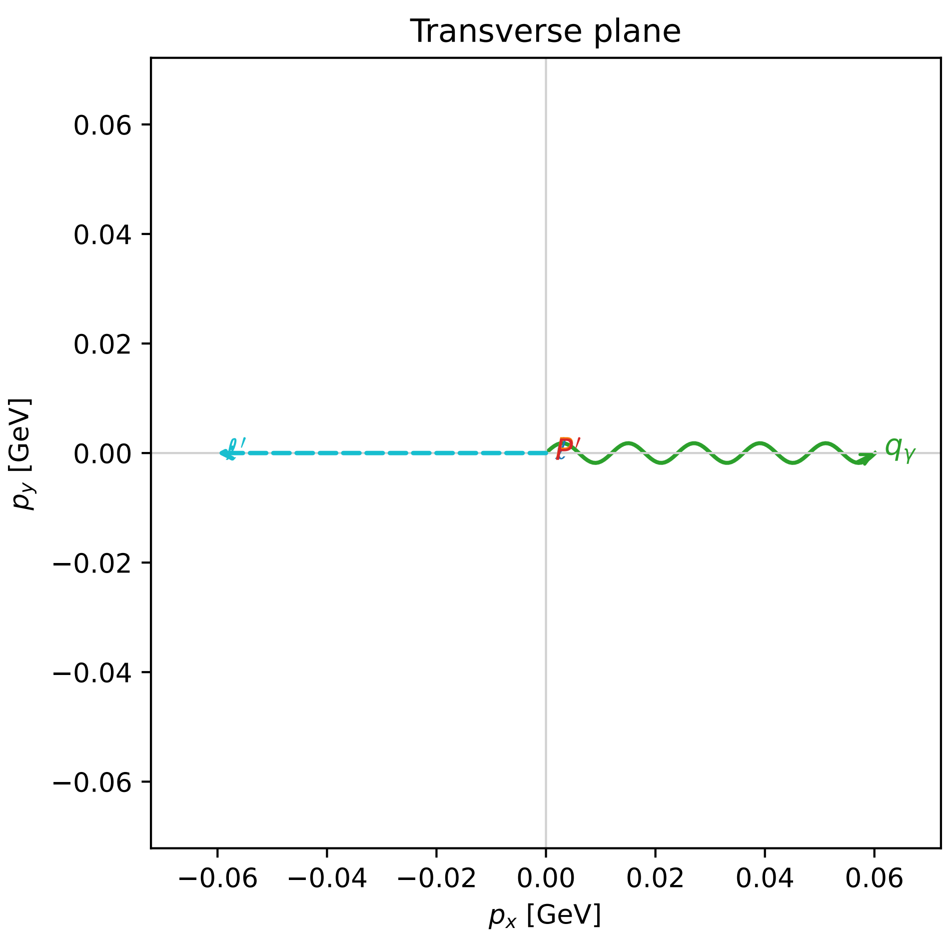
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

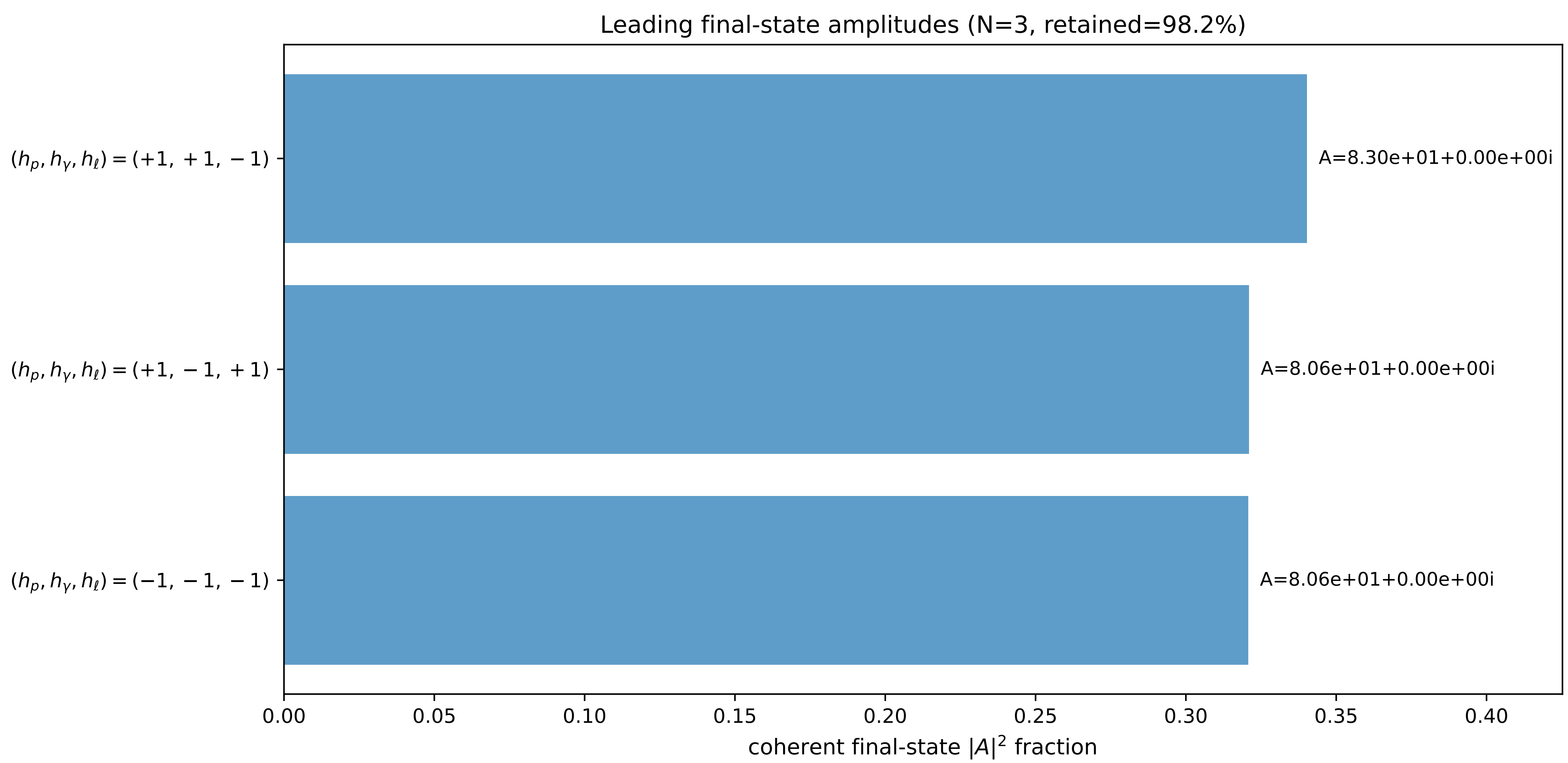


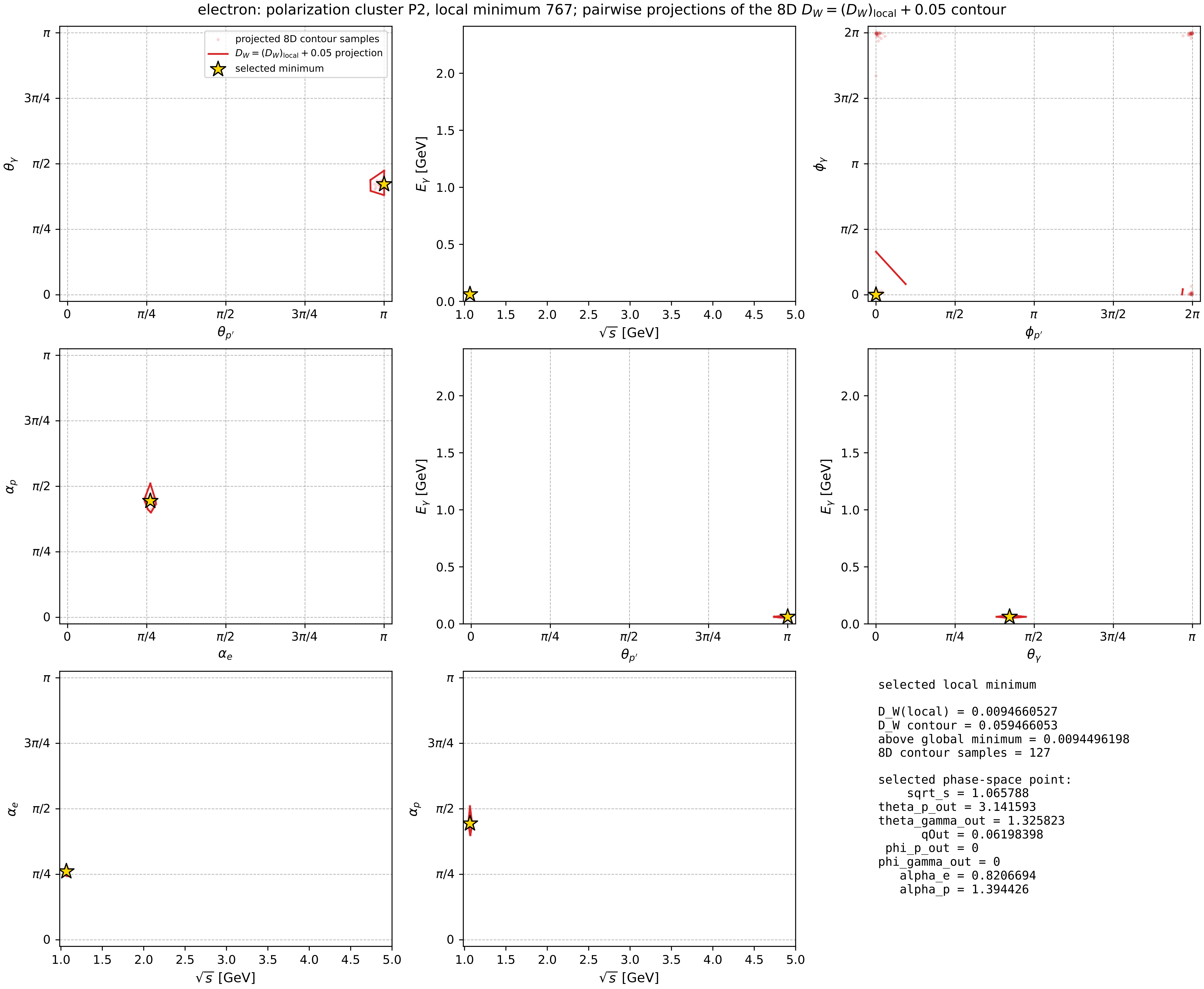
dw_mixing_angles_polarization_02_local_minimum_767 D_W=0.00946605
region: polarization_cluster_02_local_minimum_767
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.82066942 rad
incoming lepton: 0.681732|+> +0.731602|->
proton mixing angle: alpha_p=1.3944256 rad
incoming proton: 0.175458|+> +0.984487|->

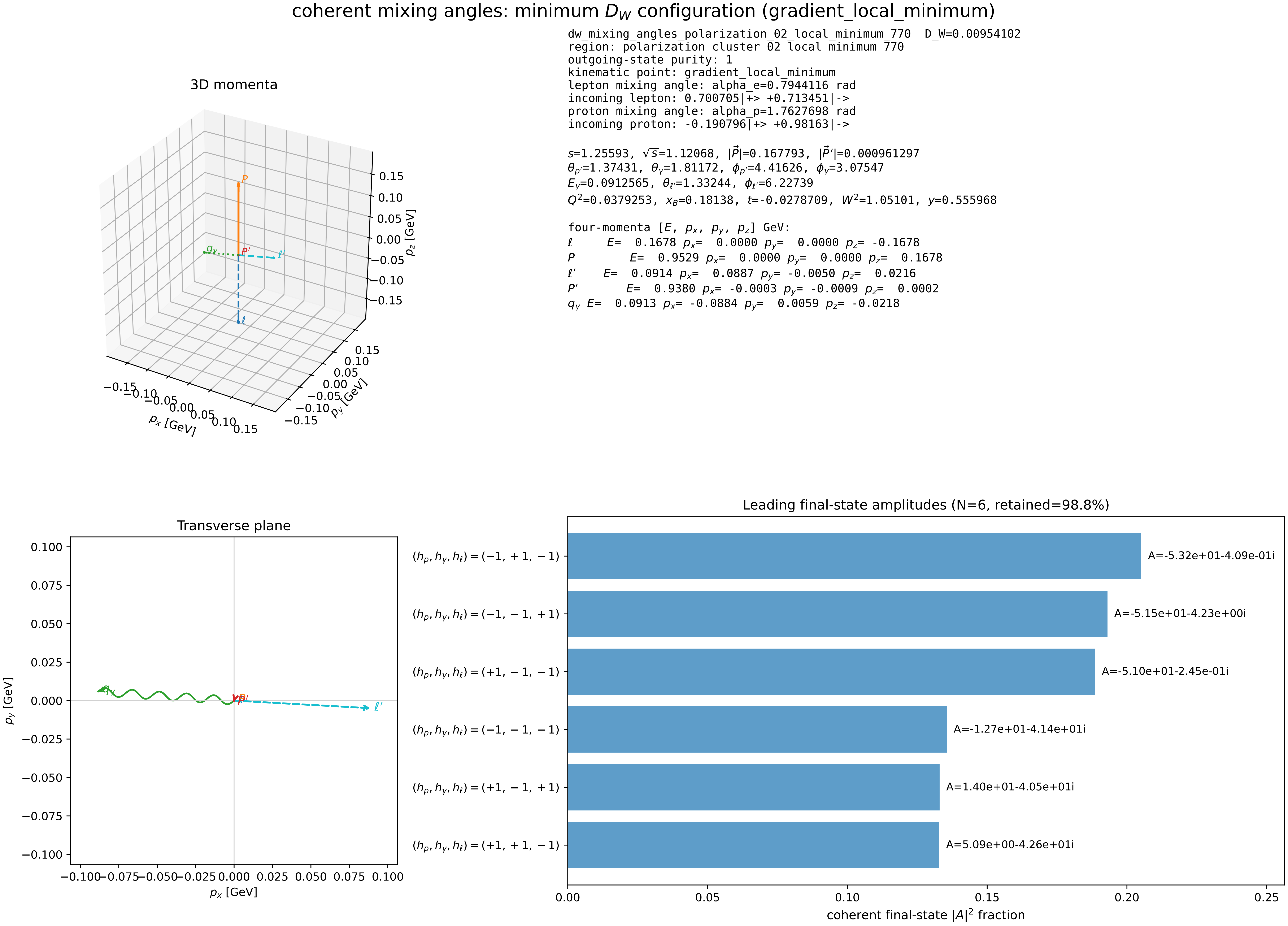
s=1.1359, $\sqrt{s}$=1.06579, $|\vec{P}|$=0.120126, $|\vec{P}'|$=0.0396205
$\theta_{p'}$=3.14159, θ_γ=1.32582, $\phi_{p'}$=0, ϕ_γ=0
E_γ=0.061984, $\theta_{\ell'}$=1.18266, $\phi_{\ell'}$=3.14159
Q^2=0.0215155, x_B=0.154685, t=-0.0254725, W^2=0.997421, y=0.543202

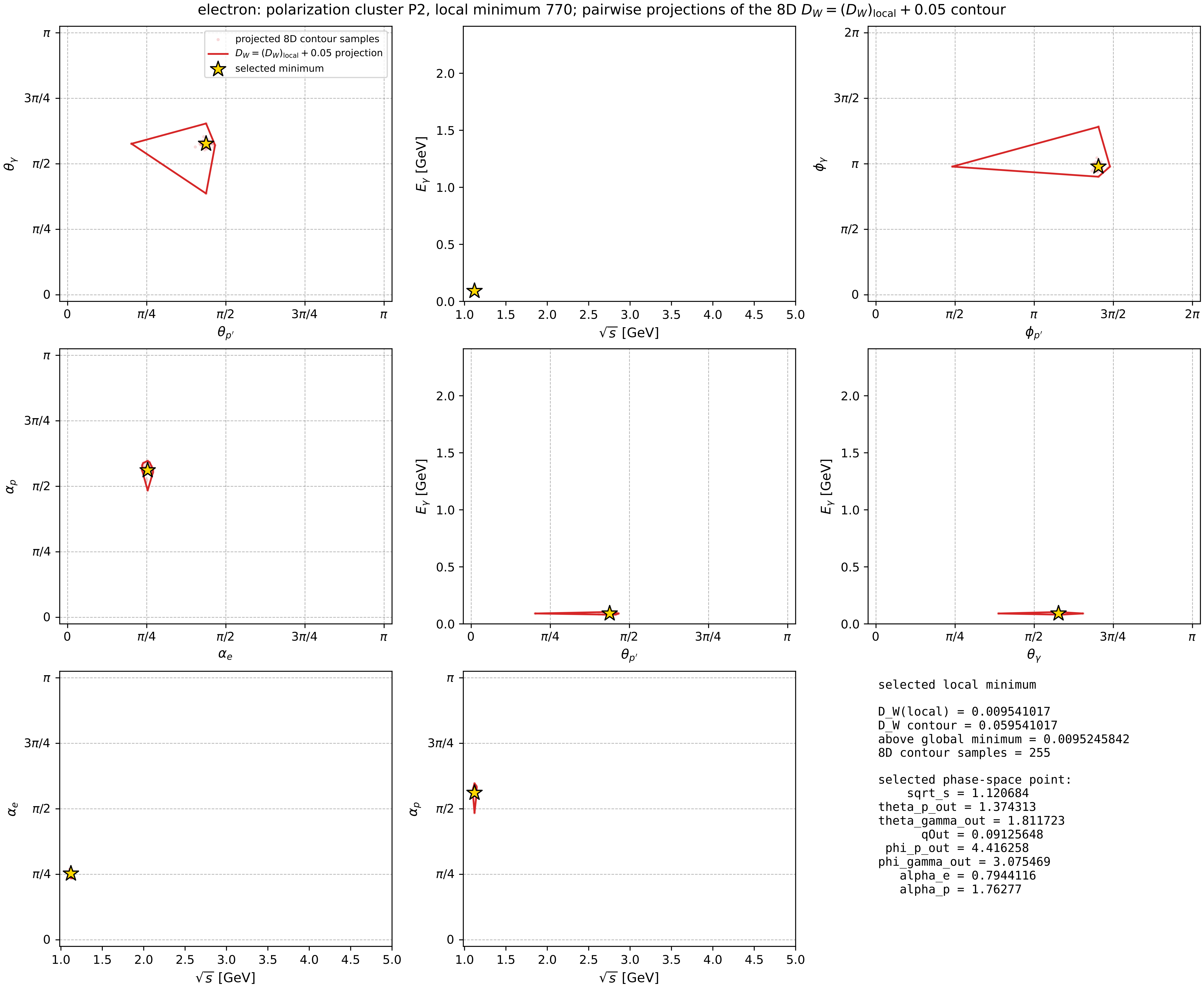
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1201 p_x= 0.0000 p_y= 0.0000 p_z= -0.1201
P E= 0.9457 p_x= 0.0000 p_y= 0.0000 p_z= 0.1201
ℓ' E= 0.0650 p_x= -0.0601 p_y= -0.0000 p_z= 0.0246
P' E= 0.9388 p_x= 0.0000 p_y= 0.0000 p_z= -0.0396
q_γ E= 0.0620 p_x= 0.0601 p_y= 0.0000 p_z= 0.0150





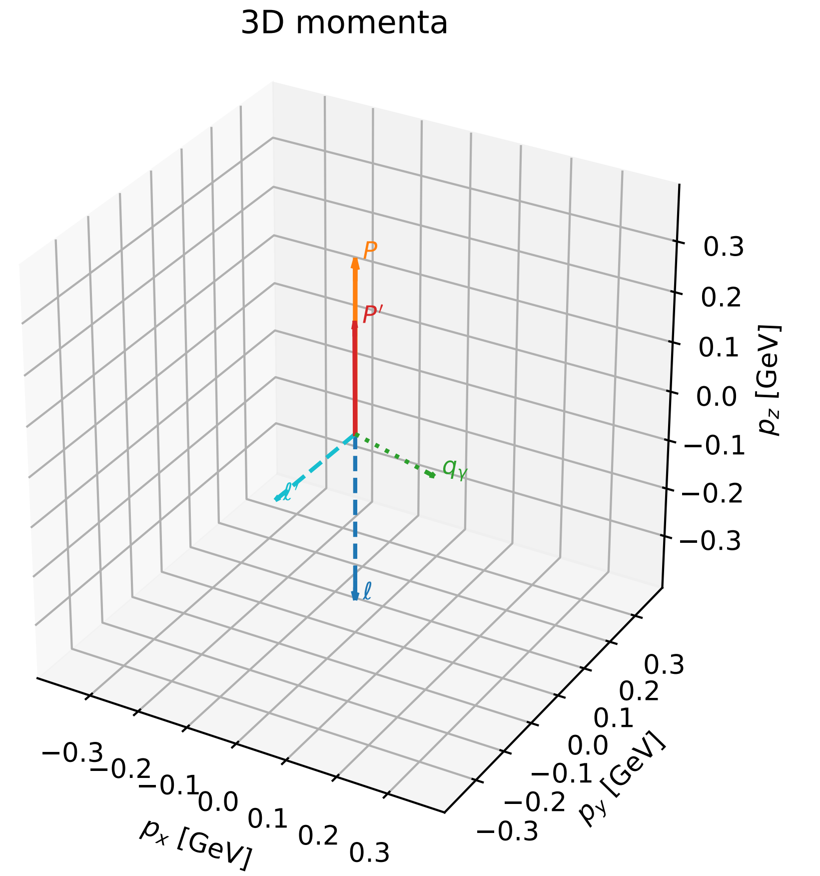

electron: polarization cluster P2, local minimum 767; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0094660527
D_W contour = 0.059466053
above global minimum = 0.0094496198
8D contour samples = 127
selected phase-space point:
sqrt_s = 1.065788
theta_p_out = 3.141593
theta_gamma_out = 1.325823
qOut = 0.06198398
phi_p_out = 0
phi_gamma_out = 0
alpha_e = 0.8206694
alpha_p = 1.394426


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_770 D_W=0.00954102
region: polarization_cluster_02_local_minimum_770
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7944116 rad
incoming lepton: 0.700705|+> +0.713451|->
proton mixing angle: alpha_p=1.7627698 rad
incoming proton: -0.190796|+> +0.98163|->
s=1.25593, $\sqrt{s}$=1.12068, $|\vec{P}|$=0.167793, $|\vec{P}'|$=0.000961297
$\theta_{p'}$=1.37431, θ_γ=1.81172, $\phi_{p'}$=4.41626, ϕ_γ=3.07547
E_γ=0.0912565, $\theta_{\ell'}$=1.33244, $\phi_{\ell'}$=6.22739
Q^2=0.0379253, x_B=0.18138, t=-0.0278709, W^2=1.05101, y=0.555968
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1678 p_x= 0.0000 p_y= 0.0000 p_z= -0.1678
P E= 0.9529 p_x= 0.0000 p_y= 0.0000 p_z= 0.1678
ℓ' E= 0.0914 p_x= 0.0887 p_y= -0.0050 p_z= 0.0216
P' E= 0.9380 p_x= -0.0003 p_y= -0.0009 p_z= 0.0002
q_γ E= 0.0913 p_x= -0.0884 p_y= 0.0059 p_z= -0.0218
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=98.8%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-5.32e+01-4.09e-01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-5.15e+01-4.23e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-5.10e+01-2.45e-01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-1.27e+01-4.14e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=1.40e+01-4.05e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=5.09e+00-4.26e+01i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P2, local minimum 770; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
0
$\pi/4$
$\pi/2$
$3\pi/4$
π
$3\pi/2$
2π
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
selected local minimum
D_W(local) = 0.009541017
D_W contour = 0.059541017
above global minimum = 0.0095245842
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.120684
theta_p_out = 1.374313
theta_gamma_out = 1.811723
qOut = 0.09125648
phi_p_out = 4.416258
phi_gamma_out = 3.075469
alpha_e = 0.7944116
alpha_p = 1.76277

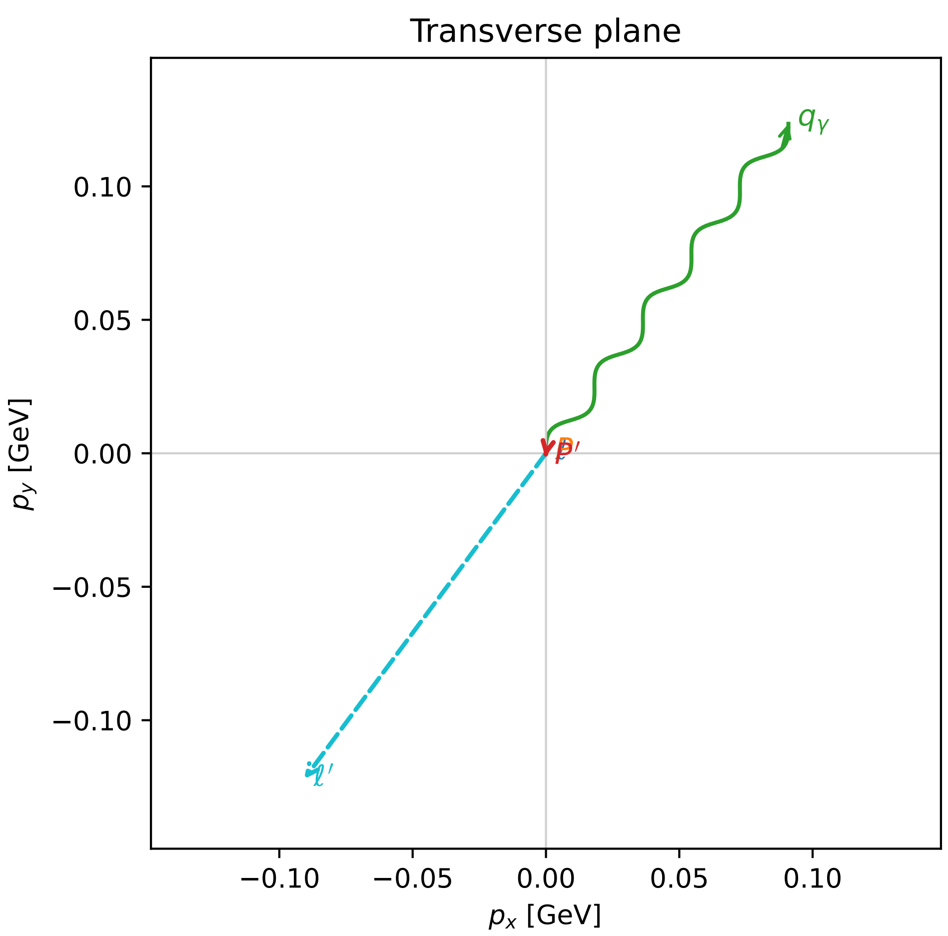
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

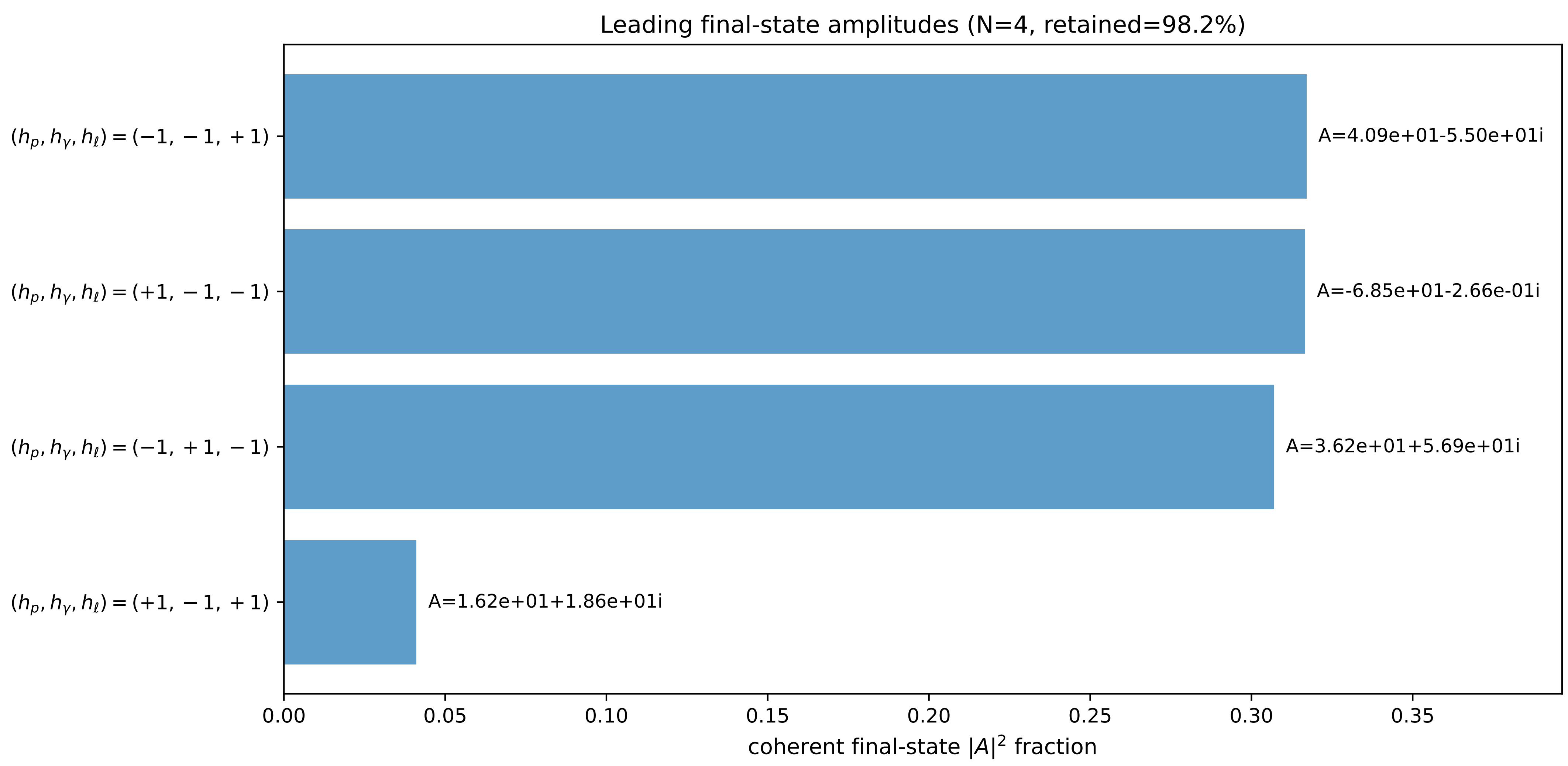


dw_mixing_angles_polarization_02_local_minimum_771 D_W=0.00957182
region: polarization_cluster_02_local_minimum_771
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75644172 rad
incoming lepton: 0.727283|+> +0.686338|->
proton mixing angle: alpha_p=1.3921329 rad
incoming proton: 0.177714|+> +0.984082|->

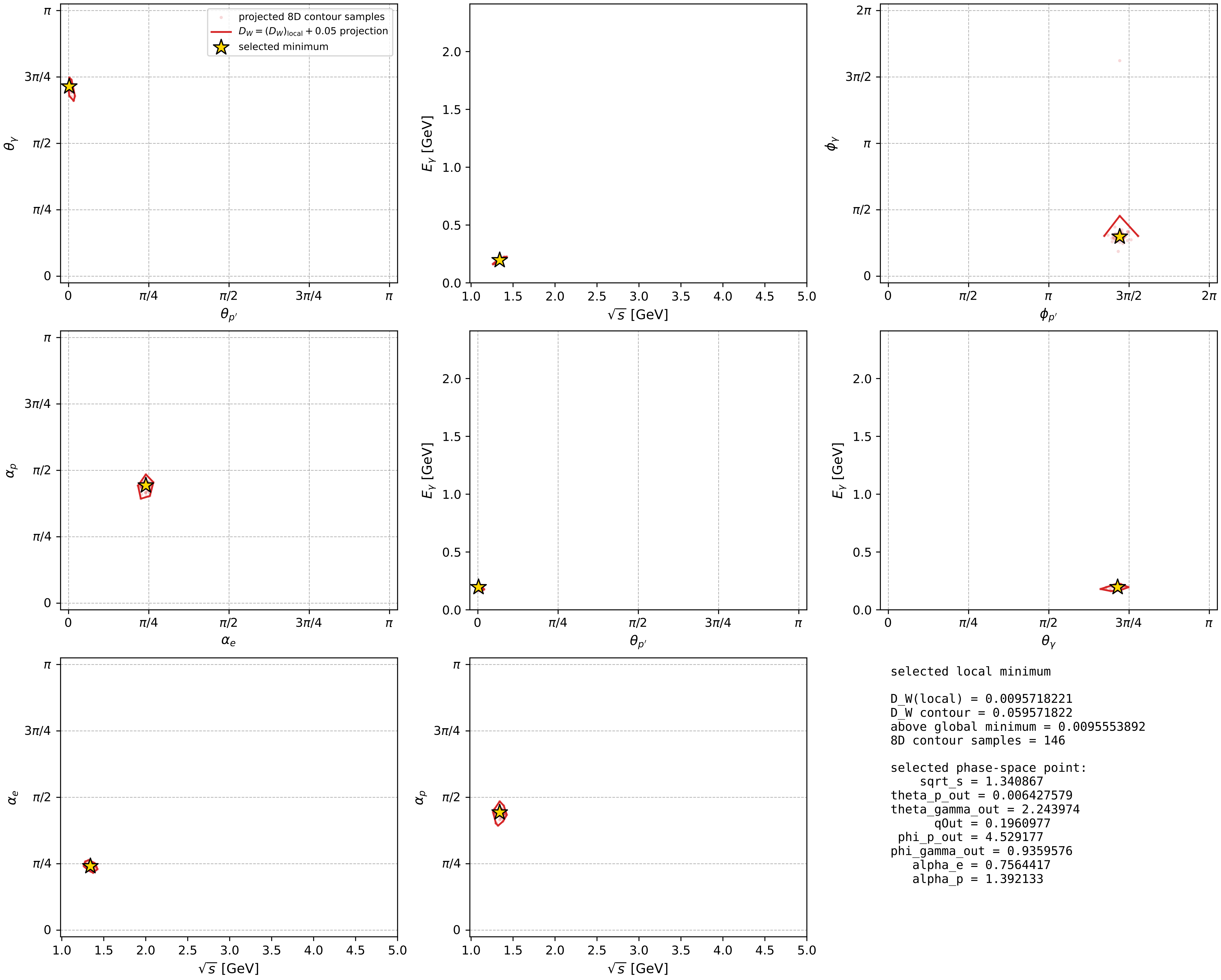
s=1.79792, $\sqrt{s}$=1.34087, $|\vec{P}|$=0.342346, $|\vec{P}'|$=0.220738
$\theta_{p'}$=0.00642758, θ_γ=2.24397, $\phi_{p'}$=4.52918, ϕ_γ=0.935958
E_γ=0.196098, $\theta_{\ell'}$=2.14553, $\phi_{\ell'}$=4.07348
Q^2=0.0566059, x_B=0.115782, t=-0.0135736, W^2=1.31214, y=0.532525

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3423 p_x= 0.0000 p_y= 0.0000 p_z= -0.3423
P E= 0.9985 p_x= 0.0000 p_y= 0.0000 p_z= 0.3423
ℓ' E= 0.1811 p_x= -0.0907 p_y= -0.1221 p_z= -0.0985
P' E= 0.9636 p_x= -0.0003 p_y= -0.0014 p_z= 0.2207
q_γ E= 0.1961 p_x= 0.0909 p_y= 0.1234 p_z= -0.1223

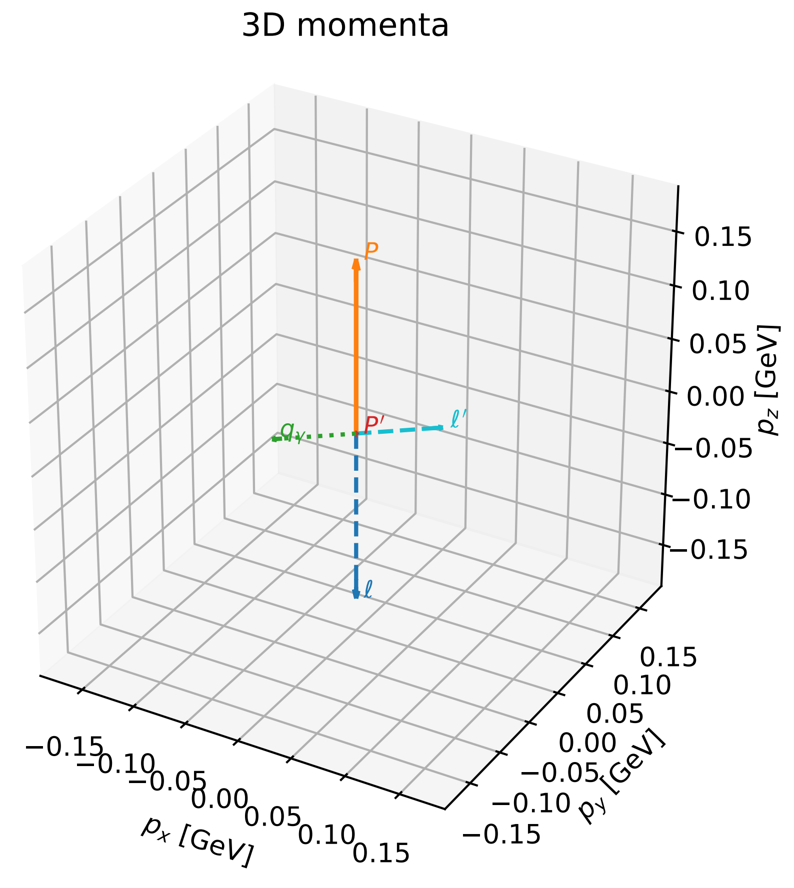




electron: polarization cluster P2, local minimum 771; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



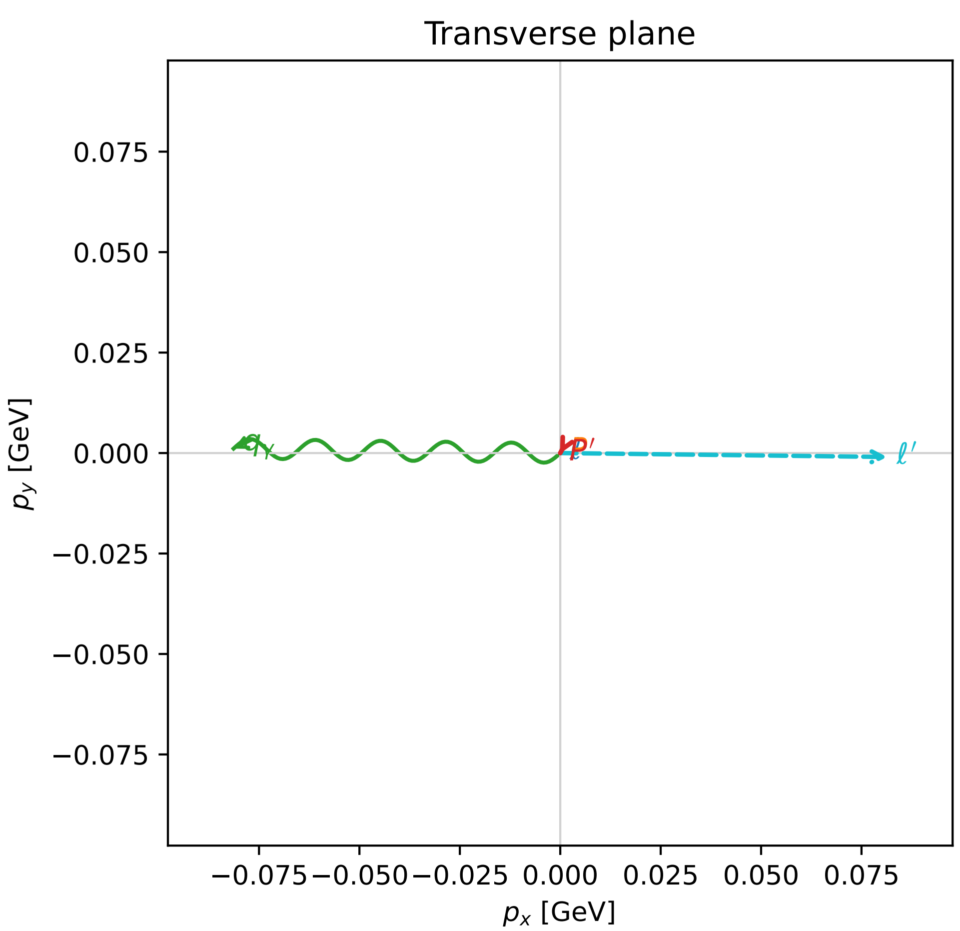
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

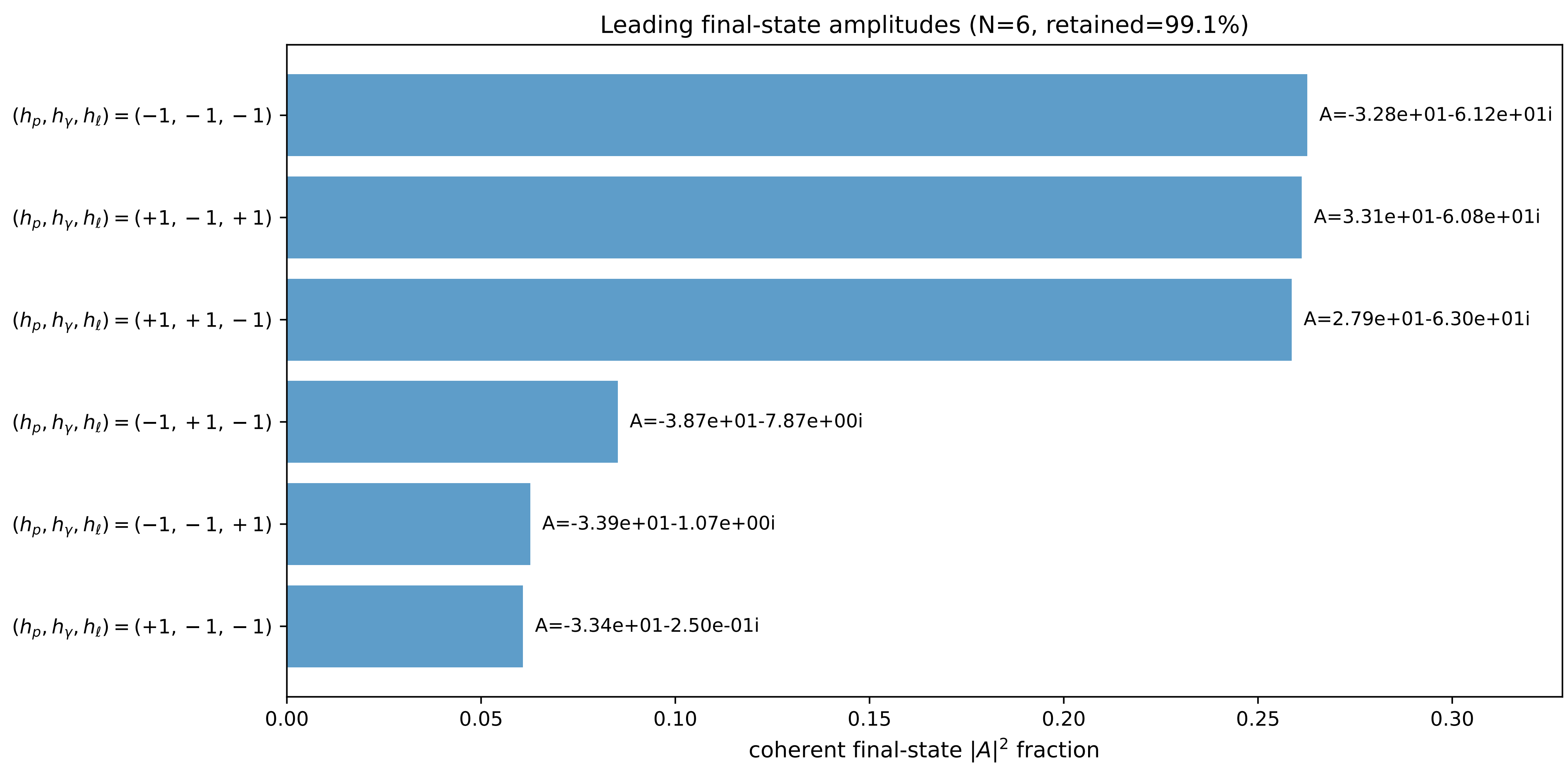


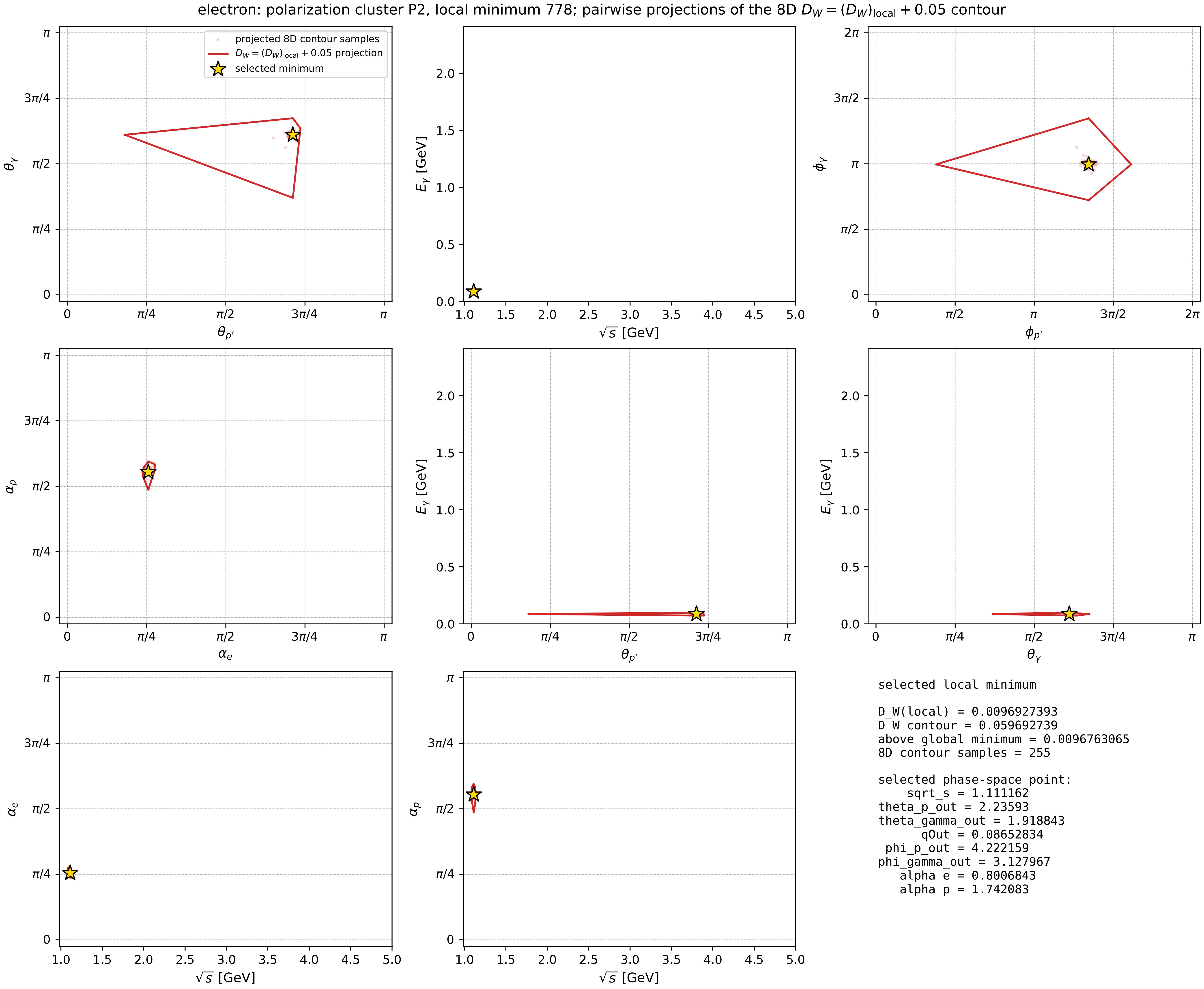

```
dw_mixing_angles_polarization_02_local_minimum_778  D_W=0.00969274
region: polarization_cluster_02_local_minimum_778
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80068434 rad
incoming lepton: 0.696216|+> +0.717833|->
proton mixing angle: alpha_p=1.7420833 rad
incoming proton: -0.170451|+> +0.985366|->
```

s=1.23468, $\sqrt{s}$=1.11116, $|\vec{P}|$=0.159669, $|\vec{P}'|$=0.000188687
$\theta_{p'}$=2.23593, θ_γ=1.91884, $\phi_{p'}$=4.22216, ϕ_γ=3.12797
E_γ=0.0865283, $\theta_{\ell'}$=1.22175, $\phi_{\ell'}$=6.27118
Q^2=0.0371263, x_B=0.186157, t=-0.0253493, W^2=1.04215, y=0.562048

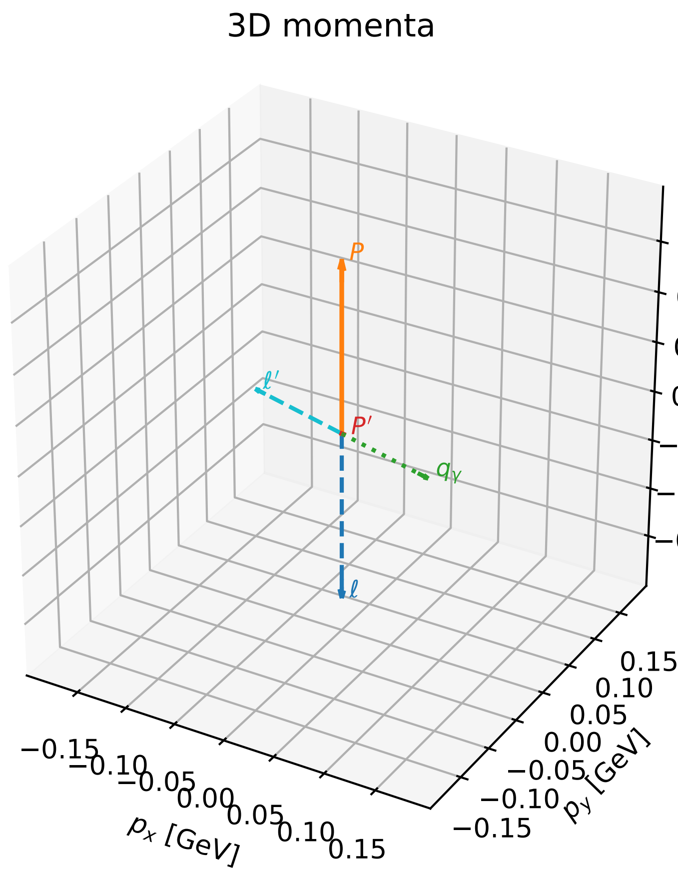
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1597 p_x= 0.0000 p_y= 0.0000 p_z= -0.1597
P E= 0.9515 p_x= 0.0000 p_y= 0.0000 p_z= 0.1597
ℓ' E= 0.0866 p_x= 0.0814 p_y= -0.0010 p_z= 0.0296
P' E= 0.9380 p_x= -0.0001 p_y= -0.0001 p_z= -0.0001
q_γ E= 0.0865 p_x= -0.0813 p_y= 0.0011 p_z= -0.0295






electron: polarization cluster P2, local minimum 778; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0096927393
D_W contour = 0.059692739
above global minimum = 0.0096763065
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.111162
theta_p_out = 2.23593
theta_gamma_out = 1.918843
qOut = 0.08652834
phi_p_out = 4.222159
phi_gamma_out = 3.127967
alpha_e = 0.8006843
alpha_p = 1.742083

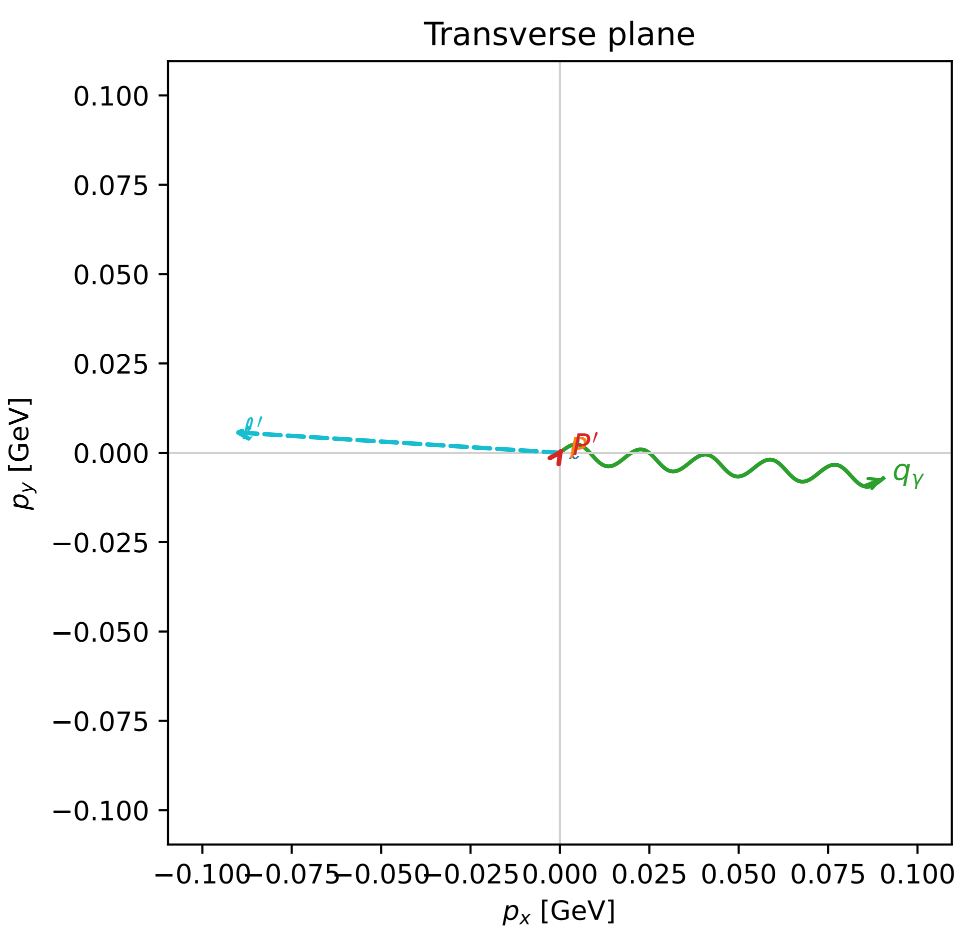
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

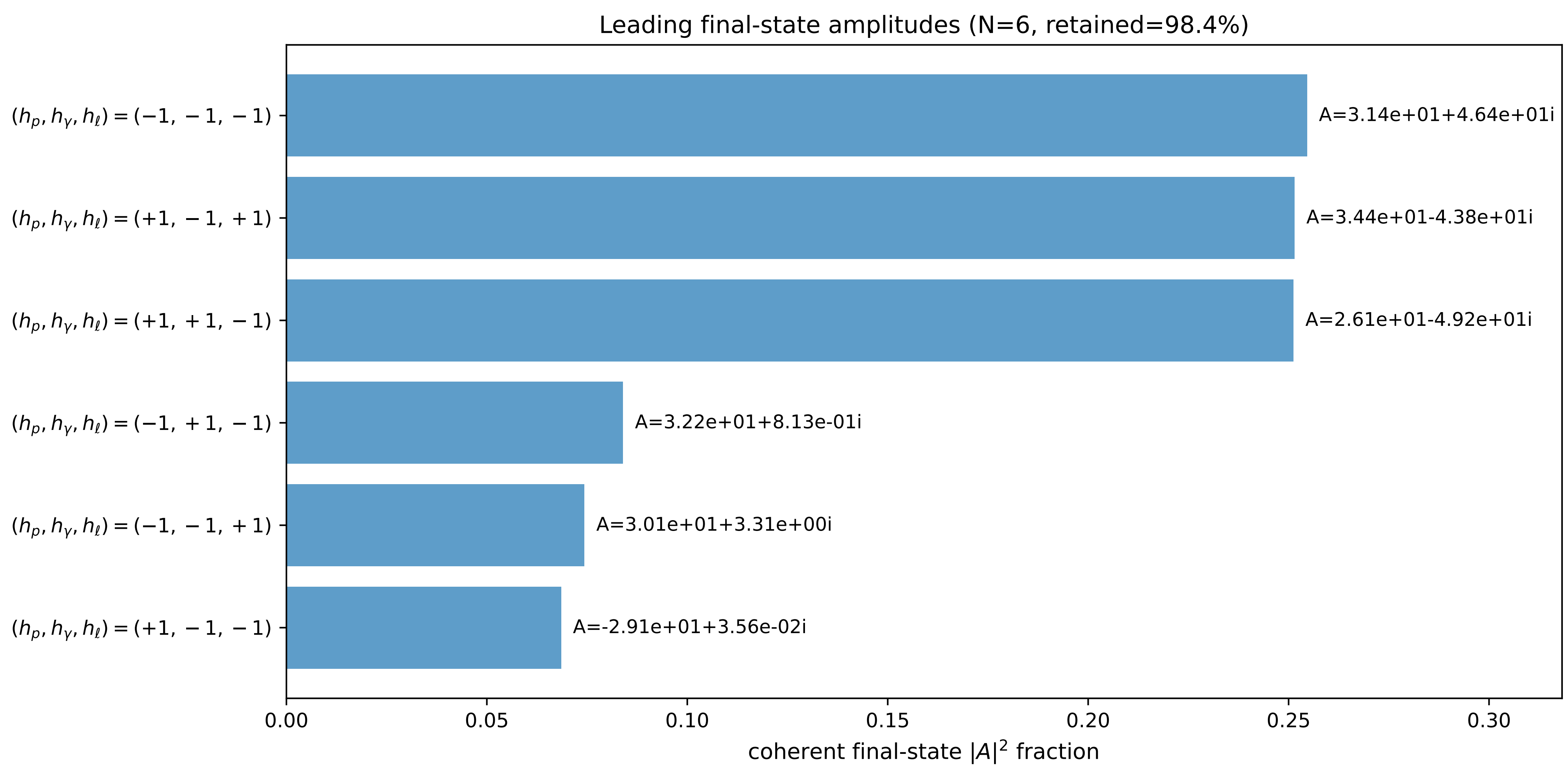


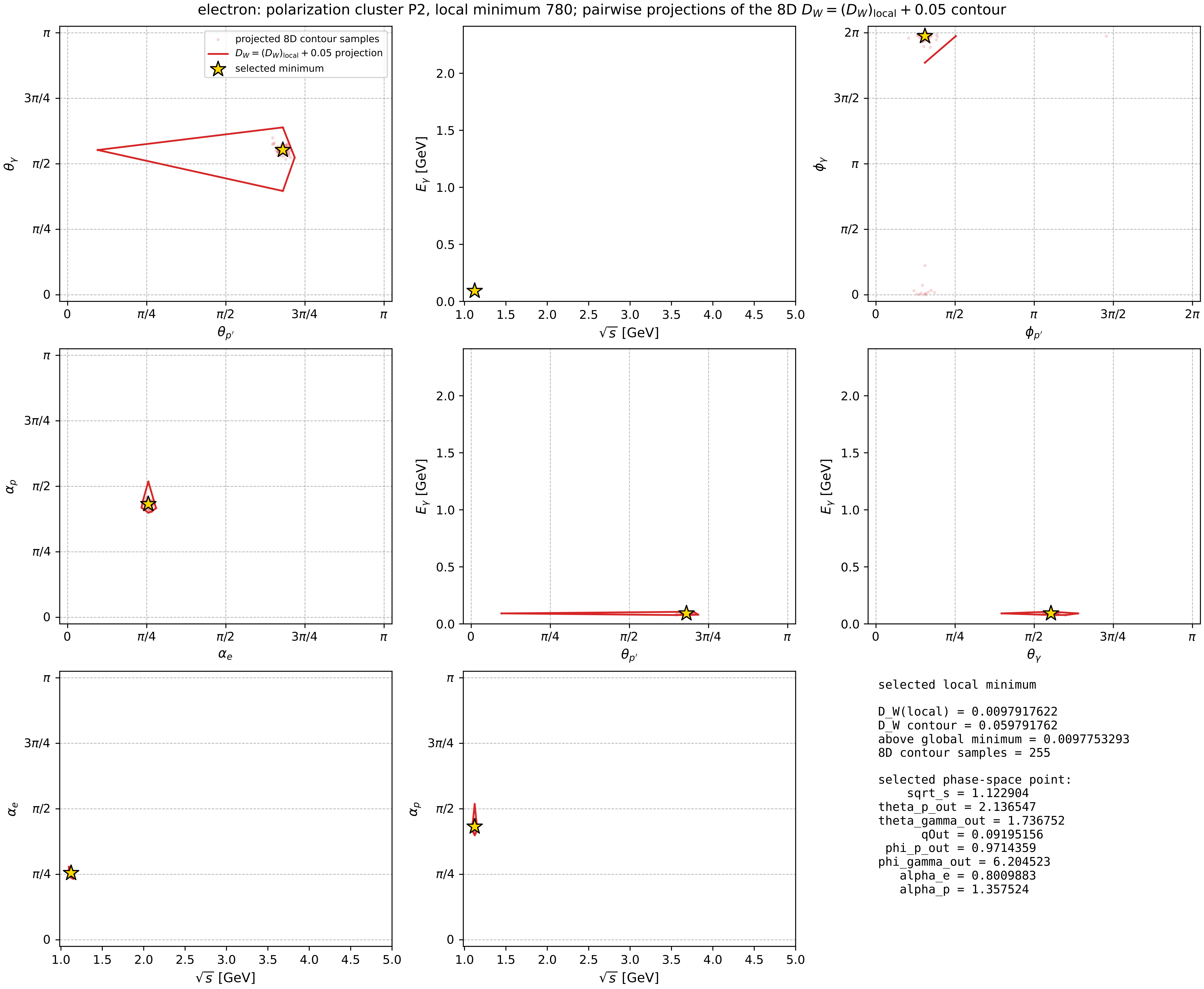
dw_mixing_angles_polarization_02_local_minimum_780 D_W=0.00979176
region: polarization_cluster_02_local_minimum_780
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80098832 rad
incoming lepton: 0.695997|+> +0.718044|->
proton mixing angle: alpha_p=1.3575236 rad
incoming proton: 0.21166|+> +0.977343|->

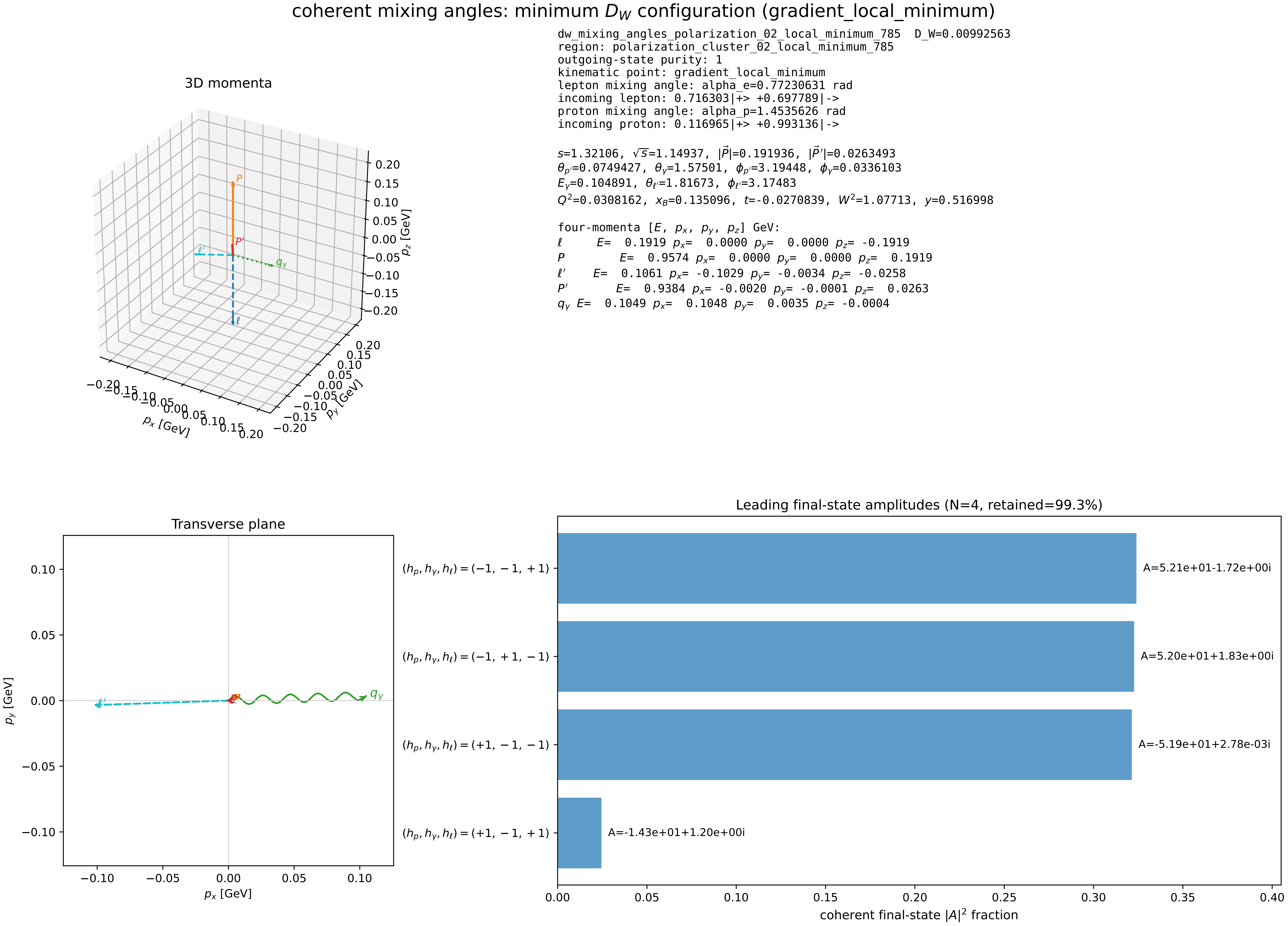
s=1.26091, $\sqrt{s}$=1.1229, $|\vec{P}|$=0.169679, $|\vec{P}'|$=0.00195255
$\theta_{p'}$=2.13655, θ_γ=1.73675, $\phi_{p'}$=0.971436, ϕ_γ=6.20452
E_γ=0.0919516, $\theta_{\ell'}$=1.39521, $\phi_{\ell'}$=3.07855
Q^2=0.037053, x_B=0.17697, t=-0.0289184, W^2=1.05216, y=0.54944

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1697 p_x= 0.0000 p_y= 0.0000 p_z= -0.1697
P E= 0.9532 p_x= 0.0000 p_y= 0.0000 p_z= 0.1697
ℓ' E= 0.0929 p_x= -0.0913 p_y= 0.0058 p_z= 0.0162
P' E= 0.9380 p_x= 0.0009 p_y= 0.0014 p_z= -0.0010
q_γ E= 0.0920 p_x= 0.0904 p_y= -0.0071 p_z= -0.0152

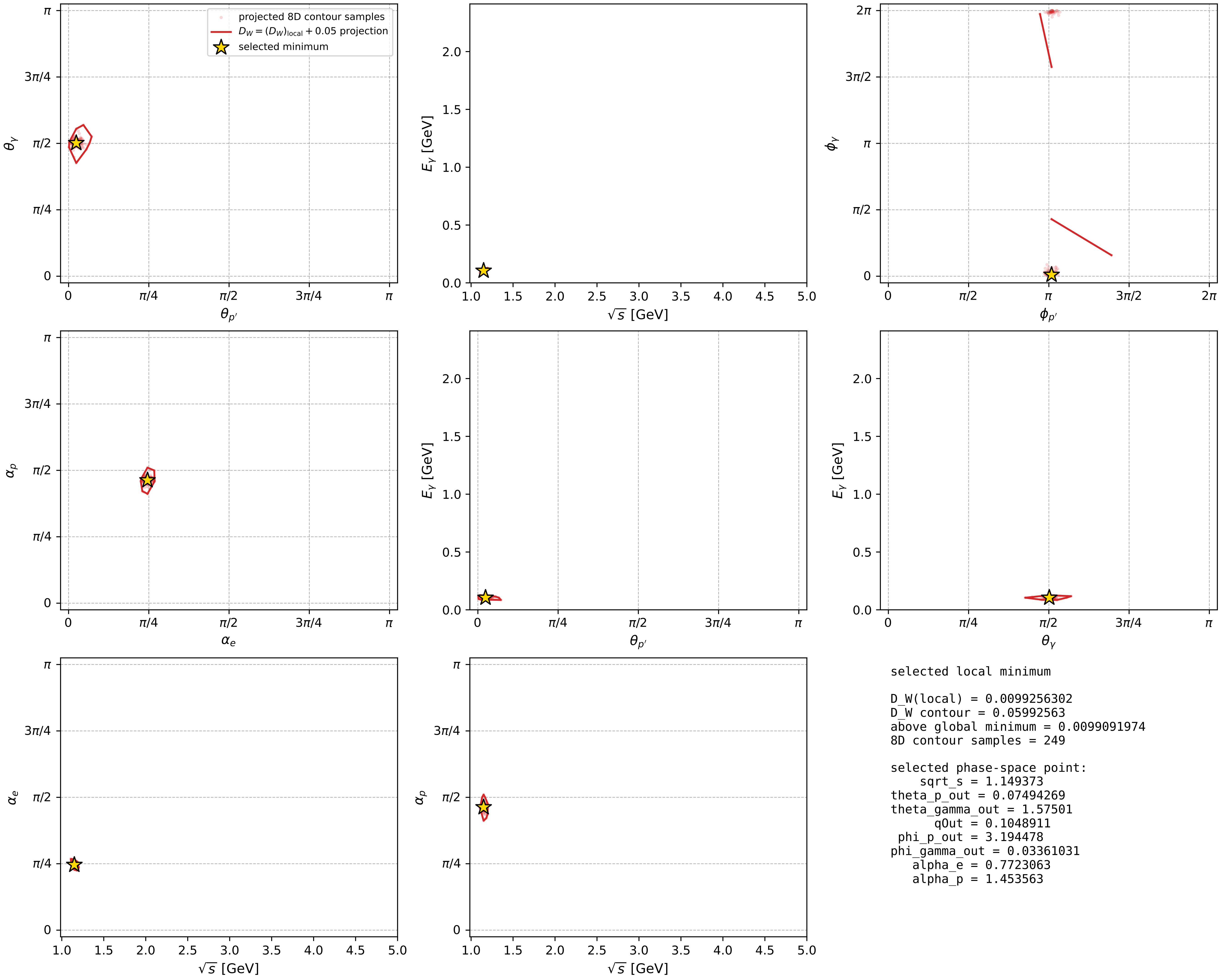




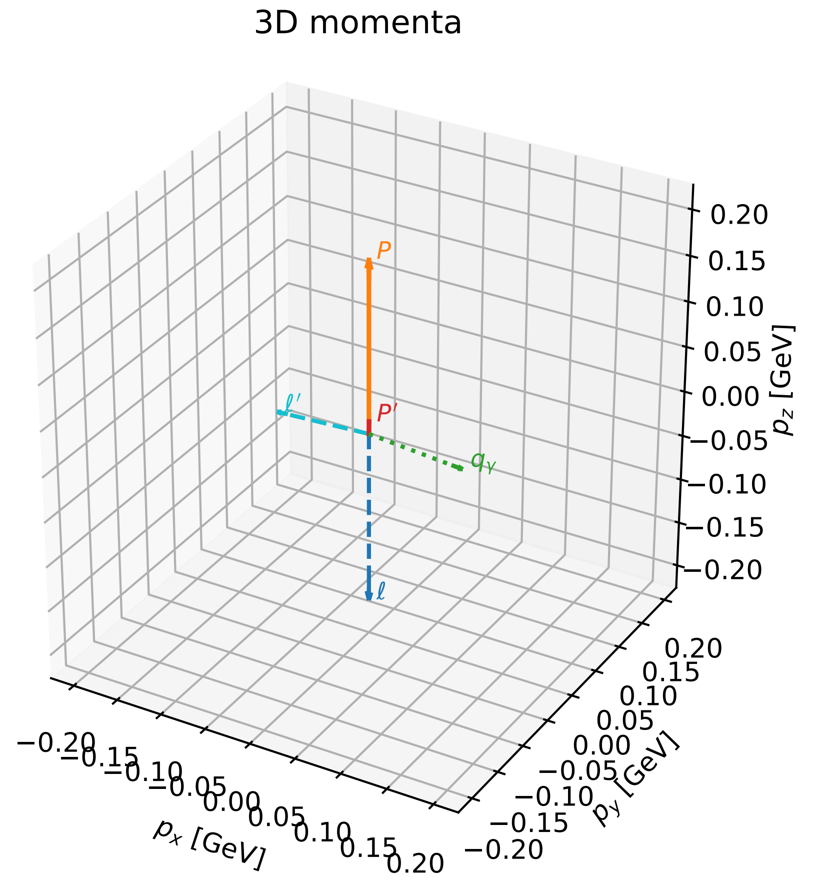

electron: polarization cluster P2, local minimum 780; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0097917622
D_W contour = 0.059791762
above global minimum = 0.0097753293
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.122904
theta_p_out = 2.136547
theta_gamma_out = 1.736752
q0out = 0.09195156
phi_p_out = 0.9714359
phi_gamma_out = 6.204523
alpha_e = 0.8009883
alpha_p = 1.357524


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
P
P'
ℓ'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_02_local_minimum_785 D_W=0.00992563
region: polarization_cluster_02_local_minimum_785
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77230631 rad
incoming lepton: 0.716303|+> +0.697789|->
proton mixing angle: alpha_p=1.4535626 rad
incoming proton: 0.116965|+> +0.993136|->
s=1.32106, $\sqrt{s}$=1.14937, $|\vec{P}|$=0.191936, $|\vec{P}'|$=0.0263493
$\theta_{p'}$=0.0749427, θ_γ=1.57501, $\phi_{p'}$=3.19448, ϕ_γ=0.0336103
E_γ=0.104891, $\theta_{\ell'}$=1.81673, $\phi_{\ell'}$=3.17483
Q^2=0.0308162, x_B=0.135096, t=-0.0270839, W^2=1.07713, y=0.516998
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1919 p_x= 0.0000 p_y= 0.0000 p_z= -0.1919
P E= 0.9574 p_x= 0.0000 p_y= 0.0000 p_z= 0.1919
ℓ' E= 0.1061 p_x= -0.1029 p_y= -0.0034 p_z= -0.0258
P' E= 0.9384 p_x= -0.0020 p_y= -0.0001 p_z= 0.0263
q_γ E= 0.1049 p_x= 0.1048 p_y= 0.0035 p_z= -0.0004
Transverse plane
ℓ'
q_γ
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=99.3%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=5.21e+01-1.72e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=5.20e+01+1.83e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-5.19e+01+2.78e-03i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=-1.43e+01+1.20e+00i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 785; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



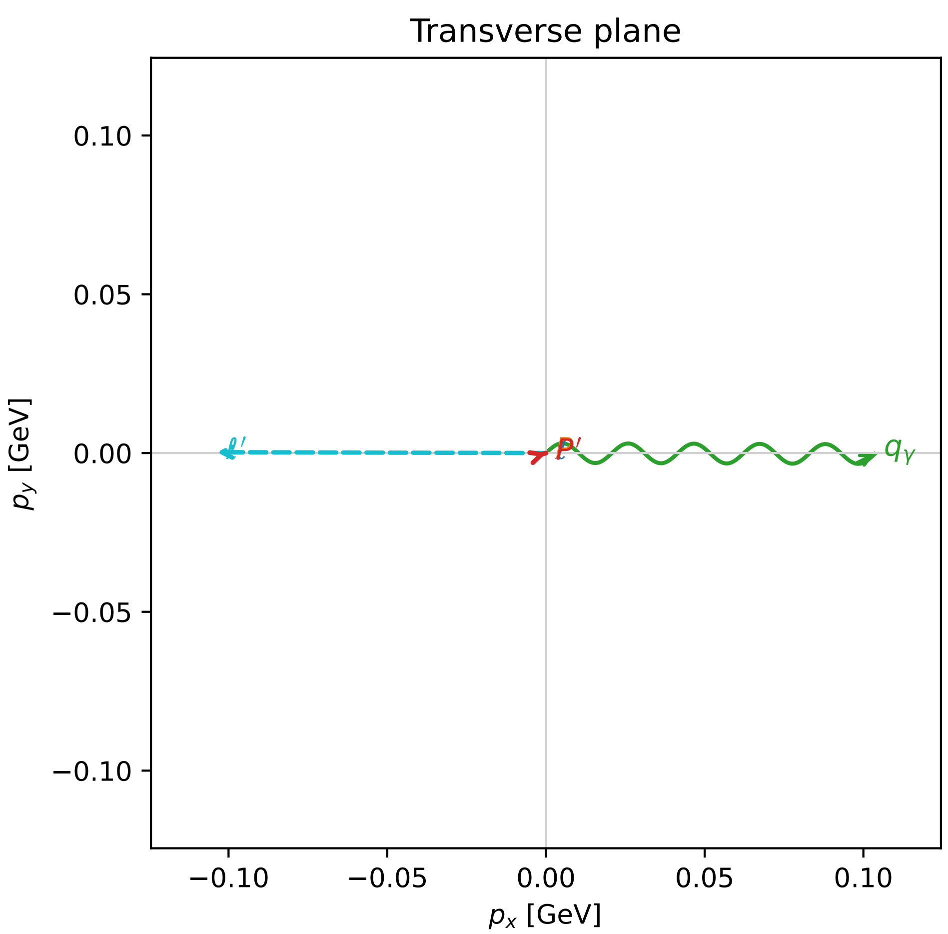
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

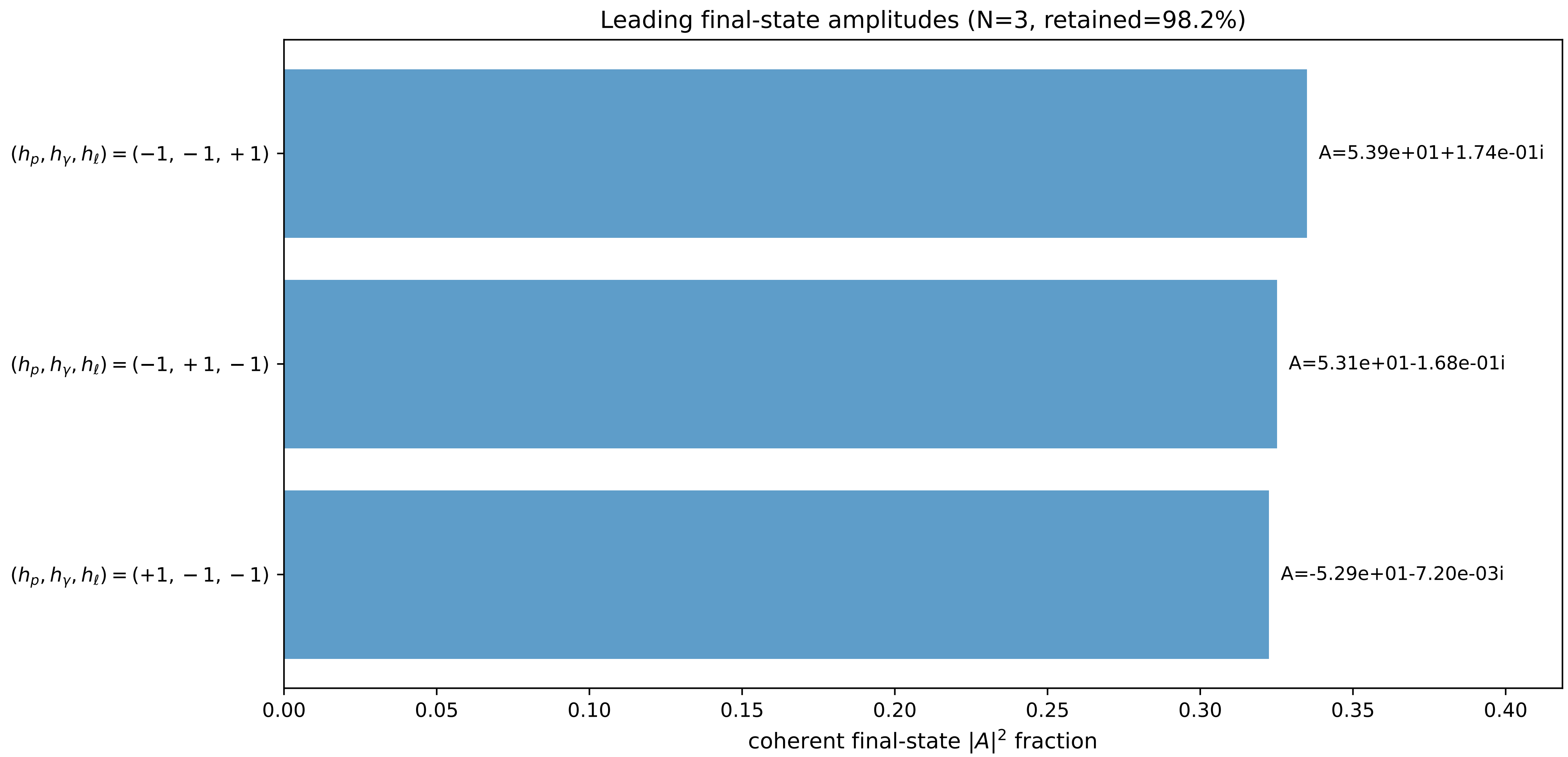


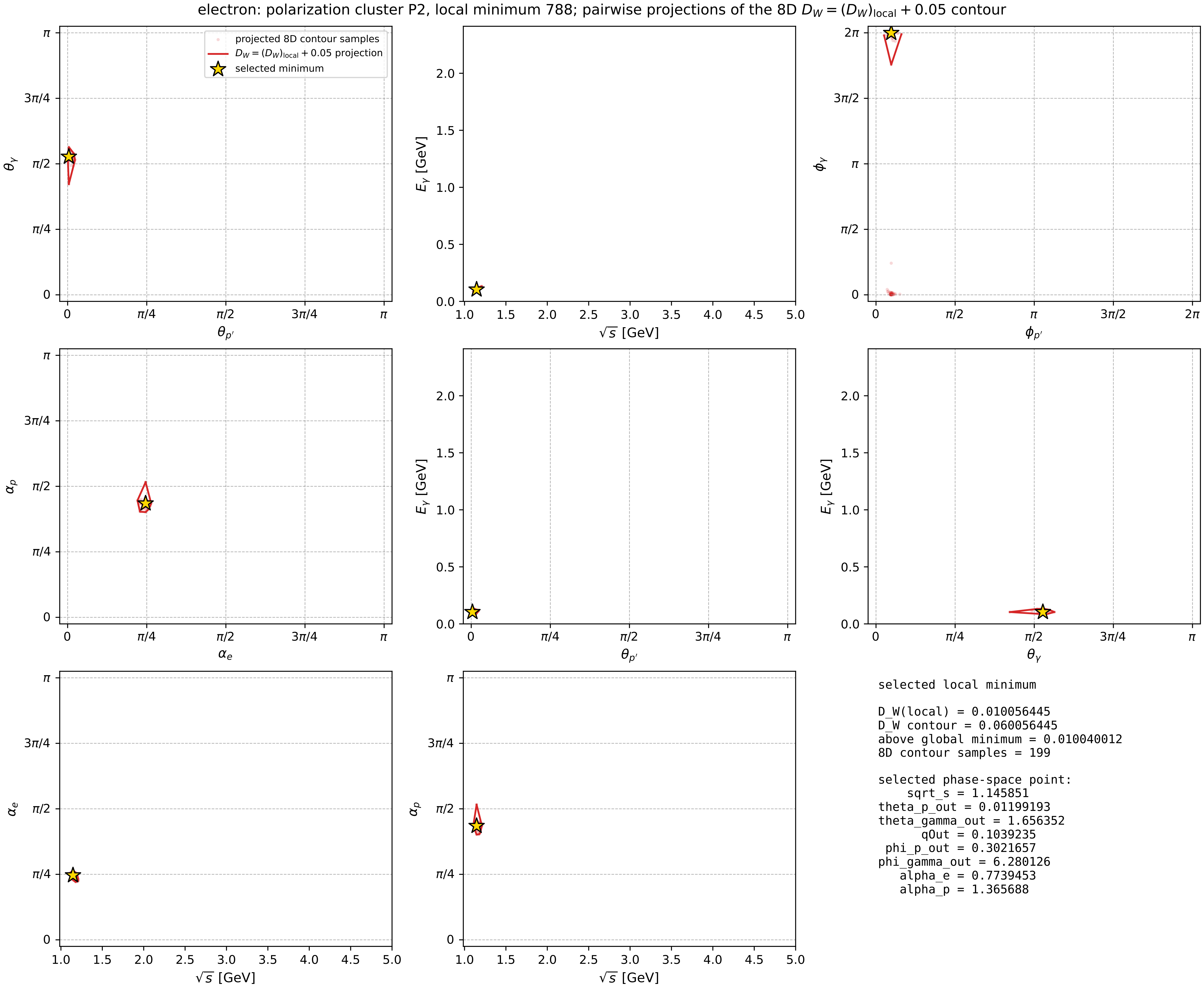

```
dw_mixing_angles_polarization_02_local_minimum_788  D_W=0.0100564
region: polarization_cluster_02_local_minimum_788
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77394525 rad
incoming lepton: 0.715159|+> +0.698962|->
proton mixing angle: alpha_p=1.3656881 rad
incoming proton: 0.203673|+> +0.979039|->
```

s=1.31297, $\sqrt{s}$=1.14585, $|\vec{P}|$=0.188999, $|\vec{P}'|$=0.0138893
$\theta_{p'}$=0.0119919, θ_γ=1.65635, $\phi_{p'}$=0.302166, ϕ_γ=6.28013
E_γ=0.103924, $\theta_{\ell'}$=1.61905, $\phi_{\ell'}$=3.13902
Q^2=0.037352, x_B=0.160621, t=-0.0303122, W^2=1.07504, y=0.536901

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1890 p_x= 0.0000 p_y= 0.0000 p_z= -0.1890
P E= 0.9569 p_x= 0.0000 p_y= 0.0000 p_z= 0.1890
ℓ' E= 0.1038 p_x= -0.1037 p_y= 0.0003 p_z= -0.0050
P' E= 0.9381 p_x= 0.0002 p_y= 0.0000 p_z= 0.0139
q_γ E= 0.1039 p_x= 0.1035 p_y= -0.0003 p_z= -0.0089







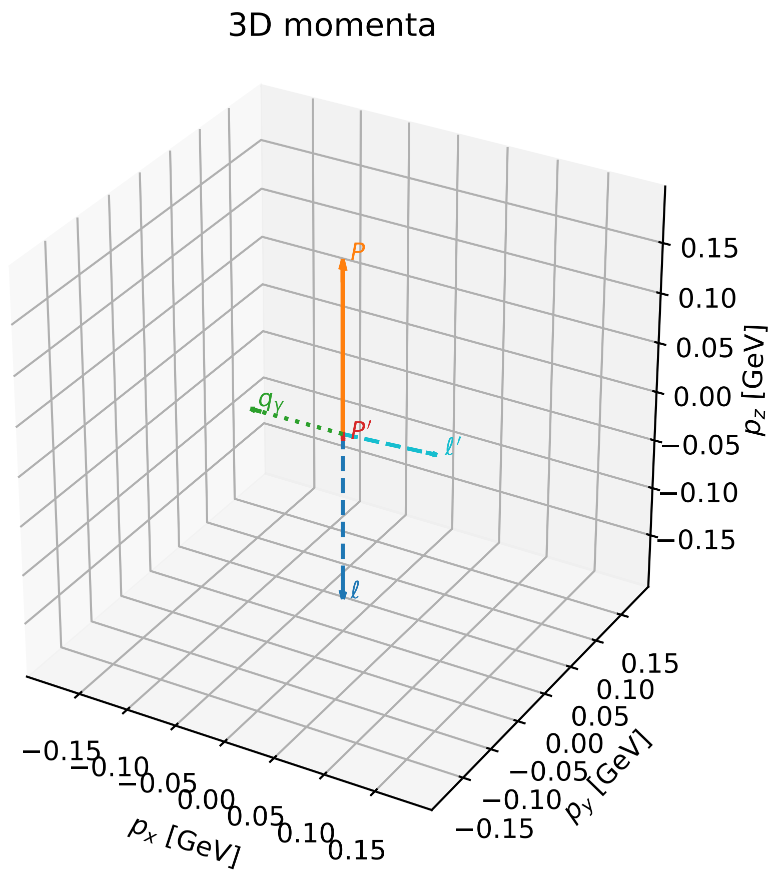
electron: polarization cluster P2, local minimum 788; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.010056445
D_W contour = 0.060056445
above global minimum = 0.010040012
8D contour samples = 199
selected phase-space point:
sqrt_s = 1.145851
theta_p_out = 0.01199193
theta_gamma_out = 1.656352
qOut = 0.1039235
phi_p_out = 0.3021657
phi_gamma_out = 6.280126
alpha_e = 0.7739453
alpha_p = 1.365688

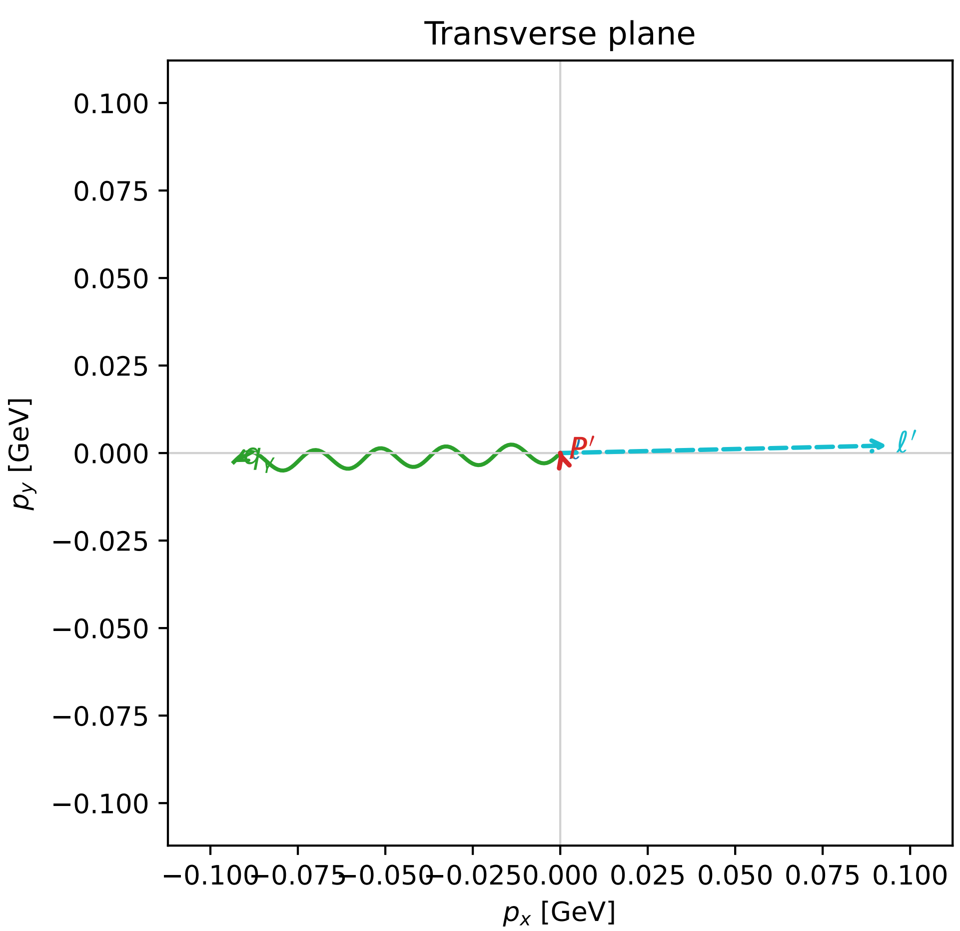
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

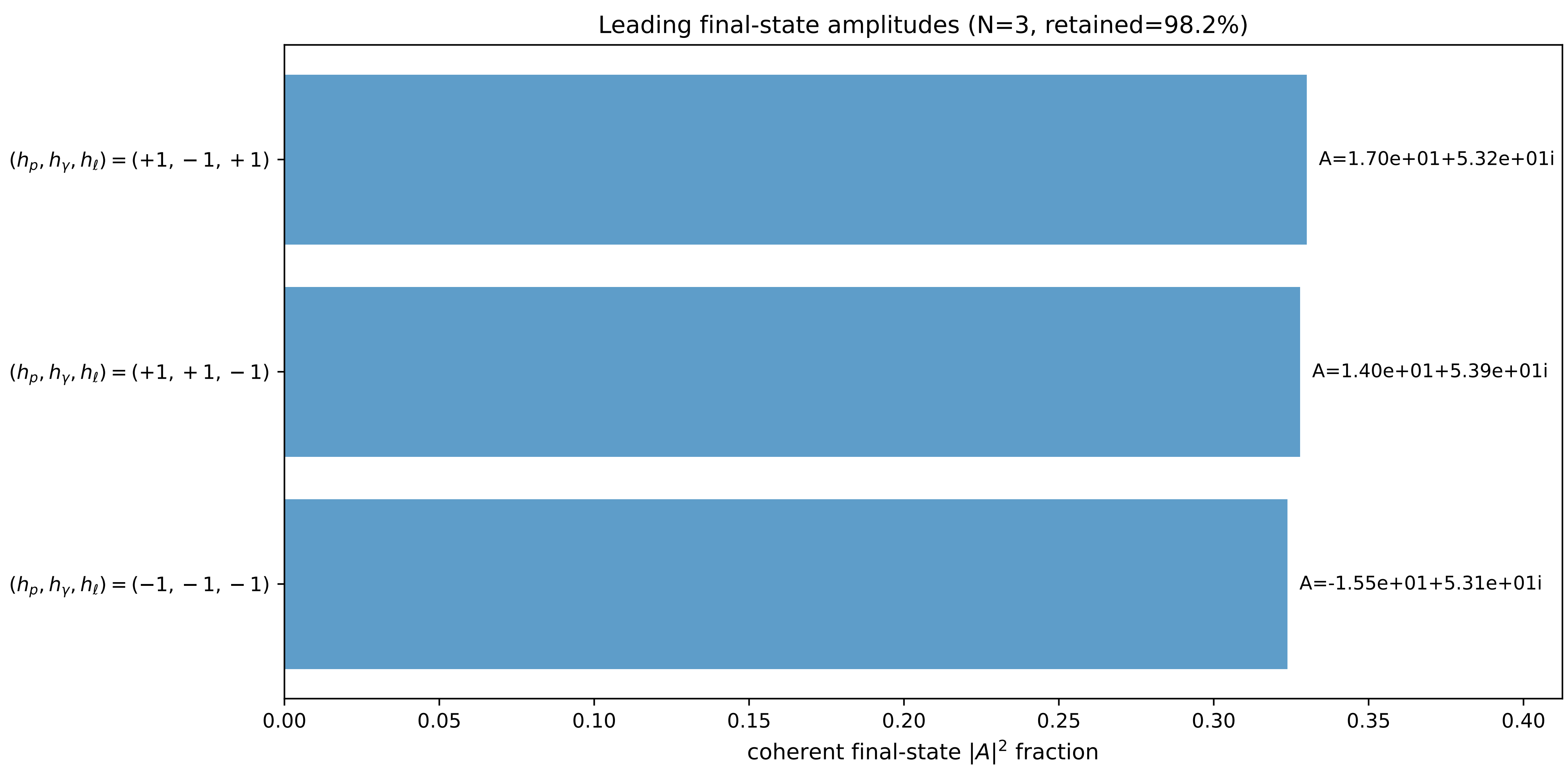
dw_mixing_angles_polarization_02_local_minimum_790 D_W=0.0101139
region: polarization_cluster_02_local_minimum_790
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77782162 rad
incoming lepton: 0.712444|+> +0.701729|->
proton mixing angle: alpha_p=1.7730066 rad
incoming proton: -0.200835|+> +0.979625|->

s=1.26556, $\sqrt{s}$=1.12497, $|\vec{P}|$=0.171433, $|\vec{P}'|$=0.00518528
$\theta_{p'}$=3.05207, θ_γ=1.56744, $\phi_{p'}$=1.85608, ϕ_γ=3.16934
E_γ=0.0933554, $\theta_{\ell'}$=1.51895, $\phi_{\ell'}$=0.0229511
Q^2=0.0337555, x_B=0.161607, t=-0.0309459, W^2=1.05496, y=0.541524

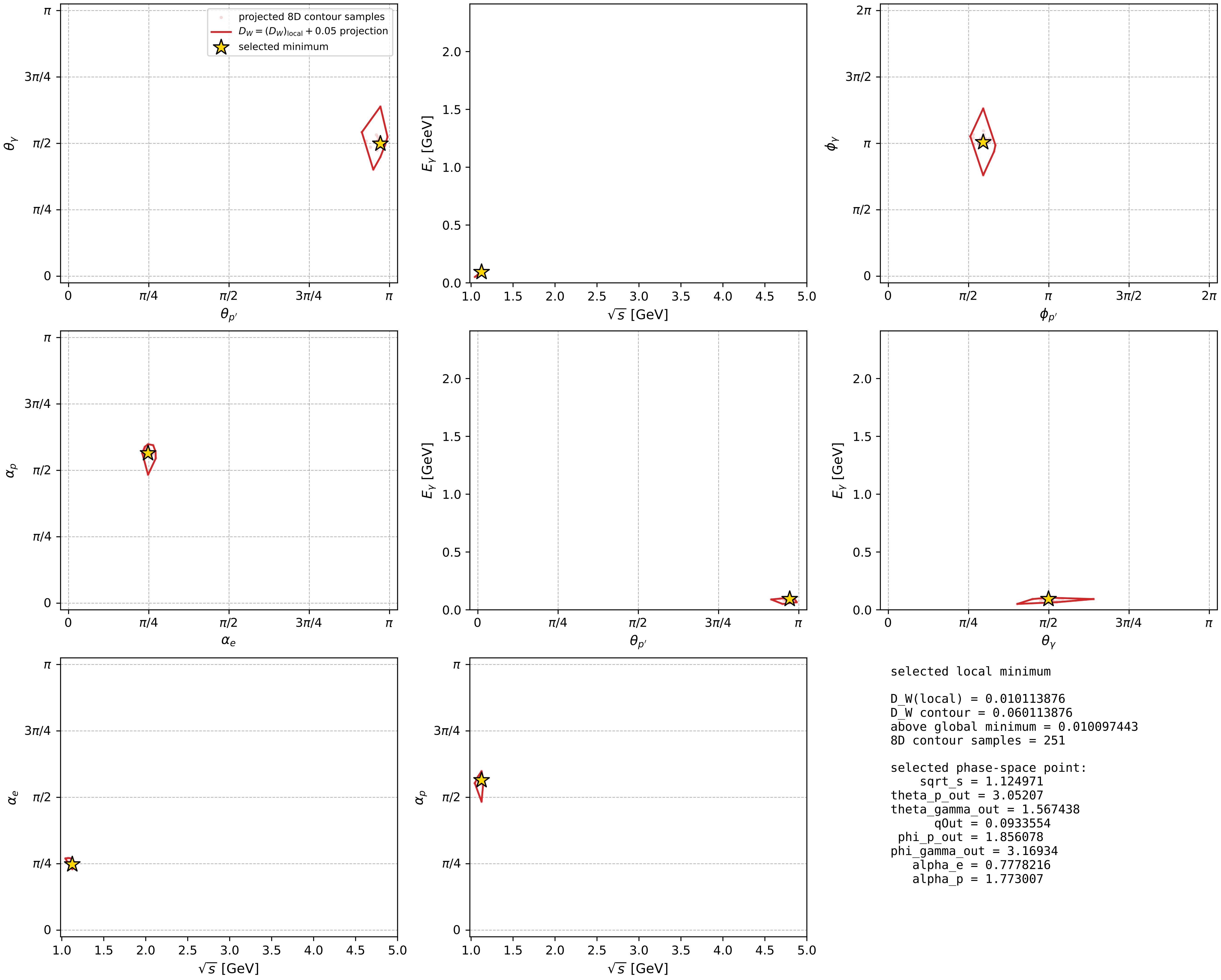
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1714 p_x= 0.0000 p_y= 0.0000 p_z= -0.1714
P E= 0.9535 p_x= 0.0000 p_y= 0.0000 p_z= 0.1714
ℓ' E= 0.0936 p_x= 0.0934 p_y= 0.0021 p_z= 0.0049
P' E= 0.9380 p_x= -0.0001 p_y= 0.0004 p_z= -0.0052
q_γ E= 0.0934 p_x= -0.0933 p_y= -0.0026 p_z= 0.0003



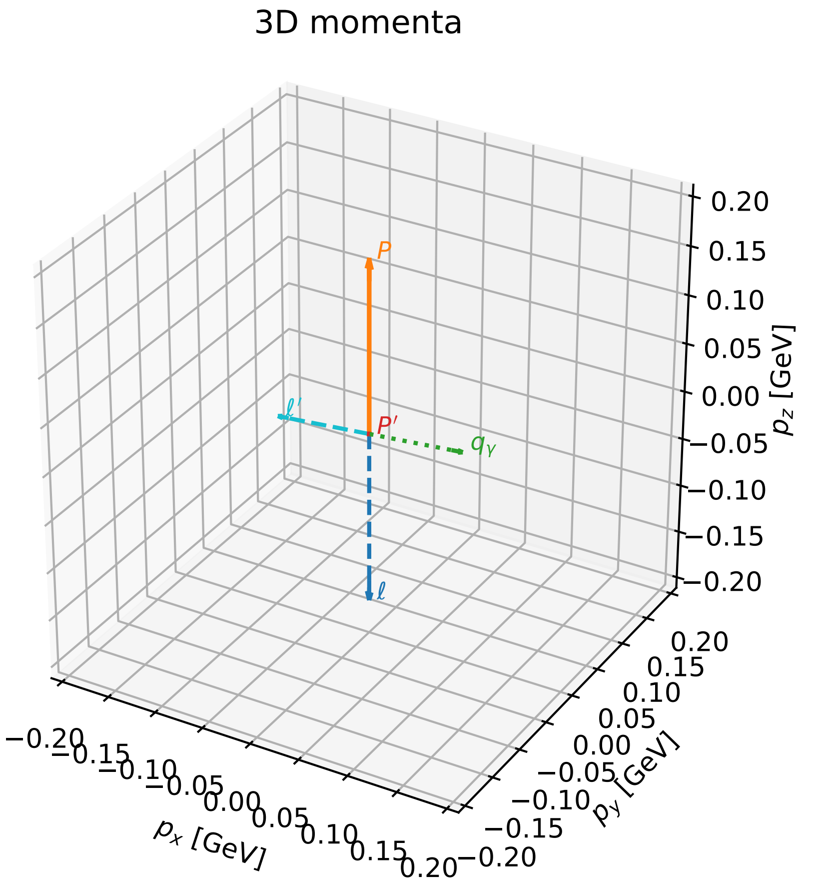




electron: polarization cluster P2, local minimum 790; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



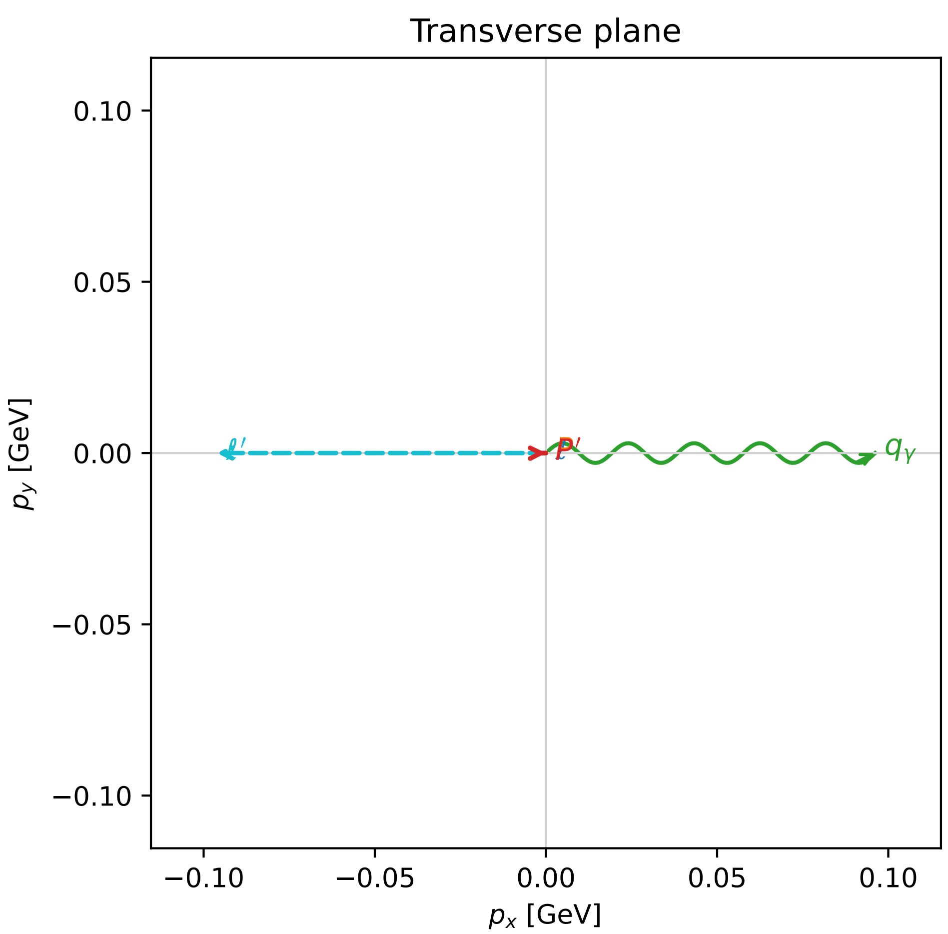
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

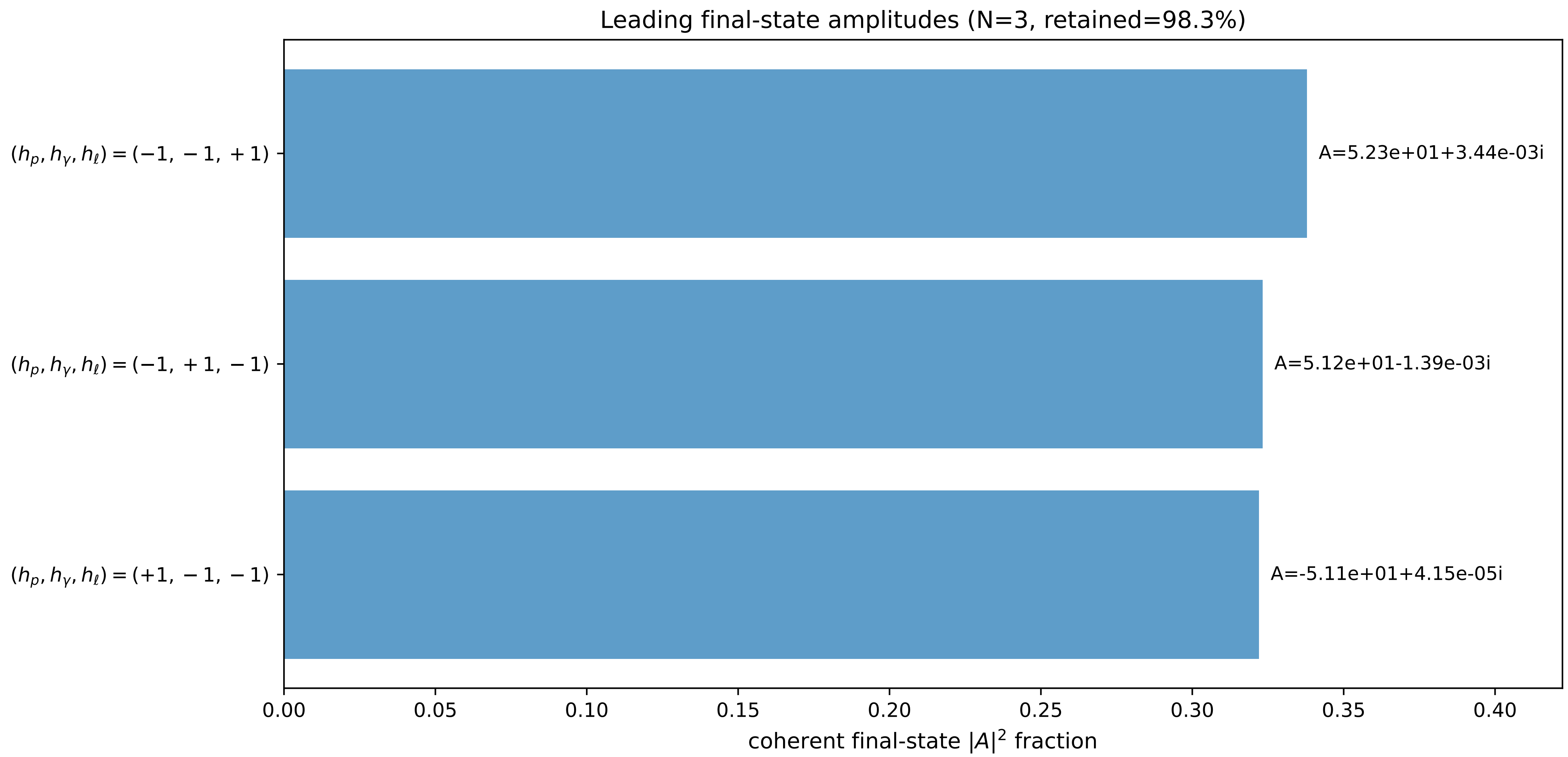


dw_mixing_angles_polarization_02_local_minimum_797 D_W=0.0103184
region: polarization_cluster_02_local_minimum_797
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.76992886 rad
incoming lepton: 0.71796|+> +0.696084|->
proton mixing angle: alpha_p=1.3669543 rad
incoming proton: 0.202433|+> +0.979296|->

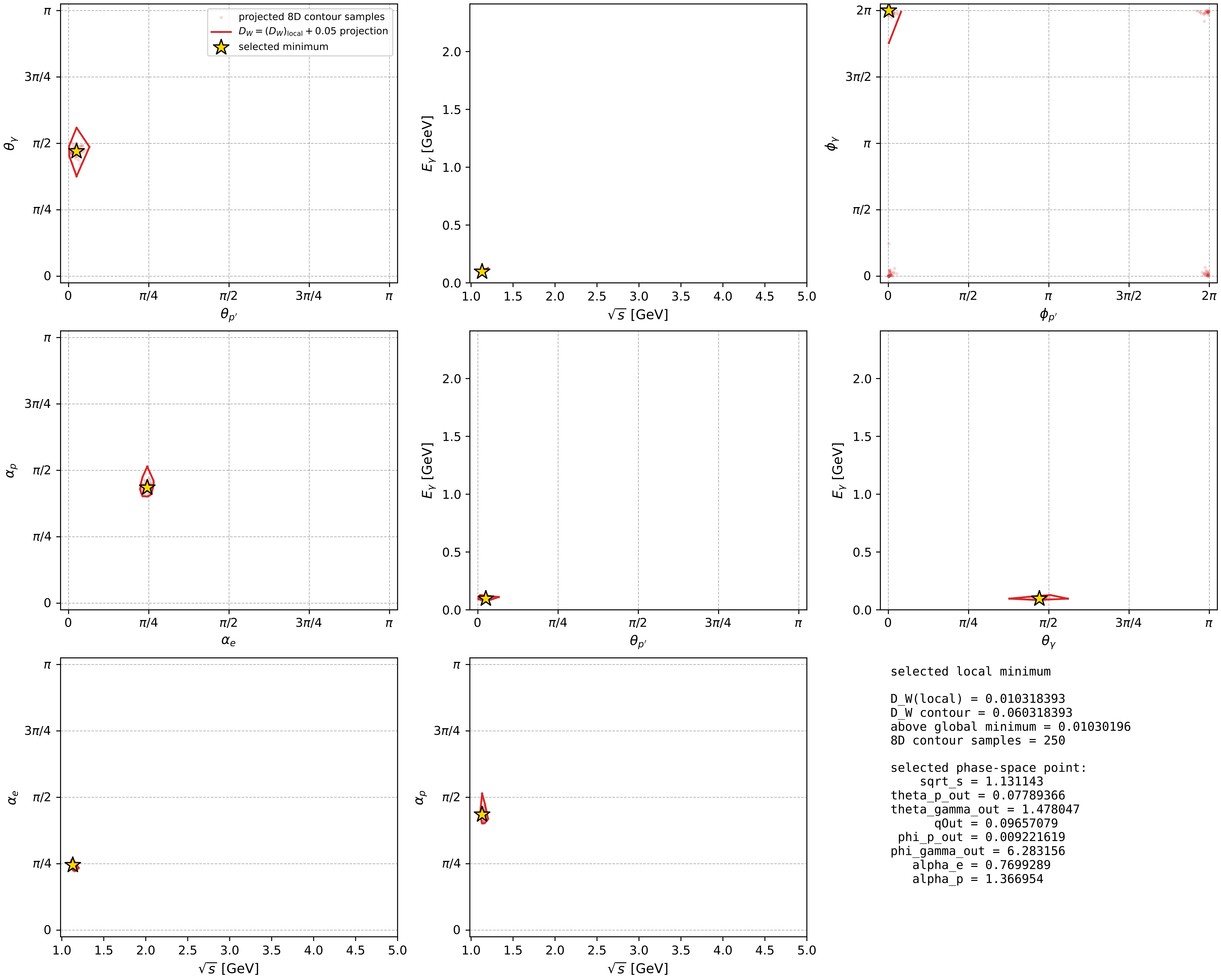
s=1.27948, $\sqrt{s}$=1.13114, $|\vec{P}|$=0.176653, $|\vec{P}'|$=9.29299e-07
$\theta_{p'}$=0.0778937, θ_γ=1.47805, $\phi_{p'}$=0.00922162, ϕ_γ=6.28316
E_γ=0.0965708, $\theta_{\ell'}$=1.66355, $\phi_{\ell'}$=3.14156
Q^2=0.0309589, x_B=0.145946, t=-0.030934, W^2=1.06101, y=0.530791

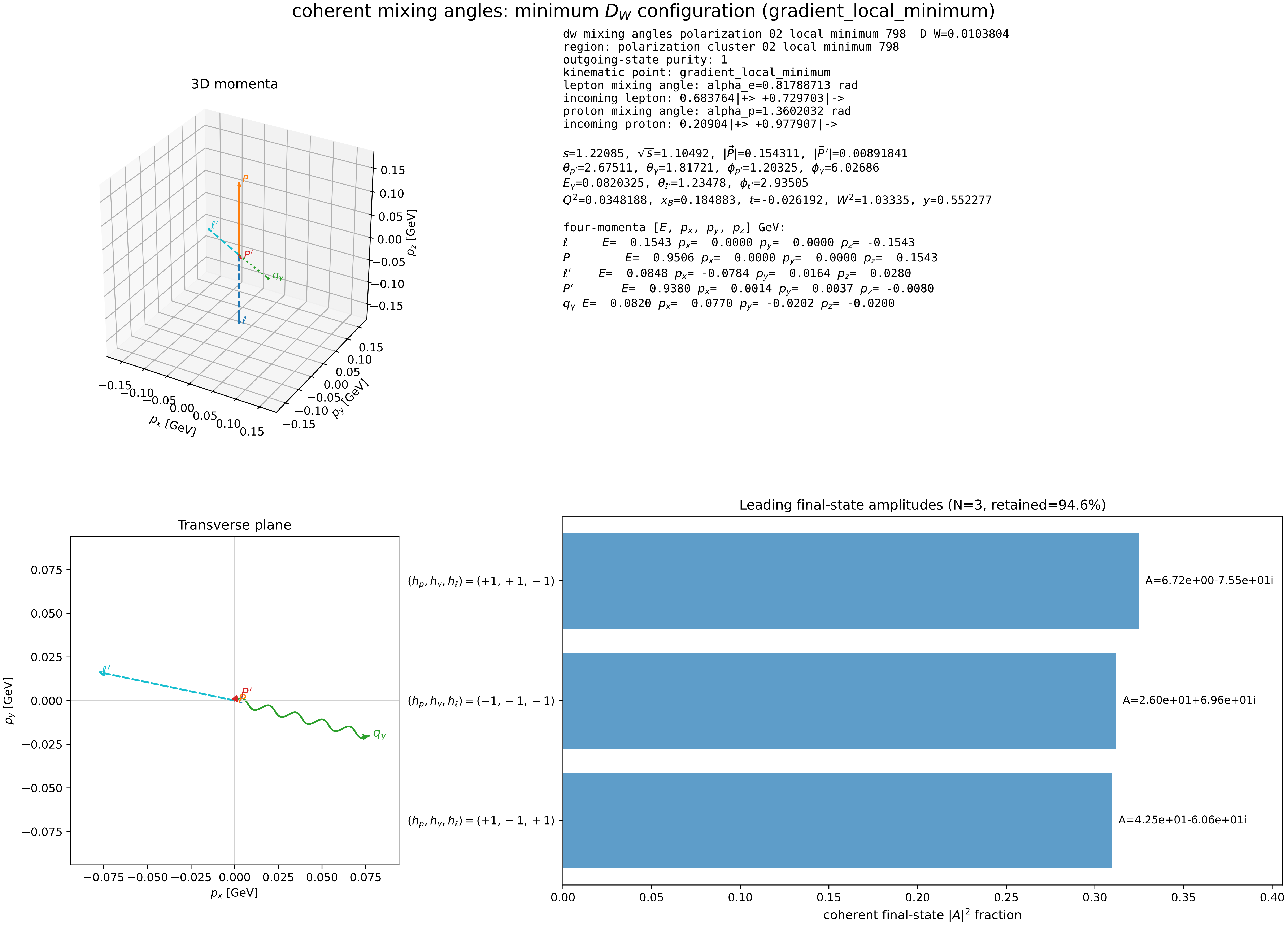
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1767 p_x= 0.0000 p_y= 0.0000 p_z= -0.1767
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1767
ℓ' E= 0.0966 p_x= -0.0962 p_y= 0.0000 p_z= -0.0089
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0000
q_γ E= 0.0966 p_x= 0.0962 p_y= -0.0000 p_z= 0.0089



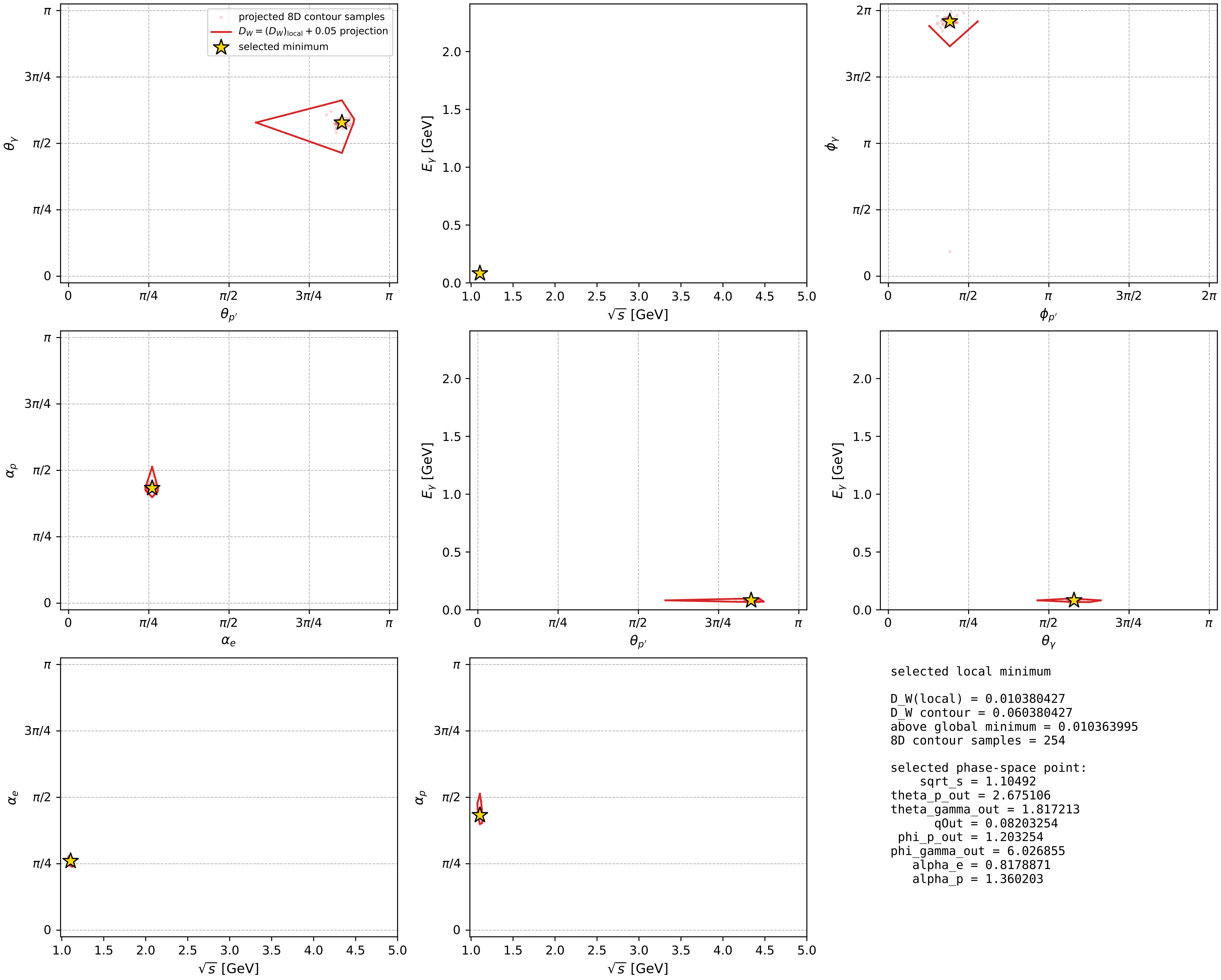


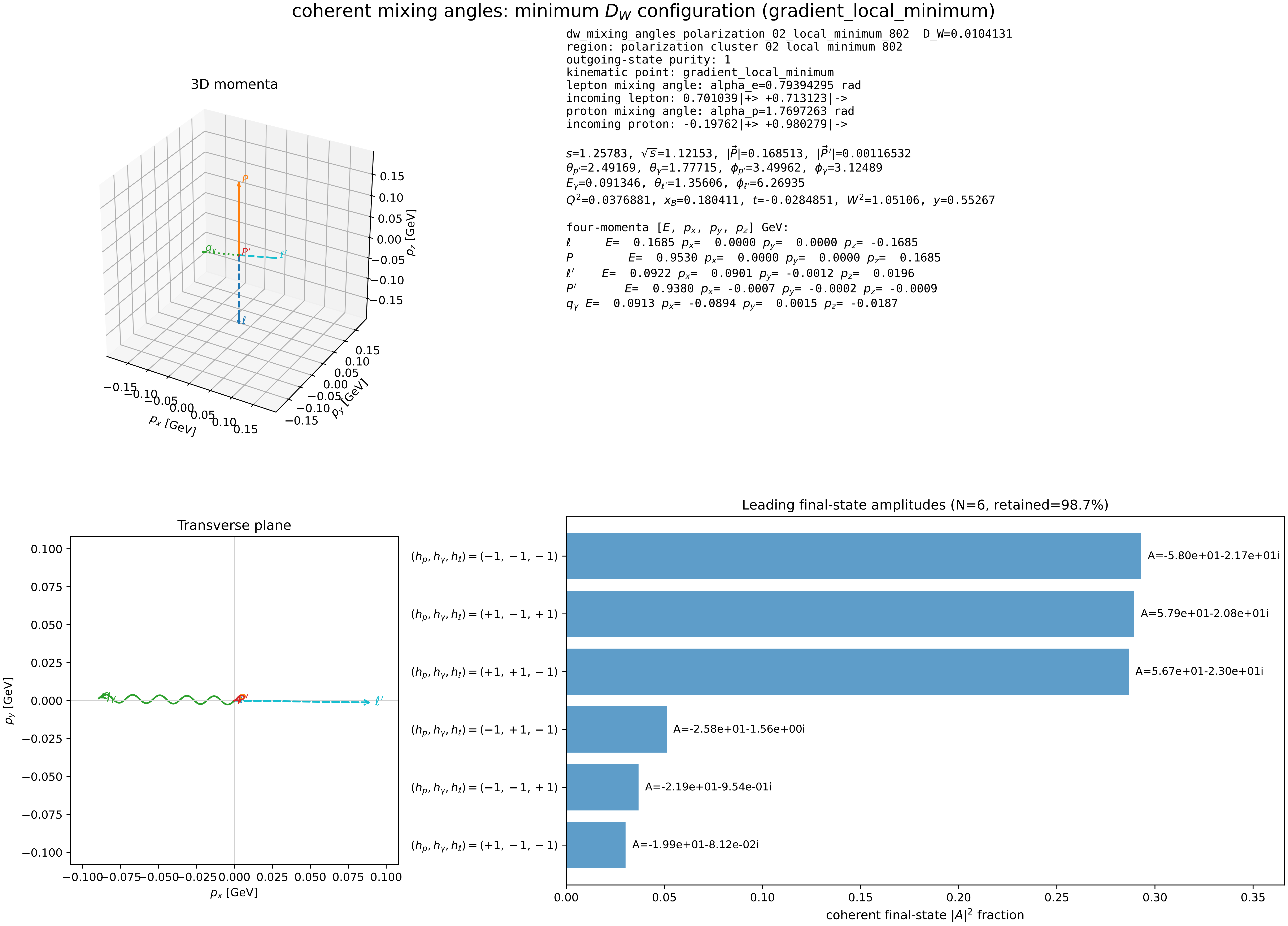
electron: polarization cluster P2, local minimum 797; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



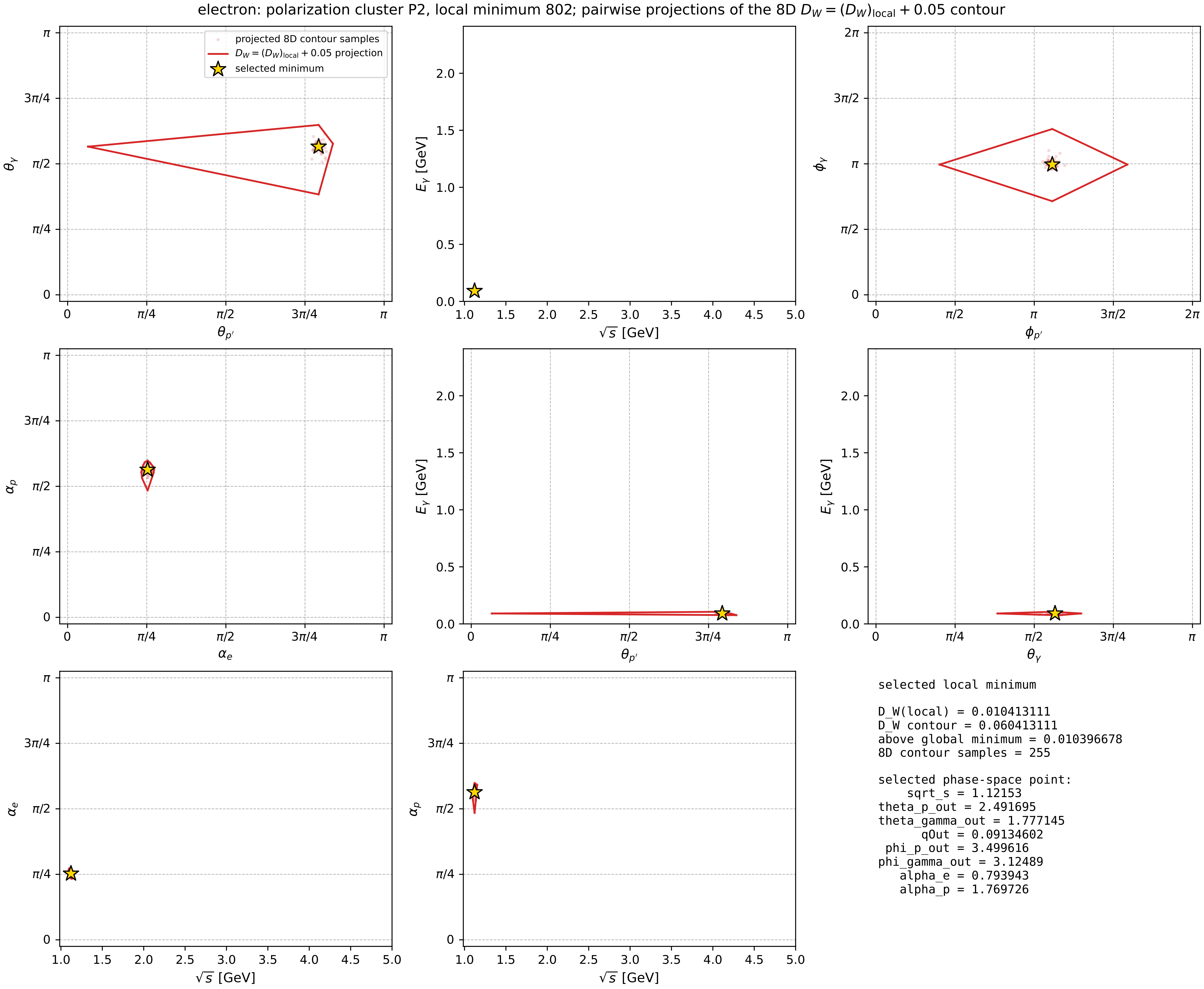

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_02_local_minimum_798 D_W=0.0103804
region: polarization_cluster_02_local_minimum_798
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.81788713 rad
incoming lepton: 0.683764|+> +0.729703|->
proton mixing angle: alpha_p=1.3602032 rad
incoming proton: 0.20904|+> +0.977907|->
s=1.22085, $\sqrt{s}$=1.10492, $|\vec{P}|$=0.154311, $|\vec{P}'|$=0.00891841
$\theta_{p'}$=2.67511, θ_γ=1.81721, $\phi_{p'}$=1.20325, ϕ_γ=6.02686
E_γ=0.0820325, $\theta_{\ell'}$=1.23478, $\phi_{\ell'}$=2.93505
Q^2=0.0348188, x_B=0.184883, t=-0.026192, W^2=1.03335, y=0.552277
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1543 p_x= 0.0000 p_y= 0.0000 p_z= -0.1543
P E= 0.9506 p_x= 0.0000 p_y= 0.0000 p_z= 0.1543
ℓ' E= 0.0848 p_x= -0.0784 p_y= 0.0164 p_z= 0.0280
P' E= 0.9380 p_x= 0.0014 p_y= 0.0037 p_z= -0.0080
q_γ E= 0.0820 p_x= 0.0770 p_y= -0.0202 p_z= -0.0200
Transverse plane
p_x [GeV]
p_y [GeV]
ℓ'
P'
q_γ
Leading final-state amplitudes (N=3, retained=94.6%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=6.72e+00-7.55e+01i
A=2.60e+01+6.96e+01i
A=4.25e+01-6.06e+01i
coherent final-state $|A|^2$ fraction

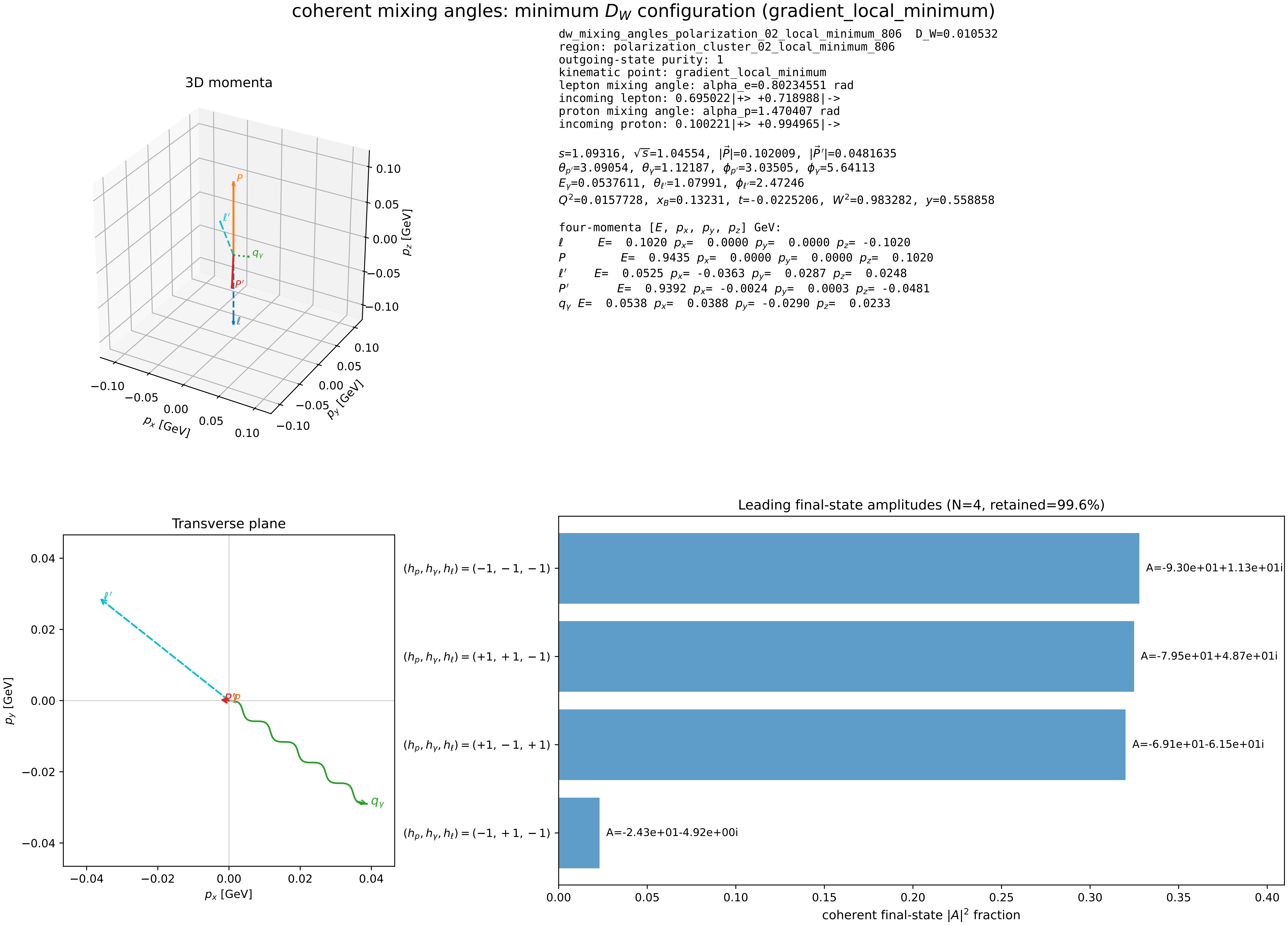
electron: polarization cluster P2, local minimum 798; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



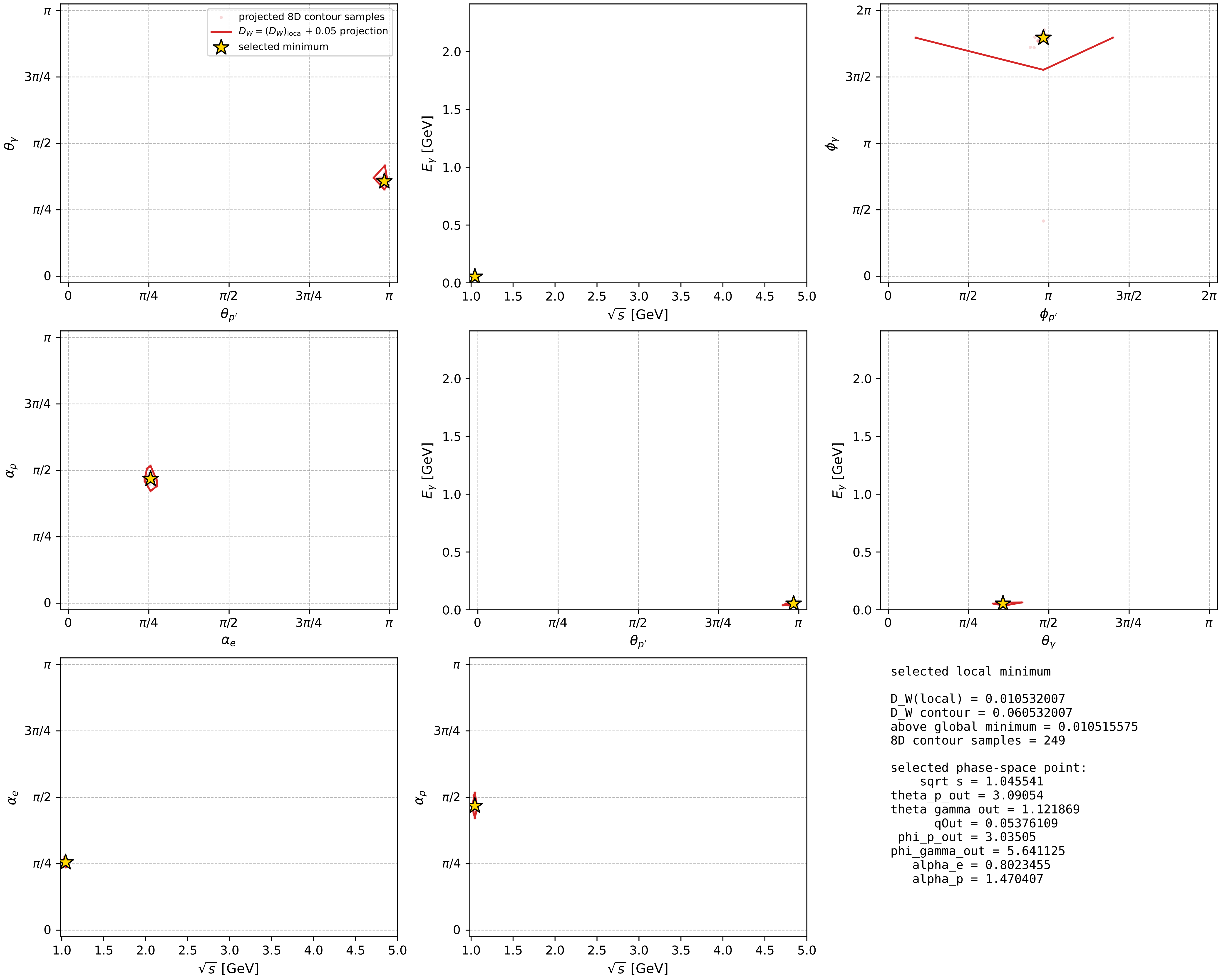


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_02_local_minimum_802 D_W=0.0104131
region: polarization_cluster_02_local_minimum_802
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.79394295 rad
incoming lepton: 0.701039|+> +0.713123|->
proton mixing angle: alpha_p=1.7697263 rad
incoming proton: -0.19762|+> +0.980279|->
s=1.25783, $\sqrt{s}$=1.12153, $|\vec{P}|$=0.168513, $|\vec{P}'|$=0.00116532
$\theta_{p'}$=2.49169, θ_γ=1.77715, $\phi_{p'}$=3.49962, ϕ_γ=3.12489
E_γ=0.091346, $\theta_{\ell'}$=1.35606, $\phi_{\ell'}$=6.26935
Q^2=0.0376881, x_B=0.180411, t=-0.0284851, W^2=1.05106, y=0.55267
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1685 p_x= 0.0000 p_y= 0.0000 p_z= -0.1685
P E= 0.9530 p_x= 0.0000 p_y= 0.0000 p_z= 0.1685
ℓ' E= 0.0922 p_x= 0.0901 p_y= -0.0012 p_z= 0.0196
P' E= 0.9380 p_x= -0.0007 p_y= -0.0002 p_z= -0.0009
q_γ E= 0.0913 p_x= -0.0894 p_y= 0.0015 p_z= -0.0187
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=6, retained=98.7%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-5.80e+01-2.17e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=5.79e+01-2.08e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=5.67e+01-2.30e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-2.58e+01-1.56e+00i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-2.19e+01-9.54e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-1.99e+01-8.12e-02i
coherent final-state $|A|^2$ fraction

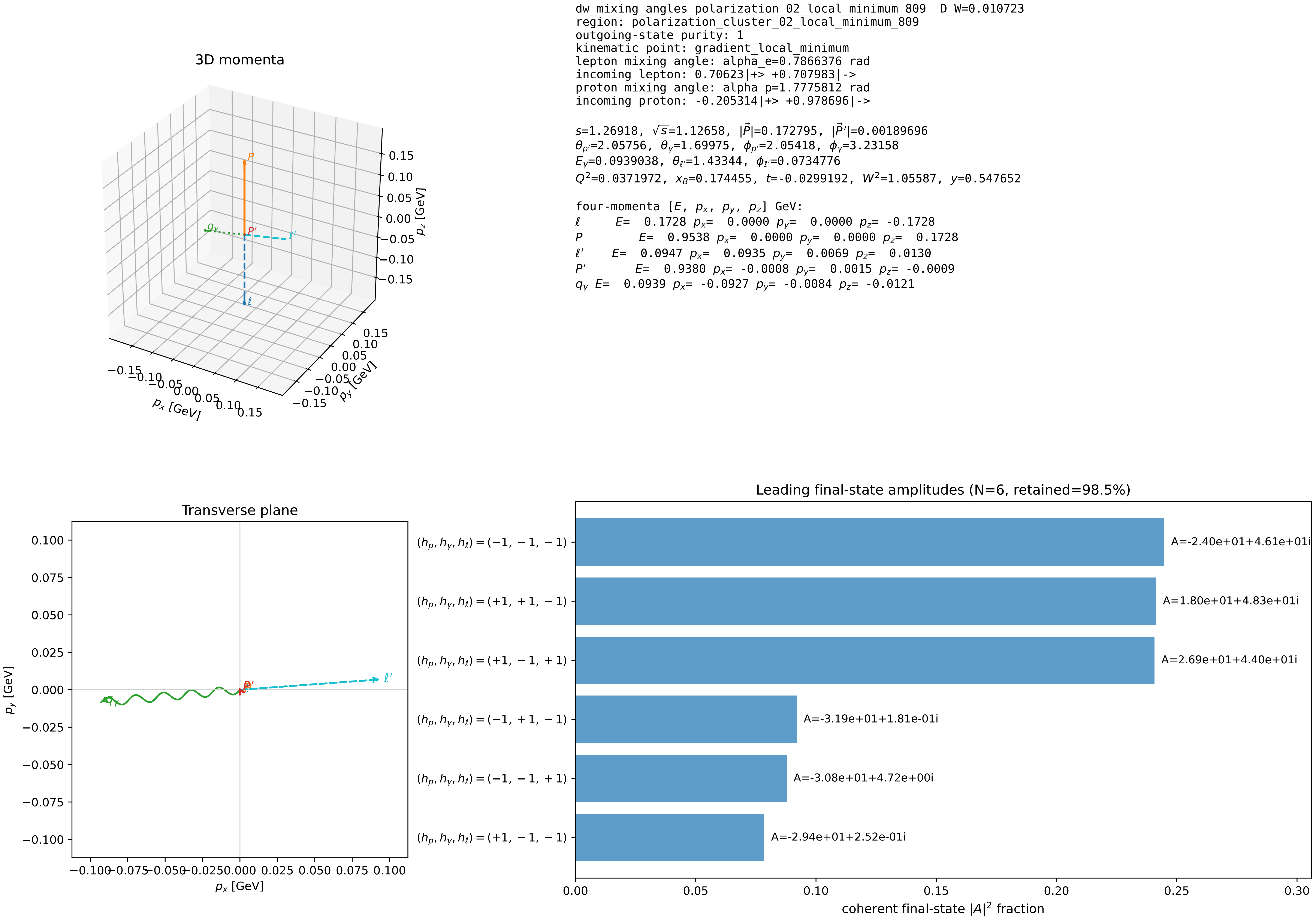

electron: polarization cluster P2, local minimum 802; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.010413111
D_W contour = 0.060413111
above global minimum = 0.010396678
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.12153
theta_p_out = 2.491695
theta_gamma_out = 1.777145
qOut = 0.09134602
phi_p_out = 3.499616
phi_gamma_out = 3.12489
alpha_e = 0.793943
alpha_p = 1.769726

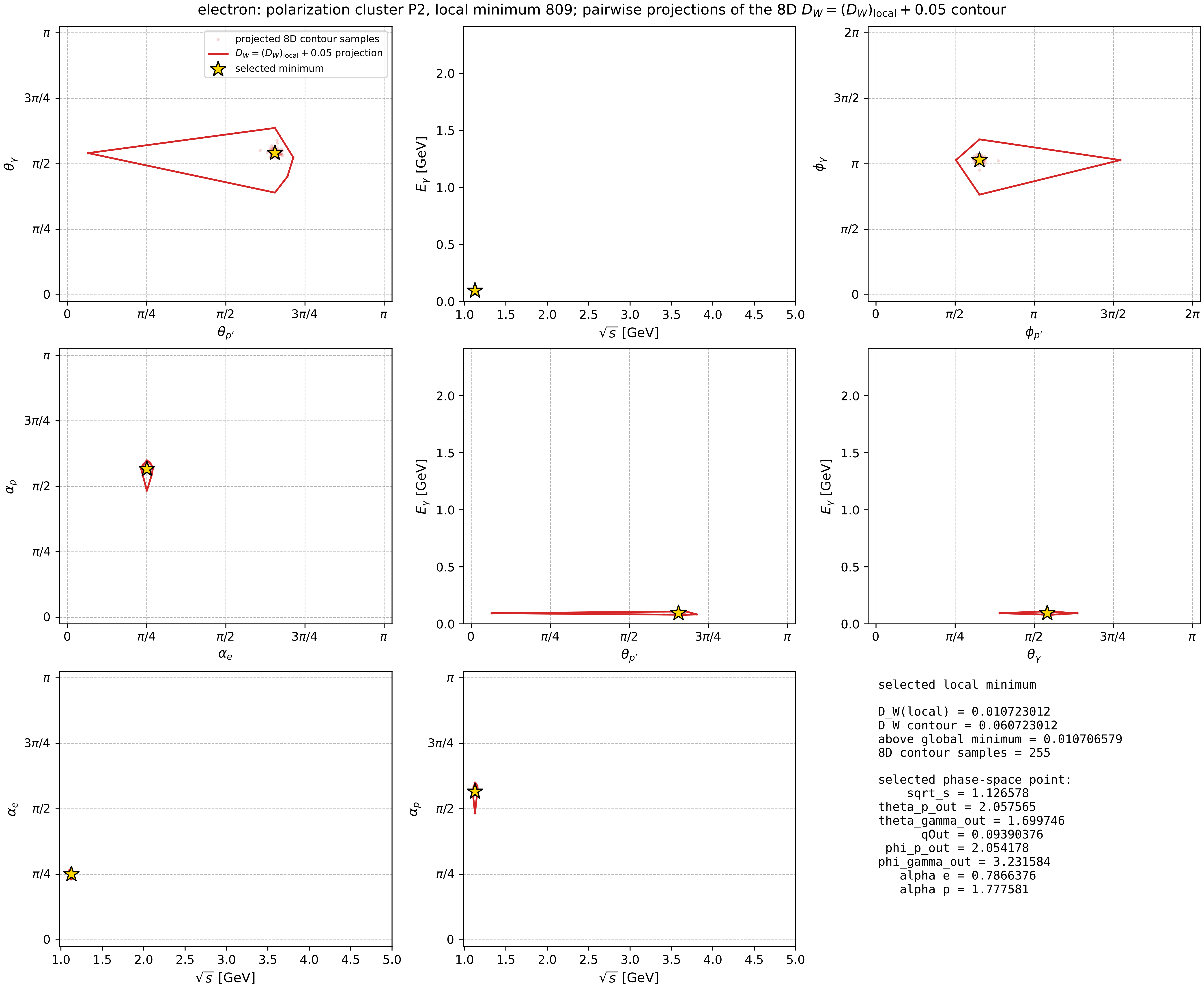

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_806 D_W=0.010532
region: polarization_cluster_02_local_minimum_806
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80234551 rad
incoming lepton: 0.695022|+> +0.718988|->
proton mixing angle: alpha_p=1.470407 rad
incoming proton: 0.100221|+> +0.994965|->
s=1.09316, $\sqrt{s}$=1.04554, $|\vec{P}|$=0.102009, $|\vec{P}'|$=0.0481635
$\theta_{p'}$=3.09054, θ_γ=1.12187, $\phi_{p'}$=3.03505, ϕ_γ=5.64113
E_γ=0.0537611, $\theta_{\ell'}$=1.07991, $\phi_{\ell'}$=2.47246
Q^2=0.0157728, x_B=0.13231, t=-0.0225206, W^2=0.983282, y=0.558858
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1020 p_x= 0.0000 p_y= 0.0000 p_z= -0.1020
P E= 0.9435 p_x= 0.0000 p_y= 0.0000 p_z= 0.1020
ℓ' E= 0.0525 p_x= -0.0363 p_y= 0.0287 p_z= 0.0248
P' E= 0.9392 p_x= -0.0024 p_y= 0.0003 p_z= -0.0481
q_γ E= 0.0538 p_x= 0.0388 p_y= -0.0290 p_z= 0.0233
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=99.6%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-9.30e+01+1.13e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-7.95e+01+4.87e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-6.91e+01-6.15e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-2.43e+01-4.92e+00i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 806; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

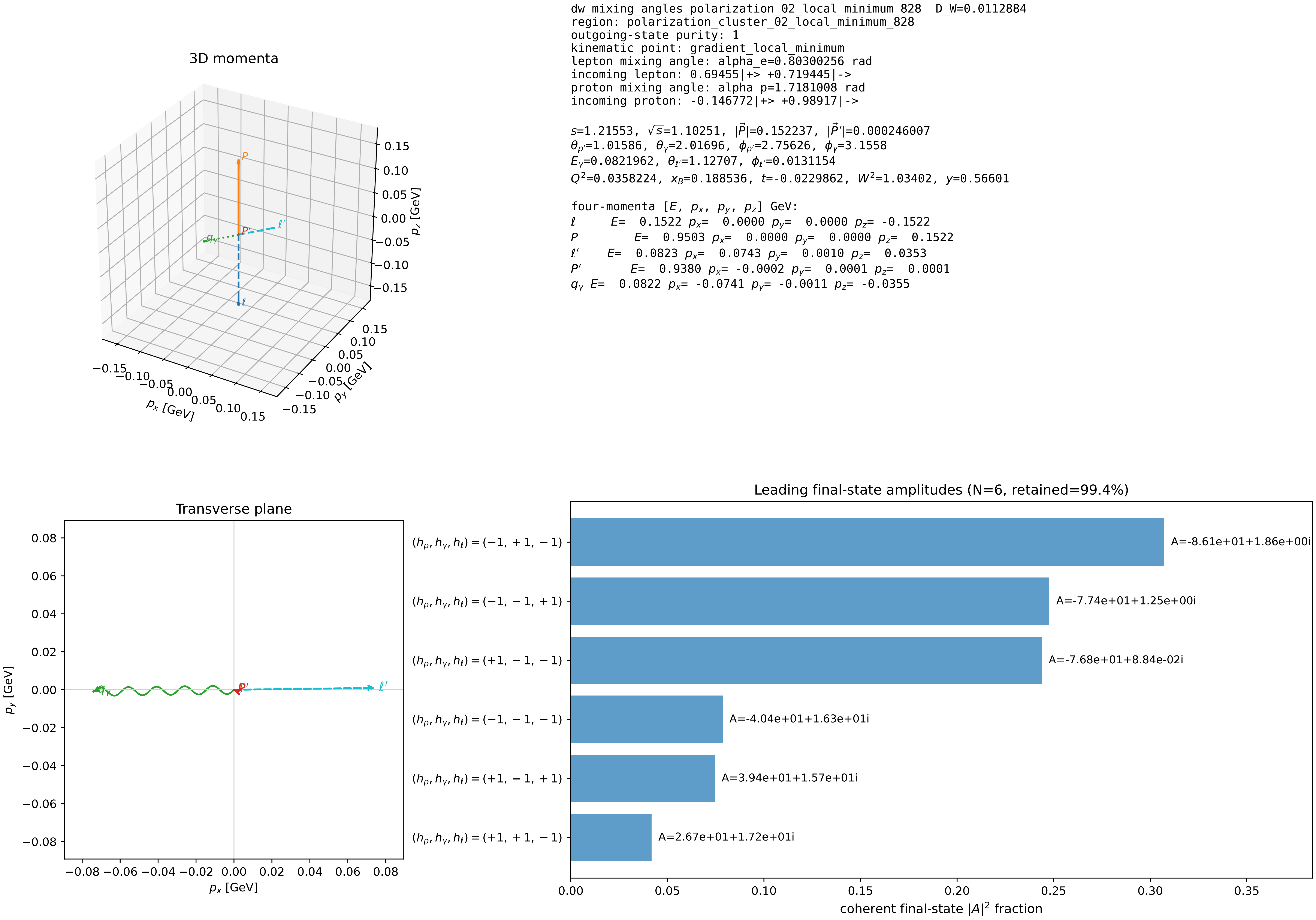


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

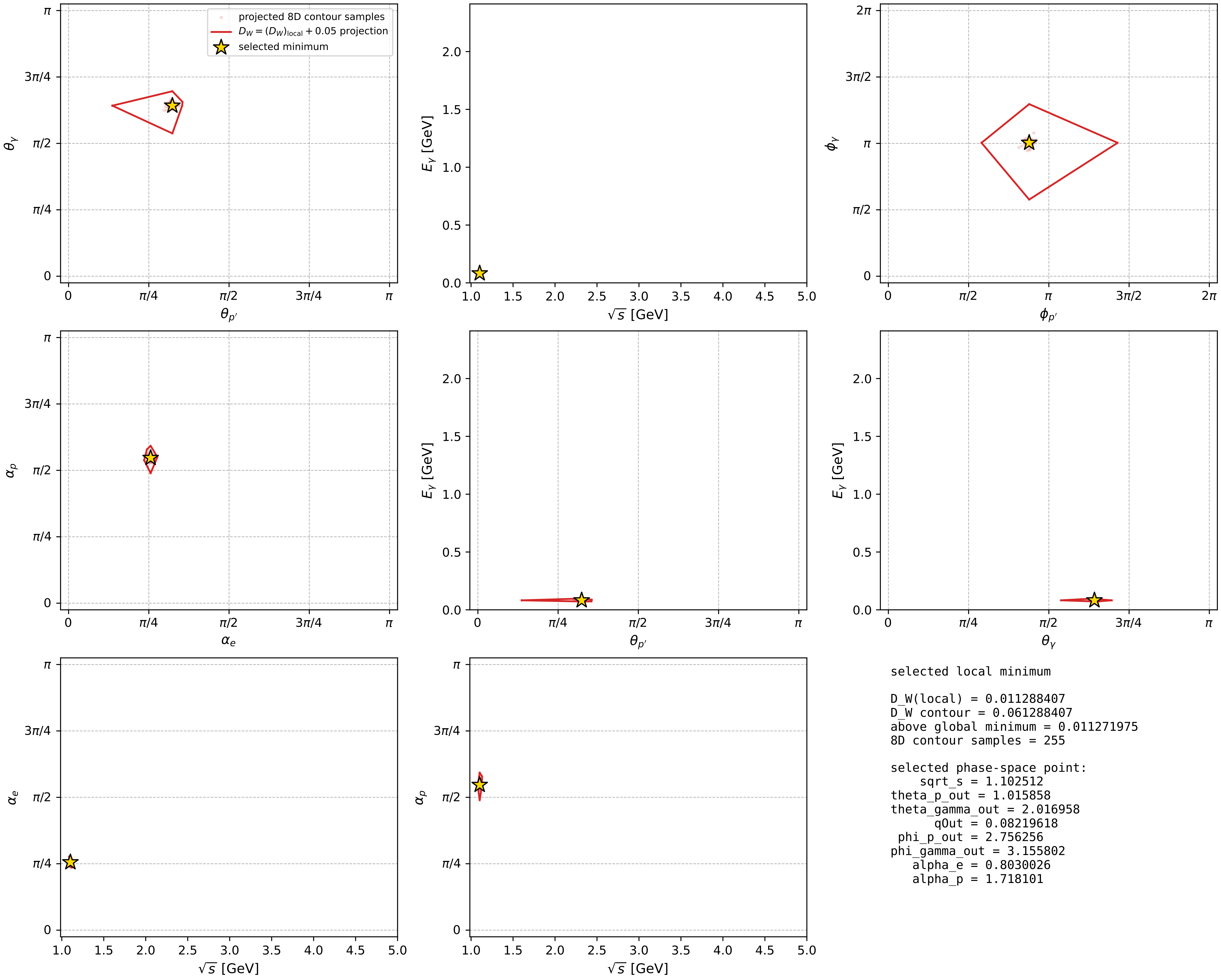



electron: polarization cluster P2, local minimum 809; pairwise projections of the 8D $D_W = (D_W)_{\rm local} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\rm local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
E_γ [GeV]
$\theta_{p'}$
E_γ [GeV]
θ_γ
α_e
$\sqrt{s}$ [GeV]
α_p
$\sqrt{s}$ [GeV]
selected local minimum
D_W(local) = 0.010723012
D_W contour = 0.060723012
above global minimum = 0.010706579
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.126578
theta_p_out = 2.057565
theta_gamma_out = 1.699746
qOut = 0.09390376
phi_p_out = 2.054178
phi_gamma_out = 3.231584
alpha_e = 0.7866376
alpha_p = 1.777581

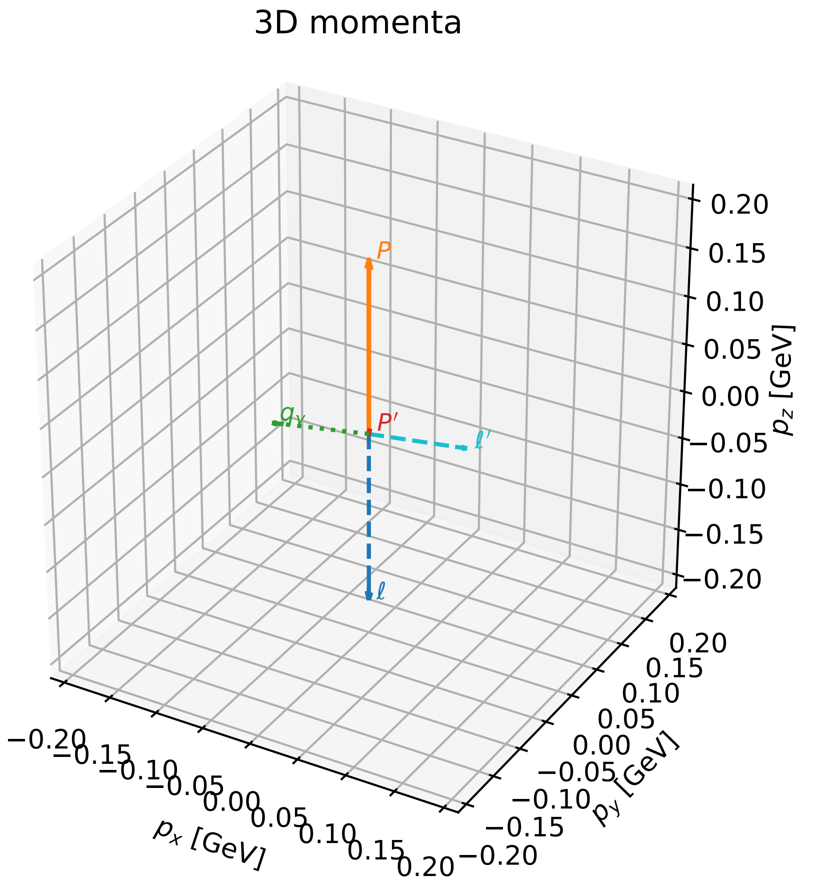
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



electron: polarization cluster P2, local minimum 828; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



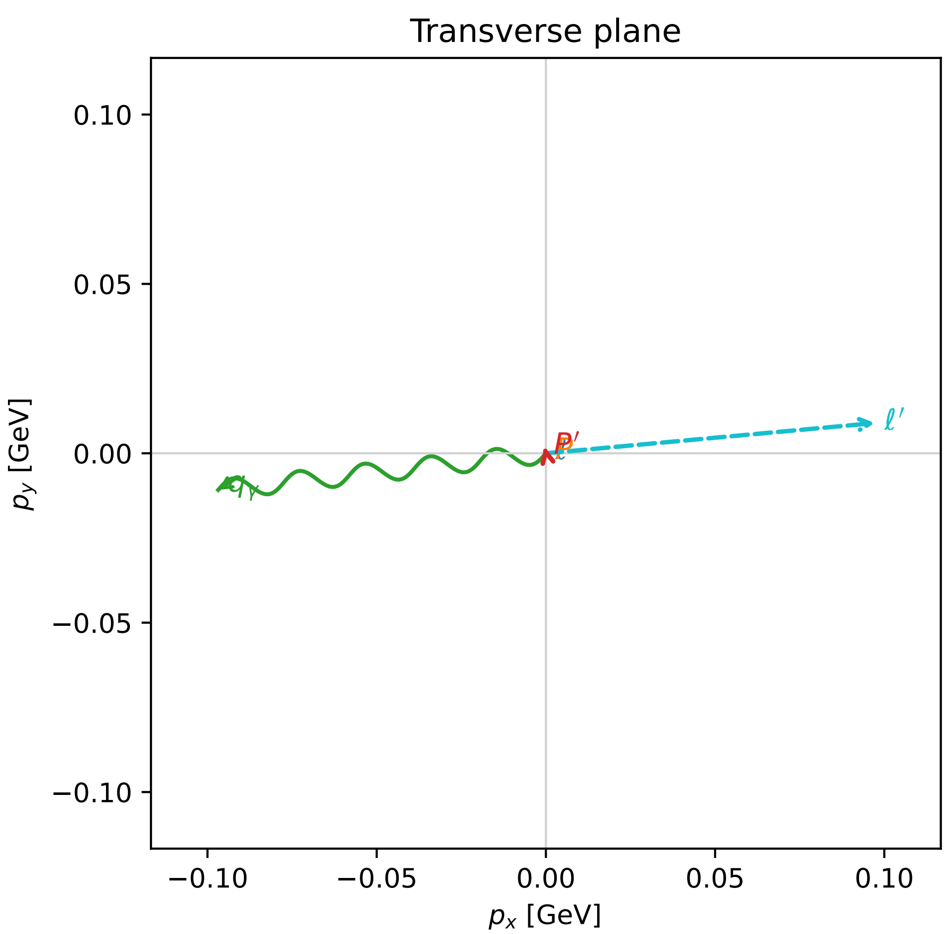
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

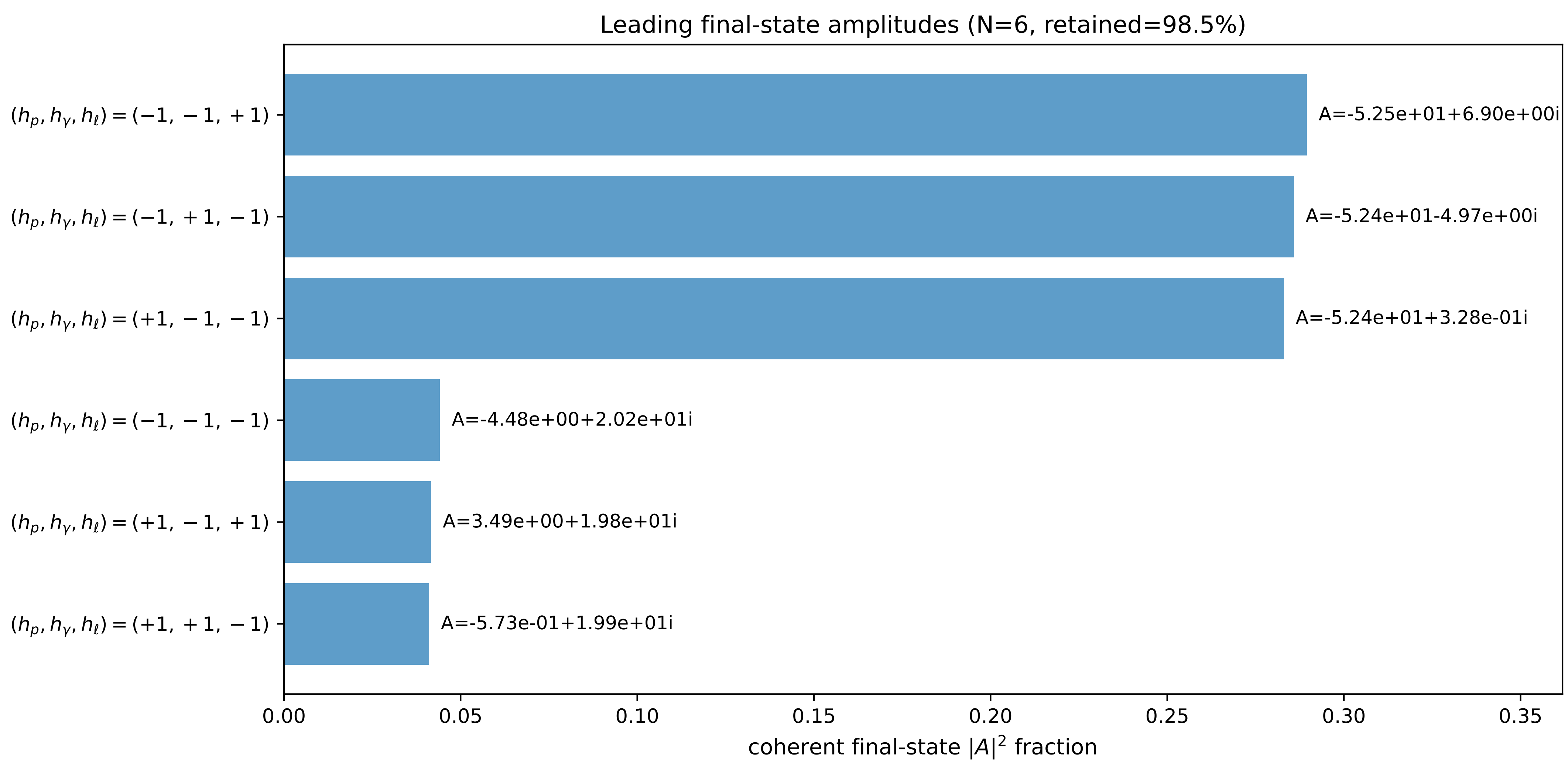



```
dw_mixing_angles_polarization_02_local_minimum_833  D_W=0.011459
region: polarization_cluster_02_local_minimum_833
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77778514 rad
incoming lepton: 0.712469|+> +0.701703|->
proton mixing angle: alpha_p=1.7767472 rad
incoming proton: -0.204498|+> +0.978867|->
```

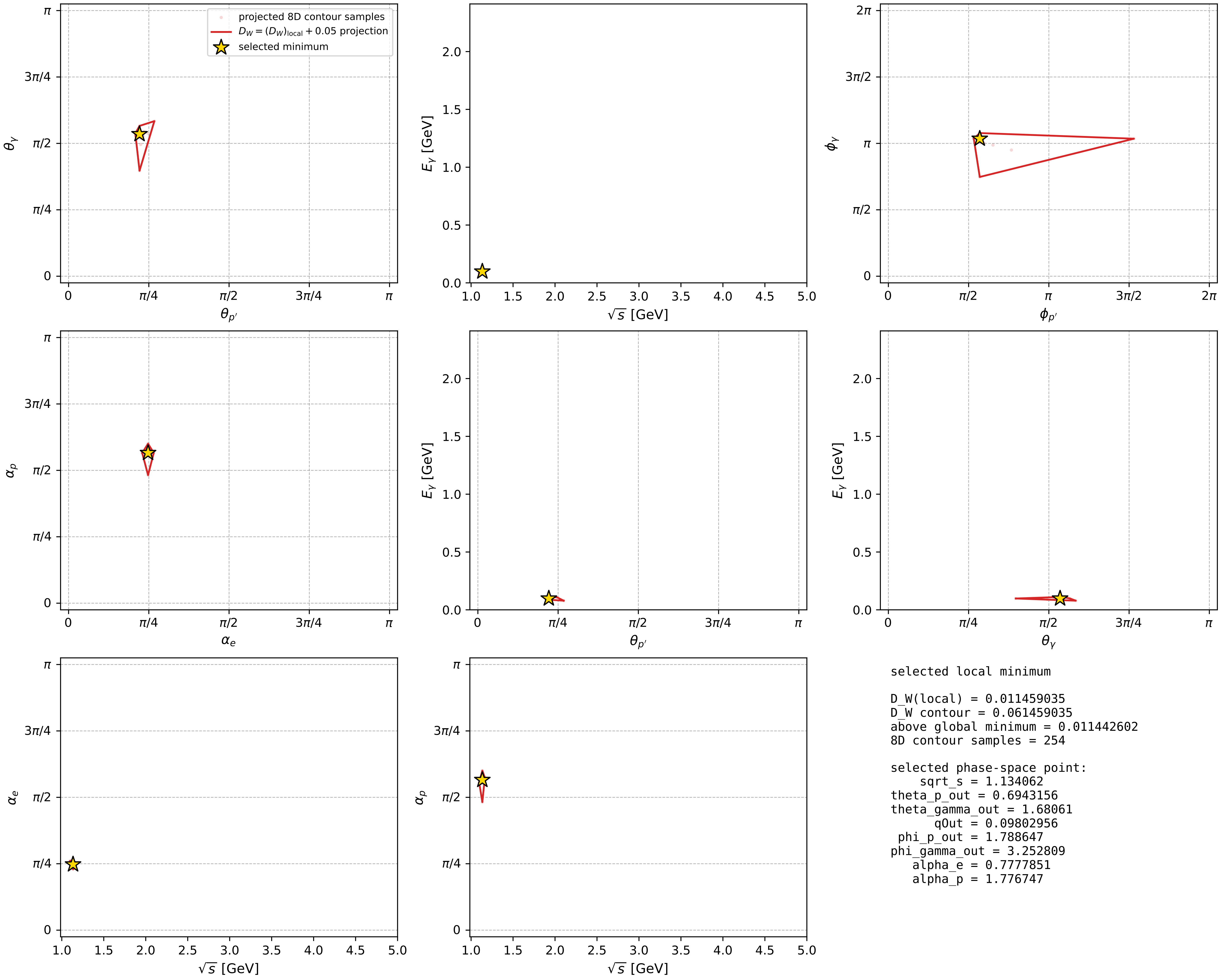
s=1.2861, $\sqrt{s}$=1.13406, $|\vec{P}|$=0.179113, $|\vec{P}'|$=0.0030043
$\theta_{p'}$=0.694316, θ_γ=1.68061, $\phi_{p'}$=1.78865, ϕ_γ=3.25281
E_γ=0.0980296, $\theta_{\ell'}$=1.48465, $\phi_{\ell'}$=0.0916434
Q^2=0.038137, x_B=0.171749, t=-0.0309764, W^2=1.06376, y=0.546585

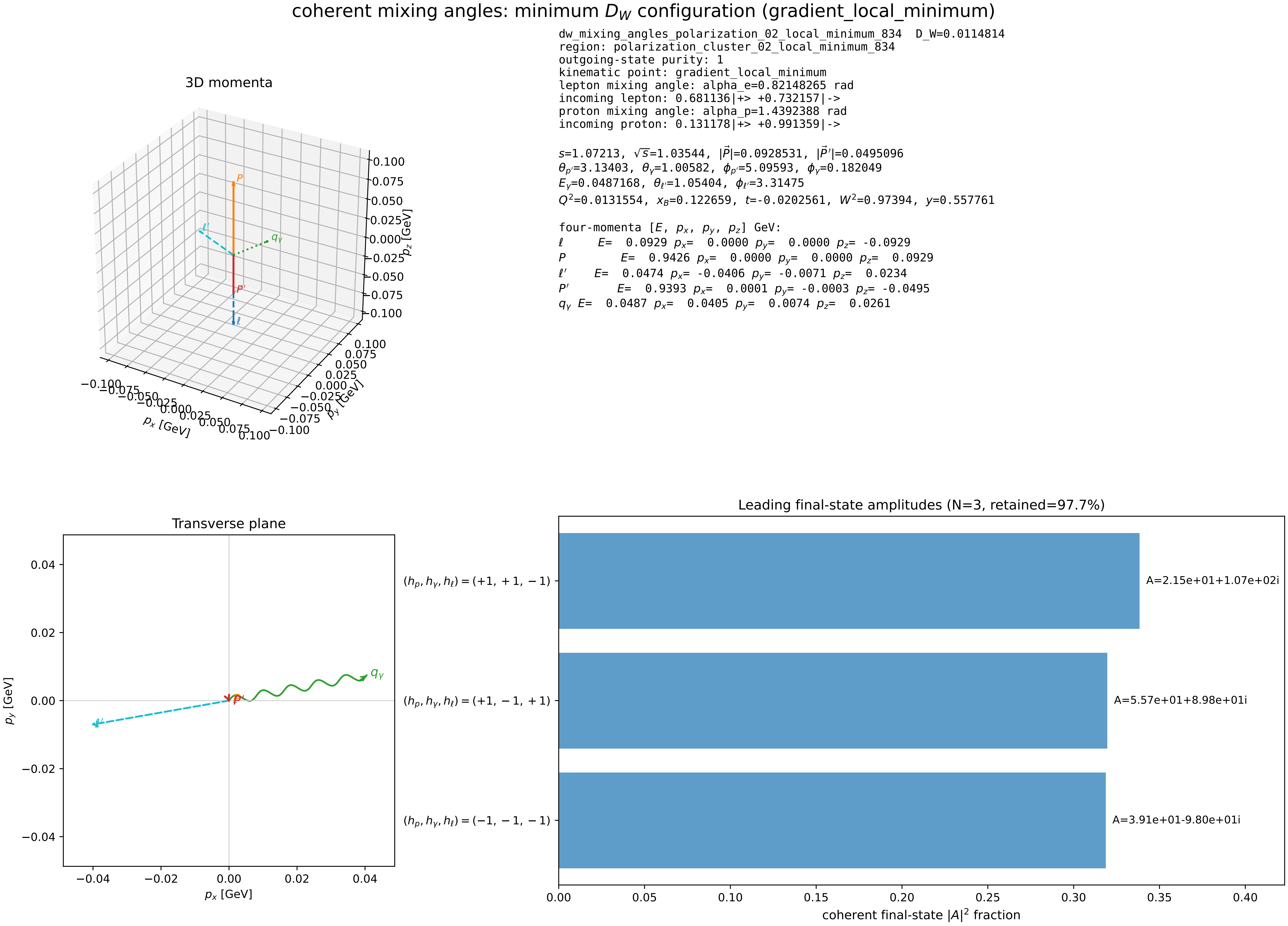
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1791 p_x= 0.0000 p_y= 0.0000 p_z= -0.1791
P E= 0.9549 p_x= 0.0000 p_y= 0.0000 p_z= 0.1791
ℓ' E= 0.0980 p_x= 0.0973 p_y= 0.0089 p_z= 0.0084
P' E= 0.9380 p_x= -0.0004 p_y= 0.0019 p_z= 0.0023
q_γ E= 0.0980 p_x= -0.0968 p_y= -0.0108 p_z= -0.0107

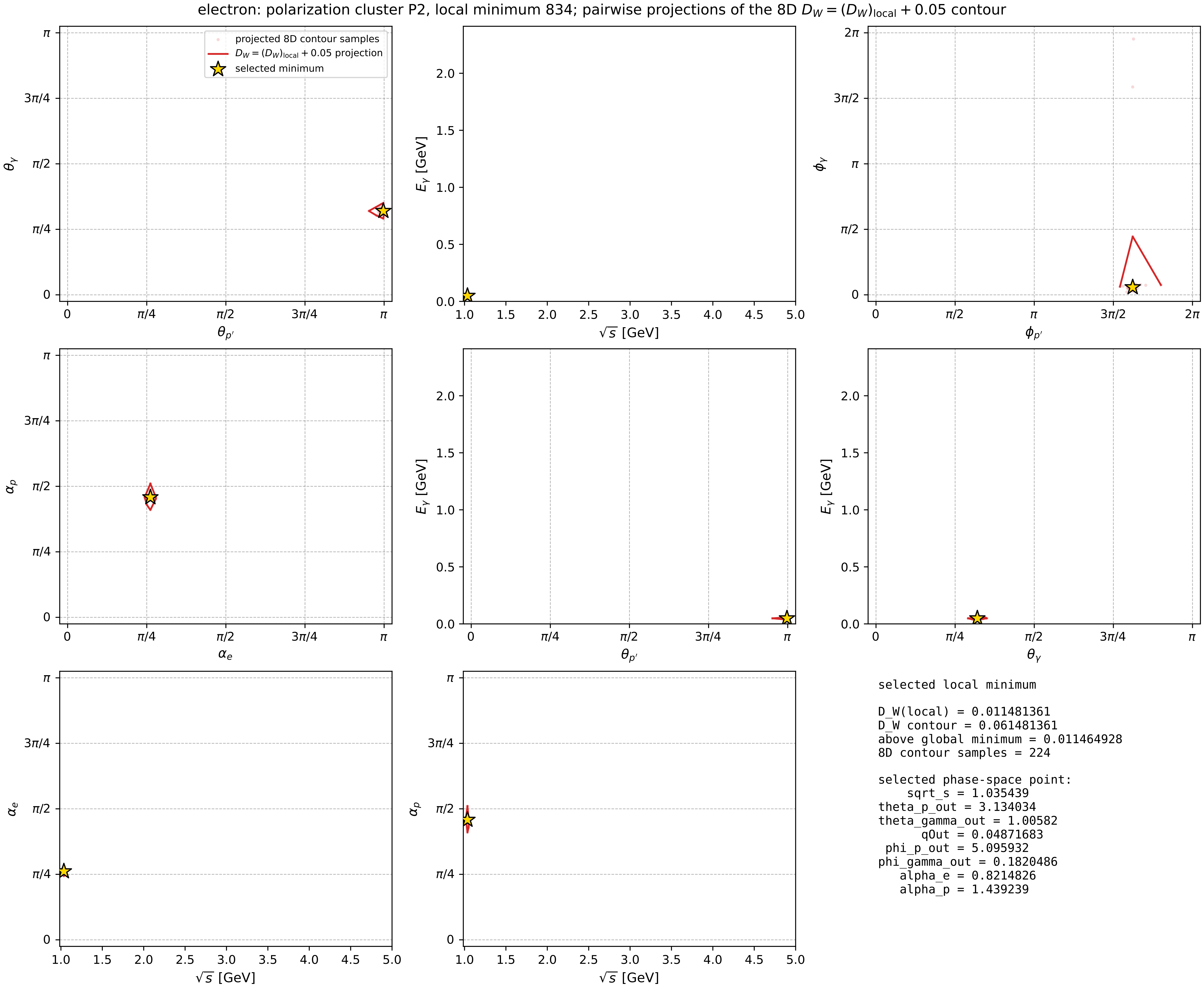


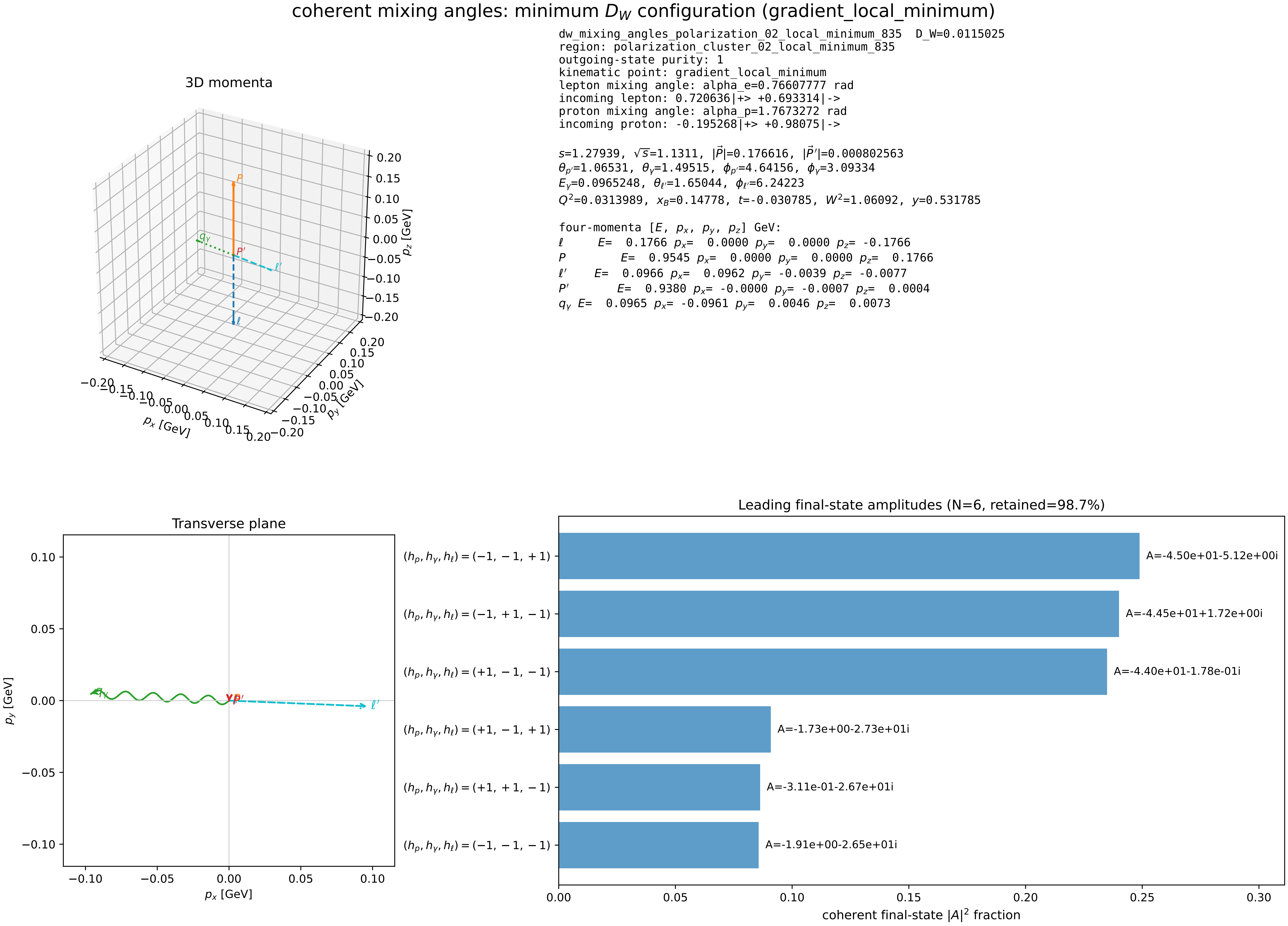


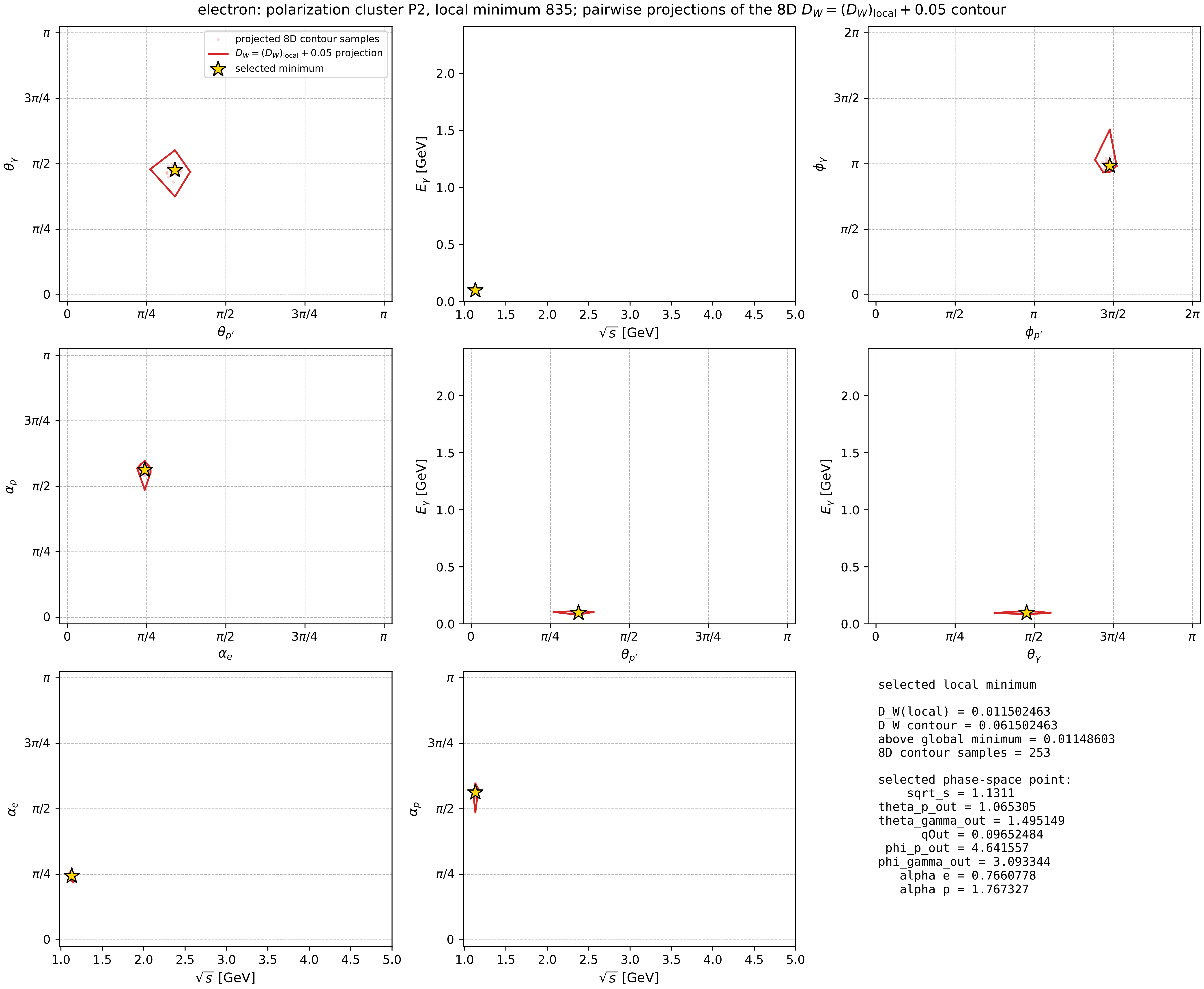
electron: polarization cluster P2, local minimum 833; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



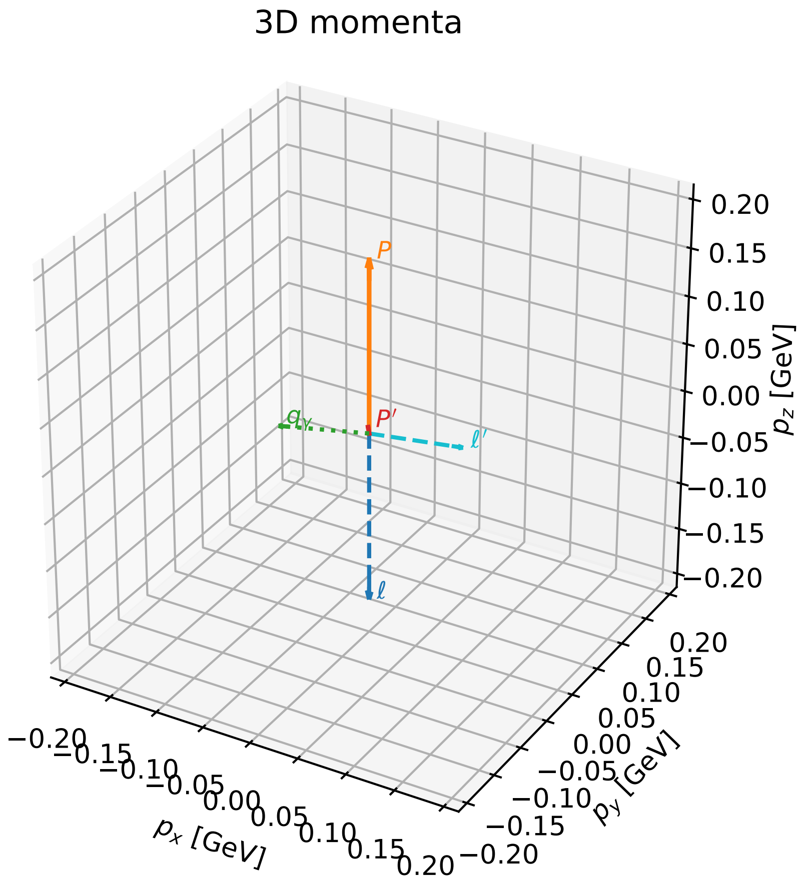

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_834 D_W=0.0114814
region: polarization_cluster_02_local_minimum_834
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.82148265 rad
incoming lepton: 0.681136|+> +0.732157|->
proton mixing angle: alpha_p=1.4392388 rad
incoming proton: 0.131178|+> +0.991359|->
s=1.07213, $\sqrt{s}$=1.03544, $|\vec{P}|$=0.0928531, $|\vec{P}'|$=0.0495096
$\theta_{p'}$=3.13403, θ_γ=1.00582, $\phi_{p'}$=5.09593, ϕ_γ=0.182049
E_γ=0.0487168, $\theta_{\ell'}$=1.05404, $\phi_{\ell'}$=3.31475
Q^2=0.0131554, x_B=0.122659, t=-0.0202561, W^2=0.97394, y=0.557761
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0929 p_x= 0.0000 p_y= 0.0000 p_z= -0.0929
P E= 0.9426 p_x= 0.0000 p_y= 0.0000 p_z= 0.0929
ℓ' E= 0.0474 p_x= -0.0406 p_y= -0.0071 p_z= 0.0234
P' E= 0.9393 p_x= 0.0001 p_y= -0.0003 p_z= -0.0495
q_γ E= 0.0487 p_x= 0.0405 p_y= 0.0074 p_z= 0.0261
3D momenta
P
ℓ'
q_γ
P'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=97.7%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=2.15e+01+1.07e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=5.57e+01+8.98e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
A=3.91e+01-9.80e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 834; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.011481361
D_W contour = 0.061481361
above global minimum = 0.011464928
8D contour samples = 224
selected phase-space point:
sqrt_s = 1.035439
theta_p_out = 3.134034
theta_gamma_out = 1.00582
qOut = 0.04871683
phi_p_out = 5.095932
phi_gamma_out = 0.1820486
alpha_e = 0.8214826
alpha_p = 1.439239


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_835 D_W=0.0115025
region: polarization_cluster_02_local_minimum_835
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.76607777 rad
incoming lepton: 0.720636|+> +0.693314|->
proton mixing angle: alpha_p=1.7673272 rad
incoming proton: -0.195268|+> +0.98075|->
s=1.27939, $\sqrt{s}$=1.1311, $|\vec{P}|$=0.176616, $|\vec{P}'|$=0.000802563
$\theta_{p'}$=1.06531, θ_γ=1.49515, $\phi_{p'}$=4.64156, ϕ_γ=3.09334
E_γ=0.0965248, $\theta_{\ell'}$=1.65044, $\phi_{\ell'}$=6.24223
Q^2=0.0313989, x_B=0.14778, t=-0.030785, W^2=1.06092, y=0.531785
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1766 p_x= 0.0000 p_y= 0.0000 p_z= -0.1766
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1766
ℓ' E= 0.0966 p_x= 0.0962 p_y= -0.0039 p_z= -0.0077
P' E= 0.9380 p_x= -0.0000 p_y= -0.0007 p_z= 0.0004
q_γ E= 0.0965 p_x= -0.0961 p_y= 0.0046 p_z= 0.0073
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=98.7%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-4.50e+01-5.12e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-4.45e+01+1.72e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-4.40e+01-1.78e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-1.73e+00-2.73e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-3.11e-01-2.67e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-1.91e+00-2.65e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 835; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.011502463
D_W contour = 0.061502463
above global minimum = 0.01148603
8D contour samples = 253
selected phase-space point:
sqrt_s = 1.1311
theta_p_out = 1.065305
theta_gamma_out = 1.495149
qOut = 0.09652484
phi_p_out = 4.641557
phi_gamma_out = 3.093344
alpha_e = 0.7660778
alpha_p = 1.767327

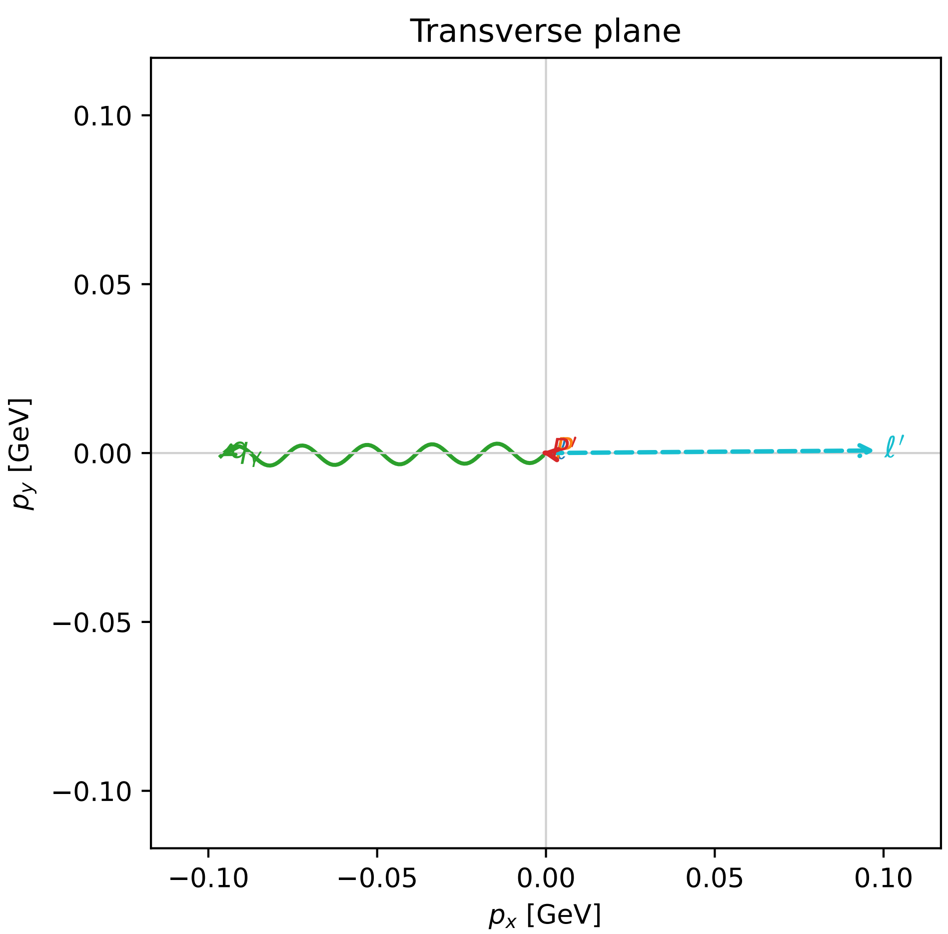
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

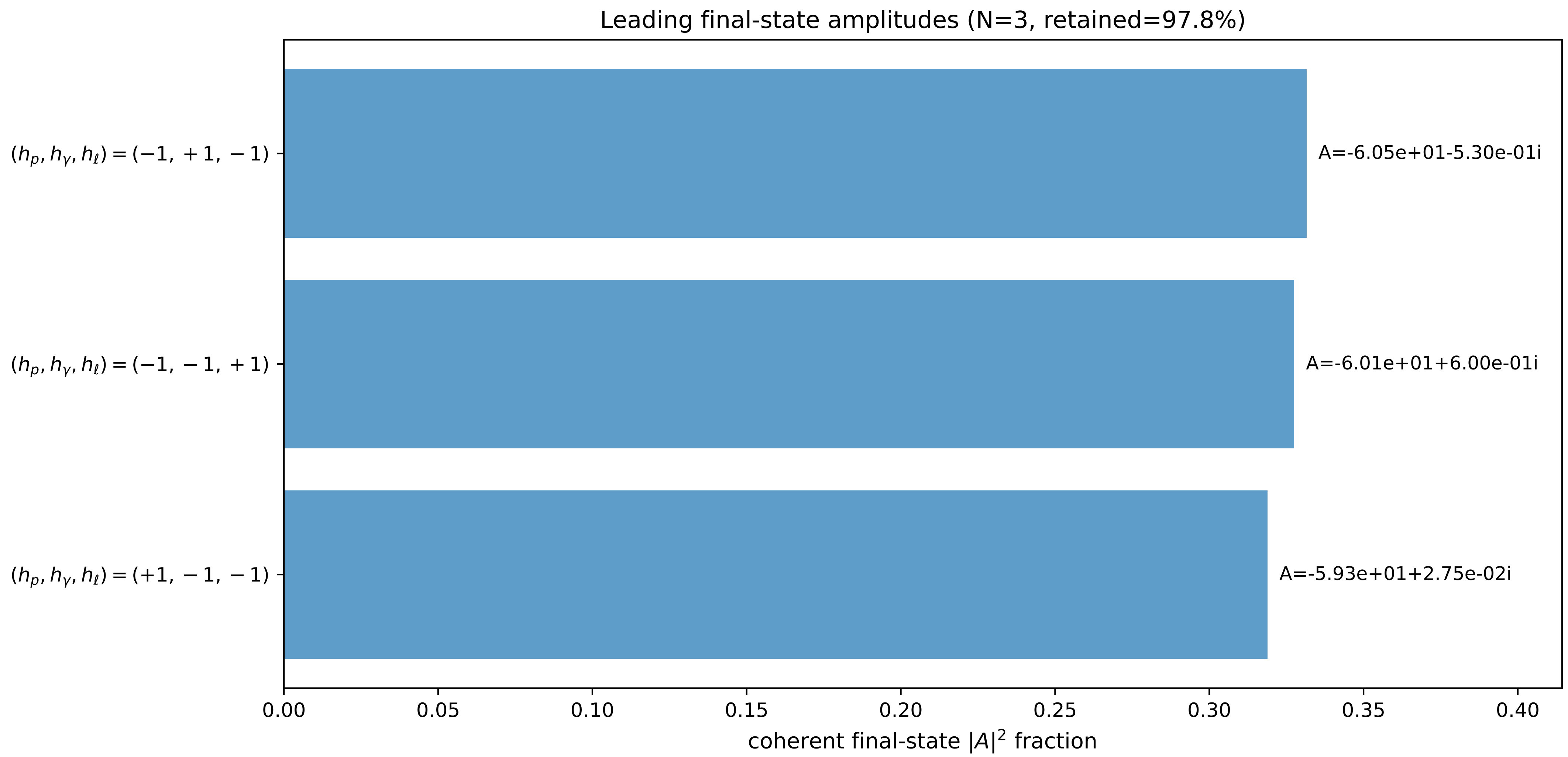



```
dw_mixing_angles_polarization_02_local_minimum_837  D_W=0.0115901
region: polarization_cluster_02_local_minimum_837
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78523567 rad
incoming lepton: 0.707222|+> +0.706992|->
proton mixing angle: alpha_p=1.7803765 rad
incoming proton: -0.208049|+> +0.978118|->
```

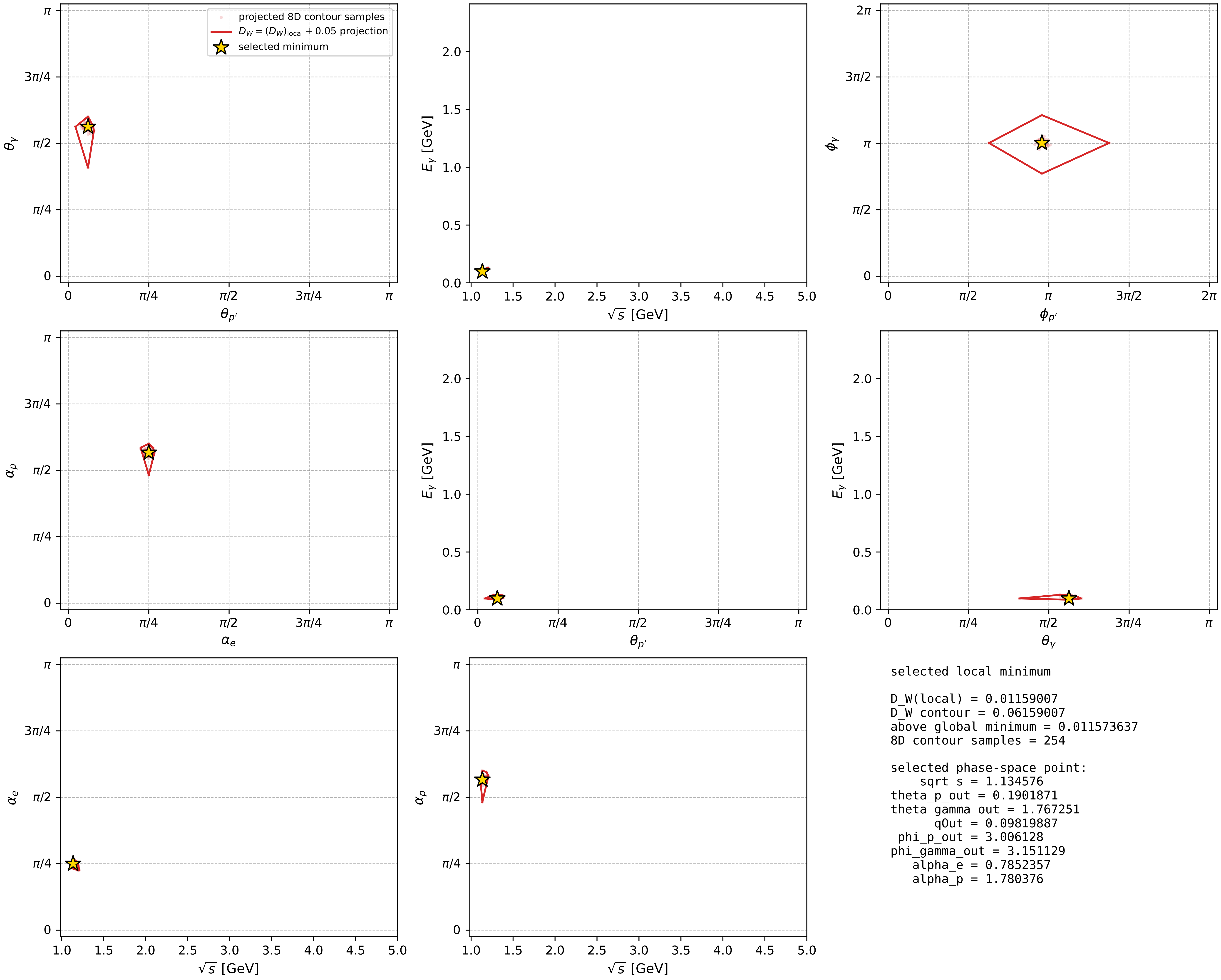
s=1.28726, $\sqrt{s}$=1.13458, $|\vec{P}|$=0.179546, $|\vec{P}'|$=0.00639123
$\theta_{p'}$=0.190187, θ_γ=1.76725, $\phi_{p'}$=3.00613, ϕ_γ=3.15113
E_γ=0.0981989, $\theta_{\ell'}$=1.43934, $\phi_{\ell'}$=0.00774615
Q^2=0.0399477, x_B=0.178192, t=-0.0297348, W^2=1.06408, y=0.550253

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1795 p_x= 0.0000 p_y= 0.0000 p_z= -0.1795
P E= 0.9550 p_x= 0.0000 p_y= 0.0000 p_z= 0.1795
ℓ' E= 0.0984 p_x= 0.0975 p_y= 0.0008 p_z= 0.0129
P' E= 0.9380 p_x= -0.0012 p_y= 0.0002 p_z= 0.0063
q_γ E= 0.0982 p_x= -0.0963 p_y= -0.0009 p_z= -0.0192

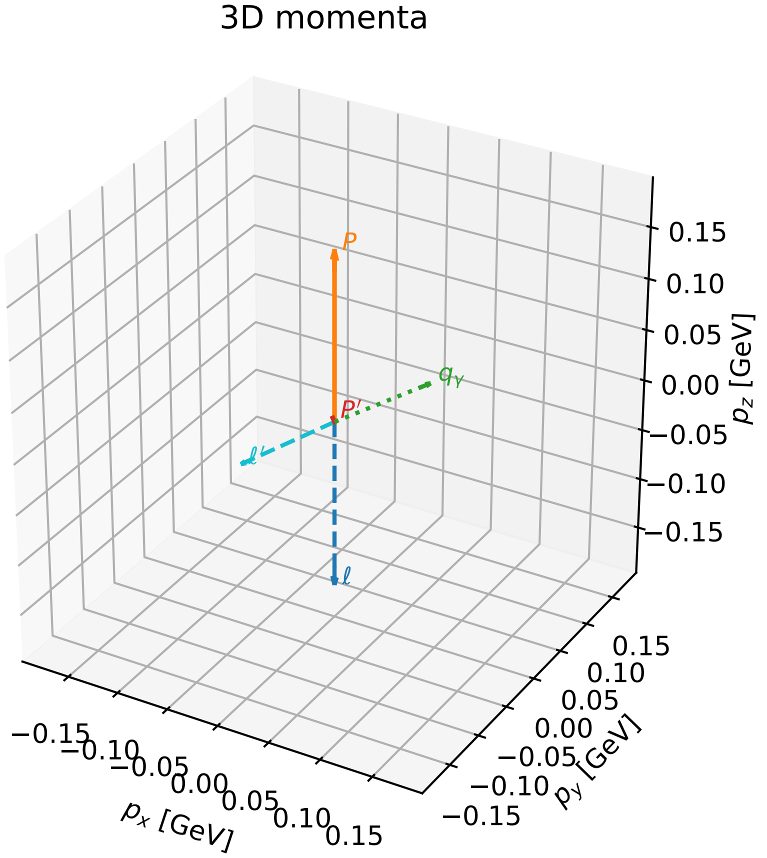




electron: polarization cluster P2, local minimum 837; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



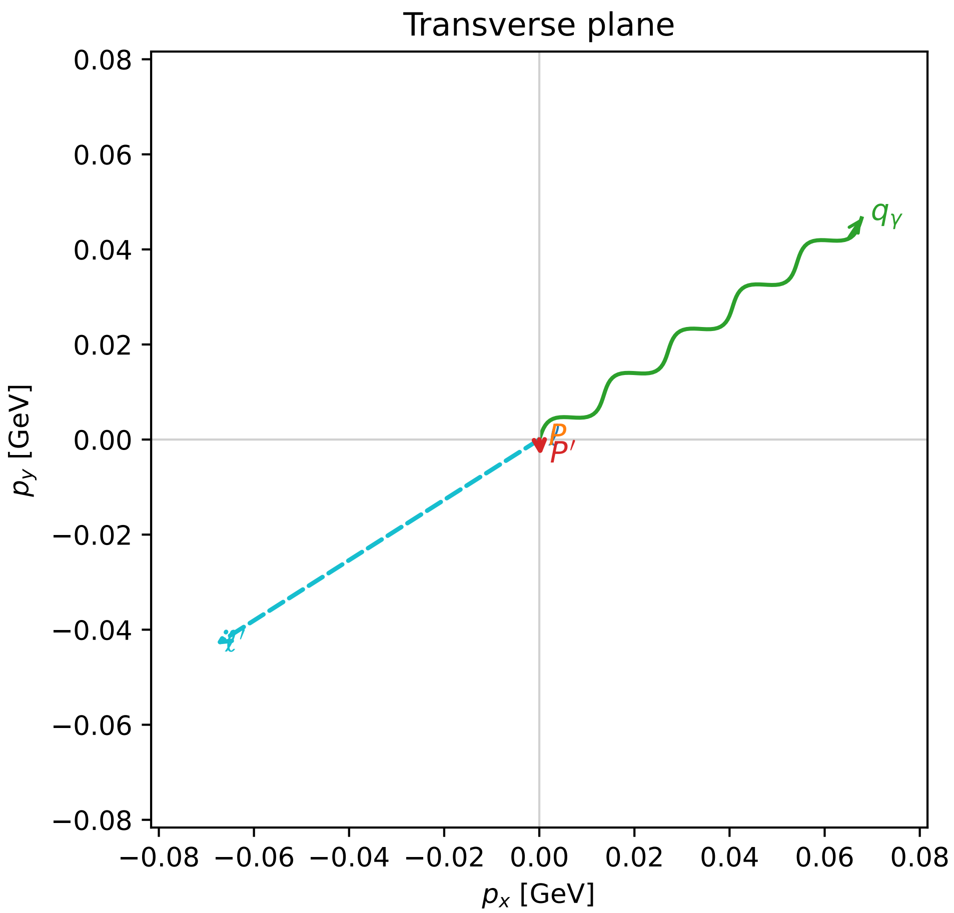
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

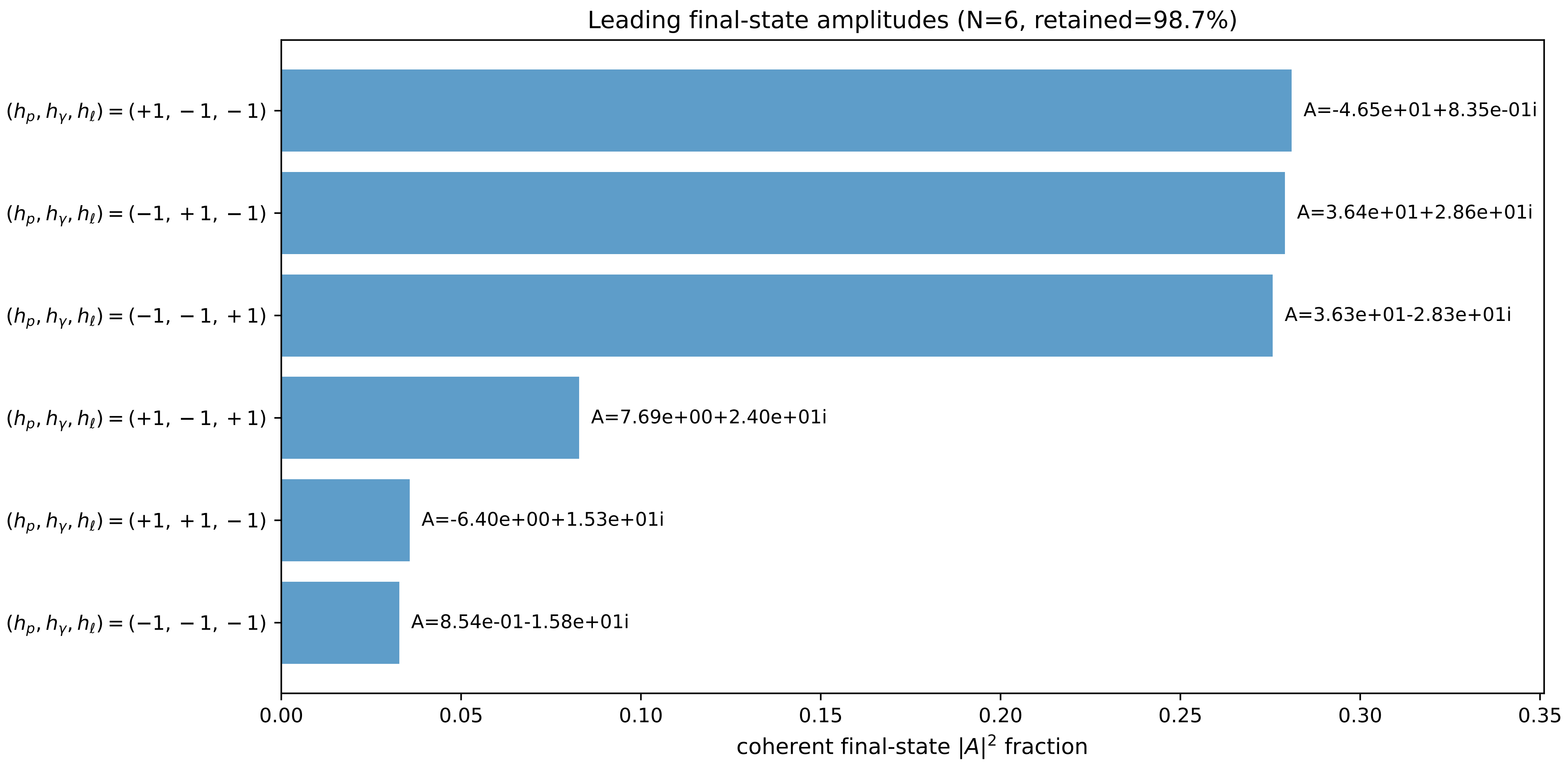


dw_mixing_angles_polarization_02_local_minimum_838 D_W=0.0115987
region: polarization_cluster_02_local_minimum_838
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.74876805 rad
incoming lepton: 0.732528|+> +0.680737|->
proton mixing angle: alpha_p=1.3917413 rad
incoming proton: 0.1781|+> +0.984012|->

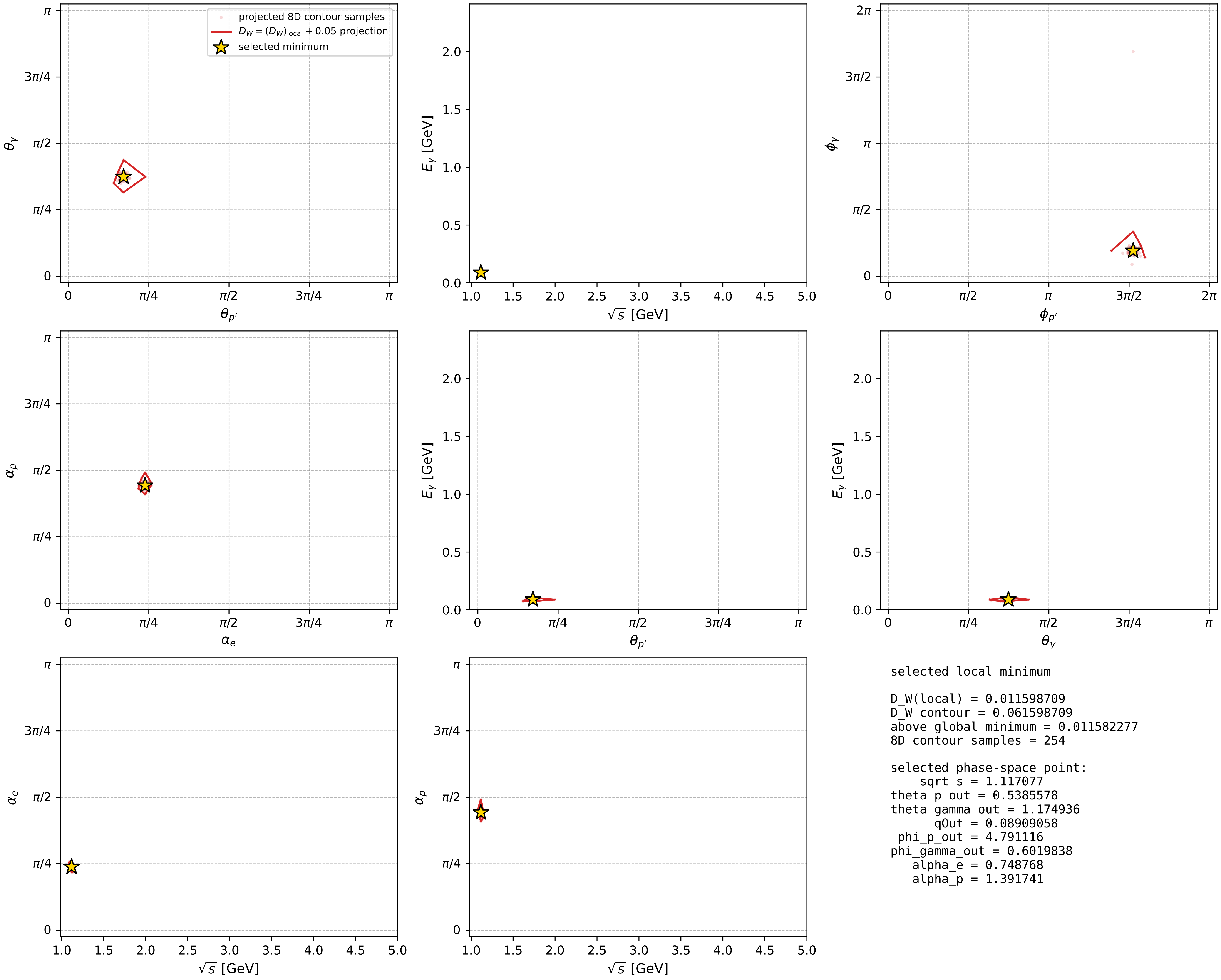
s=1.24786, $\sqrt{s}$=1.11708, $|\vec{P}|$=0.164722, $|\vec{P}'|$=0.00663877
$\theta_{p'}$=0.538558, θ_γ=1.17494, $\phi_{p'}$=4.79112, ϕ_γ=0.601984
E_γ=0.0890906, $\theta_{\ell'}$=2.03222, $\phi_{\ell'}$=3.70695
Q^2=0.0164423, x_B=0.0896191, t=-0.0250947, W^2=1.04687, y=0.498534

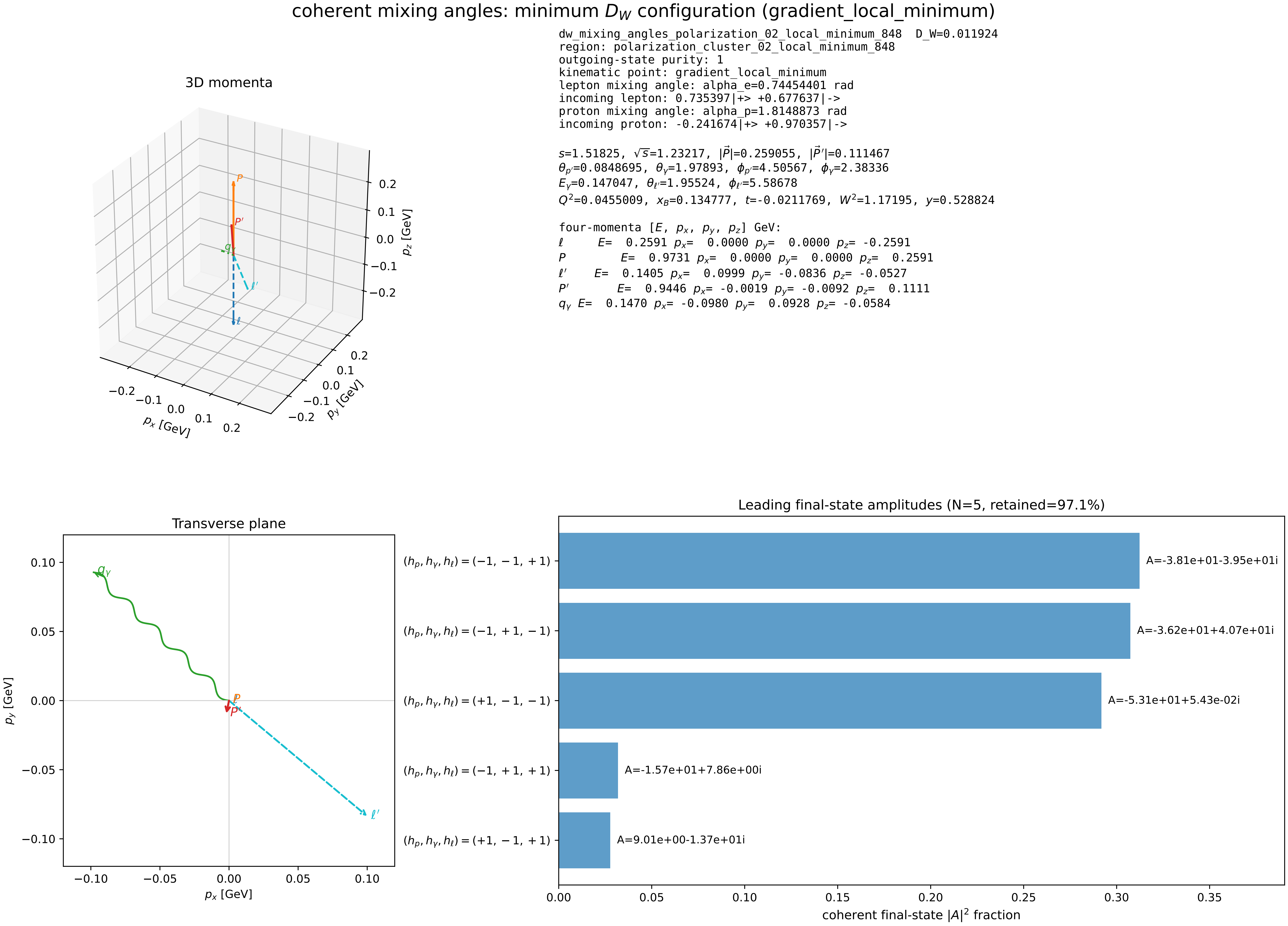
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1647 p_x= 0.0000 p_y= 0.0000 p_z= -0.1647
P E= 0.9524 p_x= 0.0000 p_y= 0.0000 p_z= 0.1647
ℓ' E= 0.0900 p_x= -0.0680 p_y= -0.0432 p_z= -0.0401
P' E= 0.9380 p_x= 0.0003 p_y= -0.0034 p_z= 0.0057
q_γ E= 0.0891 p_x= 0.0678 p_y= 0.0465 p_z= 0.0344

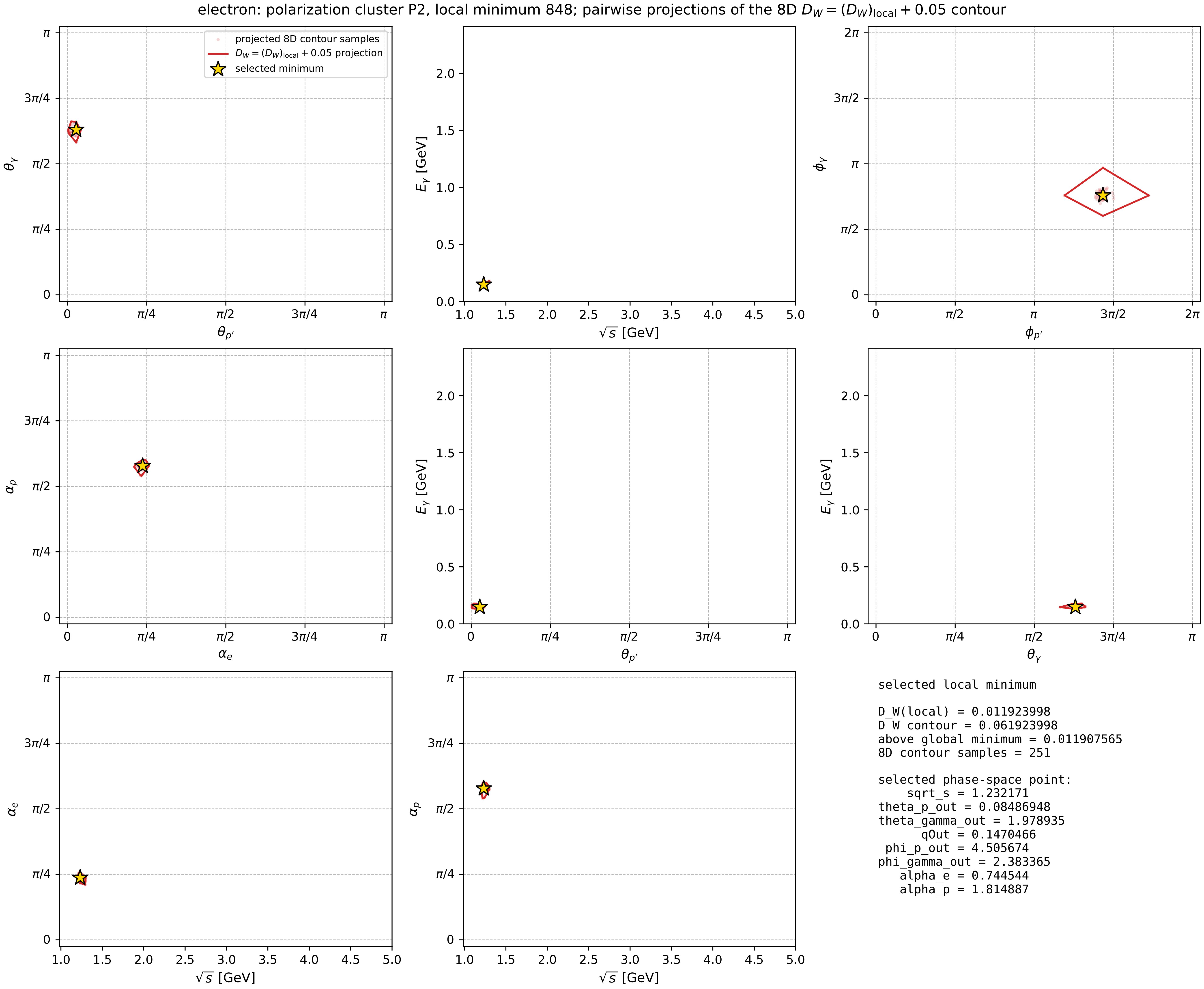




electron: polarization cluster P2, local minimum 838; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

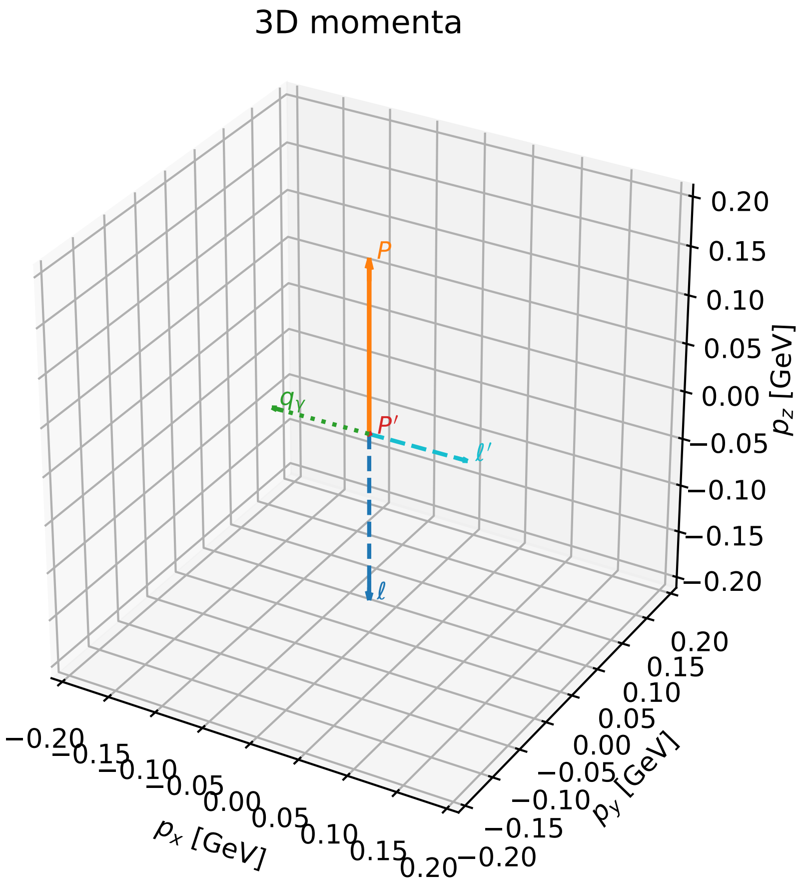



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_02_local_minimum_848 D_W=0.011924
region: polarization_cluster_02_local_minimum_848
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.74454401 rad
incoming lepton: 0.735397|+> +0.677637|->
proton mixing angle: alpha_p=1.8148873 rad
incoming proton: -0.241674|+> +0.970357|->
s=1.51825, $\sqrt{s}$=1.23217, $|\vec{P}|$=0.259055, $|\vec{P}'|$=0.111467
$\theta_{p'}$=0.0848695, θ_γ=1.97893, $\phi_{p'}$=4.50567, ϕ_γ=2.38336
E_γ=0.147047, $\theta_{\ell'}$=1.95524, $\phi_{\ell'}$=5.58678
Q^2=0.0455009, x_B=0.134777, t=-0.0211769, W^2=1.17195, y=0.528824
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2591 p_x= 0.0000 p_y= 0.0000 p_z= -0.2591
P E= 0.9731 p_x= 0.0000 p_y= 0.0000 p_z= 0.2591
ℓ' E= 0.1405 p_x= 0.0999 p_y= -0.0836 p_z= -0.0527
P' E= 0.9446 p_x= -0.0019 p_y= -0.0092 p_z= 0.1111
q_γ E= 0.1470 p_x= -0.0980 p_y= 0.0928 p_z= -0.0584
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=5, retained=97.1%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-3.81e+01-3.95e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-3.62e+01+4.07e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-5.31e+01+5.43e-02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-1.57e+01+7.86e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=9.01e+00-1.37e+01i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P2, local minimum 848; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.011923998
D_W contour = 0.061923998
above global minimum = 0.011907565
8D contour samples = 251
selected phase-space point:
sqrt_s = 1.232171
theta_p_out = 0.08486948
theta_gamma_out = 1.978935
qOut = 0.1470466
phi_p_out = 4.505674
phi_gamma_out = 2.383365
alpha_e = 0.744544
alpha_p = 1.814887

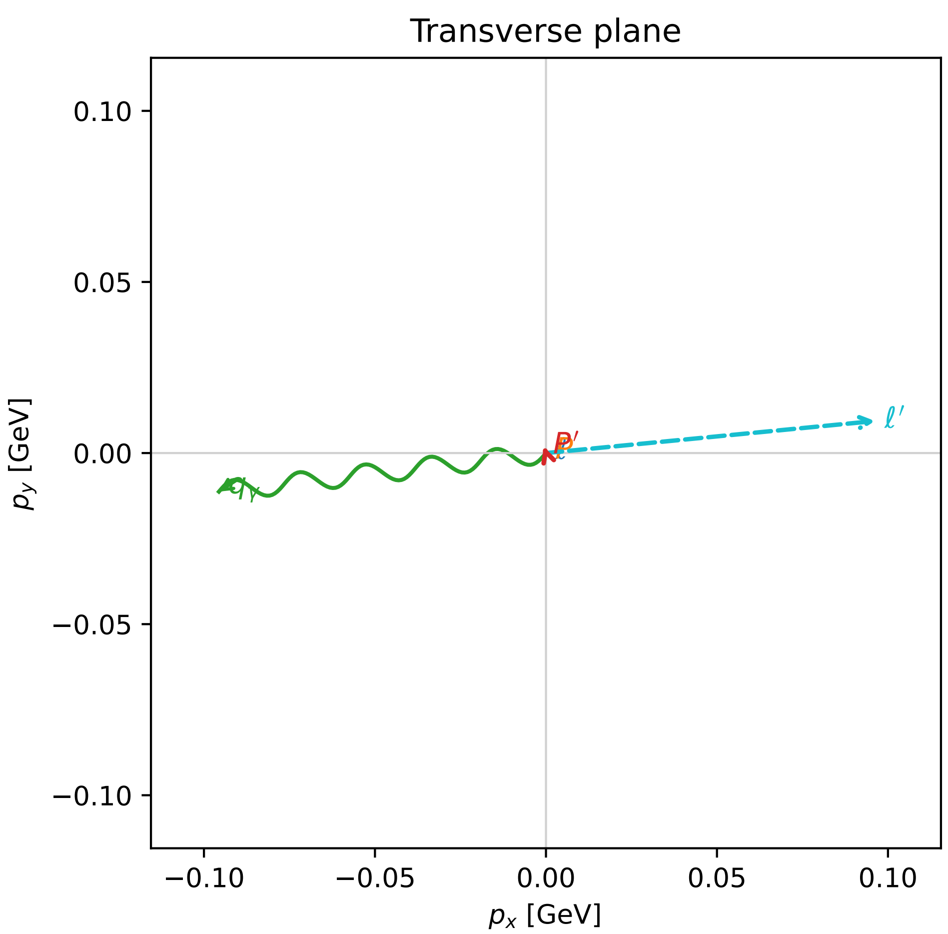
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

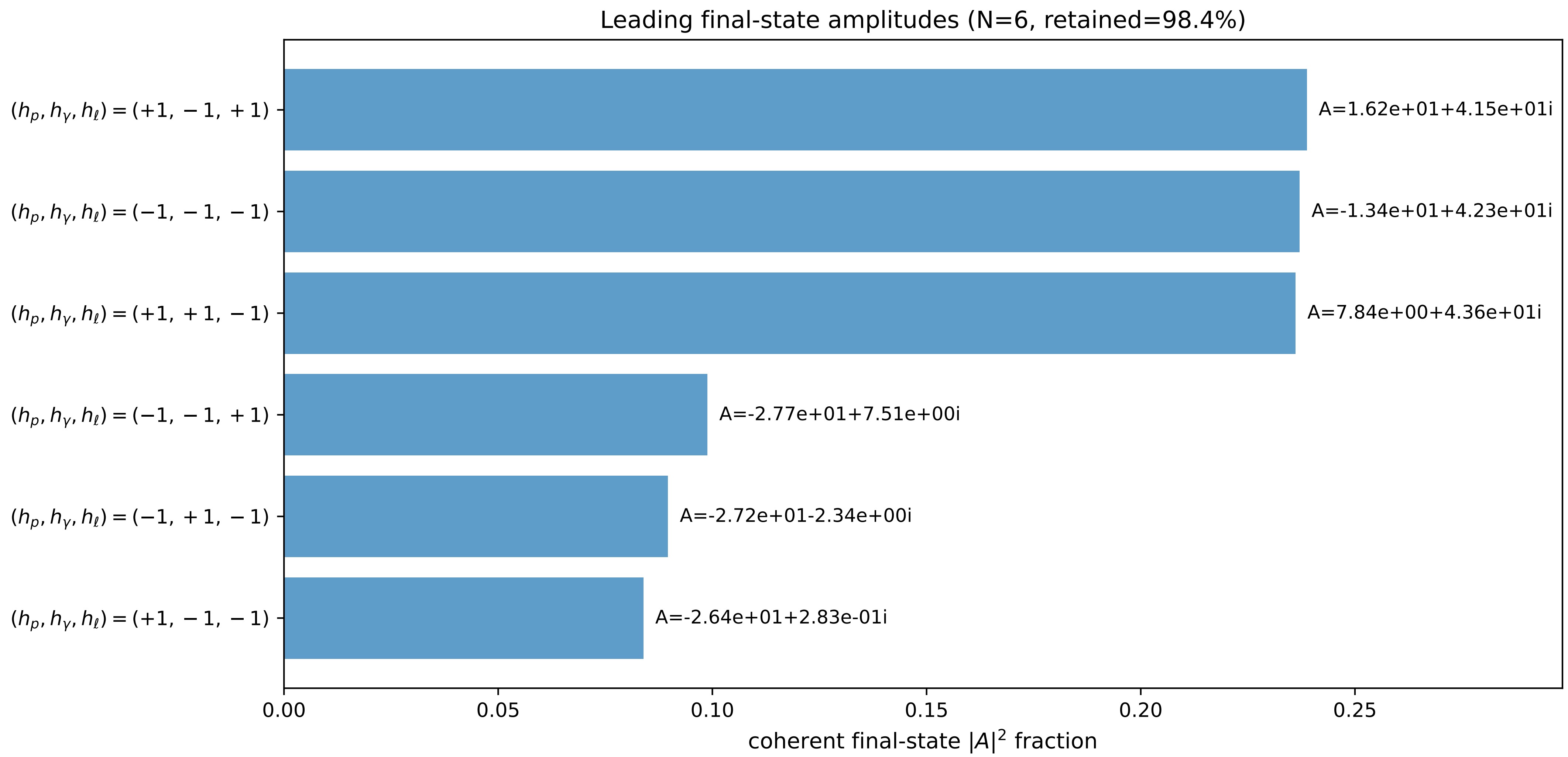


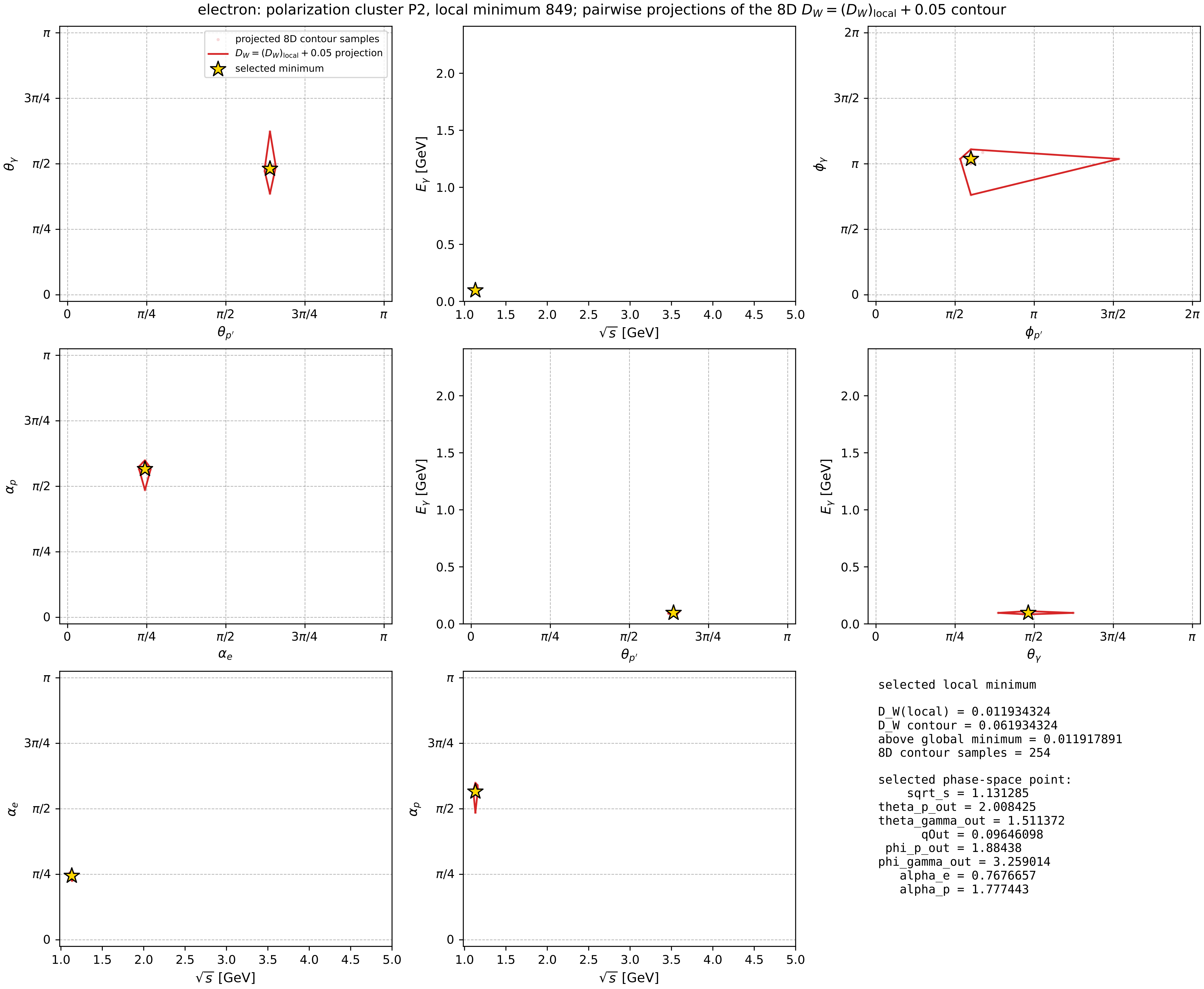
dw_mixing_angles_polarization_02_local_minimum_849 D_W=0.0119343
region: polarization_cluster_02_local_minimum_849
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7676657 rad
incoming lepton: 0.719534|+> +0.694458|->
proton mixing angle: alpha_p=1.7774429 rad
incoming proton: -0.205179|+> +0.978724|->

s=1.27981, $\sqrt{s}$=1.13129, $|\vec{P}|$=0.176773, $|\vec{P}'|$=0.00221567
$\theta_{p'}$=2.00842, θ_γ=1.51137, $\phi_{p'}$=1.88438, ϕ_γ=3.25901
E_γ=0.096461, $\theta_{\ell'}$=1.62029, $\phi_{\ell'}$=0.0970648
Q^2=0.0325372, x_B=0.152446, t=-0.031313, W^2=1.06074, y=0.533635

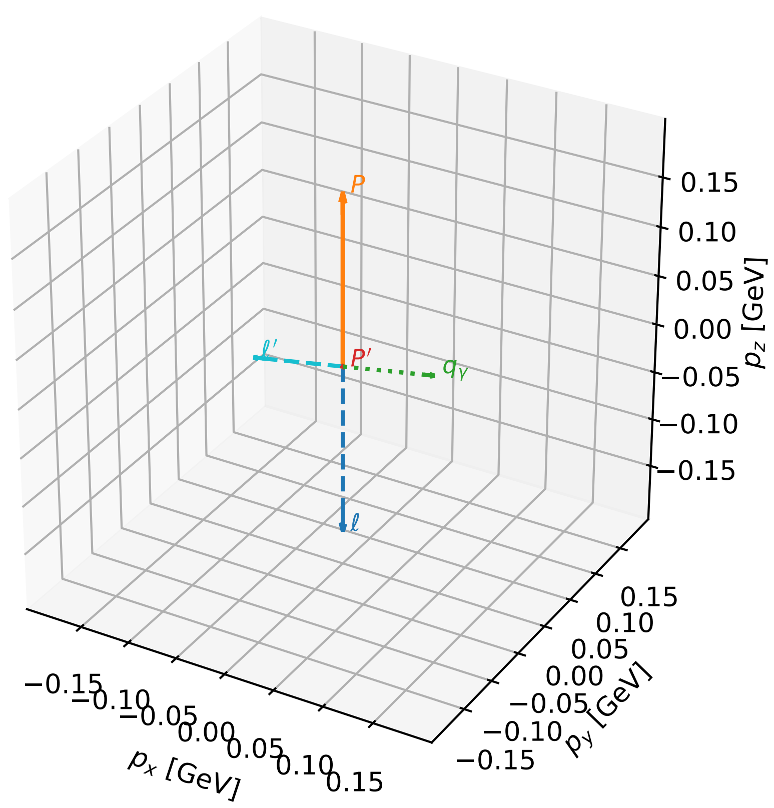
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1768 p_x= 0.0000 p_y= 0.0000 p_z= -0.1768
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1768
ℓ' E= 0.0968 p_x= 0.0962 p_y= 0.0094 p_z= -0.0048
P' E= 0.9380 p_x= -0.0006 p_y= 0.0019 p_z= -0.0009
q_γ E= 0.0965 p_x= -0.0956 p_y= -0.0113 p_z= 0.0057






electron: polarization cluster P2, local minimum 849; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
$\theta_{p'}$
E_γ [GeV]
θ_γ
E_γ [GeV]
$\sqrt{s}$ [GeV]
α_e
$\sqrt{s}$ [GeV]
α_p
selected local minimum
D_W(local) = 0.011934324
D_W contour = 0.061934324
above global minimum = 0.011917891
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.131285
theta_p_out = 2.008425
theta_gamma_out = 1.511372
qOut = 0.09646098
phi_p_out = 1.88438
phi_gamma_out = 3.259014
alpha_e = 0.7676657
alpha_p = 1.777443

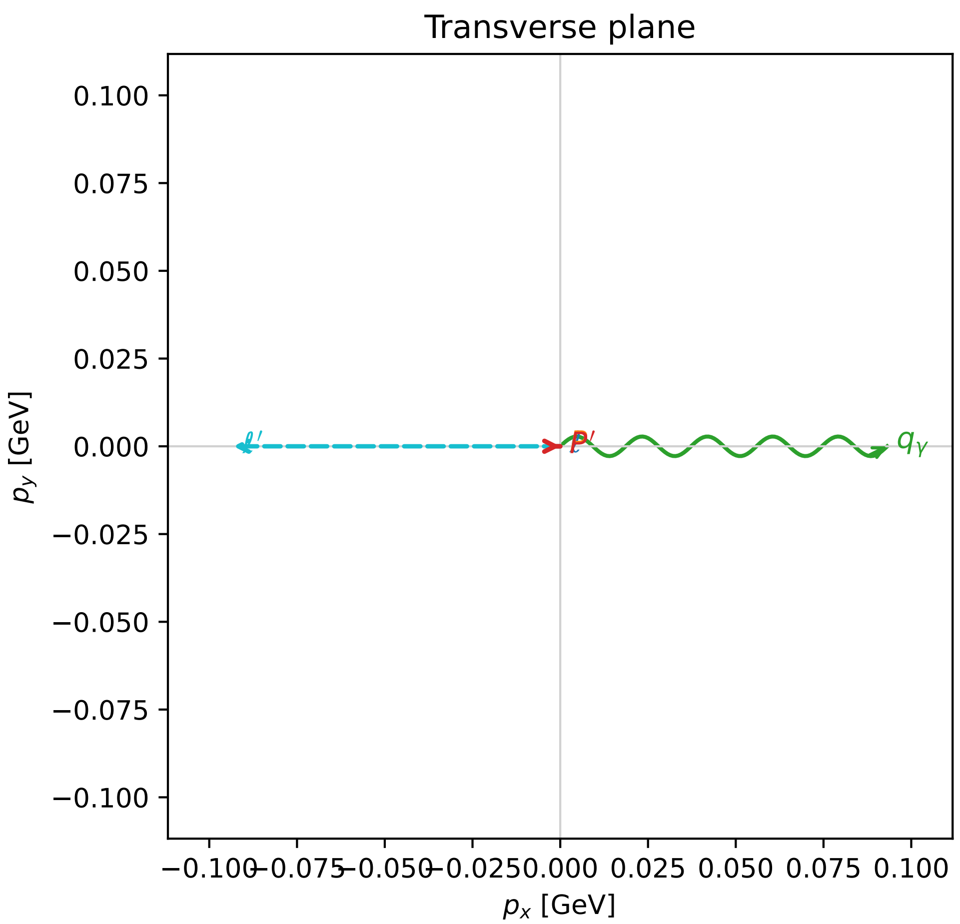
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

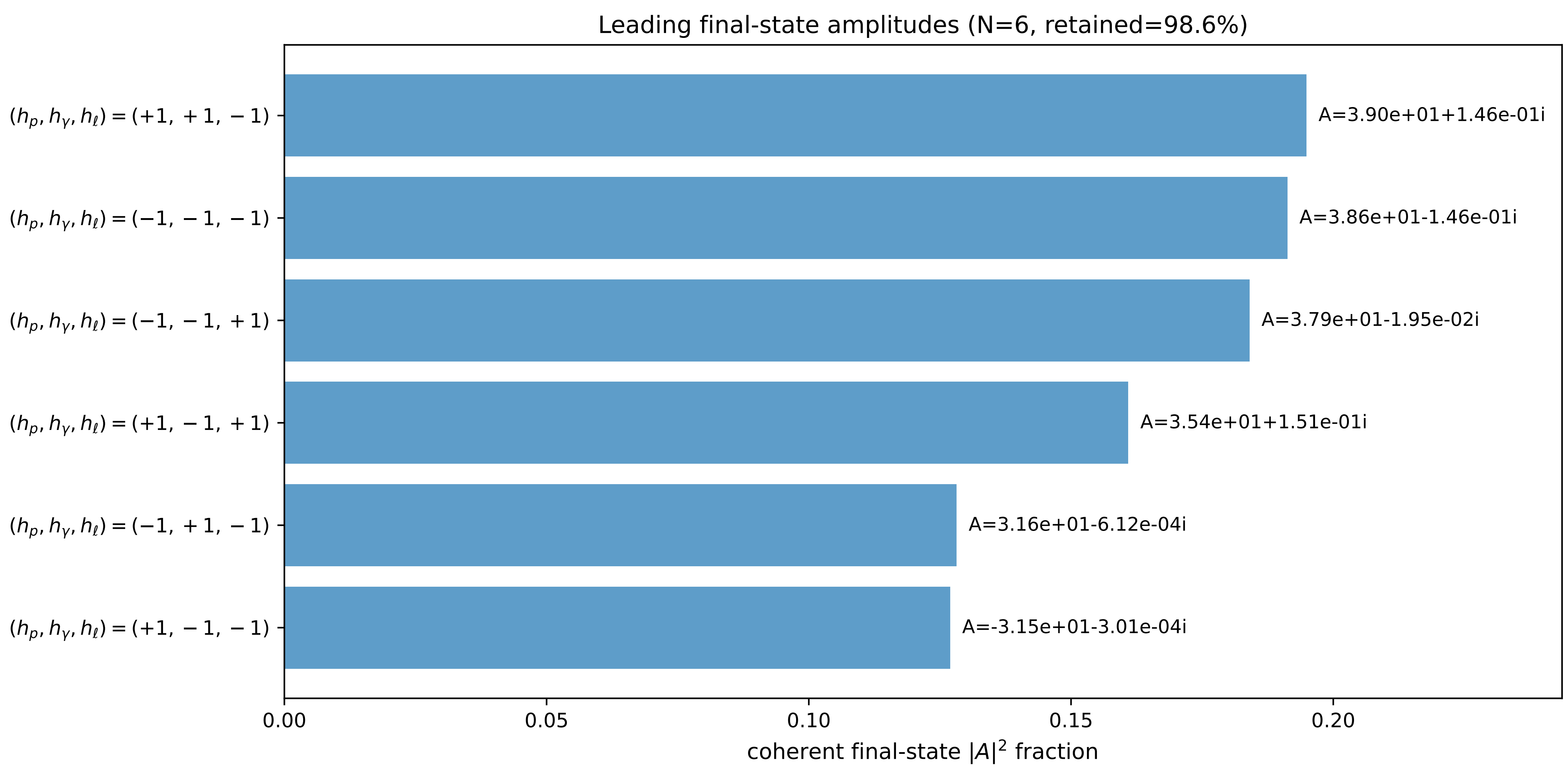


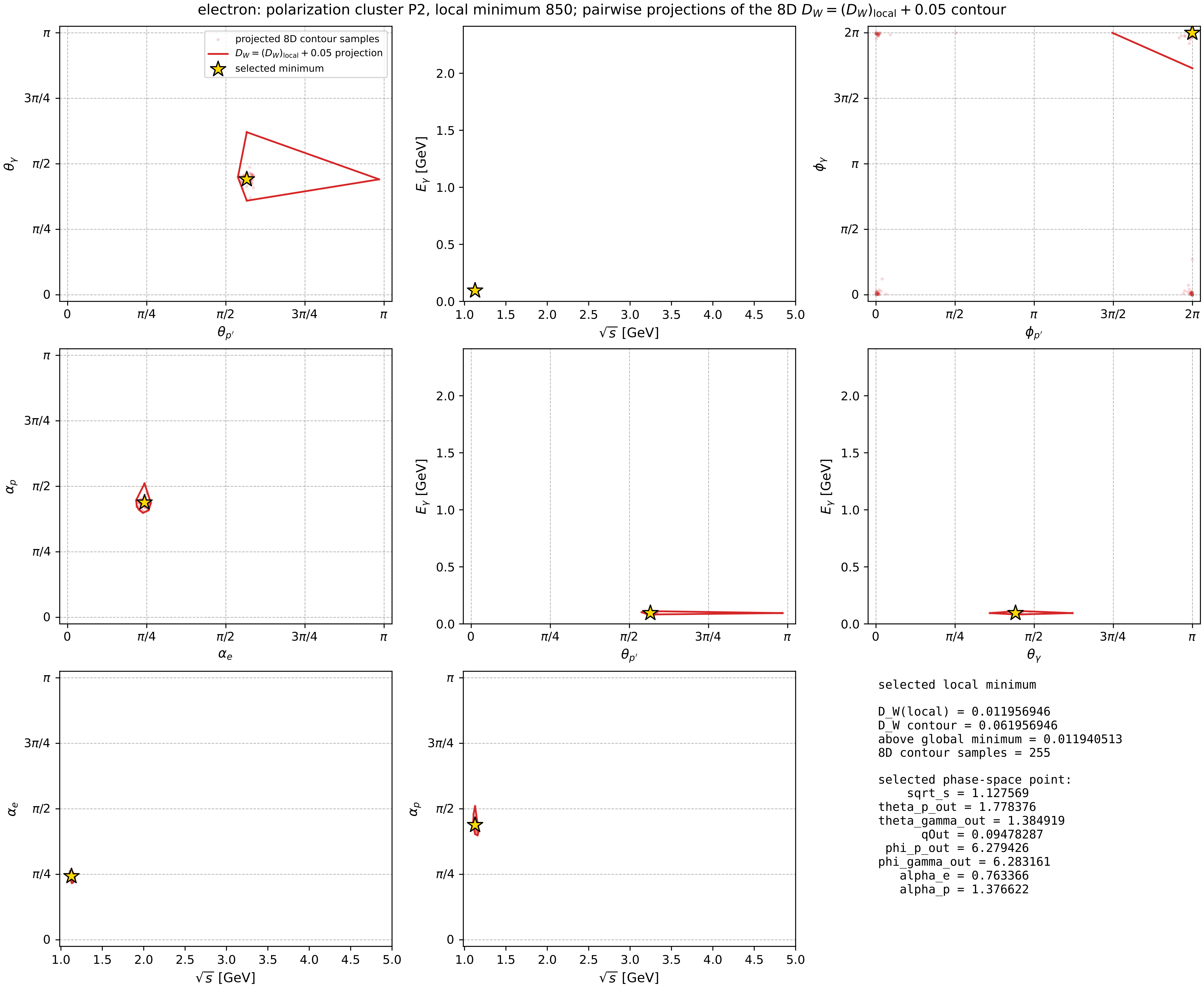

```
dw_mixing_angles_polarization_02_local_minimum_850  D_W=0.0119569
region: polarization_cluster_02_local_minimum_850
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.76336599 rad
incoming lepton: 0.722513|+> +0.691357|->
proton mixing angle: alpha_p=1.3766225 rad
incoming proton: 0.192956|+> +0.981207|->
```

s=1.27141, $\sqrt{s}$=1.12757, $|\vec{P}|$=0.173633, $|\vec{P}'|$=1.532e-06
$\theta_{p'}$=1.77838, θ_γ=1.38492, $\phi_{p'}$=6.27943, ϕ_γ=6.28316
E_γ=0.0947829, $\theta_{\ell'}$=1.75667, $\phi_{\ell'}$=3.14157
Q^2=0.0268325, x_B=0.131118, t=-0.0298945, W^2=1.05766, y=0.522631

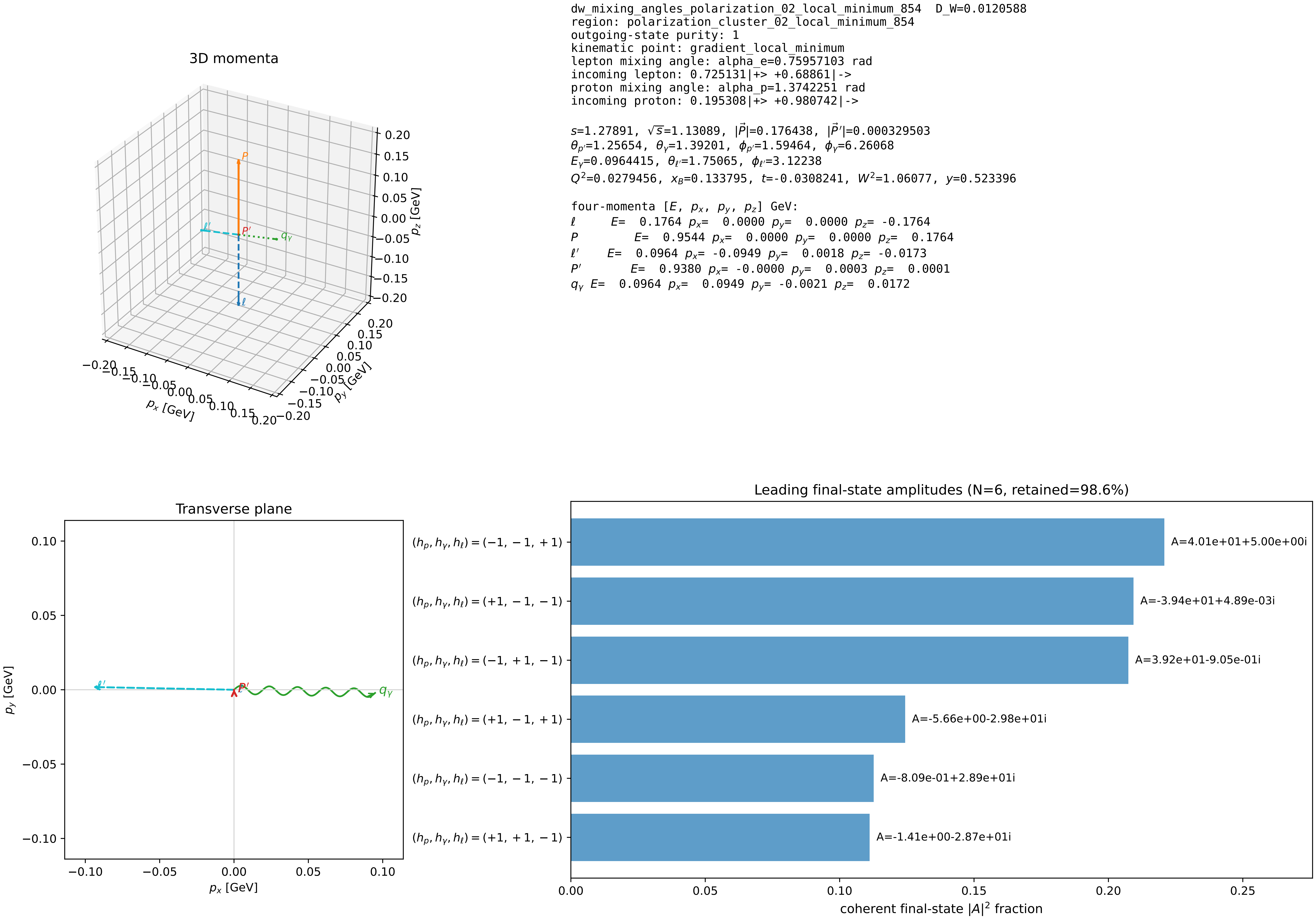
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1736 p_x= 0.0000 p_y= 0.0000 p_z= -0.1736
P E= 0.9539 p_x= 0.0000 p_y= 0.0000 p_z= 0.1736
ℓ' E= 0.0948 p_x= -0.0932 p_y= 0.0000 p_z= -0.0175
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.0948 p_x= 0.0932 p_y= -0.0000 p_z= 0.0175



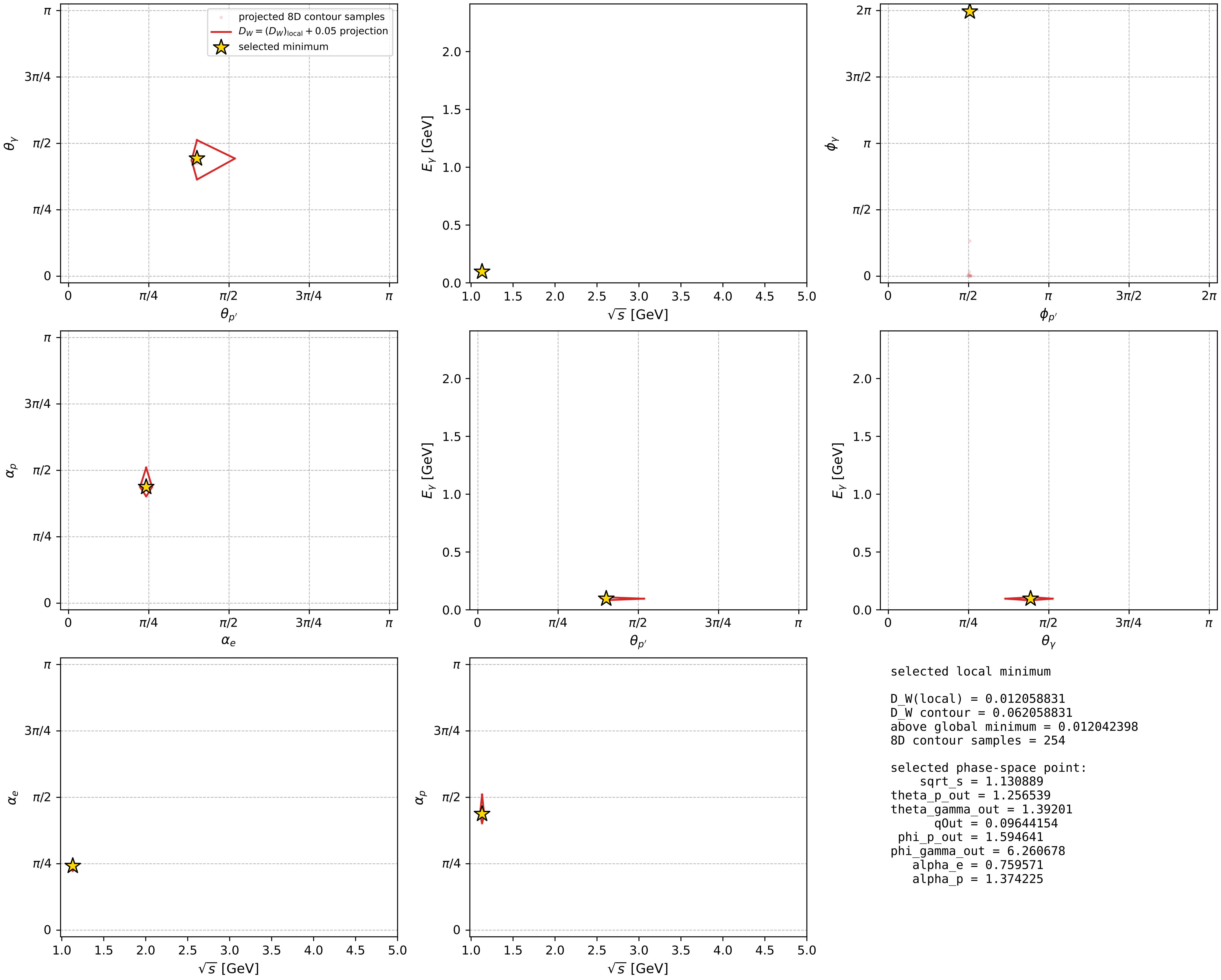



electron: polarization cluster P2, local minimum 850; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
θ_γ
selected local minimum
D_W(local) = 0.011956946
D_W contour = 0.061956946
above global minimum = 0.011940513
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.127569
theta_p_out = 1.778376
theta_gamma_out = 1.384919
qOut = 0.09478287
phi_p_out = 6.279426
phi_gamma_out = 6.283161
alpha_e = 0.763366
alpha_p = 1.376622

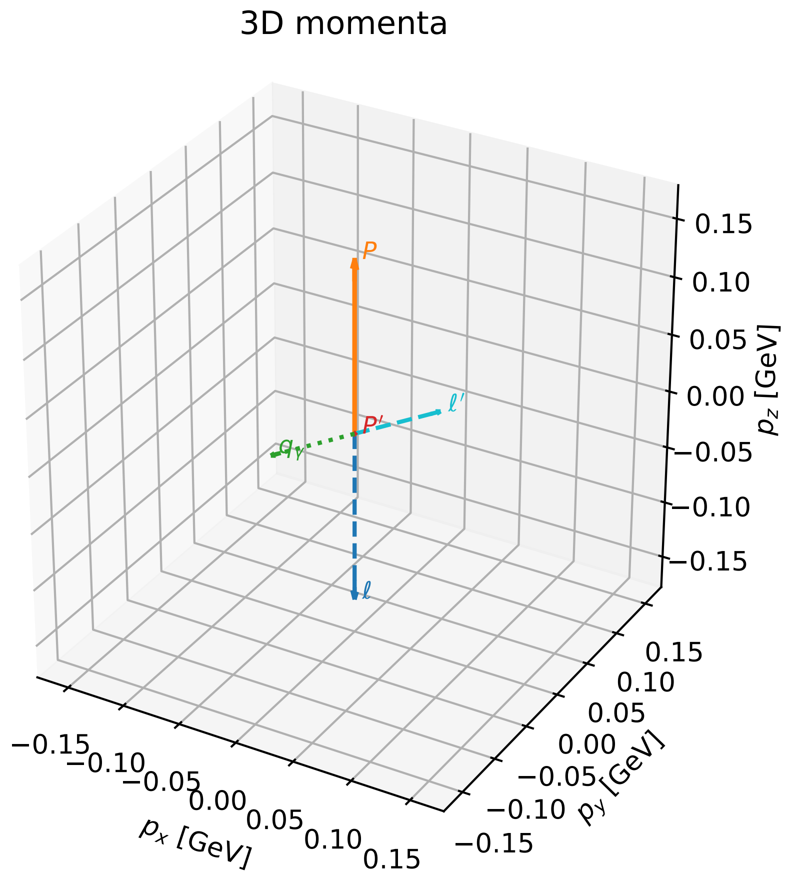
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



electron: polarization cluster P2, local minimum 854; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



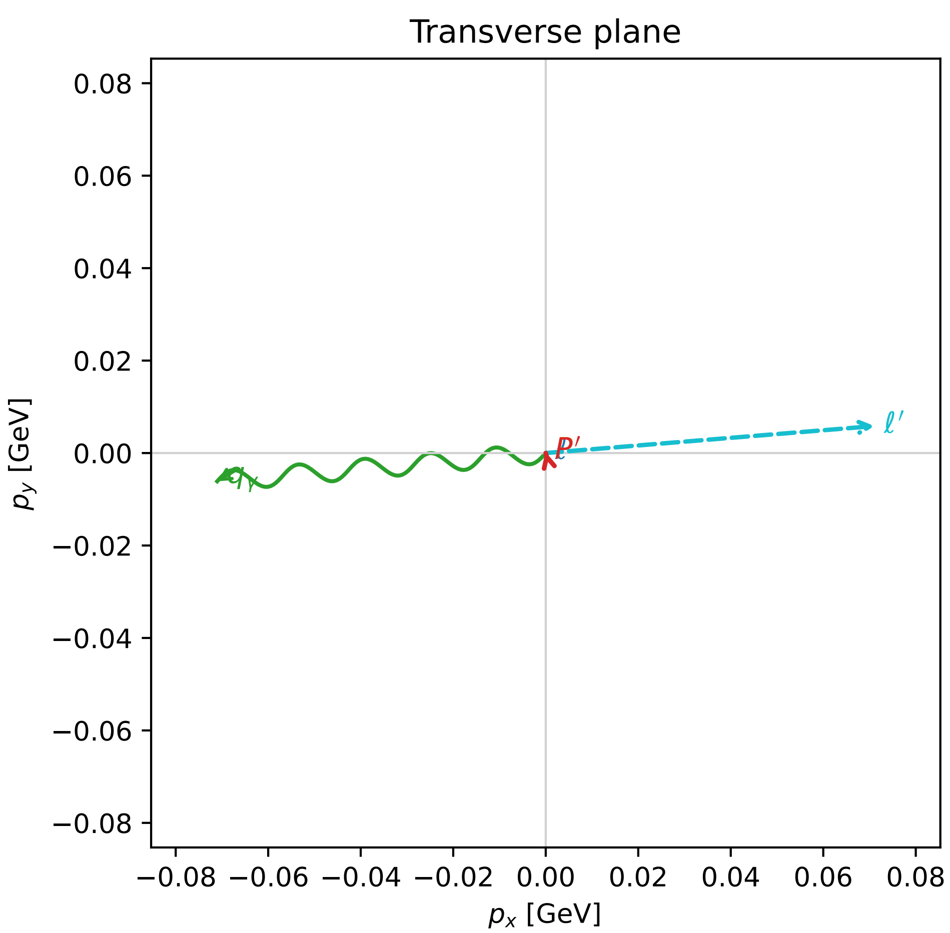
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

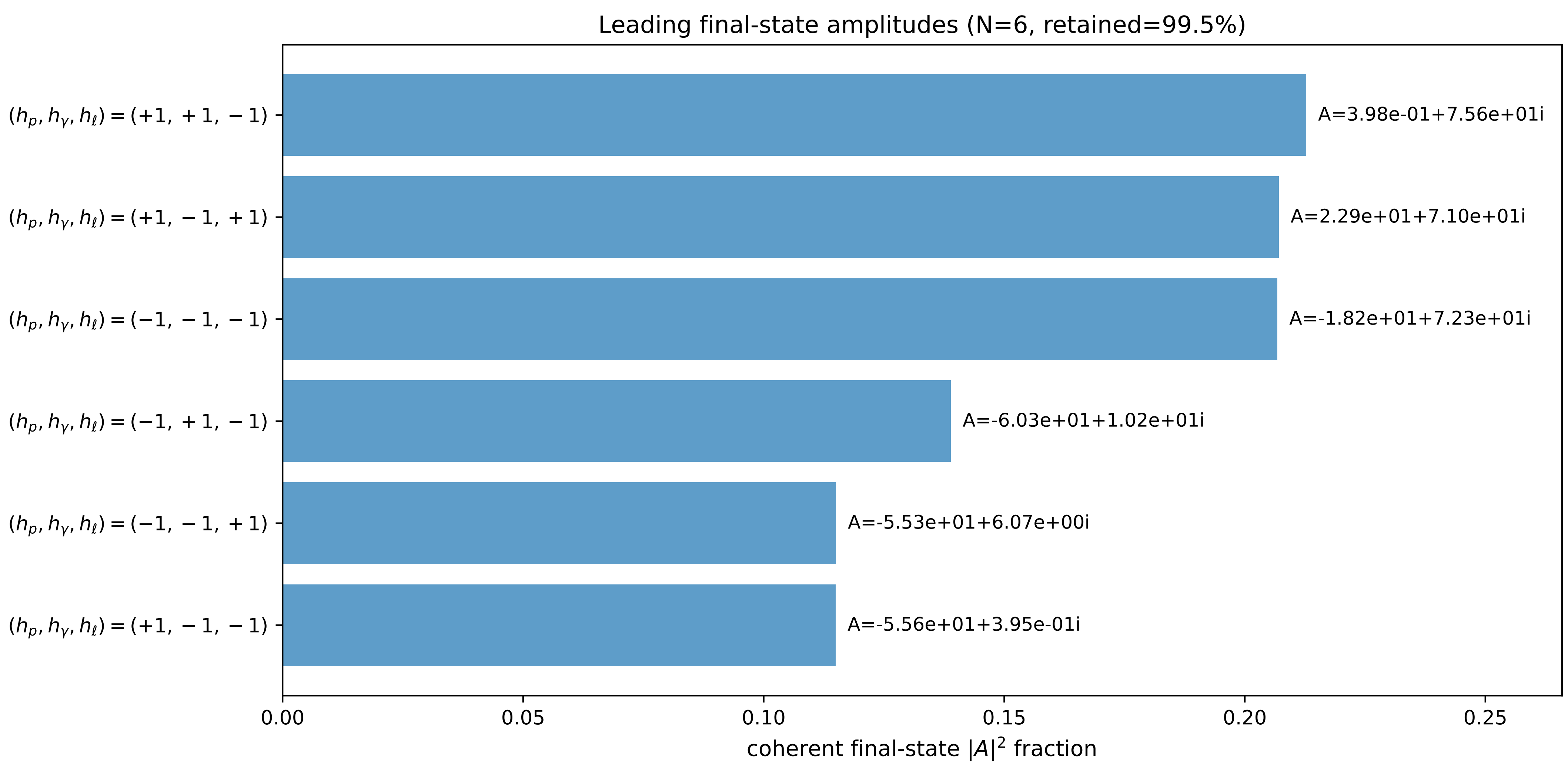


dw_mixing_angles_polarization_02_local_minimum_861 D_W=0.0121888
region: polarization_cluster_02_local_minimum_861
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80158699 rad
incoming lepton: 0.695567|+> +0.718461|->
proton mixing angle: alpha_p=1.6983263 rad
incoming proton: -0.127185|+> +0.991879|->

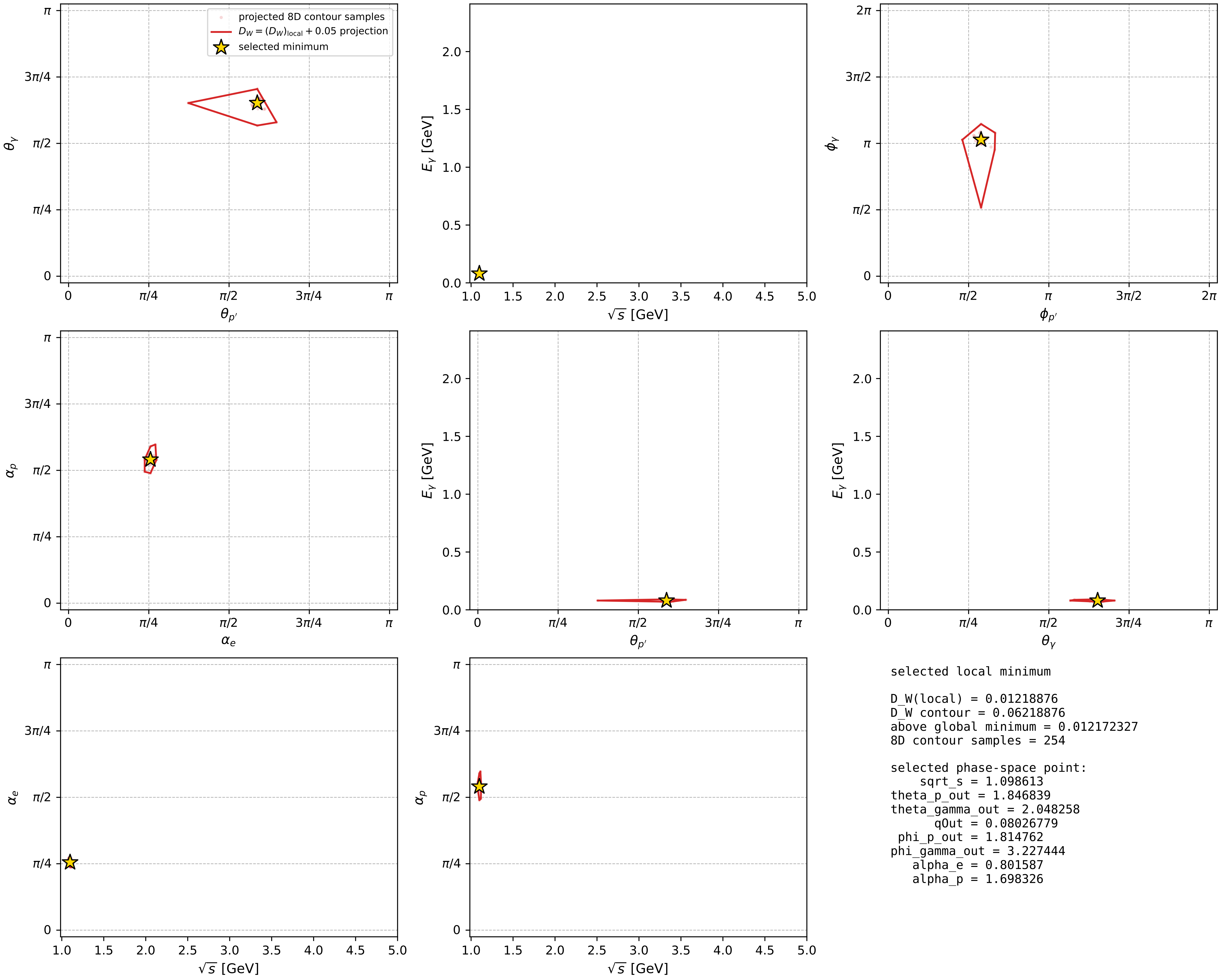
s=1.20695, $\sqrt{s}$=1.09861, $|\vec{P}|$=0.148871, $|\vec{P}'|$=0.000288619
$\theta_{p'}$=1.84684, θ_γ=2.04826, $\phi_{p'}$=1.81476, ϕ_γ=3.22744
E_γ=0.0802678, $\theta_{\ell'}$=1.09272, $\phi_{\ell'}$=0.0820072
Q^2=0.0349275, x_B=0.18829, t=-0.0220484, W^2=1.03041, y=0.567088

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1489 p_x= 0.0000 p_y= 0.0000 p_z= -0.1489
P E= 0.9497 p_x= 0.0000 p_y= 0.0000 p_z= 0.1489
ℓ' E= 0.0803 p_x= 0.0711 p_y= 0.0058 p_z= 0.0370
P' E= 0.9380 p_x= -0.0001 p_y= 0.0003 p_z= -0.0001
q_γ E= 0.0803 p_x= -0.0710 p_y= -0.0061 p_z= -0.0369

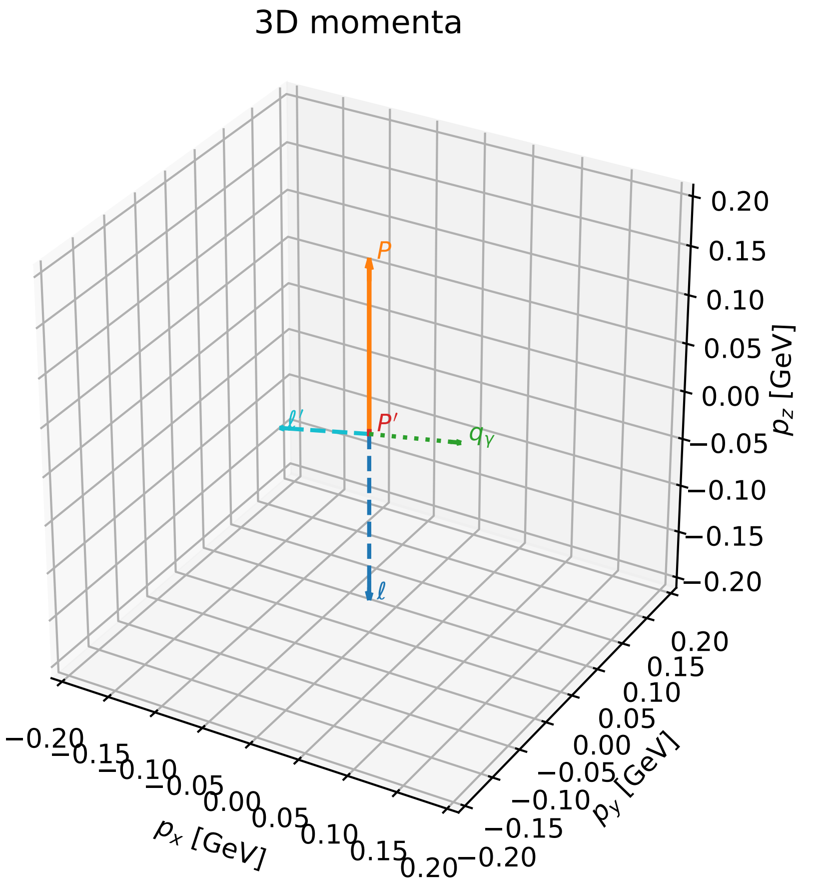




electron: polarization cluster P2, local minimum 861; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



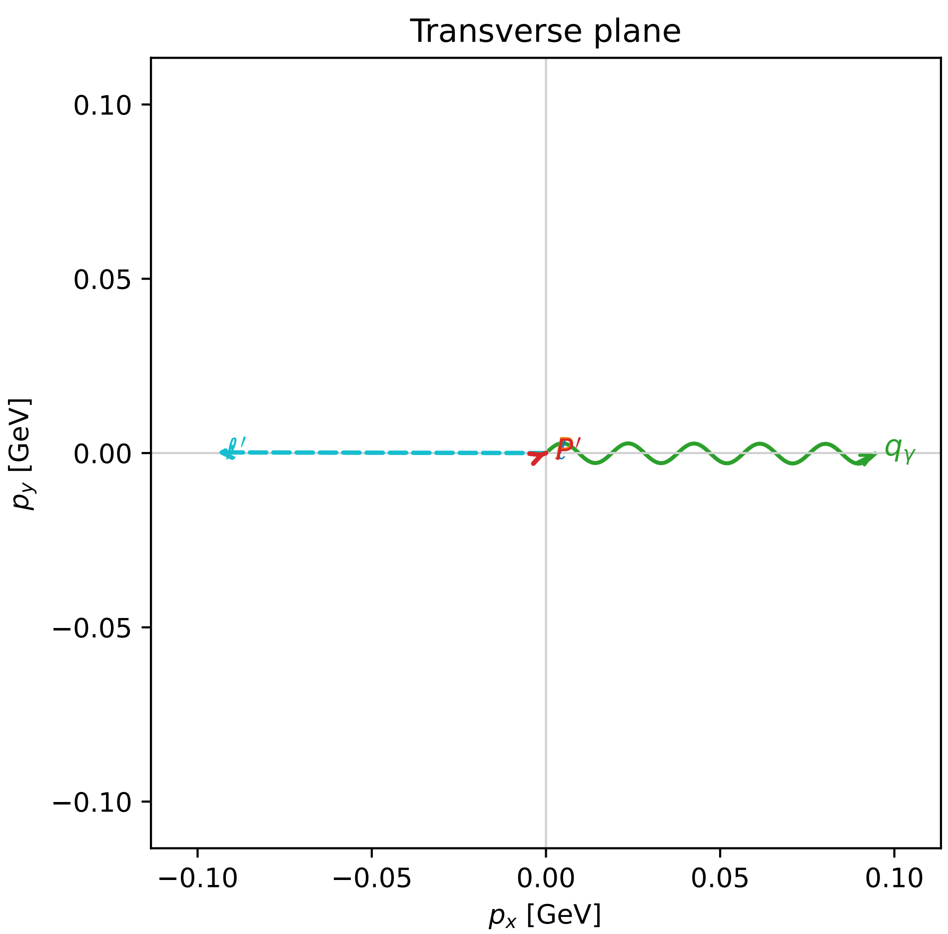
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

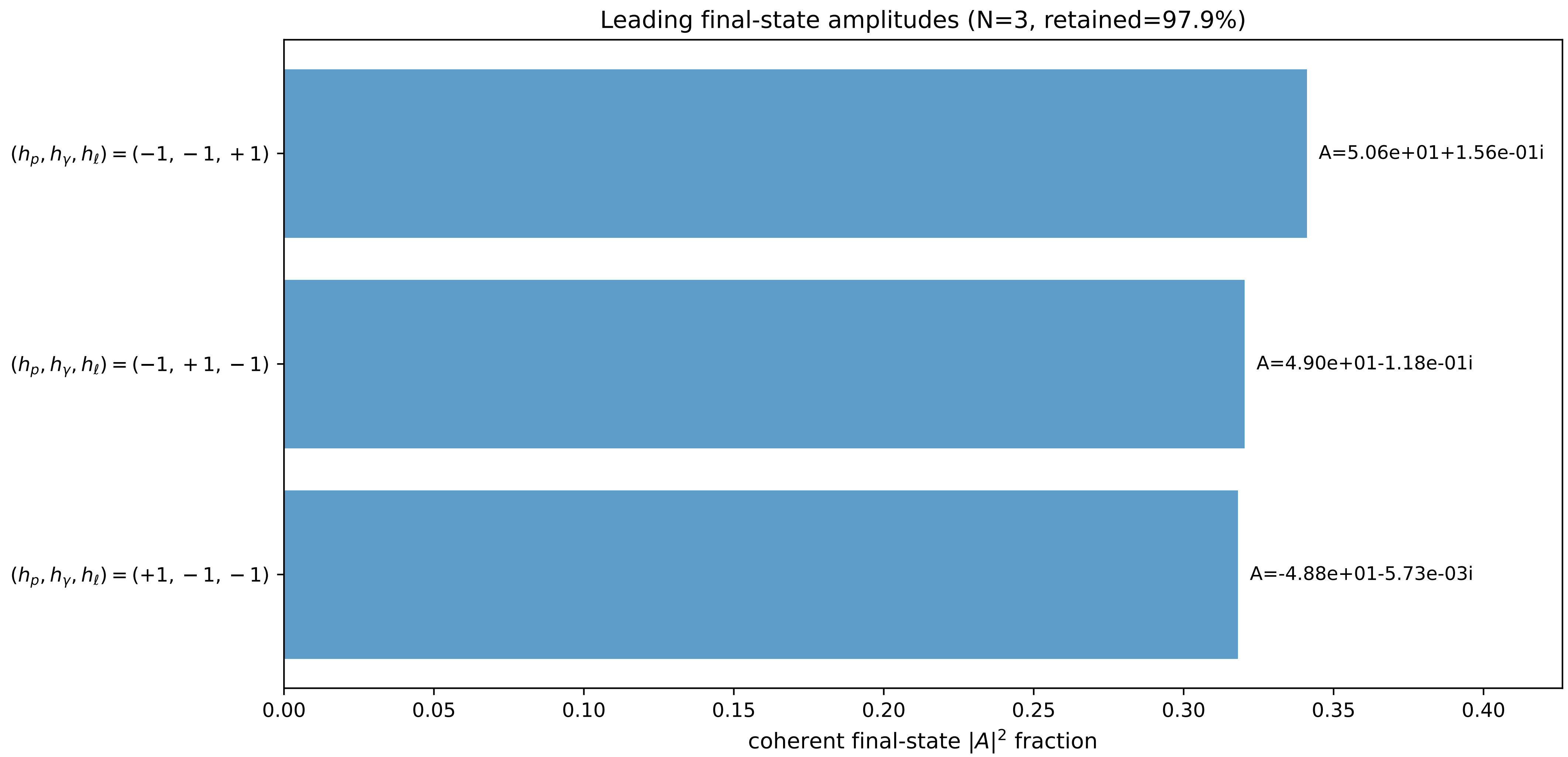


dw_mixing_angles_polarization_02_local_minimum_872 D_W=0.0124982
region: polarization_cluster_02_local_minimum_872
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75872549 rad
incoming lepton: 0.725713|+> +0.687997|->
proton mixing angle: alpha_p=1.3797863 rad
incoming proton: 0.189851|+> +0.981813|->

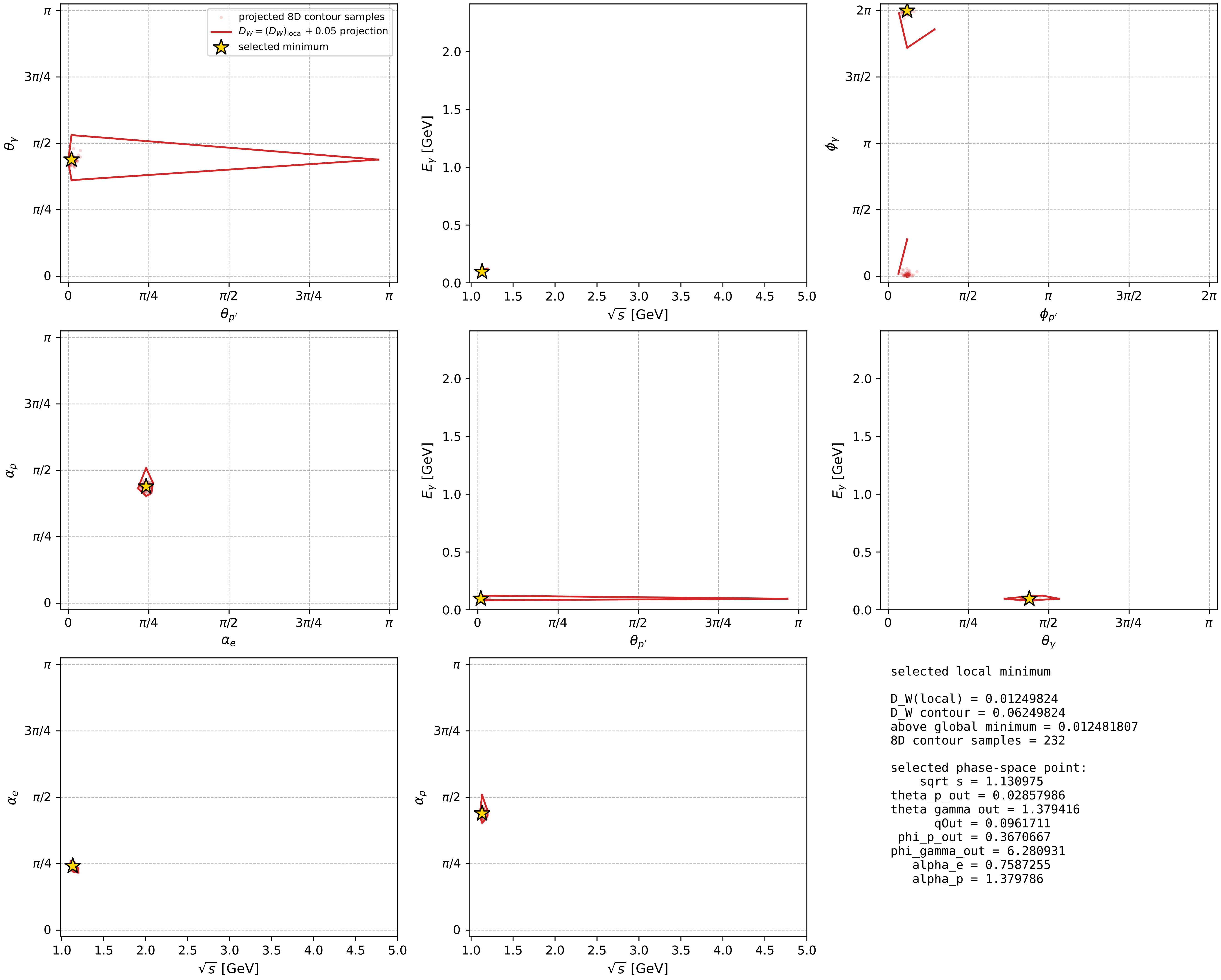
s=1.2791, $\sqrt{s}$=1.13098, $|\vec{P}|$=0.176511, $|\vec{P}'|$=0.00273129
$\theta_{p'}$=0.0285799, θ_γ=1.37942, $\phi_{p'}$=0.367067, ϕ_γ=6.28093
E_γ=0.0961711, $\theta_{\ell'}$=1.78973, $\phi_{\ell'}$=3.13964
Q^2=0.0267504, x_B=0.129195, t=-0.0299289, W^2=1.06015, y=0.518595

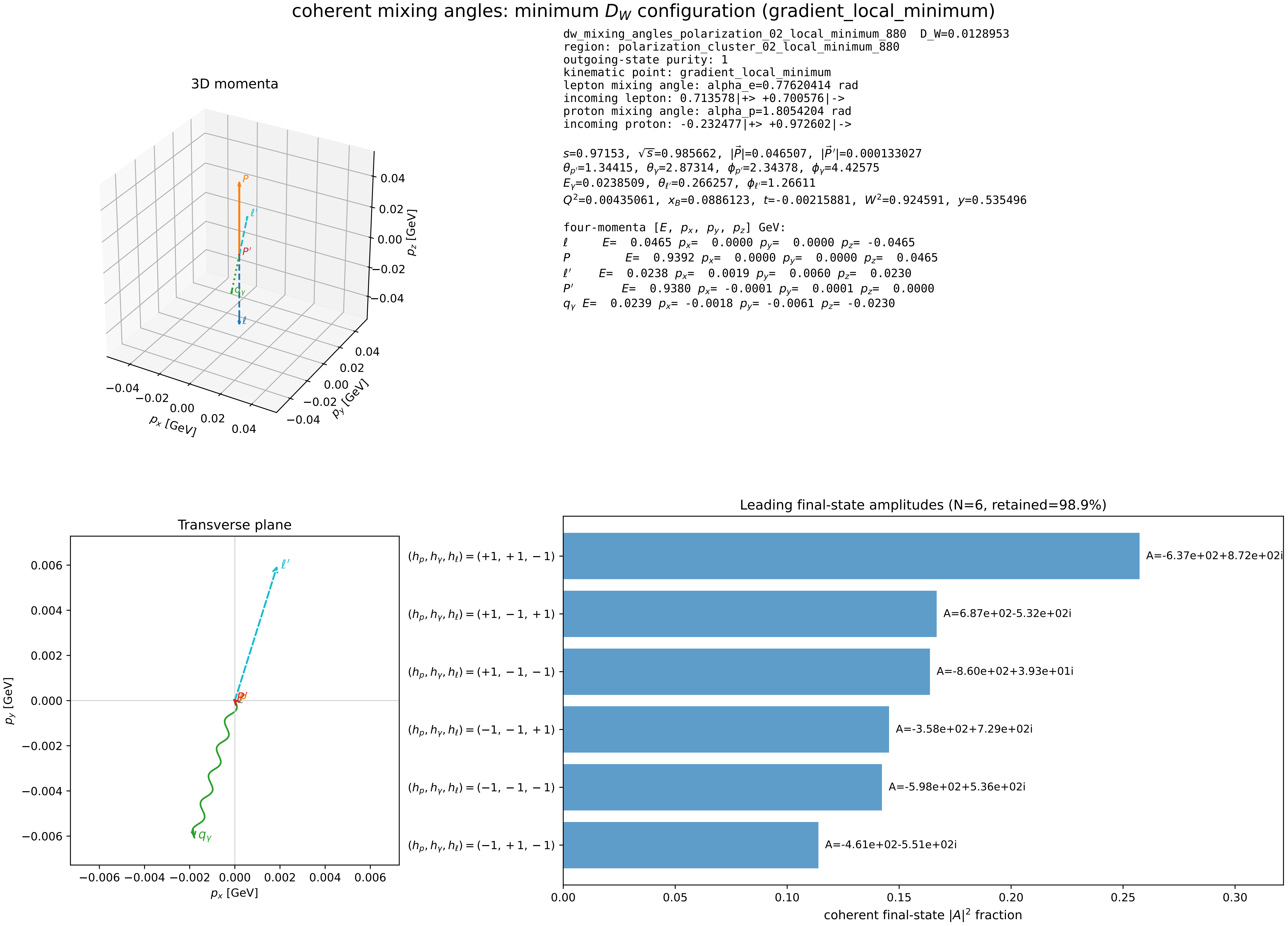
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1765 p_x= 0.0000 p_y= 0.0000 p_z= -0.1765
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1765
ℓ' E= 0.0968 p_x= -0.0945 p_y= 0.0002 p_z= -0.0210
P' E= 0.9380 p_x= 0.0001 p_y= 0.0000 p_z= 0.0027
q_γ E= 0.0962 p_x= 0.0944 p_y= -0.0002 p_z= 0.0183

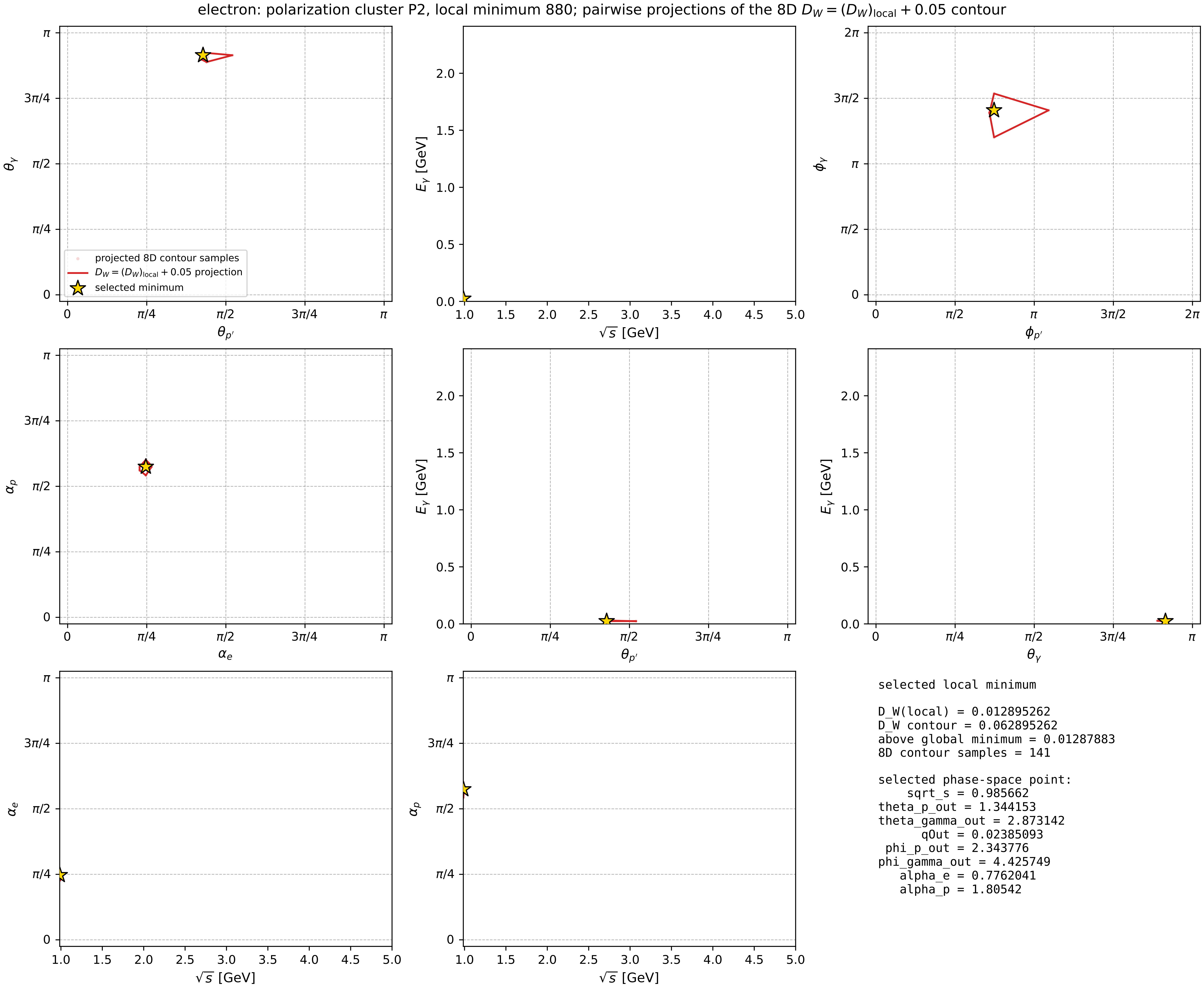




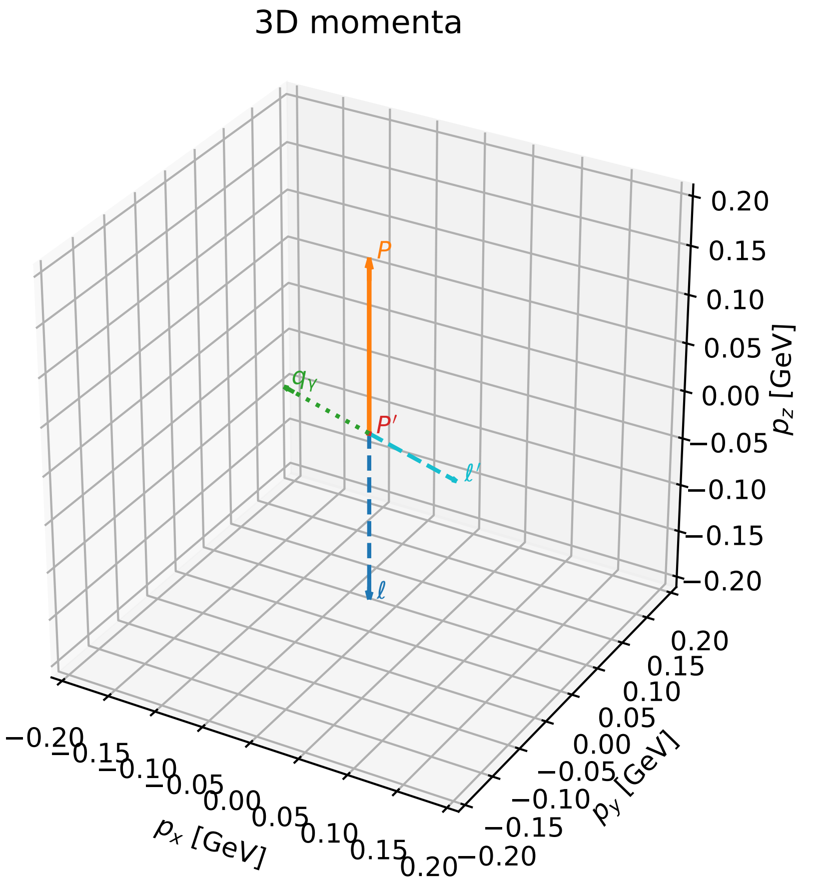
electron: polarization cluster P2, local minimum 872; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_880 D_W=0.0128953
region: polarization_cluster_02_local_minimum_880
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77620414 rad
incoming lepton: 0.713578|+> +0.700576|->
proton mixing angle: alpha_p=1.8054204 rad
incoming proton: -0.232477|+> +0.972602|->
s=0.97153, $\sqrt{s}$=0.985662, $|\vec{P}|$=0.046507, $|\vec{P}'|$=0.000133027
$\theta_{p'}$=1.34415, θ_γ=2.87314, $\phi_{p'}$=2.34378, ϕ_γ=4.42575
E_γ=0.0238509, $\theta_{\ell'}$=0.266257, $\phi_{\ell'}$=1.26611
Q^2=0.00435061, x_B=0.0886123, t=-0.00215881, W^2=0.924591, y=0.535496
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0465 p_x= 0.0000 p_y= 0.0000 p_z= -0.0465
P E= 0.9392 p_x= 0.0000 p_y= 0.0000 p_z= 0.0465
ℓ' E= 0.0238 p_x= 0.0019 p_y= 0.0060 p_z= 0.0230
P' E= 0.9380 p_x= -0.0001 p_y= 0.0001 p_z= 0.0000
q_γ E= 0.0239 p_x= -0.0018 p_y= -0.0061 p_z= -0.0230
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=98.9%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-6.37e+02+8.72e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=6.87e+02-5.32e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-8.60e+02+3.93e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-3.58e+02+7.29e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-5.98e+02+5.36e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-4.61e+02-5.51e+02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 880; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.012895262
D_W contour = 0.062895262
above global minimum = 0.01287883
8D contour samples = 141
selected phase-space point:
sqrt_s = 0.985662
theta_p_out = 1.344153
theta_gamma_out = 2.873142
qOut = 0.02385093
phi_p_out = 2.343776
phi_gamma_out = 4.425749
alpha_e = 0.7762041
alpha_p = 1.80542

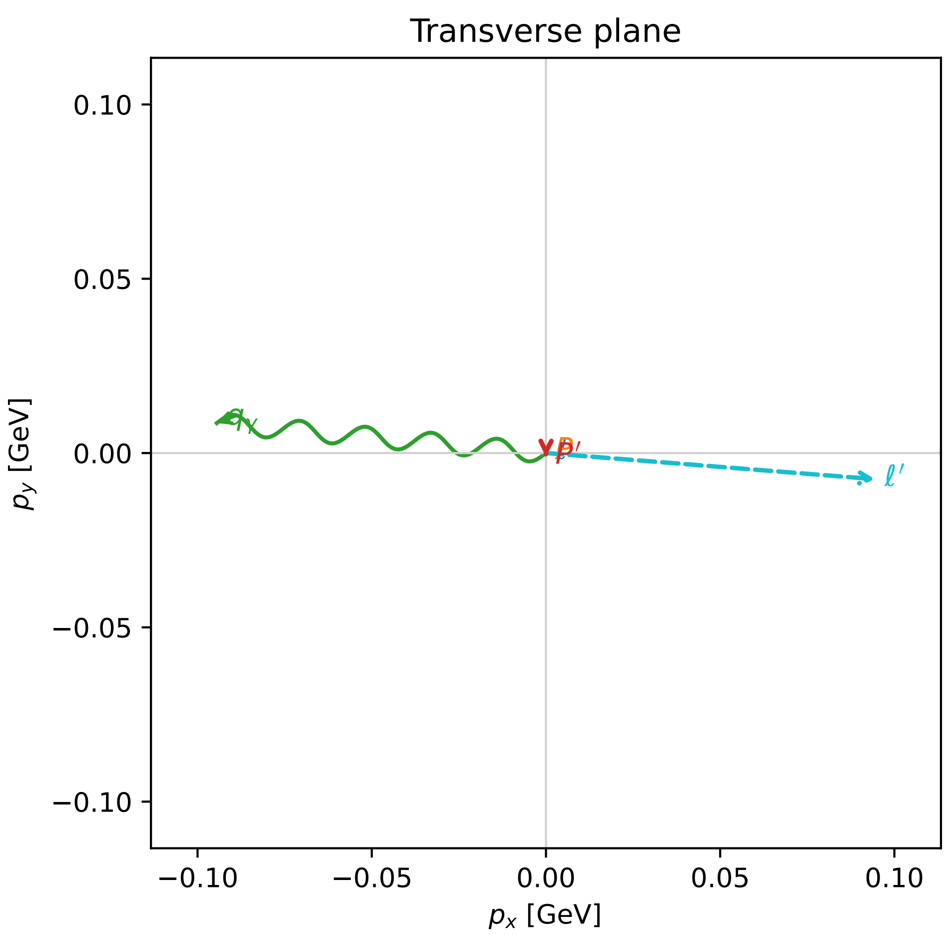
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

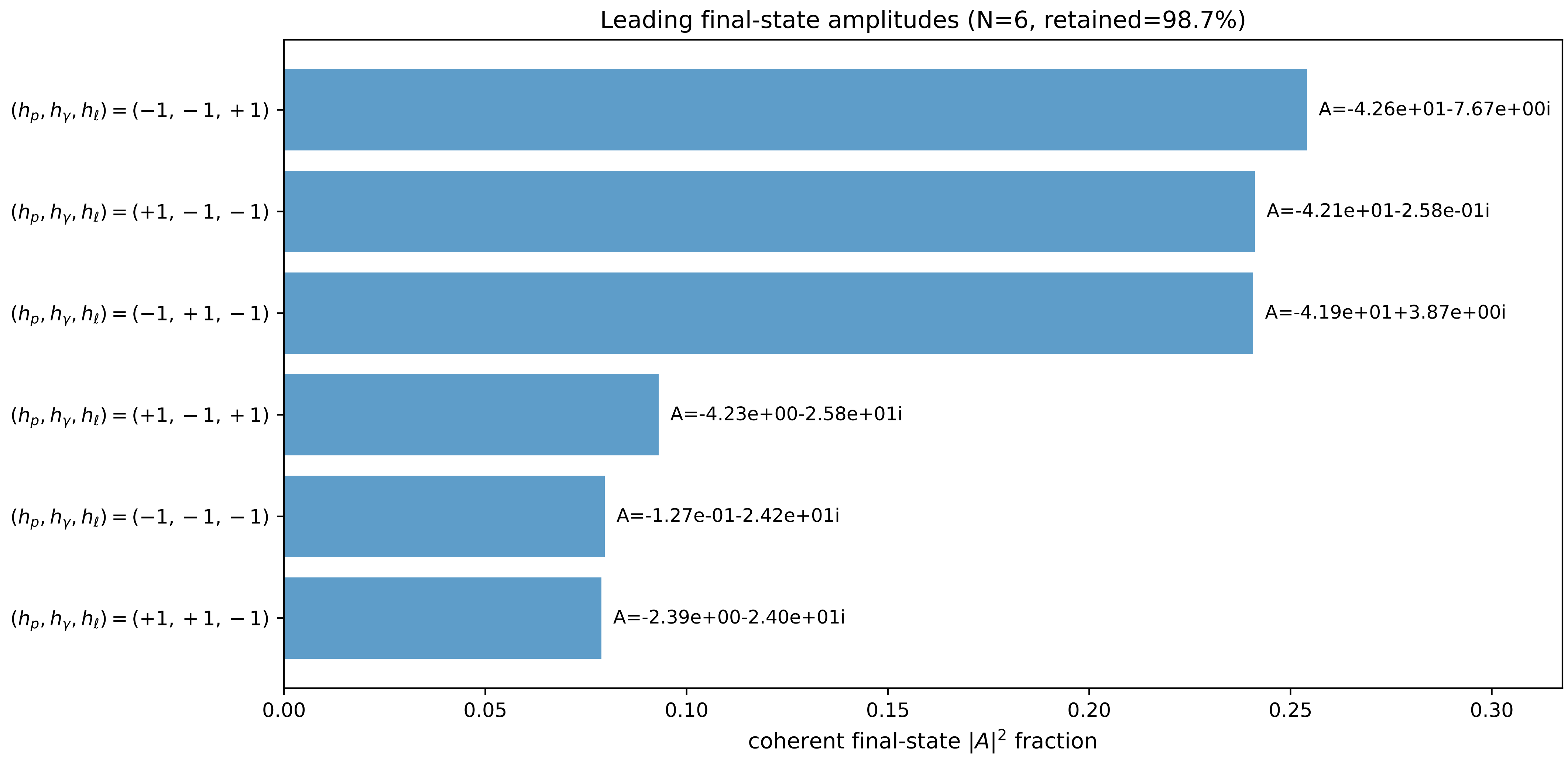



```
dw_mixing_angles_polarization_02_local_minimum_881  D_W=0.0129005
region: polarization_cluster_02_local_minimum_881
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7539679 rad
incoming lepton: 0.728978|+> +0.684537|->
proton mixing angle: alpha_p=1.7571472 rad
incoming proton: -0.185274|+> +0.982687|->
```

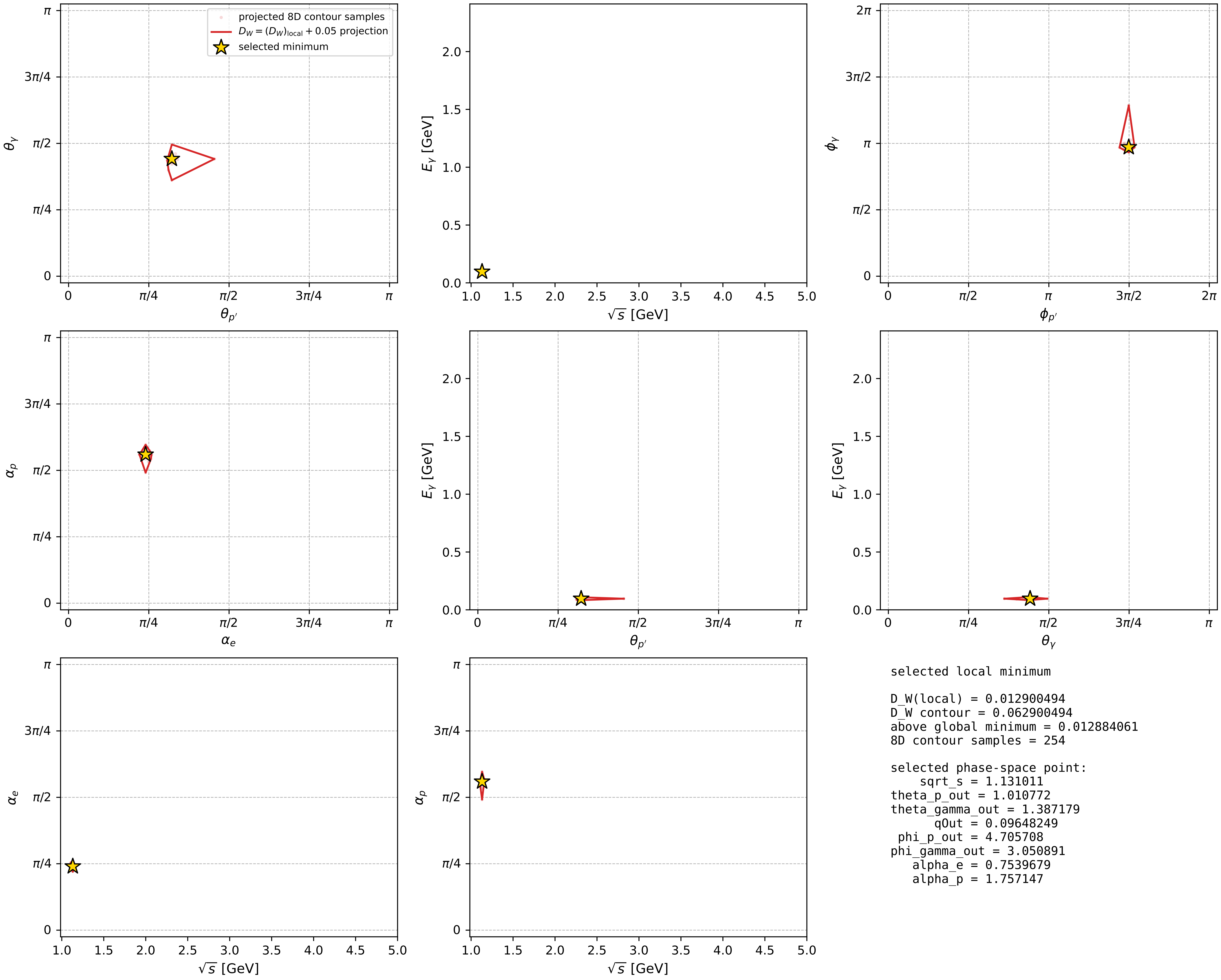
s=1.27919, $\sqrt{s}$=1.13101, $|\vec{P}|$=0.176541, $|\vec{P}'|$=0.00128881
$\theta_{p'}$=1.01077, θ_γ=1.38718, $\phi_{p'}$=4.70571, ϕ_γ=3.05089
E_γ=0.0964825, $\theta_{\ell'}$=1.76155, $\phi_{\ell'}$=6.20396
Q^2=0.0276199, x_B=0.132397, t=-0.0306555, W^2=1.06084, y=0.522397

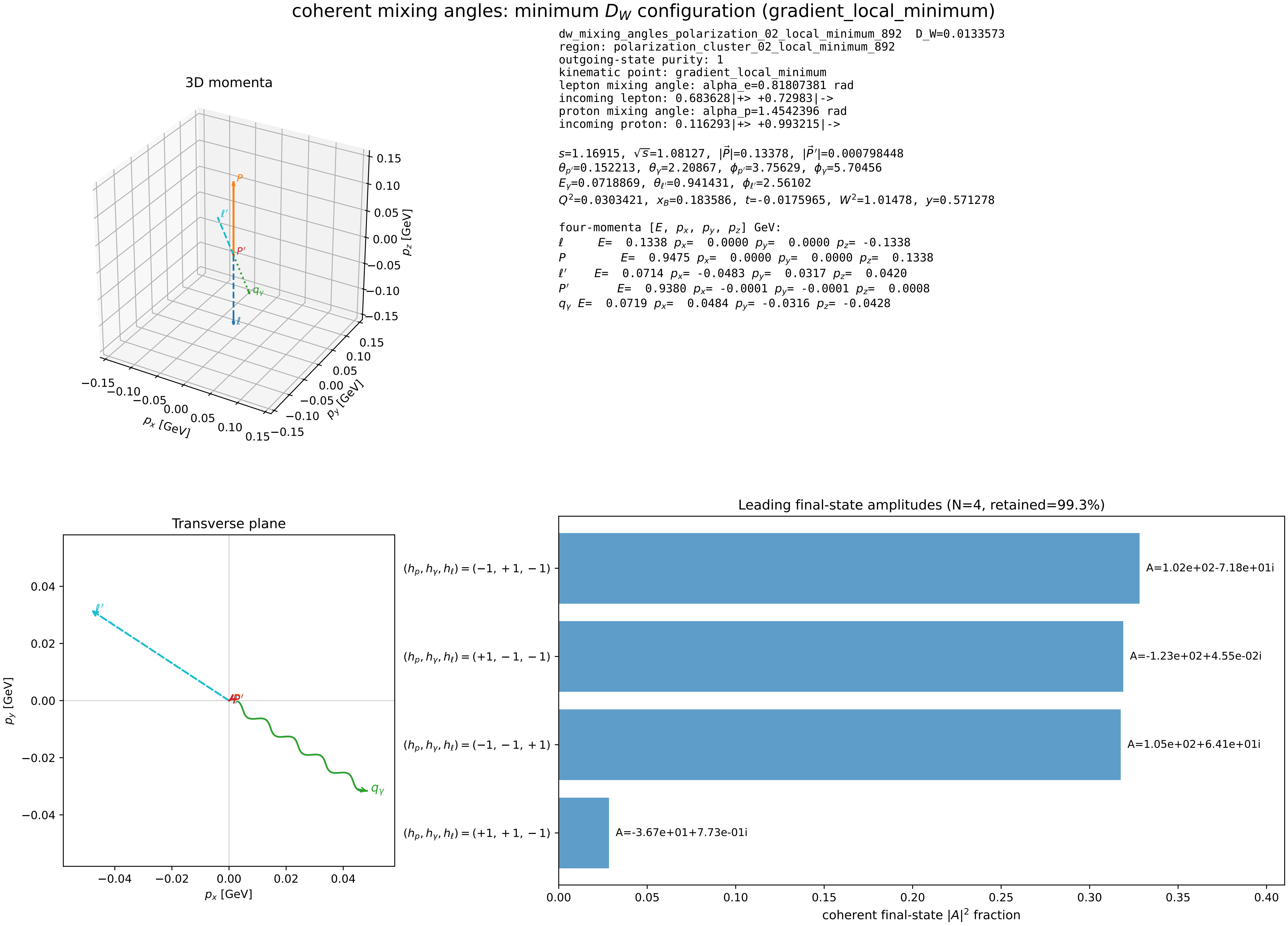
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1765 p_x= 0.0000 p_y= 0.0000 p_z= -0.1765
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1765
ℓ' E= 0.0965 p_x= 0.0945 p_y= -0.0075 p_z= -0.0183
P' E= 0.9380 p_x= -0.0000 p_y= -0.0011 p_z= 0.0007
q_γ E= 0.0965 p_x= -0.0945 p_y= 0.0086 p_z= 0.0176

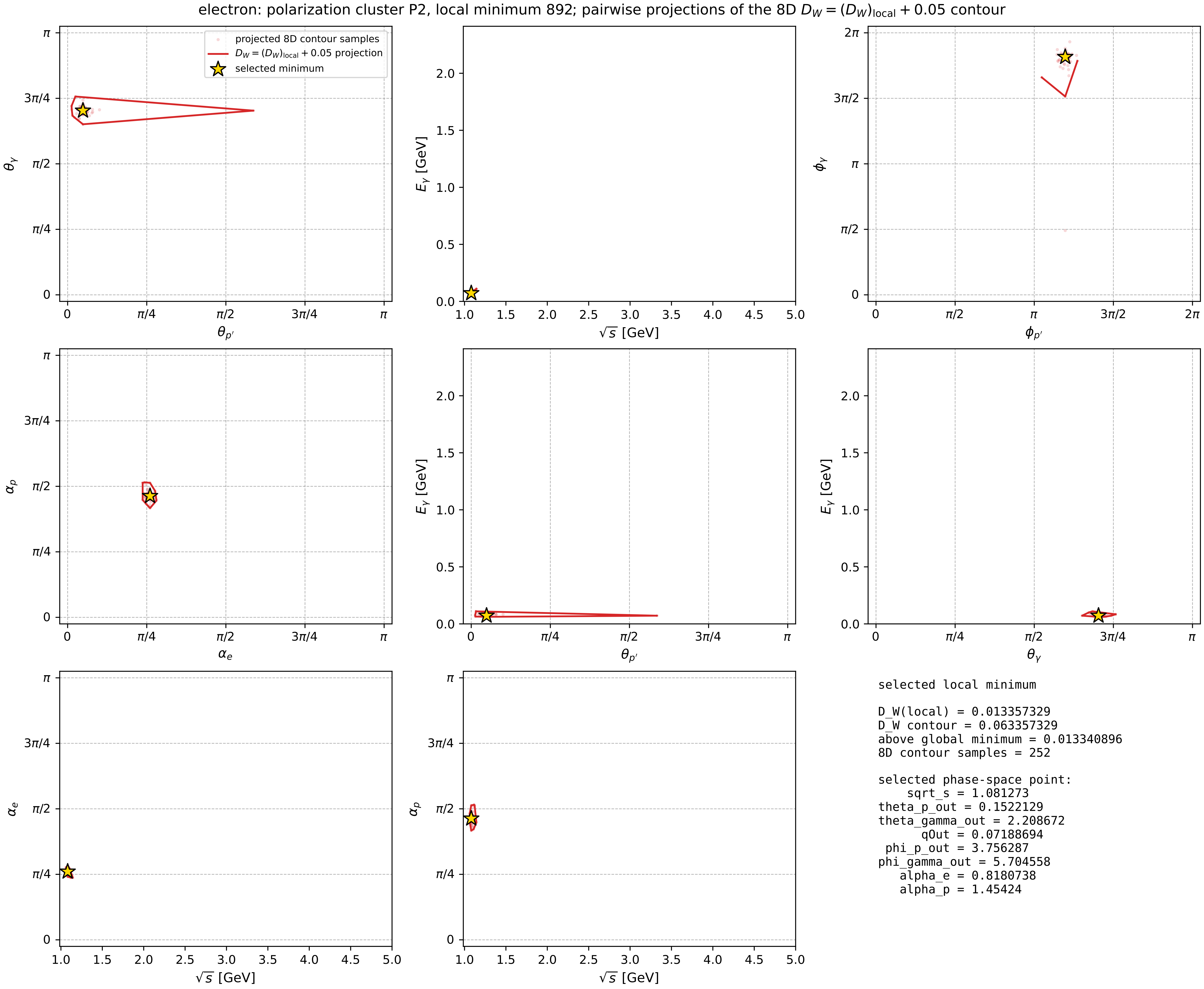




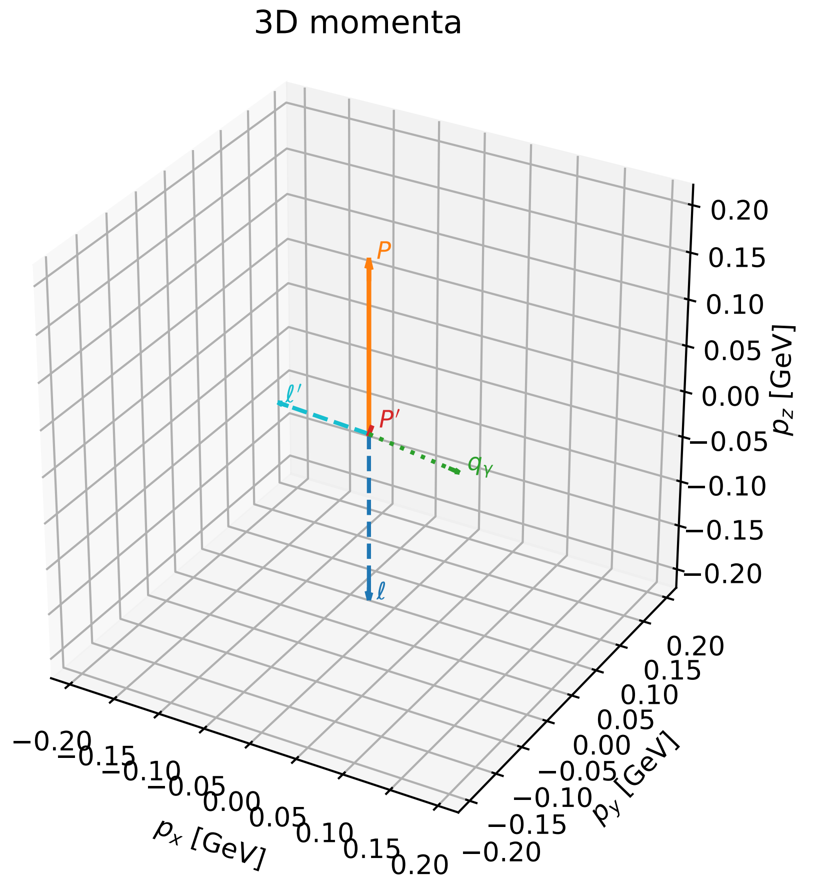
electron: polarization cluster P2, local minimum 881; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_892 D_W=0.0133573
region: polarization_cluster_02_local_minimum_892
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.81807381 rad
incoming lepton: 0.683628|+> +0.72983|->
proton mixing angle: alpha_p=1.4542396 rad
incoming proton: 0.116293|+> +0.993215|->
s=1.16915, $\sqrt{s}$=1.08127, $|\vec{P}|$=0.13378, $|\vec{P}'|$=0.000798448
$\theta_{p'}$=0.152213, θ_γ=2.20867, $\phi_{p'}$=3.75629, ϕ_γ=5.70456
E_γ=0.0718869, $\theta_{\ell'}$=0.941431, $\phi_{\ell'}$=2.56102
Q^2=0.0303421, x_B=0.183586, t=-0.0175965, W^2=1.01478, y=0.571278
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1338 p_x= 0.0000 p_y= 0.0000 p_z= -0.1338
P E= 0.9475 p_x= 0.0000 p_y= 0.0000 p_z= 0.1338
ℓ' E= 0.0714 p_x= -0.0483 p_y= 0.0317 p_z= 0.0420
P' E= 0.9380 p_x= -0.0001 p_y= -0.0001 p_z= 0.0008
q_γ E= 0.0719 p_x= 0.0484 p_y= -0.0316 p_z= -0.0428
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=4, retained=99.3%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=1.02e+02-7.18e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-1.23e+02+4.55e-02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=1.05e+02+6.41e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-3.67e+01+7.73e-01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 892; pairwise projections of the 8D $D_W = (D_W)_{\rm local} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\rm local} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.013357329
D_W contour = 0.063357329
above global minimum = 0.013340896
8D contour samples = 252
selected phase-space point:
sqrt_s = 1.081273
theta_p_out = 0.1522129
theta_gamma_out = 2.208672
qOut = 0.07188694
phi_p_out = 3.756287
phi_gamma_out = 5.704558
alpha_e = 0.8180738
alpha_p = 1.45424

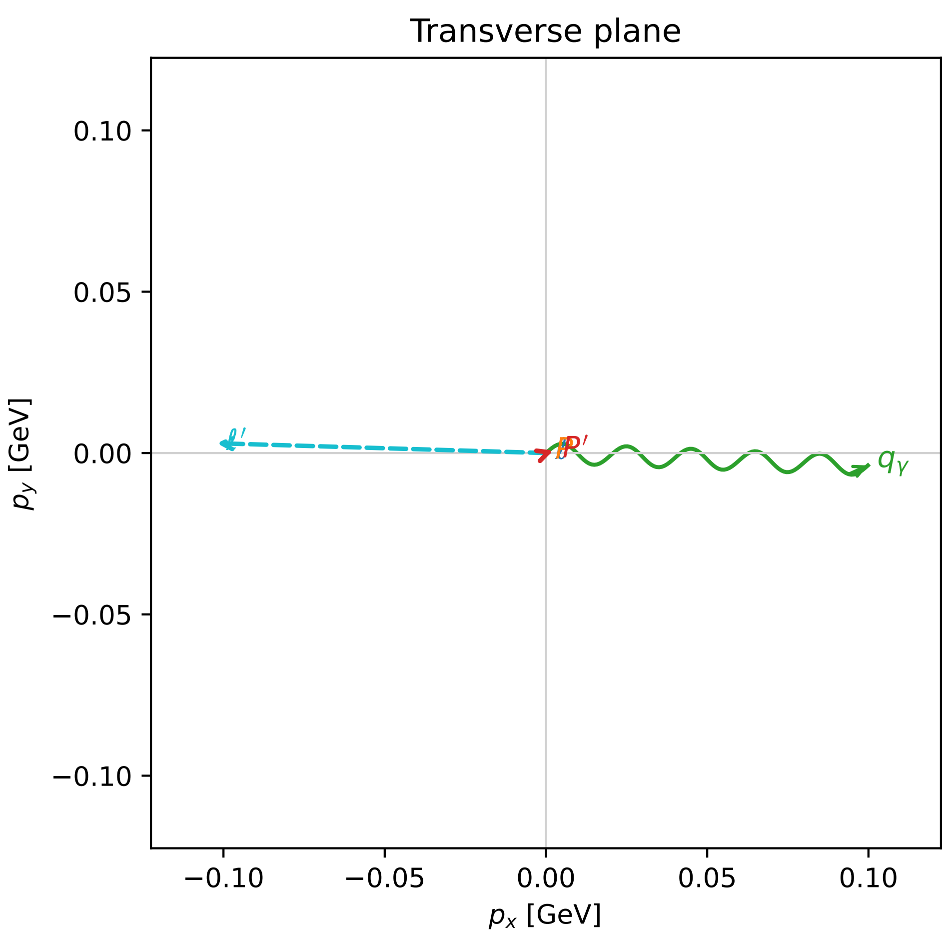
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

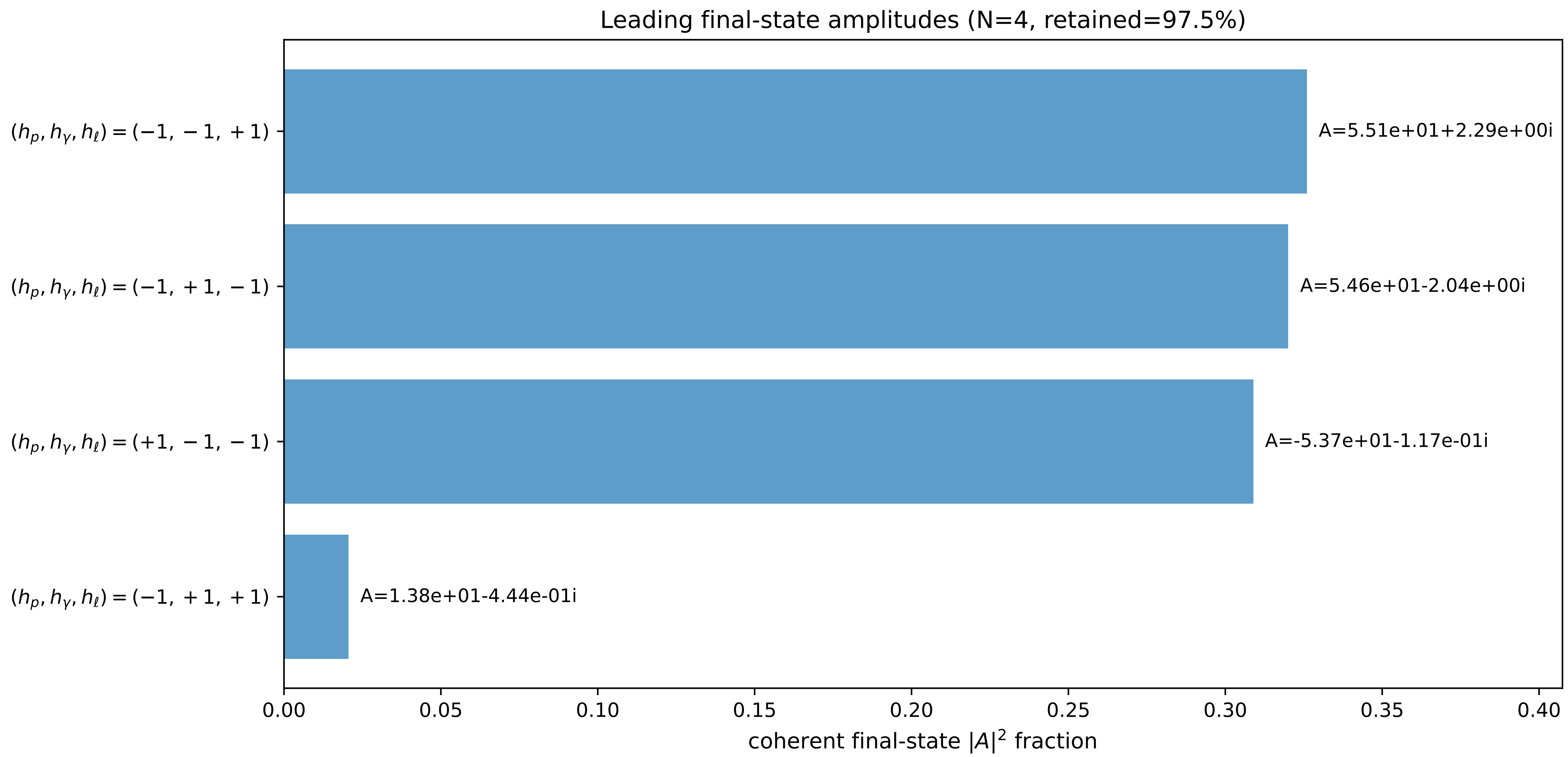



```
dw_mixing_angles_polarization_02_local_minimum_897  D_W=0.0135839
region: polarization_cluster_02_local_minimum_897
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78359061 rad
incoming lepton: 0.708384|+> +0.705827|->
proton mixing angle: alpha_p=1.3418577 rad
incoming proton: 0.226944|+> +0.973908|->
```

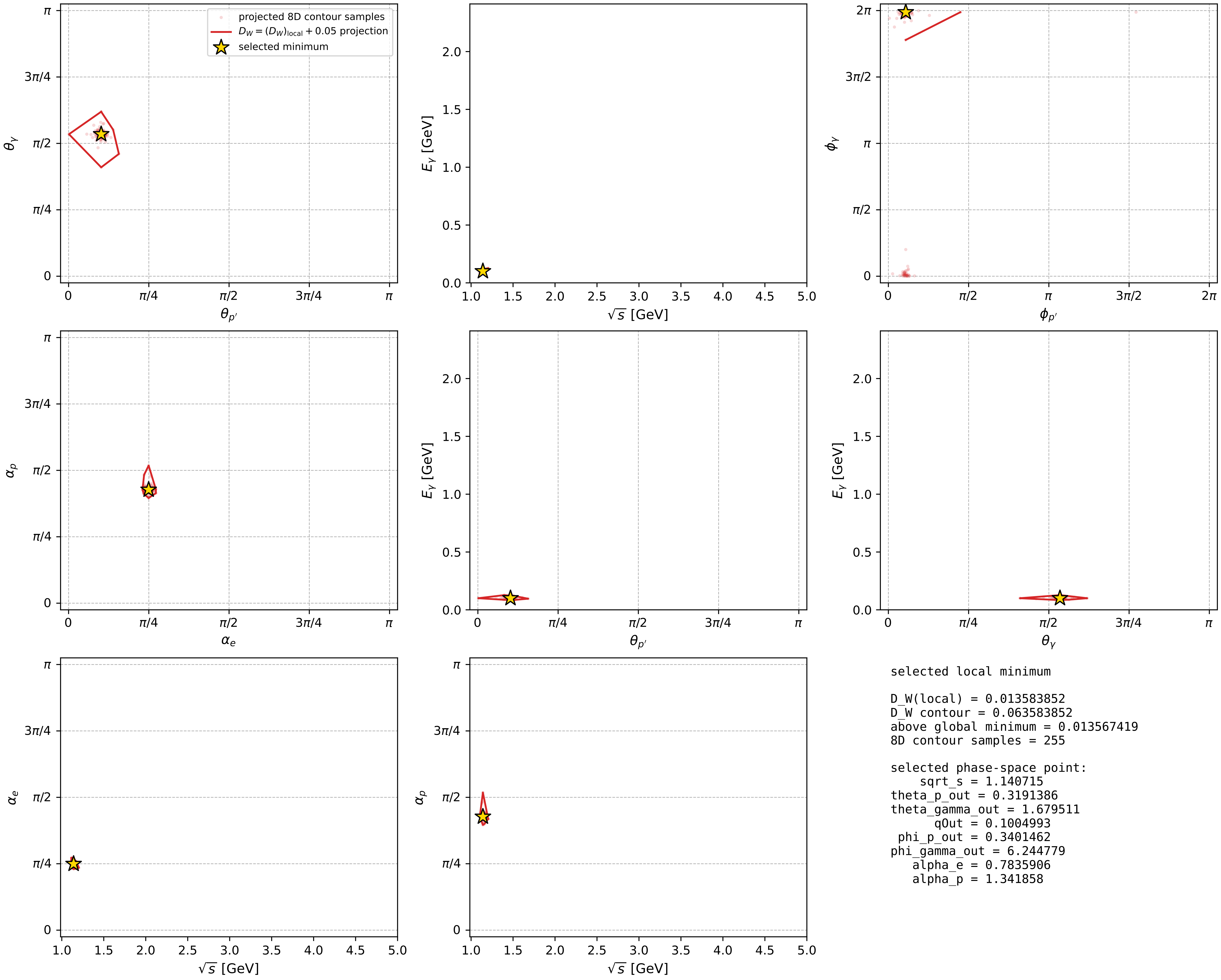
s=1.30123, $\sqrt{s}$=1.14072, $|\vec{P}|$=0.184703, $|\vec{P}'|$=0.00756985
$\theta_{p'}$=0.319139, θ_γ=1.67951, $\phi_{p'}$=0.340146, ϕ_γ=6.24478
E_γ=0.100499, $\theta_{\ell'}$=1.53442, $\phi_{\ell'}$=3.11178
Q^2=0.0391204, x_B=0.172049, t=-0.0311939, W^2=1.0681, y=0.539596

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1847 p_x= 0.0000 p_y= 0.0000 p_z= -0.1847
P E= 0.9560 p_x= 0.0000 p_y= 0.0000 p_z= 0.1847
ℓ' E= 0.1022 p_x= -0.1021 p_y= 0.0030 p_z= 0.0037
P' E= 0.9380 p_x= 0.0022 p_y= 0.0008 p_z= 0.0072
q_γ E= 0.1005 p_x= 0.0998 p_y= -0.0038 p_z= -0.0109

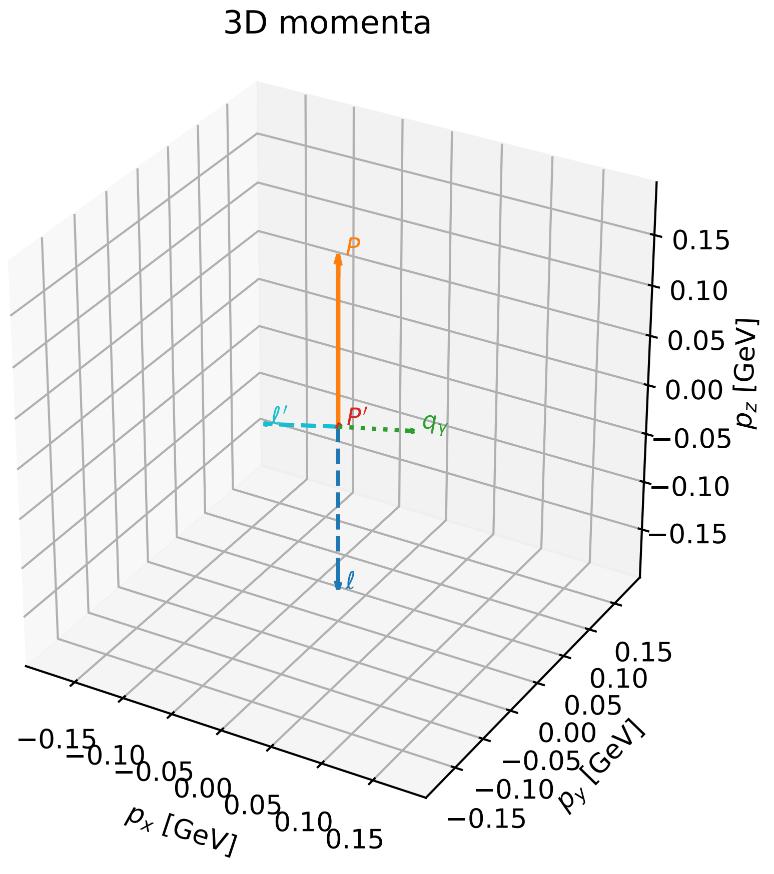




electron: polarization cluster P2, local minimum 897; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



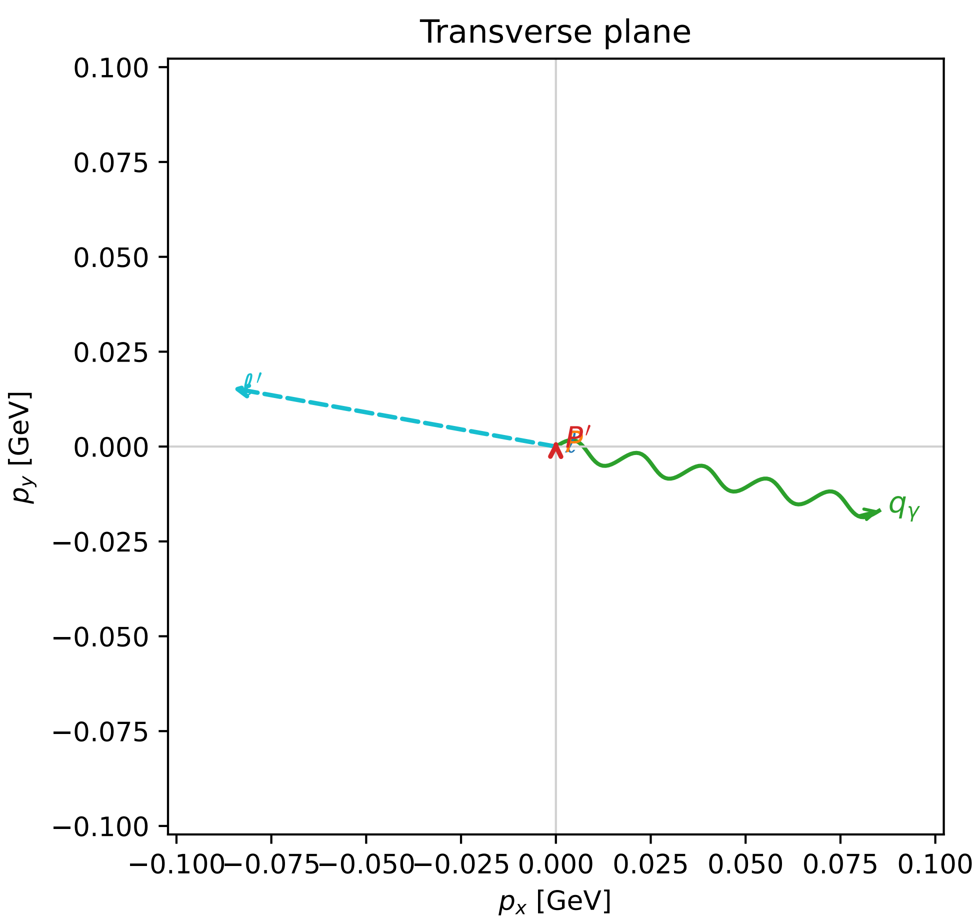
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

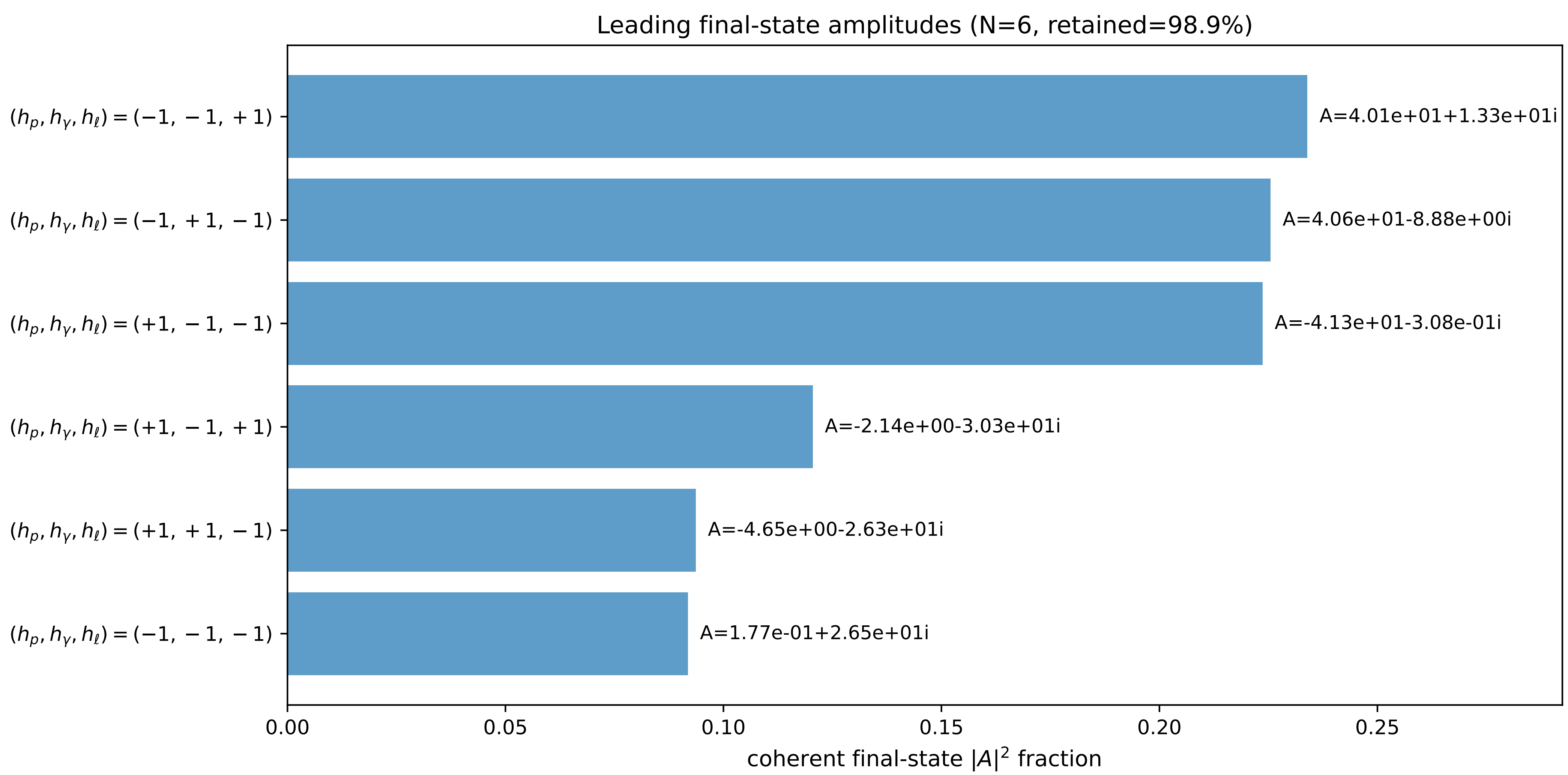


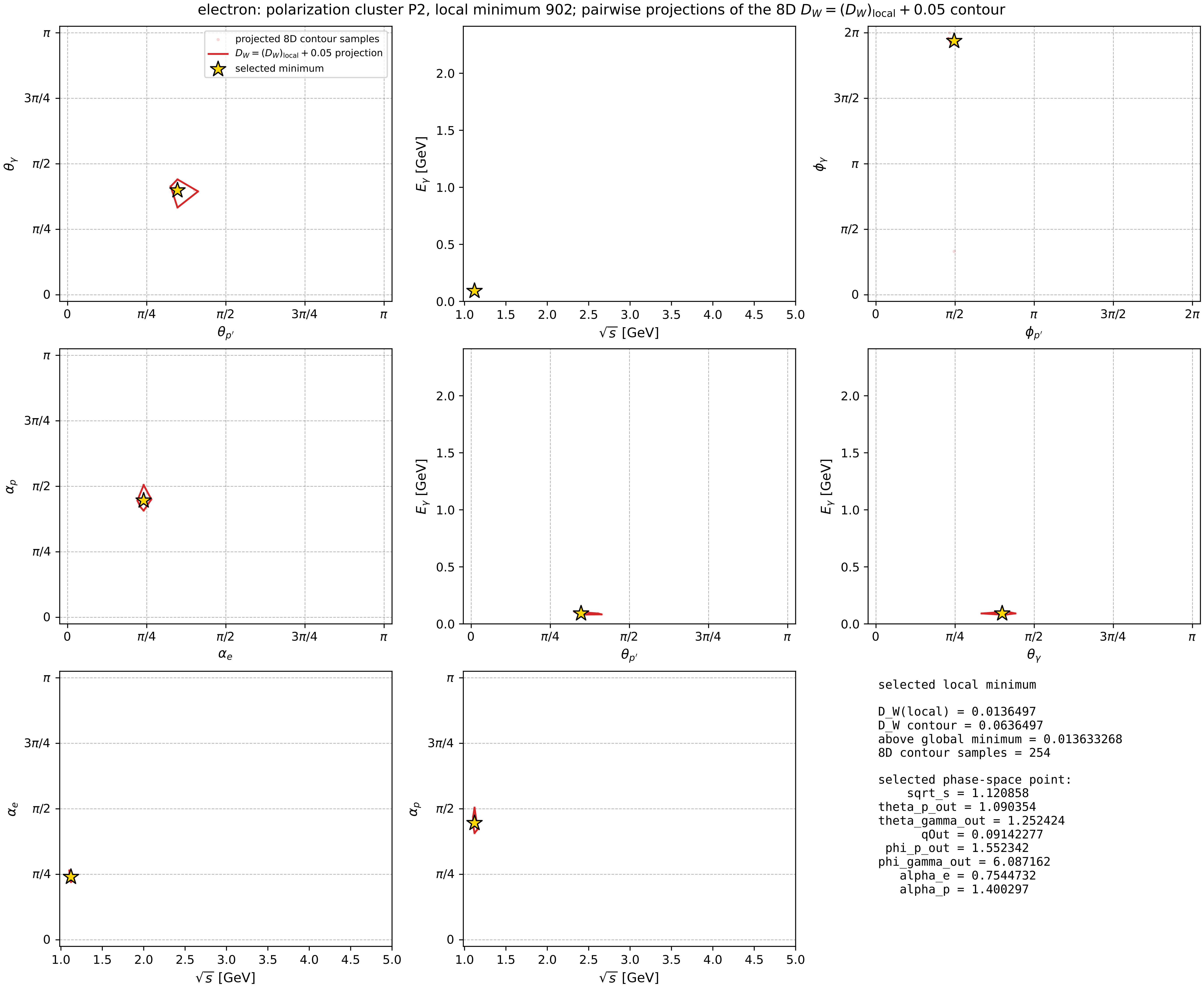
dw_mixing_angles_polarization_02_local_minimum_902 D_W=0.0136497
region: polarization_cluster_02_local_minimum_902
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7544732 rad
incoming lepton: 0.728632|+> +0.684905|->
proton mixing angle: alpha_p=1.4002975 rad
incoming proton: 0.169674|+> +0.9855|->

s=1.25632, $\sqrt{s}$=1.12086, $|\vec{P}|$=0.167941, $|\vec{P}'|$=0.00172684
$\theta_{p'}$=1.09035, θ_γ=1.25242, $\phi_{p'}$=1.55234, ϕ_γ=6.08716
E_γ=0.0914228, $\theta_{\ell'}$=1.89834, $\phi_{\ell'}$=2.96298
Q^2=0.0208304, x_B=0.108299, t=-0.0277168, W^2=1.05135, y=0.510896

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1679 p_x= 0.0000 p_y= 0.0000 p_z= -0.1679
P E= 0.9529 p_x= 0.0000 p_y= 0.0000 p_z= 0.1679
ℓ' E= 0.0914 p_x= -0.0852 p_y= 0.0154 p_z= -0.0294
P' E= 0.9380 p_x= 0.0000 p_y= 0.0015 p_z= 0.0008
q_γ E= 0.0914 p_x= 0.0852 p_y= -0.0169 p_z= 0.0286

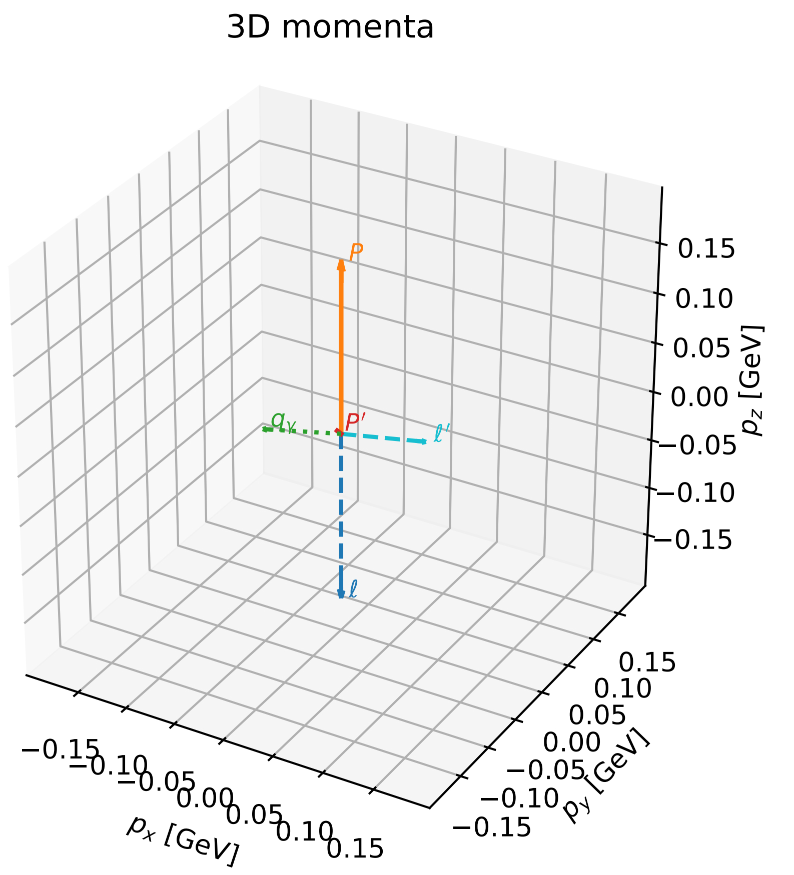






electron: polarization cluster P2, local minimum 902; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0136497
D_W contour = 0.0636497
above global minimum = 0.013633268
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.120858
theta_p_out = 1.090354
theta_gamma_out = 1.252424
qOut = 0.09142277
phi_p_out = 1.552342
phi_gamma_out = 6.087162
alpha_e = 0.7544732
alpha_p = 1.400297

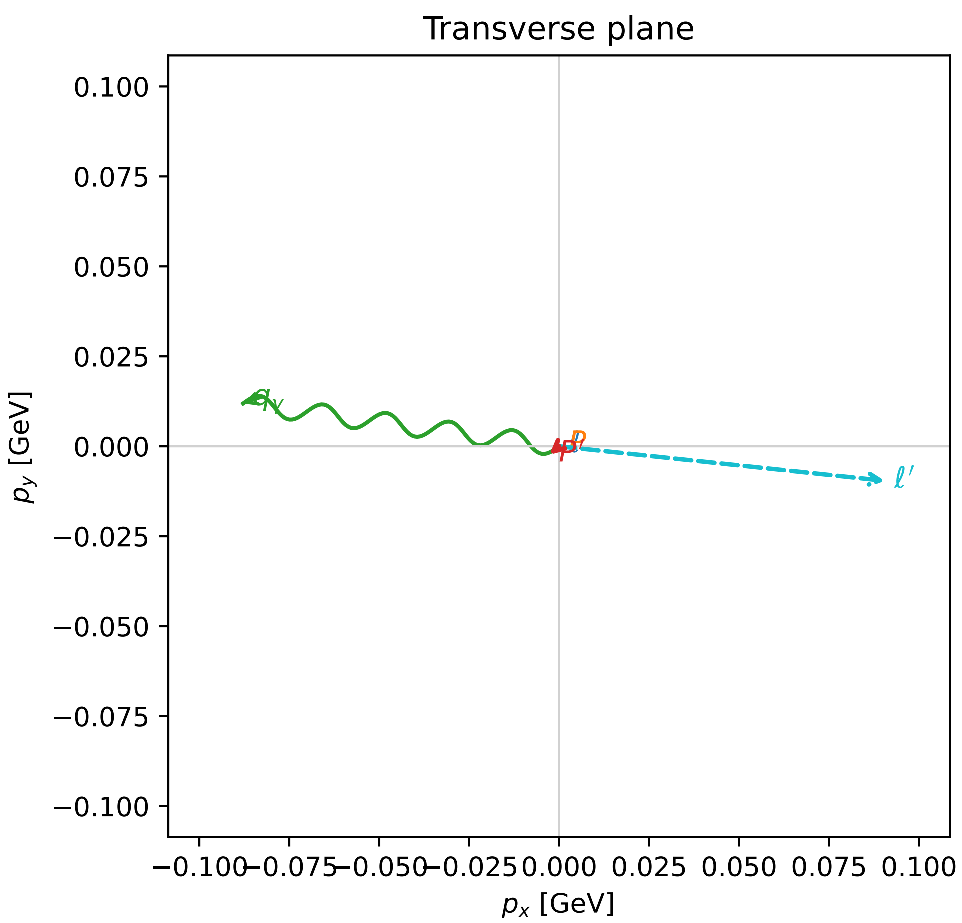
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

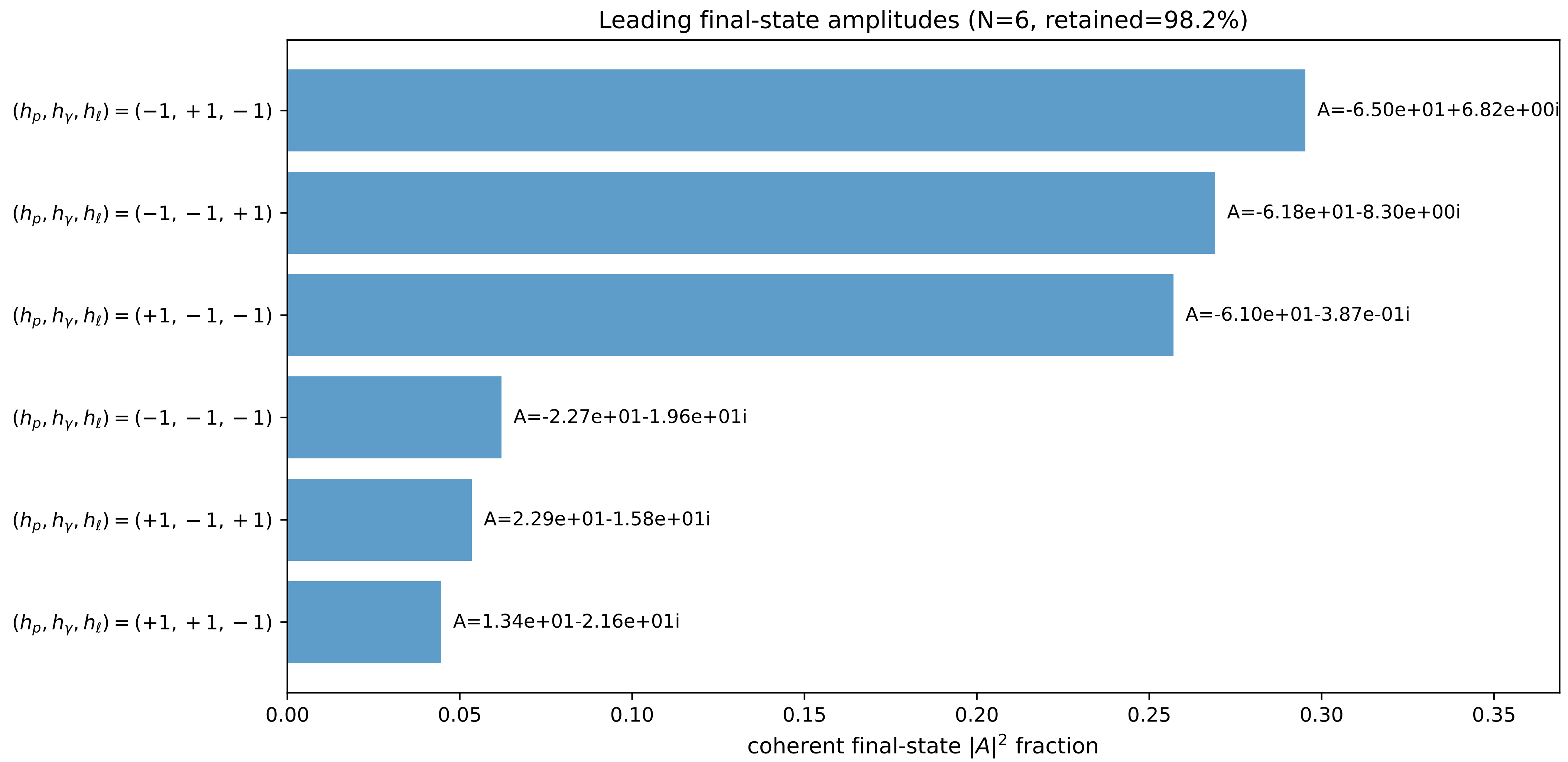


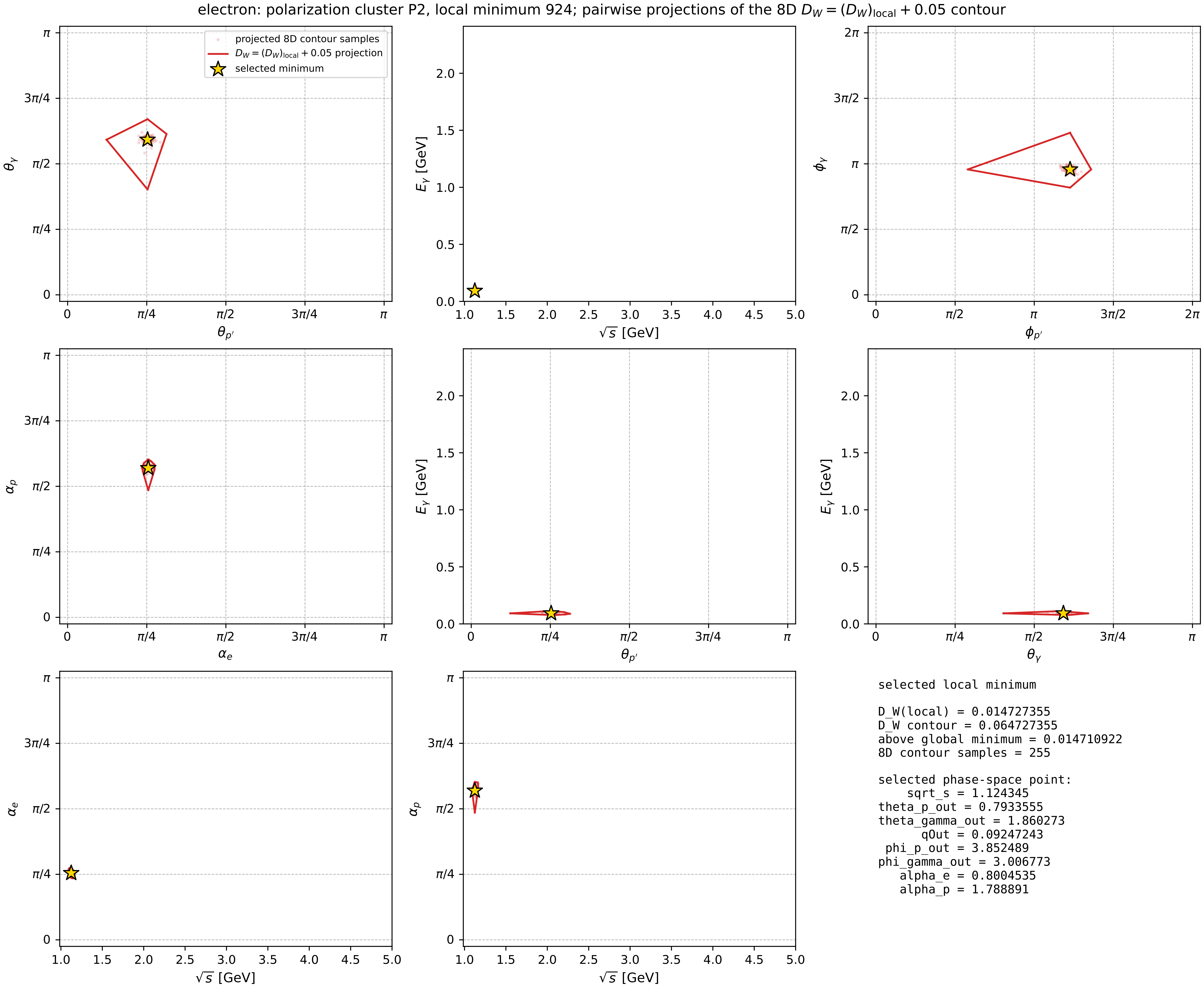
dw_mixing_angles_polarization_02_local_minimum_924 D_W=0.0147274
region: polarization_cluster_02_local_minimum_924
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80045347 rad
incoming lepton: 0.696381|+> +0.717672|->
proton mixing angle: alpha_p=1.7888906 rad
incoming proton: -0.216369|+> +0.976312|->

s=1.26415, $\sqrt{s}$=1.12435, $|\vec{P}|$=0.170902, $|\vec{P}'|$=0.00499047
$\theta_{p'}$=0.793356, θ_γ=1.86027, $\phi_{p'}$=3.85249, ϕ_γ=3.00677
E_γ=0.0924724, $\theta_{\ell'}$=1.32437, $\phi_{\ell'}$=6.17762
Q^2=0.0399071, x_B=0.187222, t=-0.027798, W^2=1.05309, y=0.554645

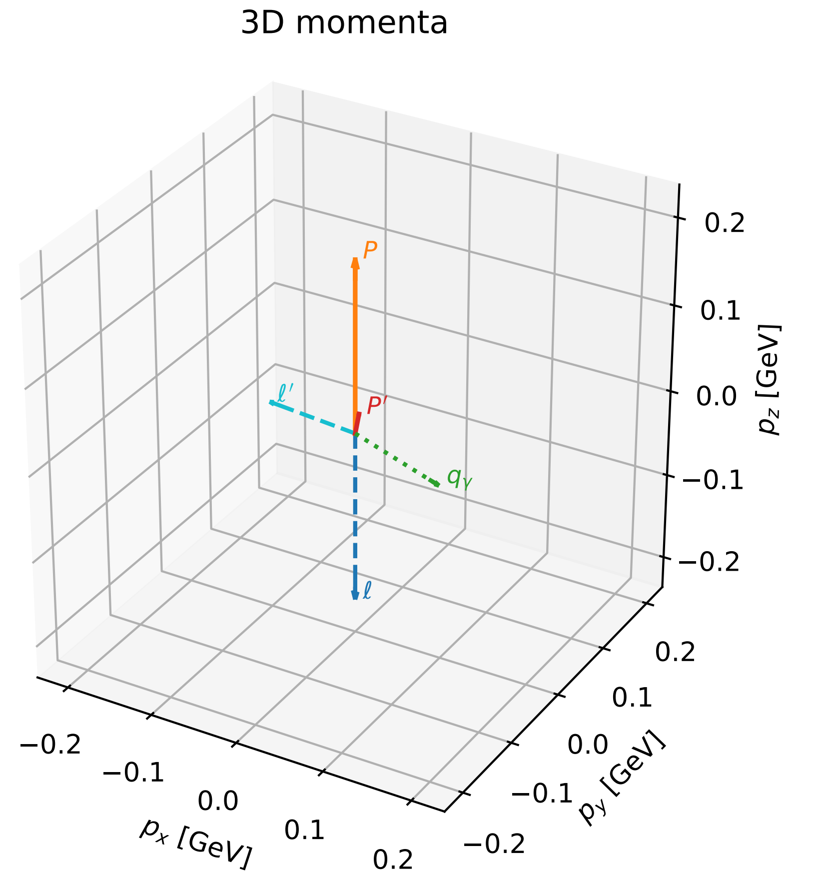
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1709 p_x= 0.0000 p_y= 0.0000 p_z= -0.1709
P E= 0.9534 p_x= 0.0000 p_y= 0.0000 p_z= 0.1709
ℓ' E= 0.0939 p_x= 0.0905 p_y= -0.0096 p_z= 0.0229
P' E= 0.9380 p_x= -0.0027 p_y= -0.0023 p_z= 0.0035
q_γ E= 0.0925 p_x= -0.0878 p_y= 0.0119 p_z= -0.0264






electron: polarization cluster P2, local minimum 924; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
E_γ [GeV]
$\theta_{p'}$
E_γ [GeV]
θ_γ
α_e
$\sqrt{s}$ [GeV]
α_p
$\sqrt{s}$ [GeV]
selected local minimum
D_W(local) = 0.014727355
D_W contour = 0.064727355
above global minimum = 0.014710922
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.124345
theta_p_out = 0.7933555
theta_gamma_out = 1.860273
qOut = 0.09247243
phi_p_out = 3.852489
phi_gamma_out = 3.006773
alpha_e = 0.8004535
alpha_p = 1.788891

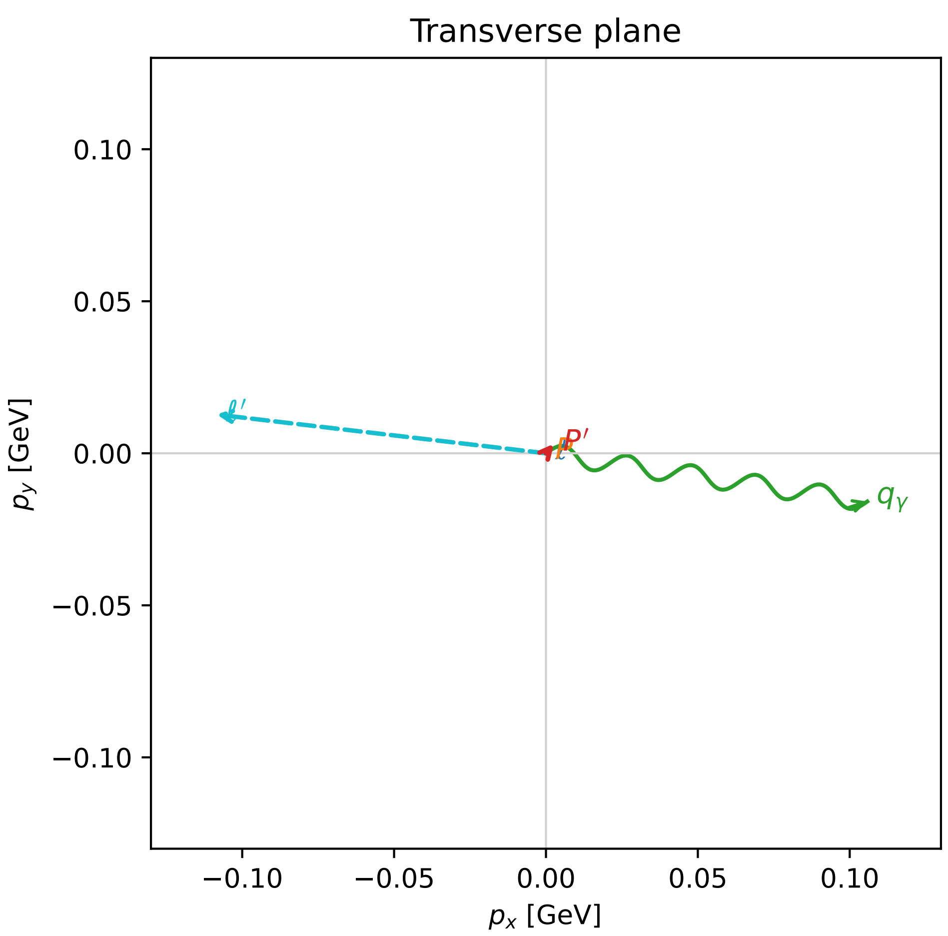
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

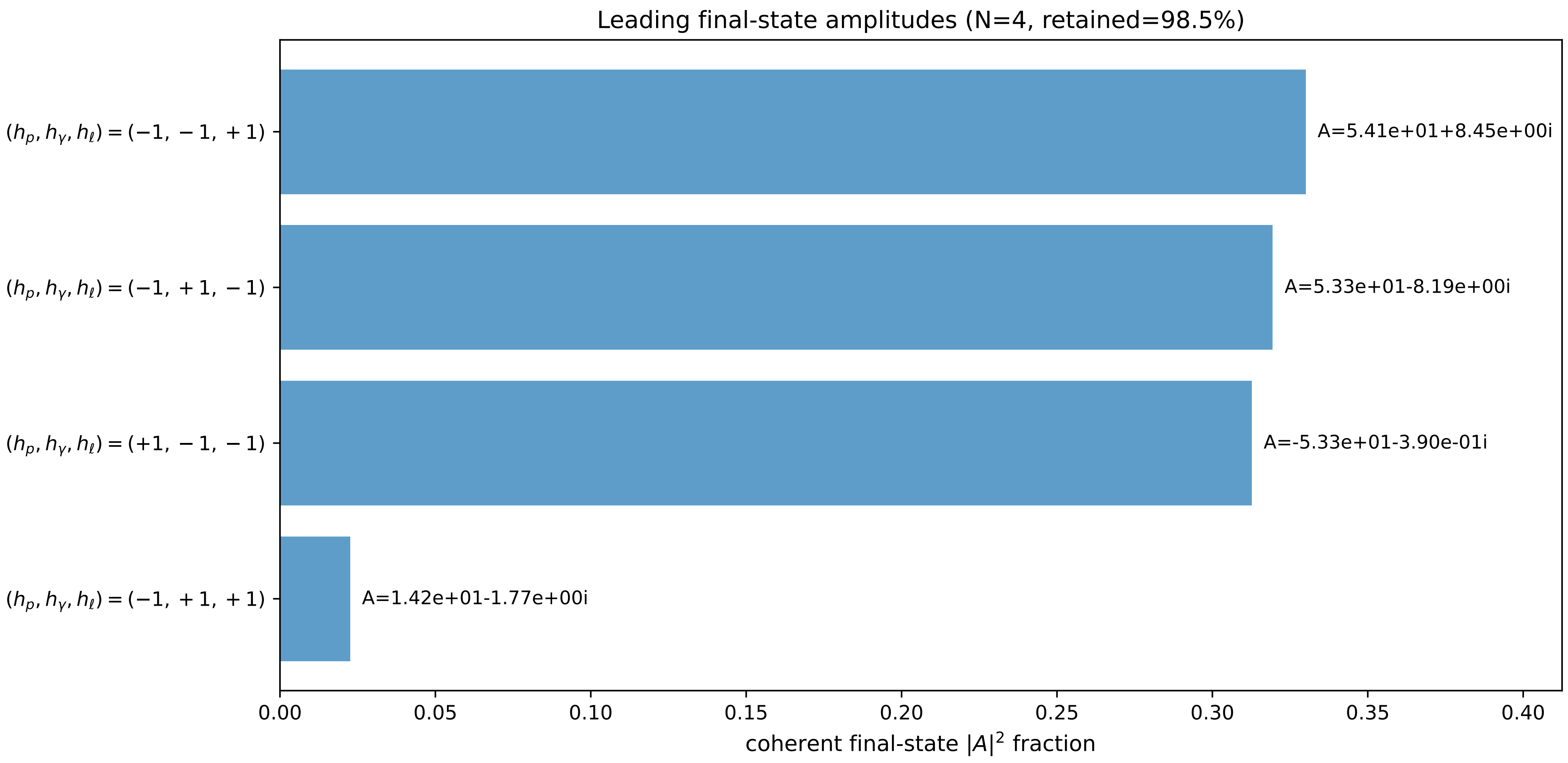


dw_mixing_angles_polarization_02_local_minimum_926 D_W=0.0148006
region: polarization_cluster_02_local_minimum_926
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77651717 rad
incoming lepton: 0.713359|+> +0.700799|->
proton mixing angle: alpha_p=1.340248 rad
incoming proton: 0.228511|+> +0.973541|->

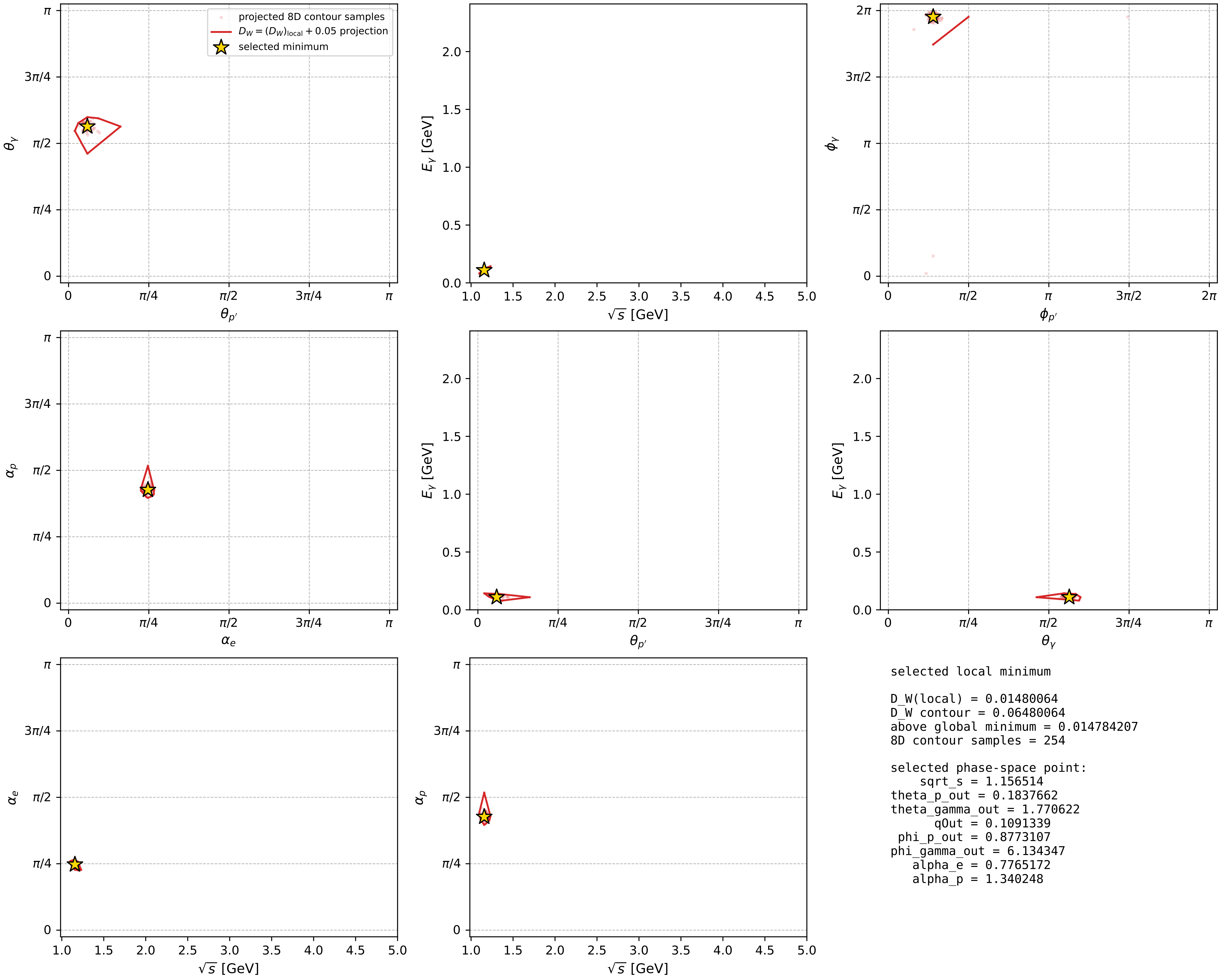
s=1.33753, $\sqrt{s}$=1.15651, $|\vec{P}|$=0.197871, $|\vec{P}'|$=0.0221792
$\theta_{p'}$=0.183766, θ_γ=1.77062, $\phi_{p'}$=0.877311, ϕ_γ=6.13435
E_γ=0.109134, $\theta_{\ell'}$=1.57211, $\phi_{\ell'}$=3.02453
Q^2=0.0431258, x_B=0.173605, t=-0.0305999, W^2=1.08513, y=0.542765

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1979 p_x= 0.0000 p_y= 0.0000 p_z= -0.1979
P E= 0.9586 p_x= 0.0000 p_y= 0.0000 p_z= 0.1979
ℓ' E= 0.1091 p_x= -0.1084 p_y= 0.0127 p_z= -0.0001
P' E= 0.9383 p_x= 0.0026 p_y= 0.0031 p_z= 0.0218
q_γ E= 0.1091 p_x= 0.1058 p_y= -0.0159 p_z= -0.0217

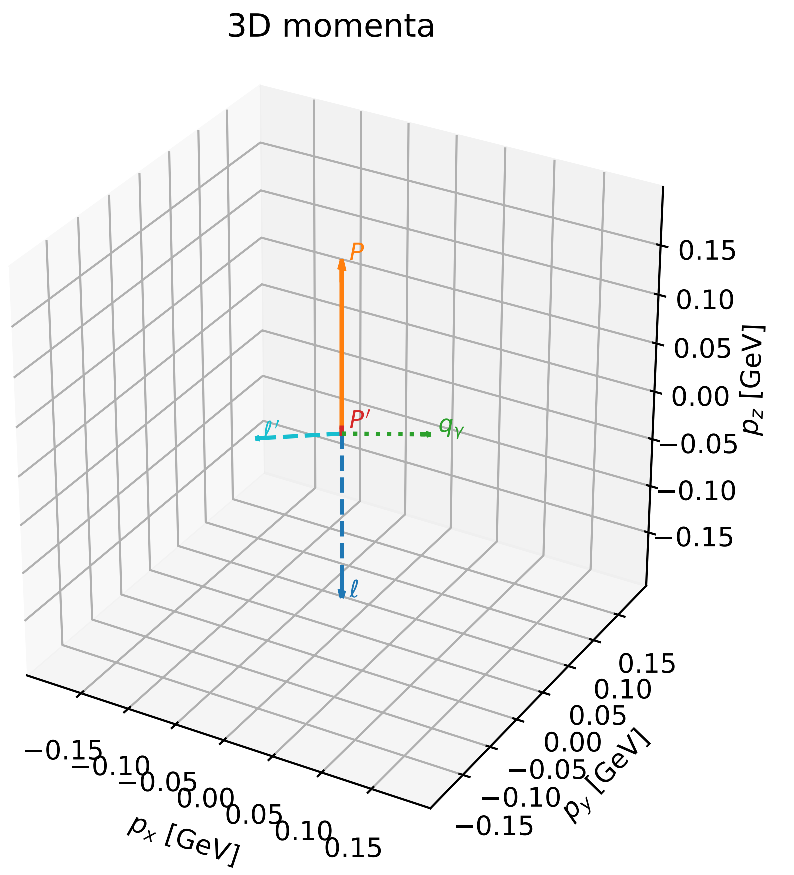




electron: polarization cluster P2, local minimum 926; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



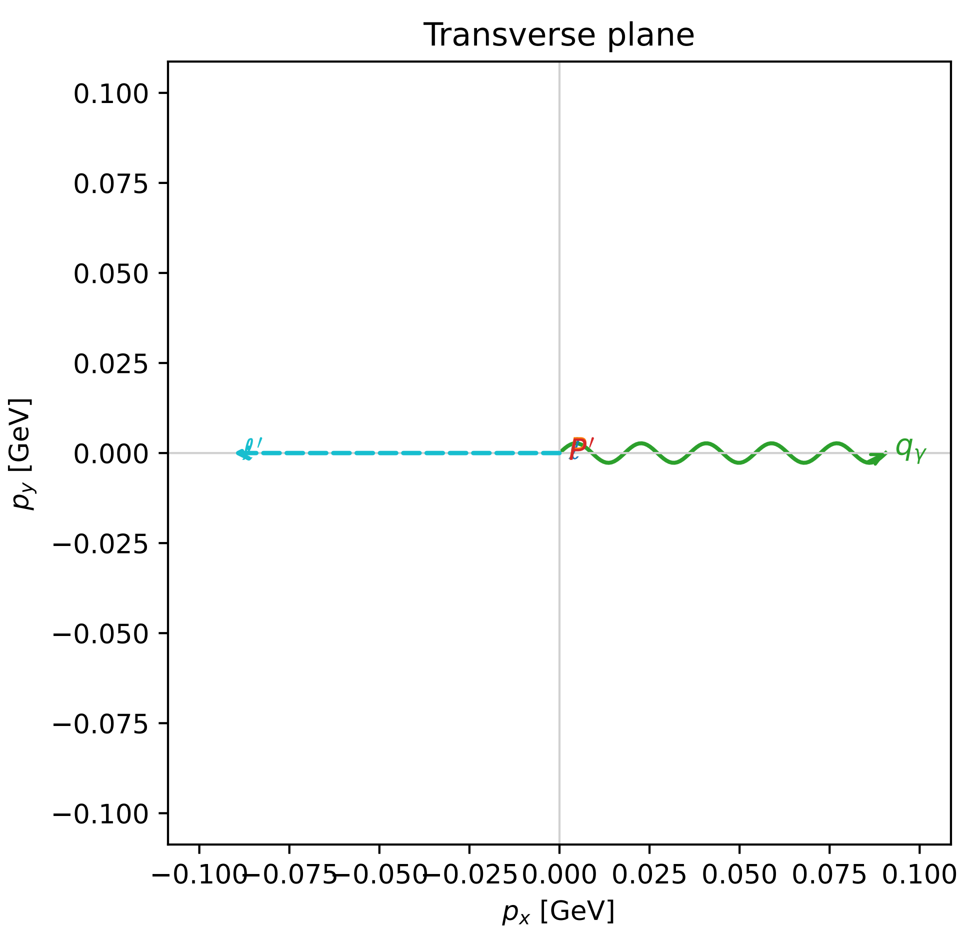
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

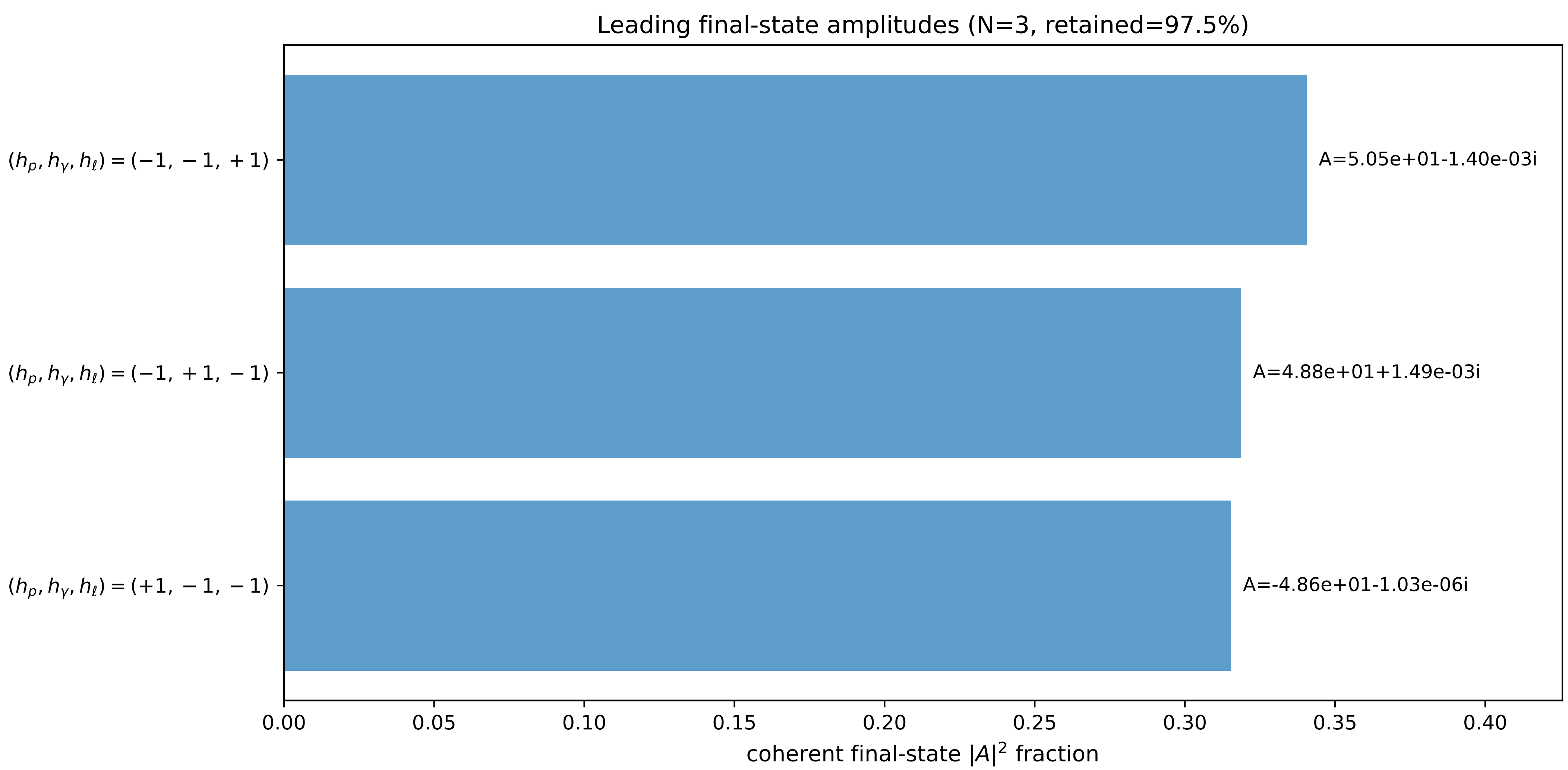



```
dw_mixing_angles_polarization_02_local_minimum_928  D_W=0.0148606
region: polarization_cluster_02_local_minimum_928
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75409916 rad
incoming lepton: 0.728889|+> +0.684632|->
proton mixing angle: alpha_p=1.4076007 rad
incoming proton: 0.162472|+> +0.986713|->
```

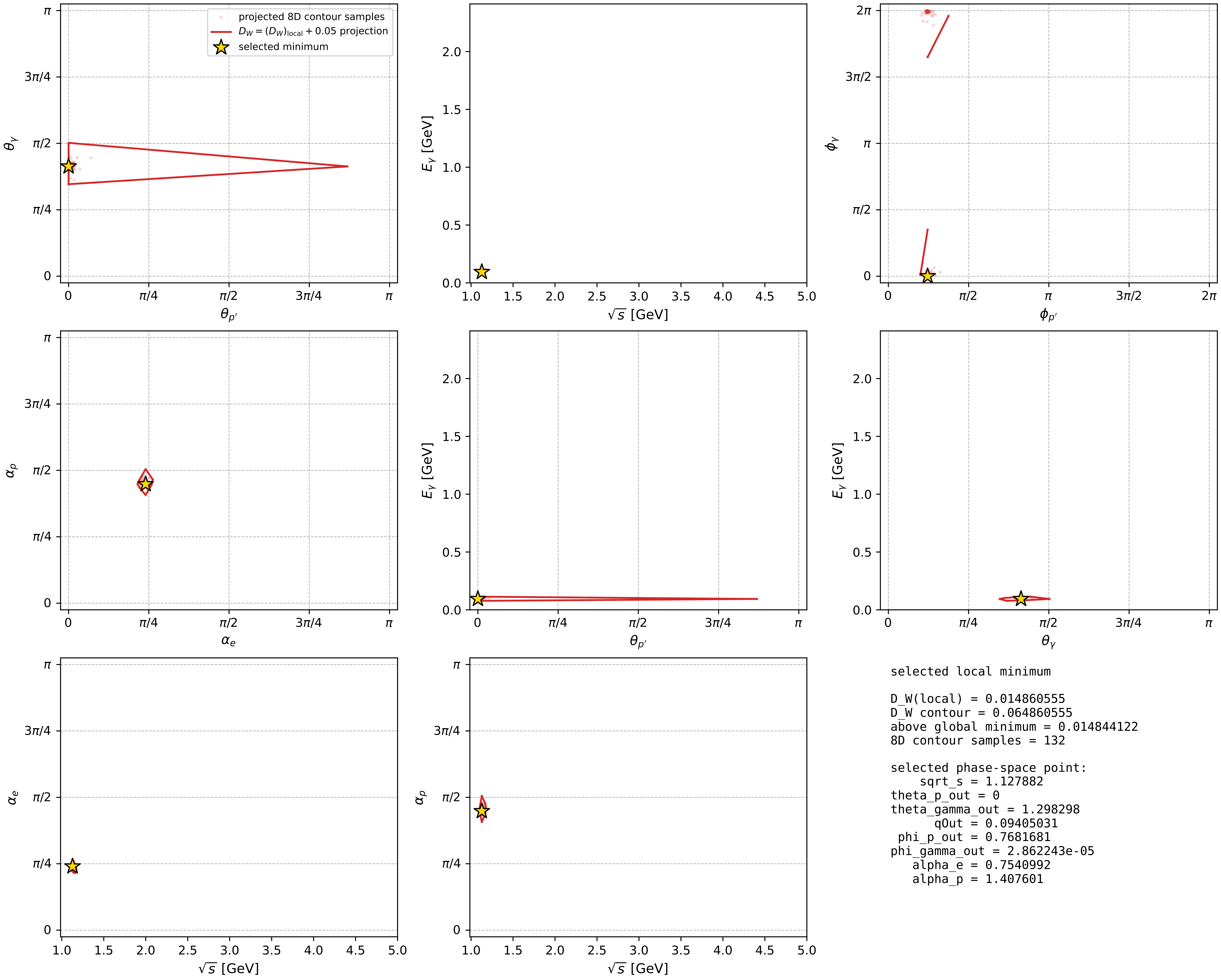
s=1.27212, $\sqrt{s}$=1.12788, $|\vec{P}|$=0.173898, $|\vec{P}'|$=0.00591441
$\theta_{p'}$=0, θ_γ=1.2983, $\phi_{p'}$=0.768168, ϕ_γ=2.86224e-05
E_γ=0.0940503, $\theta_{\ell'}$=1.90278, $\phi_{\ell'}$=3.14162
Q^2=0.0224623, x_B=0.113101, t=-0.0279635, W^2=1.05599, y=0.506292

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1739 p_x= 0.0000 p_y= 0.0000 p_z= -0.1739
P E= 0.9540 p_x= 0.0000 p_y= 0.0000 p_z= 0.1739
ℓ' E= 0.0958 p_x= -0.0906 p_y= -0.0000 p_z= -0.0312
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0059
q_γ E= 0.0941 p_x= 0.0906 p_y= 0.0000 p_z= 0.0253

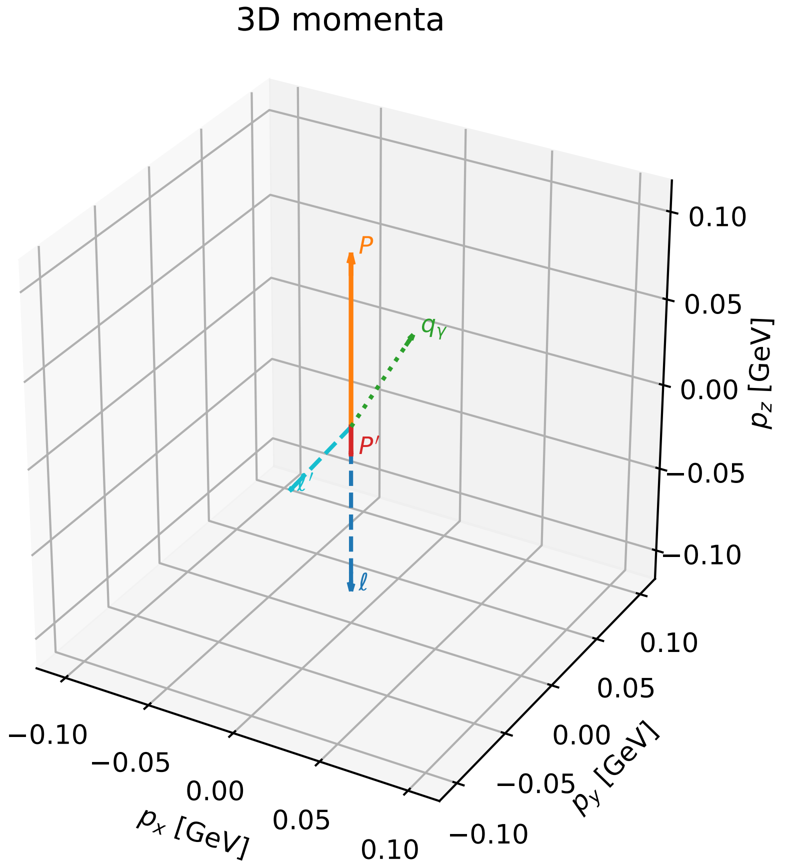




electron: polarization cluster P2, local minimum 928; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




```
dw_mixing_angles_polarization_02_local_minimum_957  D_W=0.0164699
region: polarization_cluster_02_local_minimum_957
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75489786 rad
incoming lepton: 0.728342|+> +0.685214|->
proton mixing angle: alpha_p=1.4239829 rad
incoming proton: 0.146287|+> +0.989242|->
```

s=1.08428, $\sqrt{s}$=1.04129, $|\vec{P}|$=0.0981618, $|\vec{P}'|$=0.0159796

$\theta_{p'}$=3.14159, θ_γ=0.612408, $\phi_{p'}$=1.11358, ϕ_γ=0.862165

E_γ=0.0576426, $\theta_{\ell'}$=2.32592, $\phi_{\ell'}$=4.00376

Q^2=0.00281082, x_B=0.0249914, t=-0.0130034, W^2=0.989505, y=0.550168

four-momenta [E, p_x, p_y, p_z] GeV:

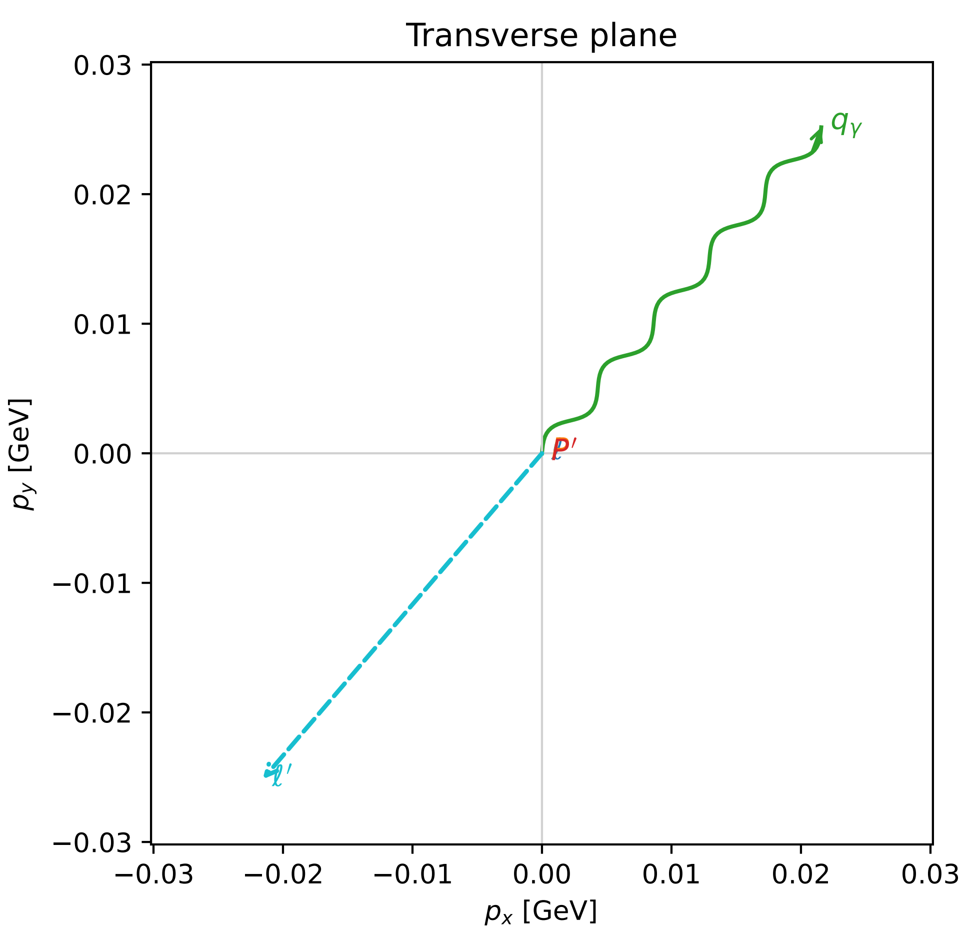
ℓ E= 0.0982 p_x= 0.0000 p_y= 0.0000 p_z= -0.0982

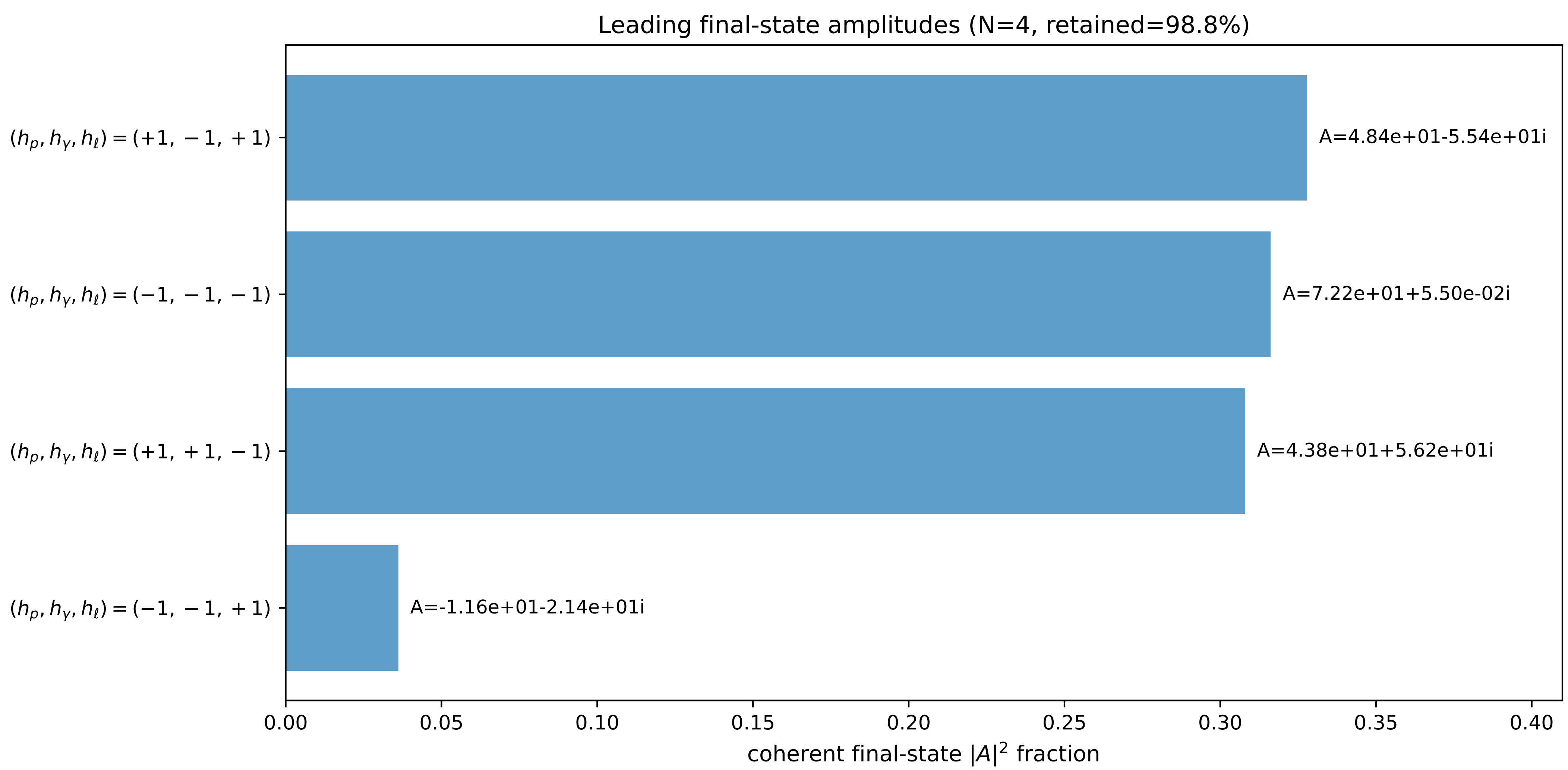
P E= 0.9431 p_x= 0.0000 p_y= 0.0000 p_z= 0.0982

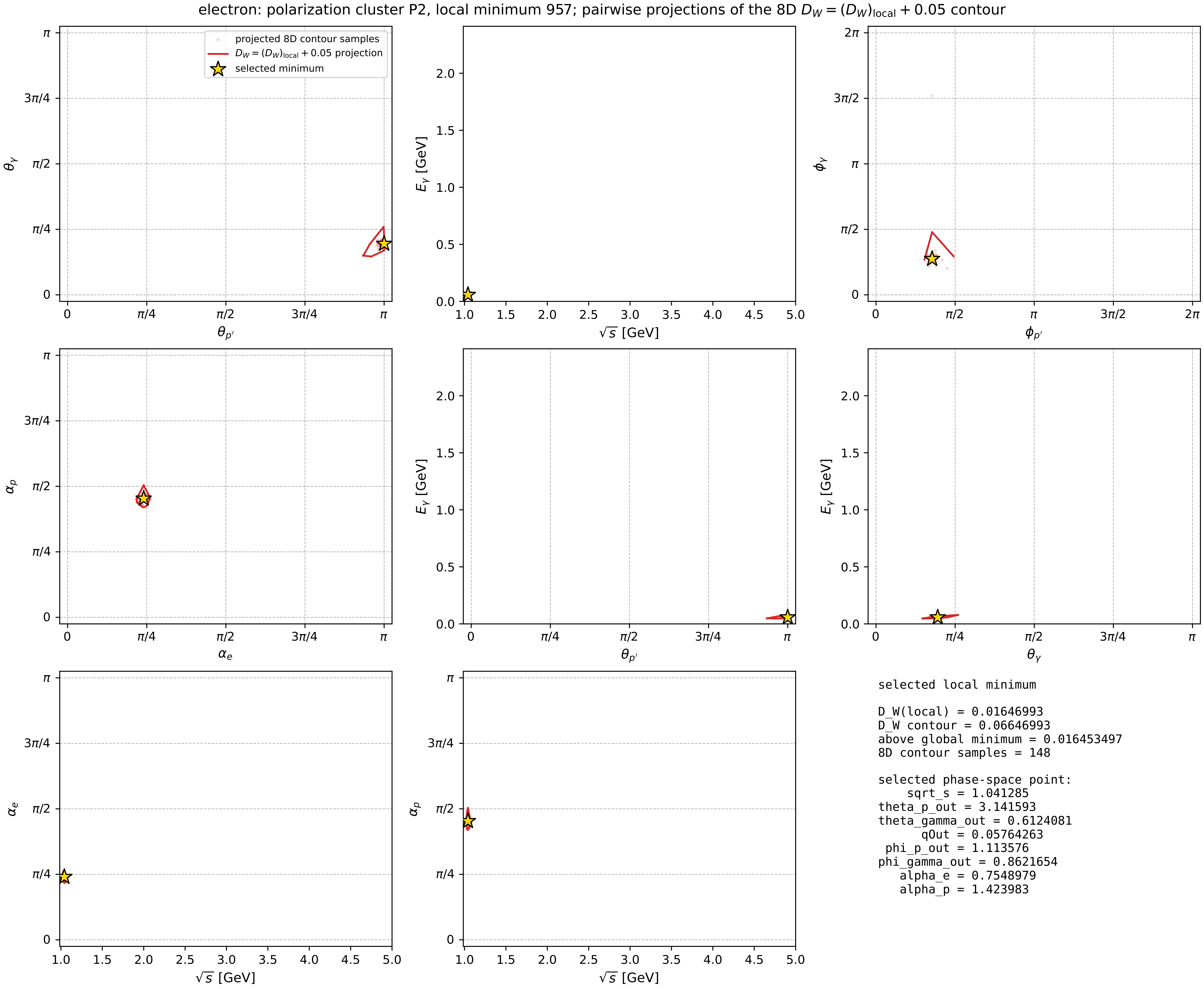
ℓ' E= 0.0455 p_x= -0.0216 p_y= -0.0252 p_z= -0.0312

P' E= 0.9381 p_x= 0.0000 p_y= 0.0000 p_z= -0.0160

q_γ E= 0.0576 p_x= 0.0216 p_y= 0.0252 p_z= 0.0472

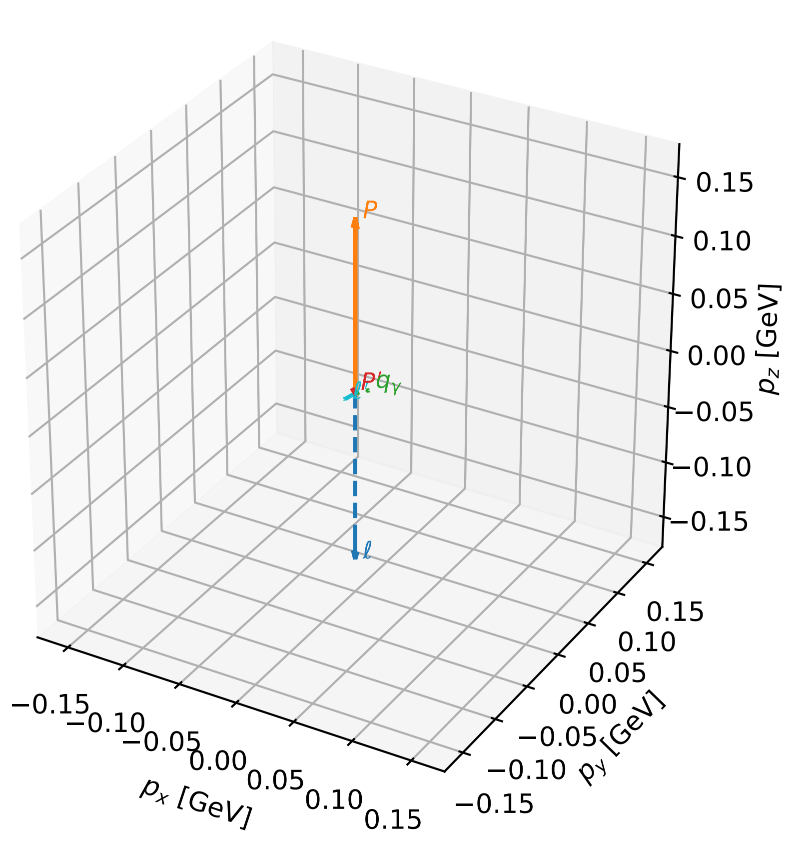





electron: polarization cluster P2, local minimum 957; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
E_γ [GeV]
$\theta_{p'}$
E_γ [GeV]
θ_γ
α_e
$\sqrt{s}$ [GeV]
α_p
$\sqrt{s}$ [GeV]
selected local minimum
D_W(local) = 0.01646993
D_W contour = 0.06646993
above global minimum = 0.016453497
8D contour samples = 148
selected phase-space point:
sqrt_s = 1.041285
theta_p_out = 3.141593
theta_gamma_out = 0.6124081
qOut = 0.05764263
phi_p_out = 1.113576
phi_gamma_out = 0.8621654
alpha_e = 0.7548979
alpha_p = 1.423983

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta



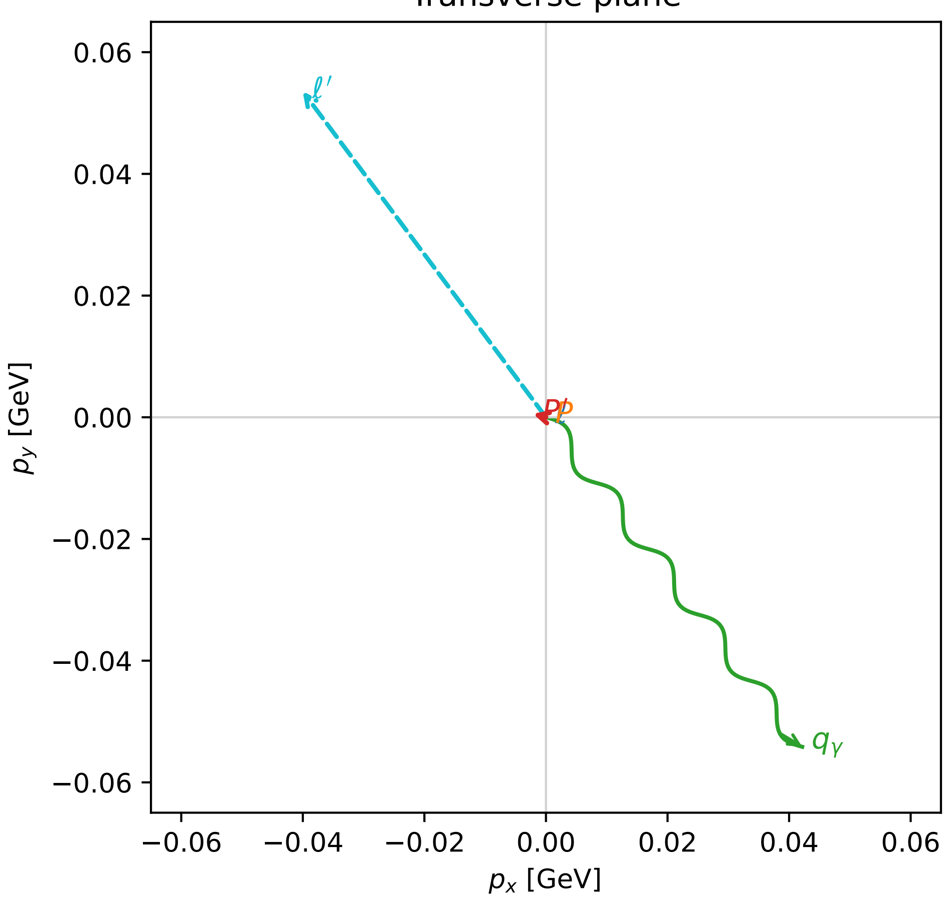

```
dw_mixing_angles_polarization_02_local_minimum_959  D_W=0.0165047
region: polarization_cluster_02_local_minimum_959
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.76584195 rad
incoming lepton: 0.720799|+> +0.693144|->
proton mixing angle: alpha_p=1.4220552 rad
incoming proton: 0.148193|+> +0.988958|->
```

s=1.20583, $\sqrt{s}$=1.0981, $|\vec{P}|$=0.14843, $|\vec{P}'|$=0.00293437
$\theta_{p'}$=0.857981, θ_γ=1.02532, $\phi_{p'}$=2.8769, ϕ_γ=5.37395
E_γ=0.0802828, $\theta_{\ell'}$=2.14828, $\phi_{\ell'}$=2.21241
Q^2=0.0107589, x_B=0.0666374, t=-0.0213344, W^2=1.03054, y=0.49528

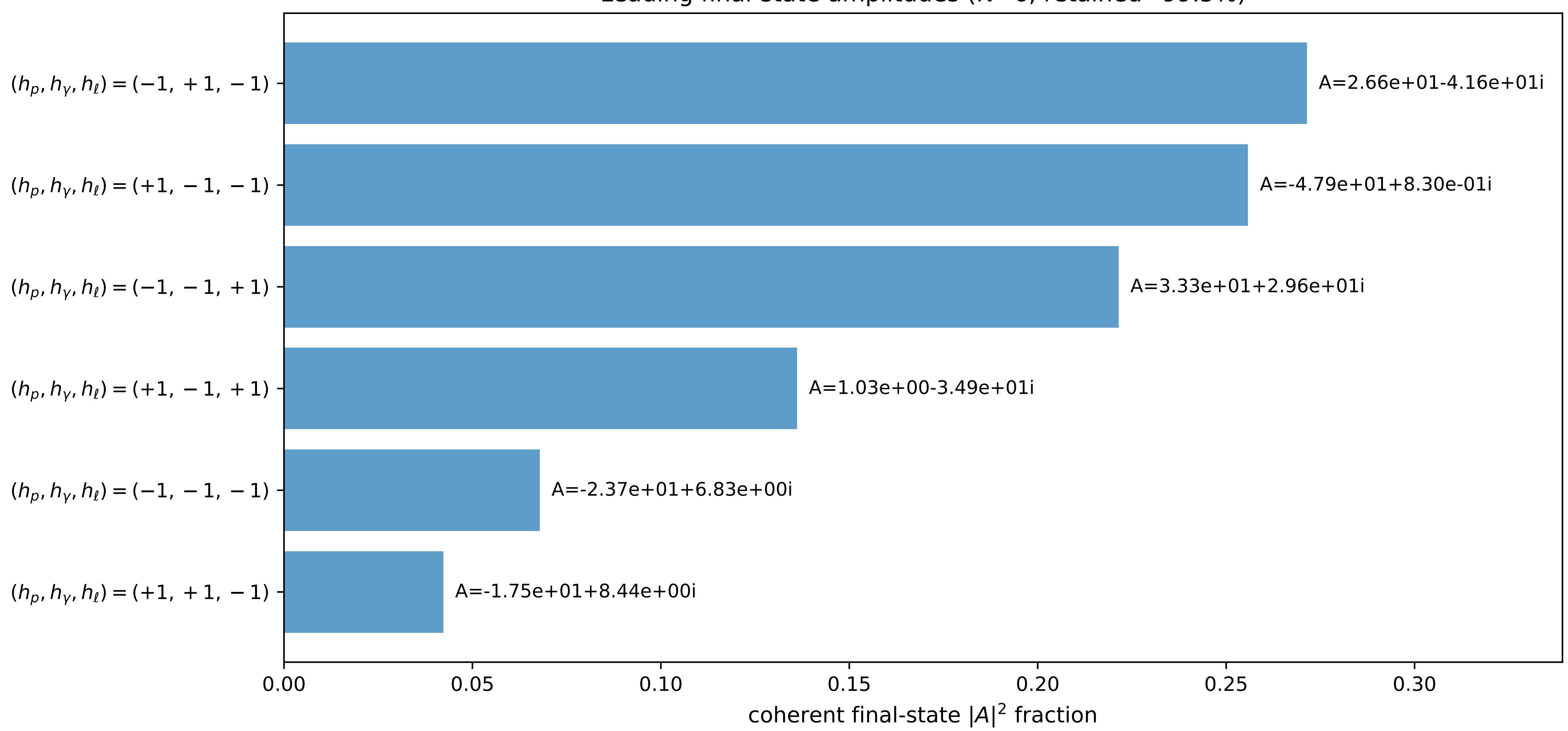
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1484	p_x= 0.0000	p_y= 0.0000	p_z= -0.1484
P	E= 0.9497	p_x= 0.0000	p_y= 0.0000	p_z= 0.1484
ℓ'	E= 0.0798	p_x= -0.0400	p_y= 0.0536	p_z= -0.0436
P'	E= 0.9380	p_x= -0.0021	p_y= 0.0006	p_z= 0.0019
q_γ	E= 0.0803	p_x= 0.0422	p_y= -0.0542	p_z= 0.0417

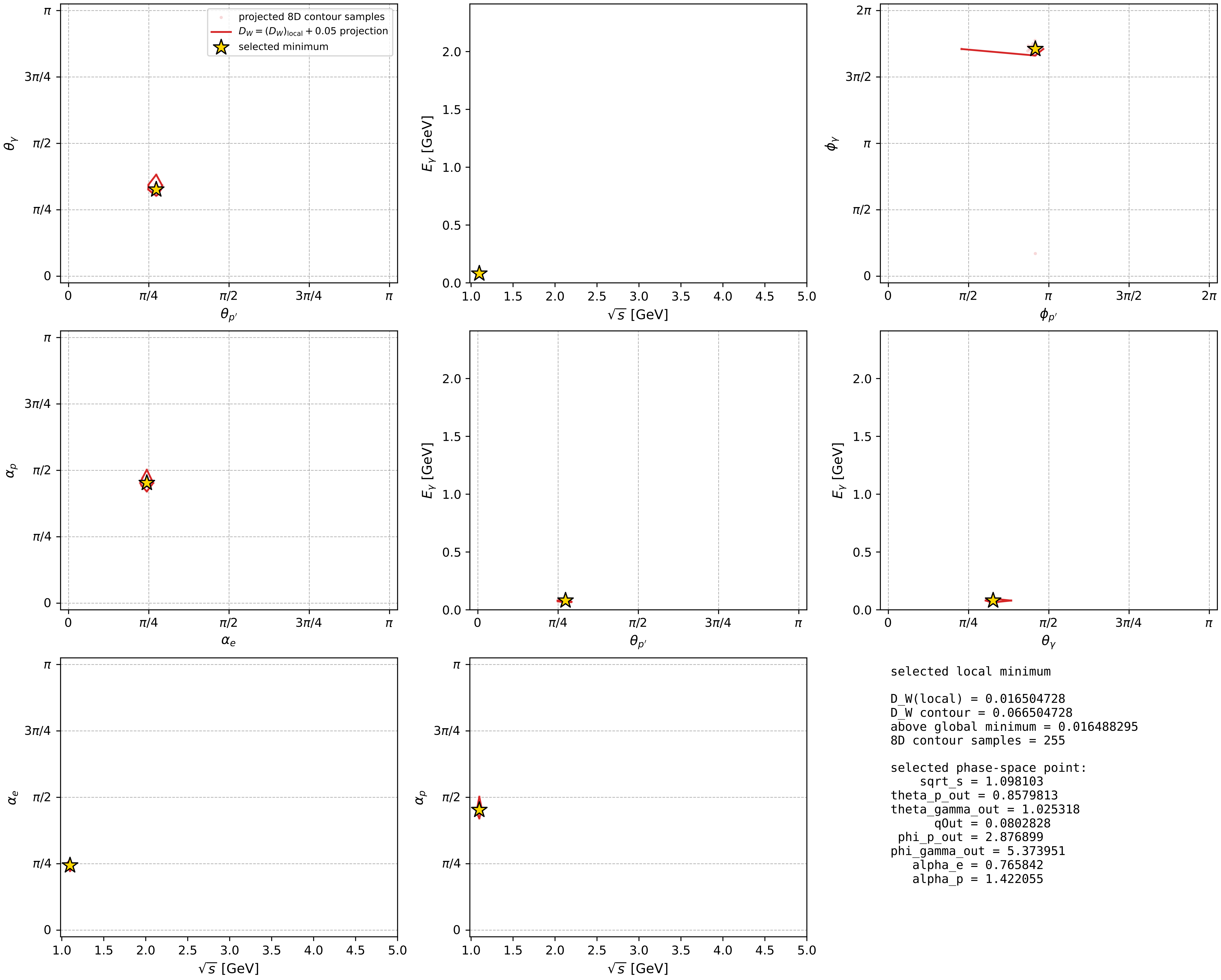
Transverse plane



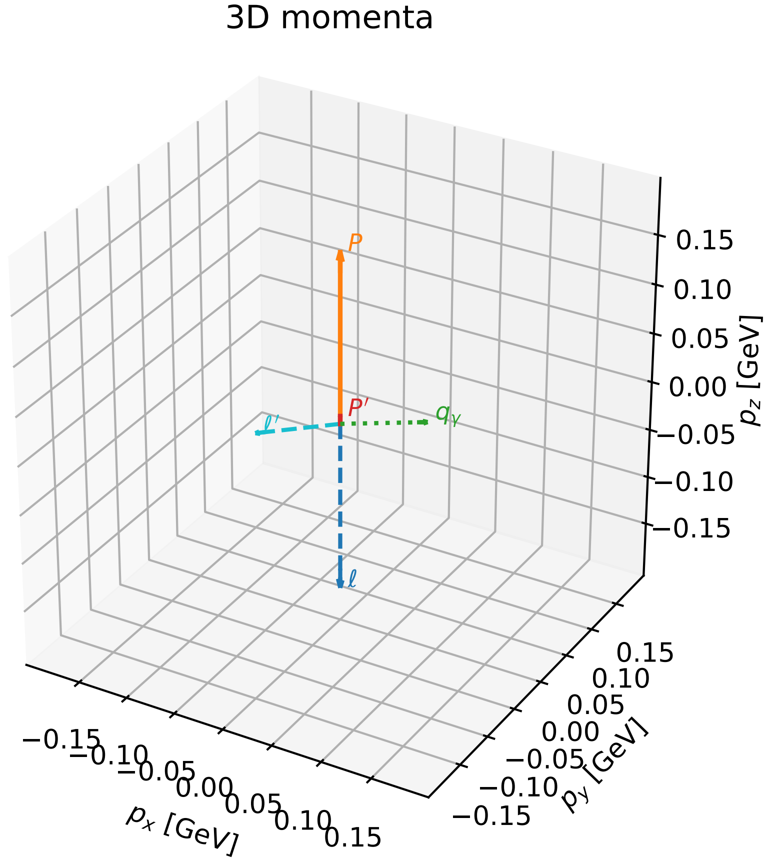
Leading final-state amplitudes (N=6, retained=99.5%)



electron: polarization cluster P2, local minimum 959; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



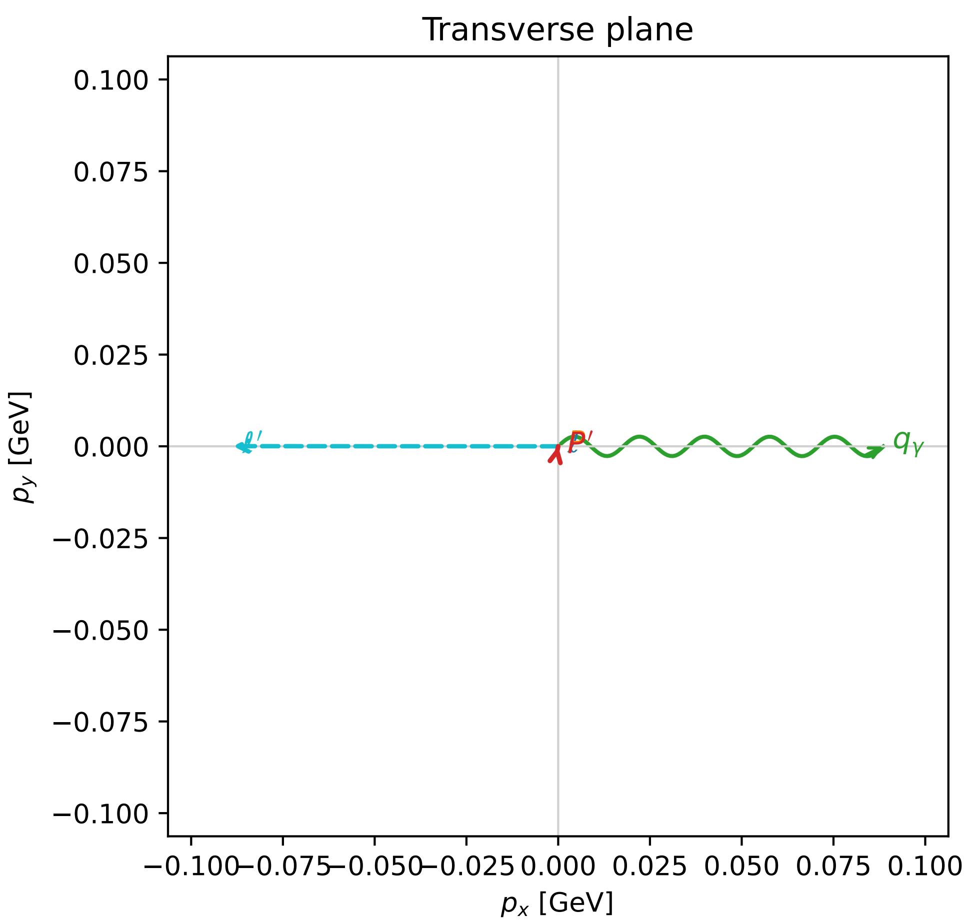
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

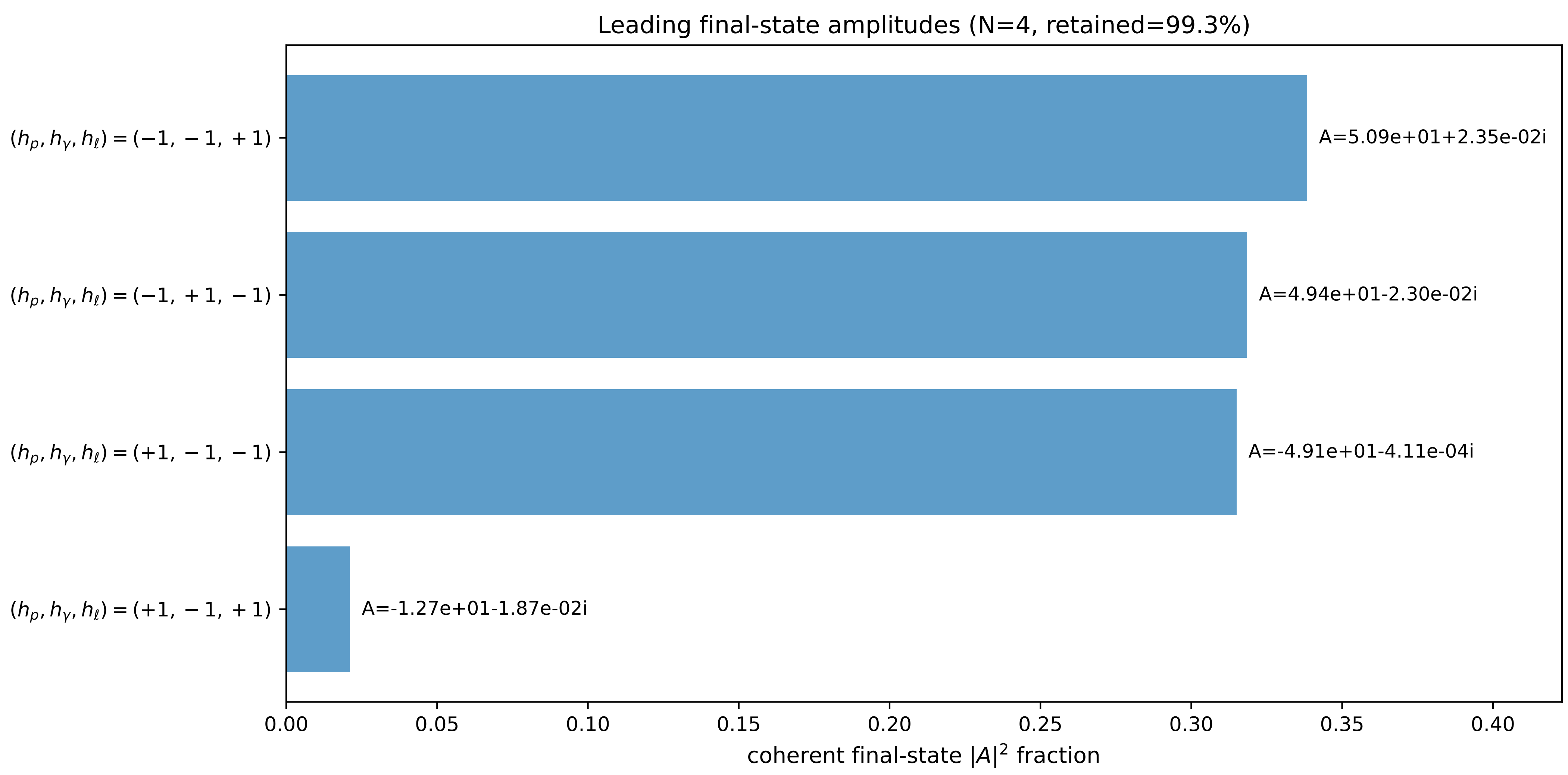


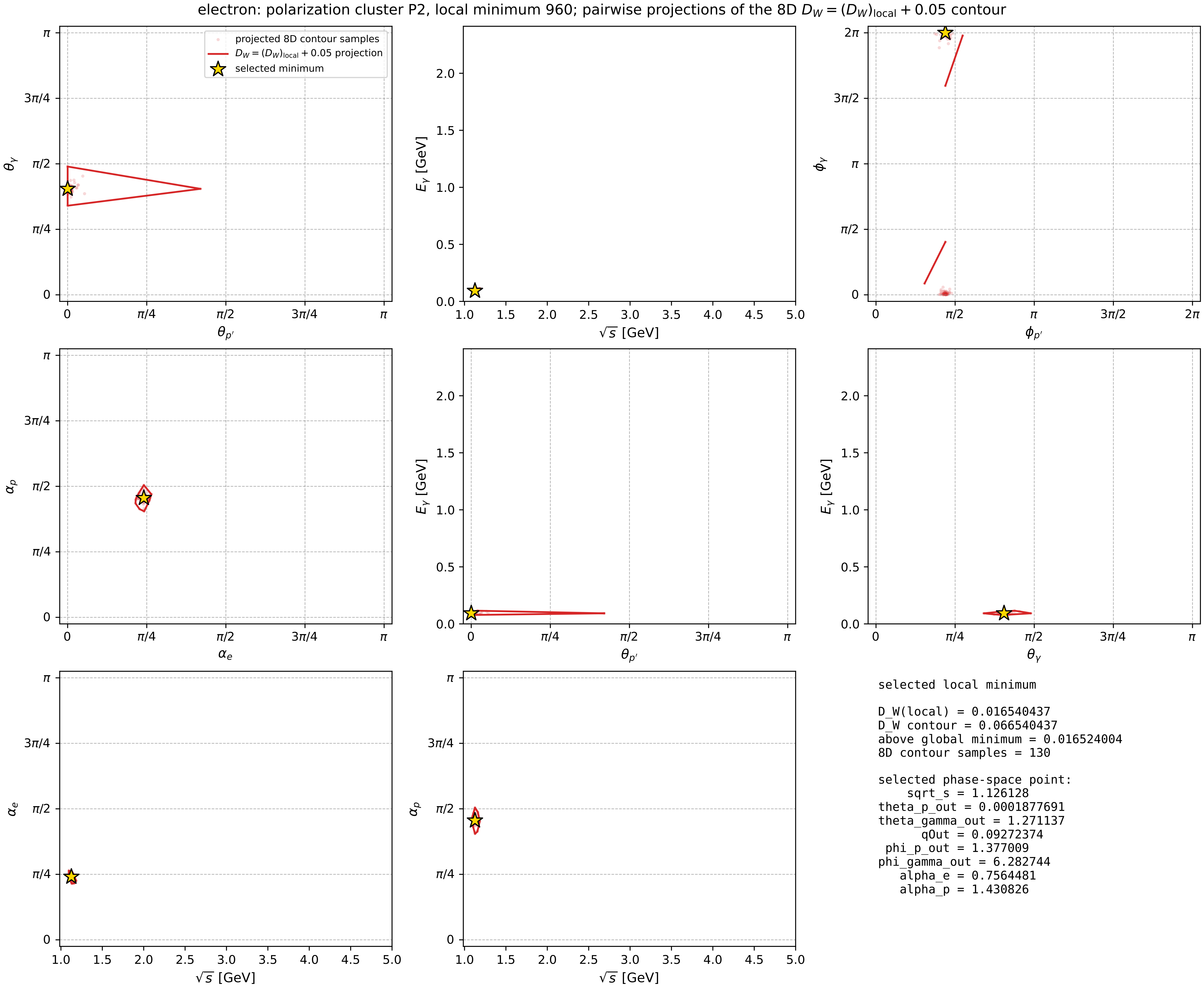

```
dw_mixing_angles_polarization_02_local_minimum_960  D_W=0.0165404
region: polarization_cluster_02_local_minimum_960
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75644806 rad
incoming lepton: 0.727278|+> +0.686343|->
proton mixing angle: alpha_p=1.4308263 rad
incoming proton: 0.139513|+> +0.99022|->
```

s=1.26817, $\sqrt{s}$=1.12613, $|\vec{P}|$=0.172414, $|\vec{P}'|$=0.00793895
$\theta_{p'}$=0.000187769, θ_γ=1.27114, $\phi_{p'}$=1.37701, ϕ_γ=6.28274
E_γ=0.0927237, $\theta_{\ell'}$=1.95007, $\phi_{\ell'}$=3.14117
Q^2=0.0207101, x_B=0.106626, t=-0.026806, W^2=1.05337, y=0.500183

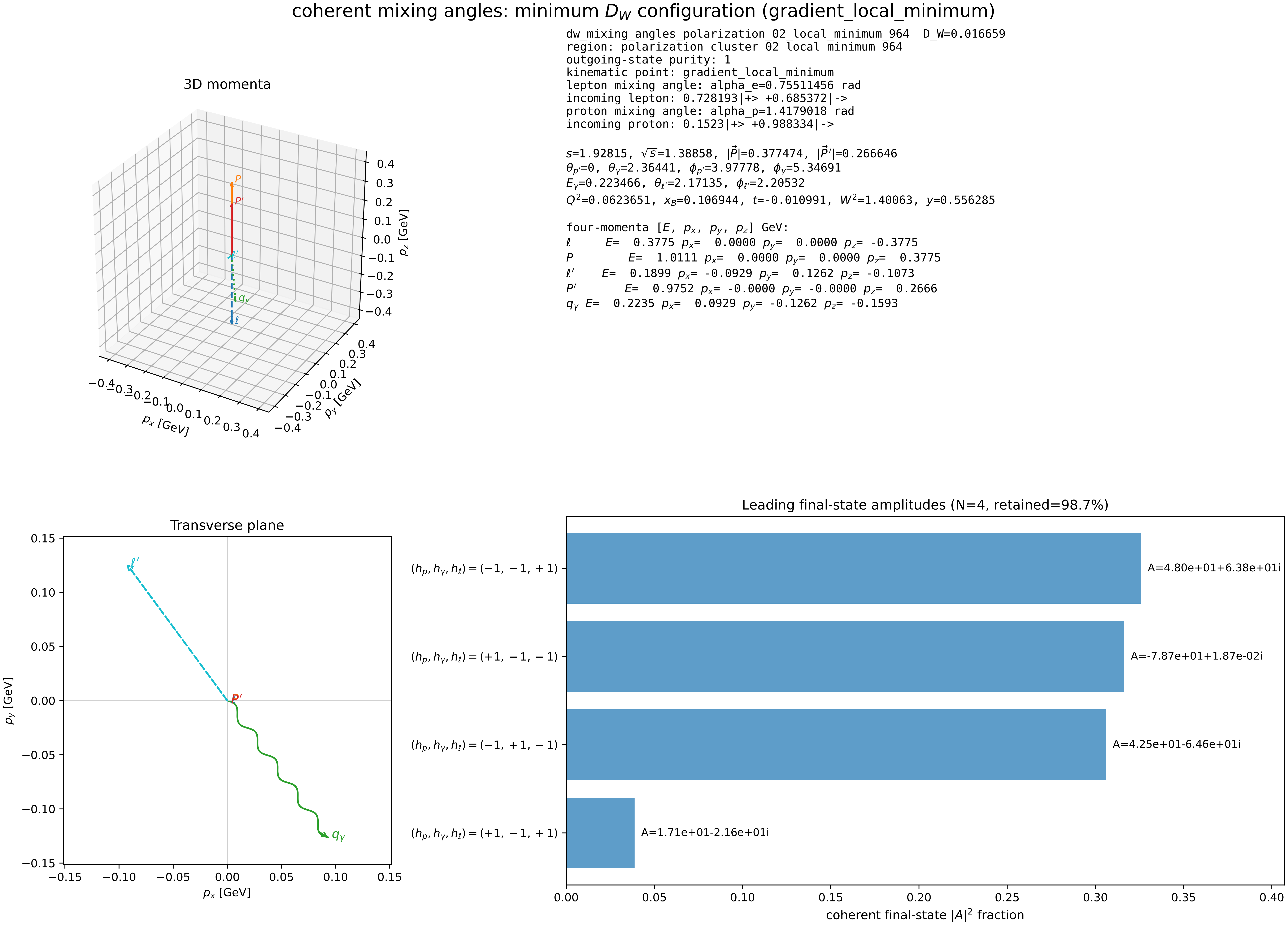
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1724 p_x= 0.0000 p_y= 0.0000 p_z= -0.1724
P E= 0.9537 p_x= 0.0000 p_y= 0.0000 p_z= 0.1724
ℓ' E= 0.0954 p_x= -0.0886 p_y= 0.0000 p_z= -0.0353
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0079
q_γ E= 0.0927 p_x= 0.0886 p_y= -0.0000 p_z= 0.0274



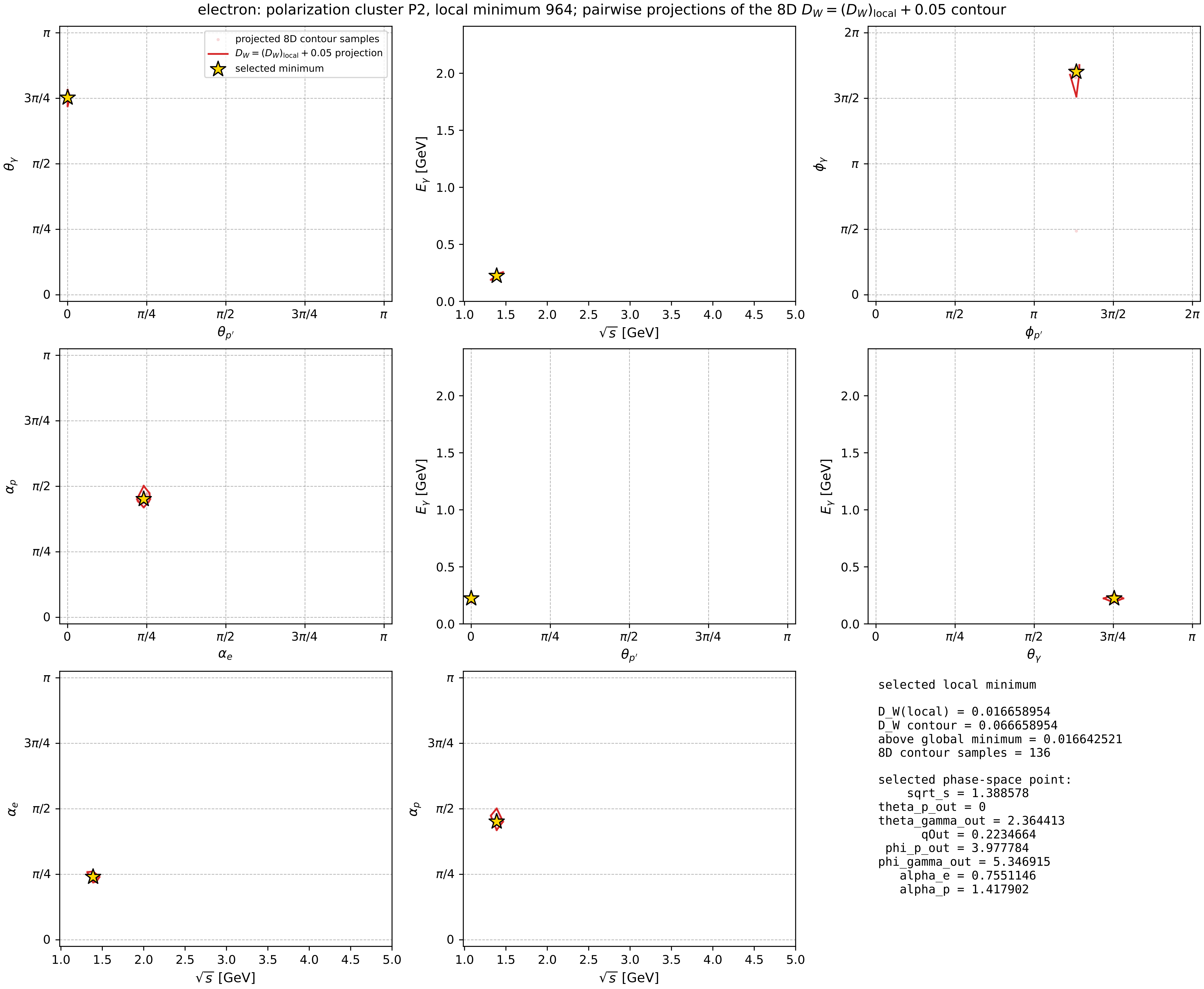




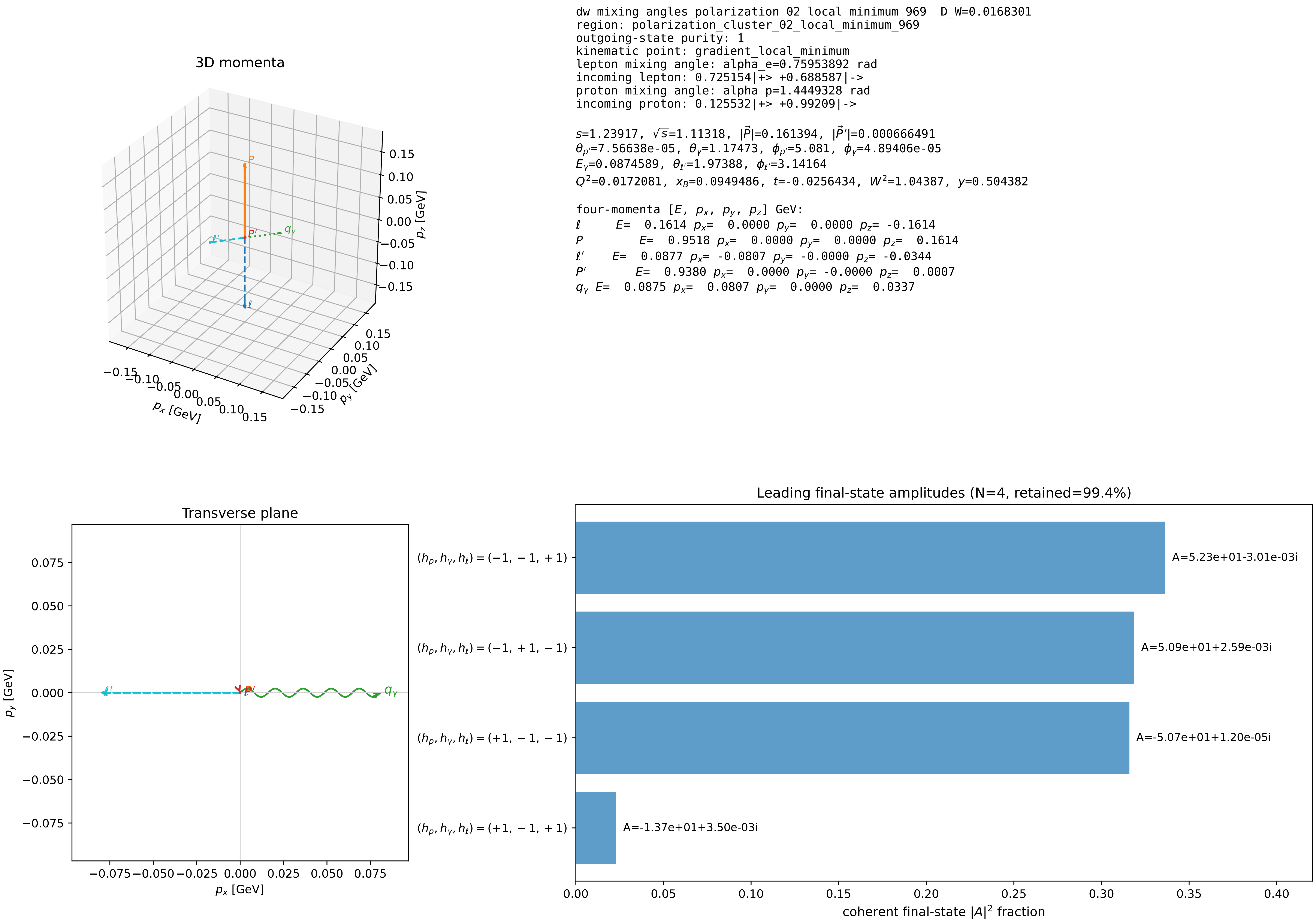
electron: polarization cluster P2, local minimum 960; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.016540437
D_W contour = 0.066540437
above global minimum = 0.016524004
8D contour samples = 130
selected phase-space point:
sqrt_s = 1.126128
theta_p_out = 0.0001877691
theta_gamma_out = 1.271137
qOut = 0.09272374
phi_p_out = 1.377009
phi_gamma_out = 6.282744
alpha_e = 0.7564481
alpha_p = 1.430826

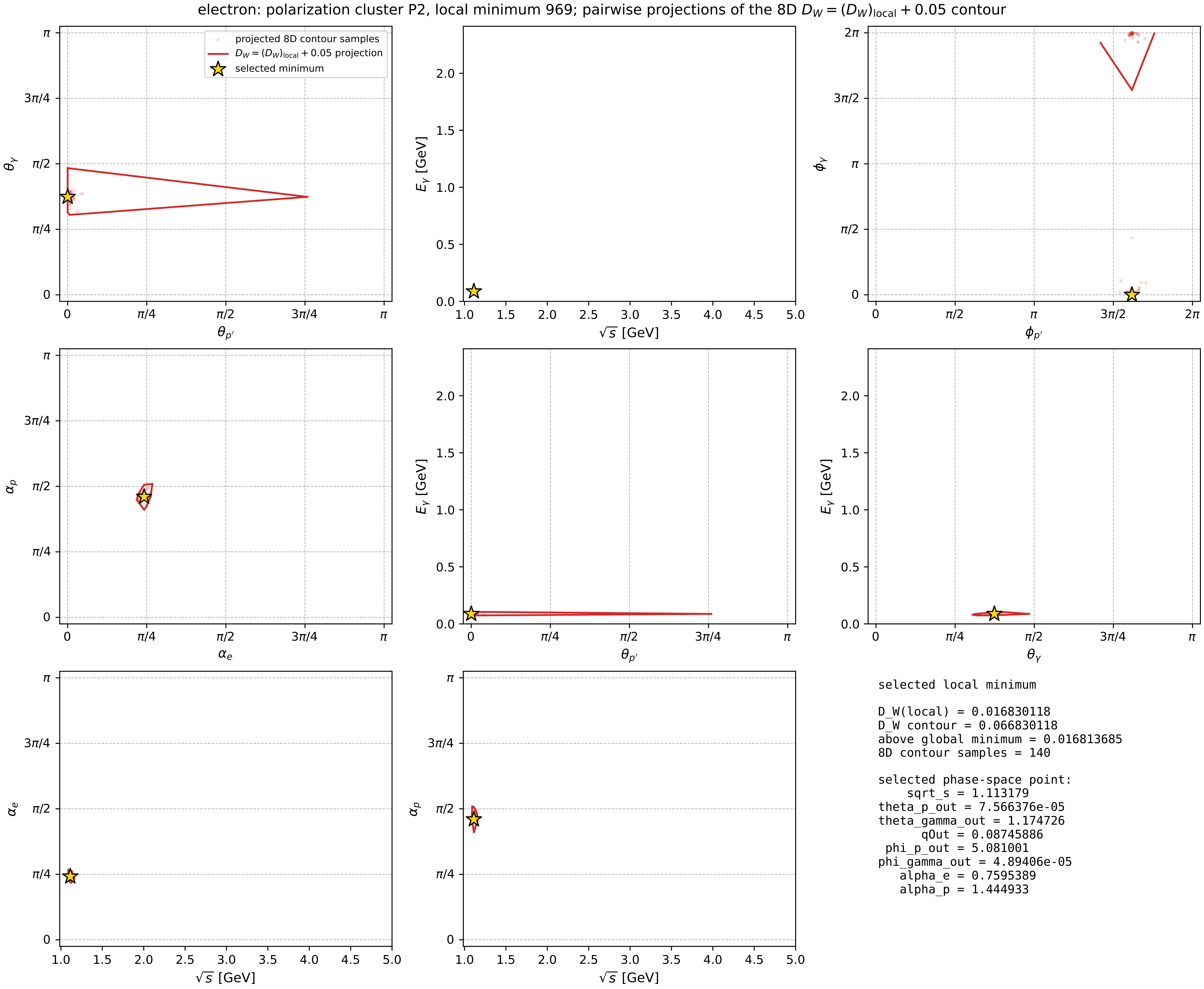


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_02_local_minimum_964 D_W=0.016659
region: polarization_cluster_02_local_minimum_964
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75511456 rad
incoming lepton: 0.728193|+> +0.685372|->
proton mixing angle: alpha_p=1.4179018 rad
incoming proton: 0.1523|+> +0.988334|->
s=1.92815, $\sqrt{s}$=1.38858, $|\vec{P}|$=0.377474, $|\vec{P}'|$=0.266646
$\theta_{p'}$=0, θ_γ=2.36441, $\phi_{p'}$=3.97778, ϕ_γ=5.34691
E_γ=0.223466, $\theta_{\ell'}$=2.17135, $\phi_{\ell'}$=2.20532
Q^2=0.0623651, x_B=0.106944, t=-0.010991, W^2=1.40063, y=0.556285
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3775 p_x= 0.0000 p_y= 0.0000 p_z= -0.3775
P E= 1.0111 p_x= 0.0000 p_y= 0.0000 p_z= 0.3775
ℓ' E= 0.1899 p_x= -0.0929 p_y= 0.1262 p_z= -0.1073
P' E= 0.9752 p_x= -0.0000 p_y= -0.0000 p_z= 0.2666
q_γ E= 0.2235 p_x= 0.0929 p_y= -0.1262 p_z= -0.1593
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=98.7%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=4.80e+01+6.38e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-7.87e+01+1.87e-02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=4.25e+01-6.46e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=1.71e+01-2.16e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 964; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.016658954
D_W contour = 0.066658954
above global minimum = 0.016642521
8D contour samples = 136
selected phase-space point:
sqrt_s = 1.388578
theta_p_out = 0
theta_gamma_out = 2.364413
qOut = 0.2234664
phi_p_out = 3.977784
phi_gamma_out = 5.346915
alpha_e = 0.7551146
alpha_p = 1.417902

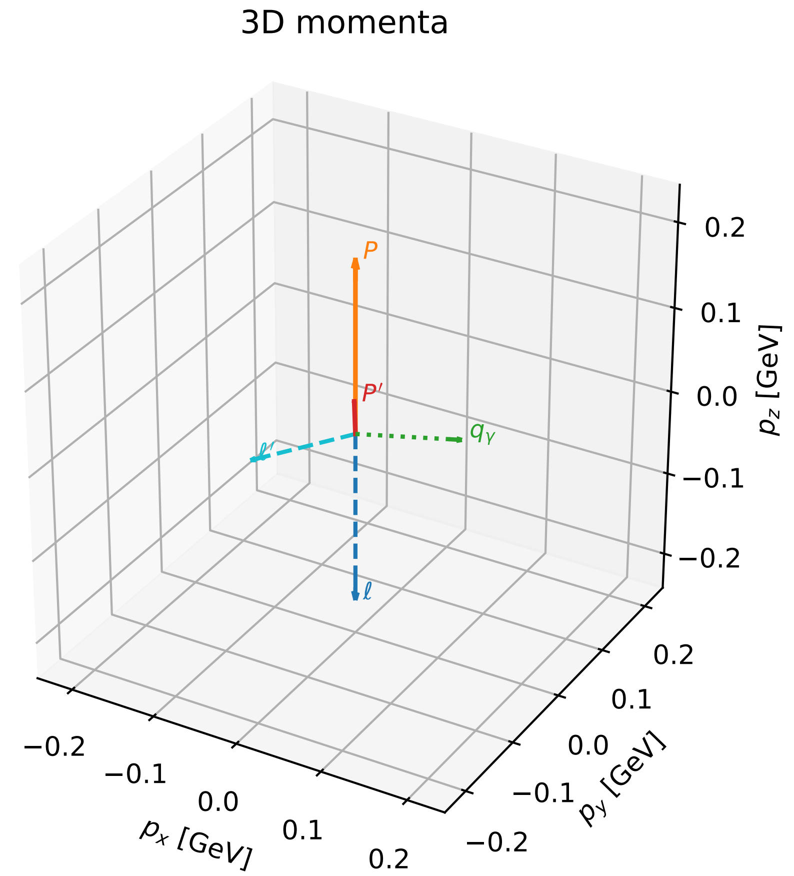
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)





electron: polarization cluster P2, local minimum 969; pairwise projections of the 8D $D_W = (D_W)_{\rm local} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\rm local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.016830118
D_W contour = 0.066830118
above global minimum = 0.016813685
8D contour samples = 140
selected phase-space point:
sqrt_s = 1.113179
theta_p_out = 7.566376e-05
theta_gamma_out = 1.174726
qOut = 0.08745886
phi_p_out = 5.081001
phi_gamma_out = 4.89406e-05
alpha_e = 0.7595389
alpha_p = 1.444933

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

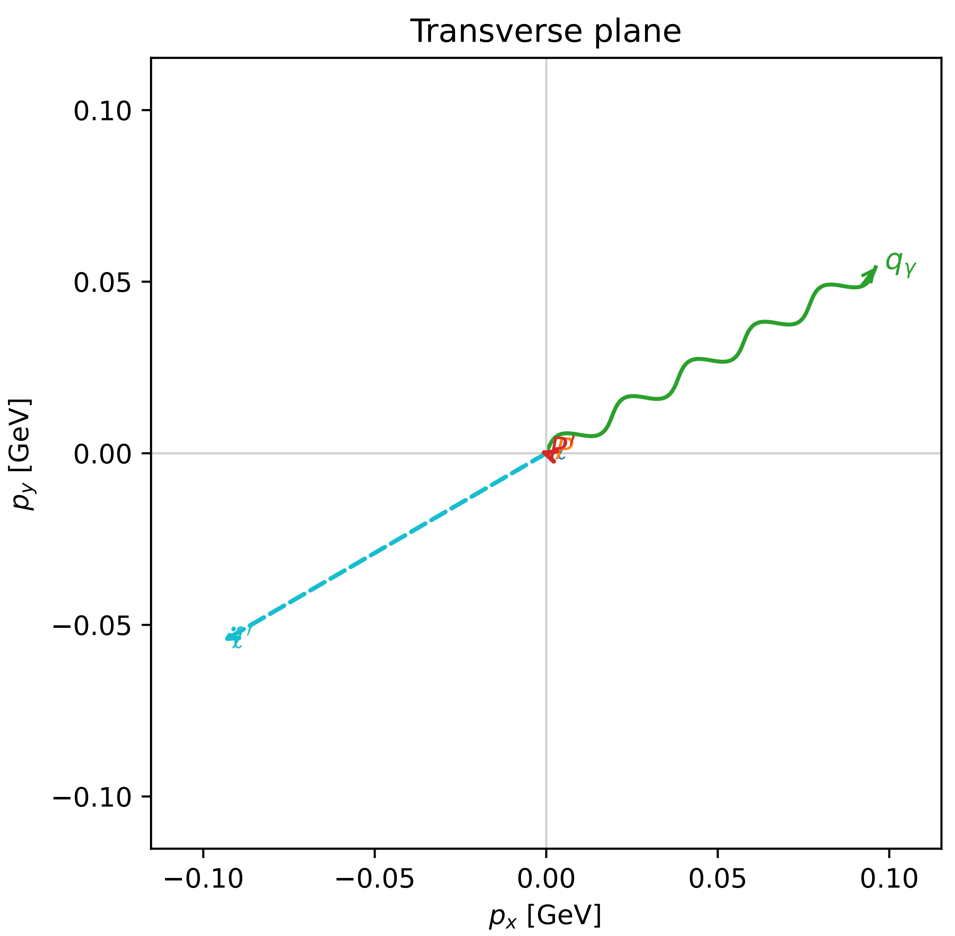


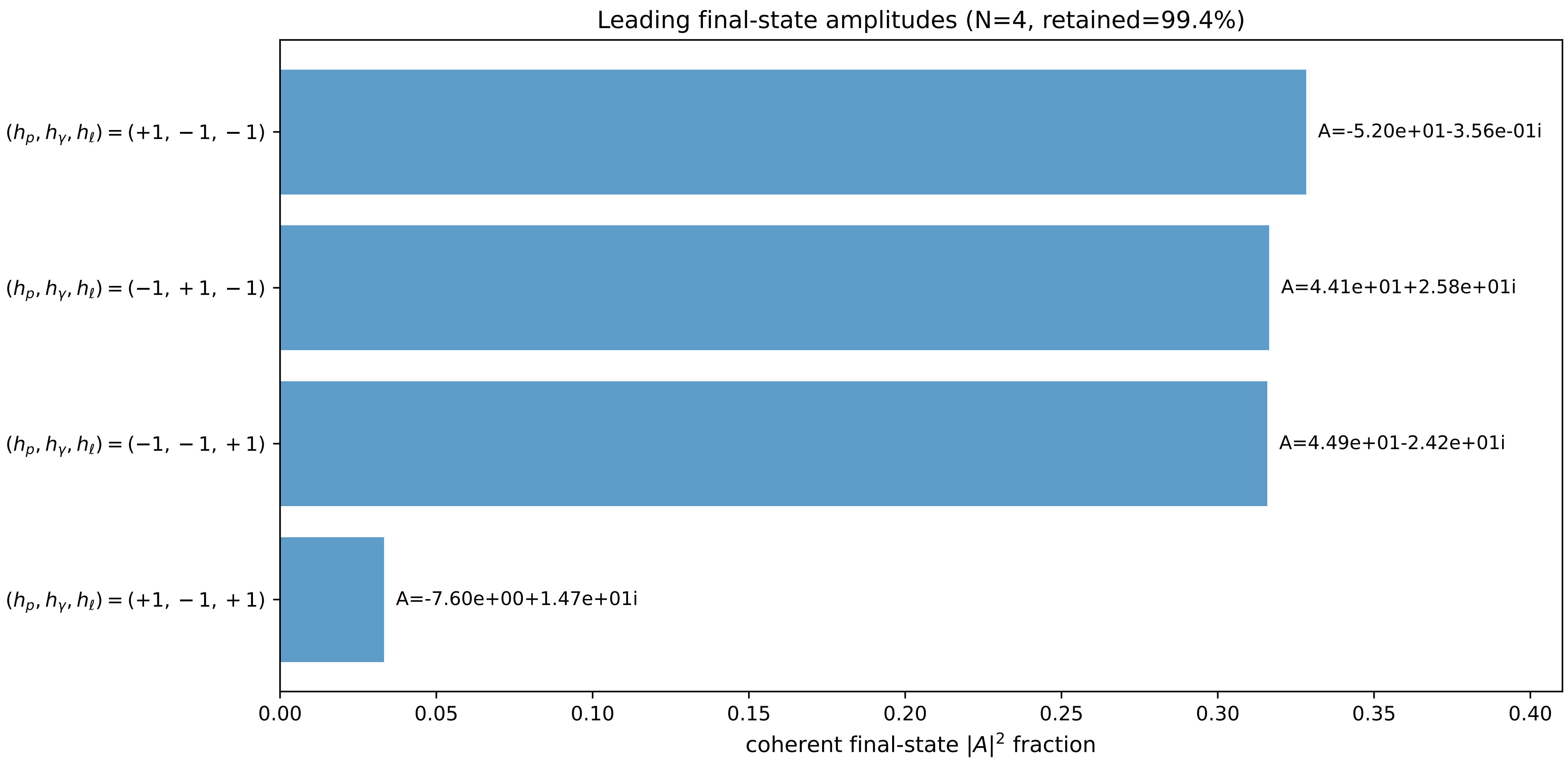

```
dw_mixing_angles_polarization_02_local_minimum_971  D_W=0.0170037
region: polarization_cluster_02_local_minimum_971
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.77990544 rad
incoming lepton: 0.71098|+> +0.703212|->
proton mixing angle: alpha_p=1.499622 rad
incoming proton: 0.0711143|+> +0.997468|->
```

s=1.35175, $\sqrt{s}$=1.16265, $|\vec{P}|$=0.202943, $|\vec{P}'|$=0.0379273
$\theta_{p'}$=0.0488704, θ_γ=1.62951, $\phi_{p'}$=2.82414, ϕ_γ=0.513739
E_γ=0.110438, $\theta_{\ell'}$=1.85127, $\phi_{\ell'}$=3.66789
Q^2=0.0332987, x_B=0.13793, t=-0.0268102, W^2=1.08796, y=0.511582

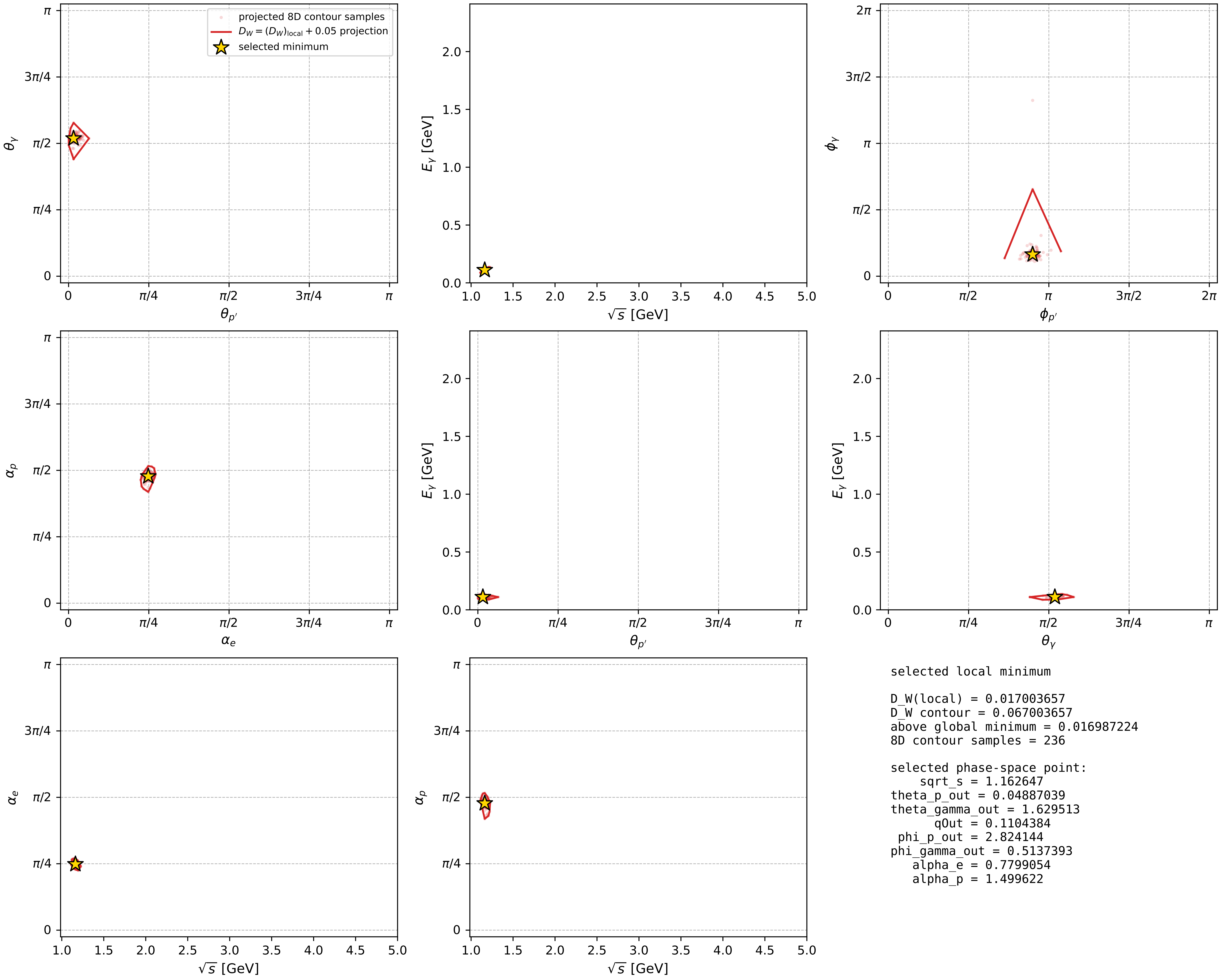
four-momenta [E, p_x, p_y, p_z] GeV:

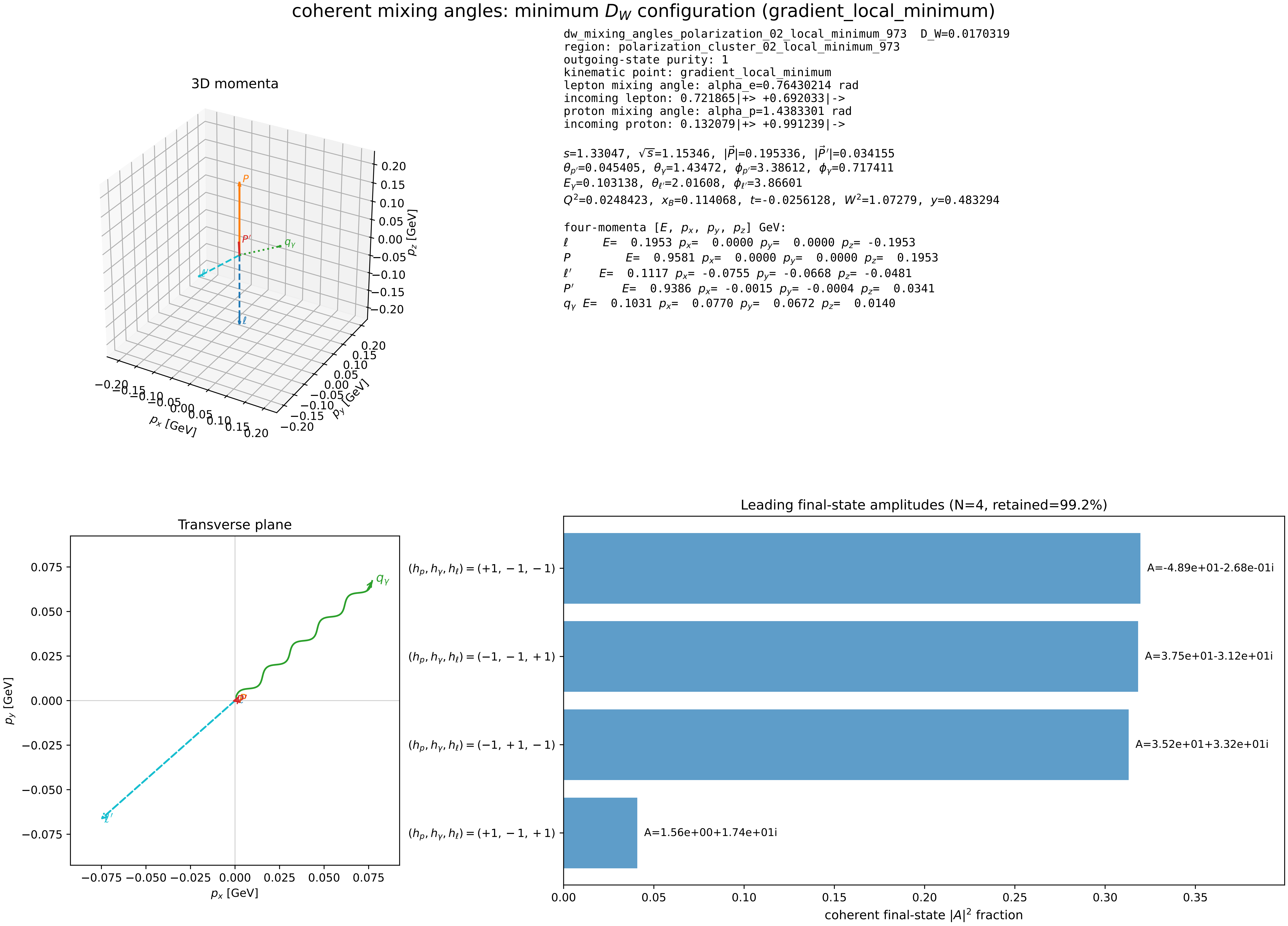
ℓ	E= 0.2029	p_x= 0.0000	p_y= 0.0000	p_z= -0.2029
P	E= 0.9597	p_x= 0.0000	p_y= 0.0000	p_z= 0.2029
ℓ'	E= 0.1134	p_x= -0.0943	p_y= -0.0548	p_z= -0.0314
P'	E= 0.9388	p_x= -0.0018	p_y= 0.0006	p_z= 0.0379
q_γ	E= 0.1104	p_x= 0.0960	p_y= 0.0542	p_z= -0.0065



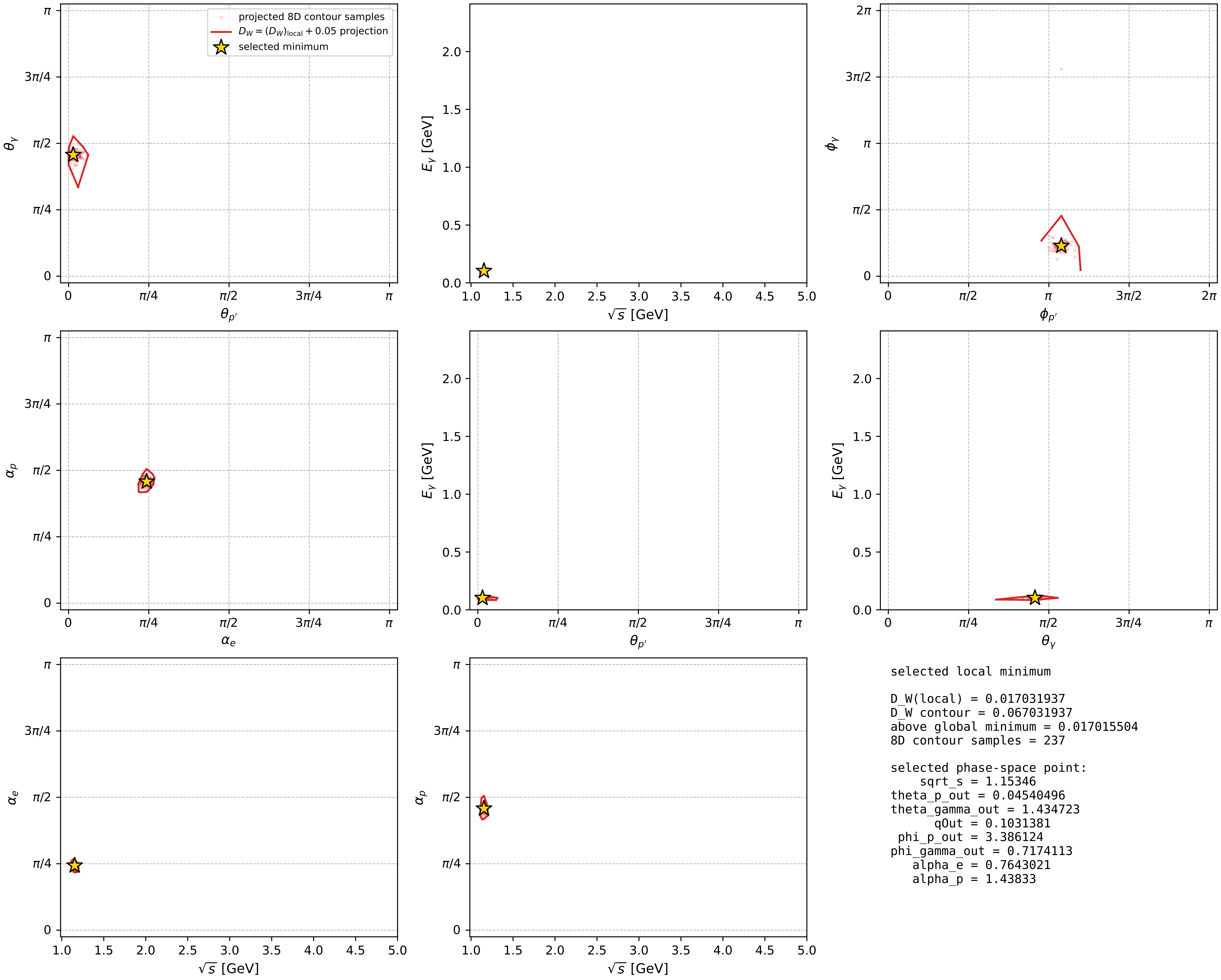


electron: polarization cluster P2, local minimum 971; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

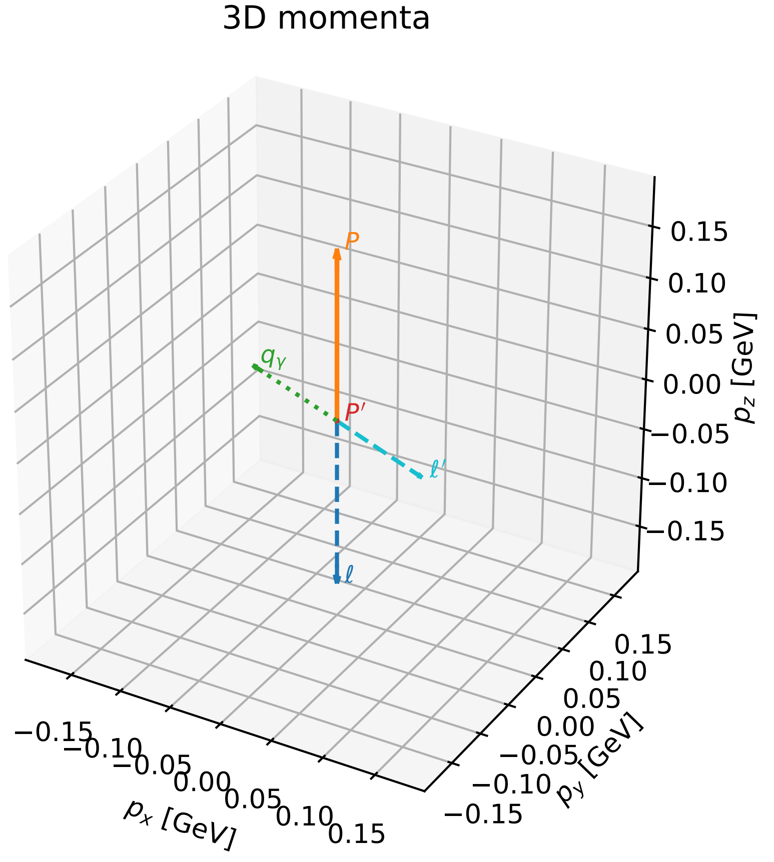



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_973 D_W=0.0170319
region: polarization_cluster_02_local_minimum_973
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.76430214 rad
incoming lepton: 0.721865|+> +0.692033|->
proton mixing angle: alpha_p=1.4383301 rad
incoming proton: 0.132079|+> +0.991239|->
s=1.33047, $\sqrt{s}$=1.15346, $|\vec{P}|$=0.195336, $|\vec{P}'|$=0.034155
$\theta_{p'}$=0.045405, θ_γ=1.43472, $\phi_{p'}$=3.38612, ϕ_γ=0.717411
E_γ=0.103138, $\theta_{\ell'}$=2.01608, $\phi_{\ell'}$=3.86601
Q^2=0.0248423, x_B=0.114068, t=-0.0256128, W^2=1.07279, y=0.483294
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1953 p_x= 0.0000 p_y= 0.0000 p_z= -0.1953
P E= 0.9581 p_x= 0.0000 p_y= 0.0000 p_z= 0.1953
ℓ' E= 0.1117 p_x= -0.0755 p_y= -0.0668 p_z= -0.0481
P' E= 0.9386 p_x= -0.0015 p_y= -0.0004 p_z= 0.0341
q_γ E= 0.1031 p_x= 0.0770 p_y= 0.0672 p_z= 0.0140
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=99.2%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-4.89e+01-2.68e-01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=3.75e+01-3.12e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=3.52e+01+3.32e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=1.56e+00+1.74e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 973; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



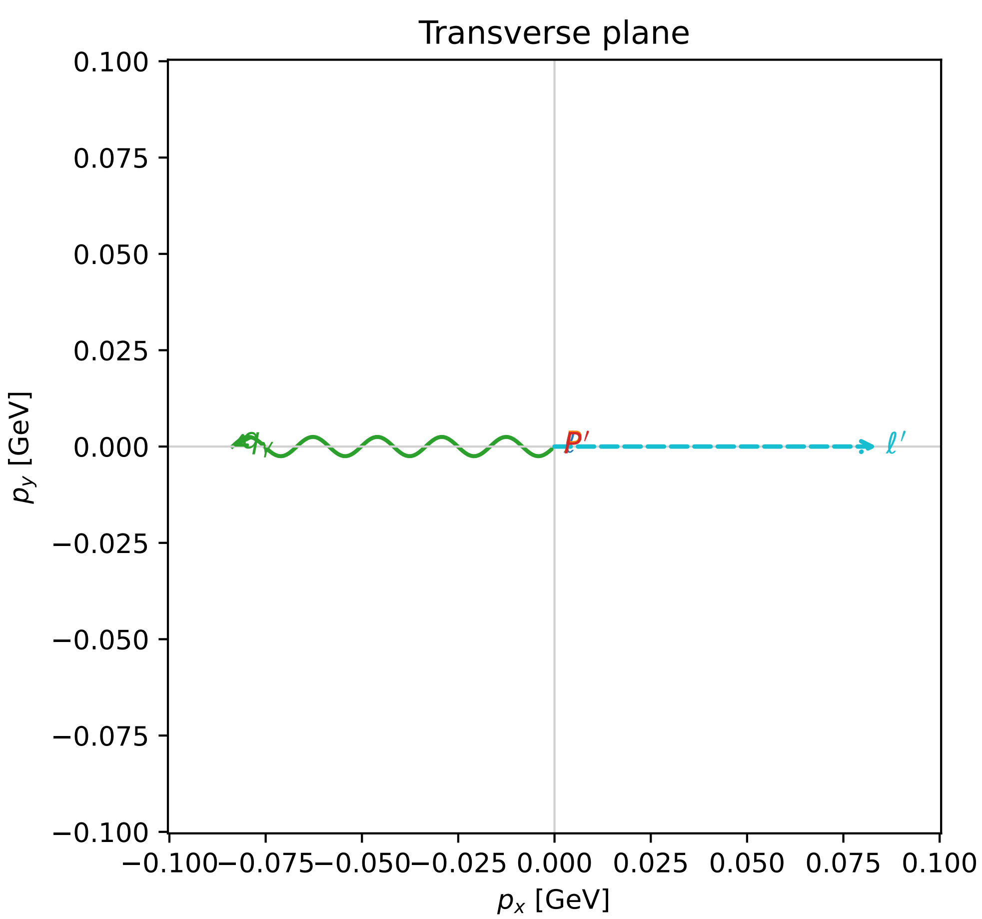
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

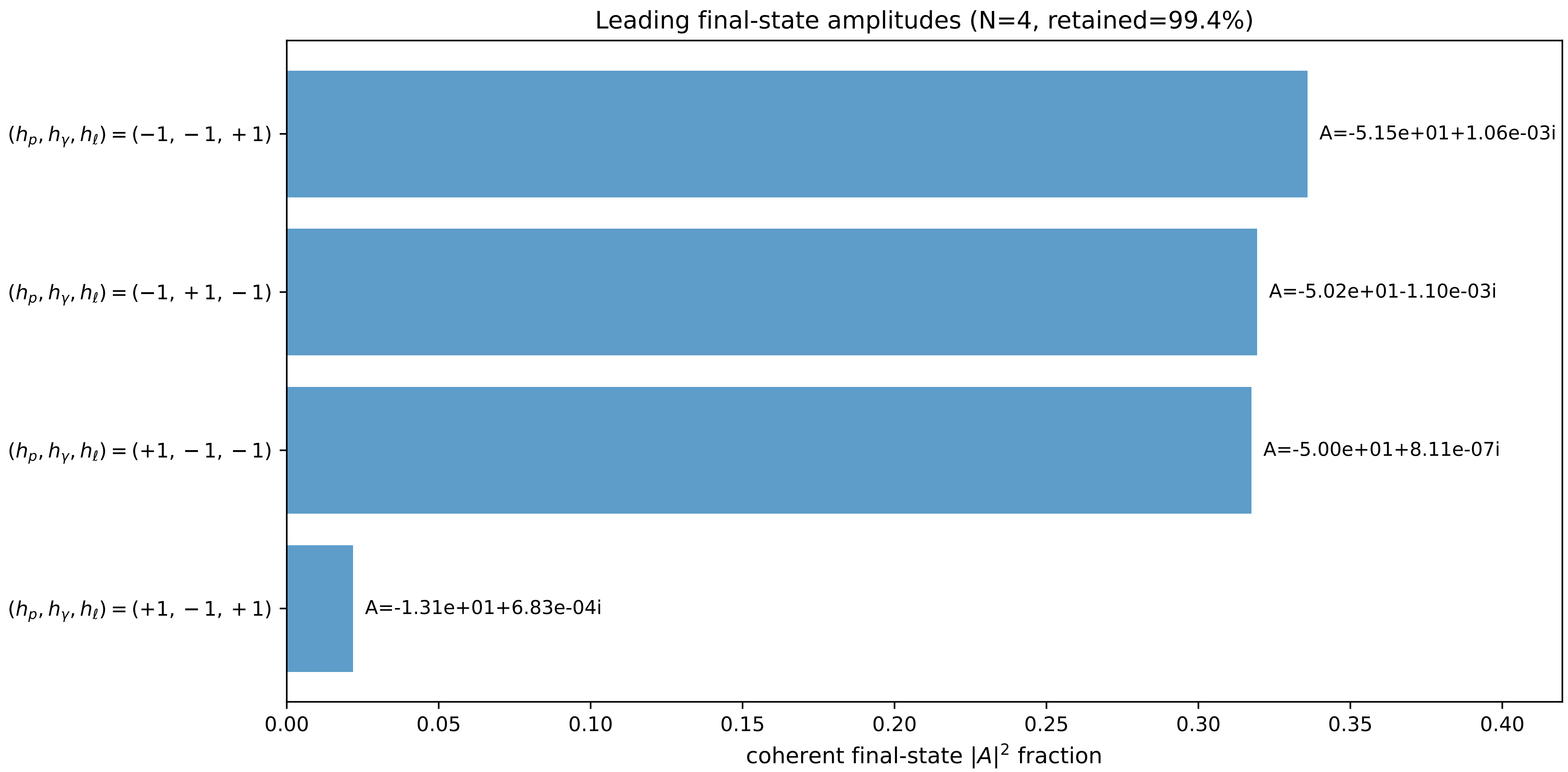


dw_mixing_angles_polarization_02_local_minimum_975 D_W=0.0171918
region: polarization_cluster_02_local_minimum_975
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75562897 rad
incoming lepton: 0.72784|+> +0.685747|->
proton mixing angle: alpha_p=1.6974719 rad
incoming proton: -0.126337|+> +0.991987|->

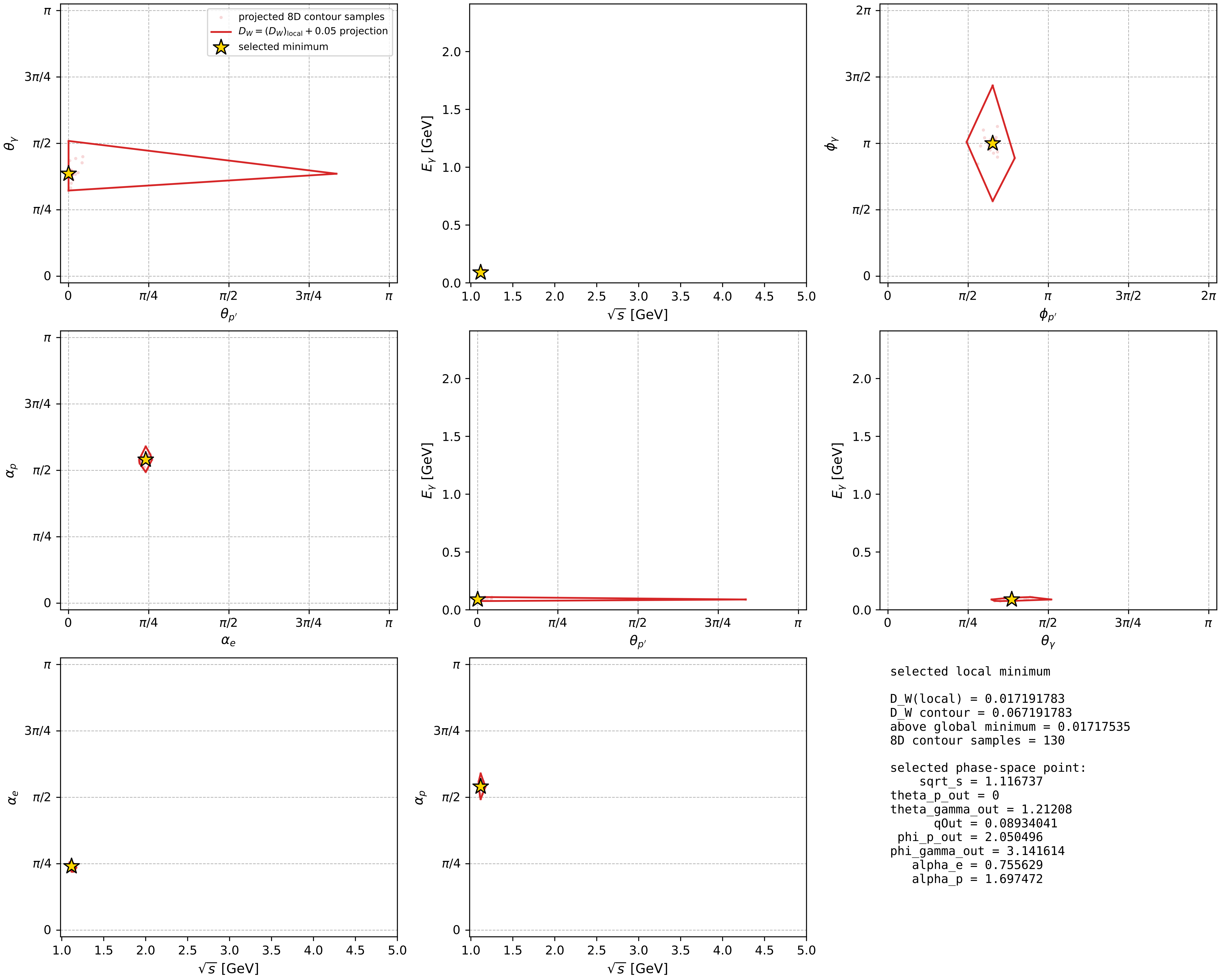
s=1.2471, $\sqrt{s}$=1.11674, $|\vec{P}|$=0.164433, $|\vec{P}'|$=0.000155304
$\theta_{p'}$=0, θ_γ=1.21208, $\phi_{p'}$=2.0505, ϕ_γ=3.14161
E_γ=0.0893404, $\theta_{\ell'}$=1.93114, $\phi_{\ell'}$=2.09702e-05
Q^2=0.0190331, x_B=0.101985, t=-0.0267824, W^2=1.04744, y=0.508162

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1644 p_x= 0.0000 p_y= 0.0000 p_z= -0.1644
P E= 0.9523 p_x= 0.0000 p_y= 0.0000 p_z= 0.1644
ℓ' E= 0.0894 p_x= 0.0837 p_y= 0.0000 p_z= -0.0315
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= 0.0002
q_γ E= 0.0893 p_x= -0.0837 p_y= -0.0000 p_z= 0.0314



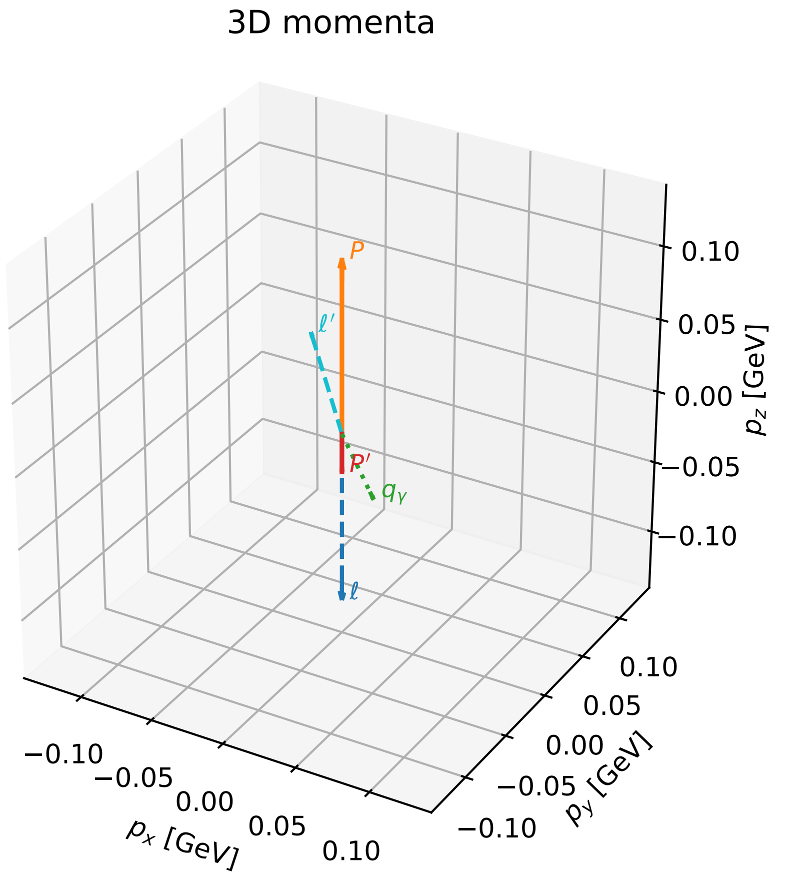


electron: polarization cluster P2, local minimum 975; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.017191783
D_W contour = 0.067191783
above global minimum = 0.01717535
8D contour samples = 130

selected phase-space point:
    sqrt_s = 1.116737
theta_p_out = 0
theta_gamma_out = 1.21208
      qOut = 0.08934041
 phi_p_out = 2.050496
phi_gamma_out = 3.141614
   alpha_e = 0.755629
   alpha_p = 1.697472
```

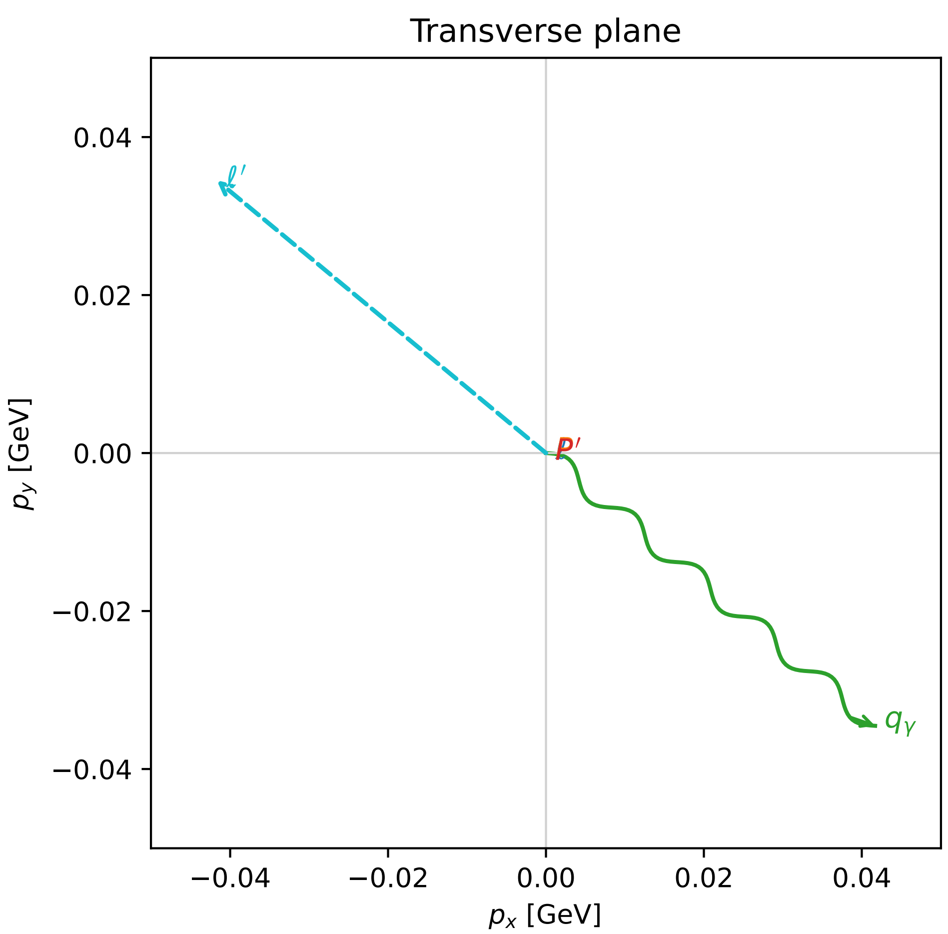
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

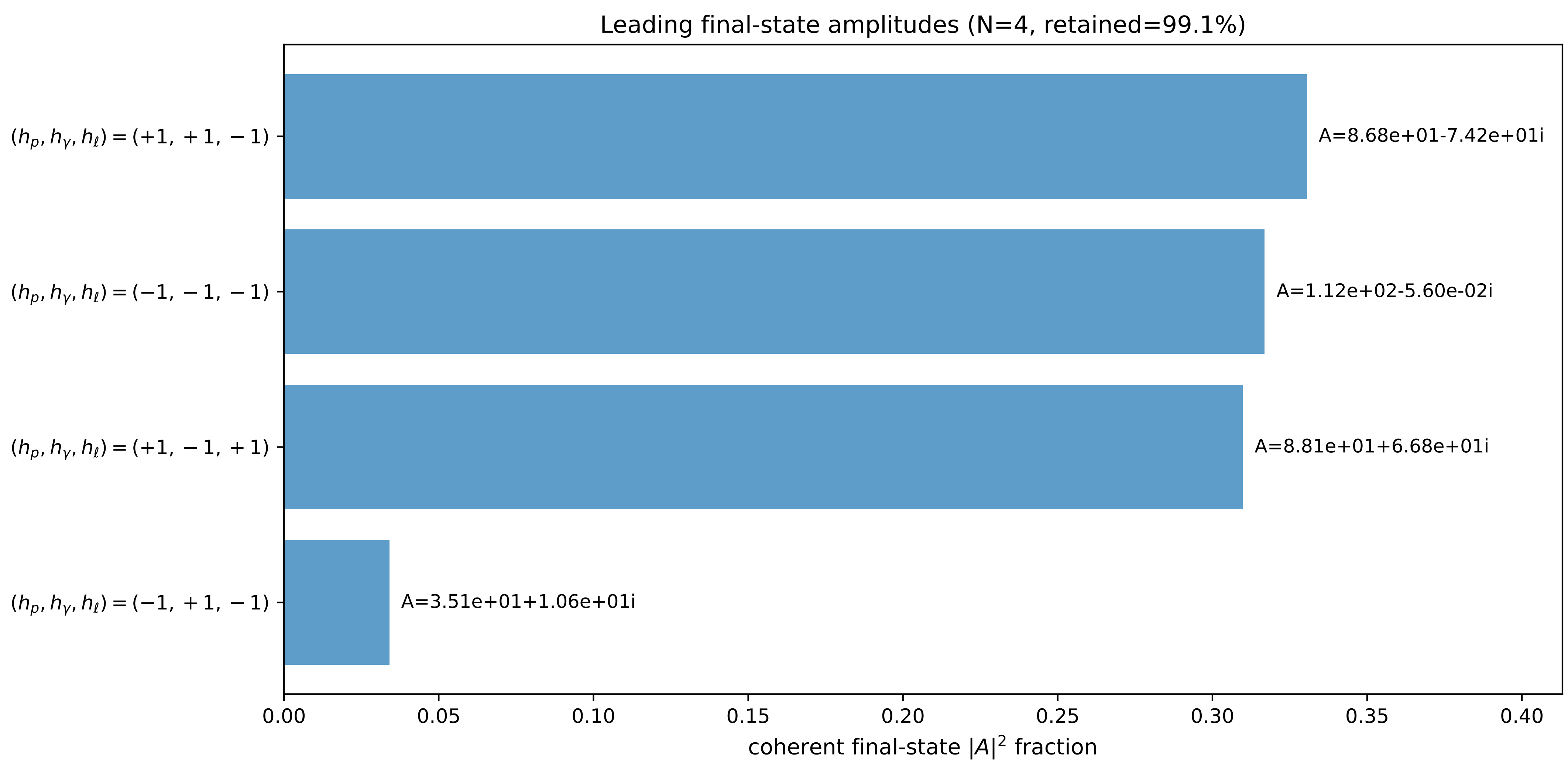


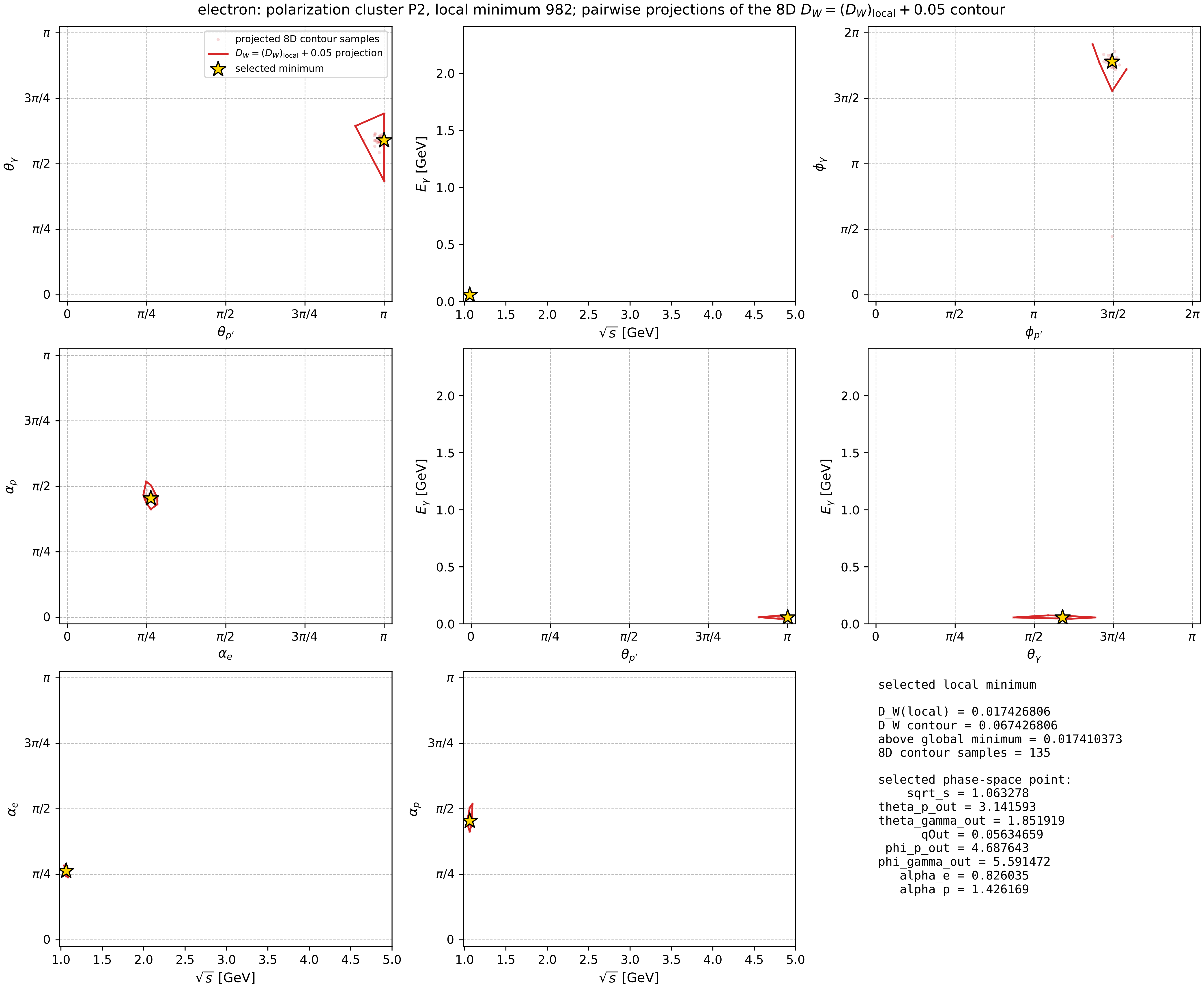

```
dw_mixing_angles_polarization_02_local_minimum_982  D_W=0.0174268
region: polarization_cluster_02_local_minimum_982
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.82603495 rad
incoming lepton: 0.677796|+> +0.73525|->
proton mixing angle: alpha_p=1.4261693 rad
incoming proton: 0.144123|+> +0.98956|->
```

s=1.13056, $\sqrt{s}$=1.06328, $|\vec{P}|$=0.117897, $|\vec{P}'|$=0.0264328
$\theta_{p'}$=3.14159, θ_γ=1.85192, $\phi_{p'}$=4.68764, ϕ_γ=5.59147
E_γ=0.0563466, $\theta_{\ell'}$=0.910208, $\phi_{\ell'}$=2.44988
Q^2=0.026084, x_B=0.199106, t=-0.0207818, W^2=0.984766, y=0.522528

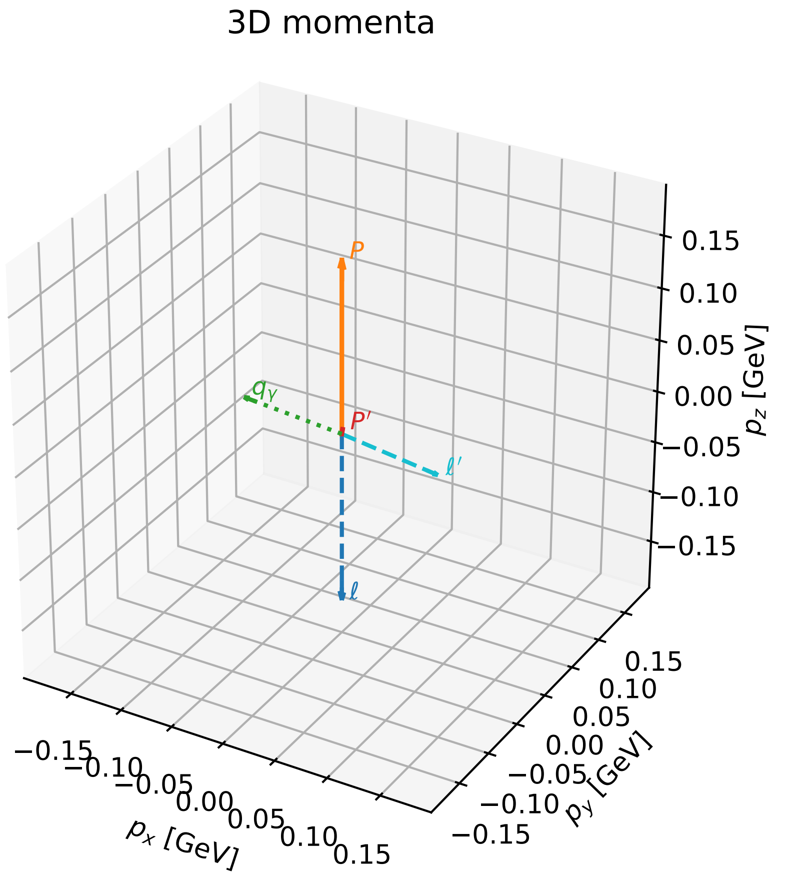
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1179 p_x= 0.0000 p_y= 0.0000 p_z= -0.1179
P E= 0.9454 p_x= 0.0000 p_y= 0.0000 p_z= 0.1179
ℓ' E= 0.0686 p_x= -0.0417 p_y= 0.0345 p_z= 0.0421
P' E= 0.9384 p_x= -0.0000 p_y= -0.0000 p_z= -0.0264
q_γ E= 0.0563 p_x= 0.0417 p_y= -0.0345 p_z= -0.0156






electron: polarization cluster P2, local minimum 982; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.017426806
D_W contour = 0.067426806
above global minimum = 0.017410373
8D contour samples = 135
selected phase-space point:
sqrt_s = 1.063278
theta_p_out = 3.141593
theta_gamma_out = 1.851919
qOut = 0.05634659
phi_p_out = 4.687643
phi_gamma_out = 5.591472
alpha_e = 0.826035
alpha_p = 1.426169

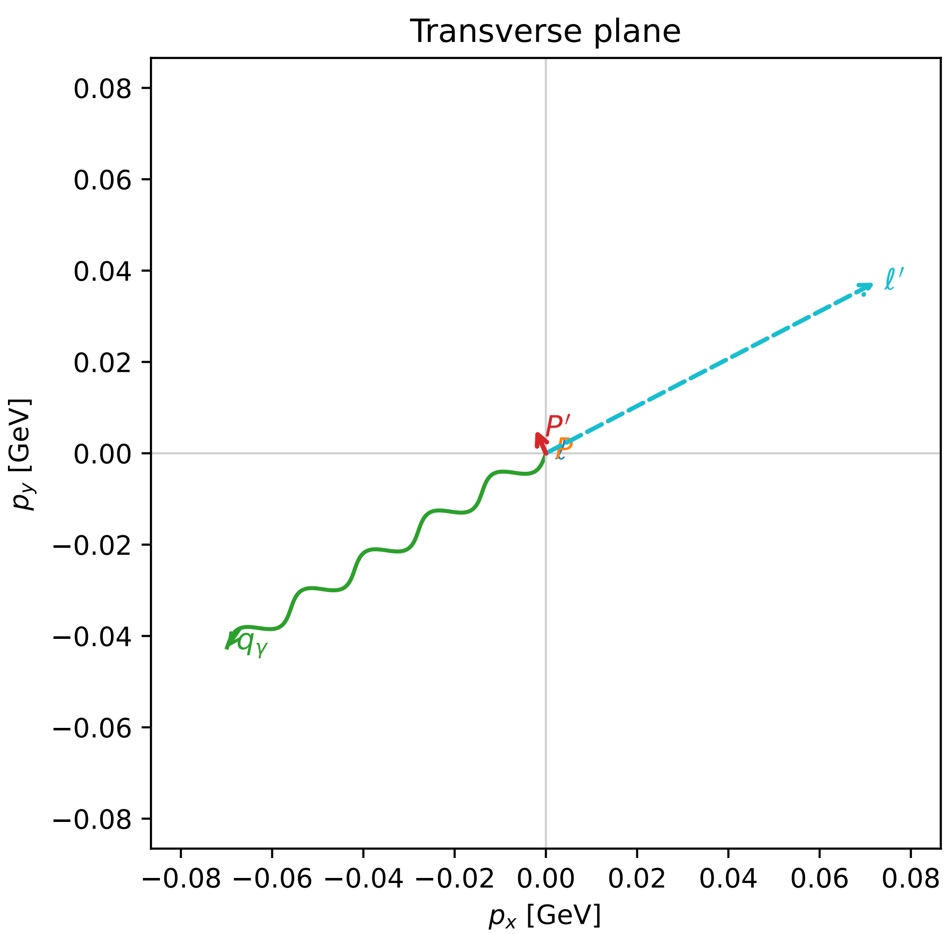
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

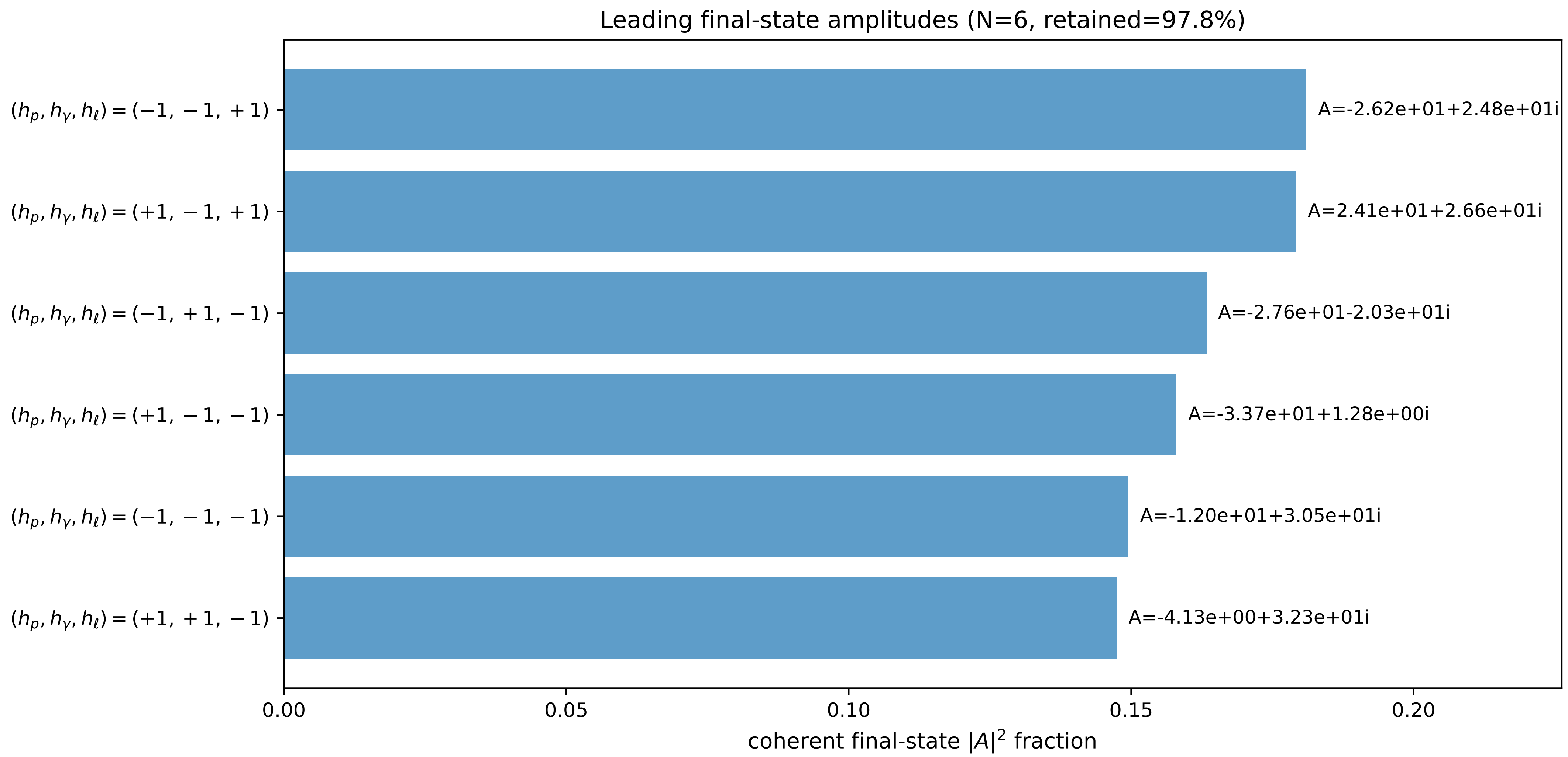


dw_mixing_angles_polarization_02_local_minimum_987 D_W=0.0177589
region: polarization_cluster_02_local_minimum_987
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.73328267 rad
incoming lepton: 0.742981|+> +0.669312|->
proton mixing angle: alpha_p=1.7773997 rad
incoming proton: -0.205137|+> +0.978733|->

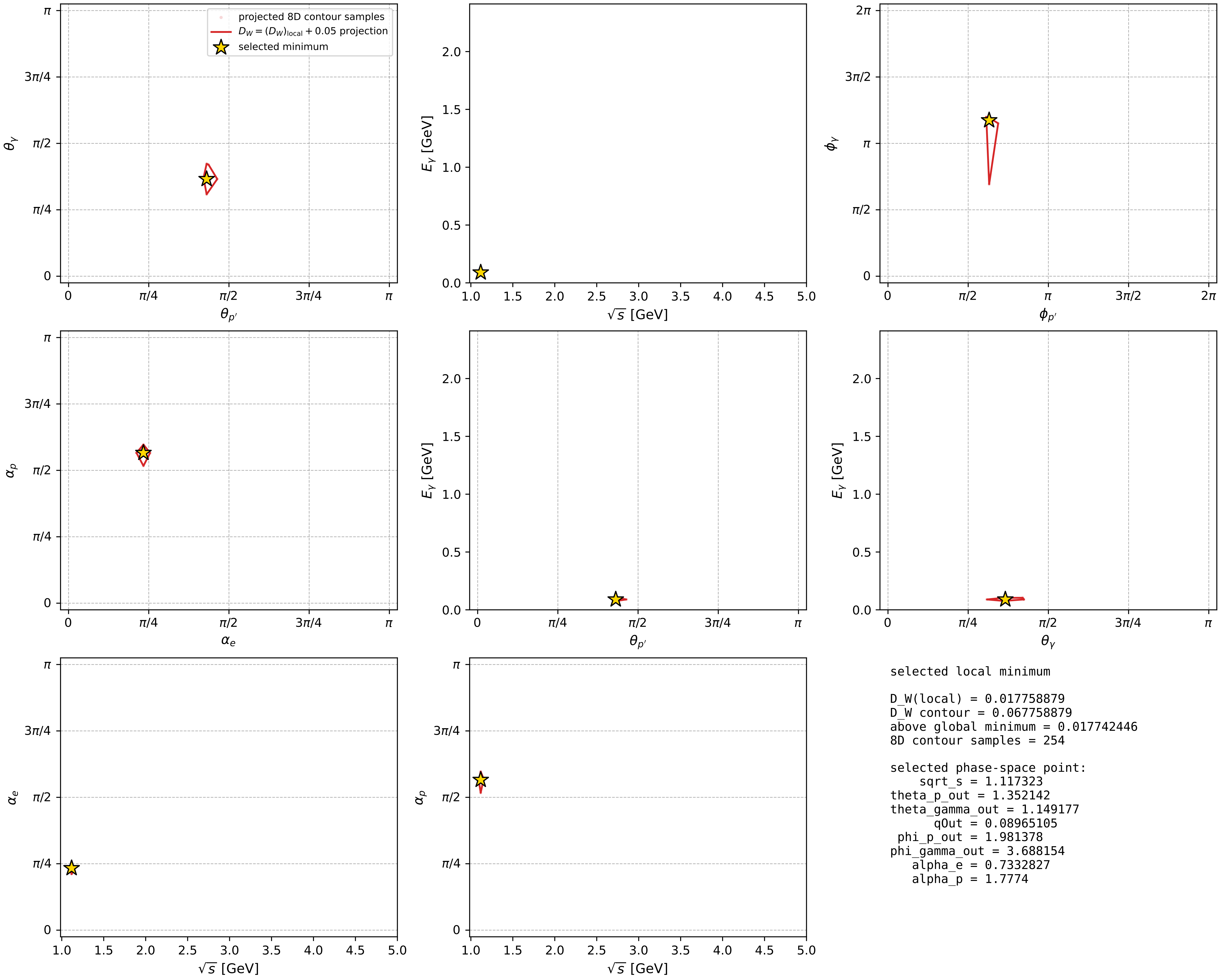
s=1.24841, $\sqrt{s}$=1.11732, $|\vec{P}|$=0.164932, $|\vec{P}'|$=0.00577067
$\theta_{p'}$=1.35214, θ_γ=1.14918, $\phi_{p'}$=1.98138, ϕ_γ=3.68815
E_γ=0.0896511, $\theta_{\ell'}$=2.00777, $\phi_{\ell'}$=0.477796
Q^2=0.017058, x_B=0.0920661, t=-0.0266163, W^2=1.04807, y=0.502705

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1649 p_x= 0.0000 p_y= 0.0000 p_z= -0.1649
P E= 0.9524 p_x= 0.0000 p_y= 0.0000 p_z= 0.1649
ℓ' E= 0.0897 p_x= 0.0721 p_y= 0.0374 p_z= -0.0379
P' E= 0.9380 p_x= -0.0022 p_y= 0.0052 p_z= 0.0013
q_γ E= 0.0897 p_x= -0.0699 p_y= -0.0425 p_z= 0.0367



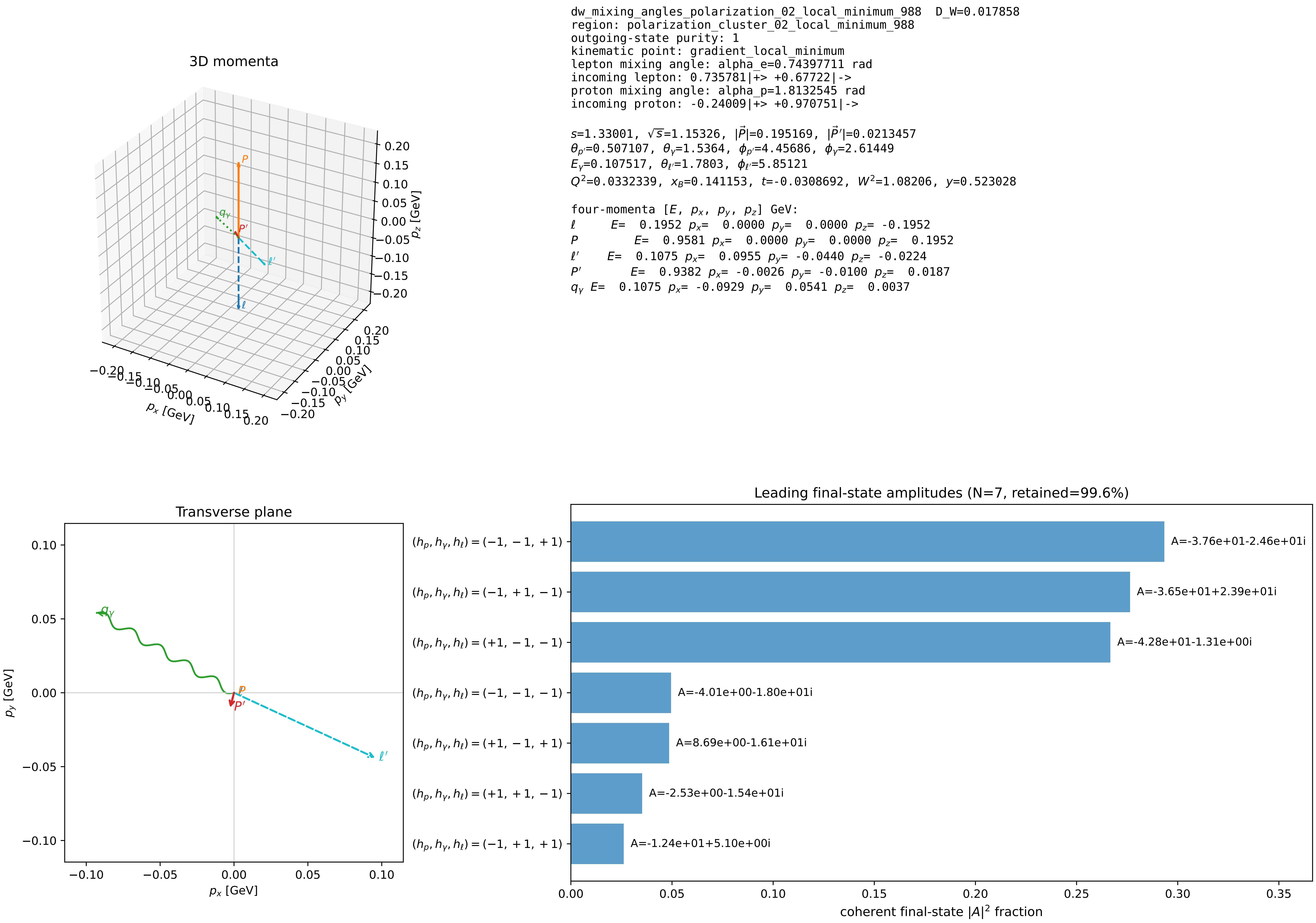


electron: polarization cluster P2, local minimum 987; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

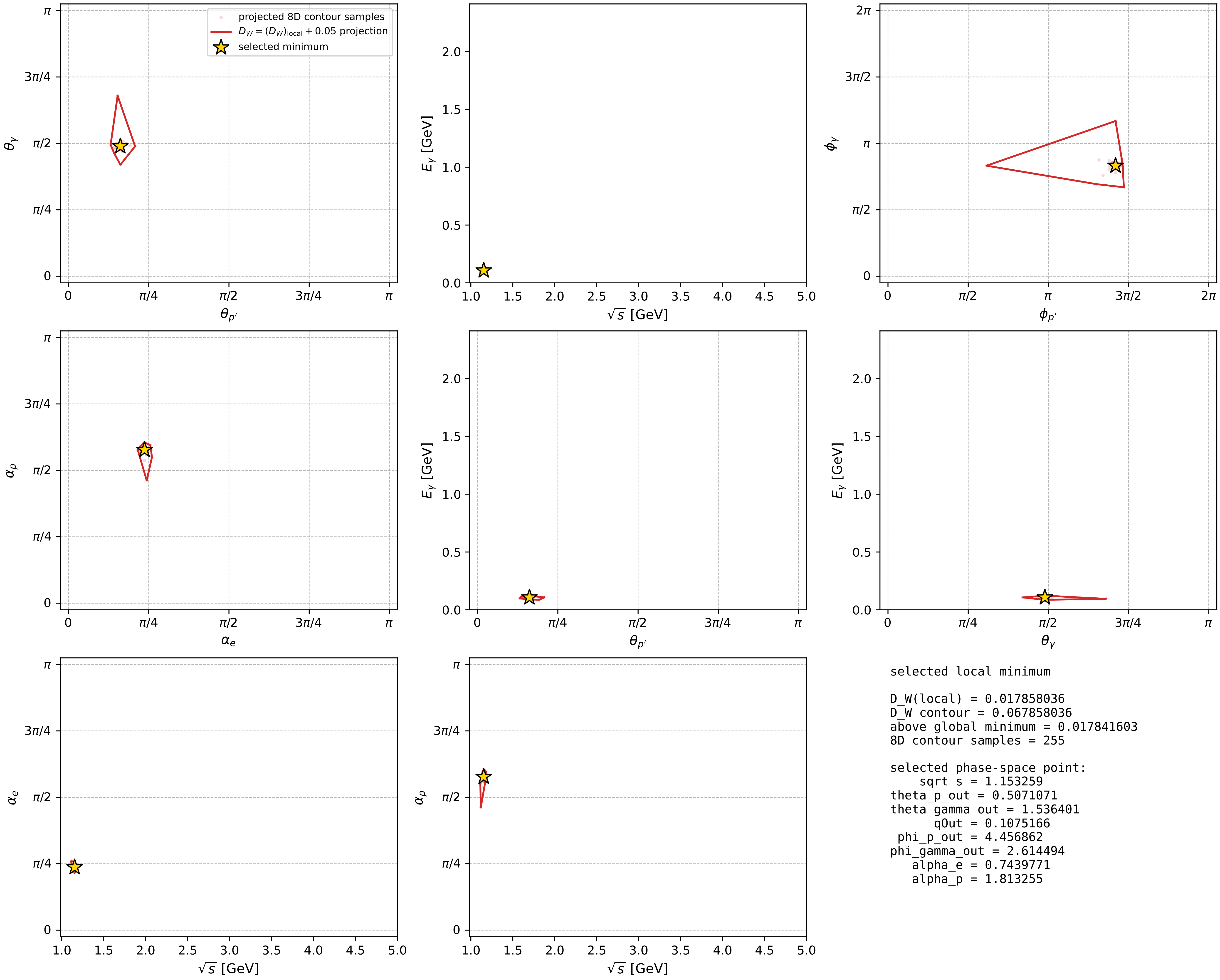



```
selected local minimum

D_W(local) = 0.017758879
D_W contour = 0.067758879
above global minimum = 0.017742446
8D contour samples = 254

selected phase-space point:
    sqrt_s = 1.117323
theta_p_out = 1.352142
theta_gamma_out = 1.149177
        qOut = 0.08965105
 phi_p_out = 1.981378
phi_gamma_out = 3.688154
   alpha_e = 0.7332827
   alpha_p = 1.7774
```

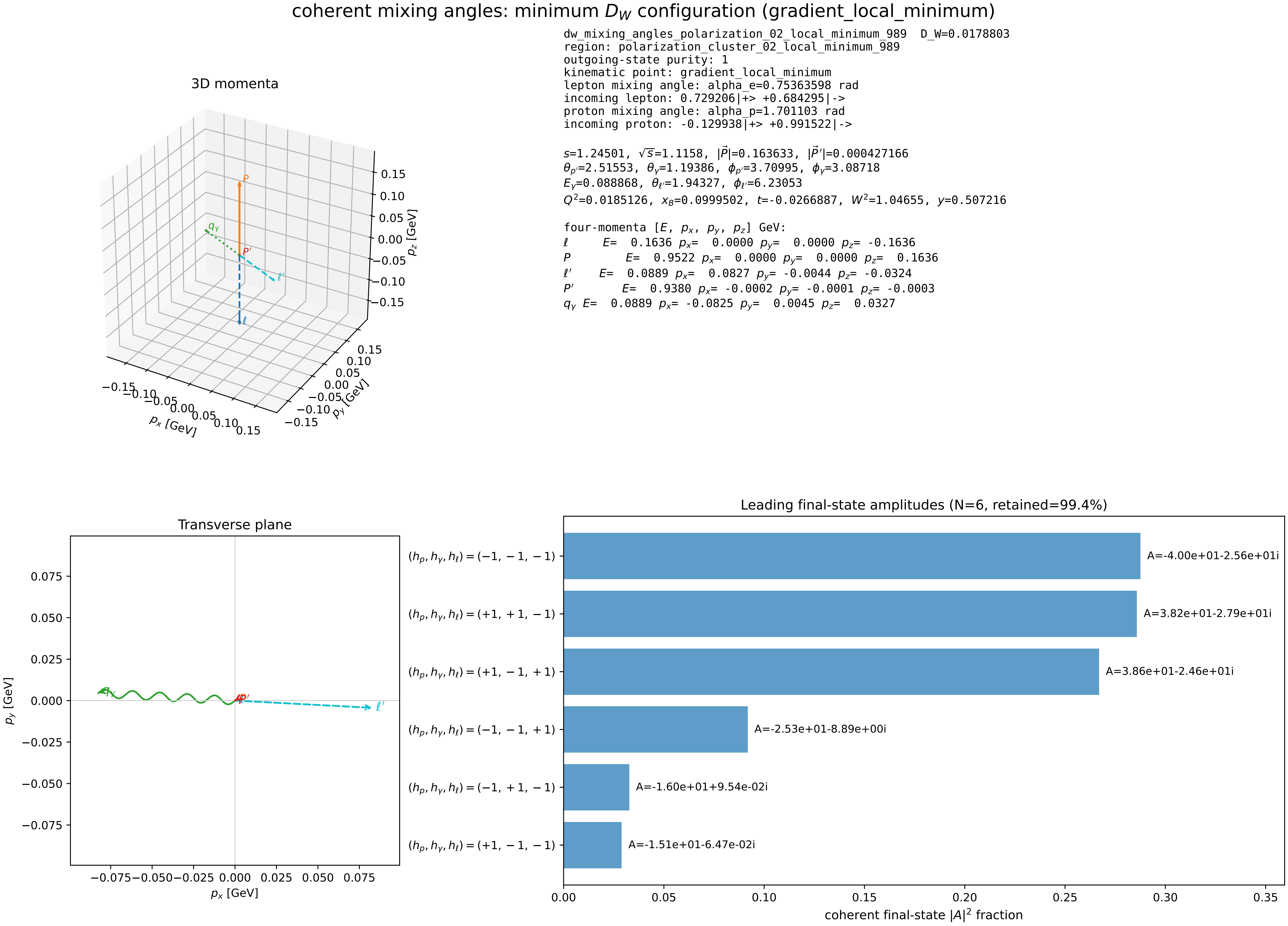
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



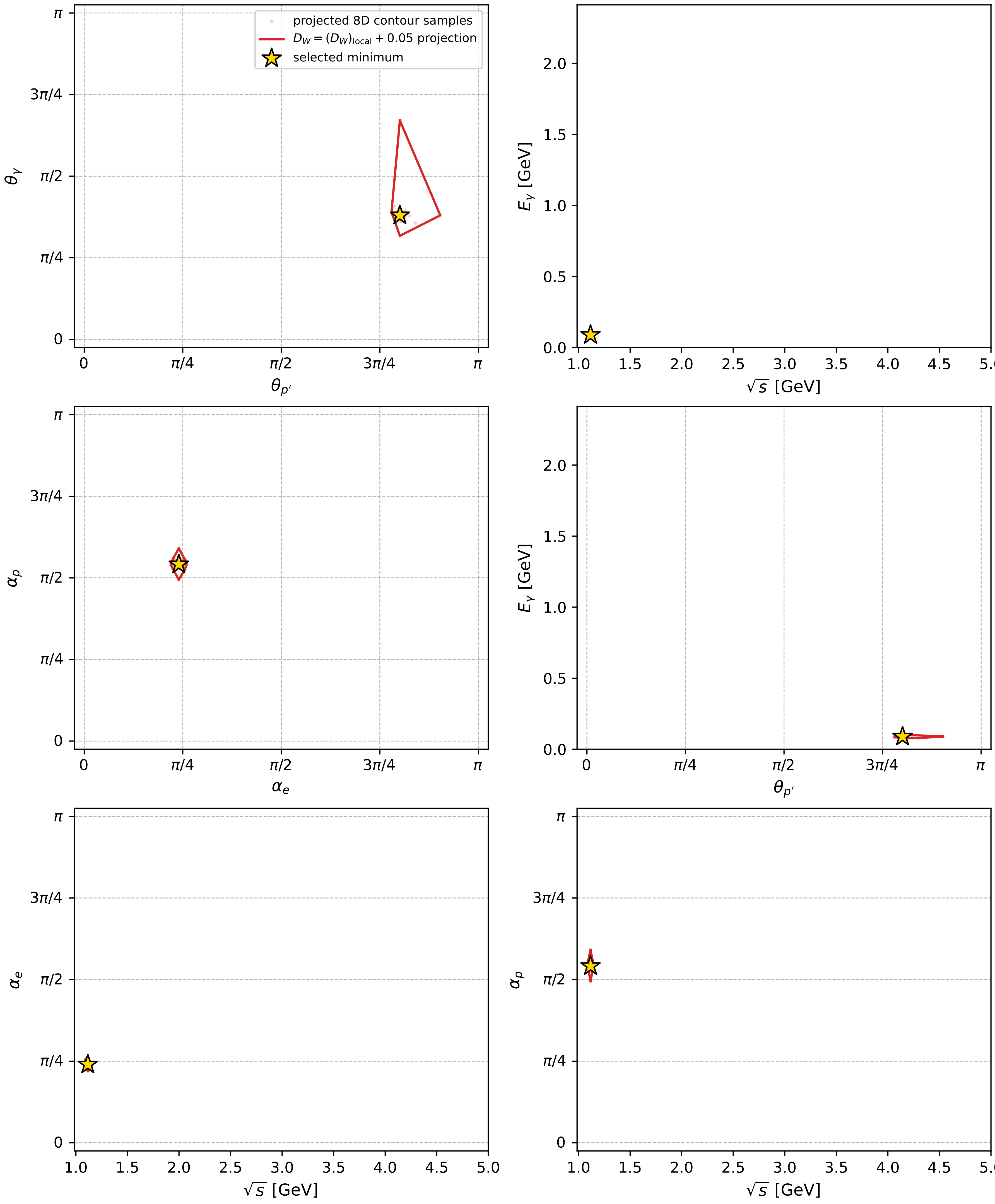
electron: polarization cluster P2, local minimum 988; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.017858036
D_W contour = 0.067858036
above global minimum = 0.017841603
8D contour samples = 255

selected phase-space point:
    sqrt_s = 1.153259
theta_p_out = 0.5071071
theta_gamma_out = 1.536401
      qOut = 0.1075166
 phi_p_out = 4.456862
phi_gamma_out = 2.614494
   alpha_e = 0.7439771
   alpha_p = 1.813255
```

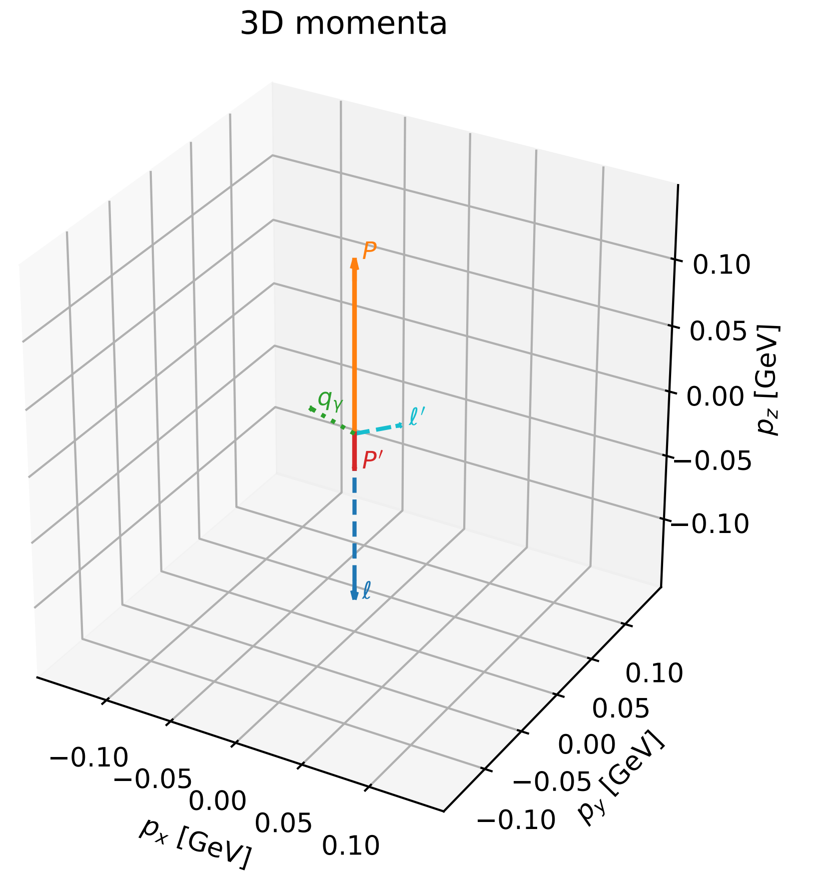

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_02_local_minimum_989 D_W=0.0178803
region: polarization_cluster_02_local_minimum_989
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75363598 rad
incoming lepton: 0.729206|+> +0.684295|->
proton mixing angle: alpha_p=1.701103 rad
incoming proton: -0.129938|+> +0.991522|->
s=1.24501, $\sqrt{s}$=1.1158, $|\vec{P}|$=0.163633, $|\vec{P}'|$=0.000427166
$\theta_{p'}$=2.51553, θ_γ=1.19386, $\phi_{p'}$=3.70995, ϕ_γ=3.08718
E_γ=0.088868, $\theta_{\ell'}$=1.94327, $\phi_{\ell'}$=6.23053
Q^2=0.0185126, x_B=0.0999502, t=-0.0266887, W^2=1.04655, y=0.507216
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1636 p_x= 0.0000 p_y= 0.0000 p_z= -0.1636
P E= 0.9522 p_x= 0.0000 p_y= 0.0000 p_z= 0.1636
ℓ' E= 0.0889 p_x= 0.0827 p_y= -0.0044 p_z= -0.0324
P' E= 0.9380 p_x= -0.0002 p_y= -0.0001 p_z= -0.0003
q_γ E= 0.0889 p_x= -0.0825 p_y= 0.0045 p_z= 0.0327
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=6, retained=99.4%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-4.00e+01-2.56e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=3.82e+01-2.79e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=3.86e+01-2.46e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-2.53e+01-8.89e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.60e+01+9.54e-02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-1.51e+01-6.47e-02i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 989; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.017880275
D_W contour = 0.067880275
above global minimum = 0.017863842
8D contour samples = 255

selected phase-space point:
    sqrt_s = 1.1158
theta_p_out = 2.51553
theta_gamma_out = 1.193861
       qOut = 0.08886795
 phi_p_out = 3.709945
phi_gamma_out = 3.087176
   alpha_e = 0.753636
   alpha_p = 1.701103
```

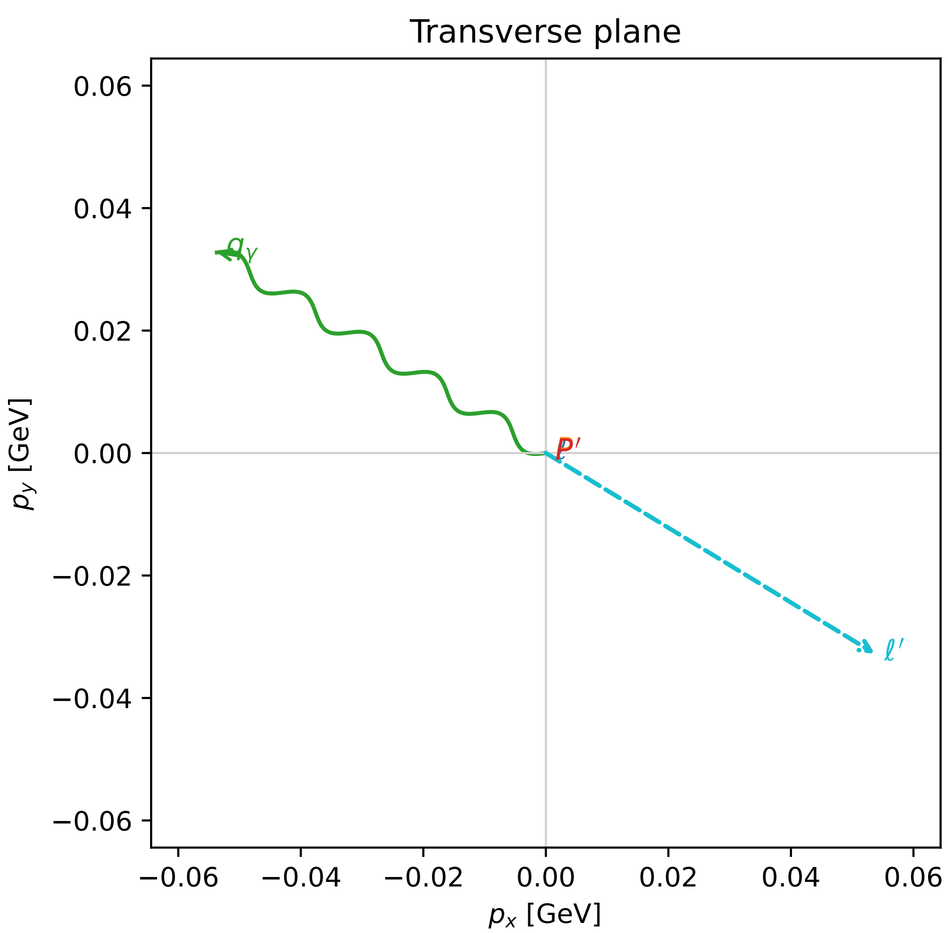
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

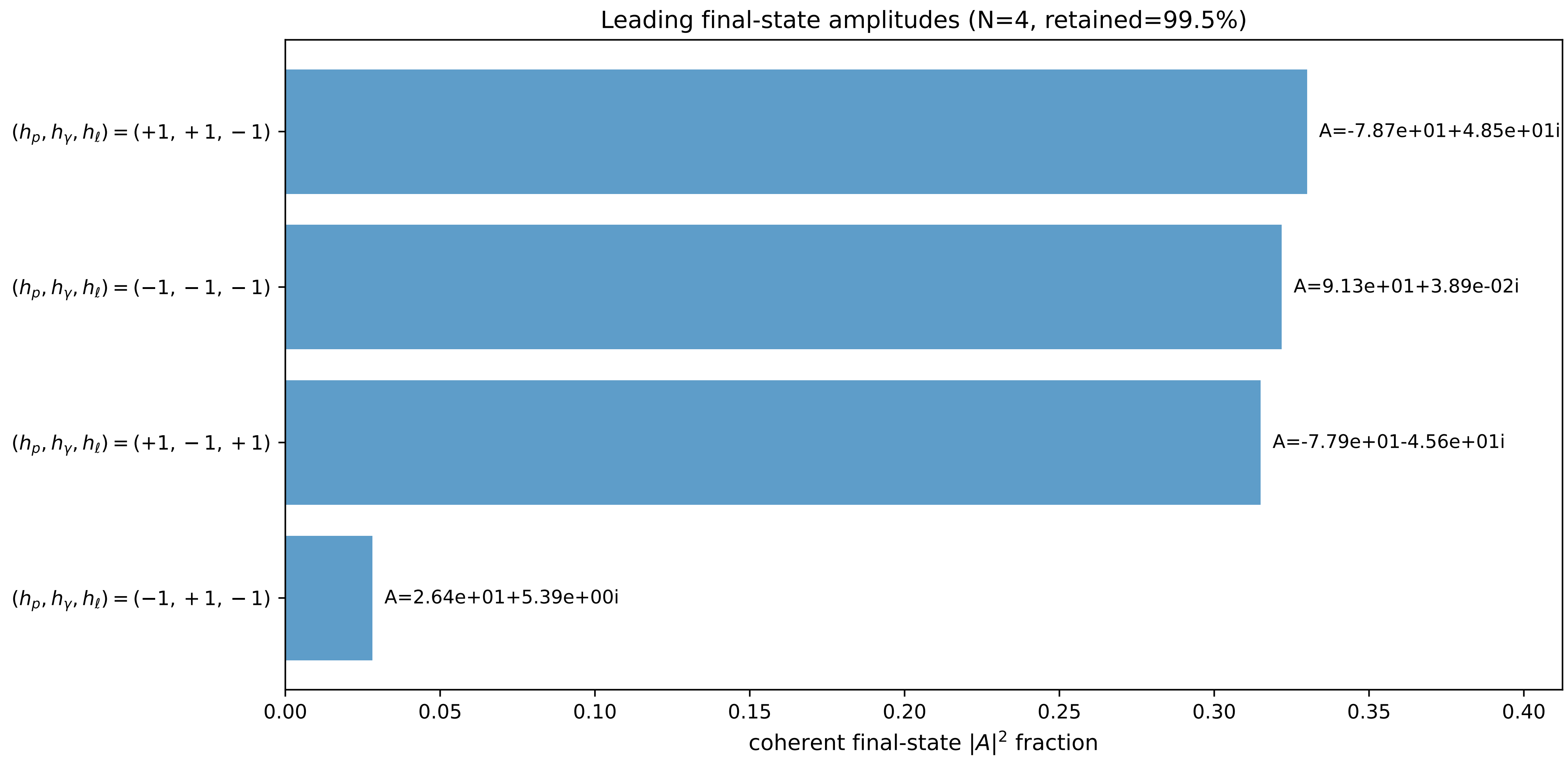


dw_mixing_angles_polarization_02_local_minimum_990 D_W=0.017934
region: polarization_cluster_02_local_minimum_990
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.8026649 rad
incoming lepton: 0.694793|+> +0.71921|->
proton mixing angle: alpha_p=1.6852833 rad
incoming proton: -0.114237|+> +0.993454|->

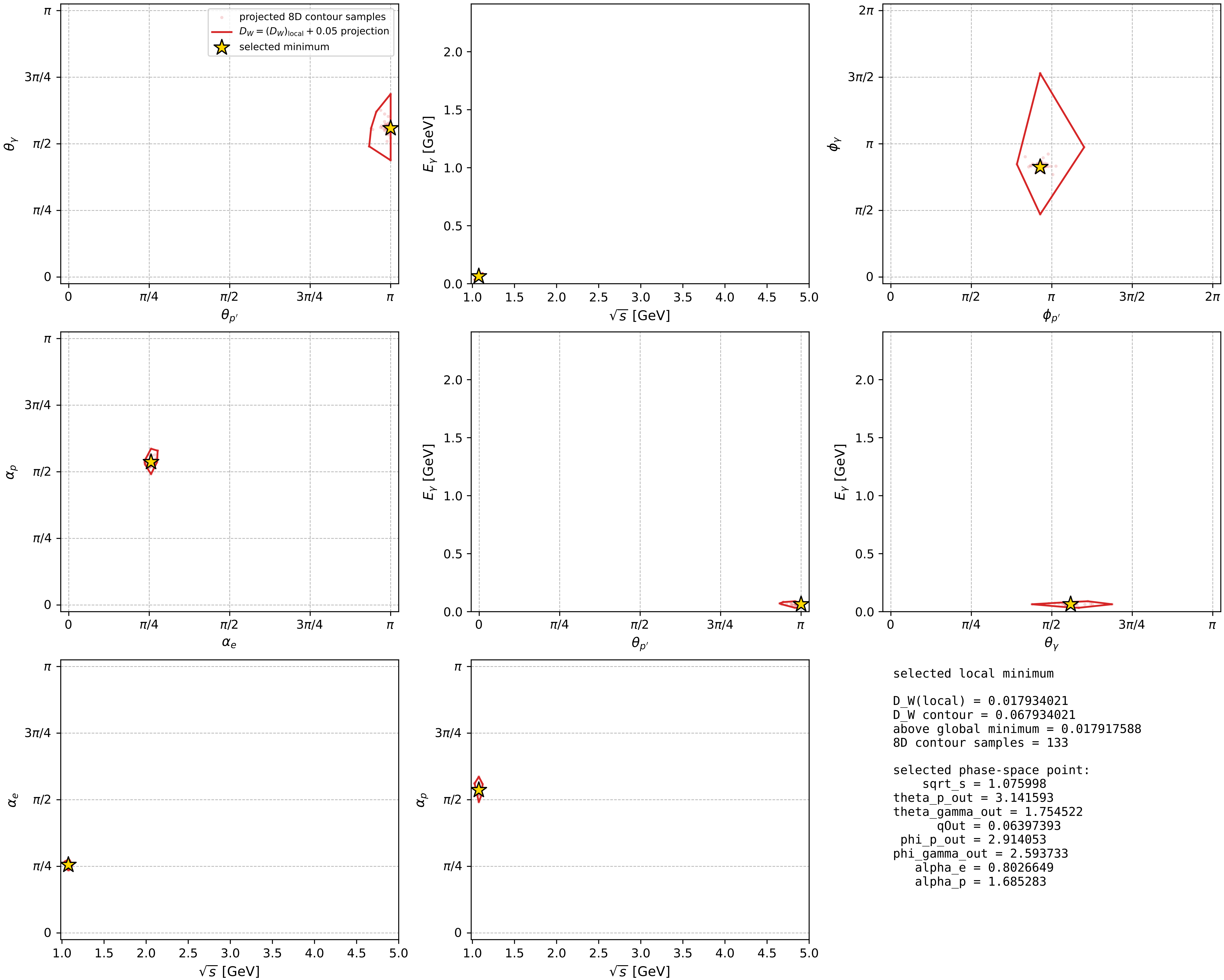
s=1.15777, $\sqrt{s}$=1.076, $|\vec{P}|$=0.129147, $|\vec{P}'|$=0.0266191
$\theta_{p'}$=3.14159, θ_γ=1.75452, $\phi_{p'}$=2.91405, ϕ_γ=2.59373
E_γ=0.0639739, $\theta_{\ell'}$=1.02376, $\phi_{\ell'}$=5.73533
Q^2=0.0289164, x_B=0.19491, t=-0.0241915, W^2=0.999285, y=0.533802

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1291 p_x= 0.0000 p_y= 0.0000 p_z= -0.1291
P E= 0.9468 p_x= 0.0000 p_y= 0.0000 p_z= 0.1291
ℓ' E= 0.0736 p_x= 0.0537 p_y= -0.0328 p_z= 0.0383
P' E= 0.9384 p_x= -0.0000 p_y= 0.0000 p_z= -0.0266
q_γ E= 0.0640 p_x= -0.0537 p_y= 0.0328 p_z= -0.0117

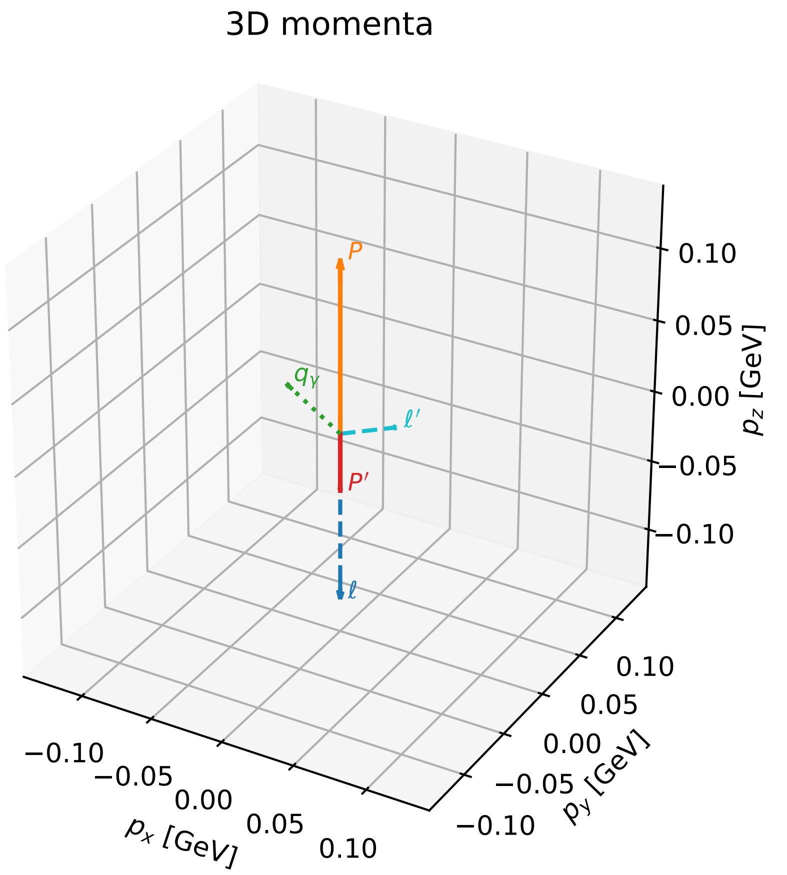




electron: polarization cluster P2, local minimum 990; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



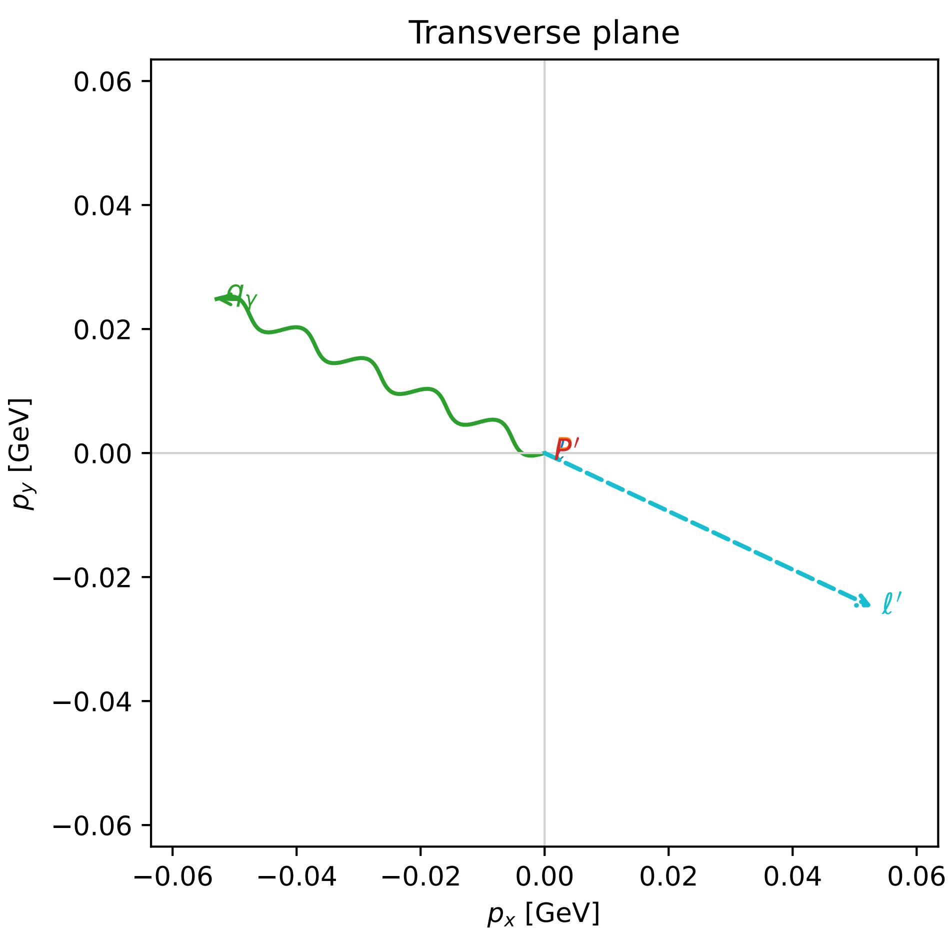
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

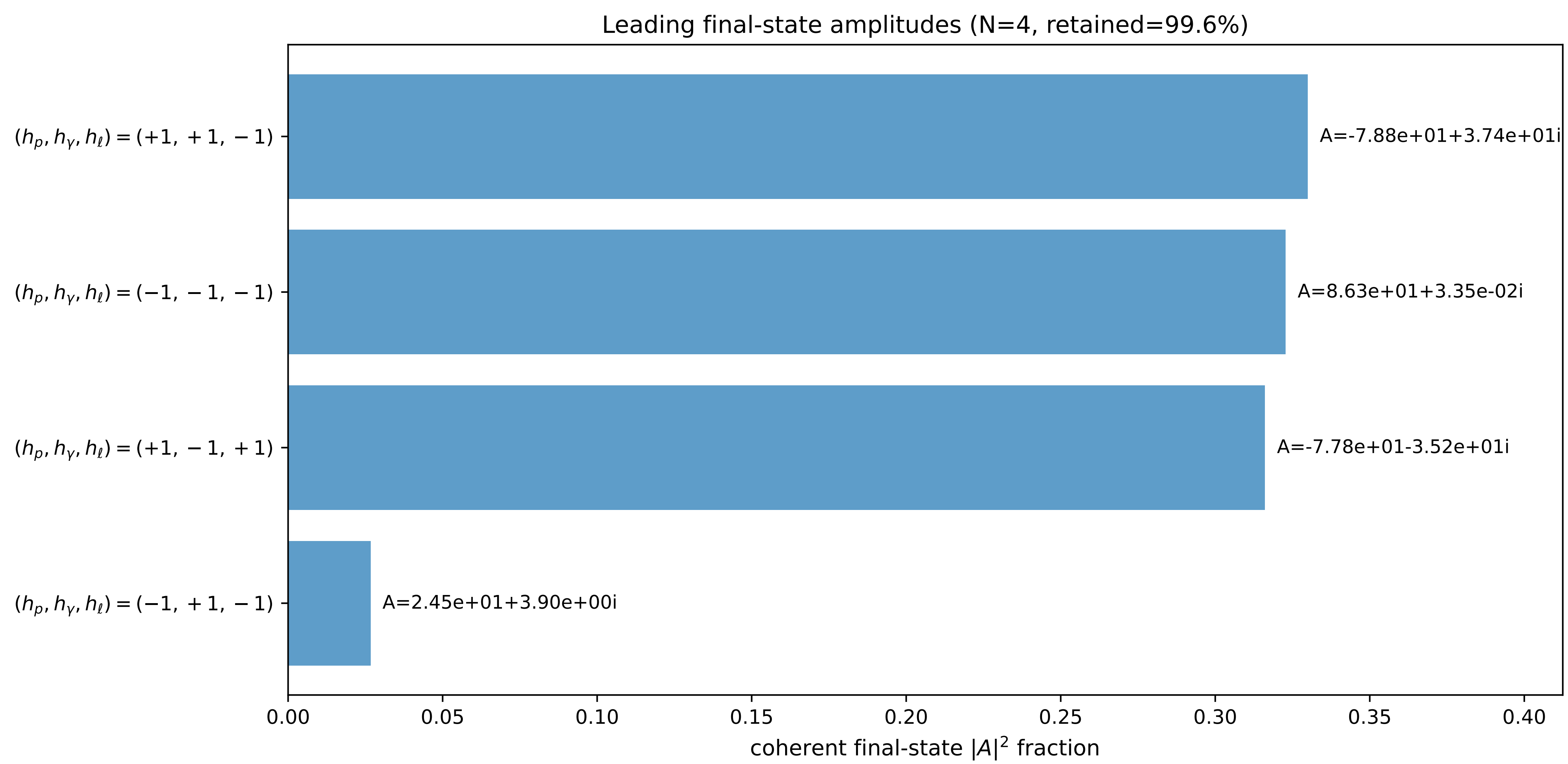


dw_mixing_angles_polarization_02_local_minimum_993 D_W=0.0180227
region: polarization_cluster_02_local_minimum_993
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.80125755 rad
incoming lepton: 0.695804|+> +0.718232|->
proton mixing angle: alpha_p=1.6781322 rad
incoming proton: -0.10713|+> +0.994245|->

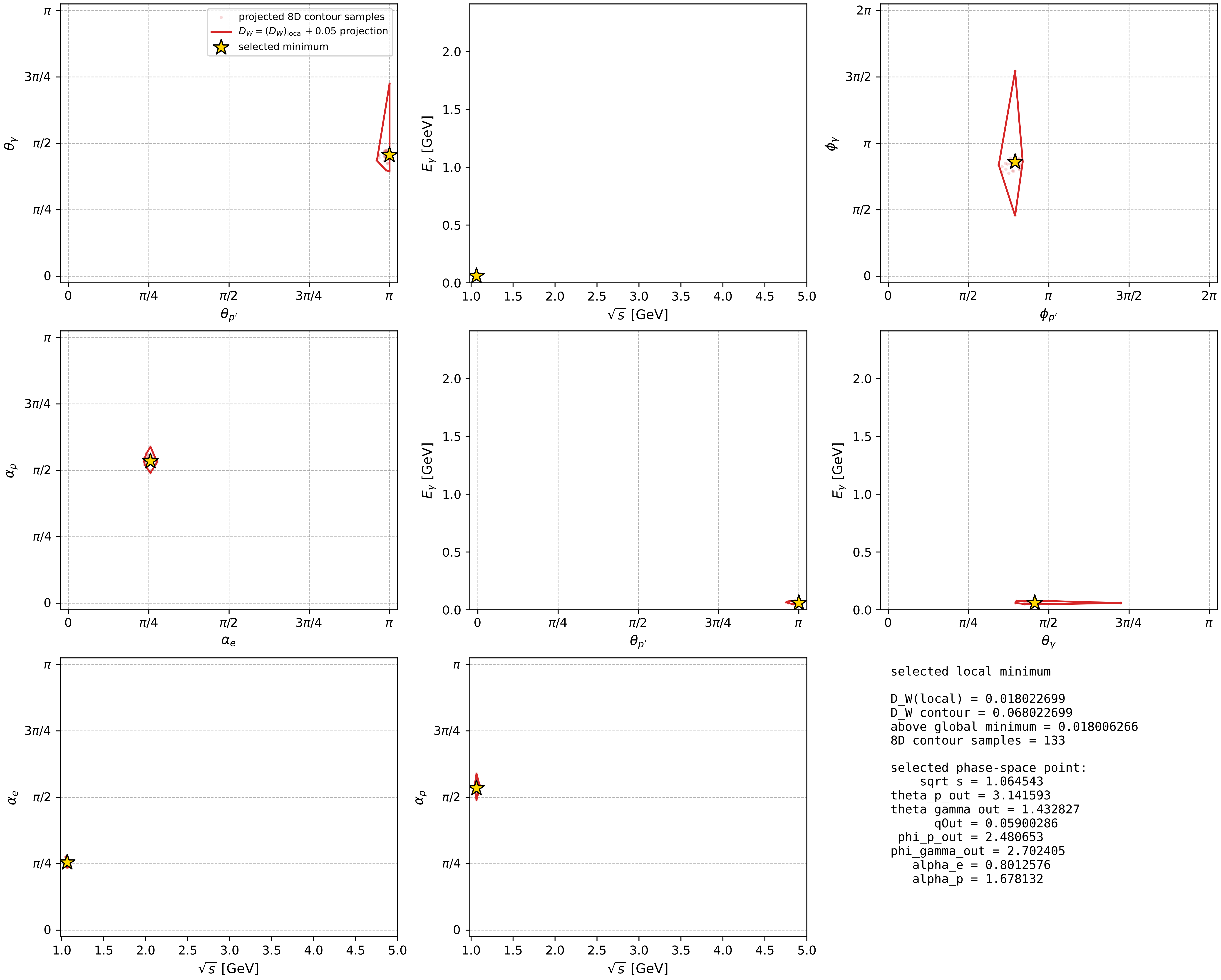
s=1.13325, $\sqrt{s}$=1.06454, $|\vec{P}|$=0.119021, $|\vec{P}'|$=0.0402126
$\theta_{p'}$=3.14159, θ_γ=1.43283, $\phi_{p'}$=2.48065, ϕ_γ=2.7024
E_γ=0.0590029, $\theta_{\ell'}$=1.06855, $\phi_{\ell'}$=5.844
Q^2=0.0235125, x_B=0.174223, t=-0.0253109, W^2=0.991288, y=0.532566

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1190 p_x= 0.0000 p_y= 0.0000 p_z= -0.1190
P E= 0.9455 p_x= 0.0000 p_y= 0.0000 p_z= 0.1190
ℓ' E= 0.0667 p_x= 0.0529 p_y= -0.0248 p_z= 0.0321
P' E= 0.9389 p_x= -0.0000 p_y= 0.0000 p_z= -0.0402
q_γ E= 0.0590 p_x= -0.0529 p_y= 0.0248 p_z= 0.0081

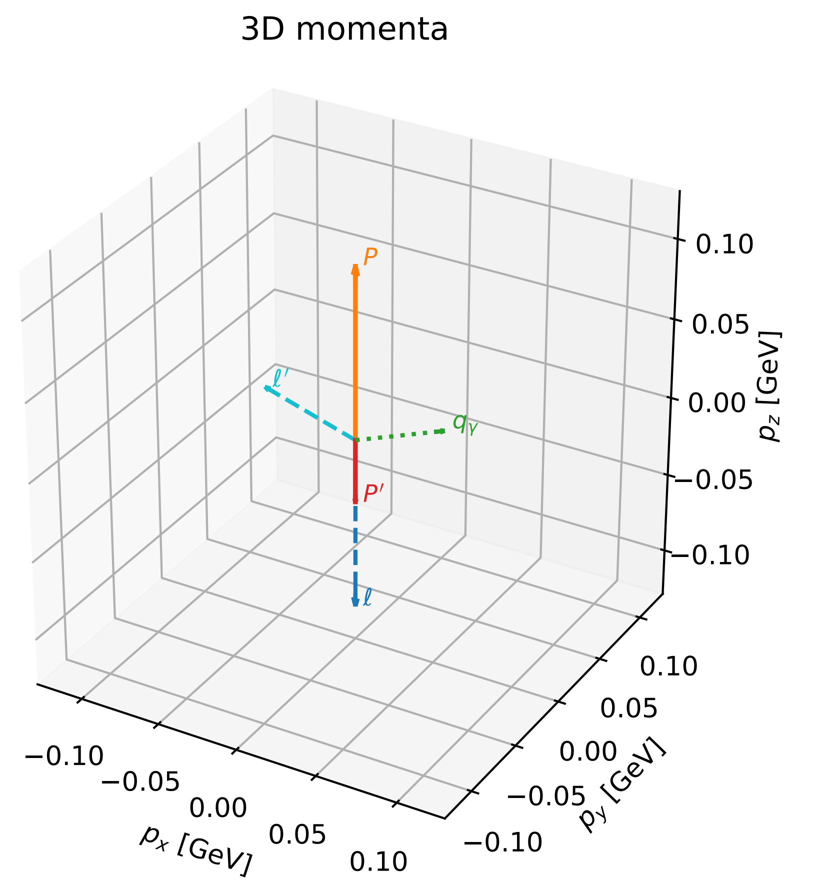




electron: polarization cluster P2, local minimum 993; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



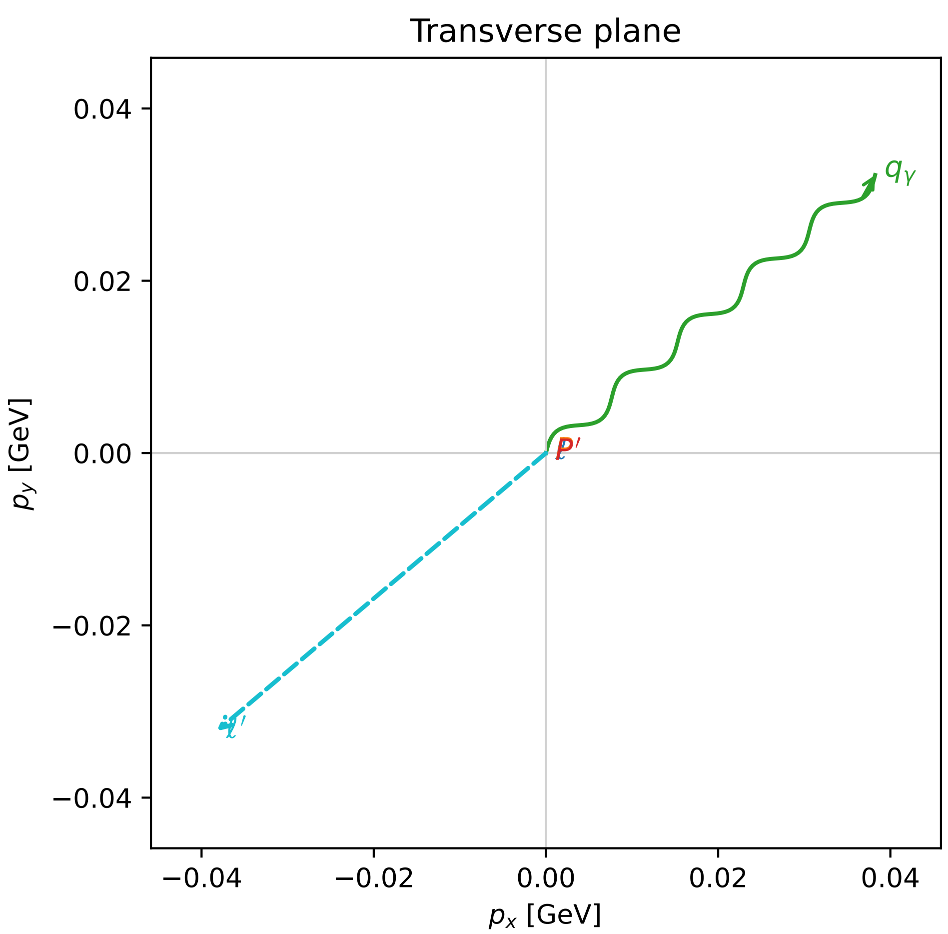
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

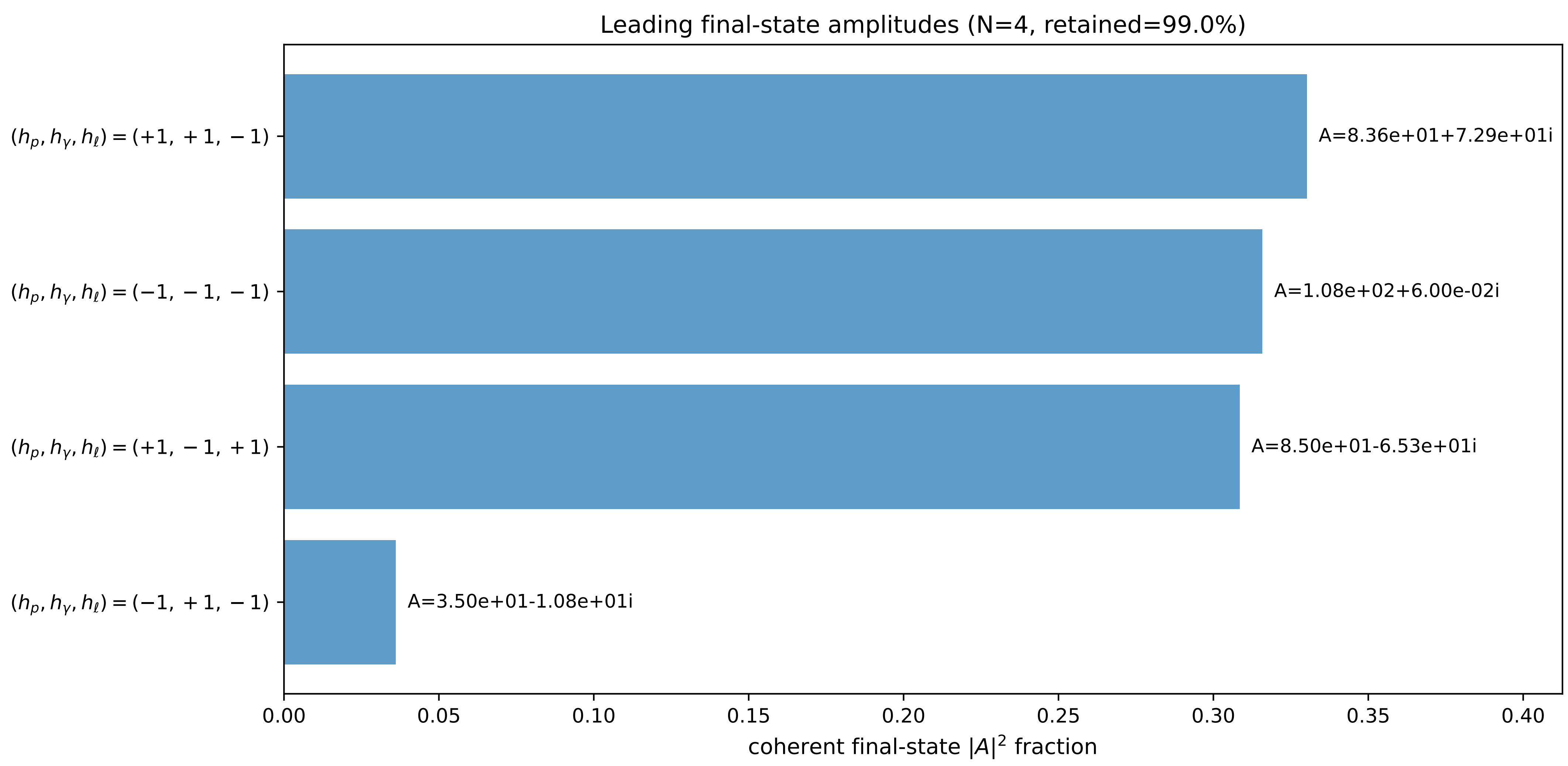


dw_mixing_angles_polarization_02_local_minimum_1000 D_W=0.0183785
region: polarization_cluster_02_local_minimum_1000
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.827528 rad
incoming lepton: 0.676698|+> +0.736261|->
proton mixing angle: alpha_p=1.42067 rad
incoming proton: 0.149563|+> +0.988752|->

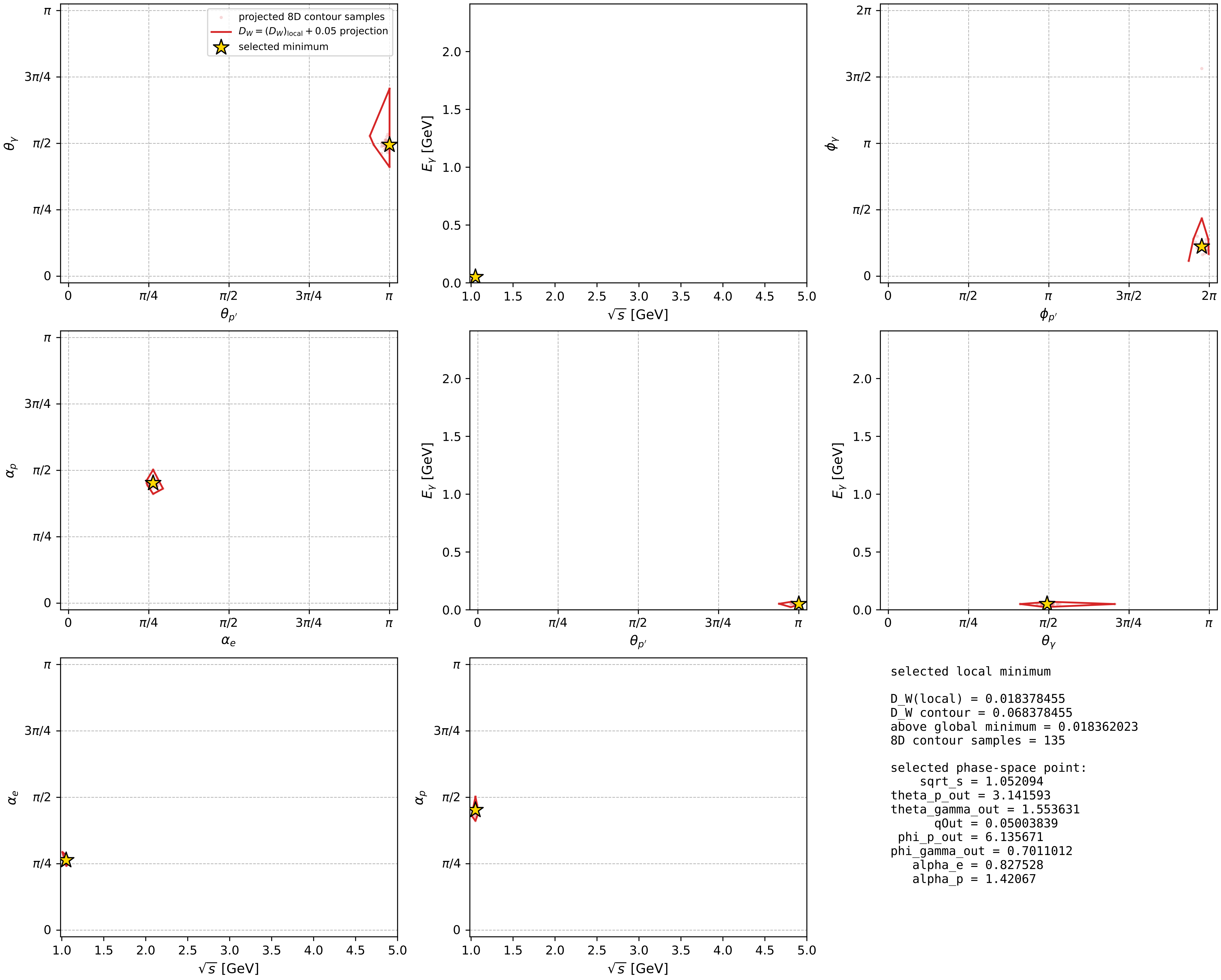
s=1.1069, $\sqrt{s}$=1.05209, $|\vec{P}|$=0.107906, $|\vec{P}'|$=0.0395095
$\theta_{p'}$=3.14159, θ_γ=1.55363, $\phi_{p'}$=6.13567, ϕ_γ=0.701101
E_γ=0.0500384, $\theta_{\ell'}$=0.913029, $\phi_{\ell'}$=3.84269
Q^2=0.0219854, x_B=0.189515, t=-0.0217027, W^2=0.973867, y=0.510922

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1079 p_x= 0.0000 p_y= 0.0000 p_z= -0.1079
P E= 0.9442 p_x= 0.0000 p_y= 0.0000 p_z= 0.1079
ℓ' E= 0.0632 p_x= -0.0382 p_y= -0.0323 p_z= 0.0387
P' E= 0.9388 p_x= 0.0000 p_y= -0.0000 p_z= -0.0395
q_γ E= 0.0500 p_x= 0.0382 p_y= 0.0323 p_z= 0.0009

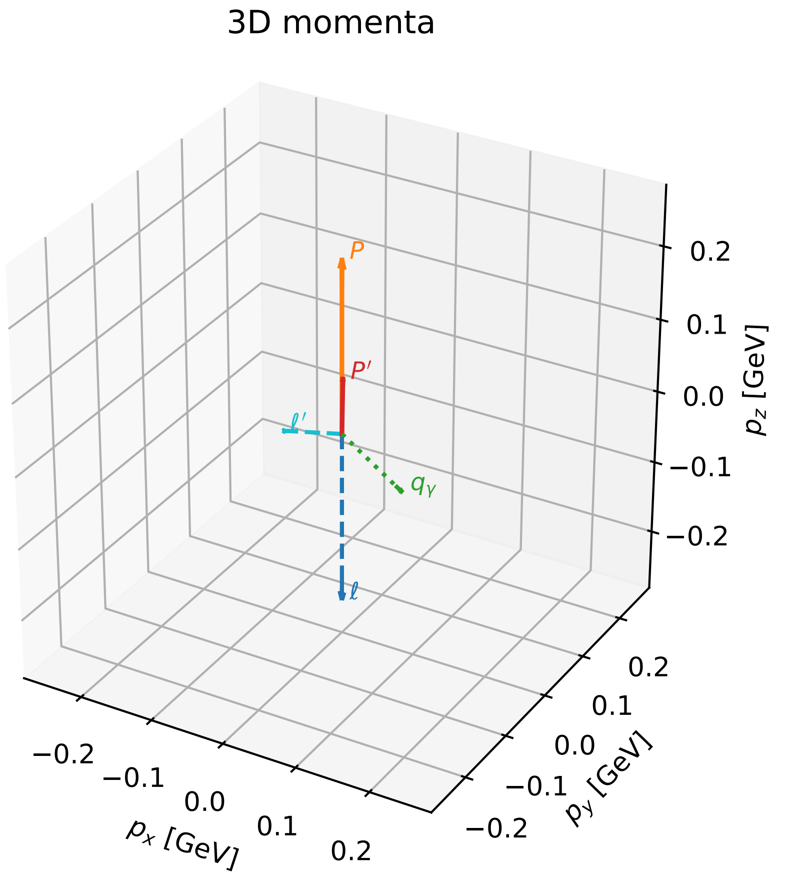




electron: polarization cluster P2, local minimum 1000; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



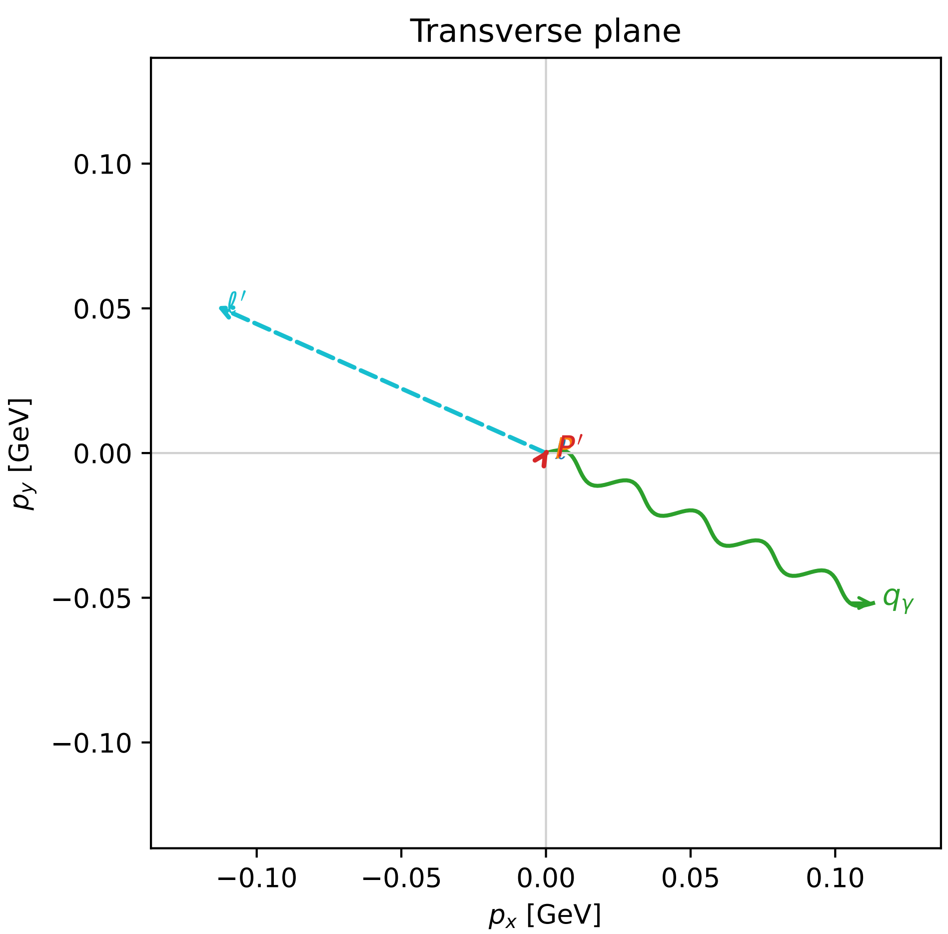
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

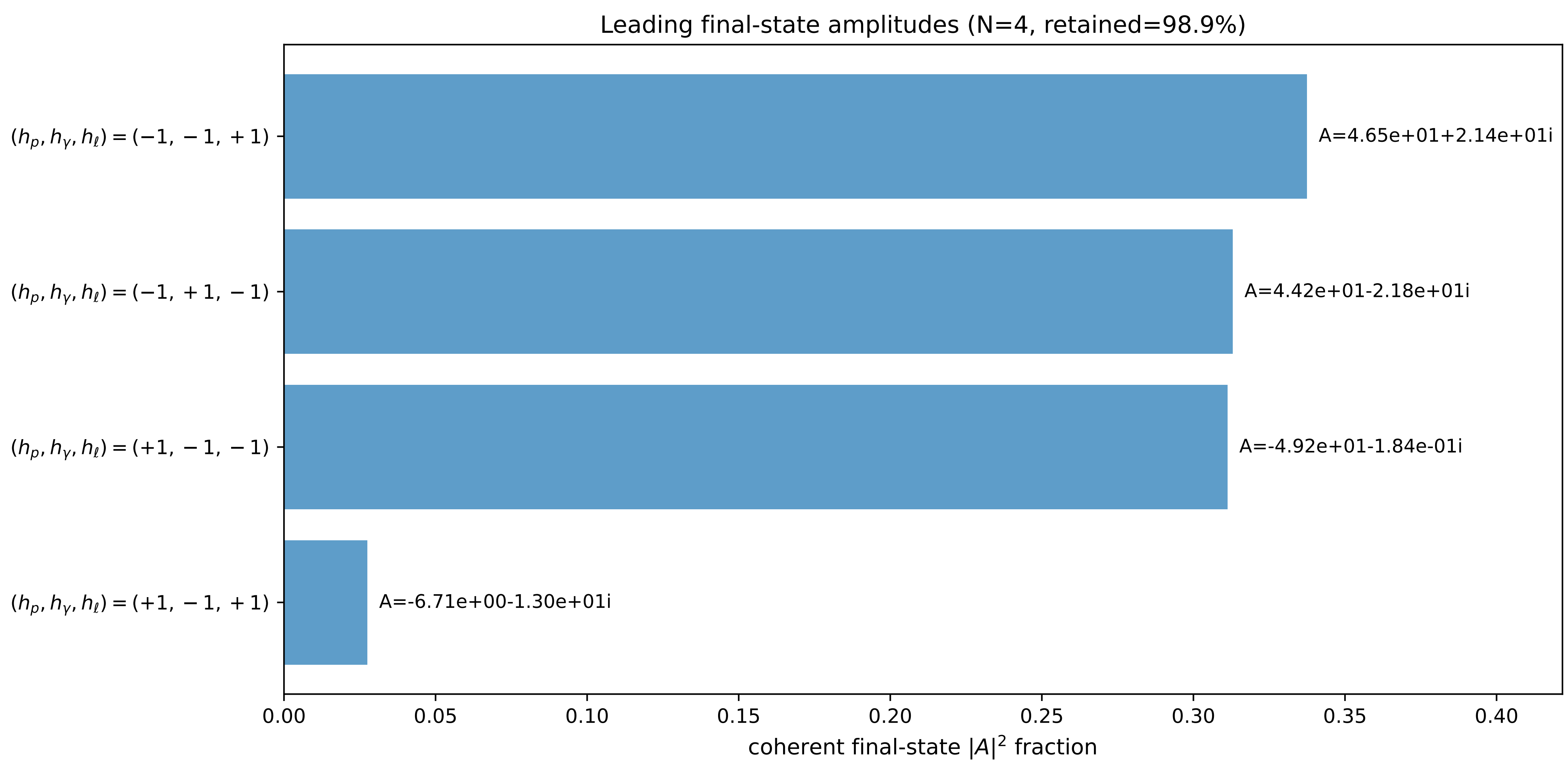



```
dw_mixing_angles_polarization_02_local_minimum_1007  D_W=0.0187718
region: polarization_cluster_02_local_minimum_1007
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75008963 rad
incoming lepton: 0.731628|+> +0.681704|->
proton mixing angle: alpha_p=1.4205603 rad
incoming proton: 0.149671|+> +0.988736|->
```

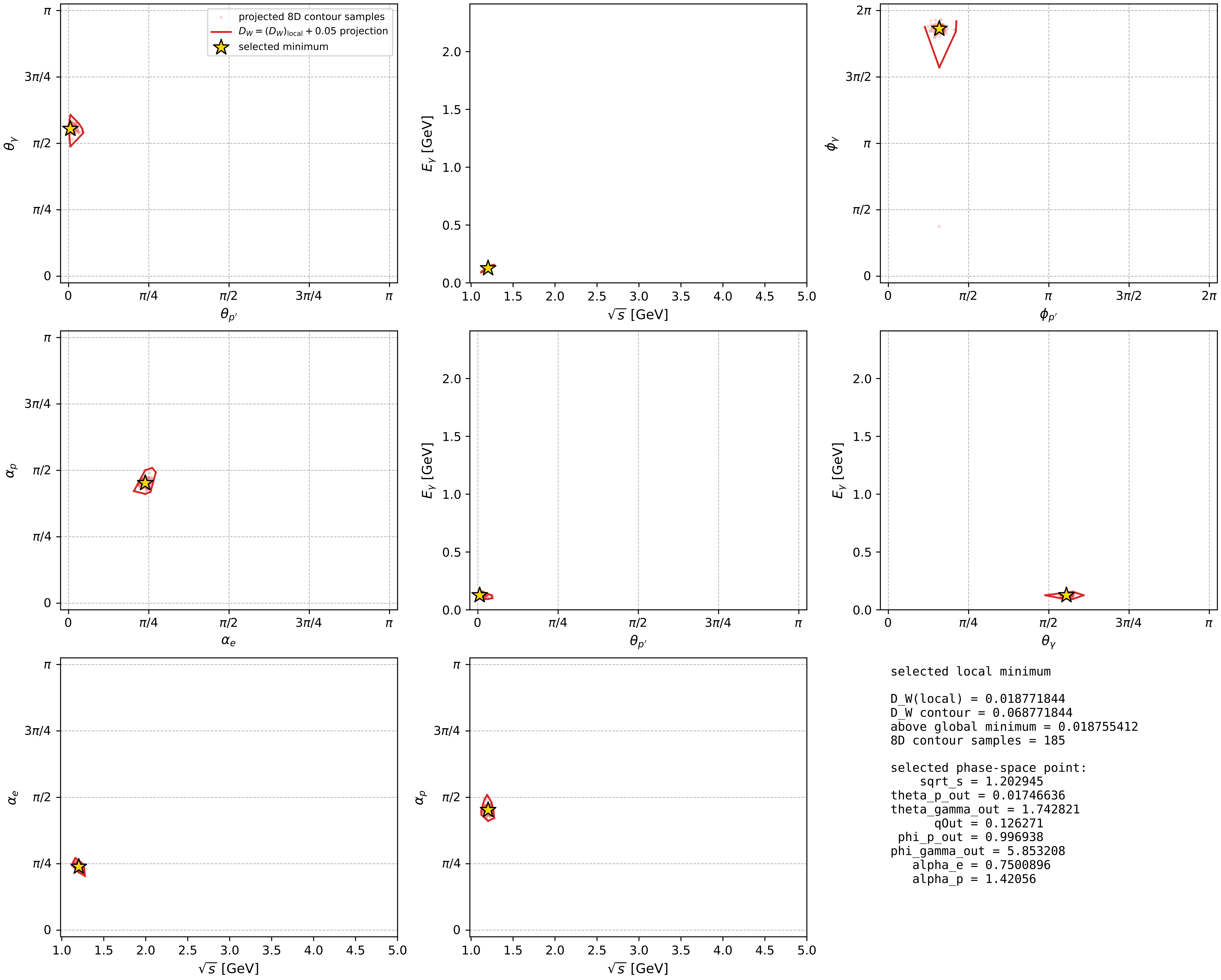
s=1.44708, $\sqrt{s}$=1.20294, $|\vec{P}|$=0.235768, $|\vec{P}'|$=0.0752668
$\theta_{p'}$=0.0174664, θ_γ=1.74282, $\phi_{p'}$=0.996938, ϕ_γ=5.85321
E_γ=0.126271, $\theta_{\ell'}$=1.97731, $\phi_{\ell'}$=2.72206
Q^2=0.0386742, x_B=0.138356, t=-0.0250815, W^2=1.1207, y=0.49279

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2358 p_x= 0.0000 p_y= 0.0000 p_z= -0.2358
P E= 0.9672 p_x= 0.0000 p_y= 0.0000 p_z= 0.2358
ℓ' E= 0.1357 p_x= -0.1138 p_y= 0.0508 p_z= -0.0536
P' E= 0.9410 p_x= 0.0007 p_y= 0.0011 p_z= 0.0753
q_γ E= 0.1263 p_x= 0.1131 p_y= -0.0519 p_z= -0.0216

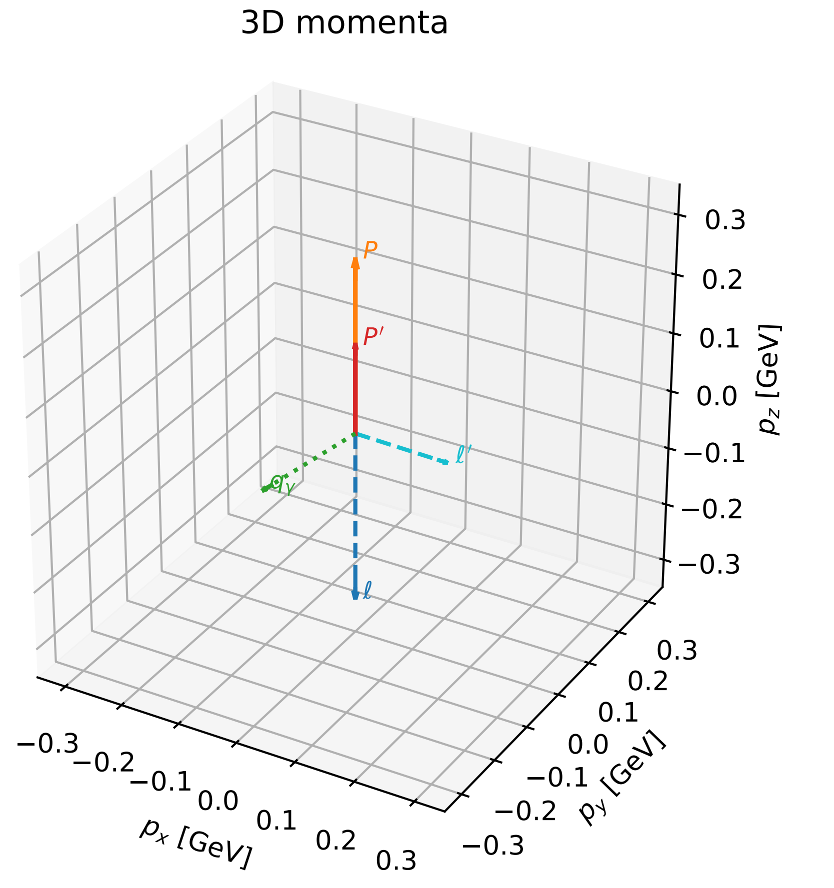




electron: polarization cluster P2, local minimum 1007; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



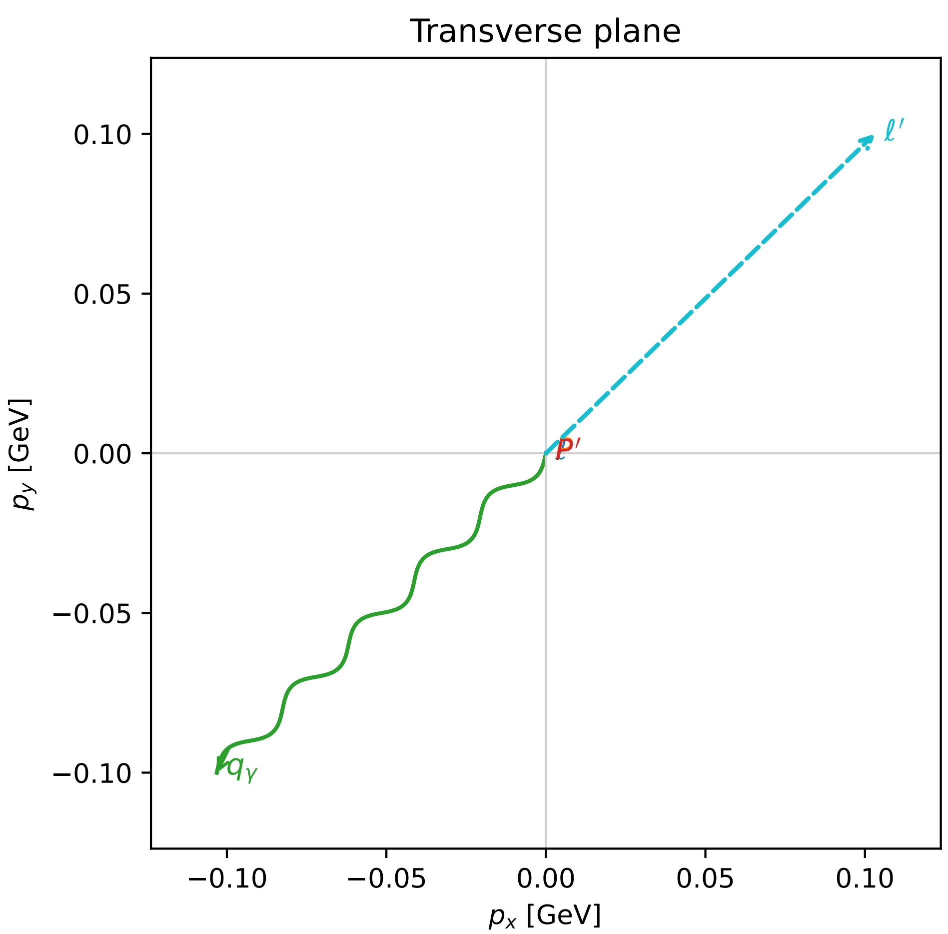
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

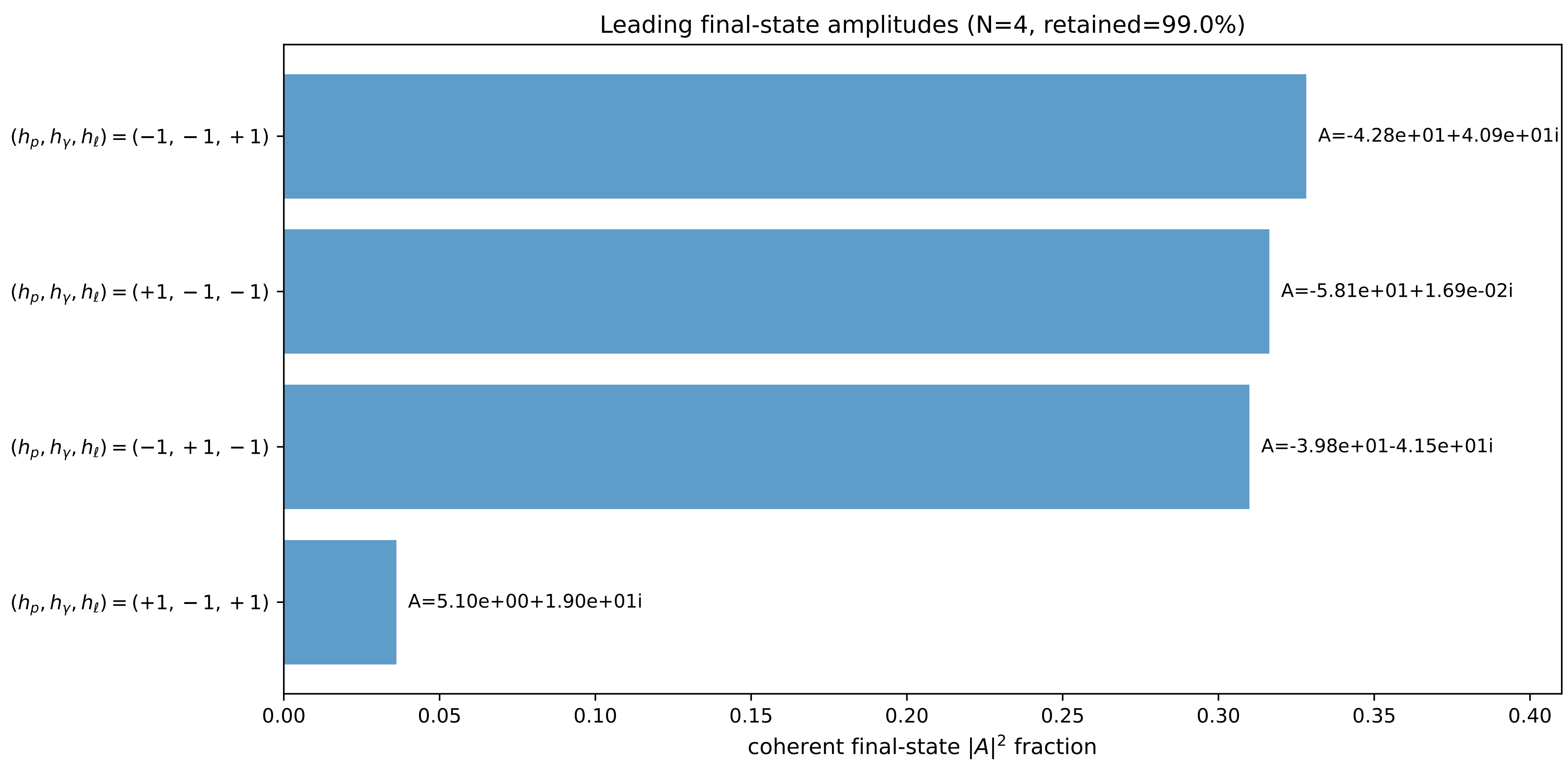


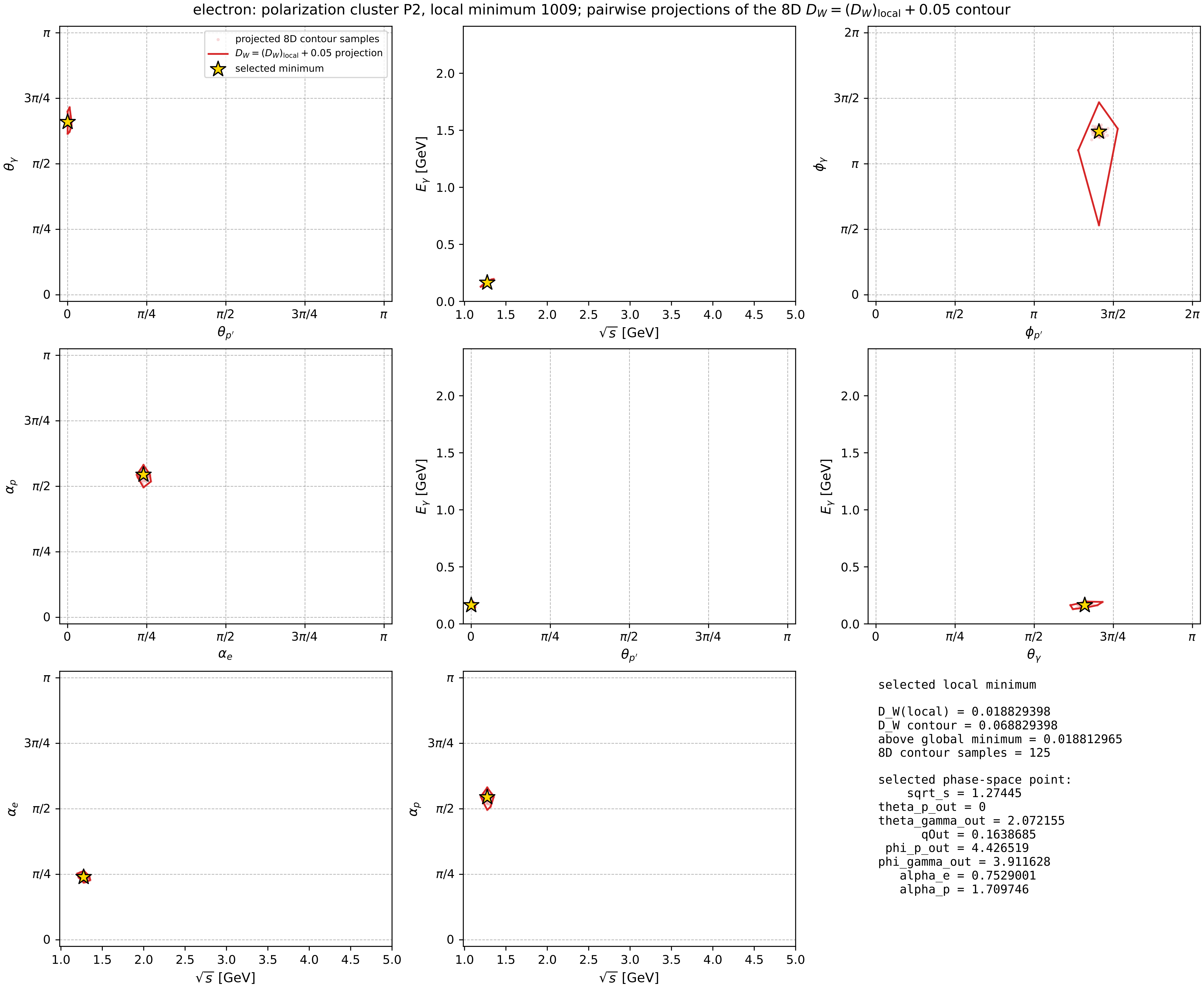
dw_mixing_angles_polarization_02_local_minimum_1009 D_W=0.0188294
region: polarization_cluster_02_local_minimum_1009
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75290014 rad
incoming lepton: 0.729709|+> +0.683758|->
proton mixing angle: alpha_p=1.7097462 rad
incoming proton: -0.138503|+> +0.990362|->

s=1.62422, $\sqrt{s}$=1.27445, $|\vec{P}|$=0.292039, $|\vec{P}'|$=0.150449
$\theta_{p'}$=0, θ_γ=2.07215, $\phi_{p'}$=4.42652, ϕ_γ=3.91163
E_γ=0.163869, $\theta_{\ell'}$=2.03356, $\phi_{\ell'}$=0.770035
Q^2=0.0519249, x_B=0.134183, t=-0.0189964, W^2=1.21489, y=0.519856

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2920 p_x= 0.0000 p_y= 0.0000 p_z= -0.2920
P E= 0.9824 p_x= 0.0000 p_y= 0.0000 p_z= 0.2920
ℓ' E= 0.1606 p_x= 0.1032 p_y= 0.1000 p_z= -0.0717
P' E= 0.9500 p_x= -0.0000 p_y= -0.0000 p_z= 0.1504
q_γ E= 0.1639 p_x= -0.1032 p_y= -0.1000 p_z= -0.0788

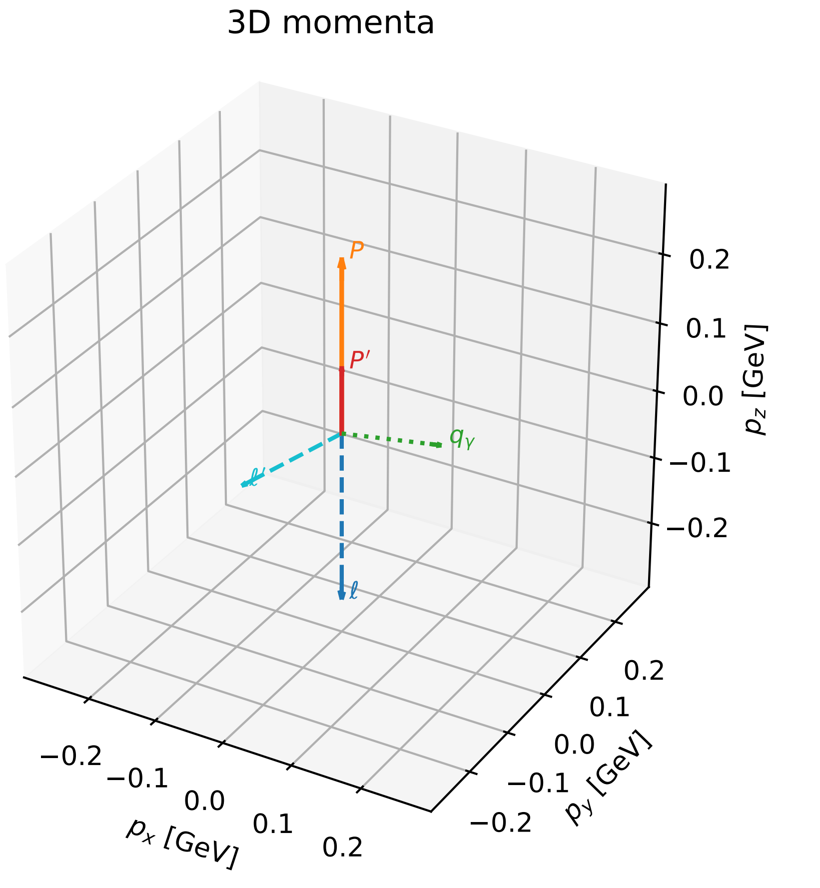






electron: polarization cluster P2, local minimum 1009; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.018829398
D_W contour = 0.068829398
above global minimum = 0.018812965
8D contour samples = 125
selected phase-space point:
sqrt_s = 1.27445
theta_p_out = 0
theta_gamma_out = 2.072155
qOut = 0.1638685
phi_p_out = 4.426519
phi_gamma_out = 3.911628
alpha_e = 0.7529001
alpha_p = 1.709746

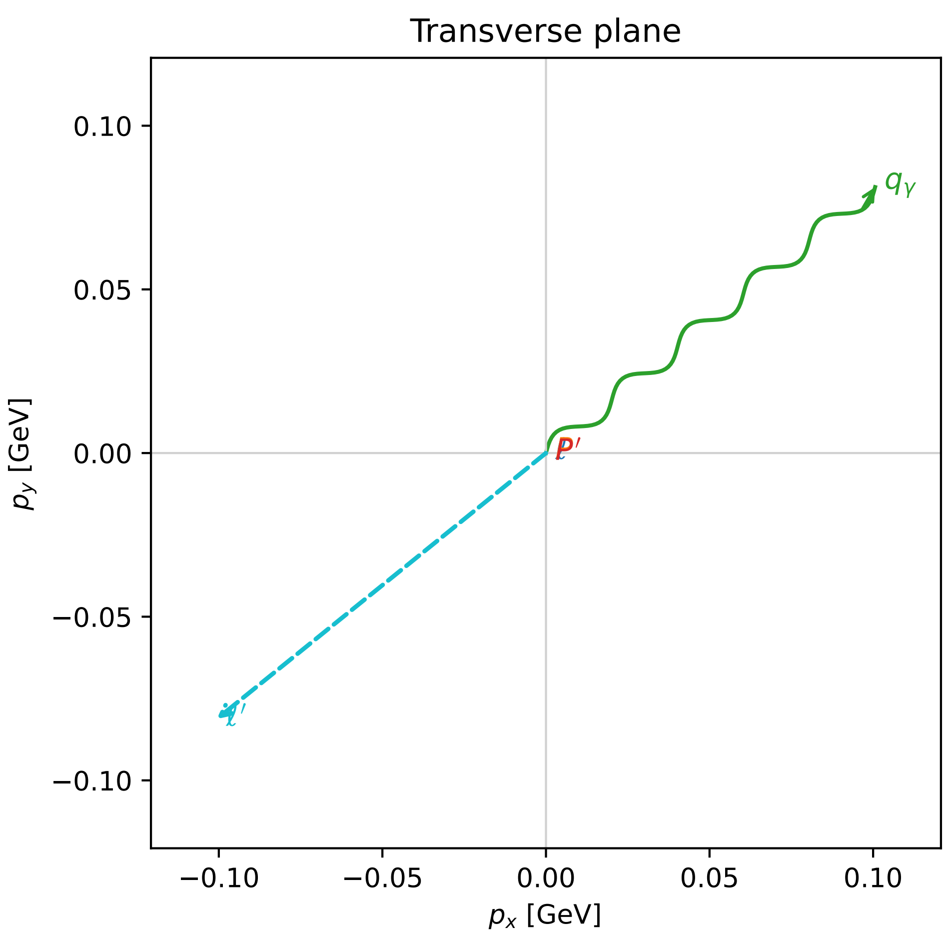
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

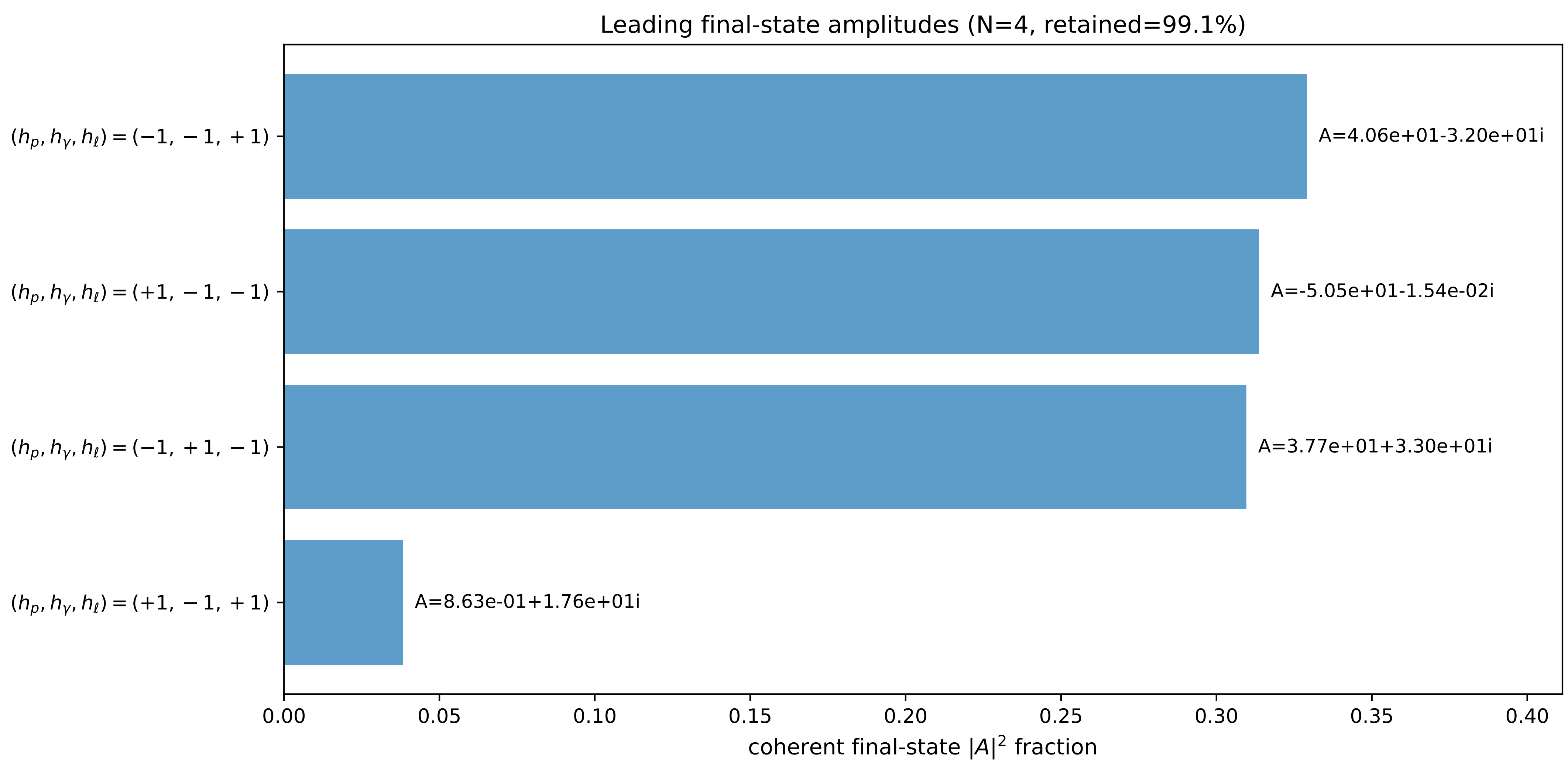


dw_mixing_angles_polarization_02_local_minimum_1044 D_W=0.0213107
region: polarization_cluster_02_local_minimum_1044
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.75455545 rad
incoming lepton: 0.728576|+> +0.684965|->
proton mixing angle: alpha_p=1.4304809 rad
incoming proton: 0.139855|+> +0.990172|->

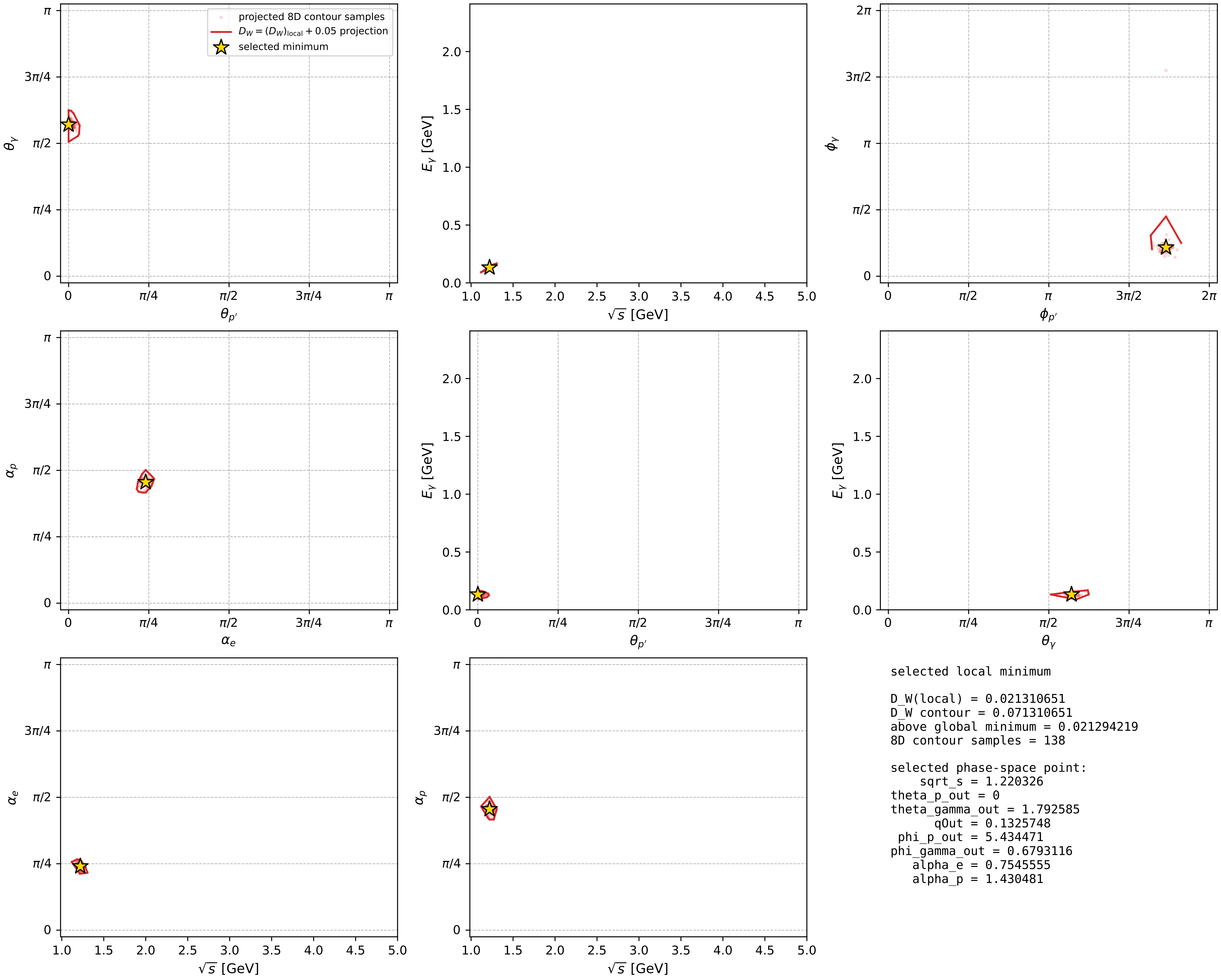
s=1.4892, $\sqrt{s}$=1.22033, $|\vec{P}|$=0.249667, $|\vec{P}'|$=0.0946948
$\theta_{p'}$=0, θ_γ=1.79259, $\phi_{p'}$=5.43447, ϕ_γ=0.679312
E_γ=0.132575, $\theta_{\ell'}$=2.0398, $\phi_{\ell'}$=3.8209
Q^2=0.0396727, x_B=0.134406, t=-0.0232386, W^2=1.13534, y=0.484401

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2497 p_x= 0.0000 p_y= 0.0000 p_z= -0.2497
P E= 0.9707 p_x= 0.0000 p_y= 0.0000 p_z= 0.2497
ℓ' E= 0.1450 p_x= -0.1006 p_y= -0.0813 p_z= -0.0655
P' E= 0.9428 p_x= 0.0000 p_y= -0.0000 p_z= 0.0947
q_γ E= 0.1326 p_x= 0.1006 p_y= 0.0813 p_z= -0.0292

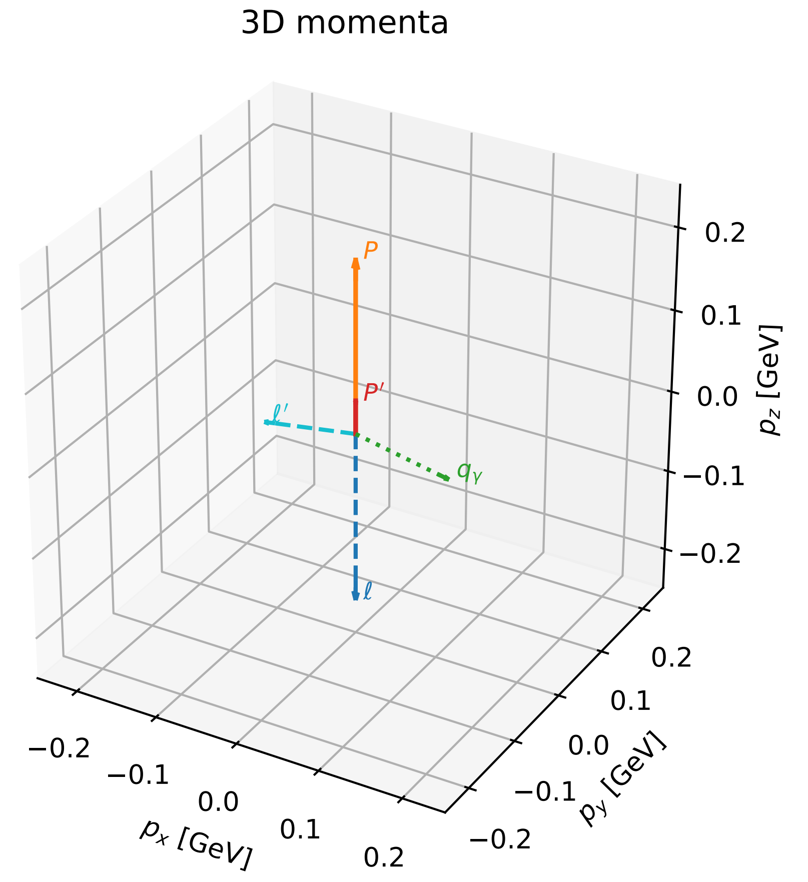




electron: polarization cluster P2, local minimum 1044; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



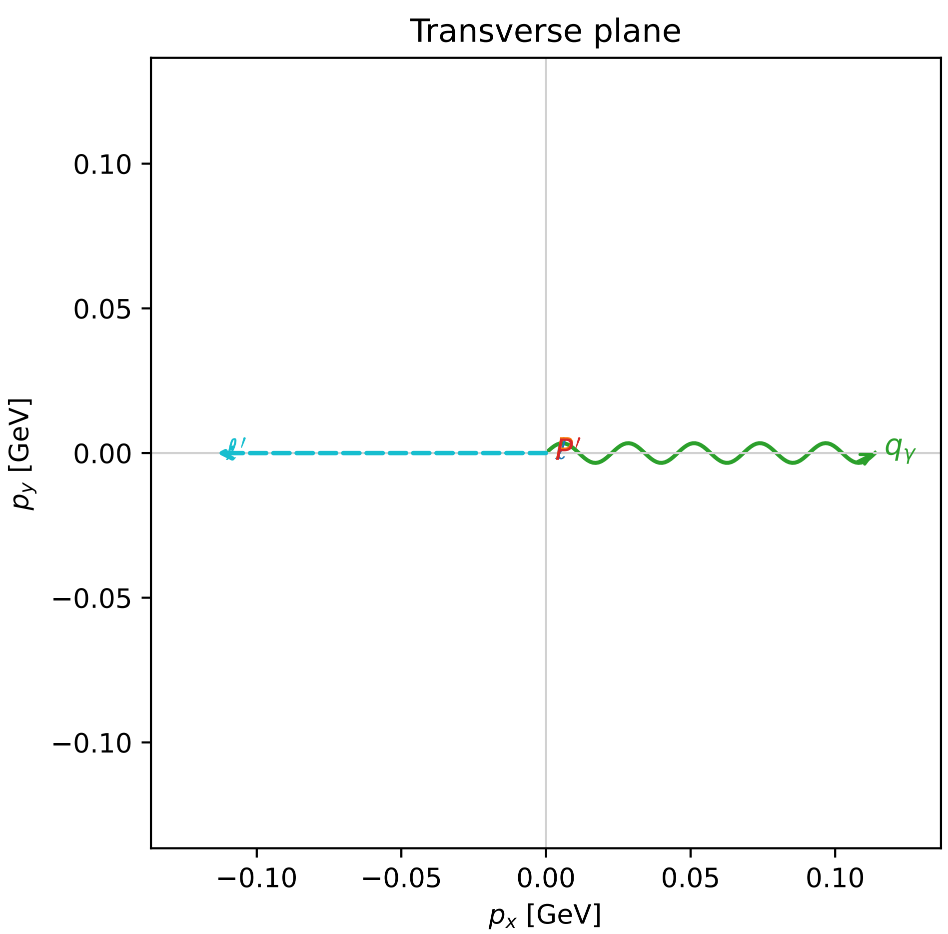
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

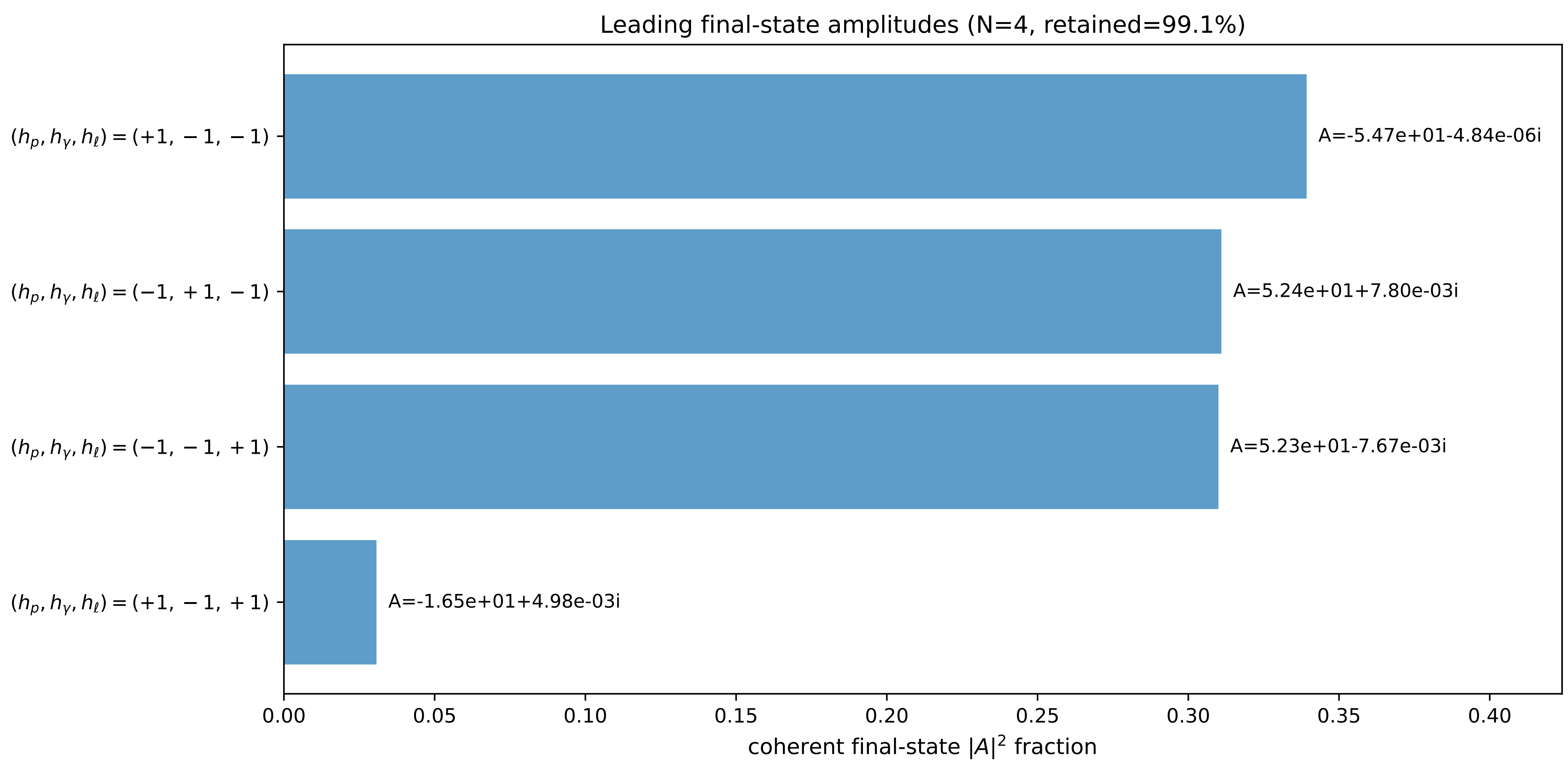


dw_mixing_angles_polarization_02_local_minimum_1057 D_W=0.0235395
region: polarization_cluster_02_local_minimum_1057
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78880565 rad
incoming lepton: 0.704693|+> +0.709512|->
proton mixing angle: alpha_p=1.5549813 rad
incoming proton: 0.0158144|+> +0.999875|->

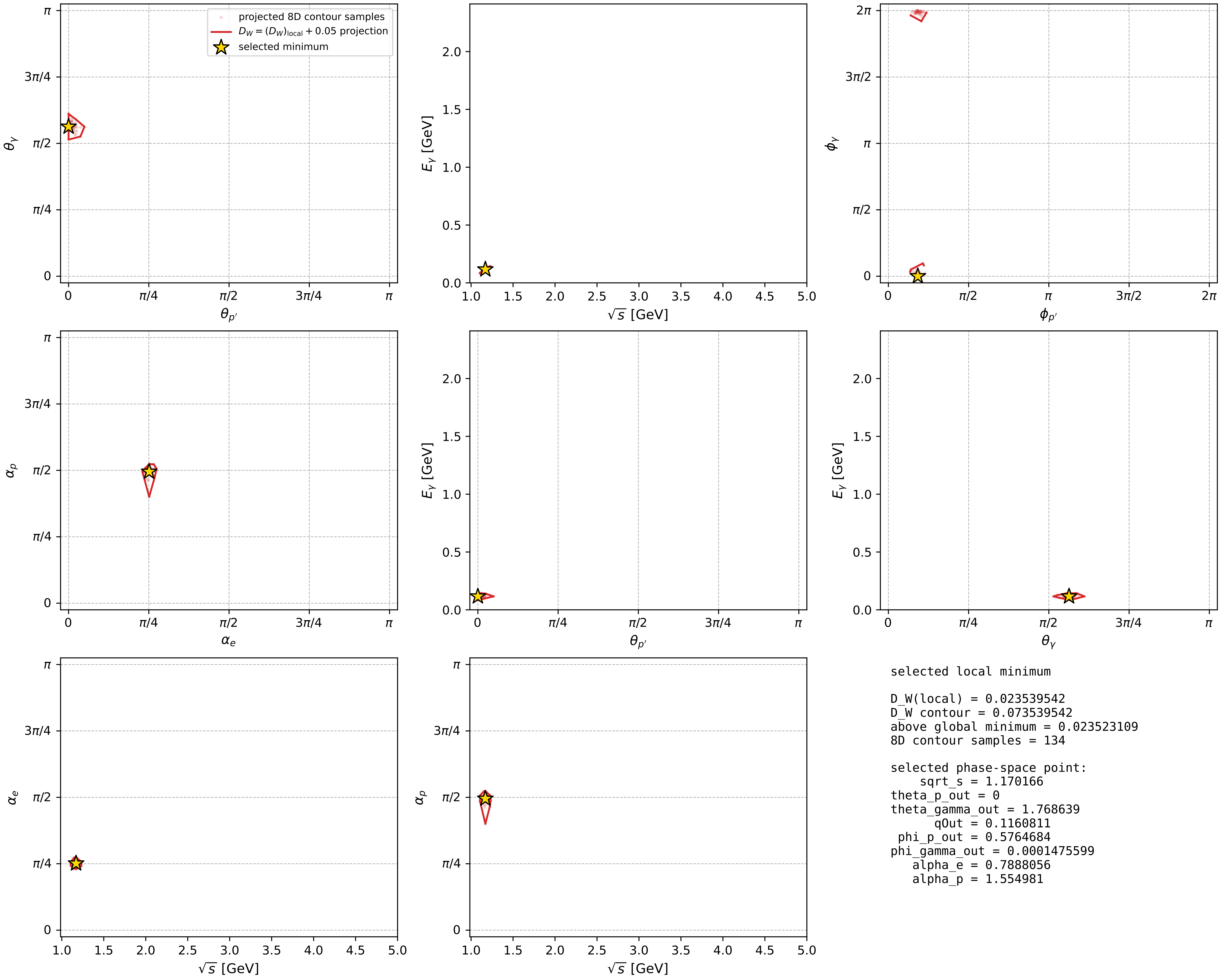
s=1.36929, $\sqrt{s}$=1.17017, $|\vec{P}|$=0.209134, $|\vec{P}'|$=0.0406376
$\theta_{p'}$=0, θ_γ=1.76864, $\phi_{p'}$=0.576468, ϕ_γ=0.00014756
E_γ=0.116081, $\theta_{\ell'}$=1.72612, $\phi_{\ell'}$=3.14174
Q^2=0.0407318, x_B=0.156325, t=-0.0279003, W^2=1.09967, y=0.532357

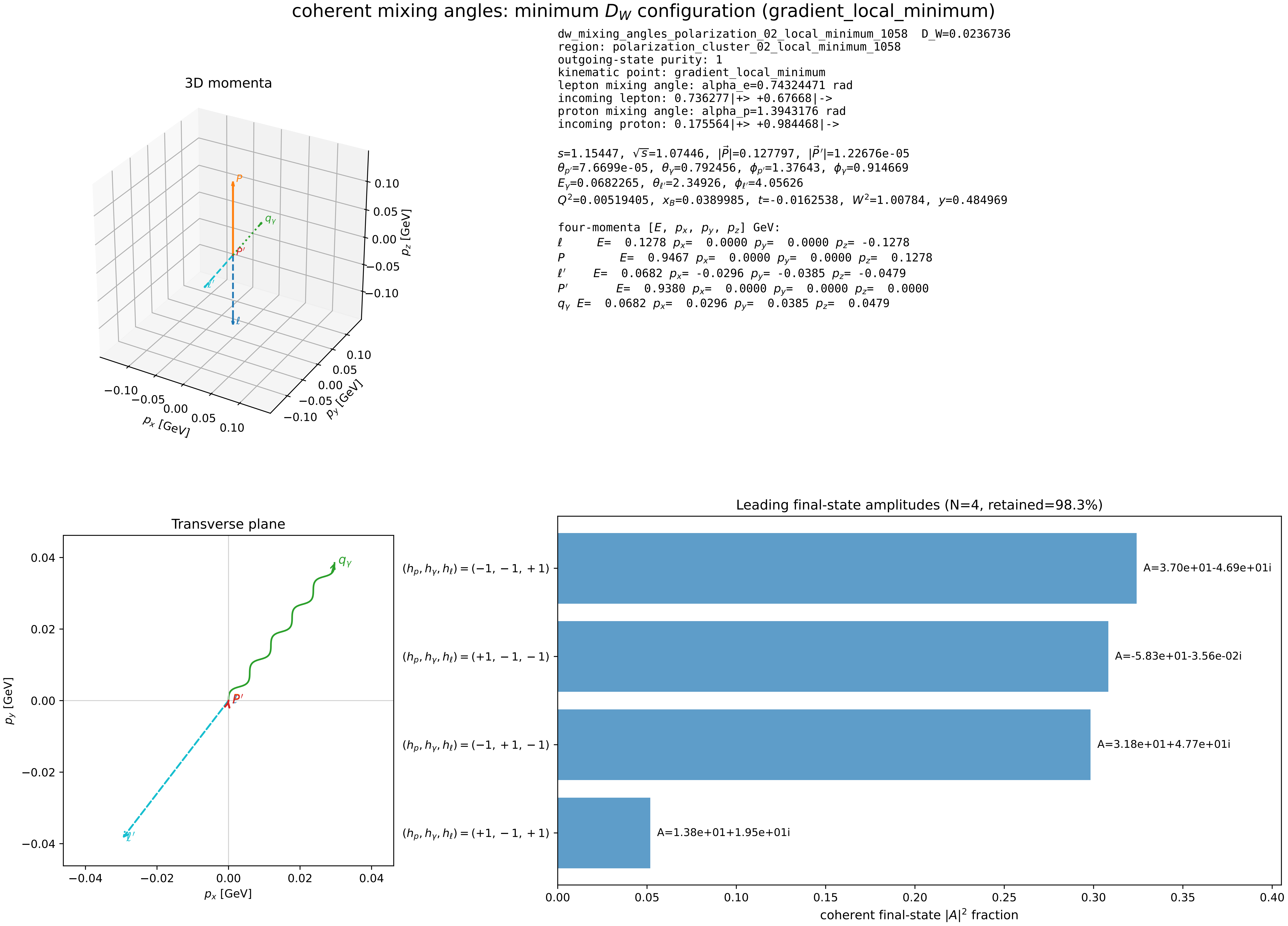
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2091 p_x= 0.0000 p_y= 0.0000 p_z= -0.2091
P E= 0.9610 p_x= 0.0000 p_y= 0.0000 p_z= 0.2091
ℓ' E= 0.1152 p_x= -0.1138 p_y= -0.0000 p_z= -0.0178
P' E= 0.9389 p_x= 0.0000 p_y= 0.0000 p_z= 0.0406
q_γ E= 0.1161 p_x= 0.1138 p_y= 0.0000 p_z= -0.0228



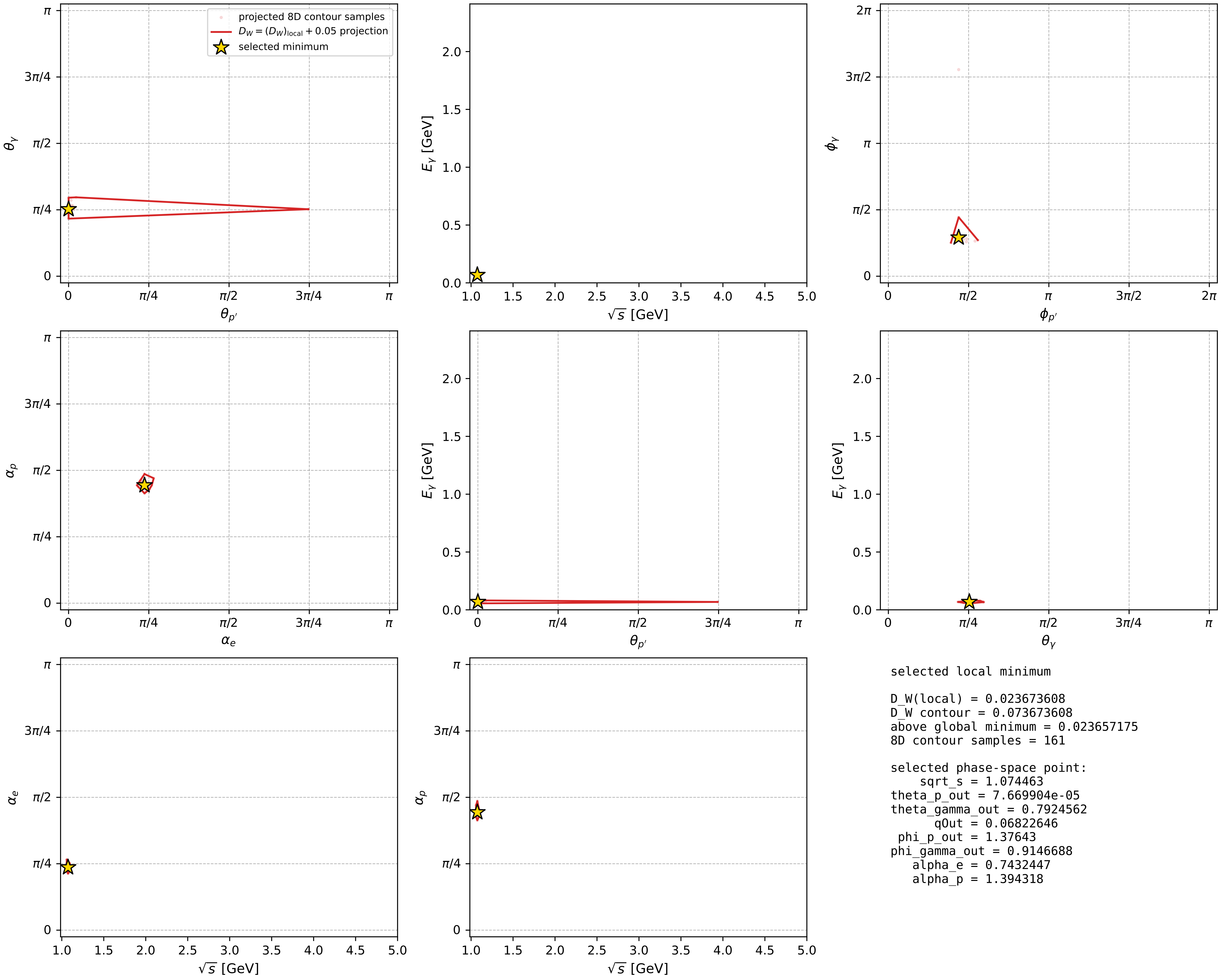


electron: polarization cluster P2, local minimum 1057; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

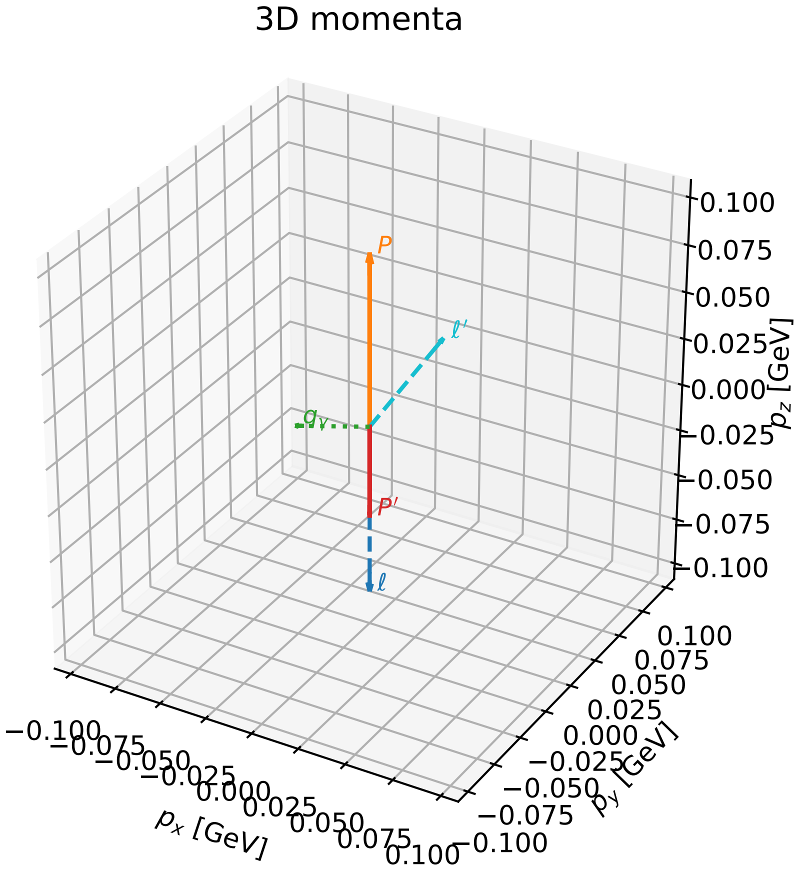



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ'
ℓ
dw_mixing_angles_polarization_02_local_minimum_1058 D_W=0.0236736
region: polarization_cluster_02_local_minimum_1058
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.74324471 rad
incoming lepton: 0.736277|+> +0.67668|->
proton mixing angle: alpha_p=1.3943176 rad
incoming proton: 0.175564|+> +0.984468|->
s=1.15447, $\sqrt{s}$=1.07446, $|\vec{P}|$=0.127797, $|\vec{P}'|$=1.22676e-05
$\theta_{p'}$=7.6699e-05, θ_γ=0.792456, $\phi_{p'}$=1.37643, ϕ_γ=0.914669
E_γ=0.0682265, $\theta_{\ell'}$=2.34926, $\phi_{\ell'}$=4.05626
Q^2=0.00519405, x_B=0.0389985, t=-0.0162538, W^2=1.00784, y=0.484969
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1278 p_x= 0.0000 p_y= 0.0000 p_z= -0.1278
P E= 0.9467 p_x= 0.0000 p_y= 0.0000 p_z= 0.1278
ℓ' E= 0.0682 p_x= -0.0296 p_y= -0.0385 p_z= -0.0479
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0000
q_γ E= 0.0682 p_x= 0.0296 p_y= 0.0385 p_z= 0.0479
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=98.3%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=3.70e+01-4.69e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
A=-5.83e+01-3.56e-02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=3.18e+01+4.77e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=1.38e+01+1.95e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 1058; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



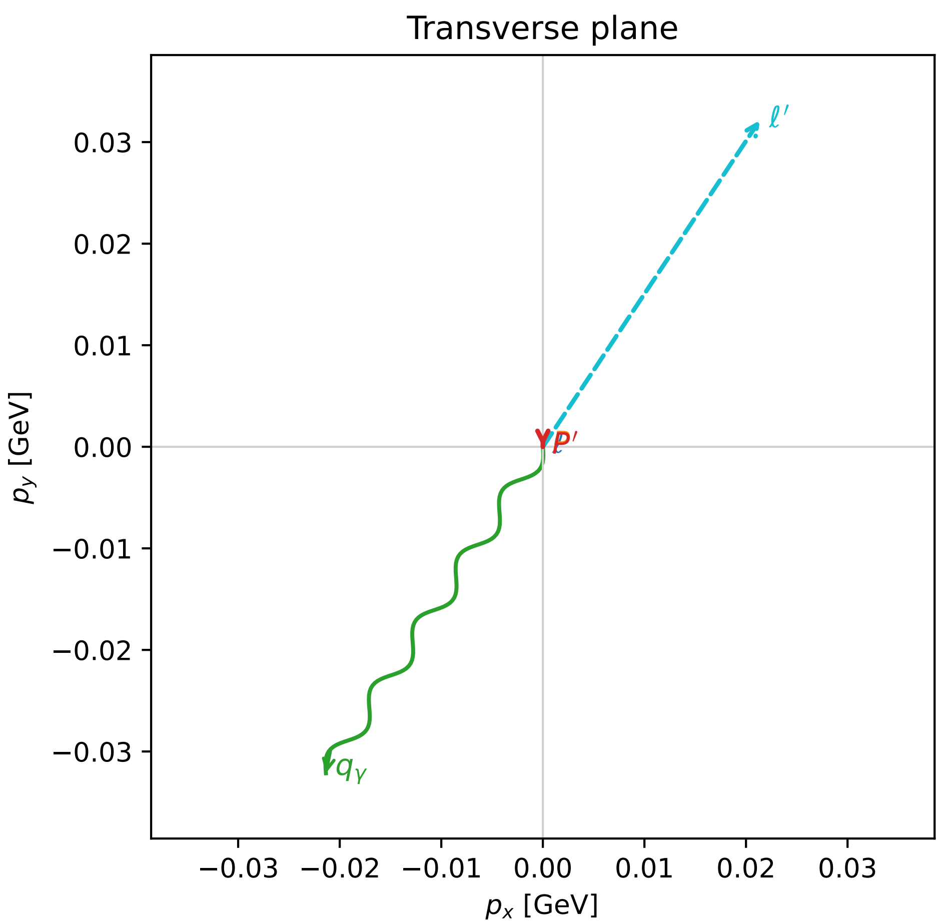
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

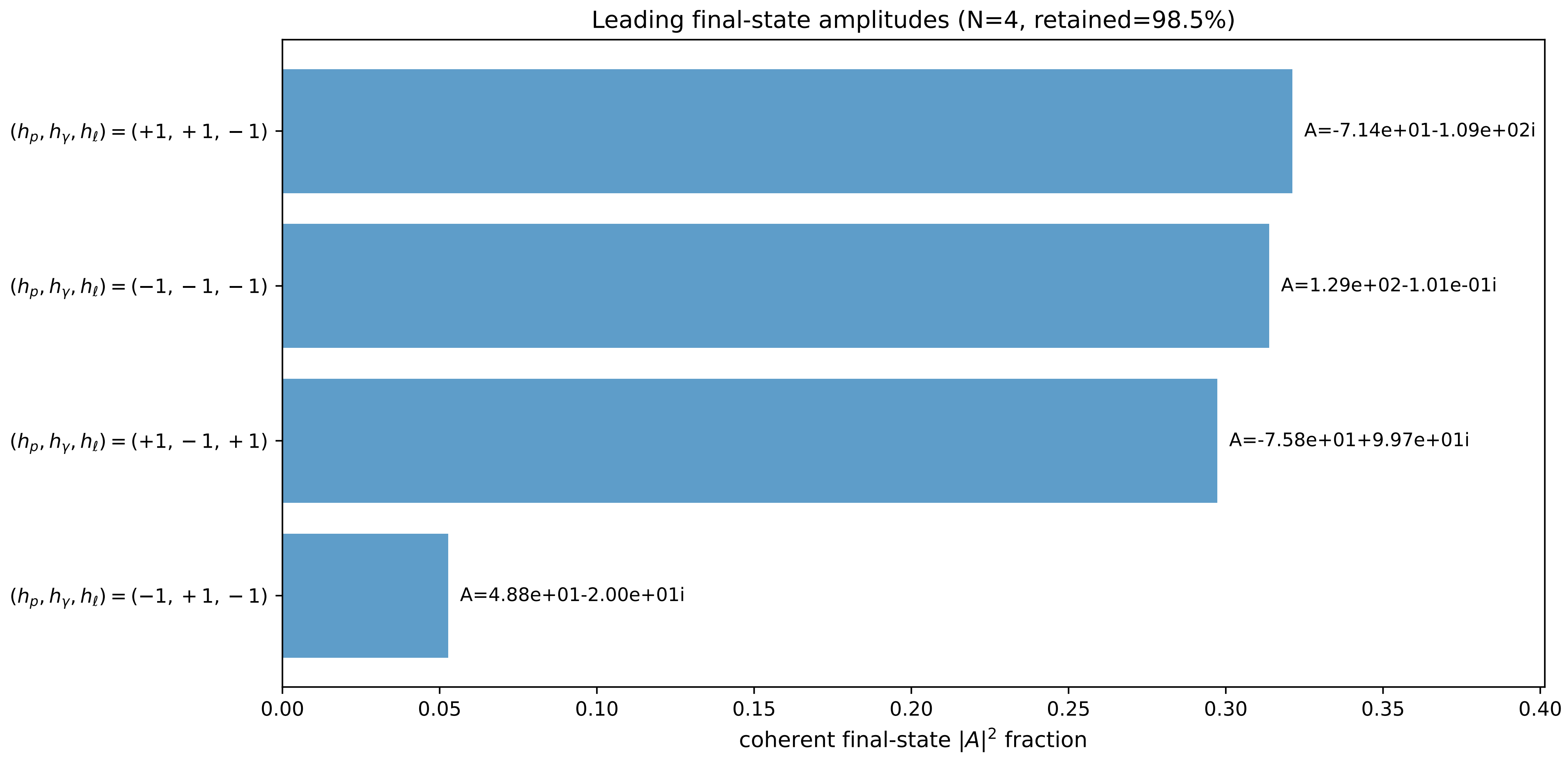


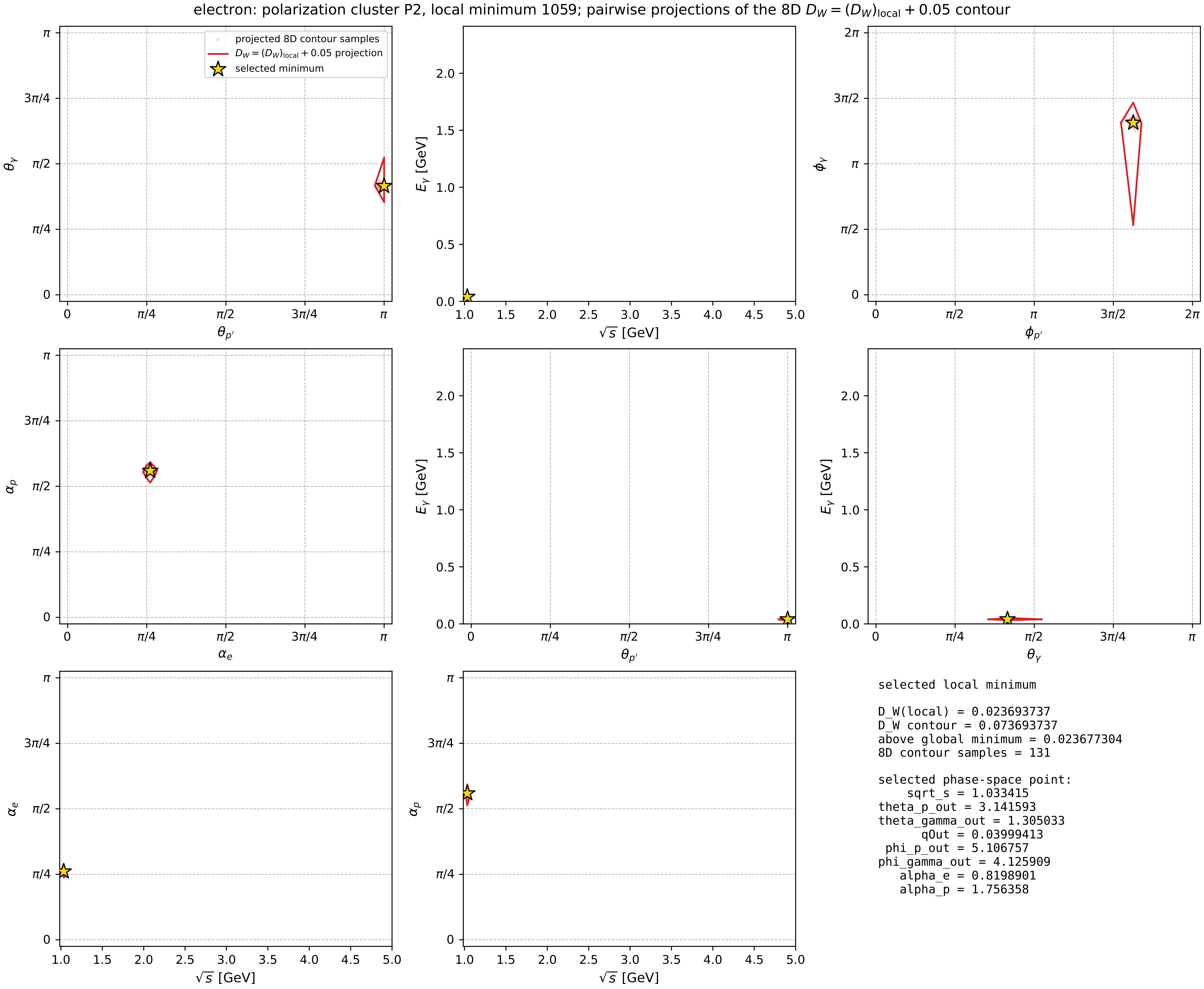
dw_mixing_angles_polarization_02_local_minimum_1059 D_W=0.0236937
region: polarization_cluster_02_local_minimum_1059
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.81989014 rad
incoming lepton: 0.682302|+> +0.731071|->
proton mixing angle: alpha_p=1.7563581 rad
incoming proton: -0.184499|+> +0.982833|->

s=1.06795, $\sqrt{s}$=1.03342, $|\vec{P}|$=0.0910091, $|\vec{P}'|$=0.0485129
$\theta_{p'}$=3.14159, θ_γ=1.30503, $\phi_{p'}$=5.10676, ϕ_γ=4.12591
E_γ=0.0399941, $\theta_{\ell'}$=0.792989, $\phi_{\ell'}$=0.984317
Q^2=0.0167773, x_B=0.180545, t=-0.0194564, W^2=0.955992, y=0.494016

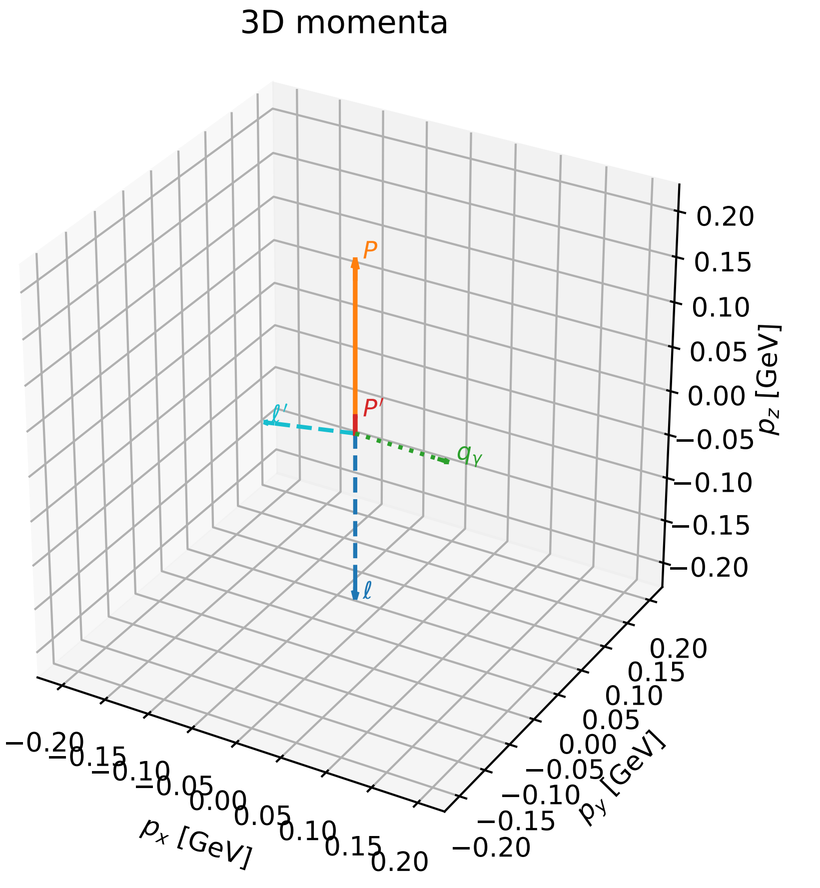
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0910 p_x= 0.0000 p_y= 0.0000 p_z= -0.0910
P E= 0.9424 p_x= 0.0000 p_y= 0.0000 p_z= 0.0910
ℓ' E= 0.0542 p_x= 0.0214 p_y= 0.0321 p_z= 0.0380
P' E= 0.9393 p_x= 0.0000 p_y= -0.0000 p_z= -0.0485
q_γ E= 0.0400 p_x= -0.0214 p_y= -0.0321 p_z= 0.0105






electron: polarization cluster P2, local minimum 1059; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.023693737
D_W contour = 0.073693737
above global minimum = 0.023677304
8D contour samples = 131
selected phase-space point:
sqrt_s = 1.033415
theta_p_out = 3.141593
theta_gamma_out = 1.305033
qOut = 0.03999413
phi_p_out = 5.106757
phi_gamma_out = 4.125909
alpha_e = 0.8198901
alpha_p = 1.756358

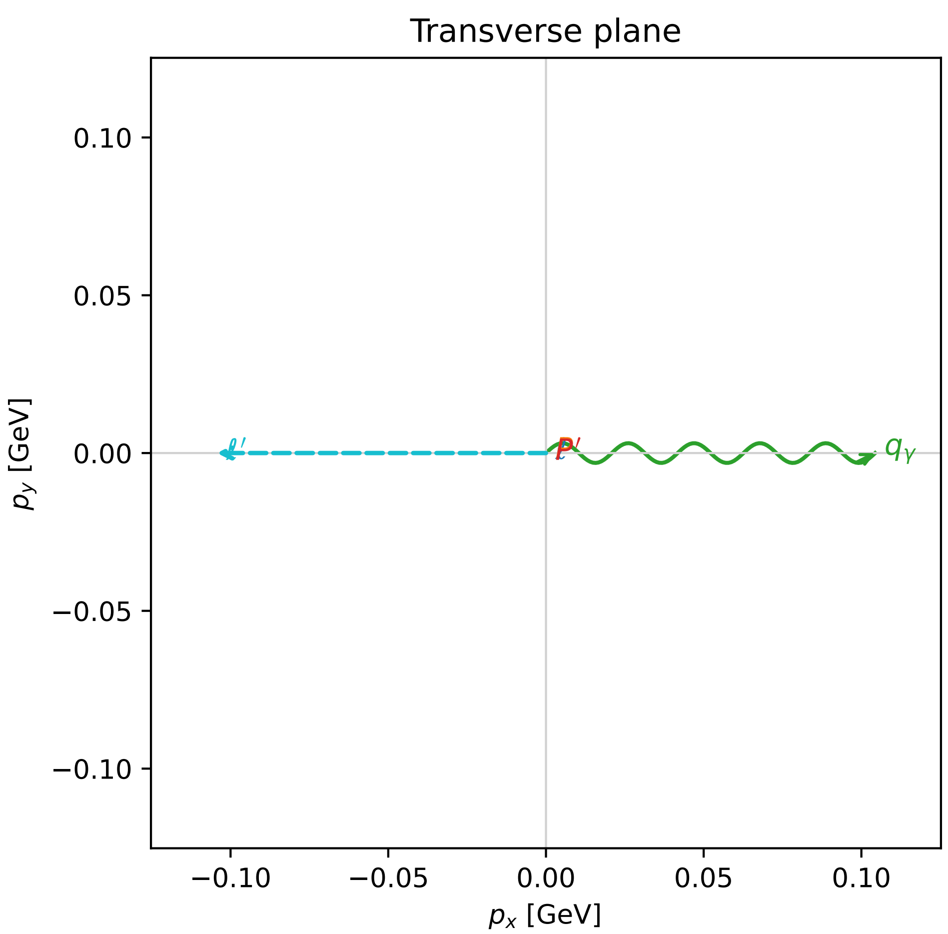
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

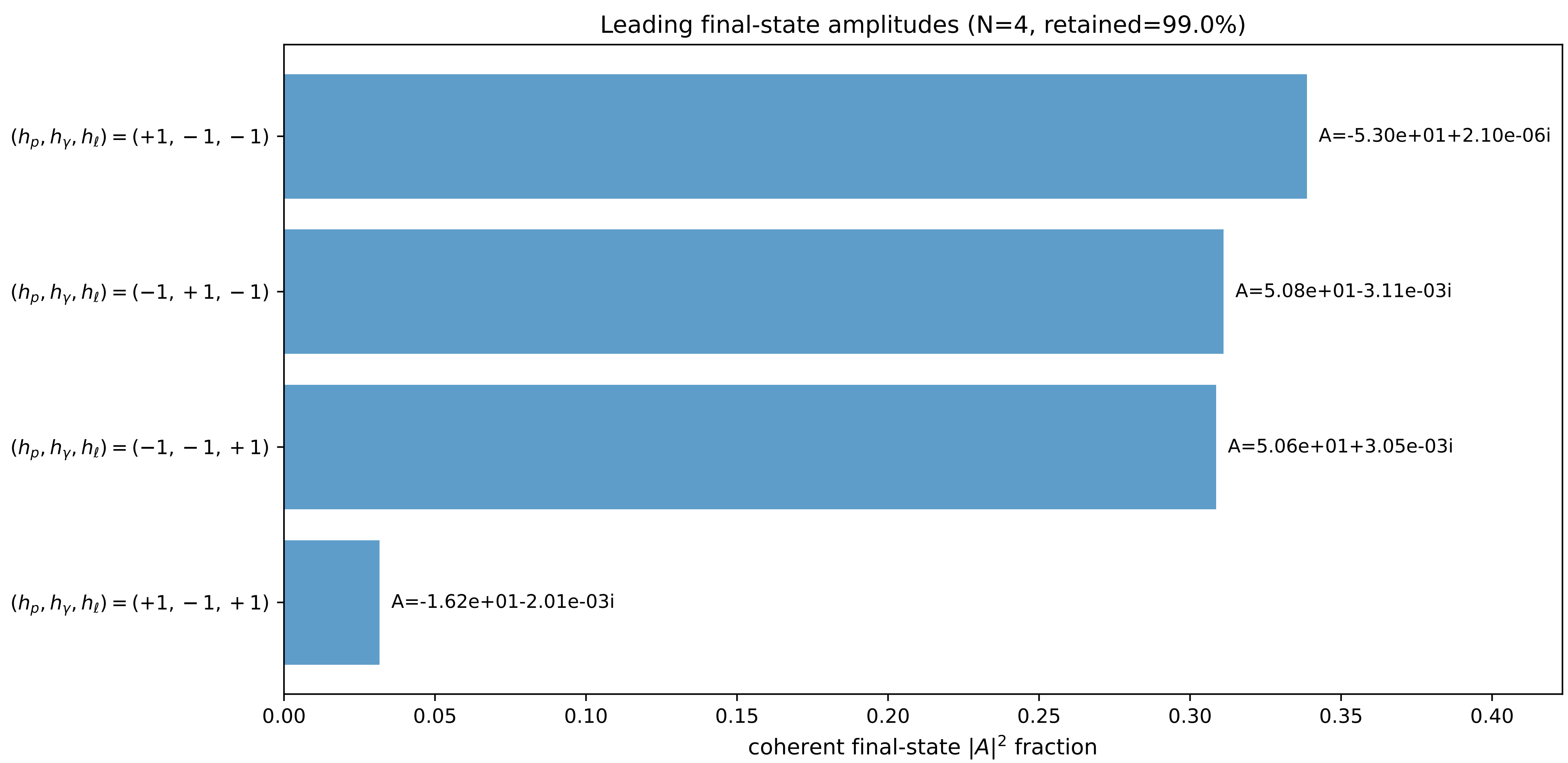


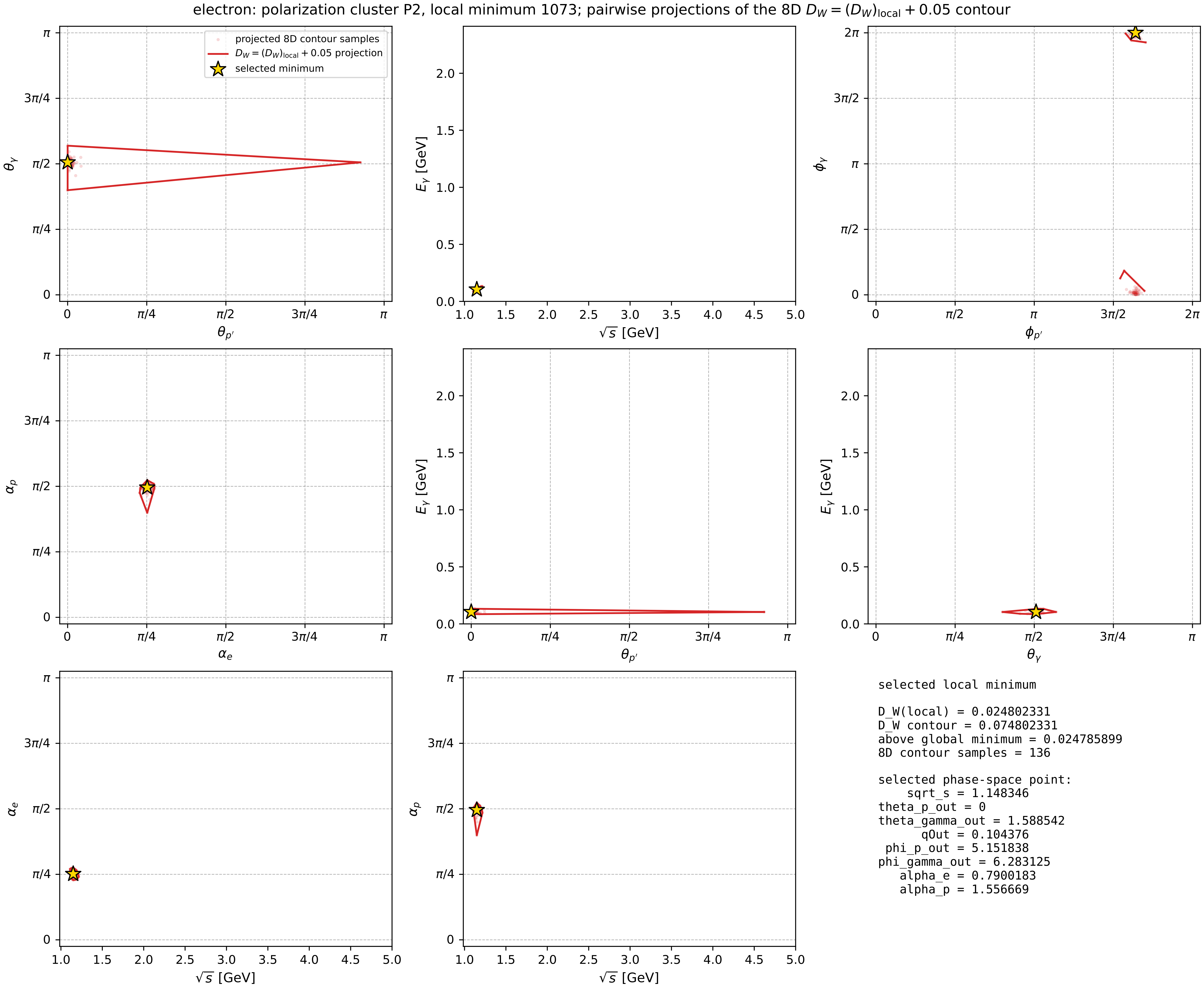
dw_mixing_angles_polarization_02_local_minimum_1073 D_W=0.0248023
region: polarization_cluster_02_local_minimum_1073
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.79001834 rad
incoming lepton: 0.703832|+> +0.710366|->
proton mixing angle: alpha_p=1.5566687 rad
incoming proton: 0.0141271|+> +0.9999|->

s=1.3187, $\sqrt{s}$=1.14835, $|\vec{P}|$=0.191081, $|\vec{P}'|$=0.0190949
$\theta_{p'}$=0, θ_γ=1.58854, $\phi_{p'}$=5.15184, ϕ_γ=6.28312
E_γ=0.104376, $\theta_{\ell'}$=1.73454, $\phi_{\ell'}$=3.14153
Q^2=0.0338335, x_B=0.147259, t=-0.0292154, W^2=1.07576, y=0.523532

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1911 p_x= 0.0000 p_y= 0.0000 p_z= -0.1911
P E= 0.9573 p_x= 0.0000 p_y= 0.0000 p_z= 0.1911
ℓ' E= 0.1058 p_x= -0.1044 p_y= 0.0000 p_z= -0.0172
P' E= 0.9382 p_x= 0.0000 p_y= -0.0000 p_z= 0.0191
q_γ E= 0.1044 p_x= 0.1044 p_y= -0.0000 p_z= -0.0019

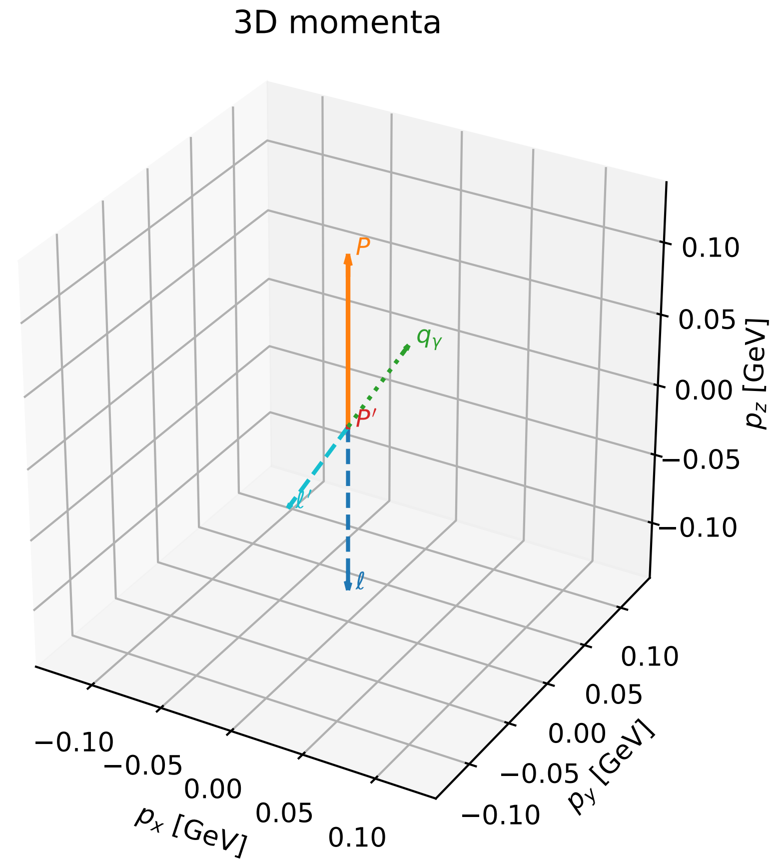






electron: polarization cluster P2, local minimum 1073; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.024802331
D_W contour = 0.074802331
above global minimum = 0.024785899
8D contour samples = 136
selected phase-space point:
sqrt_s = 1.148346
theta_p_out = 0
theta_gamma_out = 1.588542
qOut = 0.104376
phi_p_out = 5.151838
phi_gamma_out = 6.283125
alpha_e = 0.7900183
alpha_p = 1.556669

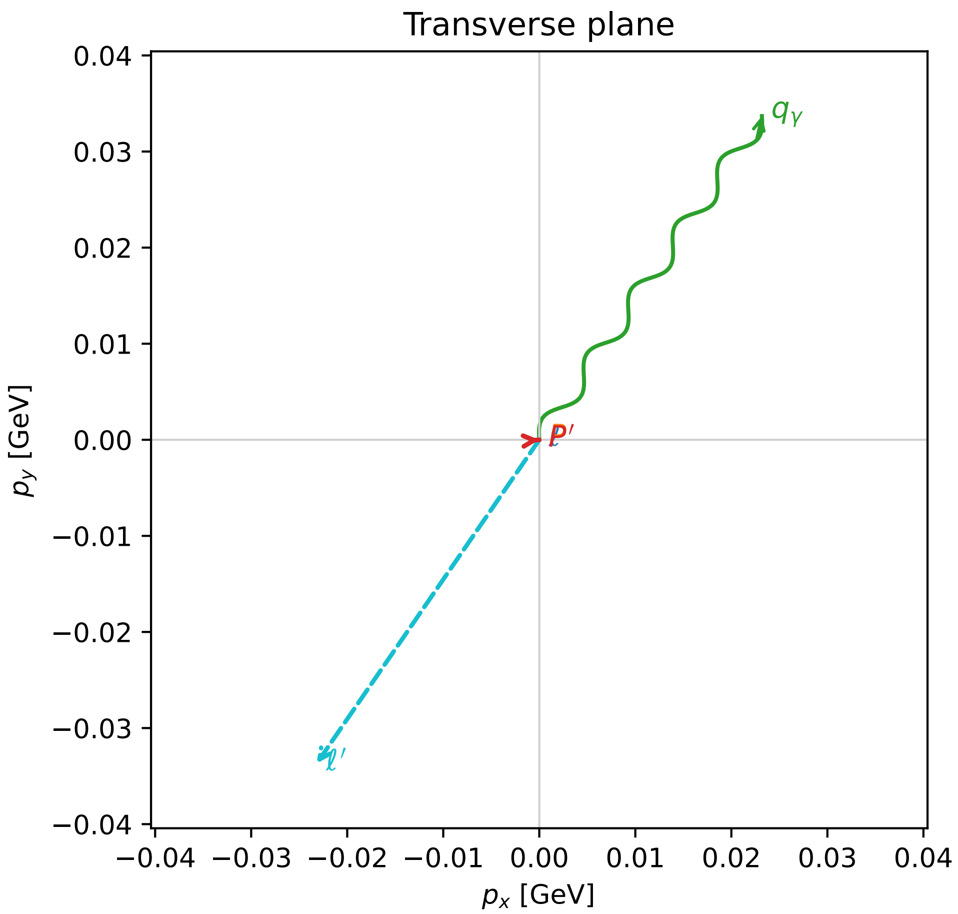
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

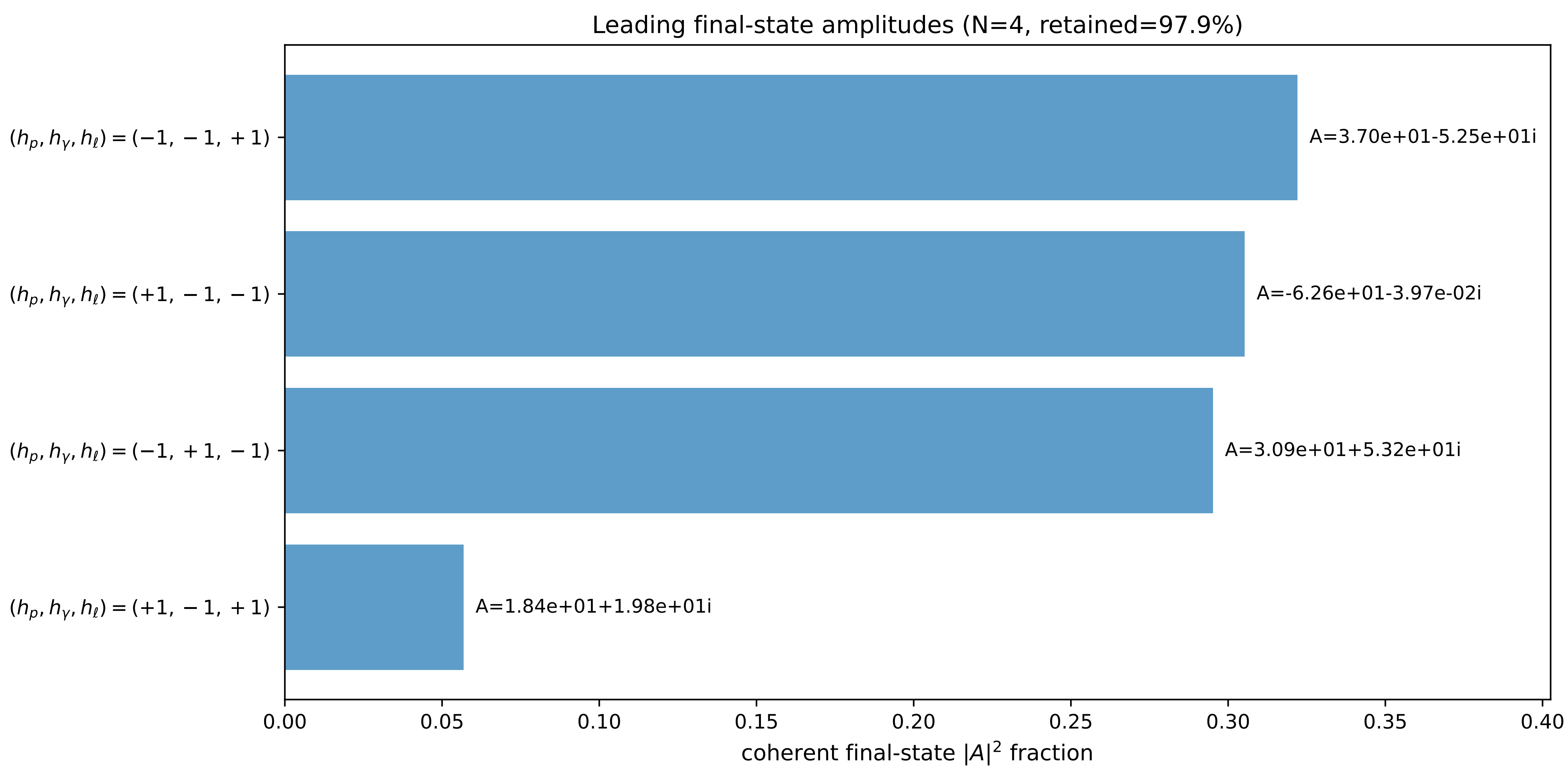


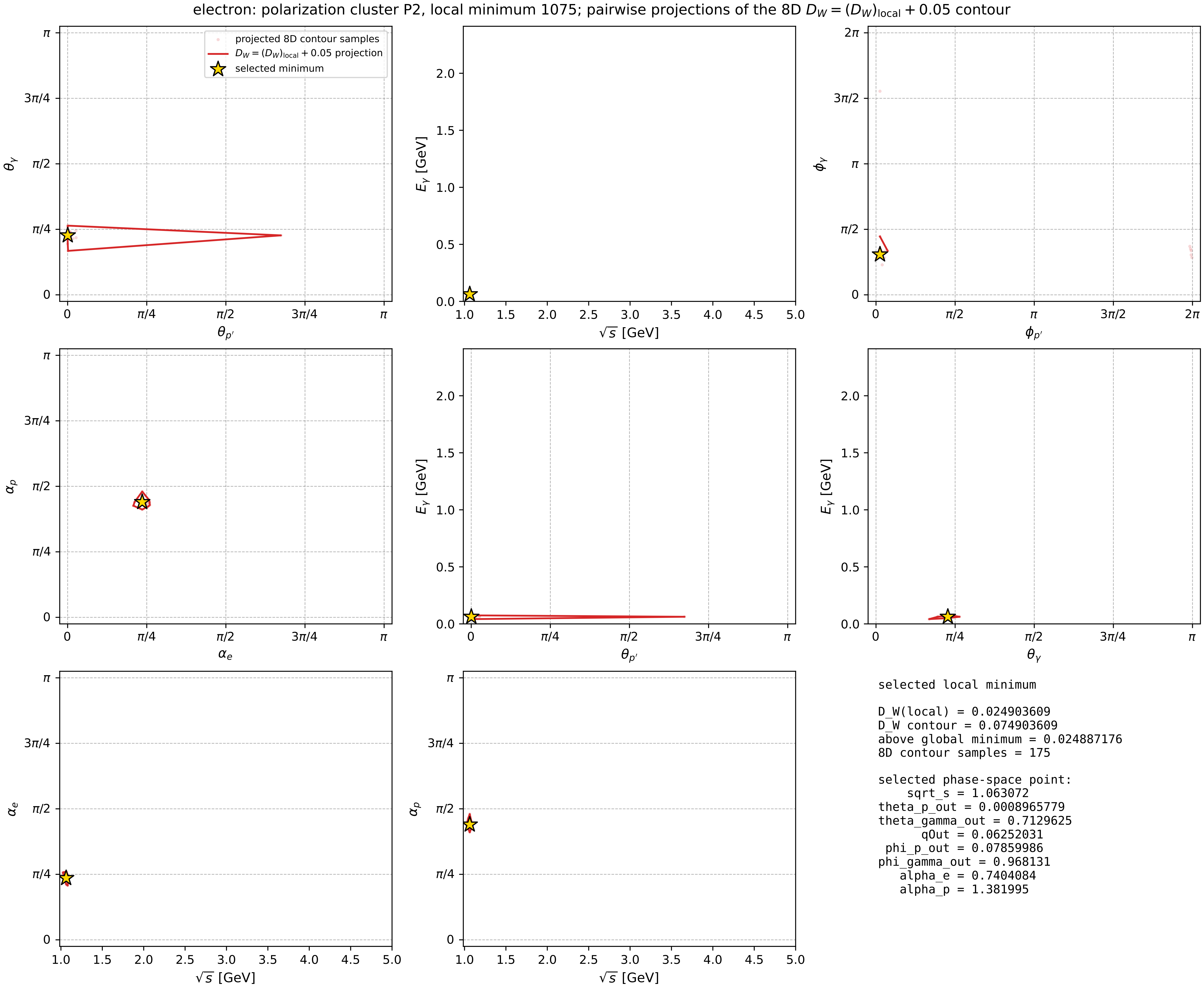
dw_mixing_angles_polarization_02_local_minimum_1075 D_W=0.0249036
region: polarization_cluster_02_local_minimum_1075
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.7404084 rad
incoming lepton: 0.738193|+> +0.674589|->
proton mixing angle: alpha_p=1.381995 rad
incoming proton: 0.187682|+> +0.98223|->

s=1.13012, $\sqrt{s}$=1.06307, $|\vec{P}|$=0.117713, $|\vec{P}'|$=3.8086e-05
$\theta_{p'}$=0.000896578, θ_γ=0.712963, $\phi_{p'}$=0.0785999, ϕ_γ=0.968131
E_γ=0.0625203, $\theta_{\ell'}$=2.42903, $\phi_{\ell'}$=4.10972
Q^2=0.00358305, x_B=0.0296445, t=-0.0137933, W^2=0.997128, y=0.482935

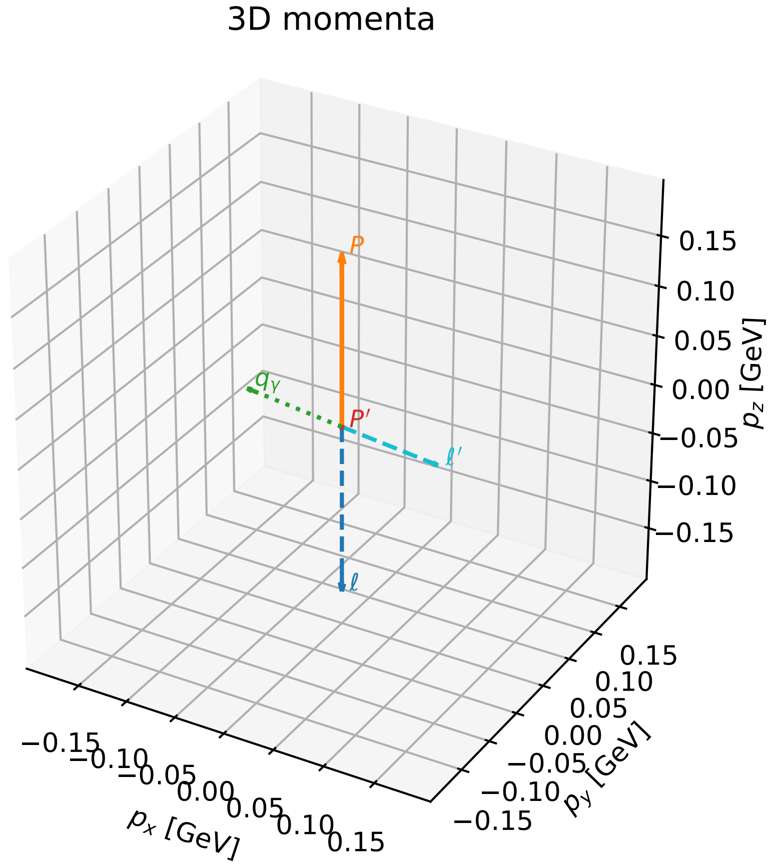
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1177 p_x= 0.0000 p_y= 0.0000 p_z= -0.1177
P E= 0.9454 p_x= 0.0000 p_y= 0.0000 p_z= 0.1177
ℓ' E= 0.0626 p_x= -0.0232 p_y= -0.0337 p_z= -0.0473
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0000
q_γ E= 0.0625 p_x= 0.0232 p_y= 0.0337 p_z= 0.0473






electron: polarization cluster P2, local minimum 1075; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.024903609
D_W contour = 0.074903609
above global minimum = 0.024887176
8D contour samples = 175
selected phase-space point:
sqrt_s = 1.063072
theta_p_out = 0.0008965779
theta_gamma_out = 0.7129625
qOut = 0.06252031
phi_p_out = 0.07859986
phi_gamma_out = 0.968131
alpha_e = 0.7404084
alpha_p = 1.381995

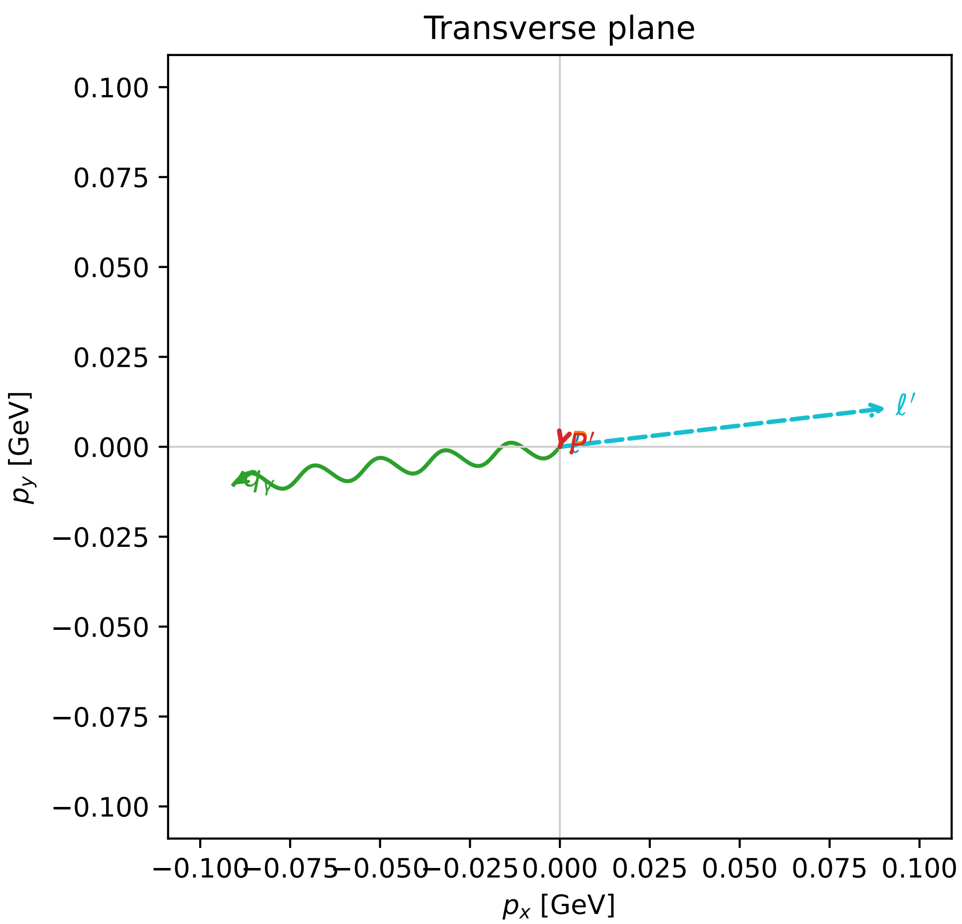
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

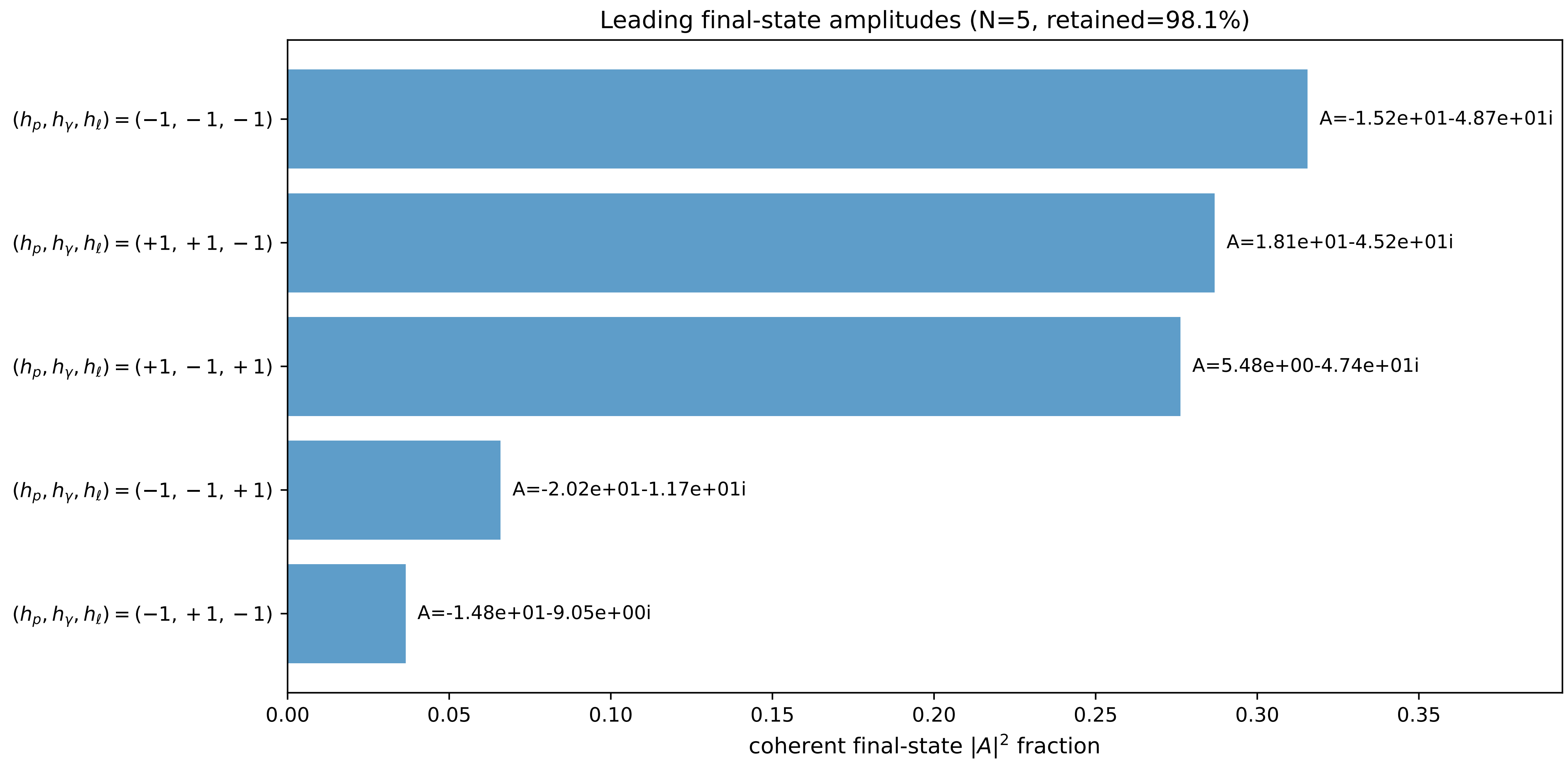


dw_mixing_angles_polarization_02_local_minimum_1076 D_W=0.0249774
region: polarization_cluster_02_local_minimum_1076
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78266076 rad
incoming lepton: 0.70904|+> +0.705168|->
proton mixing angle: alpha_p=1.5976474 rad
incoming proton: -0.0268479|+> +0.99964|->

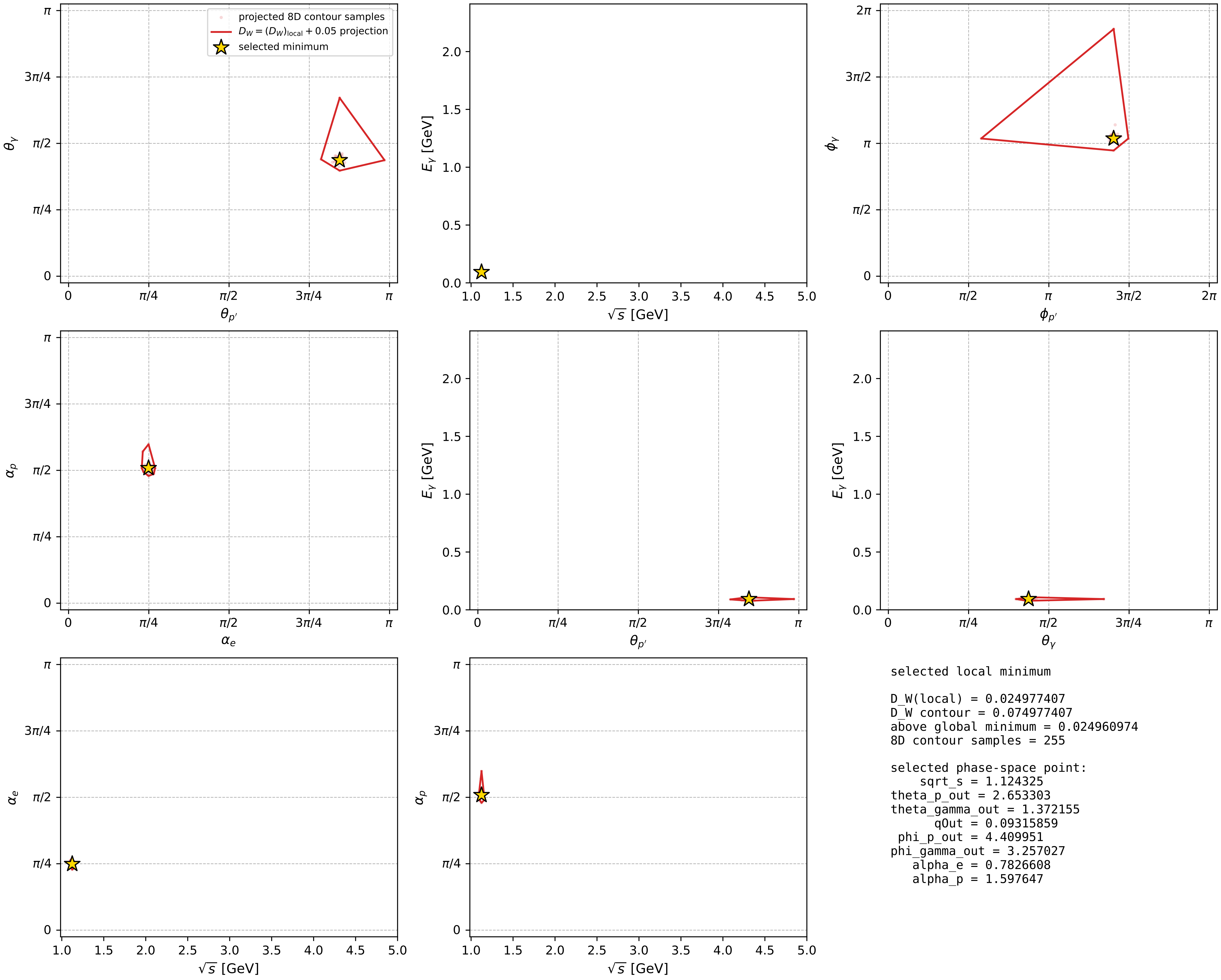
s=1.26411, $\sqrt{s}$=1.12433, $|\vec{P}|$=0.170885, $|\vec{P}'|$=0.000432685
$\theta_{p'}$=2.6533, θ_γ=1.37216, $\phi_{p'}$=4.40995, ϕ_γ=3.25703
E_γ=0.0931586, $\theta_{\ell'}$=1.76524, $\phi_{\ell'}$=0.117464
Q^2=0.0256887, x_B=0.128153, t=-0.0290942, W^2=1.05461, y=0.521657

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1709 p_x= 0.0000 p_y= 0.0000 p_z= -0.1709
P E= 0.9534 p_x= 0.0000 p_y= 0.0000 p_z= 0.1709
ℓ' E= 0.0932 p_x= 0.0908 p_y= 0.0107 p_z= -0.0180
P' E= 0.9380 p_x= -0.0001 p_y= -0.0002 p_z= -0.0004
q_γ E= 0.0932 p_x= -0.0907 p_y= -0.0105 p_z= 0.0184

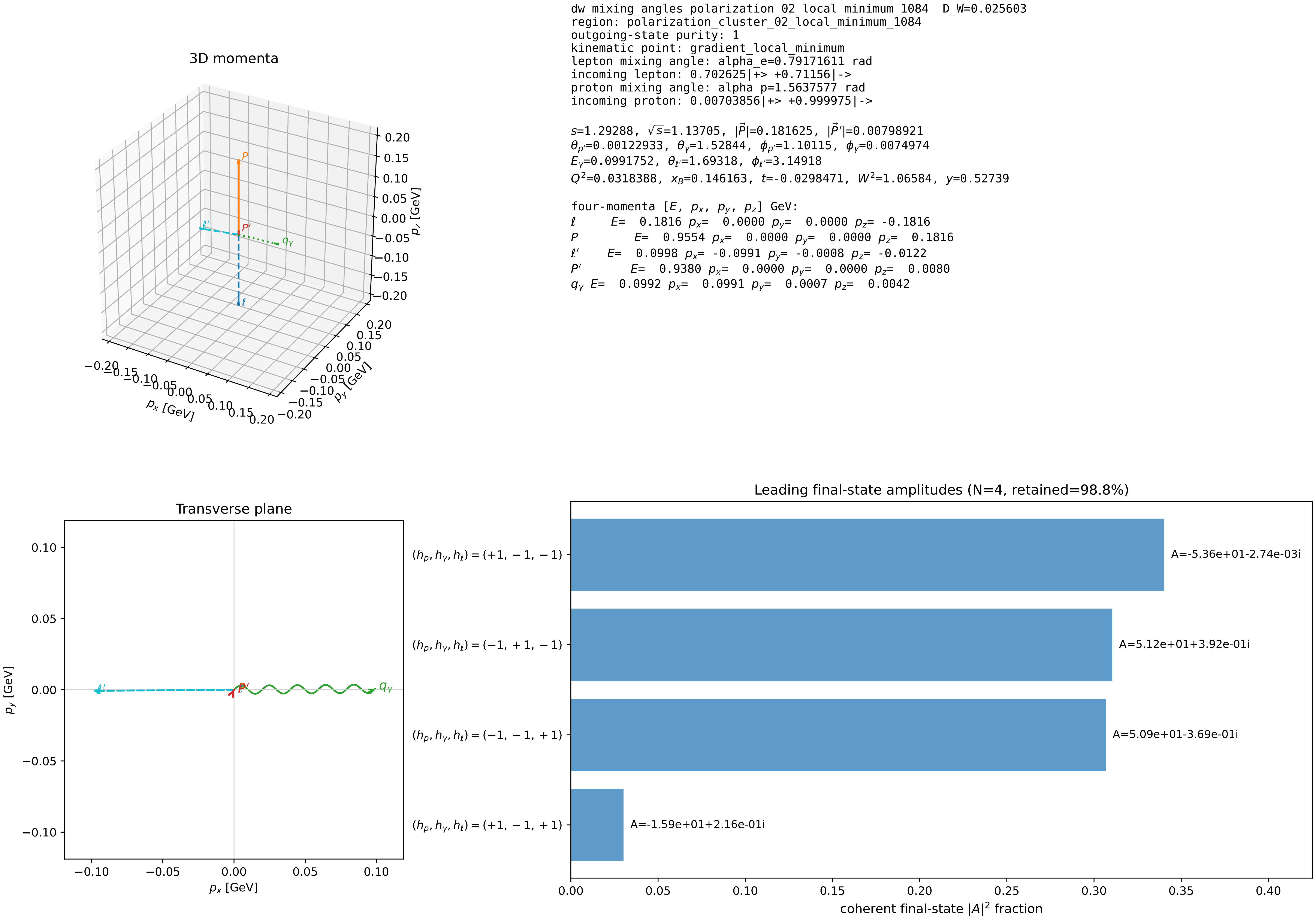




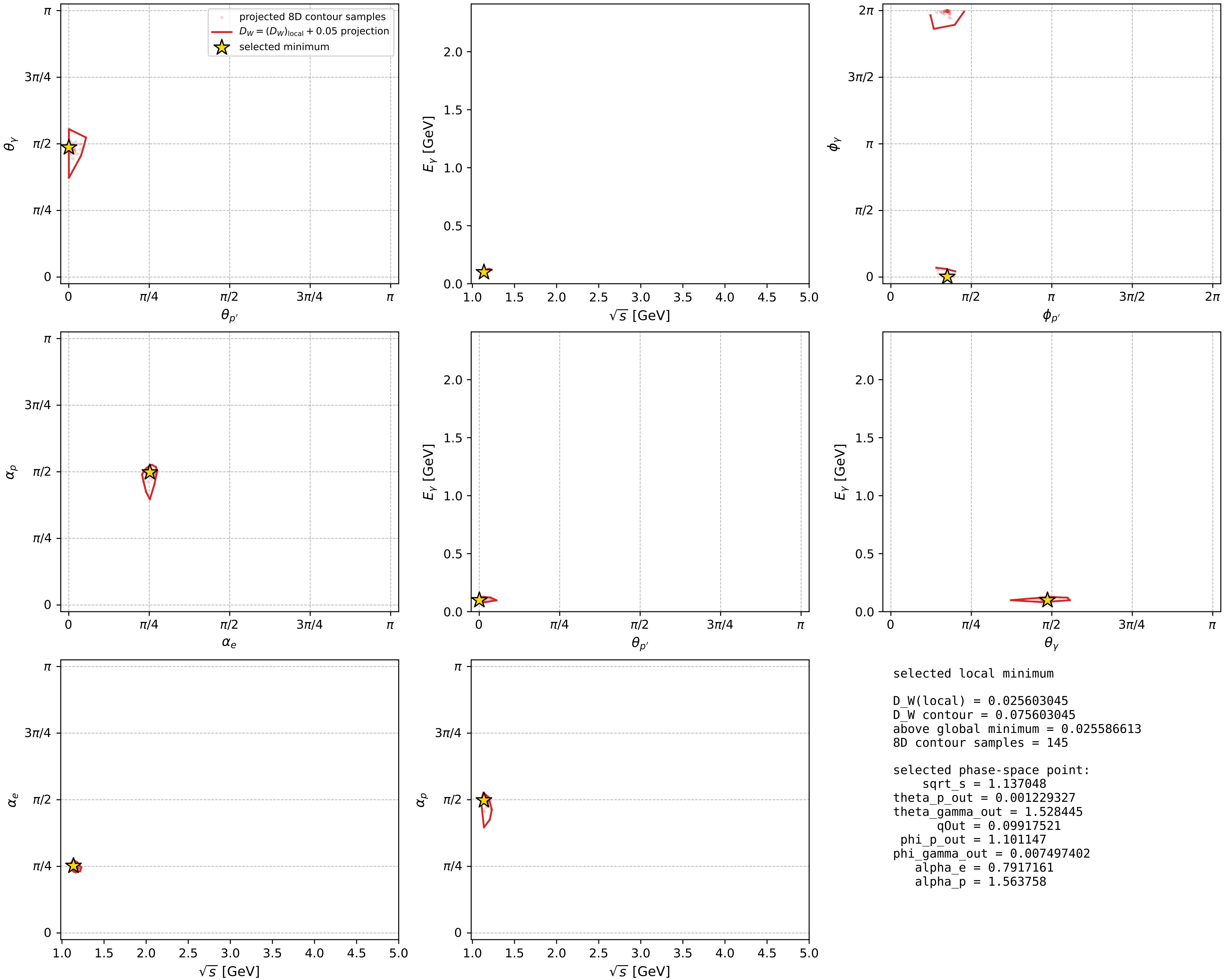
electron: polarization cluster P2, local minimum 1076; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



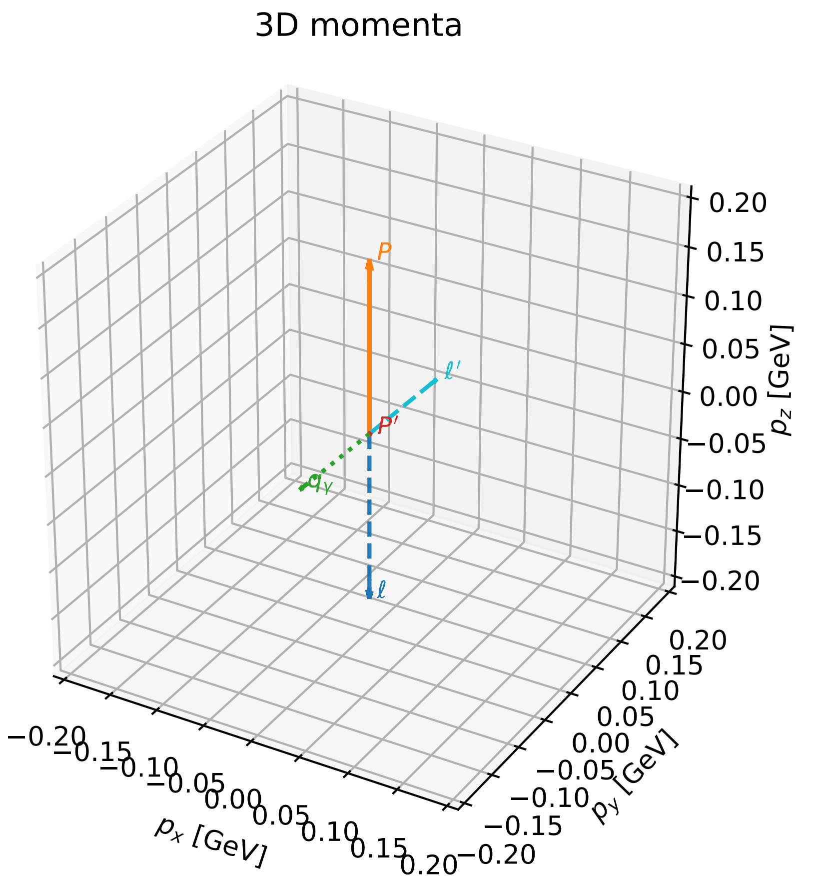
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



electron: polarization cluster P2, local minimum 1084; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



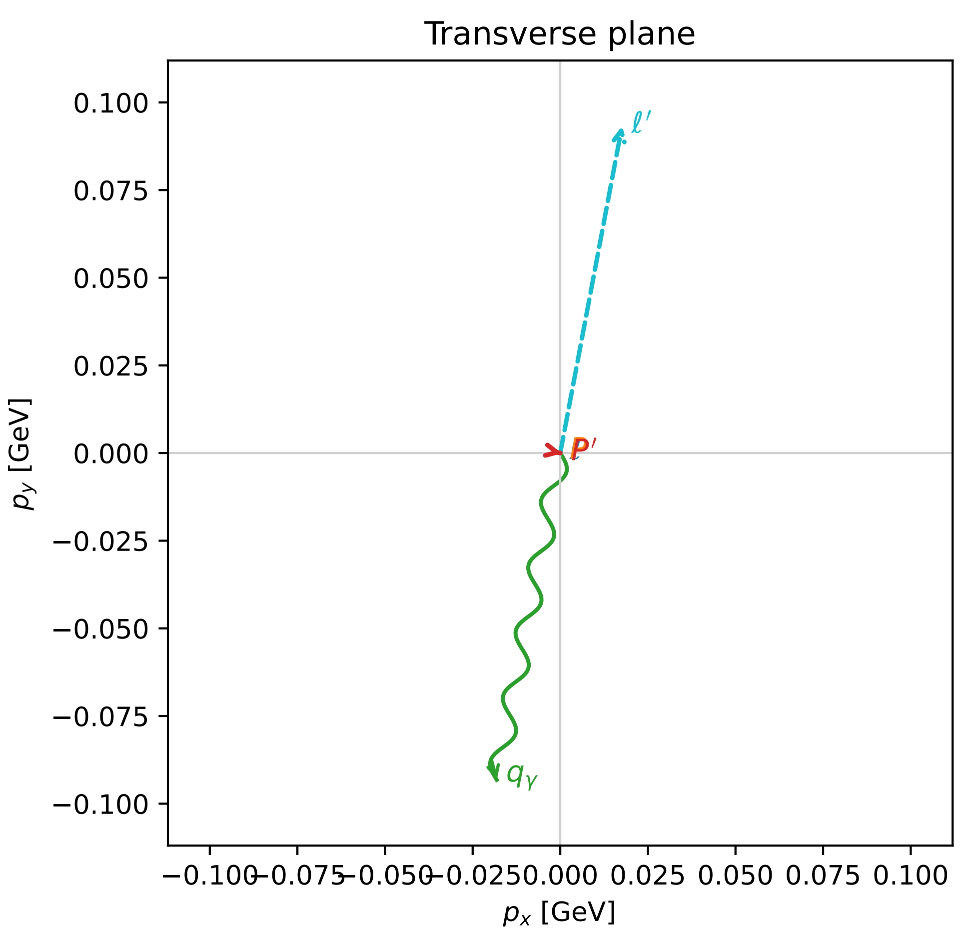
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

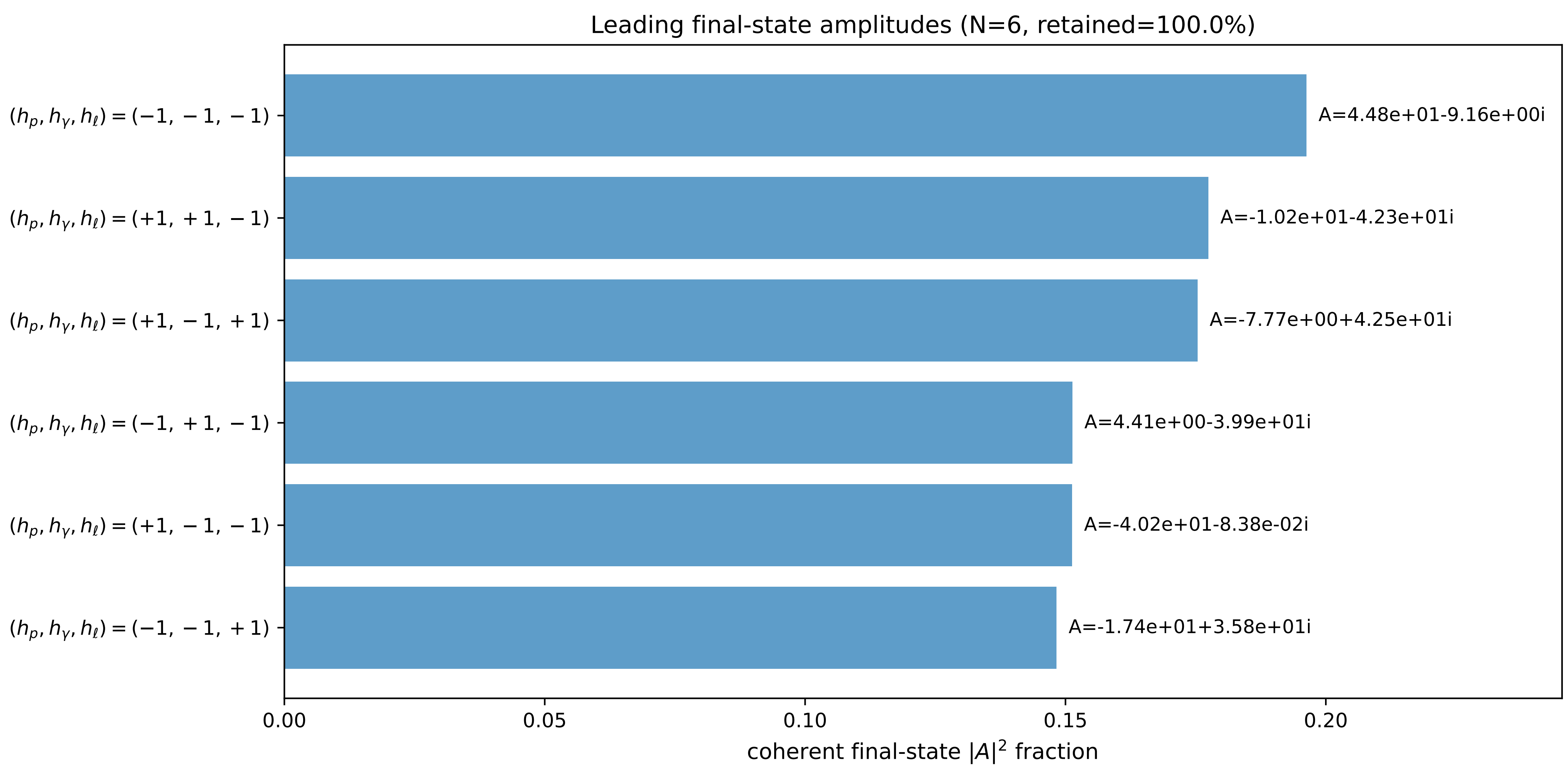


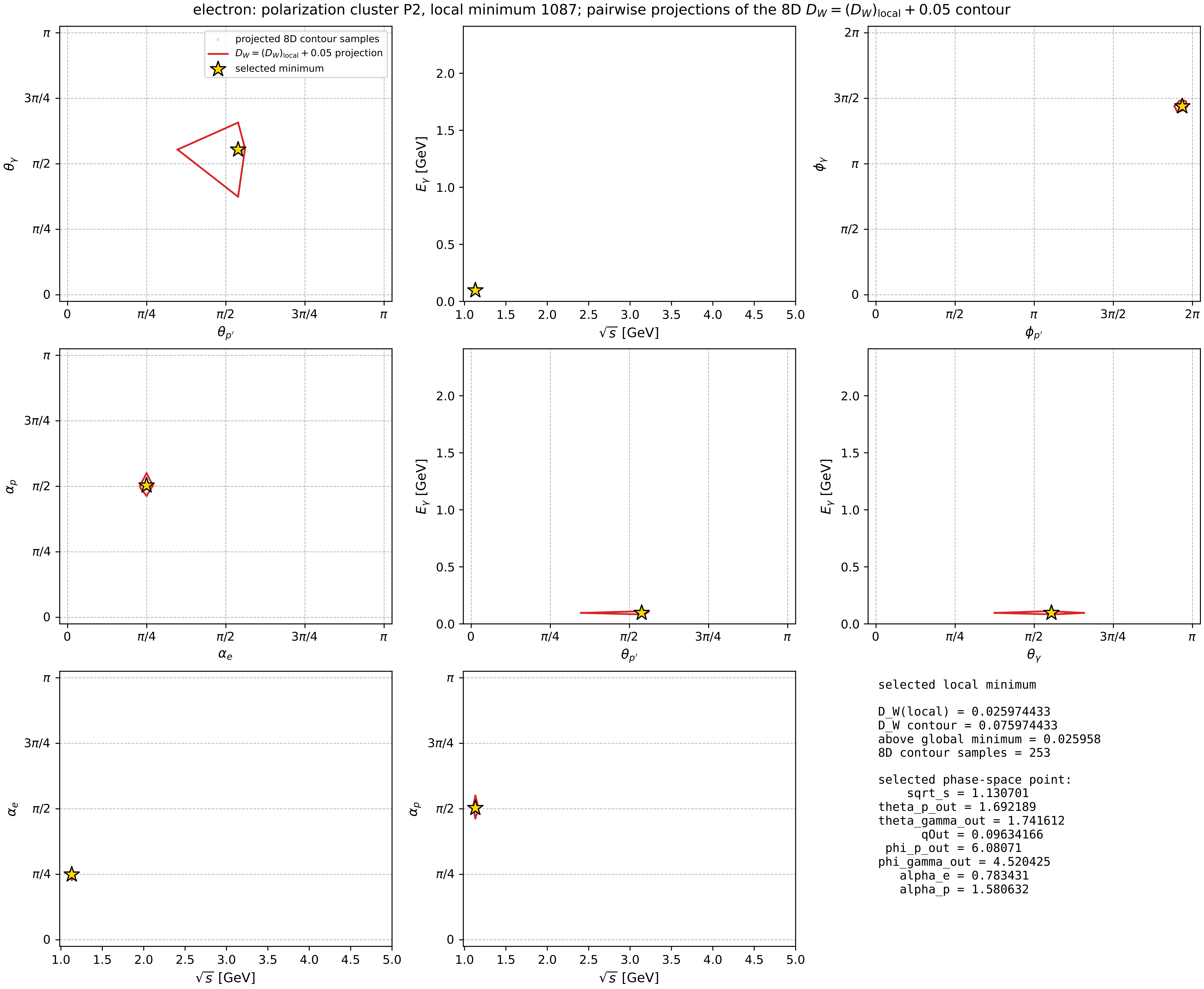
dw_mixing_angles_polarization_02_local_minimum_1087 D_W=0.0259744
region: polarization_cluster_02_local_minimum_1087
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78343104 rad
incoming lepton: 0.708496|+> +0.705714|->
proton mixing angle: alpha_p=1.5806317 rad
incoming proton: -0.00983519|+> +0.999952|->

s=1.27849, $\sqrt{s}$=1.1307, $|\vec{P}|$=0.17628, $|\vec{P}'|$=0.000488969
$\theta_{p'}$=1.69219, θ_γ=1.74161, $\phi_{p'}$=6.08071, ϕ_γ=4.52042
E_γ=0.0963417, $\theta_{\ell'}$=1.39939, $\phi_{\ell'}$=1.38394
Q^2=0.0397667, x_B=0.180347, t=-0.0308261, W^2=1.06058, y=0.553131

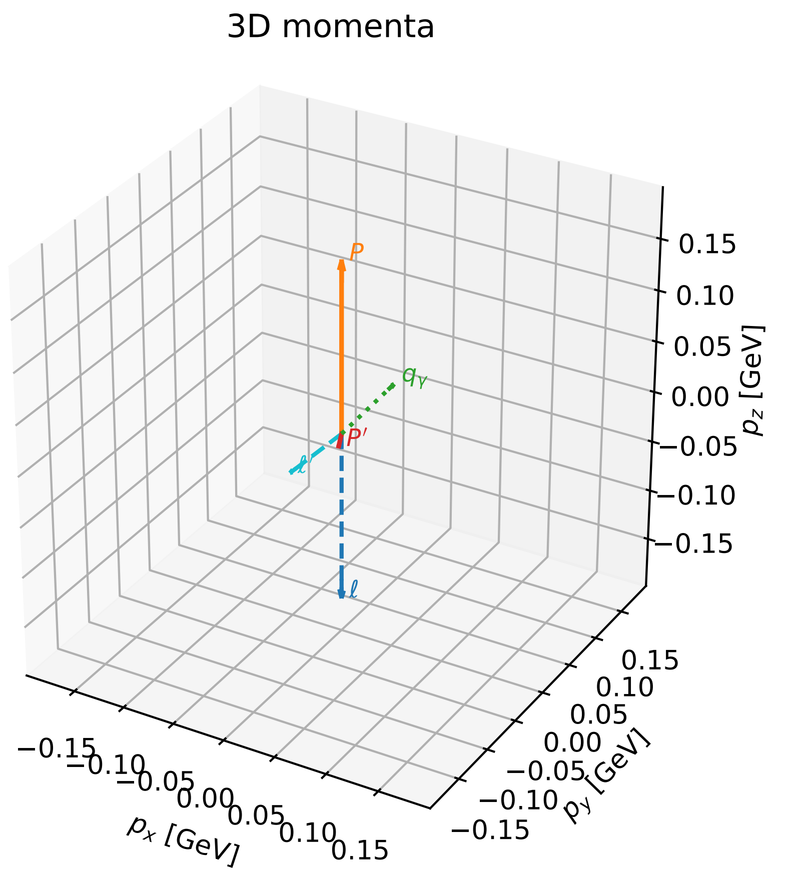
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1763 p_x= 0.0000 p_y= 0.0000 p_z= -0.1763
P E= 0.9544 p_x= 0.0000 p_y= 0.0000 p_z= 0.1763
ℓ' E= 0.0964 p_x= 0.0176 p_y= 0.0933 p_z= 0.0164
P' E= 0.9380 p_x= 0.0005 p_y= -0.0001 p_z= -0.0001
q_γ E= 0.0963 p_x= -0.0181 p_y= -0.0932 p_z= -0.0164






electron: polarization cluster P2, local minimum 1087; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.025974433
D_W contour = 0.075974433
above global minimum = 0.025958
8D contour samples = 253
selected phase-space point:
sqrt_s = 1.130701
theta_p_out = 1.692189
theta_gamma_out = 1.741612
qOut = 0.09634166
phi_p_out = 6.08071
phi_gamma_out = 4.520425
alpha_e = 0.783431
alpha_p = 1.580632

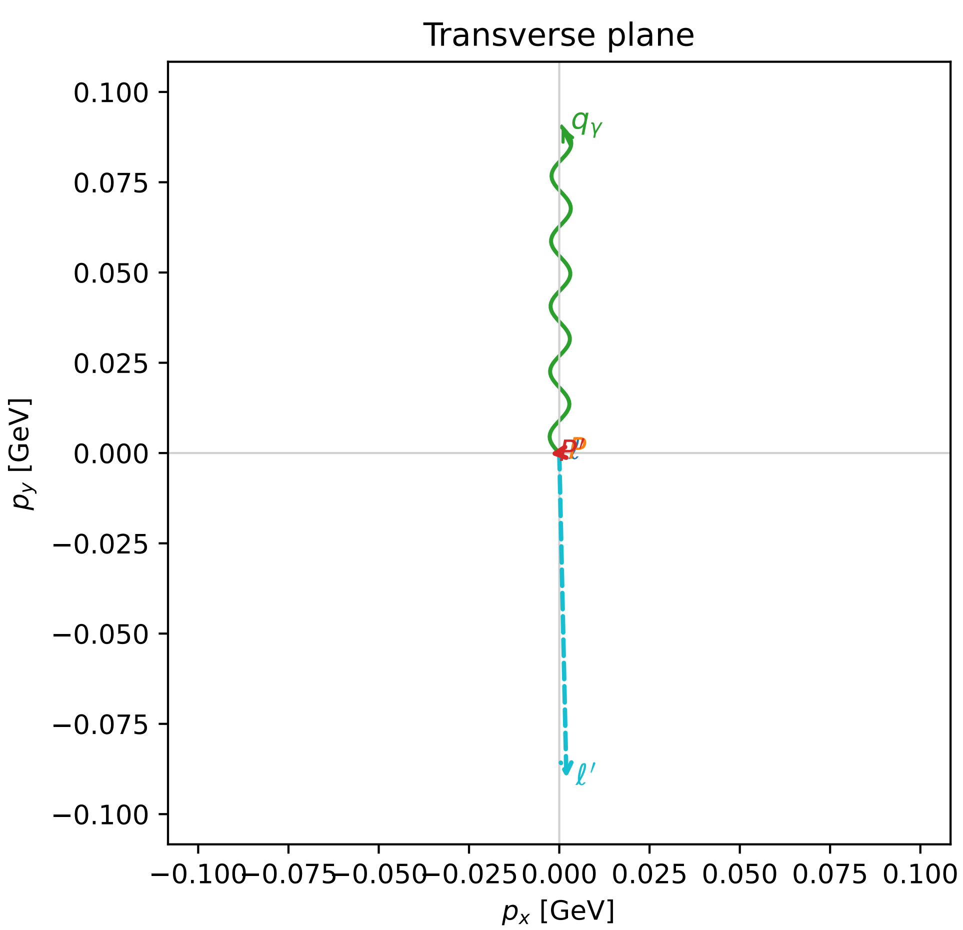
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

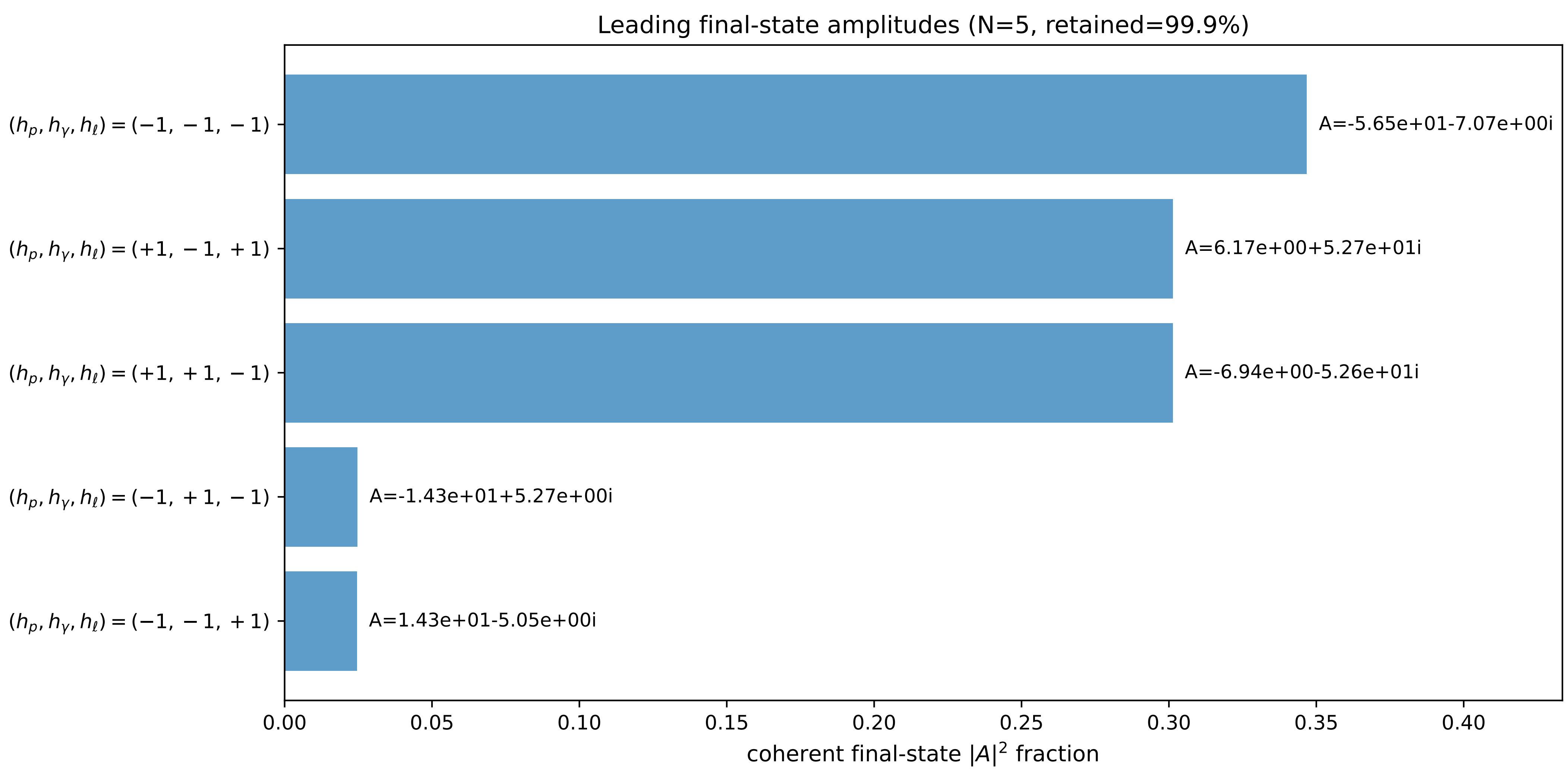


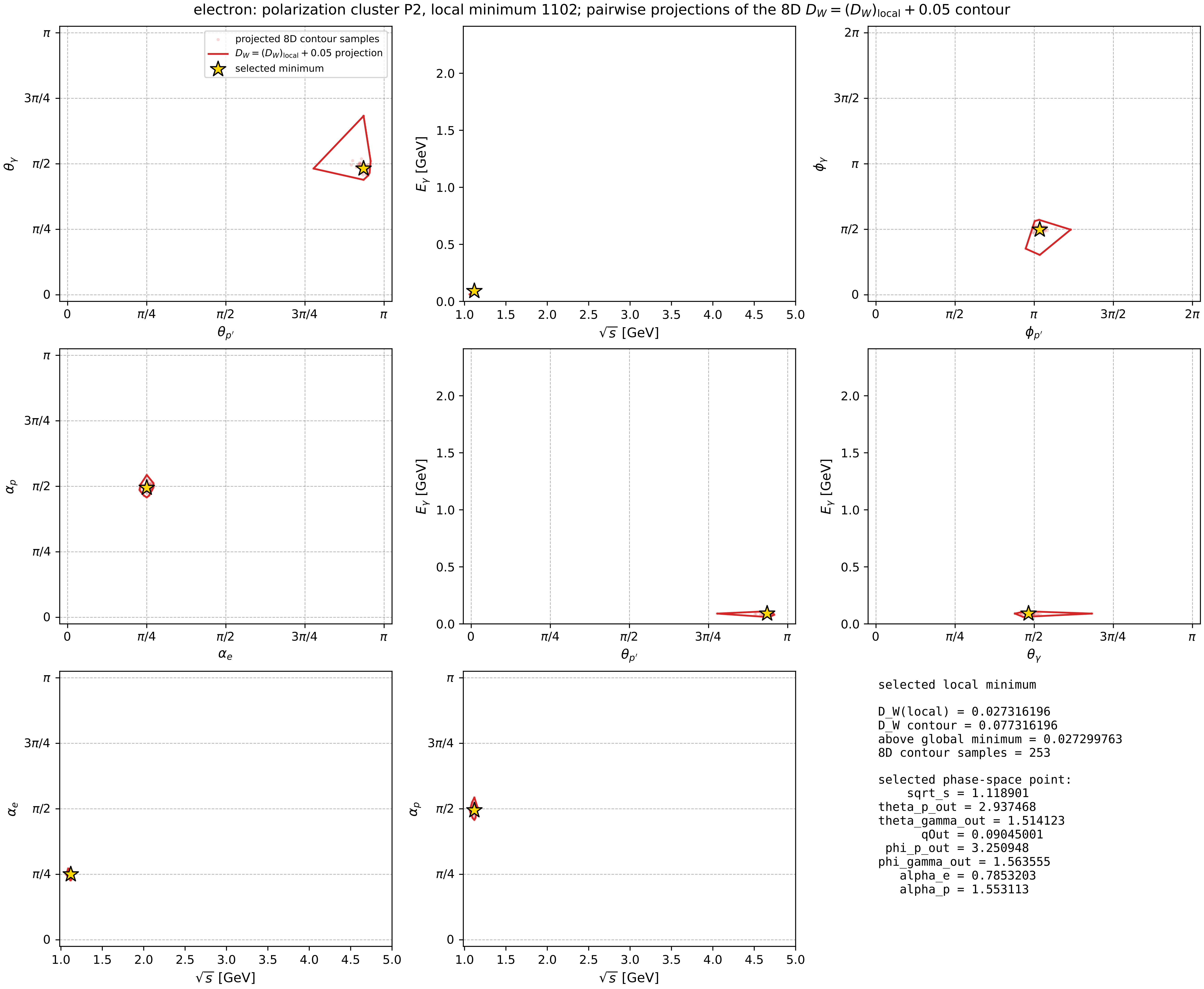

```
dw_mixing_angles_polarization_02_local_minimum_1102  D_W=0.0273162
region: polarization_cluster_02_local_minimum_1102
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.78532031 rad
incoming lepton: 0.707162|+> +0.707052|->
proton mixing angle: alpha_p=1.5531133 rad
incoming proton: 0.0176821|+> +0.999844|->
```

s=1.25194, $\sqrt{s}$=1.1189, $|\vec{P}|$=0.166277, $|\vec{P}'|$=0.0130368
$\theta_{p'}$=2.93747, θ_γ=1.51412, $\phi_{p'}$=3.25095, ϕ_γ=1.56356
E_γ=0.09045, $\theta_{\ell'}$=1.48611, $\phi_{\ell'}$=4.7343
Q^2=0.0325911, x_B=0.160961, t=-0.0318522, W^2=1.04973, y=0.544155

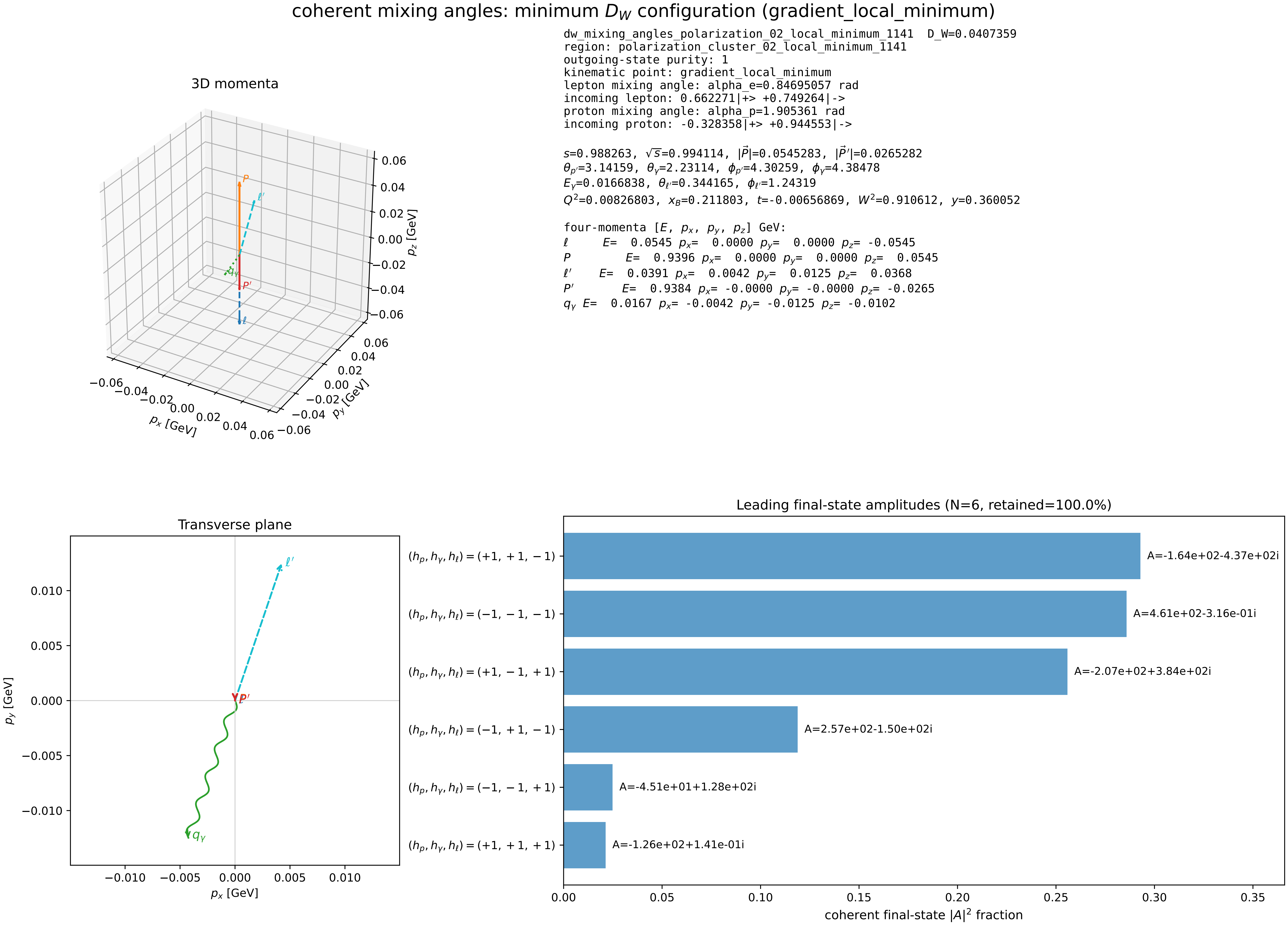
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1663 p_x= 0.0000 p_y= 0.0000 p_z= -0.1663
P E= 0.9526 p_x= 0.0000 p_y= 0.0000 p_z= 0.1663
ℓ' E= 0.0904 p_x= 0.0020 p_y= -0.0900 p_z= 0.0076
P' E= 0.9381 p_x= -0.0026 p_y= -0.0003 p_z= -0.0128
q_γ E= 0.0905 p_x= 0.0007 p_y= 0.0903 p_z= 0.0051





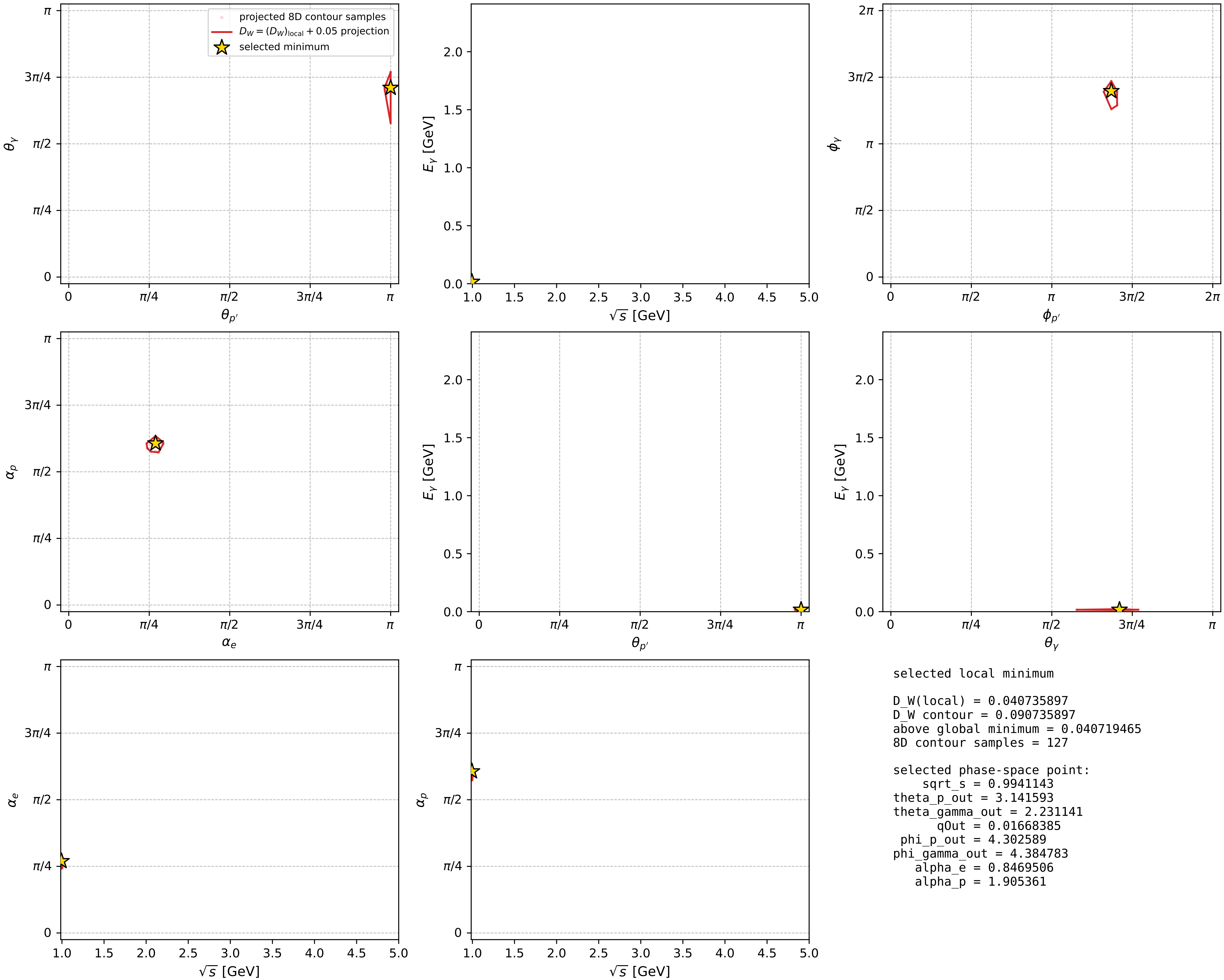


electron: polarization cluster P2, local minimum 1102; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.027316196
D_W contour = 0.077316196
above global minimum = 0.027299763
8D contour samples = 253
selected phase-space point:
sqrt_s = 1.118901
theta_p_out = 2.937468
theta_gamma_out = 1.514123
qOut = 0.09045001
phi_p_out = 3.250948
phi_gamma_out = 1.563555
alpha_e = 0.7853203
alpha_p = 1.553113



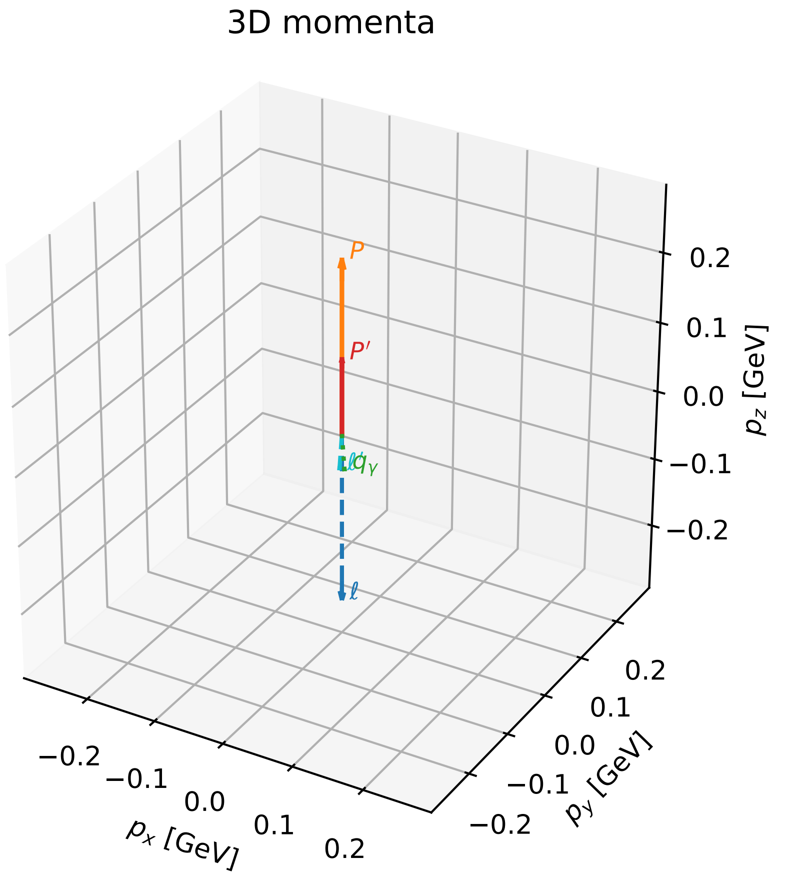
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_1141 D_W=0.0407359
region: polarization_cluster_02_local_minimum_1141
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.84695057 rad
incoming lepton: 0.662271|+> +0.749264|->
proton mixing angle: alpha_p=1.905361 rad
incoming proton: -0.328358|+> +0.944553|->
s=0.988263, $\sqrt{s}$=0.994114, $|\vec{P}|$=0.0545283, $|\vec{P}'|$=0.0265282
$\theta_{p'}$=3.14159, θ_γ=2.23114, $\phi_{p'}$=4.30259, ϕ_γ=4.38478
E_γ=0.0166838, $\theta_{\ell'}$=0.344165, $\phi_{\ell'}$=1.24319
Q^2=0.00826803, x_B=0.211803, t=-0.00656869, W^2=0.910612, y=0.360052
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0545 p_x= 0.0000 p_y= 0.0000 p_z= -0.0545
P E= 0.9396 p_x= 0.0000 p_y= 0.0000 p_z= 0.0545
ℓ' E= 0.0391 p_x= 0.0042 p_y= 0.0125 p_z= 0.0368
P' E= 0.9384 p_x= -0.0000 p_y= -0.0000 p_z= -0.0265
q_γ E= 0.0167 p_x= -0.0042 p_y= -0.0125 p_z= -0.0102
3D momenta
P
ℓ'
q_γ
P'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
−0.06 −0.04 −0.02 0.00 0.02 0.04 0.06
Transverse plane
p_x [GeV]
p_y [GeV]
−0.010 −0.005 0.000 0.005 0.010
Leading final-state amplitudes (N=6, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-1.64e+02-4.37e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=4.61e+02-3.16e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-2.07e+02+3.84e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=2.57e+02-1.50e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-4.51e+01+1.28e+02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-1.26e+02+1.41e-01i
coherent final-state $|A|^2$ fraction
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35

electron: polarization cluster P2, local minimum 1141; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.040735897
D_W contour = 0.090735897
above global minimum = 0.040719465
8D contour samples = 127

selected phase-space point:
    sqrt_s = 0.9941143
theta_p_out = 3.141593
theta_gamma_out = 2.231141
      qOut = 0.01668385
 phi_p_out = 4.302589
phi_gamma_out = 4.384783
   alpha_e = 0.8469506
   alpha_p = 1.905361
```

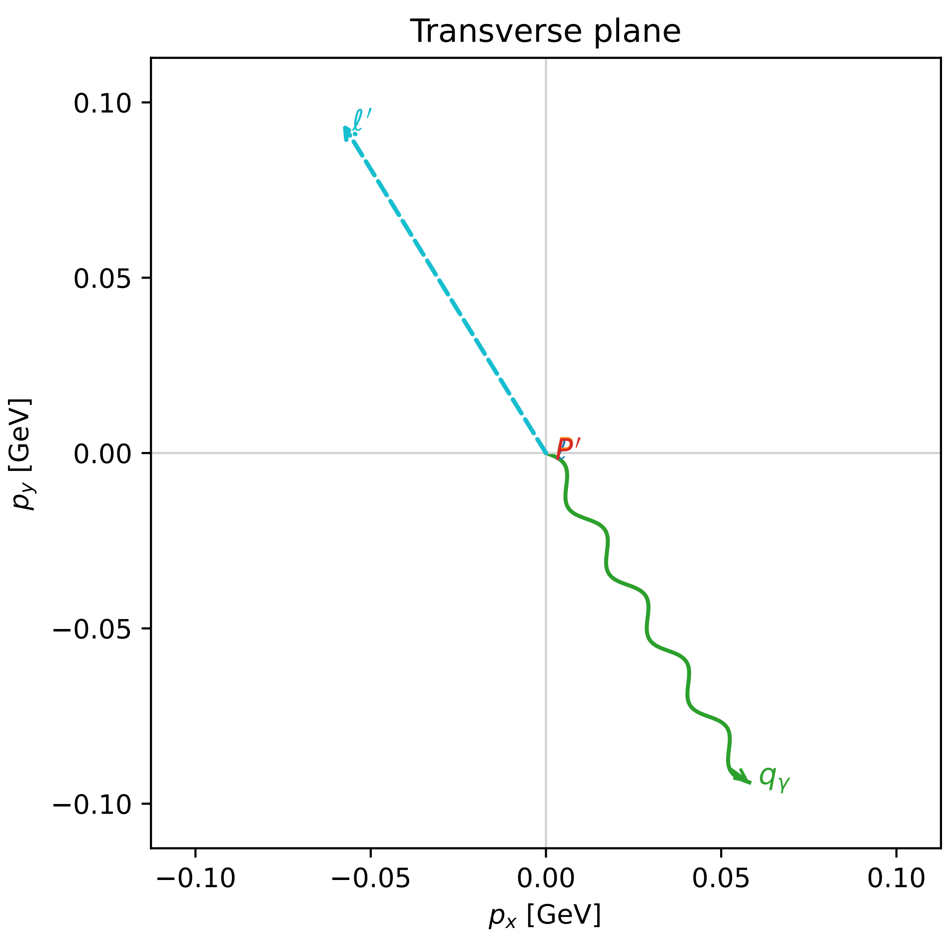
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

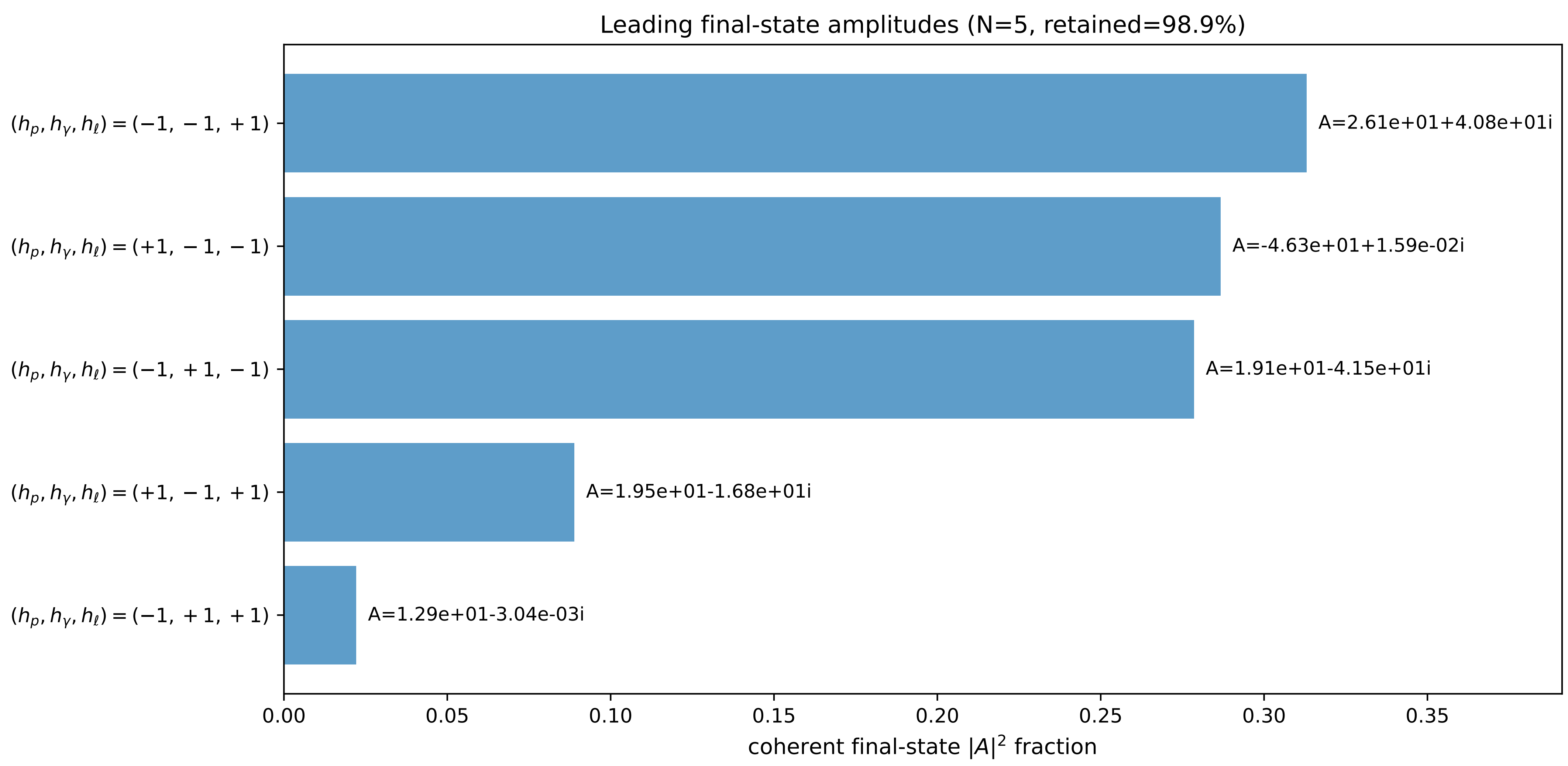


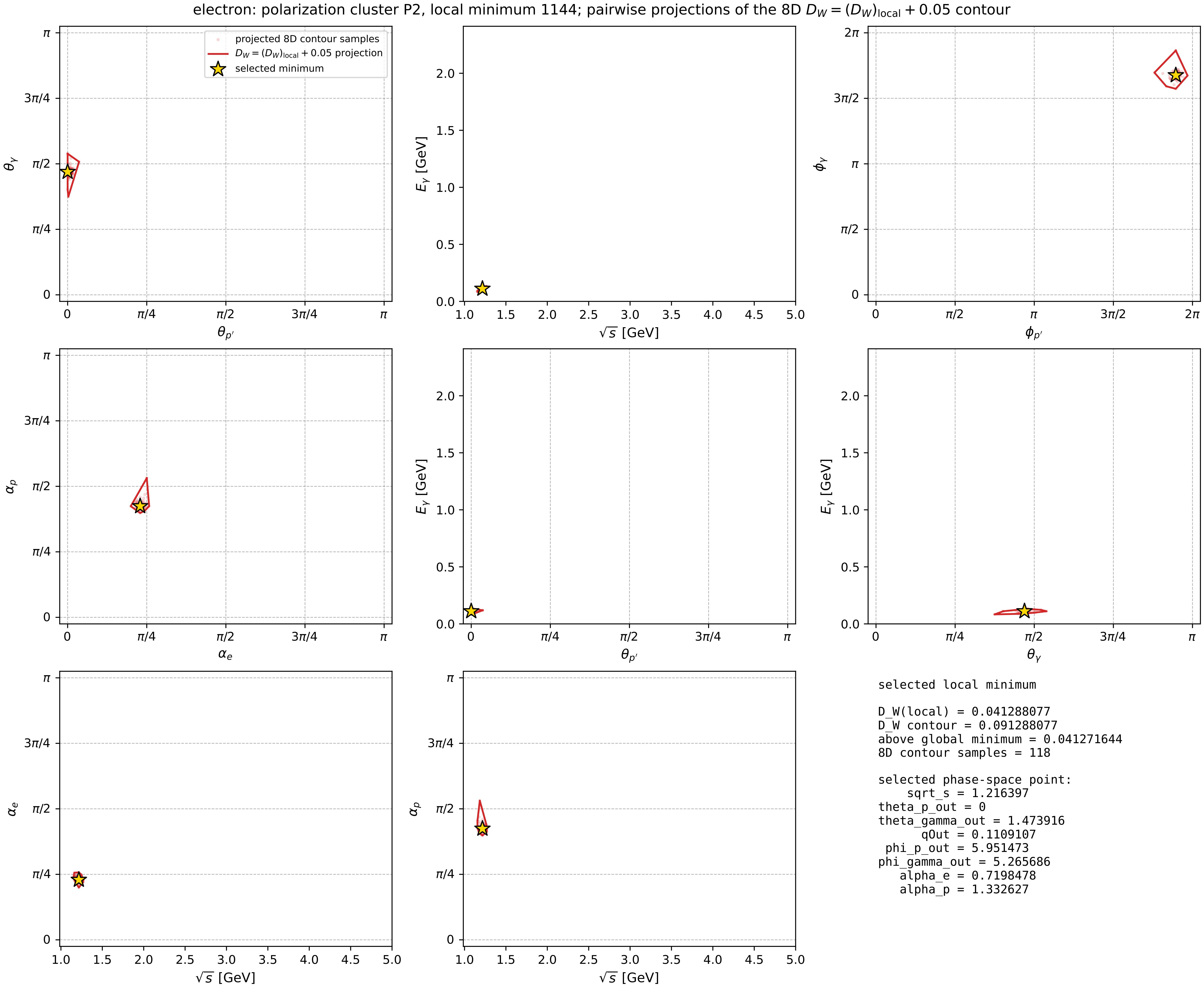
dw_mixing_angles_polarization_02_local_minimum_1144 D_W=0.0412881
region: polarization_cluster_02_local_minimum_1144
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.71984781 rad
incoming lepton: 0.751906|+> +0.65927|->
proton mixing angle: alpha_p=1.3326271 rad
incoming proton: 0.235924|+> +0.971772|->

s=1.47962, $\sqrt{s}$=1.2164, $|\vec{P}|$=0.246538, $|\vec{P}'|$=0.107017
$\theta_{p'}$=0, θ_γ=1.47392, $\phi_{p'}$=5.95147, ϕ_γ=5.26569
E_γ=0.110911, $\theta_{\ell'}$=2.38842, $\phi_{\ell'}$=2.12409
Q^2=0.0215252, x_B=0.0941417, t=-0.0188019, W^2=1.08697, y=0.38122

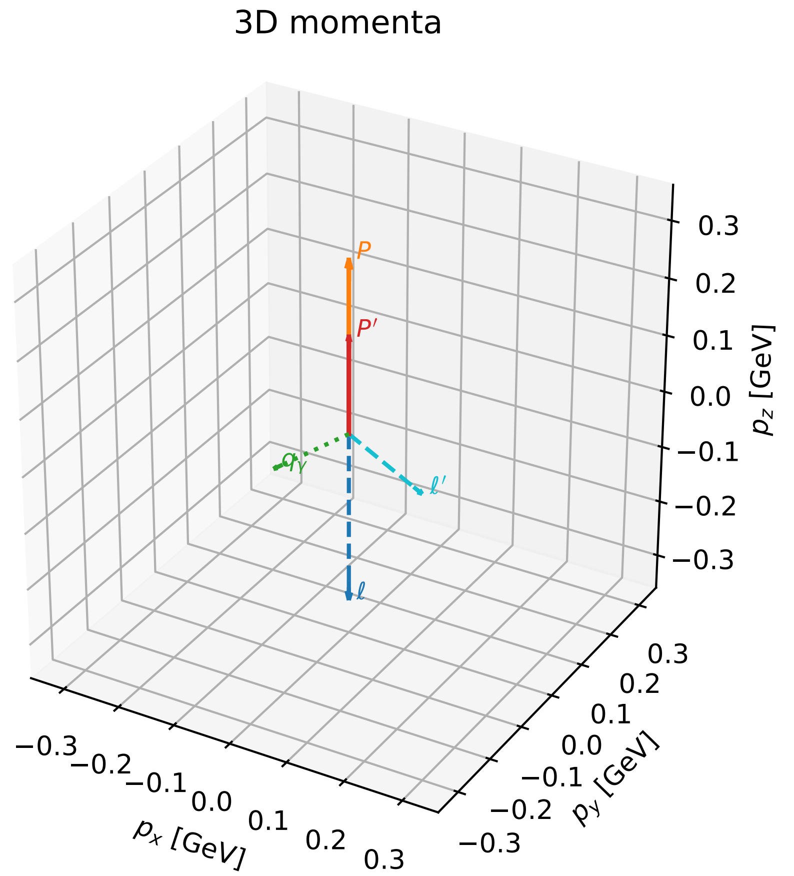
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2465 p_x= 0.0000 p_y= 0.0000 p_z= -0.2465
P E= 0.9699 p_x= 0.0000 p_y= 0.0000 p_z= 0.2465
ℓ' E= 0.1614 p_x= -0.0580 p_y= 0.0939 p_z= -0.1177
P' E= 0.9441 p_x= 0.0000 p_y= -0.0000 p_z= 0.1070
q_γ E= 0.1109 p_x= 0.0580 p_y= -0.0939 p_z= 0.0107






electron: polarization cluster P2, local minimum 1144; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.041288077
D_W contour = 0.091288077
above global minimum = 0.041271644
8D contour samples = 118
selected phase-space point:
sqrt_s = 1.216397
theta_p_out = 0
theta_gamma_out = 1.473916
qOut = 0.1109107
phi_p_out = 5.951473
phi_gamma_out = 5.265686
alpha_e = 0.7198478
alpha_p = 1.332627

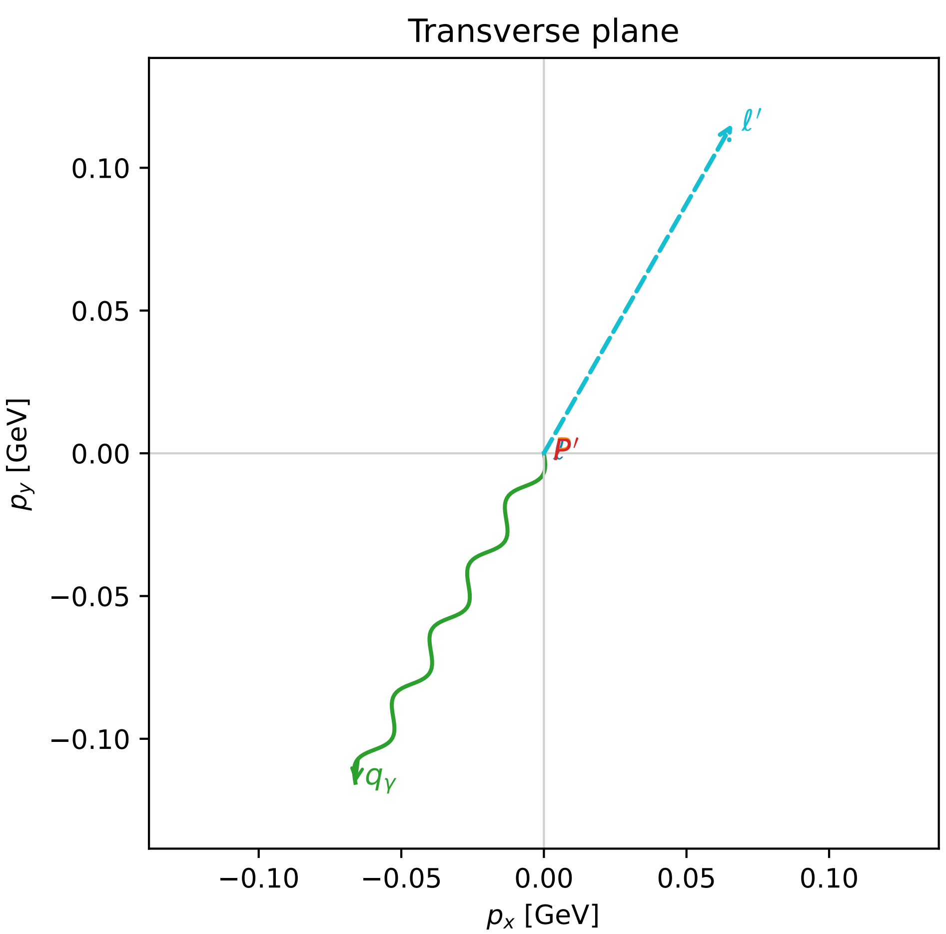
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

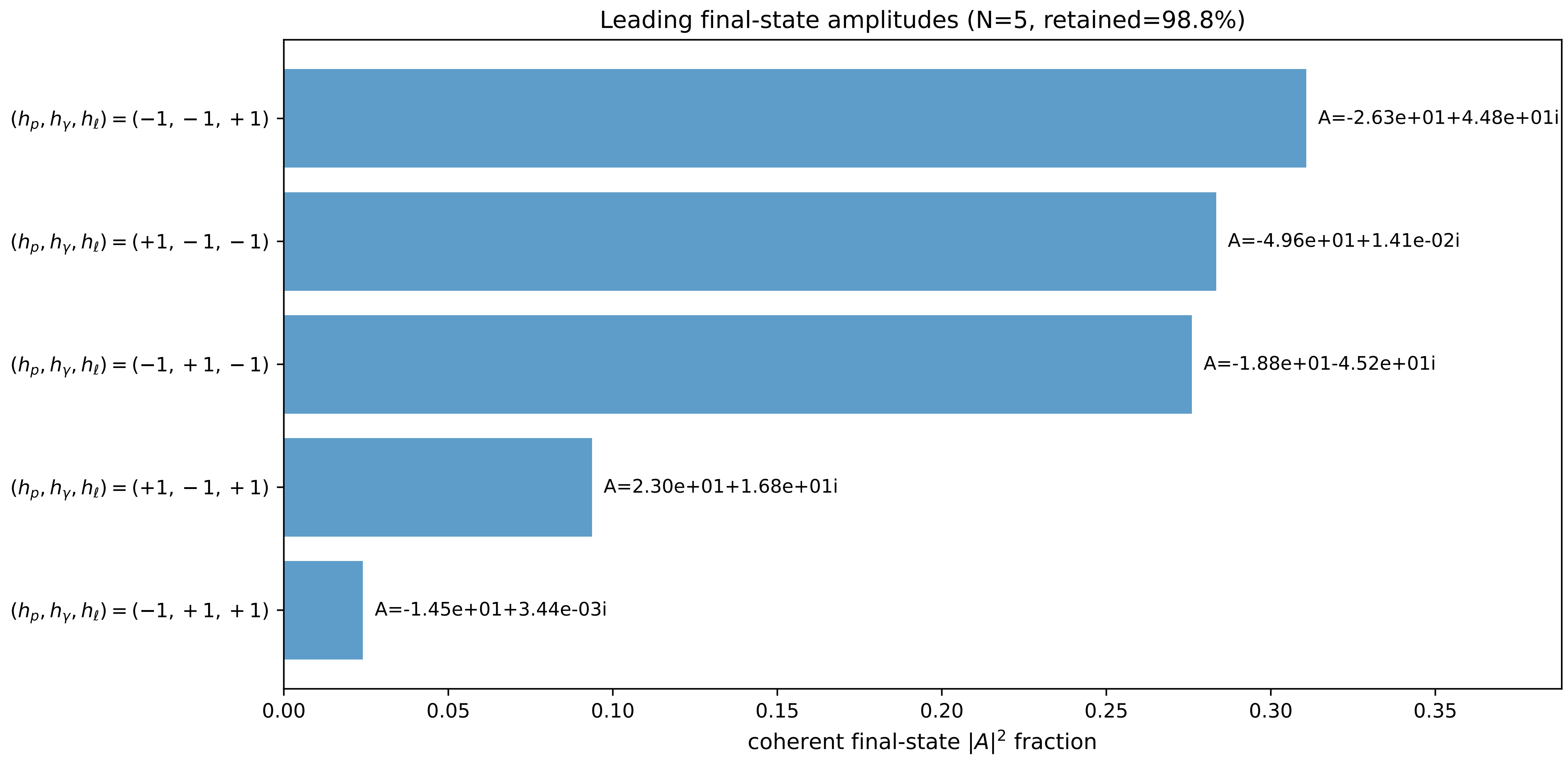


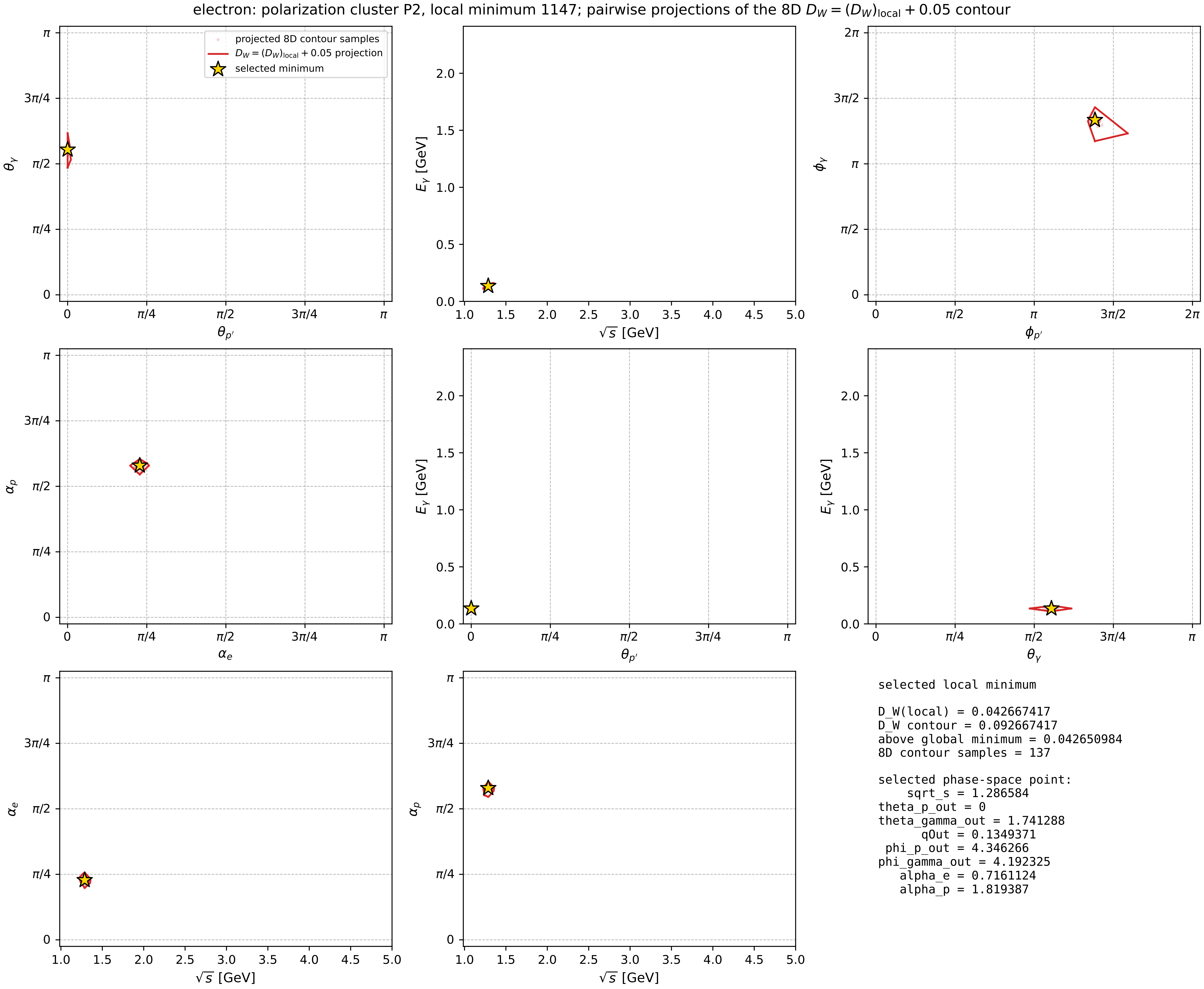

```
dw_mixing_angles_polarization_02_local_minimum_1147  D_W=0.0426674
region: polarization_cluster_02_local_minimum_1147
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=0.71611236 rad
incoming lepton: 0.754363|+> +0.656457|->
proton mixing angle: alpha_p=1.8193873 rad
incoming proton: -0.246039|+> +0.96926|->
```

s=1.6553, $\sqrt{s}$=1.28658, $|\vec{P}|$=0.301362, $|\vec{P}'|$=0.170069
$\theta_{p'}$=0, θ_γ=1.74129, $\phi_{p'}$=4.34627, ϕ_γ=4.19232
E_γ=0.134937, $\theta_{\ell'}$=2.40681, $\phi_{\ell'}$=1.05073
Q^2=0.030847, x_B=0.104247, t=-0.0162183, W^2=1.1449, y=0.381587

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3014 p_x= 0.0000 p_y= 0.0000 p_z= -0.3014
P E= 0.9852 p_x= 0.0000 p_y= 0.0000 p_z= 0.3014
ℓ' E= 0.1984 p_x= 0.0661 p_y= 0.1154 p_z= -0.1472
P' E= 0.9533 p_x= -0.0000 p_y= -0.0000 p_z= 0.1701
q_γ E= 0.1349 p_x= -0.0661 p_y= -0.1154 p_z= -0.0229






electron: polarization cluster P2, local minimum 1147; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
selected local minimum
D_W(local) = 0.042667417
D_W contour = 0.092667417
above global minimum = 0.042650984
8D contour samples = 137
selected phase-space point:
sqrt_s = 1.286584
theta_p_out = 0
theta_gamma_out = 1.741288
qOut = 0.1349371
phi_p_out = 4.346266
phi_gamma_out = 4.192325
alpha_e = 0.7161124
alpha_p = 1.819387